

Error analysis for MLB teams, 2023

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Version
2024-11-21

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Chapter 1

Preface

This book delves into the fascinating world of errors in MLB games and how they affect runs. We meticulously analyze the error-to-run conversion rates of all 30 teams. Our focus is not solely on the frequency of errors, but rather on the strategic advantage they offer, if the offense can capitalize on them! Errors, often perceived as mere mishaps, serve as catalysts for scoring runs, making them an interesting topic for analysis. In addition to the relationship to run-scoring, we ask other naturally arising questions such as, what is the frequency of the game configurations in those Retrosheet play records in which an error was committed?

This book uses a number of Python programs I've written to extract from Retrosheet records batter, runner and fielder information from it, which then gets re-encoded into game state notation. The Retrosheet event data provides both season-wide and game-by-game error totals.

- Season totals: Did the home team make more errors than their opponents? Which team did better at converting errors to runs? Which fielding positions had the most common error (e.g., third base or short-stop)?
- Game-by-game totals: For each play where an error occurred, what was the game state (before and after), which batter (and fielder positions) were involved, and a summary of each play.

Hopefully, this book will serve as a resource for fans, offering insights into the tactical nuances of the game.

Addressing such error-related questions is the goal of this work.

Each error is summarized in a short paragraph. There are approximately 2500 errors total in the MLB 2023 regular season. That’s about 2500 paragraphs. In one of the earliest written chapters, that of the Philadelphia Phillies, I used AI to help create some of these paragraphs. Eventually, I wrote a Python program that parsed the most common error cases and returned a template for that case. Only my program, and no AI, was used for the majority of the MLB teams.

When I *did* use Google’s **Advanced Gemini** to help generate some of the summaries, the prompt was often something like¹

“summarize the play, using the attached to determine the runners (if any) on base, and write the response in the following format:
 In the (top/bottom) of the Xth inning, with
 (bases empty/runners on bases ...), (team) batter (name) hit ...”

then a copy of the play information, such as

“ Play with error: `play,5,0,mauej001,32,CB*BBC>B,W+0A.1-3(E2/TH)`
 during the game: `CIN vs PHI on 20230408`
 state before error: `(1, 1, 0, 0, 3, 1, 0, 1) ”`

along with the event file document [Re2]. While **Gemini** was correct over half the time (and seemed to get better with practice), human editing (by me) was required each time. I hope, with improving machine-learning technology, the prompts can be improved and the next report like this (if there will be one) can be generated more easily using AI.

¹This prompt doesn’t make sense for play such as an error in a pick off (PO) attempt, but gives the basic idea.

Chapter 2

Game states via Retrosheet

This chapter dives into Retrosheet records and explains the notation as an introduction to the official documentation [Re2]. We also explain how to extract batter, runner and fielder information from it, which can be re-encoded into game state notation. (All these terms will be defined below.) For example, we'll see in §1.3 below that a game state is simply an 8-tuple associated in a well-defined way to a play in a baseball game.

The goal here is to use our

$$\text{play record} \longmapsto \text{game state}$$

function¹ to create a data source to investigate the frequency with which certain baseball configurations occur. For example, what is the most common error state? Addressing such questions is planned for future works.

In section 1, we review some Markov chain models for baseball, including a discussion of game states. In section 2, we give a quick overview of the Retrosheet event notation, from [Re2]. From the department or recursive definitions, for us a *play* is an event described in a Retrosheet play record as part of a Retrosheet event file associated to an MLB game. As a detailed example, §3 includes an entire game from a Retrosheet event file – the home opener for the Orioles in the 2023 season – with an explanation of each play.

¹This function is actually an ever-evolving collection of Python programs written by the author.

2.1 Markov chains and baseball

We assume some basic knowledge of probability and statistics.

2.1.1 Definitions

For the following definition, we follow D’Angelo [DA10].

Let $B_1, B_2, \dots$, be a sequence of random variables (representing batters in a randomized baseball game) taking values in a finite set, S . An element of this finite set will be referred to as a *state* (or, *game state* in the case of randomized baseball). This sequence of random variables is called a *Markov chain* if the following two conditions hold:

- For all $\ell > 0$, the conditional probability $P(B_{\ell+1} = \mathbf{s}_j \mid B_\ell = \mathbf{s}_i)$ is independent of the values of B_k , $k < \ell$.
- For each pair of indices i, j , there is a fixed probability² $p_{i,j}$ such that, for all $\ell > 0$, each time the random variable B_ℓ is in state $\mathbf{s}_i$ the probability it will be in state $\mathbf{s}_j$ equals $p_{i,j}$. That is,

$$p_{i,j} = P(B_{\ell+1} = \mathbf{s}_j \mid B_\ell = \mathbf{s}_i).$$

Some papers in the subject, such as [DA10] and [So04], assume that each of the random variables B_k is taken from the sample space of batters whose stats are consistent with actual (possibly retired, such as retired Oriole Cal Ripken) MLB players. In this case, $p_{i,j}$ would be derived from sabermetric data on that player’s at-bats.

Question: Can baseball be “randomized” such that it can be modeled by Markov chains?

2.1.2 A generalized Markov chain model

This paper considers a generalized Markov chain where there are two sample spaces (one for each team), and the Markov chain $B_1, B_2, \dots$, know which space to draw from because of the value of one of the coordinates in the game state.

²That is, the associated *transition* matrix $(p_{i,j})_{0 \leq i,j \leq |S|-1}$ is doubly stochastic.

This can be described more precisely as follows. Given are two sample spaces, S_0 and S_1 , and two lists of random variables,

$$B_1^{(0)}, B_2^{(0)} \dots \quad \text{and} \quad B_1^{(1)}, B_2^{(1)}, \dots$$

These satisfy the following properties.

- (1) Each B_i is equal to some $B_k^{(j)}$ (as explained below).
- (2) Samples for $B_k^{(0)}$ are taken from S_0 (representing a random at-bat from the visiting team), when the state indicates the visiting team is at bat.
- (3) Samples for $B_k^{(1)}$ are taken from S_1 (representing a random at-bat from the home team), when the state indicates the home team is at bat.
- (4) In order to determine where a sample should be selected from, that is to determine B_i in step (1) above, we use a simple function $c(i)$ having values in $\{(0, 0), (1, 0), (2, 0), (0, 1), (1, 1), (2, 1)\}$ (representing outs in the first coordinate and team-at-bats in the second):
 - (a) when $c(i) = (\ell, 0)$ (meaning that there are $\ell < 3$ outs and the visiting team is at-bat) then $B_i = B_k^{(0)}$, where k is the smallest value for which $B_1^{(0)}, B_2^{(0)}, \dots, B_{k-1}^{(0)}$ already appeared but $B_k^{(0)}$ has not,
 - (b) when $c(i) = (\ell, 0)$ (meaning that there are $\ell \geq 3$ outs and the visiting team is at-bat) then reset $c(i) = (0, 1)$ and let $B_i = B_k^{(1)}$, where k is the smallest value for which $B_1^{(1)}, B_2^{(1)}, \dots, B_{k-1}^{(1)}$ already appeared but $B_k^{(1)}$ has not,
 - (c) the cases $c(i) = (\ell, 1)$ with $\ell < 3$ and $c(i) = (\ell, 1)$ with $\ell \geq 3$ are handled similar to (a) and (b).

In this scenerio, we have two transition probabilities and therefore two transition matrices, $P^{(0)} = (p_{i,j}^{(0)})$ and $P^{(1)} = (p_{i,j}^{(1)})$, representing the visiting team and the home team. Transition probabilities will be examined in a future work.

Brief history of Markov chains and baseball

A number of papers discuss Markov chains and baseball, going back at least to the 1960s. We briefly examine some of these next.

Russian mathematician Andrey Markov studied Markov chains in the early 1900s. Applications to games did not wait long: in 1912, French mathematician Henri Poincaré studied Markov chains on finite groups with an aim to study card shuffling.

A large number of papers have focused on applications of Markov chains to baseball. A few of the earliest papers are mentioned below.

Starting in the late 1950s, George Lindsey, a researcher for the Royal Canadian Air Force, wrote several papers on modeling baseball using ideas from probability theory and operations research. For example, his paper [Li63], while not obviously dealing with Markov chains, introduces the 24 game states and performs computations related to the so-called runs expectancy matrix (see his Table I, page 485). From his abstract:

Two sets of data taken from a large number of major league games are used to give (1) the dependence of the probability of winning the game on the score and inning, and (2) the distribution of runs scored between the arrival of a new batter at the plate in each of twenty-four situations and the end of the half-inning. By combining the two sets of data, the situations are determined in which an intentional base on balls, a double play allowing a run to score, a sacrifice, and an attempted steal are advisable strategies.

Written in the 1970s, the paper by Cover and Keilers [CK77] is now almost 50 years old. While not working directly with Markov chains, the term “Markovian recurrence” in their §3 tells us this paper does indeed use a Markov chain model. From their abstract:

This paper studies a baseball statistic that plays the role of an offensive earned-run average (OERA). This statistic has the obvious natural interpretation and tends to evaluate strictly personal rather than team achievement. Some theoretical properties of this statistic are developed, and we give our answer to the question, “Who is the greatest hitter in baseball history?”

Another of the earlier papers is that by Bukiet, Harold, and Palacios [BHP97]. The two above-mentioned papers take the perspective (as we do in this paper), that all (“randomized”) batters are equal. The next paper seems to take a different approach. From their abstract:

We introduce a Markov chain method that considers teams made up of players with different abilities and which is not restricted to a given model for runner advancement. We apply our method to find optimal batting orders, run distributions per half inning and per game, and the expected number of games a team should win.

In the 1980s there were a number of interesting articles on this topic by Mark Pankin. Of course, Charles Pavitt has compiled a book [Pa22] devoted to discussing research papers on baseball. While his work is much broader, mathematical topics along the lines discussed here can be found in Chapter 2 of his book. Further references, as well as a short but excellent introduction to Markov chains in sabermetrics, can be found in D’Angelo [DA10].

2.1.3 Games states and weighted graphs

We return to the Markov chain model introduced in the first section.

Game states

For this paper, a *game state* of a baseball play is an 8-tuple

$$\mathbf{s} = (s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7),$$

where

- s_0 denotes the number of outs³,
- s_1 is 0 if there is no runner on 1st base, and 1 otherwise,
- s_2 is 0 if there is no runner on 2nd base, and 1 otherwise,
- s_3 is 0 if there is no runner on 3rd base, and 1 otherwise,

³This value can be represented as an integer $(\bmod 3)$, i.e. in $\mathbb{Z}/3\mathbb{Z}$, if desired.

- s_4 is the inning number,
- s_5 is the number of runs by team 0 (the visiting team),
- s_6 is the number of runs by team 1 (the home team), and
- s_7 is the team number at bat (0 for the visiting team, 1 for the home team).

For example, take the state $s = (2, 1, 1, 1, 3, 4, 5, 0)$. The first coordinate being $s_0 = 2$ means there are 2 outs. Similarly reading the remaining coordinates: the bases are loaded in the 3rd inning, the visiting team has 4 runs while the home team has 5, and the visiting team is at bat. More visually:

$$\begin{array}{ccccccc}
 & & 1\text{B} & 2\text{B} & 3\text{B} & & \text{vt score} & \text{ht score} \\
 (& 2, & 1, & 1, & 1, & 3, & 4, & 5, & 0 &) \\
 & \text{outs} & & & & \text{inning} & & & \text{at-bat}
 \end{array}$$

When we want to specify the play as well, we use a more precise notation. Let P be a baseball play⁴ and let P' denote the next play in the game. Define $\mathbf{s}(P)$ to be the above-defined game state *after* the play P . We have the following fact, which we call the

Conservation of Runners Law: Assume P is a play where the batter is involved⁵. The following quantity is conserved in a state transition.

- (a) If the visiting team is at-bat both plays ($s_7(P) = s_7(P') = 0$) then

$$1 + s_0(P) + s_1(P) + s_2(P) + s_3(P) + s_5(P) = s_0(P') + s_1(P') + s_2(P') + s_3(P') + s_5(P').$$

- (b) If the home team is at-bat both plays ($s_7(P) = s_7(P') = 1$) then

$$1 + s_0(P) + s_1(P) + s_2(P) + s_3(P) + s_6(P) = s_0(P') + s_1(P') + s_2(P') + s_3(P') + s_6(P').$$

⁴Equivalently, this is a play record in the Retrosheet format. We assume P is not the last play of the game.

⁵For example, P could be a triple or a walk, but is not a stolen base or a pickoff.

Example 1 Believe it or not, there are plays which do not change the state. As an example, consider the Retrosheet play

`play,1,1,hendg002,31,BBCBF,FLE7`

which occurred during the game SEA vs BAL on 2023-06-23. The state before the play was $(0, 0, 0, 0, 1, 0, 0, 1)$. The state after the play was the same! No change in the runners, the outs, the score, nor the inning. In this play, this is what happened: In the bottom of the 1st inning, with bases empty and a 3-1 count, BAL batter Gunnar Henderson hit a foul fly ball that the left fielder failed to catch, resulting in an error. Henderson is still at bat. now with a 3-2 count.

Such a play that does not change the state will be called an ∞ (*infinity*) *play* since it can be played infinitely often without ending the game.

Of course, there exist plays where the same FLE error occurred as in the pplay above, but the count was full. In such a situation, even the “extended state” discussed below doesn’t change.

There are situations where one wants *extend* this state to a 10-parameter state obtained by adding these two more parameters:

- s_8 , the ball count against the batter,
- s_9 , the strike count⁶ against the batter.

There are infinitely many possible states, but 100 plays per game is typical.

Conditions for a state reset:

(1) If $s_4 < 9$ and $s_0 = 3$ then the inning is over, so the state gets

- *outs reset* to $s_0 = 0$,
- *innings incremented*⁷ $s_4 = s_4 + 1$,
- s_7 gets toggled⁸, and
- *balls-and-strikes reset*, $s_8 = s_9 = 0$.

⁶Adding this to the state allows one to model, in principle, foul ball situations when there are 2 strikes in the count. (In this case, foul balls are “free.”)

⁷Unless it becomes $s_4 = 10$ and $s_5 \neq s_6$, in which case the game is over.

⁸That is, if s_7 was equal to 0 then it is now equal to 1, whereas if s_7 was equal to 1 then it is now equal to 0.

(2) If $s_4 \geq 9$ and $s_0 = 3$ then the state gets

- *half-inning reset* to $s_0 = 0$, and
- s_7 is toggled.

Question: What is the probability, for a given MLB season, that a given state transitions to another given state for a “randomly selected” batter?

This question has been studied by many people. See for example, the theses [Th14] and [Ur14] for readable introductions.

To construct the state graph described in Figure 2.1, we will ignore all but the first 4 state variables. Therefore, for the remainder, regard the set of all possible states as a list of 4-tuples $\mathbf{s} = (s_0, s_1, s_2, s_3)$. These states can be listed in a 8×3 array:

(0, 0, 0, 0)	(1, 0, 0, 0)	(2, 0, 0, 0)
(0, 1, 0, 0)	(1, 1, 0, 0)	(2, 1, 0, 0)
(0, 0, 1, 0)	(1, 0, 1, 0)	(2, 0, 1, 0)
(0, 0, 0, 1)	(1, 0, 0, 1)	(2, 0, 0, 1)
(0, 1, 1, 0)	(1, 1, 1, 0)	(2, 1, 1, 0)
(0, 1, 0, 1)	(1, 1, 0, 1)	(2, 1, 0, 1)
(0, 0, 1, 1)	(1, 0, 1, 1)	(2, 0, 1, 1)
(0, 1, 1, 1)	(1, 1, 1, 1)	(2, 1, 1, 1)

Some of these transitions relate to well-known baseball statistics. These reductions are mentioned in several papers but here are some examples from Ursin [Ur14]. Ursin says that the probability of the transition

- $\mathbf{s} = (s_0, 0, 0, 0) \rightarrow \mathbf{s}' = (s_0, 0, 0, 1)$ is equal to the probability of a triple,
- $\mathbf{s} = (s_0, 0, 0, 0) \rightarrow \mathbf{s}' = (s_0, 0, 1, 0)$ is equal to the probability of a double, and
- $\mathbf{s} = (s_0, 0, 0, 0) \rightarrow \mathbf{s}' = (s_0, 1, 0, 0)$ is the probability of a single or walk.

None of these is 100% mathematically correct in general, but such approximations enable sabermetricians to get a good approximation to the transition matrix and the run expectancy matrix.

The state graph

Define a graph $\Gamma = (V, E)$ to have vertices V given by the 24 states described above and edges E being those pairs of states $(\mathbf{s}, \mathbf{s}')$ where there is some legal at-bat for which state $\mathbf{s}$ transitions to state $\mathbf{s}'$. This is the *baseball states graph*, a graph with 182 edges. The vertex associated to $(2, 0, 0, 0)$ has degree 22, which is maximal. In fact, the degree of each vertex in the array above is recorded in the corresponding position of the array below.

8	9	22
11	18	18
11	18	18
11	18	18
17	18	12
17	18	12
17	18	12
20	15	8

This is depicted in Figure 2.1.

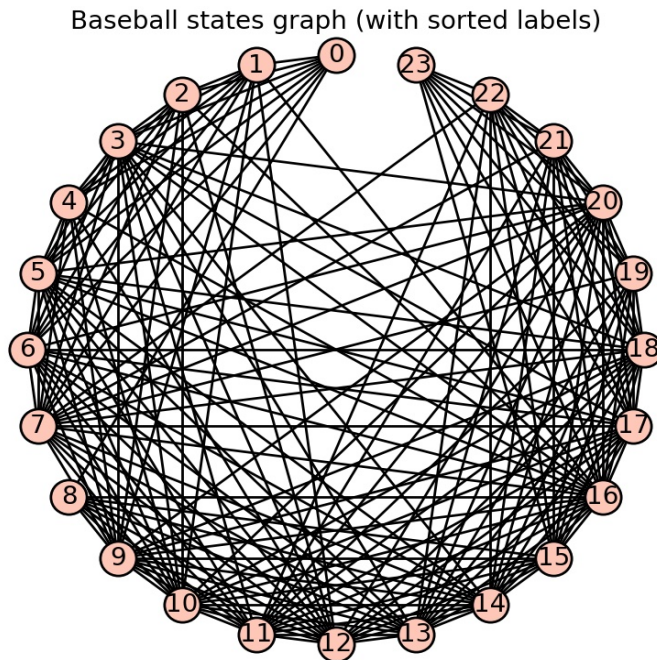


Figure 2.1: The labeled baseball states graph.

2.1.4 Frequency data for the game states

The average number of at-bats in an MLB game is in the 65-70 range, depending on the teams. Of course, each half-inning starts in the *initial state* with no outs and no one on base. We typically see 17 or 18 innings in a game. Therefore, the state $(0, 0, 0, 0)$ occurs, must occur at least that many times in a game.

From Lindsey’s 1963 article [Li63], who gathered and computed frequencies “by hand” from the years 1954-59 (I use the version of this table found in [AB01]):

Lindsey's table using data from 1954-1959

states\outs	0	1	2
(0,0,0)	24.3	17.3	13.7
(1,0,0)	6.4	7.6	7.8
(0,1,0)	1.1	2.4	2.9
(0,0,1)	0.2	0.7	1.2
(1,1,0)	1.4	2.6	3.3
(1,0,1)	0.4	1.1	1.6
(0,1,1)	0.3	0.7	0.8
(1,1,1)	0.3	0.8	1.0

To compare this with more modern data, we use an ever-evolving Python program written by the author to parse all the Retrosheet play records for a game, along with the corresponding states. The states that arise in the 81 Baltimore home games in the 2021 regular season⁹ appear, with their relative frequency, in the tables below.

From the 2021 Baltimore home games

states\outs	0	1	2
(0,0,0)	23.06	17.79	14.12
(1,0,0)	8.71	6.95	5.44
(0,1,0)	4.35	3.52	2.98
(0,0,1)	2.59	2.18	1.2
(1,1,0)	1.54	1.24	1.13
(1,0,1)	0.97	0.69	0.41
(0,1,1)	0.26	0.14	0.05
(1,1,1)	0.01	0.0	0.0

These tables are similar. However, the frequencies do vary from year to year. For example, the corresponding table from the Baltimore 2021 season had about 25% fewer empty base states and small increases in the 1 runner on base states. None-the-less, we make the following hypothesis.

Uniformity Hypothesis¹⁰: *The frequency that the empty state occurs in a*

⁹The data is collected from all the plays in the 2021 home games, which totals about 6350.

¹⁰Indeed, data from hundreds of games from various teams and seasons (in the last 10 years or so) suggests this is true.

game is over twice that of the lone runner on 1st.

Question 1 *Do errors occur uniformly? That is, if you look at enough games and enough errors, will the distribution of errors be uniform?*

Example 2 The most suprising fact, to me, that for teams such as the 2023 San Francisco Giants, the number of errors in the empty state is about the same as the number of errors with a lone runner on 1st. Based on the Uniformity Hypothesis, this is surprising because it *suggests* (but doesn't prove) that errors don't occur randomly with a uniform distribution.

In any case, note that the *most common states* are those three states with bases empty and the *least common states* are those three states with bases loaded.

2.2 Retrosheet play record notation

This section is based on the Retrosheet documentation [Re2]. Many thanks to the many Retrosheet contributors for the material included below.

For each game there are eight different record types discussed in this section¹¹. Each record type has a unique designator, which is followed by several fields separated by commas.

2.2.1 The id record type

Each game begins with a twelve character id record which identifies the date, location, and number of the game. See Example 4.

2.2.2 The version record type

The **version** statement corresponds to when the Retrosheet file was created. The current version is '2'.

2.2.3 The info record type

There are up to 30 types of **info** records, each of which contains a single piece of information, such as the temperature, the attendance, the identity of each umpire, etc. See Example 5.

¹¹Namely id, version, info, start, play, com, sub, and data.

2.2.4 The start record type

The start records identify the starting lineups for the game. Each start record has five fields.

1. The first field is the Retrosheet ID code, which is unique for each player. This 8 digit code is constructed from the first four letters of the player's last name, the first initial of his common name, and a three digit numeral.

In Example 6, `kremd001` is the Retrosheet ID for Dean Kremer.

2. The second field is the player's name.
3. The next field is either 0 (for visiting team), or 1 (for home team).
4. The next field is the position in the batting order.
5. The last field is the starting fielding position, a numeral from 1 to 10.

2.2.5 The play record type

The play record is a concise shorthand code using alpha-numeric strings to describe the events of a (modern era) baseball game. This section attempts to explain this code.

Each play record describes a specific type of baseball event, such as a base hit, an out, or a runner advance. Each such record has 6 fields (not including the name "play", which we call the 0-th field).

1. The first field is the inning.
2. The second field is either
0, for visiting team, or
1, for the home team.
3. The third field is the batter's Retrosheet ID code.
4. The fourth field is the count on the batter when this particular event occurred.

5. The fifth field is of variable length and contains all pitches to this batter in this plate appearance up to the point where the particular event occurred.

Additional notation:

- If the base number is preceded by a + sign, the pickoff throw was made by the catcher.
- The > symbol indicates a runner was going on the pitch that follows¹².
- The * symbol is before a pitch that would have been wild but was blocked by the catcher.

There is occasionally more than one event for each plate appearance, such as stolen bases, wild pitches, and balks in which the same batter remains at the plate. On these occasions the pitch sequence is interrupted by a period (such as in Example 9), and there is another play record for the resumption of the batter's plate appearance.

See Examples 8 and 9 below.

6. The sixth field describes the play which occurred. This field is variable in length and has three main portions which follow the Retrosheet scoring system. Roughly speaking, the three parts are described in Figure 2.2.

<part 1>/<part 2>.<part 3a>;<part 3b>

Figure 2.2: The three parts in the 6th field of a **play** record.

In Figure 2.2,

<part 1> is the basic description of the play,

<part 2> consists of modifiers to part 1 (such as hit location),

<part 3> describes the runner advances, different runners being separated by semicolons (;), starting with the 3rd base runner, then the 2nd

¹²Note, that a pitchout (P) often will be seen in the **play** record in conjunction with an attempted steal (<).

base runner, ending with the batter¹³.

See Example 10.

The scoring procedure description also uses a commonly known “location codes” diagram that explains clearly how each area of the playing field is designated (see Figure 2.3).

¹³More precisely, if the batter’s destination was already given in part 1, this last runner is omitted. For example, if part 1 read W (for walk) then we know that B-1 is true, so we don’t write that down again in part 3.

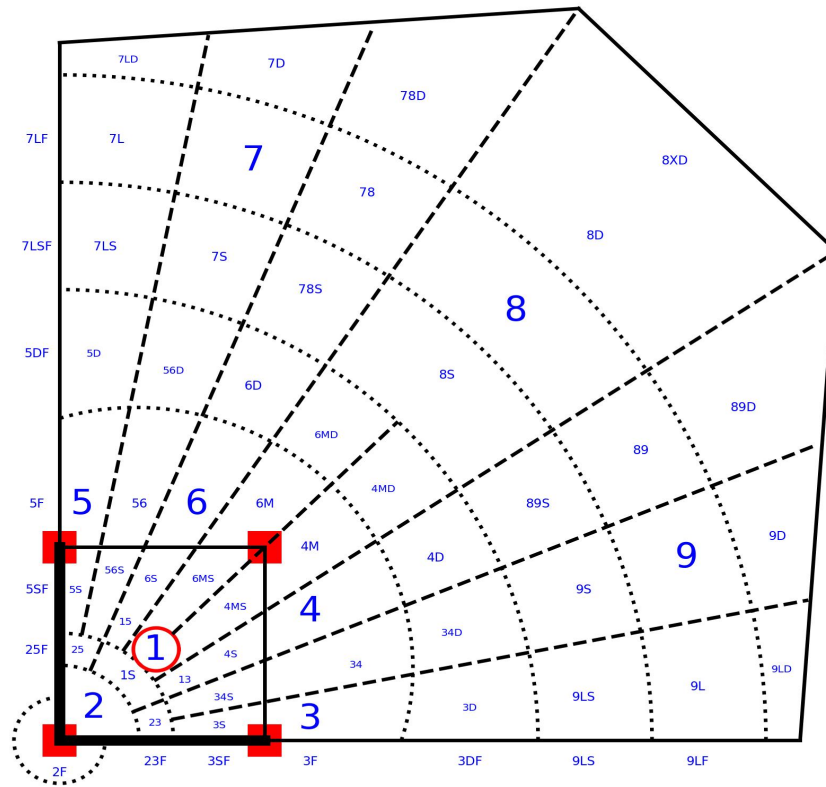
Dimensions: about 325 ft down

the 1st and 3rd base lines,

and 400 ft to deep center.

The usual 90 ft from home to 1st

(or 3rd) base, and 60.5 ft to the pitcher.



Location codes diagram

(used by retrosheet)

Figure 2.3: The **location codes diagram** (used by Retrosheet).

- a. The first portion is a description of the basic play, following standard baseball scoring notation.

Occasionally this portion ends with a + or - indicating a ball that was exceptionally hard hit or softly hit, respectively.

- b. The second portion is a modifier of the first part and is separated from it with a forward slash, "/". In fact, there may be more than

one second portion.

- c. The third portion describes the advancement of any runners, separated from the earlier parts (not involving a runner) by a period (“.”).

A successful¹⁴ advance is indicated by a dash, -. For example, 1-2 indicates the runner advanced from first to second.

An out made while advancing is indicated by an X. For example, 1X2 indicates the runner was out at second advancing from first.

2.2.6 The com record type

This record is used for those plays which require a special comment to describe the event.

For instance, this could be a replay, ejection, umpire change, protest, or a suspension (e.g., due to weather the game is suspended). Usually, there is a detailed comment at the spot in the game where the protest occurred.

2.2.7 The sub record type

Whenever any change in the lineup occurs, there is a substitute record. There are five fields in this type of record.

- a. The first field is the Retrosheet ID code.
- b. The second field is the player’s name.
- c. The third field is either 0 (visiting team) or 1 (home team).
- d. The fourth field is batting order position of the substitute.
- e. The fifth field is the position the substitute is going to play. Code used:
 - 11 is for pinch hitters,
 - 12 is for pinch runners.

¹⁴Even 1-1 is occasionally used when the runner tries to advance to second but can’t and makes it back to 1st base without being put out. See, for example, the game COL-BAL-2023-08-26-0 and the play `play,3,0,tovae001,12,*BSFFX,43/G34.3-H;1-1`.

Note that whenever a substitution occurs, it is preceded by a play record with the play event described as NP, meaning No Play. The purpose of this record is to “mark the place” of the substitution for other programs.

Example 3 *The ninth inning substitution of Felix Bautista in the game from §2.3 is indicated*

```
play,9,0,volpa001,00,.,NP
sub,bautf001,"Felix Bautista",1,0,1
play,9,0,volpa001,02,..SCS,K
```

2.2.8 The data record type

The several data records appear after all plays from the game and contain the number of earned runs allowed by each pitcher. Each record contains the pitcher’s Project Scoresheet code and the number of earned runs he allowed.

For example,

```
data,er,kremd001,4
```

means that Dean Kremer picked up 4 earned runs this game.

2.2.9 Examples

The examples below were taken from the Retrosheet event record of the first home game played by the Baltimore Orioles in the 2023 season. These examples help explain how to interpret the above-defined record types using event data from the game in §2.3.

Example 4 *The first line of the game record provided in §2.3, namely*
id, BAL198304040

should be read as follows. After the identifier, id, the first three characters identify the home team (the Baltimore Orioles). The next two are the last two digits of the year (1983). The next two are the month (April), the next two are the day (04). The last digit is the number of the game: here it is a 0 for a single game; it would be 1 for the first game of a double header, 2 for the second game.

Example 5 *An example of the info record.*

The following entries give us the temperature (55° F), the wind speed (6 mph), the wind direction (torf means towards right field), whether or not it is raining (no precipitation), and the sky (overcast).

```
info,temp,55
info,winddir,torf
info,windspeed,6
info,precip,none
info,sky,overcast
```

Example 6 For instance, `start,kremd001,"Dean Kremer",1,0,1` tells us the starting pitcher was Dean Kremer. The last three entries 1,0,1 mean

- Kremer is playing for the home team (1),
- Kremer does not bat (0), and
- Kremer is the starting pitcher (1).

Example 7 Austin Hays' first at-bat is a complicated record:

```
play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1
```

This will be decoded further in Examples 10, 13 below.

Example 8 The play record

```
play,4,0,trevj001,32,*BBCFB>*B,W.1-2
```

indicates that Jose Trevino took a 6-pitch walk (Namely, a ball blocked by the catcher (*B), a called strike, a foul ball, another ball (BBCFB). A runner went on the next pitch (>), another ball blocked by the catcher (*B). Trevino took his walk (W.1-2) and the runner on 1st advanced to 2nd.

Now for a more complicated example where a single at-bat required two lines.

Example 9 The play records

```
play,6,1,mullc002,02,.1C+1FF>P,SB2
```

```
play,6,1,mullc002,22,.1C+1FF>P.BX,63/G6.2-3
```

represent Curtis Mullins' at-bat in the 6th inning. He gets two lines because after his second pitch (the count was 0 and 2), the runner on 1st stole 2nd base. The code `.1C+1FF>P` means that before the 1st pitch, the pitcher threw to 1st base (1) in an attempted pick-off. Then there was a called strike. Then the catcher threw to 1st base in an attempted pick-off¹⁵. The next two pitches were fouled off (the first of them make the 2nd strike but the second one did

¹⁵A 90 ft throw by the catcher, compared to about 64 ft for the pitcher to 1st.

not count in the stike total). Next, the runner on 1st ran for 2nd base (>), then the pitcher threw a pitchout to the catcher (P). The runner made it to 2nd, so was awarded a stolen base (SB2). At this point, the pitch count is interrupted and a new **play** record is created for the remainder of the at-bat.

After the stolen base, the count is now 1-2 (since the pitchout was a ball). Another ball was thrown, after which Mullins got a hit (.BX), a ground ball to the shortstop (G6), who threw him out at 1st (63). However, the runner on 2nd base advanced safely to 3rd base (.2-3).

Example 10 *If the play record is as in Example 7, namely,*
`play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1`
then (in terms of the parts described in Figure 2.2)

- part 1 is 'FC4',
- part 2 is ['G34'] (there is only one modifier in this case), and
- part 3 is ['3-H', '1-2(E4)', 'B-1'] (there are three in this case).

The last part encodes the motion of the runners on base.

In the following example, a passed ball (PB) stops the **play** record for a batter and starts a new **play** record with that batter.

Example 11 *The first line of the play records*

`play,4,1,mullc002,10,1BB,PB.1-2`
`play,4,1,mullc002,32,1BB.FBFFX,8/F8`

represents Curtis Mullins' at-bat in the 4th inning. After an attempted pick-off (a throw to first), the next pitch Mullins gets is a ball (1B). The next pitch is a passed ball (B,PB) and the runner on 1st advances to 2nd (.1-2).

Now for the 2nd line. After the runner makes it to 2nd base, the play continues. Mullins fouls off the next pitch (his strike 1) then gets a ball and fouls off the next two pitches (now the count is full, 3-2). Mullins hits a fly ball to center field and is out.

Example 12 *The record*

`play,2,1,mounr001,01,FX,8/F8XD+`

is Ryan Mountcastle's first at-bat. Examining the previous event record, we see that Mountcastle was also the first Baltimore player to bat in the 2nd inning.) The last parts of the record FX,8/F8XD+ describes Ryan Mountcastle's

first at-bat. The first pitch was fowled off by the batter (the first **F**), while the next pitch was hit into play (**X**). Mountcastle sent a fly ball (the second **F**) caught (for the out) by the center fielder (**8**), the ball hit to extra-deep center field (**8XD+**). The **+** sign at the end indicates extra defensive effort.

Example 13 Continuing Example 7, Austin Hays' first at-bat is a more complicated record:

`play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1`

Last things first: **B-1** means the batter was awarded 1st base after an error by the second baseman (**E4**). First, **FC4** means fielder's choice by the 2nd baseman, so **FC4/G34** means the 1st baseman fielded a ground ball and threw the ball to the 2nd baseman. The runner on 3rd base ran home, scoring a run for Baltimore (**3-H**). At the same time, the runner on 1st ran to 2nd, during which the 2nd baseman committed an error (**1-2(E4)**).

Remark 1 Note that many code letters are ambiguous without further context. That is, depending on where they appear in the record, they could mean entirely different things. Take the code **S**. If **S** appears in the list of pitches then it denotes a swinging strike. If **S** appears in the list of play descriptions then it denotes a hit scored as a single (so the batter advances to first past). If **S** appears in the list of fielding descriptions then it denotes a location (such as **8S** is short center field, see Figure 2.3).

In fact, **S** can even occur as part of another code, such as **CS**.

Example 14 Anthony Rizzo's 3rd inning at-bat is complicated as well:

`play,3,0,rizza001,32,BBCC1B.>S,K+CS2(26)/DP`

Rizzo was pitched 2 balls and 2 called strikes (**BBCC**), after which Kremer threw to the 1st baseman (**1**). The next pitch was a ball (**B**). Now it's a 3-2 count. During the pitch, the runner on 1st tries to steal 2nd (**.>**). The pitch is a swinging strike (**S**), so Rizzo strikes out (**K**). The catcher throws to the shortstop, resulting in the runner being caught stealing, the out credited to the catcher (**CS2(26)**). These two outs are coded as a double play (**DP**).

2.3 Example of a single game event file

In this section, we include the first game from `2023BAL.EVA`, verbatim. (On April 7th, 2023, they faced the New York Yankees. The Yankees lost 6 – 7.) This helps illustrate the breakdown of a game into a sequence of game states.

2.3.1 A game record from an event file

This first home game is described in the following 179 lines.

```
id,BAL202304070
version,2
info,visteam,NYA
info,hometeam,BAL
info,site,BAL12
info,date,2023/04/07
info,number,0
info,gametype,regular
info,starttime,3:05PM
info,daynight,day
info,innings,9
info,tiebreaker,2
info,usedh,true
info,umphome,whitc901
info,ump1b,ortir901
info,ump2b,millb901
info,ump3b,drakr901
info,umplf,(none)
info,umprf,(none)
info,inputtime,2023/07/28 12:23:55
info,howscored,unknown
info,pitches,pitches
info,oscorer,jacom701
info,temp,55
info,winddir,torf
info,windspeed,6
info,fieldcond,unknown
info,precip,none
info,sky,overcast
info,timeofgame,184
info,attendance,45017
info,wp,perec004
info,lp,marir001
info,save,bautf001
start,lemad001,"DJ LeMahieu",0,1,5
```



```

start,judga001,"Aaron Judge",0,2,8
start,rizza001,"Anthony Rizzo",0,3,3
start,stanm004,"Giancarlo Stanton",0,4,10
start,torrg001,"Gleyber Torres",0,5,4
start,cabro002,"Oswaldo Cabrera",0,6,7
start,trevj001,"Jose Trevino",0,7,2
start,cordf003,"Franchy Cordero",0,8,9
start,volpa001,"Anthony Volpe",0,9,6
start,schmc002,"Clarke Schmidt",0,0,1
start,mullc002,"Cedric Mullins",1,1,8
start,rutsa001,"Adley Rutschman",1,2,2
start,santa003,"Anthony Santander",1,3,9
start,mounr001,"Ryan Mountcastle",1,4,3
start,hendg002,"Gunnar Henderson",1,5,10
start,uriar001,"Ramon Urias",1,6,5
start,fraza001,"Adam Frazier",1,7,4
start,haysa001,"Austin Hays",1,8,7
start,matej003,"Jorge Mateo",1,9,6
start,kremd001,"Dean Kremer",1,0,1
play,1,0,lemad001,32,BBCBF.FB,W
play,1,0,judga001,22,FBBFS,K
play,1,0,rizza001,00,X,S9/L9+.1-2
play,1,0,stanm004,02,CFX,7/F78
play,1,0,torrg001,11,BFX,31/G34
play,1,1,mullc002,12,FFFBFX,8/F8D
play,1,1,rutsa001,22,CBFFF.BFS,K
play,1,1,santa003,10,BX,9/L9D
play,2,0,cabro002,12,SBCX,6/P6MD
play,2,0,trevj001,01,CX,53/G5
play,2,0,cordf003,31,BBBFB,W
play,2,0,volpa001,01,CX,4/L4D
play,2,1,mounr001,01,FX,8/F8XD+
play,2,1,hendg002,10,BX,S9/L9+
play,2,1,uriar001,31,BBBCB,W.1-2
play,2,1,fraza001,01,CX,S9/G34.2-H;1-3
play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1
play,2,1,matej003,32,SBC.*BFBS,K
play,2,1,mullc002,01,SX,56(1)/F0/G56.B-1

```

```

play,3,0,lemad001,21,BCBX,S4/G4
play,3,0,judga001,22,B*BCSX,9/F9S
play,3,0,rizza001,32,BBCC1B.>S,K+CS2(26)/DP
play,3,1,rutsa001,32,BFFF.BBB,W
play,3,1,santa003,01,SX,D9/L9LS.1-3
play,3,1,mounr001,01,SX,8/SF/F8XD.3-H
play,3,1,hendg002,31,BBCBX,D8/L78XD+.2-H
play,3,1,uriar001,12,CCB.C,K
play,3,1,fraza001,32,BFBSFBFFB,W
play,3,1,haysa001,22,BFBCX,2/P2F/FL
play,4,0,stanm004,32,BBCFBX,53/G5
play,4,0,torrg001,32,BCFBFBFX,8/L89
play,4,0,cabro002,00,X,S68/G4
play,4,0,trevj001,32,*BBCFB>*B,W.1-2
play,4,0,cordf003,11,CBX,HR/F89XD.2-H;1-H
play,4,0,volpa001,02,CFS,K
play,4,1,matej003,22,CCBBFX,S8/G4
play,4,1,mullc002,10,1BB,PB.1-2
play,4,1,mullc002,32,1BB.FBFFX,8/F8
play,4,1,rutsa001,00,,NP
sub,hamii001,"Ian Hamilton",0,0,1
play,4,1,rutsa001,11,.VCX,31/G3.2-3
play,4,1,santa003,12,BCCS,K
play,5,0,lemad001,11,CBX,63/G6
play,5,0,judga001,22,CBSBS,K
play,5,0,rizza001,11,FBX,43/G34D
play,5,1,mounr001,12,CBC.S,K
play,5,1,hendg002,02,SCS,K
play,5,1,uriar001,10,BX,D9/L9LS+
play,5,1,fraza001,22,BCCBFX,8/L89
play,6,0,stanm004,10,BX,S7/L7S
play,6,0,torrg001,00,,NP
sub,gilll001,"Logan Gillaspie",1,0,1
play,6,0,torrg001,22,.FBBFFB,WP.1-2
play,6,0,torrg001,32,.FBBFFB.B,W
play,6,0,cabro002,32,*BBF*BFX,D8/L89XD+.2-H;1-H
play,6,0,trevj001,02,SFX,13/G1S-
play,6,0,cordf003,00,,NP

```

```

sub,could001,"Danny Coulombe",1,0,1
play,6,0,cordf003,00,,NP
sub,kinei001,"Isiah Kiner-Falefa",0,8,11
play,6,0,kinei001,10,..BN,BK.2-3
play,6,0,kinei001,11,..BN.FX,3/P3
play,6,0,volpa001,12,SS*BS,K
play,6,1,haysa001,00,,NP
sub,judga001,"Aaron Judge",0,2,9
play,6,1,haysa001,00,,NP
sub,kinei001,"Isiah Kiner-Falefa",0,8,8
play,6,1,haysa001,02,..CCC,K
play,6,1,matej003,10,BX,S7/L7L
play,6,1,mullc002,00,,NP
sub,marir001,"Ron Marinaccio",0,0,1
play,6,1,mullc002,02,.1C+1FF>P,SB2
play,6,1,mullc002,22,.1C+1FF>P.BX,63/G6.2-3
play,6,1,rutsa001,32,BCBBS.FFX,S7/L78D+.3-H
play,6,1,santa003,12,CSBS,K
play,7,0,lemad001,00,,NP
sub,perec004,"Cionel Perez",1,0,1
play,7,0,lemad001,02,.CFC,K
play,7,0,judga001,00,X,43/G4
play,7,0,rizza001,00,X,9/F9LD
play,7,1,mounr001,31,BBCBB,W
play,7,1,hendg002,32,BSBBC.>C,K+SB2
play,7,1,uriar001,00,,NP
sub,cordj001,"Jimmy Cordero",0,0,1
play,7,1,uriar001,00,.B,WP.2-3
play,7,1,uriar001,10,.B.X,D8/L78XD+.3-H
play,7,1,fraza001,01,CX,63/G6.2-3
play,7,1,haysa001,01,CB,WP.3-H(NR)
play,7,1,haysa001,32,CB.BSBB,W
play,7,1,matej003,12,CFB1X,43/G4
play,8,0,stanm004,12,CFBFX,D7/L7D+
play,8,0,torrg001,10,BX,S8/L8D+.2-3
play,8,0,cabro002,00,X,S9/G34.3-H;1-2
play,8,0,trevj001,00,,NP
sub,bakeb001,"Bryan Baker",1,0,1

```

```

play,8,0,trevj001,20,.BB+2X,5(2)3/GDP/G56.1-2
play,8,0,kinei001,00,>C,SB3
play,8,0,kinei001,12,>C.FBFS,K
play,8,1,mullc002,00,,NP
sub,peraw002,"Wandy Peralta",0,0,1
play,8,1,mullc002,00,.X,8/F8
play,8,1,rutsa001,32,BCBFBB,W
play,8,1,santa003,10,BX,8/L8XD+
play,8,1,mounr001,00,1X,S7/L7LS.1-2
play,8,1,hendg002,22,SBS*BS,K
play,9,0,volpa001,00,,NP
sub,mcker001,"Ryan McKenna",1,3,9
play,9,0,volpa001,00,,NP
sub,bautf001,"Felix Bautista",1,0,1
play,9,0,volpa001,02,..SCS,K
play,9,0,lemad001,22,FCBBC,K
play,9,0,judga001,30,BBB*B,W
play,9,0,rizza001,02,C*S>*B,SB2
play,9,0,rizza001,12,C*S>*B.B,WP.2-3
play,9,0,rizza001,32,C*S>*B.B.BFX,7/F56D
data,er,hamii001,1
data,er,peraw002,0
data,er,cordj001,1
data,er,schmc002,4
data,er,marir001,1
data,er,kremd001,4
data,er,perec004,1
data,er,gilll001,1
data,er,bautf001,0
data,er,bakeb001,0
data,er,could001,0

```

2.3.2 State-by-state breakdown of the game

Now we go over each play in the game and explain how the states transition into one another.

1. The starting state is $\mathbf{s}_0 = (0, 0, 0, 0, 1, 0, 0, 0)$. The play
`play,1,0,lemad001,32,BBCBF.FB,W`
indicates that DJ LeMahieu got a walk after 7 pitches (two balls,
a called strike, a ball, a foul ball, a play not involving the batter
(.), a foul ball, and ball 4). This play transitions state $\mathbf{s}_0$ to $\mathbf{s}_1 =$
 $(0, 1, 0, 0, 1, 0, 0, 0)$.
2. The play
`play,1,0,judga001,22,FBBFS,K`
where Aaron Judge strikes out on 5 pitches, transitions that state to
 $\mathbf{s}_2 = (1, 1, 0, 0, 1, 0, 0, 0)$.
3. The play
`play,1,0,rizza001,00,X,S9/L9+.1-2`
tells us that Anthony Rizzo singles to right field, and DJ LeMahieu ad-
vances to 2nd base. Therefore, state $\mathbf{s}_2$ transitions to $\mathbf{s}_3 = (1, 1, 1, 0, 1, 0, 0, 0)$.
4. The play
`play,1,0,stanm004,02,CFX,7/F78,`
where Giancarlo Stanton hits a fly ball caught by the left fielder, tran-
sitions that state to $\mathbf{s}_4 = (2, 1, 1, 0, 1, 0, 0, 0)$.
5. The play
`play,1,0,torrg001,11,BFX,31/G34`
tells us that Gleyber Torres hit a grounder fielded by the 1st base-
man, but the play drew him off the base, so 1st base was covered
by the pitcher, who put out the batter there. That out transitions
 $\mathbf{s}_4$ to $\mathbf{s}_5 = (3, 1, 1, 0, 1, 0, 0, 0)$, which is reset for the next half inning,
 $\mathbf{s}_6 = (0, 0, 0, 0, 1, 0, 0, 1)$.
6. The 6-pitch play
`play,1,1,mullc002,12,FFFBFX,8/F8D`
where Curtis Mullins hits a pop fly to (ironically) center field, transi-
tions that state to $\mathbf{s}_7 = (1, 0, 0, 0, 1, 0, 0, 1)$.
7. The play
`play,1,1,rutsa001,22,CBFFF.BFS,K,`
an 8-pitch Adley Rutschman strikeout, transitions that state to $\mathbf{s}_8 =$
 $(2, 0, 0, 0, 1, 0, 0, 1)$.

8. The play
`play,1,1,santa003,10,BX,9/L9D`
 ironically has Anthony Santander fly out to right field, so s_8 transitions to $s_9 = (3, 0, 0, 0, 1, 0, 0, 1)$, which is reset to the next inning, $s_{10} = (0, 0, 0, 0, 2, 0, 0, 0)$.
9. The Oswaldo Cabrera pop fly to the shortstop at location 6MD (see Figure 2.3) indicated in the play
`play,2,0,cabro002,12,SBCX,6/P6MD`
 transitions s_{10} to $s_{11} = (1, 0, 0, 0, 2, 0, 0, 0)$.
10. The play
`play,2,0,trevj001,01,CX,53/G5`
 where Jose Trevino hits a grounder to 3rd relayed to 1st for the putout, transitions that state to $s_{12} = (2, 0, 0, 0, 2, 0, 0, 0)$.
11. Franchy Cordero's 5-pitch walk, described by
`play,2,0,cordf003,31,BBBFB,W`
 transitions that state to $s_{13} = (2, 1, 0, 0, 2, 0, 0, 0)$.
12. The play
`play,2,0,volpa001,01,CX,4/L4D`
 where Anthony Volpe hits a line drive caught by the 2nd baseman, transitions that state to $s_{14} = (3, 1, 0, 0, 2, 0, 0, 0)$. As this is out 3, this state must be reset to the next at-bat, $s_{15} = (0, 0, 0, 0, 2, 0, 0, 1)$.
13. The play
`play,2,1,mounr001,01,FX,8/F8XD+`
 where Ryan Mountcastle flies out to deep left, transitions that state to $s_{16} = (1, 0, 0, 0, 2, 0, 0, 1)$.
14. The single by Gunner Henderson,
`play,2,1,hendg002,10,BX,S9/L9+`
 transitions that state to $s_{17} = (1, 1, 0, 0, 2, 0, 0, 1)$.
15. The play
`play,2,1,uriar001,31,BBBCB,W.1-2`
 where Ramon Urias walks (and Henderson advances to 2nd), transitions that state to $s_{18} = (1, 1, 1, 0, 2, 0, 0, 1)$.

16. The 1 RBI play by Adam Frazier,
`play,2,1,fraza001,01,CX,S9/G34.2-H;1-3`
 is “only” a single to right field, but Henderson scores and Urias advances to 3rd. As a result, the old state s_{18} transitions to the new state $s_{19} = (1, 1, 0, 1, 2, 0, 1, 1)$.
17. The Austin Hays play¹⁶
`play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1`
 transitions that state to $s_{20} = (1, 1, 1, 0, 2, 0, 2, 1)$.
18. The 7-pitch strikeout of Jorge Mateo,
`play,2,1,matej003,32,SBC.*BFBS,K`
 transitions that state to $s_{21} = (2, 1, 1, 0, 2, 0, 2, 1)$.
19. The play below describes a groundball to the 2nd baseman, who relayed the ball to the shortstop resulting in a force out (F0) of the runner coming from 1st base. The batter (Curtis Mullins) makes it to 1st base safely. The play
`play,2,1,mullc002,01,SX,56(1)/F0/G56.B-1`
 transitions state s_{21} to $s_{22} = (3, 1, 1, 0, 2, 0, 2, 1)$. This ends the inning so the state is updated to $s_{23} = (0, 0, 0, 0, 3, 0, 2, 0)$.
20. The play
`play,3,0,lemad001,21,BCBX,S4/G4`
 a single by DJ LeMahieu, transitions that state to $s_{24} = (0, 1, 0, 0, 3, 0, 2, 0)$.
21. The play
`play,3,0,judga001,22,B*BCSX,9/F9S`
 a pop fly to right field by Aaron Judge, transitions that state to $s_{25} = (1, 1, 0, 0, 3, 0, 2, 0)$.
22. The play
`play,3,0,rizza001,32,BBCC1B.>S,K+CS2(26)/DP`
 describes a 6-pitch strikeout of Anthony Rizzo, with an attempted stolen base (Aaron Judge was thrown out by the catcher at 2nd, with the tag being credited to the shortstop). As a result, s_{25} transitions to $s_{26} = (0, 0, 0, 0, 3, 0, 2, 1)$.

¹⁶This example has been broken down in Examples 10, 13 above.

23. The play
`play,3,1,rutsa001,32,BFFF.BBB,W`
tells us that Adley Rutschman walks on 7 pitches. This transitions the previous state to $s_{27} = (0, 1, 0, 0, 3, 0, 2, 1)$.
24. The double by Anthony Santander in the play below advances Rutschman to 3rd base:
`play,3,1,santa003,01,SX,D9/L9LS.1-3`
As a result, state s_{27} transitions to $s_{28} = (0, 0, 1, 1, 3, 0, 2, 1)$.
25. The sacrifice fly (SF) in the Mountcastle play
`play,3,1,mounr001,01,SX,8/SF/F8XD.3-H`
brought home Rutschman to score, so transitions that state to $s_{29} = (1, 0, 1, 0, 3, 0, 3, 1)$.
26. The 5-pitch double by Henderson that brought home Santander,
`play,3,1,hendg002,31,BBCBX,D8/L78XD+.2-H`
transitions that state to $s_{30} = (1, 0, 1, 0, 3, 0, 4, 1)$
27. Urias struck out in the next play,
`play,3,1,uriar001,12,CCB.C,K`
so s_{30} transitions $s_{31} = (2, 0, 1, 0, 3, 0, 4, 1)$.
28. The 9-pitch walk by Frazier,
`play,3,1,fraza001,32,BFBSFBFFB,W`
transitions that state to $s_{32} = (2, 1, 1, 0, 3, 0, 4, 1)$.
29. Below, the code `2/P2F/FL` indicates that Hays hit a pop fly to foul territory caught by the catcher. That play
`play,3,1,haysa001,22,BFBCX,2/P2F/FL`
transitions s_{32} to $s_{33} = (3, 1, 1, 0, 3, 0, 4, 1)$. This 3rd out ends the inning so the state is updated to $s_{34} = (0, 0, 0, 0, 4, 0, 4, 0)$.
30. The 6-pitch groundout by Stanton,
`play,4,0,stanm004,32,BBCFBX,53/G5`
transitions that state to $s_{35} = (1, 0, 0, 0, 4, 0, 4, 0)$.
31. The 8-pitch line drive by Torres was caught by the center fielder in the play
`play,4,0,torrg001,32,BCFBFBFX,8/L89`
Therefore, s_{35} transitions to $s_{36} = (2, 0, 0, 0, 4, 0, 4, 0)$.

32. The play
`play,4,0,cabro002,00,X,S68/G4`
 tells us that Carbrera makes it to first one a single, so s_{36} transitions to $s_{37} = (2, 1, 0, 0, 4, 0, 4, 0)$.
33. The Trevino play¹⁷
`play,4,0,trevj001,32,*BBCFB>*B,W.1-2`
 transitions that state to $s_{38} = (2, 1, 1, 0, 4, 0, 4, 0)$.
34. The play
`play,4,0,cordf003,11,CBX,HR/F89XD.2-H;1-H`
 describes Corderero's 3-run homer. As a result, s_{38} transitions to $s_{39} = (2, 0, 0, 0, 4, 3, 4, 0)$.
35. Volpe strikes out in the play
`play,4,0,volpa001,02,CFS,K`
 causing s_{39} to transition to $s_{40} = (3, 0, 0, 0, 4, 3, 4, 0)$. This ends the half-inning so, as usual, the state is updated to $s_{41} = (0, 0, 0, 0, 4, 3, 4, 1)$.
36. The play
`play,4,1,matej003,22,CCBBFX,S8/G4`
 describes Mateo's single, transitioning that state to $s_{42} = (0, 1, 0, 0, 4, 3, 4, 1)$.
37. Mullins is at bat and his play record is broken into two parts¹⁸. First, the play
`play,4,1,mullc002,10,1BB,PB.1-2`
 transitions that state to $s_{43} = (0, 0, 1, 0, 4, 3, 4, 1)$. Next, the play
`play,4,1,mullc002,32,1BB.FBFFX,8/F8`
 transitions that state to $s_{44} = (1, 0, 1, 0, 4, 3, 4, 1)$.
38. Rutschman comes to bat but his first play
`play,4,1,rutsa001,00,,NP`
 is a No Play, so doesn't change the state. After the substitution (pitcher Ian Hamilton for Clarke Schmidt), the play
`play,4,1,rutsa001,11,.VCX,31/G3.2-3`
 transitions that state to $s_{45} = (2, 0, 0, 1, 4, 3, 4, 1)$. In this play, `.V` is unusual and indicates an automatic ball was called by the umpire with no pitch actually thrown.

¹⁷This play is broken down into more detail in Example 8.

¹⁸This play is broken down into more detail in Example 11.

39. Santander strikes out in the play
`play,4,1,santa003,12,BCCS,K`
 This transitions $\mathbf{s}_{45}$ to the 3-out state $\mathbf{s}_{46} = (3, 0, 0, 1, 4, 3, 4, 1)$. As usual, this marks the end of the inning and the state gets reset to $\mathbf{s}_{47} = (0, 0, 0, 0, 5, 3, 4, 0)$.
40. The play
`play,5,0,lemad001,11,CBX,63/G6`
 tells us DJ Lemahieu hits a grounder to shortstop, who puts him out at first, so the new state is $\mathbf{s}_{48} = (1, 0, 0, 0, 5, 3, 4, 0)$.
41. Judge strikes out on 5 pitches in the play
`play,5,0,judga001,22,CBSBS,K`
 so transitions that state to $\mathbf{s}_{49} = (2, 0, 0, 0, 5, 3, 4, 0)$.
42. Rizzo is put out in his in-field grounder in the play
`play,5,0,rizza001,11,FBX,43/G34D`
 The new state is $\mathbf{s}_{50} = (3, 0, 0, 0, 5, 3, 4, 0)$ which is reset to $\mathbf{s}_{51} = (0, 0, 0, 0, 5, 3, 4, 1)$.
43. Mountcastle strikes out in his at bat, so the play
`play,5,1,mounr001,12,CBC.S,K`
 transitions that state to $\mathbf{s}_{52} = (1, 0, 0, 0, 5, 3, 4, 1)$.
44. Henderson also strikes out, so his play
`play,5,1,hendg002,02,SCS,K`
 transitions that state to $\mathbf{s}_{53} = (2, 0, 0, 0, 5, 3, 4, 1)$.
45. The play
`play,5,1,uriar001,10,BX,D9/L9LS+`
 describes the double by Urias hit to right field. As a result, $\mathbf{s}_{53}$ transitions to $\mathbf{s}_{54} = (2, 0, 1, 0, 5, 3, 4, 1)$.
46. In the 6-pitch play
`play,5,1,fraza001,22,BCCBFX,8/L89`
 Frazier hits a line drive caught by the center fielder, so transitions that state to $\mathbf{s}_{55} = (3, 0, 1, 0, 5, 3, 4, 1)$, which gets updated to $\mathbf{s}_{56} = (0, 0, 0, 0, 6, 3, 4, 0)$.

47. Stanton gets a single in the play
`play,6,0,stanm004,10,BX,S7/L7S`
 so transitions that state to $s_{57} = (0, 1, 0, 0, 6, 3, 4, 0)$.
48. Gleyber Torres' play consists of 4 lines. First,
`play,6,0,torrg001,00,,NP`
 demarks when pitcher Logan Gillaspie is brought in as a reliever,
`sub,gilll001,"Logan Gillaspie",1,0,1`
 and does not affect the game state. The continuation play
`play,6,0,torrg001,22,.FBBFFB,WP.1-2`
 transitions that state to $s_{58} = (0, 0, 1, 0, 6, 3, 4, 0)$. Note the above `play`
 record has a period before the first pitch, `.F` (a foul). This means the
 play is a continuation of an earlier play. But there's more to his at-bat.
 The next continuation play
`play,6,0,torrg001,32,.FBBFFB.B,W`
 transitions that state to $s_{59} = (0, 1, 1, 0, 6, 3, 4, 0)$.
49. The play, a two RBI double by Cabrera,
`play,6,0,cabro002,32,*BBF*BFX,D8/L89XD+.2-H;1-H`
 transitions that state to $s_{60} = (0, 0, 1, 0, 6, 5, 4, 0)$.
50. In his next at bat, Trevino hits an in-field grounder. His play
`play,6,0,trevj001,02,SFX,13/G1S-`
 transitions that state to $s_{61} = (1, 0, 1, 0, 6, 5, 4, 0)$. Note: The dash
 or minus sign (-) is how Retrosheet indicates that the ball was hit
 exceptionally softly. Here, Trevino's groundball to the `1S` field location,
 was fielded by the pitcher then thrown to first base for the out.
51. Franchy Cordero steps into the batter's box. His first play
`play,6,0,cordf003,00,,NP`
 is a No Play demarking the substitution of the relief pitcher Danny
 Coulombe,
`sub,could001,"Danny Coulombe",1,0,1`
 At this point, Cordero is replaced by pinch hitter Isiah Kiner-Falefa,
`sub,kinei001,"Isiah Kiner-Falefa",0,8,11`
 who is now at bat. He strikes out in his play
`play,6,0,kinei001,10,..BN,BK.2-3`
 so s_{61} transitions to $s_{62} = (1, 0, 0, 1, 6, 5, 4, 0)$. Here, the period in `..BN`
 means this play is a continuation, starting off with a ball, then N means

no pitch. Indeed, BK.2-3 tells us the pitcher balked and so the runner on 2nd base was awarded an advance to 3rd base. Finally, the play concludes with a pop fly fielded by the 1st baseman,

`play,6,0,kinei001,11,..BN.FX,3/P3`

which transitions that state to $s_{63} = (2, 0, 0, 1, 6, 5, 4, 0)$.

52. Volpe strikes out in his play

`play,6,0,volpa001,12,SS*BS,K`

which transitions that state to $s_{64} = (3, 0, 0, 1, 6, 5, 4, 0)$. This is the end of the half-inning, so the state is reset to $s_{65} = (0, 0, 0, 0, 6, 5, 4, 1)$.

53. Hays' at bat starts with a No Play

`play,6,1,haysa001,00,,NP`

due to a substitution: First, Aaron Judge has been taken out of center field and moved to right field (where Franchy Cordero started, but now is taken out of the game),

`sub,judga001,"Aaron Judge",0,2,9`

Next,

`play,6,1,haysa001,00,,NP`

indicates another substitution: Isiah Kiner-Falefa fills in for Judge at center field,

`sub,kinei001,"Isiah Kiner-Falefa",0,8,8`

Finally, Hays is out on three called strikes in the play

`play,6,1,haysa001,02,..CCC,K`

The new state is $s_{66} = (1, 0, 0, 0, 6, 5, 4, 1)$.

54. The play

`play,6,1,matej003,10,BX,S7/L7L`

transitions that state to $s_{67} = (1, 1, 0, 0, 6, 5, 4, 1)$.

55. The play

`play,6,1,mullc002,00,,NP`

`sub,marir001,"Ron Marinaccio",0,0,1`

`play,6,1,mullc002,02,.1C+1FF>P,SB2`

starts with a pitcher substitution - Ron Marinaccio enters as a reliever - and is followed by

- a called strike (C),

- an attempted pick off at 1st base by the catcher (+1),
- two pitches fouled off,
- an attempted run and a pitchout (>P), and finally
- a stolen base (SB2).

As a result, the game state is transitioned to $s_{68} = (1, 0, 1, 0, 6, 5, 4, 1)$. Mullins' at bat The 6-pitch play continued with

`play,6,1,mullc002,22,.1C+1FF>P.BX,63/G6.2-3`

This describes a grounder to shortstop, who put out Mullins at 1st but allowed the runner to advance to 3rd, so s_{68} transitions to $s_{69} = (2, 0, 0, 1, 6, 5, 4, 1)$.

56. Rutschman singles, driving home the runner on 3rd. His play
`play,6,1,rutsa001,32,BCBBS.FFX,S7/L78D+.3-H`
transitions that state to $s_{70} = (2, 1, 0, 0, 6, 5, 5, 1)$. Here the code L78D indicates where Adley Rutschman's line drive was hit - to deep left-center field, according to the location codes diagram in Figure 2.3.
57. Santander strikes out in his play
`play,6,1,santa003,12,CSBS,K`
which transitions that state to $s_{70} = (3, 1, 0, 0, 6, 5, 5, 1)$. This ends the inning so the state is updated to $s_{71} = (0, 0, 0, 0, 7, 5, 5, 0)$.
58. Lehamieu's play

`play,7,0,lemad001,00,,NP`
`sub,perec004,"Cionel Perez",1,0,1`
`play,7,0,lemad001,02,.CFC,K`

indicates a pitching substitution: Cionel Perez is the new reliever. The at-bat record ends with a strikeout, so this transitions that state to $s_{72} = (1, 0, 0, 0, 7, 5, 5, 0)$.
59. Judge is put out by a grounder to 2nd. His play
`play,7,0,judga001,00,X,43/G4`
transitions that state to $s_{73} = (2, 0, 0, 0, 7, 5, 4, 0)$.
60. Rizzo's fly ball to right field brings the third out in his play
`play,7,0,rizza001,00,X,9/F9LD`

The new state $\mathbf{s}_{74} = (3, 0, 0, 0, 7, 5, 5, 0)$ means the end of the half-inning, so gets reset to $\mathbf{s}_{75} = (0, 0, 0, 0, 7, 5, 5, 1)$.

61. Mountcastle walks on 5 pitches, so the play
`play,7,1,mounr001,31,BBCBB,W`
transitions that state to $\mathbf{s}_{76} = (0, 1, 0, 0, 7, 5, 5, 1)$.
62. While Henderson strikes out on 5 pitches in his play
`play,7,1,hendg002,32,BSBBC.>C,K+SB2`
in the process, Mountcastle steals 2nd. The new state is $\mathbf{s}_{77} = (1, 0, 1, 0, 7, 5, 5, 1)$.
Note: the notation `.>C` indicates the runner took off just before the
(called strike) pitch.
63. Urias' at bat starts with a No Play demarking a pitching substitution

```
play,7,1,uriar001,00,,NP
sub,cordj001,"Jimmy Cordero",0,0,1
```

Jimmy Cordero is the new reliever for the Yankees. The wild pitch (WP) in the continuation play

```
play,7,1,uriar001,00,.B,WP.2-3
```

allows Mountcastle to advance to 3rd. This play transitions that state to $\mathbf{s}_{79} = (1, 0, 0, 1, 7, 5, 5, 1)$. The second continuation play (an RBI double bringing Mountcastle home)

```
play,7,1,uriar001,10,.B.X,D8/L78XD+.3-H
```

transitions that state to $\mathbf{s}_{80} = (1, 0, 1, 0, 7, 5, 6, 1)$.

64. Frazier's grounder
`play,7,1,fraza001,01,CX,63/G6.2-3`
advances Urias to 3rd, but Frazier gets put out at 1st, so transitions that state to $\mathbf{s}_{81} = (2, 0, 0, 1, 7, 5, 6, 1)$.
65. In Hays' at bat, a wild pitch
`play,7,1,haysa001,01,CB,WP.3-H(NR)`
brings Urias home. This transitions that state to $\mathbf{s}_{82} = (2, 0, 0, 0, 7, 5, 7, 1)$.
Here, `3-H(NR)` indicates that Hays was not given an RBI when Urias ran home. Finally, Hays walks on 6 pitches and play
`play,7,1,haysa001,32,CB.BSBB,W`
transitions that state to $\mathbf{s}_{83} = (2, 1, 0, 0, 7, 5, 7, 1)$.

66. The play
`play,7,1,matej003,12,CFB1X,43/G4`
 describes a ground out by Mateo which transitions that state to $s_{84} = (3, 1, 0, 0, 7, 5, 7, 1)$. This ends the inning so the state is updated to $s_{85} = (0, 0, 0, 0, 8, 5, 6, 0)$.
67. Stanton's 5-pitch double in the play
`play,8,0,stanm004,12,CFBFX,D7/L7D+`
 transitions that state to $s_{86} = (0, 0, 1, 0, 8, 5, 7, 0)$.
68. In the play
`play,8,0,torrg001,10,BX,S8/L8D+.2-3`
 Torres singles, advancing Stanton to 3rd. This transitions that state to $s_{87} = (0, 1, 0, 1, 8, 5, 7, 0)$.
69. Cabrera's single brings Stanton home and advances Torres to 2nd. His play
`play,8,0,cabro002,00,X,S9/G34.3-H;1-2`
 transitions that state to $s_{88} = (0, 1, 1, 0, 8, 6, 7, 0)$.
70. Trevino's at bat starts with a No Play
`play,8,0,trevj001,00,,NP`
`sub,bakeb001,"Bryan Baker",1,0,1`
 due to a pitching substitution; Bryan Baker is the new Orioles reliever. The continuation (double) play
`play,8,0,trevj001,20,.BB+2X,5(2)3/GDP/G56.1-2`
 puts out Torres on 2nd and Trevino. The new state is $s_{89} = (2, 0, 1, 0, 8, 6, 7, 0)$. Note: BB+2X indicates that, after two balls, the catcher threw an attempted pick off to 2nd base (+2). The remainder, 5(2)3/GDP/G56, indicates a ground ball was hit to the 3rd baseman, who got the runner on 2nd base out. The ball was then thrown to 1st base, resulting in an out for the batter (making this a double play, GDP).
71. Kiner-Falefa starts his at bat. First, there's a stolen base,
`play,8,0,kinei001,00,>C,SB3`
 which transitions that state to $s_{90} = (2, 0, 0, 1, 8, 6, 7, 0)$. The continuation play

play,8,0,kinei001,12,>C.FBFS,K

describes a strike out giving the 3rd out. The new state $s_{91} = (3, 0, 0, 1, 8, 6, 7, 0)$ makes the end of the half-inning, so is reset to $s_{75} = (0, 0, 0, 0, 8, 6, 6, 1)$.

72. After replacing the Yankees pitcher by Wandy Peralta, the plays

play,8,1,mullc002,00,,NP
sub,peraw002,"Wandy Peralta",0,0,1
play,8,1,mullc002,00,.X,8/F8

tells us Mullins hit a fly ball to center field. This transitioned that state to $s_{92} = (1, 0, 0, 0, 8, 6, 7, 1)$.

73. Rutschman walks on 6 pitches in the play
play,8,1,rutsa001,32,BCBFBB,W
resulting in the new state $s_{93} = (1, 1, 0, 0, 8, 6, 7, 1)$.

74. Santander line dribes to center, so his play
play,8,1,santa003,10,BX,8/L8XD+
transitions that state to $s_{94} = (2, 1, 0, 0, 8, 6, 7, 1)$.

75. The play
play,8,1,mounr001,00,1X,S7/L7LS.1-2
describes a single by Mountcastle, advancing Rutschman to 2nd, so
transitions that state to $s_{95} = (2, 1, 1, 0, 8, 6, 7, 1)$.

76. Finally, Henderson strikes out and his play
play,8,1,hendg002,22,SBS*BS,K
transitions that state to $s_{96} = (3, 1, 1, 0, 8, 6, 7, 1)$. This ends the inning
so the state is updated to $s_{97} = (0, 0, 0, 0, 9, 6, 7, 0)$.

77. After replacing the pitcher by Felix Bautista and the right fielder by
Ryan McKenna,

play,9,0,volpa001,00,,NP
sub,mcker001,"Ryan McKenna",1,3,9
play,9,0,volpa001,00,,NP
sub,bautf001,"Felix Bautista",1,0,1
play,9,0,volpa001,02,..SCS,K

the 3-pitch strikeout of Volpe transitioned that state to $\mathbf{s}_{98} = (1, 0, 0, 0, 9, 6, 7, 0)$.

78. Lemahieu also strikes out, so his play

`play,9,0,lemad001,22,FCBBC,K`

transitions that state to $\mathbf{s}_{99} = (2, 0, 0, 0, 9, 6, 7, 0)$.

79. The play

`play,9,0,judga001,30,BBB*B,W`

indicates Judge walks on 4 pitches, so transitions that state to $\mathbf{s}_{100} = (2, 1, 0, 0, 9, 6, 7, 0)$.

80. Anthony Rizzo's 9th inning at-bat breaks down into several plays. The play

`play,9,0,rizza001,02,C*S>*B,SB2`

describes Judge stealing 2nd, which transitions to the new state $\mathbf{s}_{101} = (2, 0, 1, 0, 9, 6, 7, 0)$. The wild pitch

`play,9,0,rizza001,12,C*S>*B.B,WP.2-3`

advances Judge to 3rd base, so transitions that state to $\mathbf{s}_{102} = (2, 0, 0, 1, 9, 6, 7, 0)$.

Finally, Rizzo fly out to left field. That play

`play,9,0,rizza001,32,C*S>*B.B.BFX,7/F56D`

transitions that state to $\mathbf{s}_{103} = (3, 0, 0, 1, 9, 6, 7, 0)$ and ends the inning and the game.

2.3.3 The inning-by-inning states

The entire game can be described using a chain of game states. See Figures 2.4 to 2.12.

0	(0, 0, 0, 0, 1, 0, 0, 0)
1	(0, 1, 0, 0, 1, 0, 0, 0)
2	(1, 1, 0, 0, 1, 0, 0, 0)
3	(1, 1, 1, 0, 1, 0, 0, 0)
4	(2, 1, 1, 0, 1, 0, 0, 0)
5	(0, 0, 0, 0, 1, 0, 0, 1)
6	(1, 0, 0, 0, 1, 0, 0, 1)
7	(2, 0, 0, 0, 1, 0, 0, 1)

Figure 2.4: First inning of game in §2.3. Recall the first coordinate of the state records the outs and the last coordinate records who is at bat. Since the home team is at bat in the last line of the table, once their third out has just been reached, the inning must be incremented, the `team_at_bat` gets toggled, and the outs are reset to 0.

8	(0, 0, 0, 0, 2, 0, 0, 0)
9	(1, 0, 0, 0, 2, 0, 0, 0)
10	(2, 0, 0, 0, 2, 0, 0, 0)
11	(2, 1, 0, 0, 2, 0, 0, 0)
12	(0, 0, 0, 0, 2, 0, 0, 1)
13	(1, 0, 0, 0, 2, 0, 0, 1)
14	(1, 1, 0, 0, 2, 0, 0, 1)
15	(1, 1, 1, 0, 2, 0, 0, 1)
16	(1, 1, 0, 1, 2, 0, 1, 1)
17	(1, 1, 1, 0, 2, 0, 2, 1)
18	(2, 1, 1, 0, 2, 0, 2, 1)

Figure 2.5: Second inning of game in §2.3

19	(0, 0, 0, 0, 3, 0, 2, 0)
20	(0, 1, 0, 0, 3, 0, 2, 0)
21	(1, 1, 0, 0, 3, 0, 2, 0)
22	(0, 0, 0, 0, 3, 0, 2, 1)
23	(0, 1, 0, 0, 3, 0, 2, 1)
24	(0, 0, 1, 1, 3, 0, 2, 1)
25	(1, 0, 1, 0, 3, 0, 3, 1)
26	(1, 0, 1, 0, 3, 0, 4, 1)
27	(2, 0, 1, 0, 3, 0, 4, 1)
28	(2, 1, 1, 0, 3, 0, 4, 1)

Figure 2.6: Third inning of game in §2.3

29	(0, 0, 0, 0, 4, 0, 4, 0)
30	(1, 0, 0, 0, 4, 0, 4, 0)
31	(2, 0, 0, 0, 4, 0, 4, 0)
32	(2, 1, 0, 0, 4, 0, 4, 0)
33	(2, 1, 1, 0, 4, 0, 4, 0)
34	(2, 0, 0, 0, 4, 3, 4, 0)
35	(0, 0, 0, 0, 4, 3, 4, 1)
36	(0, 1, 0, 0, 4, 3, 4, 1)
37	(0, 0, 1, 0, 4, 3, 4, 1)
38	(1, 0, 1, 0, 4, 3, 4, 1)
39	(2, 0, 0, 1, 4, 3, 4, 1)

Figure 2.7: Fourth inning of game in §2.3

40	(0, 0, 0, 0, 5, 3, 4, 0)
41	(1, 0, 0, 0, 5, 3, 4, 0)
42	(2, 0, 0, 0, 5, 3, 4, 0)
43	(0, 0, 0, 0, 5, 3, 4, 1)
44	(1, 0, 0, 0, 5, 3, 4, 1)
45	(2, 0, 0, 0, 5, 3, 4, 1)
46	(2, 0, 1, 0, 5, 3, 4, 1)

Figure 2.8: Fifth inning of game in §2.3

47	(0, 0, 0, 0, 6, 3, 4, 0)
48	(0, 1, 0, 0, 6, 3, 4, 0)
49	(0, 0, 1, 0, 6, 3, 4, 0)
50	(0, 1, 1, 0, 6, 3, 4, 0)
51	(0, 0, 1, 0, 6, 5, 4, 0)
52	(1, 0, 1, 0, 6, 5, 4, 0)
53	(1, 0, 0, 1, 6, 5, 4, 0)
54	(2, 0, 0, 1, 6, 5, 4, 0)
55	(0, 0, 0, 0, 6, 5, 4, 1)
56	(1, 0, 0, 0, 6, 5, 4, 1)
57	(1, 1, 0, 0, 6, 5, 4, 1)
58	(1, 0, 1, 0, 6, 5, 4, 1)
59	(2, 0, 0, 1, 6, 5, 4, 1)
60	(2, 1, 0, 0, 6, 5, 5, 1)

Figure 2.9: Sixth inning of game in §2.3

61	(0, 0, 0, 0, 7, 5, 5, 0)
62	(1, 0, 0, 0, 7, 5, 5, 0)
63	(2, 0, 0, 0, 7, 5, 5, 0)
64	(0, 0, 0, 0, 7, 5, 5, 1)
65	(0, 1, 0, 0, 7, 5, 5, 1)
66	(1, 0, 1, 0, 7, 5, 5, 1)
67	(1, 0, 0, 1, 7, 5, 5, 1)
68	(1, 0, 1, 0, 7, 5, 6, 1)
69	(2, 0, 0, 1, 7, 5, 6, 1)
70	(2, 0, 0, 0, 7, 5, 7, 1)
71	(2, 1, 0, 0, 7, 5, 7, 1)

Figure 2.10: Seventh inning of game in §2.3

72	(0, 0, 0, 0, 8, 5, 7, 0)
73	(0, 0, 1, 0, 8, 5, 7, 0)
74	(0, 1, 0, 1, 8, 5, 7, 0)
75	(0, 1, 1, 0, 8, 6, 7, 0)
76	(2, 0, 1, 0, 8, 6, 7, 0)
77	(2, 0, 0, 1, 8, 6, 7, 0)
78	(0, 0, 0, 0, 8, 6, 7, 1)
79	(1, 0, 0, 0, 8, 6, 7, 1)
80	(1, 1, 0, 0, 8, 6, 7, 1)
81	(2, 1, 0, 0, 8, 6, 7, 1)
82	(2, 1, 1, 0, 8, 6, 7, 1)

Figure 2.11: Eighth inning of game in §2.3

83	(0, 0, 0, 0, 9, 6, 7, 0)
84	(1, 0, 0, 0, 9, 6, 7, 0)
85	(2, 0, 0, 0, 9, 6, 7, 0)
86	(2, 1, 0, 0, 9, 6, 7, 0)
87	(2, 0, 1, 0, 9, 6, 7, 0)

Figure 2.12: Top of ninth inning of game in §2.3

You see how common the situation with one out and only one man on 2nd base is, in reference to the game state frequency tables above.

2.4 Game states transitions

In this section, we use a sequence of Python programs to parse Retrosheet event records and extract the data needed to determine the frequencies of transitions from one game state to the next. The goal here is to compute, using a season's worth of home games (usually 81 in total), extract the Retrosheet data necessary to compute the frequencies of the state transitions and then derive the transition probabilities from this counting data.

Let $B_1, B_2, \dots$, be a sequence of random variables (representing batters in a randomized baseball game) taking values in the finite set S of baseball states. We assume, as before, that the following two conditions hold:

- For all $\ell > 0$, the conditional probability $P(B_{\ell+1} = \mathbf{s}_j \mid B_\ell = \mathbf{s}_i)$ is independent of the values of B_k , $k < \ell$.
- The associated *transition matrix* $(p_{i,j})_{0 \leq i,j \leq |S|-1}$ is doubly stochastic, where

$$p_{i,j} = P(B_{\ell+1} = \mathbf{s}_j \mid B_\ell = \mathbf{s}_i).$$

There are infinitely many possible states, but a typical game has about 100 states. As a game typically has 17 or 18 innings, divide 100 by 17 or 18 to get the typical number of states per side per inning. The most common states are (in rough order of decreasing frequency),

$$\begin{aligned}
&(0, 0, 0, 0), (1, 0, 0, 0), (2, 0, 0, 0), (1, 1, 0, 0), (2, 1, 0, 0), \\
&(0, 1, 0, 0), (2, 0, 1, 0), (1, 0, 1, 0), (2, 1, 1, 0), (1, 1, 1, 0), \\
&(0, 0, 1, 0), (2, 0, 0, 1), (0, 1, 1, 0).
\end{aligned}$$

The three most common states are (1) batter starts with no outs and bases empty, batter ends play with one out and bases empty, (2) batter starts with 1 out and bases empty, batter ends play with 2 outs and bases empty, (3) batter starts with 2 outs and bases empty, batter ends play (and inning) with out number 3.

Question: What is the probability, for a given MLB season, that a given state transitions to another given state for a “randomly selected” batter?

To find the transition probabilities, often simplifying assumptions are made so that one can apply individual batter statistics. Here we try a different approach, one that relies on the games themselves and computes the frequencies of the state transitions and then derived the transition probabilities from this counting data.

2.4.1 The state transition frequencies

It is also possible to compute the frequency of a pair of states

$$(\mathbf{s}, \mathbf{s}'),$$

where some play in a 2017-2023 Orioles home game resulted in the state transition

$$\mathbf{s} \rightarrow \mathbf{s}'.$$

From this data, one could compute an approximate state transition matrix of the system, if desired.

First, a technical note on the assumptions. For the transition probabilities derived from Retrosheet home game event files for a given season, the associated sample space for the home team is disjoint from that of the visiting team. As a result, if we are to use a Markov chain model, a random variable (a random batter, in this case) is taken from the sample space corresponding

to which team is at bat. Therefore, we use a modified Markov chain model with two sample spaces.

2.4.2 Oriole home games, 2023

Examples of these numbers can be found in the table in Figures 2.13.

Figure 2.13, show the most common state transitions for the Baltimore Orioles during their 2023 home games.

percentage	transitioning states
15.7	$((0, 0, 0, 0), (1, 0, 0, 0))$
12.8	$((1, 0, 0, 0), (2, 0, 0, 0))$
9.70	$((2, 0, 0, 0), (0, 0, 0, 0))$
5.38	$((0, 0, 0, 0), (0, 1, 0, 0))$
4.74	$((2, 1, 0, 0), (0, 0, 0, 0))$
3.77	$((1, 0, 0, 0), (1, 1, 0, 0))$
3.42	$((2, 0, 0, 0), (2, 1, 0, 0))$
3.24	$((1, 1, 0, 0), (2, 1, 0, 0))$
2.57	$((0, 1, 0, 0), (1, 1, 0, 0))$
2.20	$((2, 0, 1, 0), (0, 0, 0, 0))$
1.84	$((2, 1, 1, 0), (0, 0, 0, 0))$
1.20	$((2, 1, 0, 0), (2, 1, 1, 0))$
1.18	$((1, 1, 0, 0), (1, 1, 1, 0))$
1.13	$((0, 1, 0, 0), (0, 1, 1, 0))$
1.09	$((1, 0, 1, 0), (2, 0, 1, 0))$
1.07	$((0, 0, 0, 0), (0, 0, 1, 0))$

Figure 2.13: State transition frequency table, for the most common transitions, taken from the Orioles home games in 2023.

Note that the table in Figure 2.13 supports the

35% Rule: About 35 percent of the time, a batter is up with bases empty and his play results in an out.

Figure 2.14 shows the probabilities of the 30 top transitions in 2023.

probability in game	transitioning states
0.1592	((0, 0, 0, 0), (1, 0, 0, 0))
0.1286	((1, 0, 0, 0), (2, 0, 0, 0))
0.09995	((2, 0, 0, 0), (0, 0, 0, 0))
0.04880	((0, 0, 0, 0), (0, 1, 0, 0))
0.04392	((2, 1, 0, 0), (0, 0, 0, 0))
0.03500	((1, 0, 0, 0), (1, 1, 0, 0))
0.03147	((1, 1, 0, 0), (2, 1, 0, 0))
0.03113	((2, 0, 0, 0), (2, 1, 0, 0))
0.02524	((0, 1, 0, 0), (1, 1, 0, 0))
0.02440	((2, 1, 1, 0), (0, 0, 0, 0))
0.02070	((2, 0, 1, 0), (0, 0, 0, 0))
0.01380	((2, 1, 1, 0), (2, 1, 1, 0))
0.01346	((1, 1, 1, 0), (2, 1, 1, 0))
0.01178	((2, 1, 0, 0), (2, 1, 1, 0))
0.01111	((1, 0, 0, 0), (1, 0, 0, 0))
0.01077	((0, 1, 0, 0), (0, 1, 1, 0))
0.01060	((1, 1, 0, 0), (1, 1, 1, 0))
0.01060	((0, 0, 0, 0), (0, 0, 0, 0))
0.01026	((0, 0, 0, 0), (0, 0, 1, 0))
0.01010	((1, 0, 1, 0), (2, 0, 1, 0))
0.009591	((2, 0, 0, 0), (2, 0, 0, 0))
0.009086	((0, 1, 1, 0), (1, 1, 1, 0))
0.008582	((2, 1, 0, 1), (0, 0, 0, 0))
0.007404	((2, 0, 0, 1), (0, 0, 0, 0))
0.007235	((1, 1, 1, 0), (1, 1, 1, 0))
0.006899	((2, 1, 0, 0), (2, 1, 0, 0))
0.006899	((2, 0, 0, 0), (2, 0, 1, 0))
0.006731	((1, 1, 0, 0), (0, 0, 0, 0))
0.006562	((2, 0, 1, 0), (2, 0, 1, 0))
0.006562	((0, 0, 1, 0), (1, 0, 1, 0))

Figure 2.14: State transition frequency table, for 30 most common transitions, taken from the Orioles home games in 2023

In Figure 2.14 a home run corresponds to one of these transitions:

$$((0, 0, 0, 0), (0, 0, 0, 0)), ((1, 0, 0, 0), (1, 0, 0, 0)), ((2, 0, 0, 0), (2, 0, 0, 0)).$$

One of these occurs with probability $0.01111 + 0.01060 + 0.00959 = 0.0313$, or about 3.1% of the time in 2023. Oddly, such a HR was about as frequent as a 2-out single!

Chapter 3

Defensive metrics

This chapter is devoted to comparing various defensive metrics, in particular the total team errors.

3.1 UZR, SDI, and all that

First, we list a few of the most commonly used and respected defensive metrics in sabermetrics.

- UZR (Ultimate Zone Rating): This is Mitchel Lichtman's comprehensive metric that attempts to quantify how many runs a player saved or cost their team with their defense. It considers a wide range of factors, including range, errors, arm strength (for outfielders), and double play ability (for infielders).
- DRS (Defensive Runs Saved): Similar to UZR, John Dewan's DRS aims to measure how many runs a player saved compared to an average defender. It uses a different methodology than UZR.
- OAA (Outs Above Average): This metric uses Statcast data to measure how many outs a player made compared to an average defender on similar plays. It takes into account the difficulty of the play, based on factors like the hang time and distance of the ball. OAA is an especially good metric for outfielders.
- RED (Runs Effectively Defended): Chris Dial's zone rating system.

- TZR (Total Zone Rating) Sean Smith’s defensive statistic is calculated exclusively using play-by-play data available from Retrosheet [TZR].
- SDI (SABR Defensive Index). According to [BA], the SDI is a

composite measure of fielding based on five fielding metrics and computed by a committee working under the auspices of the Society for American Baseball Research. Four of the metrics comprising it are zone rating statistic; Chris Dial’s RED, John Dewan’s DRS, Sean Smith’s TZR, and Mitchel Lichtman’s UZR. The fifth metric, Michael Humphrey’s *Defensive Regression Analysis*, is based on conventional fielding indices but uses statistical procedures to rid them of bias.

See also <https://sabr.org/sdi>. For 2023, American league teams with positive SDI (in increasing order):

MIN, SEA, CLE, TBA, NYA, BAL, TOR, TEX.

IN 2023, National league teams with positive SDI (in increasing order):

ATL, COL, CHN, LAN, SDN, ARI, MIL.

- DRA (Defensive Regression Analysis), by Michael Humphreys, starts with publicly available defensive stats (putouts, assists, etc.), then uses regression analysis to determine how those traditional stats relate to the number of runs allowed [Hu05], [Hu11]. Based on the regression analysis, DRA develops a formula to estimate a player’s defensive runs saved.

While these are useful, most are proprietary fielding metrics. The DER, invented in the 1980s by Bill James and explained below, is not.

3.2 Defensive efficiency ratio

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. High DER is ideal.

The average DER for all the MLB teams was 0.691 in 2023 [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$. Therefore, any team with DER less than .680 will be one standard deviation below from the MLB average (which is bad), while any team with DER more than .702 will be one standard deviation above from the MLB average (which is good).

- Teams with $DER < 0.680$: SFN, OAK, BOS, COL, SLN
- Teams with $DER > 0.702$: SDN, TEX, NYA, LAN, MIL.

The formula for the DER is:

$$DER = \frac{\text{Outs on Balls in Play}}{\text{Total Balls in Play}}$$

The formula for the DER is:

$$\begin{aligned} DER &= \frac{\text{Outs on Balls in Play}}{\text{Total Balls in Play}} \\ &= 1 - \left(\frac{H + ROE - HR}{PA - BB - SO - HBP - HR} \right), \end{aligned}$$

where:

- **Outs on Balls in Play** refers to the total number of outs the defense makes on balls that are hit into play (excluding strikeouts, walks, and home runs). The expression $(H + ROE - HR)$ in the numerator represents the total number of times the defense has failed to convert a batted ball into an out. This includes hits and errors that allow runners to reach base, while subtracting home runs since these are automatic runs and not converted outs.
 - H: Hits allowed by the defense.
 - ROE: Reached on Error; this counts the number of times a batter reaches base due to an error.
 - HR: Home Runs allowed by the defense.

- **Total Balls in Play** includes all batted balls that are not strikeouts, walks, or home runs. The expression $(PA - BB - SO - HBP - HR)$ in the denominator calculates the total number of batted ball opportunities where an out could potentially be made. This excludes walks (BB), strikeouts (SO), hit by pitches (HBP), and home runs (HR), since these do not result in a batted ball that the defense can handle.
 - PA: Plate Appearances; the total number of times a batter comes to the plate.
 - BB: Walks; free passes given to batters.
 - SO: Strikeouts; when a batter fails to hit the ball and receives three strikes.
 - HBP: Hit by Pitch; when a batter is hit by a pitch and awarded first base.

Explanation of the overall Calculation: The fraction gives the proportion of batted ball opportunities that resulted in hits or errors (not converted to outs). By subtracting this fraction from 1, you get the percentage of batted balls that were converted into outs.

Interpretation:

- A higher DER (closer to 1) indicates a more effective defense, as it means they are converting a higher percentage of batted balls into outs.
- A lower DER (closer to 0) indicates poorer defensive performance, suggesting that more batted balls are resulting in hits or reaching base.

If you look at the DER values for the 30 teams in the MLB and compare them to the E values for the same teams¹, the two lists of numbers are not well-correlated (the Pearson correlation coefficient is about -0.39). The negative correlation *suggests* that a lower DER value results in a larger number of errors, there are several examples of MLB teams where this is not true. A scatterplot of DER vs E is depicted in Figure 2.1.

¹For example, see baseball-reference.com [BR] for this data.

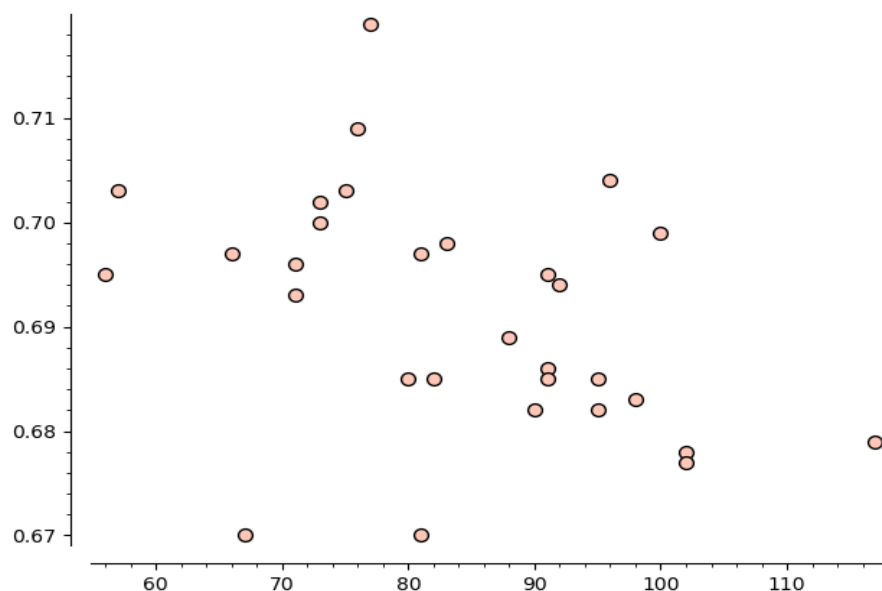


Figure 3.1: A scatterplot of DER vs E for 2023 MLB teams.

We normalize the values x of DER by replacing x by $(x - \mu)/\sigma$, where μ is the average DER value for the 2023 MLB teams and σ is the standard deviation of the DER values. To keep “positive is good”, we instead normalize the values x of E by replacing x by $(\mu - x)/\sigma$, where μ is the average E value for the 2023 MLB teams and σ is the standard deviation of the E values. With the MLB teams ordered alphabetically², a plot of these normalized values is depicted in Figure 3.2. Note how the first team (the Arizona Diamondbacks) have both their values above the x-axis, indicating that they are better than average in both team errors and team DER.

²The teams “Arizona Diamondbacks,” . . . , “Washington Nationals” are represented by 0, . . . , 29 along the x -axis.

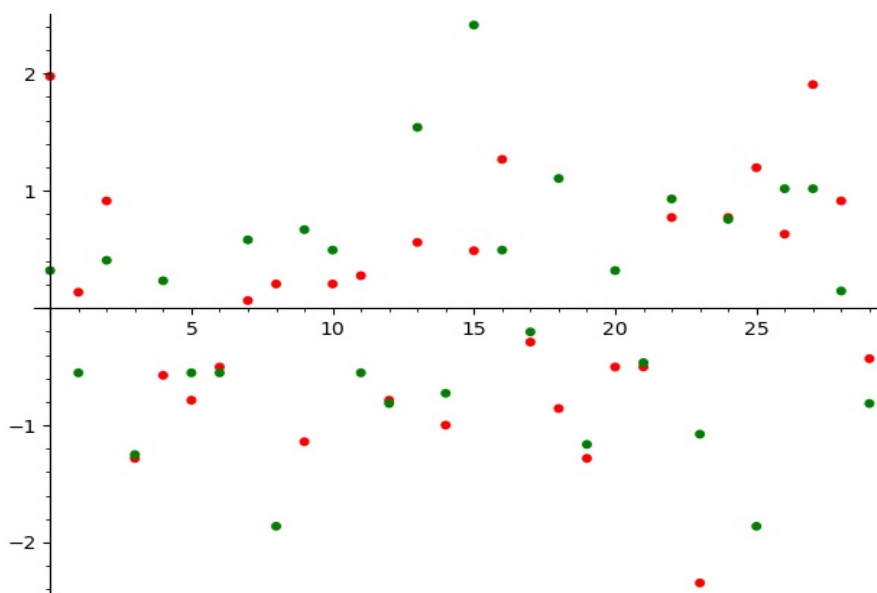


Figure 3.2: A plot of normalized DER vs the negative of normalized E for 2023 MLB teams. E is in red and DER is in green.

We look at errors more in the next section.

3.3 Errors

Error totals is another team fielding performance metric. Low E is good.

The average number of errors made over the regular season in 2023 was 83.9, while the standard deviation was $\sigma = 14.1$. For comparison, the highest value was $E = 117$ for the SFN (the Giants), over twice that of the lowest value, which was $E = 56$ for ARI (the D-backs).

Therefore, any team with E less than 70 will be one standard deviation below from the MLB average (which is good), while any team with E more than 97 will be one standard deviation above from the MLB average (which is bad).

- MB teams with $E < 70$ in 2023: SLN, MIN, TEX, ARI
- MB teams with $E > 97$ in 2023: SFN, OAK, BOS, DET, MIA.

Let X be any MLB team. We compile

- E , total errors by team X , home and away,
- E_h , total errors by team X at home
- E_1 , total errors made by a visiting team playing X on the X 's home ballpark,
- E_0 , total errors made by X on their home ballpark.

The plot is depicted in Figure 3.3.

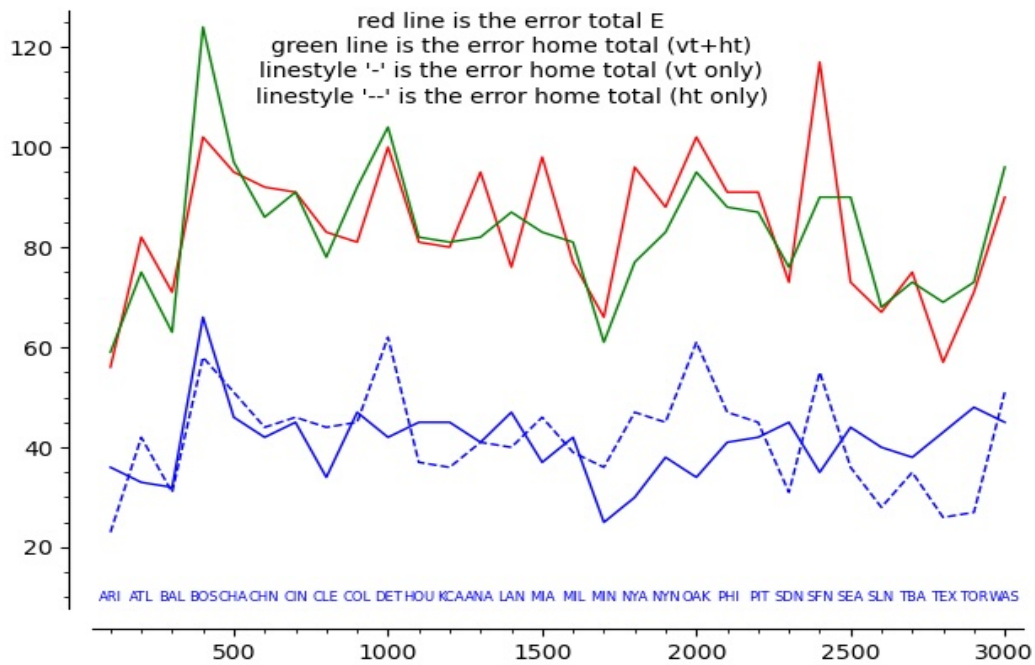


Figure 3.3: Error comparison for 2023 MLB.

Chapter 4

Arizona Diamondbacks (ARI)

Based in Phoenix, the Arizona Diamondbacks compete as a member of the National League West Division. One of the newest teams to MLB, the franchise began play in 1998 and are the youngest team to win a World Series (in 2001). They have also won 5 NLWD titles, the latest in 2011.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Phoenix against the Diamondbacks. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (DER or DefEff) is one measure of the fielding performance of a team (see §3.2 for more details). The DER of the 2023 ARI team is .695 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the D-backs is only slightly above the MLB average.

In ARI 2023, there were 59 errors in the 81 home games, 23 by the D-backs and the rest (36) by visiting teams. The D-backs had an exceptionally good year in 2023, with a relatively low error total. Indeed, the highest value was $E = 117$ for the SFN (the Giants), over twice that if the lowest value, which was $E = 56$ for these D-backs.

4.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

4.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and error-to-run rate.

Runs and errors	totals
Errors by vt:	36
Runs by ht because of errors by vt:	12
Errors by ht:	23
Runs by vt because of errors by ht:	5

This is a good error-to-run conversion rate: $12/36 = 33.3\%$ on the part of the Diamondbacks (compared to $5/23 = 21.7\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 ARI season (home and away) has about $R/PA = 746/6124$ (12.2%) likelihood of resulting in a run. Therefore, a randomly selected error is (for the D-backs) nearly three times as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 59, so the expected number of errors (by either team) in a randomly selected ARI 2023 home game is about 0.73.

Errors by position	vt	ht
errors due to pitcher:	4	4
to catcher:	8	2
to 1st baseman:	3	2
to 2nd baseman:	3	3
to 3rd baseman:	6	1
to shortstop:	8	9
to left fielder:	0	0
to center fielder:	0	1
to right fielder:	1	0

4.1.2 Error total by state

This section tabulates the totals by game state.

Errors by state	vt	ht	totals
bases empty	13	7	20
(lone) runner on 1st	10	9	19
(lone) runner on 2nd	1	3	4
(lone) runner on 3rd	0	0	0
two runners on base	11	4	15
bases loaded	1	0	1
total	36	23	59

A table of frequencies of most common errors by the Diamondbacks ARI 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	5	21.7%
(1, 0, 0, 0)	1	4.3%
(2, 0, 0, 0)	1	4.3%
(0, 1, 0, 0)	5	21.7%
(1, 1, 0, 0)	3	13%
(2, 1, 0, 0)	1	4.3%
(0, 0, 1, 0)	1	4.3%
(1, 1, 1, 0)	3	13%
(0, 1, 0, 1)	1	4.3%

A table of frequencies of most common errors by a visiting team in ARI 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	19.4%
(1, 0, 0, 0)	3	8.3%
(2, 0, 0, 0)	3	8.3%
(0, 1, 0, 0)	5	13.9%
(1, 1, 0, 0)	3	8.3%
(2, 1, 0, 0)	2	5.6%
(0, 1, 1, 0)	1	2.8%
(2, 1, 1, 0)	5	13.9%
(1, 0, 1, 1)	3	8.3%
(1, 1, 1, 1)	1	2.8%

For comparison, Figure 4.1 gives part of Lindsey’s table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 D-backs homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `ARI2023.evn`):

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

Figure 4.1: Lindsey's frequency table

state	ARI2023 percentages
(0, 0, 0, 0)	23.8%
(1, 0, 0, 0)	15.7%
(2, 0, 0, 0)	12.5%
(0, 1, 0, 0)	6.1%
(1, 1, 0, 0)	6.9%
(2, 1, 0, 0)	7.3%
(0, 0, 1, 0)	1.9%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	4.3%
(0, 0, 0, 1)	0.5%
(1, 0, 0, 1)	1.2%
(2, 0, 0, 1)	2.3%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.1%
(2, 1, 1, 0)	2.6%

4.1.3 Some interesting errors

Some of the most interesting error plays in the ARI 2023 home games:

- The game PIT vs ARI on 2023-07-07 had 3 errors but the series against the pirates had 7.
- The game CHN vs ARI on 2023-09-16 had 3 errors, including a double stolen base.
- The game SFN vs ARI on 2023-09-19 only had 2 errors but there were in the same play and by the same fielder!

4.1.4 Relative team rankings

We have 81 home games between ARI and some visiting MLB team. A number of teams did not play ARI in the regular season.

However, of those that did, the home team (ht) ARI *lost* more games than won against¹:

ATL (2/3), BAL (2/3), BOS (2/3), MIA (2/3), SDN (2/3),
SEA (2/3), SLN (2/3), TBA (2/3), HOU (3/3), NYN (3/3)

The teams that *split* their series were:

CIN (2/4), PHI (2/4),

and *won* more than lost against

COL (2/7), SFN (2/6), CHN (0/3), TEX (0/2), CLE (1/3),
KCA (1/3), MIL (1/3), PIT (1/3), WAS (1/3).

The teams that did not play in any of the 81 home games in ARI 2023 were:

ANA, CHA, DET, LAN, MIN, NYA, OAK, TOR.

4.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for ARI 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Gabriel Moreno	105	.339
3 (1B)	Christian Walker	153	.333
4 (2B)	Ketel Marte	146	.358
5 (3B)	Emmanuel Rivera	65	.314
6 (SS)	Geraldo Perdomo	117	.356
7 (LF)	Lourdes Gurriel Jr.	95	.309
8 (CF)	Alek Thomas	118	.273
9 (RF)	Jake McCarthy	85	.318

¹Visiting teams (vt) are listed with that team's W/T record in parentheses (W = numbers of games vt won against ht at ARI, T = Total home games vt played vs ht at ARI).

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 ARI home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 59 errors in the 81 home games, 23 by the Diamondbacks and the rest (36) by visiting teams.

4.2.1 April

game	LAN vs ARI on 20230406
state before	(1, 1, 1, 0, 3, 1, 0, 0)
1. error	play,3,0,muncm001,22,CCBBX,36(1)/FO/G34.2-H(NR)(UR);BX1(6E1)
state after	(2, 1, 0, 0, 3, 2, 0, 1)
fielder	1 (pitcher on the home team)

Summary 1 *In the top of the 3rd inning, with runners on 1st and 2nd, LAN batter Max Muncy hit a ground ball to the 1st/2nd base gap. At 2nd, the shortstop (with an assist from the first baseman) forced out the runner from 1st. The runner on 2nd to score (unearned, no RBI) while an error by the pitcher (with an assist from the shortstop) allowed the batter to safely reach 1st.*

game	MIL vs ARI on 20230410
state before	(0, 1, 0, 0, 1, 0, 1, 1)
2. error	play,1,1,gurrl001,11,BCX,FC6/G6.1-2(E6);B-1
state after	(0, 1, 1, 0, 1, 0, 1, 1)
fielder	6 (shortstop on the visiting team)

Summary 2 *In the bottom of the 1st inning, with a runner on 1st, ARI batter Lourdes Gurriel hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for MIL vs ARI on 2023-04-12

position	innings	player	position	innings	player
1	1	Janson Junk	1	1	Drey Jameson
1	5	Hoby Milner	1	5	Kevin Ginkel
1	6	Gus Varland	1	5	Kyle Nelson
1	8	Javy Guerra	1	7	Andrew Chafin
2	1	William Contreras	1	7	Luis Frias
3	1	Luke Voit	1	8	Miguel Castro
4	1	Brice Turang	1	9	Scott McGough
5	1	Mike Brosseau	2	1	Jose Herrera
6	1	Willy Adames	3	1	Christian Walker
7	1	Christian Yelich	4	1	Ketel Marte
8	1	Garrett Mitchell	5	1	Josh Rojas
9	1	Joey Wiemer	6	1	Geraldo Perdomo
			7	1	Corbin Carroll
			8	1	Alek Thomas
			9	1	Jake McCarthy

	game	MIL vs ARI on 20230412
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
3.	error	play,1,1,rojaj001,02,CCX,C/E2.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 3 This game had 3 errors. The final score was 3-7.

In the bottom of the 1st inning, with bases empty, ARI batter Josh Rojas had a 0-2 count. Catcher's interference allowed the batter to take 1st.

	game	MIL vs ARI on 20230412
	state before	(0, 1, 0, 0, 4, 0, 4, 1)
4.	error	play,4,1,herrj003,11,BCB,0A.1-2(E2/TH)
	state after	(0, 0, 1, 0, 4, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 4 *In the bottom of the 4th inning, with a runner on 1st, ARI batter Jose Herrera had a 1-1 count. The next pitch was a ball, after which a throwing error by the catcher allowed the runner on 1st to advance to 2nd. Herrera is still at bat, now with a 1-2 count.*

	game	MIL vs ARI on 20230412
	state before	(2, 0, 0, 0, 5, 0, 5, 1)
5.	error	play,5,1,mccaj003,00,X,E5/TH/BG2.B-1
	state after	(2, 1, 0, 0, 5, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 5 *In the bottom of the 5th inning, with bases empty, ARI batter Jake McCarthy hit a bunt to very shallow infield. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	SDN vs ARI on 20230420
	state before	(1, 1, 0, 0, 6, 5, 5, 0)
6.	error	play,6,0,cruzn002,01,.FX,E6/G6D.1-2;B-1
	state after	(1, 1, 1, 0, 6, 5, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 6 This game had 2 errors. The final score was 7-5. *In the top of the 6th inning, with a runner on 1st, SDN batter Nelson Cruz hit a ground ball to deep short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SDN vs ARI on 20230420
	state before	(0, 1, 0, 0, 6, 7, 5, 1)
7.	error	play,6,1,gurrl001,32,CB1B*BC1,P01(E1/TH).1-2
	state after	(0, 0, 1, 0, 6, 7, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 7 *In the bottom of the 6th inning, with a runner on 1st, ARI batter Lourdes Gurriel has a full count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed the runner on 1st to advance to 2nd. Gurriel is still at bat with a full count.*

	game	KCA vs ARI on 20230424
	state before	(1, 1, 0, 1, 8, 4, 4, 1)
8.	error	play,8,1,ahmen001,12,.FFBX,FC3/G3S-.3-H;1-3(E3/TH);B-2
	state after	(1, 0, 1, 1, 8, 4, 5, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 8 *In the bottom of the 8th inning, with runners on 1st and 3rd, ARI batter Nick Ahmed hit a soft ground ball into a fielder's choice at shallow 1st. The runner on 3rd scored (1 RBI). A throwing error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	KCA vs ARI on 20230425
	state before	(1, 0, 1, 1, 8, 5, 2, 1)
9.	error	play,8,1,rivee001,21,.B>C.BX,S6/G6M.3-H;2-H(E6/TH)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 8, 5, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 9 *In the bottom of the 8th inning, with runners on 2nd and 3rd, ARI batter Emmanuel Rivera hit a ground ball single to short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	KCA vs ARI on 20230426
	state before	(0, 1, 0, 0, 9, 0, 2, 0)
10.	error	play,9,0,olive001,02,CC1X,E6/G78S.1-2;B-1
	state after	(0, 1, 1, 0, 9, 0, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 10 *In the top of the 9th inning, with a runner on 1st, KCA batter Edward Olivares hit a ground ball to shallow left center field. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

4.2.2 May

	game	WAS vs ARI on 20230505
	state before	(0, 1, 0, 0, 4, 1, 1, 1)
11.	error	play,4,1,moreg001,02,TFFX,S1/G1+.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 4, 1, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 11 *In the bottom of the 4th inning, with a runner on 1st, ARI batter Gabriel Moreno hit a hard ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	WAS vs ARI on 20230506
	state before	(1, 0, 1, 1, 8, 2, 4, 1)
12.	error	play,8,1,moreg001,00,X,E6/G6M.3-H;2-3;B-1
	state after	(1, 1, 0, 1, 8, 2, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 12 *In the bottom of the 8th inning, with runners on 2nd and 3rd, ARI batter Gabriel Moreno hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	WAS vs ARI on 20230507
	state before	(0, 1, 0, 0, 8, 5, 8, 0)
13.	error	play,8,0,abrac001,10,BX,S4/BG34.1-3(E4/TH)
	state after	(0, 1, 0, 1, 8, 5, 8, 0)
	fielder	4 (2nd baseman on the home team)

Summary 13 This game had 2 errors. The final score was 9-8.

In the top of the 8th inning, with a runner on 1st, WAS batter C. J. Abrams hit a bunt single to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	WAS vs ARI on 20230507
	state before	(1, 0, 0, 0, 9, 9, 8, 0)
14.	error	play,9,0,calla002,22,BCBCX,E1/TH/G6MS-.BX3(95)
	state after	(2, 0, 0, 0, 9, 9, 8, 0)
	fielder	1 (pitcher on the home team)

Summary 14 *In the top of the 9th inning, with bases empty, WAS batter Alex Call hit a soft ground ball to shallow short. A throwing error by the pitcher allowed the batter to reach 2nd. However, when the batter attempted to advance to 3rd, he was tagged out by the third baseman (with an assist from the right fielder).*

	game	MIA vs ARI on 20230508
	state before	(1, 1, 0, 0, 6, 1, 2, 1)
15.	error	play,6,1,smitp002,21,BF*BX,S9/L9+.1-3(E9)
	state after	(1, 1, 0, 1, 6, 1, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 15 *In the bottom of the 6th inning, with a runner on 1st, ARI batter Pavin Smith hit a hard line drive single to right field. An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

16.	game	SFN vs ARI on 20230511
	state before	(0, 1, 0, 1, 9, 4, 0, 0)
	error	play,9,0,wade1001,22,CBBSFX,FC4/G4.3-H;1-3(E4/TH);B-2
	state after	(0, 0, 1, 1, 9, 5, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 16 *In the top of the 9th inning, with runners on 1st and 3rd, SFN batter LaMonte Wade hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

17.	game	BOS vs ARI on 20230527
	state before	(0, 0, 0, 0, 6, 2, 1, 0)
	error	play,6,0,duraj001,00,X,S1/BG2.B-2(E1/TH)
	state after	(0, 0, 1, 0, 6, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 17 *In the top of the 6th inning, with bases empty, BOS batter Jarren Duran hit a bunt single to very shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

Visiting/Home team rosters for COL vs ARI on 2023-05-31

position	innings	player	position	innings	player
1	1	Dinelson Lamet	1	1	Tommy Henry
1	4	Peter Lambert	1	8	Austin Adams
1	6	Jake Bird	1	9	Jose Ruiz
1	8	Brent Suter	2	1	Gabriel Moreno
2	1	Austin Wynns	3	1	Christian Walker
3	1	Elehuris Montero	4	1	Ketel Marte
4	1	Alan Trejo	5	1	Josh Rojas
5	1	Ryan McMahon	6	1	Geraldo Perdomo
6	1	Ezequiel Tovar	7	1	Lourdes Gurriel Jr.
7	1	Nolan Jones	8	1	Corbin Carroll
8	1	Brenton Doyle	9	1	Jake McCarthy
9	1	Randal Grichuk			

	game	COL vs ARI on 20230531
	state before	(0, 1, 0, 0, 3, 0, 5, 0)
18.	error	play,3,0,treja001,01,CX,FC4/G4-.1-2(E4);B-1
	state after	(0, 1, 1, 0, 3, 0, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 18 This game had 3 errors. The final score was 0-6.

In the top of the 3rd inning, with a runner on 1st, COL batter Alan Trejo hit a soft ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	COL vs ARI on 20230531
	state before	(2, 1, 1, 0, 4, 0, 5, 1)
19.	error	play,4,1,carrc005,11,CB2,P02(E1/TH).2-3;1-2
	state after	(2, 0, 1, 1, 4, 0, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 19 *In the bottom of the 4th inning, with runners on 1st and 2nd, ARI batter Corbin Carroll has a 1-1 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base. Carroll is still at bat with a 1-1 count.*

	game	COL vs ARI on 20230531
	state before	(2, 1, 0, 0, 5, 0, 6, 1)
20.	error	play,5,1,mccaj003,32,FFBB*B>X,E6/TH/G6.1-3;B-2
	state after	(2, 0, 1, 1, 5, 0, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 20 *In the bottom of the 5th inning, with a runner on 1st, ARI batter Jake McCarthy hit a ground ball to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

4.2.3 June

	game	ATL vs ARI on 20230602
	state before	(0, 0, 0, 0, 6, 1, 3, 1)
21.	error	play,6,1,walkc002,01,CX,C/E2.B-1
	state after	(0, 1, 0, 0, 6, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 21 *In the bottom of the 6th inning, with bases empty, ARI batter Christian Walker had a 0-1 count. Catcher's interference allowed the batter to take 1st.*

	game	ATL vs ARI on 20230604
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
22.	error	play,2,0,ozunm001,02,F1*S>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 22 *In the top of the 2nd inning, with a runner on 1st, ATL batter Marcell Ozuna had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Ozuna is still at bat, with a 1-2 count.*

	game	PHI vs ARI on 20230613
	state before	(1, 0, 1, 0, 1, 2, 0, 0)
23.	error	play,1,0,realj001,01,CX,E5/TH/G5.2-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 1, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 23 *In the top of the 1st inning, with a runner on 2nd, PHI batter J.T. Realmuto hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for PHI vs ARI on 2023-06-14

position	innings	player	position	innings	player
1	1	Ranger Suarez	1	1	Merrill Kelly
1	10	Jose Alvarado	1	7	Kyle Nelson
1	8	Gregory Soto	1	8	Austin Adams
1	8	Seranthony Dominguez	1	9	Scott McGough
1	9	Craig Kimbrel	2	1	Carson Kelly
2	1	J.T. Realmuto	3	1	Christian Walker
3	1	Kody Clemens	4	1	Geraldo Perdomo
4	1	Bryson Stott	5	1	Emmanuel Rivera
5	1	Edmundo Sosa	6	1	Nick Ahmed
6	1	Trea Turner	7	1	Lourdes Gurriel Jr.
7	1	Kyle Schwarber	8	1	Jake McCarthy
8	1	Brandon Marsh	8	9	Corbin Carroll
9	1	Nick Castellanos	9	1	Pavin Smith
			9	9	Jake McCarthy

24.	game	PHI vs ARI on 20230614
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
	error	play,1,0,harpb003,11,SBX,8/L8XD+.2-3(E6/TH);1-2
	state after	(2, 0, 1, 1, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 24 This game had 3 errors, including one in the 10th inning. The final score was 4-3.

In the top of the 1st inning, with runners on 1st and 2nd, PHI batter Bryce Harper hit a hard line drive to extra deep center field. The center fielder put out the batter. A throwing error by the shortstop allowed the runners on 1st and 2nd to each advance one base.

25.	game	PHI vs ARI on 20230614
	state before	(1, 1, 0, 0, 3, 2, 0, 1)
	error	play,3,1,rivee001,12,C*SB1F1X,FC1/G1.1X3(1E4);B-1
	state after	(1, 1, 0, 1, 3, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 25 *In the bottom of the 3rd inning, with a runner on 1st, ARI batter Emmanuel Rivera hit a ground ball into a fielder's choice at shallow infield. An error by the second baseman (with an assist from the pitcher) allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

26.	game	PHI vs ARI on 20230614
	state before	(1, 0, 1, 0, 10, 3, 3, 0)
	error	play,10,0,turnt001,12,FB*SX,E9/F34D.2-3;B-2
	state after	(1, 0, 1, 1, 10, 3, 3, 0)
	fielder	9 (2nd baseman on the visiting team)

Summary 26 *In the top of the 10th inning, with a runner on 2nd, PHI batter Trea Turner hit a fly ball to the 1st/2nd base gap. An error by the right fielder allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd base.*

	game	PHI vs ARI on 20230615
	state before	(2, 1, 0, 0, 7, 5, 4, 1)
27.	error	play,7,1,perdg001,01,C1>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 5, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 27 *In the bottom of the 7th inning, with a runner on 1st, ARI batter Gerardo Perdomo had a 0-1 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Perdomo is still at bat, now with a 0-2 count.*

	game	CLE vs ARI on 20230617
	state before	(2, 0, 0, 0, 1, 1, 0, 1)
28.	error	play,1,1,carrc005,12,CBCFX,E3/G34.B-1
	state after	(2, 1, 0, 0, 1, 1, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 28 *In the bottom of the 1st inning, with bases empty, ARI batter Corbin Carroll hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	CLE vs ARI on 20230618
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
29.	error	play,1,0,kwans001,32,CFBFB.B.X,E6/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 29 *In the top of the 1st inning, with bases empty, CLE batter Steven Kwan hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs ARI on 20230627
	state before	(1, 1, 0, 0, 6, 4, 7, 0)
30.	error	play,6,0,ralel001,11,C1BX,FC6/G4M.1-2(E6);B-1
	state after	(1, 1, 1, 0, 6, 4, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 30 *In the top of the 6th inning, with a runner on 1st, TBA batter Luke Raley hit a ground ball into a fielder's choice at 2nd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

4.2.4 July

Visiting/Home team rosters for PIT vs ARI on 2023-07-07

position	innings	player	position	innings	player
1	1	Rich Hill	1	1	Zac Gallen
1	5	Yohan Ramirez	1	8	Justin Martinez
1	7	Dauri Moreta	1	9	Miguel Castro
2	1	Austin Hedges	2	1	Gabriel Moreno
2	8	Henry Davis	3	1	Christian Walker
3	1	Carlos Santana	4	1	Ketel Marte
4	1	Nick Gonzales	5	1	Evan Longoria
5	1	Jared Triolo	6	1	Nick Ahmed
6	1	Tucupita Marcano	7	1	Lourdes Gurriel Jr.
7	1	Bryan Reynolds	7	9	Corbin Carroll
8	1	Jack Suwinski	8	1	Corbin Carroll
9	1	Henry Davis	8	9	Alek Thomas
9	8	Josh Palacios	9	1	Jake McCarthy

	game	PIT vs ARI on 20230707
	state before	(0, 1, 0, 0, 4, 0, 4, 1)
31.	error	play,4,1,martk001,10,B1,SB2.1-3(E3/TH)
	state after	(0, 0, 0, 1, 4, 0, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 31 This game had 2 errors but 3 stolen bases. The final score was 3-7.

In the bottom of the 4th inning, with a runner on 1st, ARI batter Ketel Marte had a 1-0 count. During the next pitch, which was a no pitch, the runner on 1st stole 2nd. A throwing error by the first baseman allowed that runner to advance to 3rd. Marte is still at bat with a 1-0 count.

	game	PIT vs ARI on 20230707
	state before	(2, 1, 1, 0, 6, 1, 5, 1)
32.	error	play,6,1,walkc002,00,>C,SB3;SB2.2-H(E2/TH) (NR)
	state after	(2, 1, 0, 0, 6, 1, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 32 *In the bottom of the 6th inning, with runners on 1st and 2nd, ARI batter Christian Walker had a 0-0 count. During the next pitch, which was a ball, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Walker is still at bat, now with a 0-1 count.*

	game	PIT vs ARI on 20230708
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
33.	error	play,2,0,santc002,01,FX,E6/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 33 This game had 2 errors, with one in the 10th inning. The final score was 2-3.

In the top of the 2nd inning, with bases empty, PIT batter Carlos Santana hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.

	game	PIT vs ARI on 20230708
	state before	(0, 0, 1, 0, 10, 2, 1, 1)
34.	error	play,10,1,mccaj003,00,...X,FC/SH/BG1S-.2-3;B-1(E1/TH)
	state after	(0, 1, 0, 1, 10, 2, 1, 1)
	fielder	1 (shortstop on the home team)

Summary 34 *In the bottom of the 10th inning, with a runner on 2nd, ARI batter Jake McCarthy hit a sacrifice bunt into a fielder's choice at very shallow infield. The runner on 2nd advanced to 3rd while a throwing error by the pitcher allowed the batter to reach 1st.*

	game	PIT vs ARI on 20230709
	state before	(2, 1, 1, 0, 1, 0, 0, 1)
35.	error	play,1,1,gurrl001,00,.>C,SB3;SB2.2-H(E2/TH)(NR)(UR);1-3
	state after	(2, 0, 0, 1, 1, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 35 This game had 2 errors, The final score was 4-2.

In the bottom of the 1st inning, with runners on 1st and 2nd, ARI batter Lourdes Gurriel had a 0-0 count. During the next pitch, which was a called strike, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Gurriel is still at bat, now with a 0-1 count.

	game	PIT vs ARI on 20230709
	state before	(0, 0, 0, 0, 7, 3, 2, 1)
36.	error	play,7,1,rivee001,22,CCBBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 7, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 36 *In the bottom of the 7th inning, with bases empty, ARI batter Emmanuel Rivera hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SLN vs ARI on 20230724
	state before	(0, 0, 0, 0, 6, 5, 3, 1)
37.	error	play,6,1,gurrl001,00,X,E5/G56.B-1
	state after	(0, 1, 0, 0, 6, 5, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 37 *In the bottom of the 6th inning, with bases empty, ARI batter Lourdes Gurriel hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SLN vs ARI on 20230725
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
38.	error	play,2,0,oneit001,01,1CX,FC6/G6.1-2(E6);B-1
	state after	(2, 1, 1, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 38 *In the top of the 2nd inning, with a runner on 1st, SLN batter Tyler O'Neill hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SEA vs ARI on 20230728
	state before	(1, 0, 1, 1, 7, 5, 0, 1)
39.	error	play,7,1,martk001,12,.CBSB,WP.3-H(NR);2-H(E2/TH)(NR)
	state after	(1, 0, 0, 0, 7, 5, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 39 *In the bottom of the 7th inning, with runners on 2nd and 3rd, ARI batter Ketel Marte hat a 1-2 count. During the next pitch which was a wild pitch (ball), the runner on 3rd scored (no RBI). A throwing error by the catcher allowed the runner on 2nd to also score (unearned, no RBI). Marte is still at bat, now with a 2-2 count.*

	game	SEA vs ARI on 20230730
	state before	(0, 0, 0, 0, 4, 3, 0, 1)
40.	error	play,4,1,walkc002,11,BFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 4, 3, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 40 *In the bottom of the 4th inning, with bases empty, ARI batter Christian Walker hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

4.2.5 August

	game	LAN vs ARI on 20230808
	state before	(0, 1, 0, 0, 5, 1, 0, 0)
41.	error	play,5,0,rosaa003,00,X,6(1)/F0/G6M.B-2(E6/TH)
	state after	(1, 0, 1, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 41 This game had 2 errors. The final score was 5-4.
In the top of the 5th inning, with a runner on 1st, LAN batter Amed

Rosario hit a ground ball to short. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	LAN vs ARI on 20230808
	state before	(0, 0, 0, 0, 7, 4, 0, 0)
42.	error	play,7,0,rosaa003,22,.FSBBFFX,E6/G6D.B-1
	state after	(0, 1, 0, 0, 7, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 42 *In the top of the 7th inning, with bases empty, LAN batter Amed Rosario hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TEX vs ARI on 20230821
	state before	(1, 1, 0, 0, 8, 1, 0, 0)
43.	error	play,8,0,semim001,12,BCCF1,P01(E1/TH).1-3
	state after	(1, 0, 0, 1, 8, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 43 *In the top of the 8th inning, with a runner on 1st, TEX batter Marcus Semien has a 1-2 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner from 1st to advance safely to 3rd. Semien is still at bat with a 1-2 count.*

	game	TEX vs ARI on 20230822
	state before	(2, 1, 1, 0, 2, 0, 2, 1)
44.	error	play,2,1,carrc005,32,SF*BB.B>X,S1/G3S.2-H;1-3(E1/TH)
	state after	(2, 1, 0, 1, 2, 0, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 44 *In the bottom of the 2nd inning, with runners on 1st and 2nd, ARI batter Corbin Carroll hit a ground ball single to shallow 1st. The runner from 2nd scored (1 RBI) while a throwing error by the pitcher allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	CIN vs ARI on 20230825
	state before	(1, 1, 1, 1, 6, 4, 5, 1)
45.	error	play,6,1,walkc002,02,SS.FX,6/P6D.3-H;2-3(E6/TH);1-2
	state after	(2, 0, 1, 1, 6, 4, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 45 *In the bottom of the 6th inning, with bases loaded, ARI batter Christian Walker hit a pop fly to deep short. The shortstop put out the batter. The runner on 3rd scored (1 RBI) while a throwing error by the shortstop allowed the runners on 1st and 2nd to each advance one base.*

Visiting/Home team rosters for CIN vs ARI on 2023-08-26

position	innings	player	position	innings	player
1	1	Fernando Cruz	1	1	Zach Davies
1	10	Alexis Diaz	1	10	Miguel Castro
1	11	Lucas Sims	1	10	Nabil Crismatt
1	2	Sam Moll	1	6	Scott McGough
1	3	Ben Lively	1	7	Kyle Nelson
1	9	Ian Gibaut	1	8	Kevin Ginkel
2	1	Tyler Stephenson	1	9	Joe Mantiply
2	9	Luke Maile	1	9	Paul Sewald
3	1	Christian Encarnacion-Strand	2	1	Jose Herrera
4	1	Matt McLain	3	1	Christian Walker
5	1	Noelvi Marte	4	1	Ketel Marte
6	1	Elly De La Cruz	5	1	Jace Peterson
7	1	Spencer Steer	6	1	Geraldo Perdomo
8	1	TJ Friedl	7	1	Lourdes Gurriel Jr.
9	1	Will Benson	8	1	Alek Thomas
			9	1	Corbin Carroll

	game	CIN vs ARI on 20230826
	state before	(1, 0, 0, 0, 3, 0, 1, 1)
46.	error	play,3,1,perdg001,12,CBFX,E4/G4.B-1
	state after	(1, 1, 0, 0, 3, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 46 This game had 3 errors, including one in the 10th inning. The final score was 8-7.

In the bottom of the 3rd inning, with bases empty, ARI batter Gerardo Perdomo hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.

	game	CIN vs ARI on 20230826
	state before	(2, 0, 0, 0, 4, 0, 1, 1)
47.	error	play,4,1,thoma004,21,.BCBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 4, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 47 *In the bottom of the 4th inning, with bases empty, ARI batter Alek Thomas hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CIN vs ARI on 20230826
	state before	(1, 1, 1, 0, 10, 7, 5, 1)
48.	error	play,10,1,phamt001,01,CX,E4/G89S.2-H(NR)(UR);1-3;B-1
	state after	(1, 1, 0, 1, 10, 7, 6, 1)
	fielder	4 (shortstop on the visiting team)

Summary 48 *In the bottom of the 10th inning, with runners on 1st and 2nd, ARI batter Tommy Pham hit a ground ball to shallow right center field. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

4.2.6 September

	game	BAL vs ARI on 20230901
	state before	(1, 0, 0, 0, 8, 2, 4, 1)
49.	error	play,8,1,longe001,00,X,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 8, 2, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 49 *In the bottom of the 8th inning, with bases empty, ARI batter Evan Longoria hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	BAL vs ARI on 20230903
	state before	(2, 0, 0, 0, 4, 4, 2, 0)
50.	error	play,4,0,fraza001,01,CX,S8/L89.B-2(E8)
	state after	(2, 0, 1, 0, 4, 4, 2, 0)
	fielder	8 (center field on the home team)

Summary 50 *In the top of the 4th inning, with bases empty, BAL batter Adam Frazier hit a line drive single to right center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	COL vs ARI on 20230905
	state before	(2, 0, 1, 0, 3, 1, 1, 0)
51.	error	play,3,0,jonen002,32,BBCFBFF>B,W+SB3.2-H(E2/TH) (NR)
	state after	(2, 1, 0, 0, 3, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 51 *In the top of the 3rd inning, with a runner on 2nd, COL batter Nolan Jones had a full count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI). Jones took 1st on a walk.*

Visiting/Home team rosters for CHN vs ARI on 2023-09-16

position	innings	player	position	innings	player
1	1	Kyle Hendricks	1	1	Zach Davies
1	10	Marcus Stroman	1	10	Andrew Saalfrank
1	11	Drew Smyly	1	12	Joe Mantiplay
1	12	Drew Smyly	1	5	Luis Frias
1	13	Hayden Wesneski	1	6	Miguel Castro
1	6	Jose Cuas	1	7	Ryan Thompson
1	7	Mark Leiter Jr.	1	8	Kevin Ginkel
1	8	Julian Merryweather	1	9	Paul Sewald
2	1	Yan Gomes	2	1	Gabriel Moreno
3	1	Cody Bellinger	3	1	Christian Walker
4	1	Nico Hoerner	3	12	Emmanuel Rivera
5	1	Nick Madrigal	4	1	Ketel Marte
5	10	Miles Mastrobuoni	5	1	Jace Peterson
5	11	Patrick Wisdom	5	12	Geraldo Perdomo
6	1	Dansby Swanson	6	1	Geraldo Perdomo
7	1	Ian Happ	6	12	Jordan Lawlar
8	1	Mike Tauchman	7	1	Lourdes Gurriel Jr.
8	12	Christopher Morel	7	12	Corbin Carroll
8	6	Pete Crow-Armstrong	8	1	Alek Thomas
9	1	Seiya Suzuki	9	1	Corbin Carroll
			9	12	Jake McCarthy

	game	CHN vs ARI on 20230916
	state before	(0, 0, 0, 0, 7, 3, 3, 0)
52.	error	play,7,0,happi001,02,.CFX,E3/G3.B-1
	state after	(0, 1, 0, 0, 7, 3, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 52 This game had 3 errors, one in the 10th inning. The final score was 6-7.

In the top of the 7th inning, with bases empty, CHN batter Ian Happ hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.

	game	CHN vs ARI on 20230916
	state before	(0, 0, 0, 0, 9, 3, 3, 1)
53.	error	play,9,1,petej002,01,CX,E6/G4.B-1
	state after	(0, 1, 0, 0, 9, 3, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 53 *In the bottom of the 9th inning, with bases empty, ARI batter Jace Peterson hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	CHN vs ARI on 20230916
	state before	(0, 1, 1, 0, 10, 3, 3, 0)
54.	error	play,10,0,bellc002,02,CF>B,SB3;SB2.2-H(E2/TH)(NR)(UR);1-3
	state after	(0, 0, 0, 1, 10, 4, 3, 0)
	fielder	2 (shortstop on the visiting team)

Summary 54 *In the top of the 10th inning, with runners on 1st and 2nd, CHN batter Cody Bellinger had a 0-2 count. During the next pitch, which was a ball, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Bellinger is still at bat, now with a 1-2 count.*

	game	CHN vs ARI on 20230917
	state before	(0, 0, 0, 0, 4, 2, 3, 1)
55.	error	play,4,1,moreg001,22,BCCBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 2, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 55 *In the bottom of the 4th inning, with bases empty, ARI batter Gabriel Moreno hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

56.	game	SFN vs ARI on 20230919
	state before	(2, 1, 1, 0, 2, 2, 3, 1)
	error	play,2,1,phamt001,00,>B,SB3;SB2.2-H(E5)(NR);1-H(E5/TH)(NR)
	state after	(2, 0, 0, 0, 2, 2, 5, 1)
	fielder	5,5 (3rd baseman on the visiting team)

Summary 56 This play had 2 errors by the same fielder and also had 2 stolen bases by different runners. The final score was 4-8.

In the bottom of the 2nd inning, with runners on 1st and 2nd, ARI batter Tommy Pham had a 0-0 count. During the next pitch, which was a ball, the runners on 1st and 2nd stole the next base. An error by the third baseman allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). A throwing error by the same third baseman allowed the runner from 1st (now on 2nd) to also score (unearned, no RBI). Pham is still at bat, now with a 1-0 count.

57.	game	SFN vs ARI on 20230920
	state before	(1, 1, 0, 0, 3, 1, 1, 1)
	error	play,3,1,martk001,20,BB>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 57 *In the bottom of the 3rd inning, with a runner on 1st, ARI batter Ketel Marte had a 2-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Marte is still at bat, now with a 3-0 count.*

	game	HOU vs ARI on 20230929
	state before	(1, 0, 0, 0, 9, 2, 0, 1)
58.	error	play,9,1,thoma004,00,X,E4/G34.B-1
	state after	(1, 1, 0, 0, 9, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 58 *In the bottom of the 9th inning, with bases empty, ARI batter Alek Thomas hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

4.2.7 October

	game	HOU vs ARI on 20231001
	state before	(1, 1, 1, 0, 1, 2, 0, 0)
59.	error	play,1,0,branm003,32,BBCBCFFFX,FC3/G3.2-H(NR)(UR);1-3(E3/TH);B-1
	state after	(1, 1, 0, 1, 1, 3, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 59 *In the top of the 1st inning, with runners on 1st and 2nd, HOU batter Michael Brantley hit a ground ball into a fielder's choice at 1st. The runner on 2nd scored (unearned, no RBI) while a throwing error by the firstbaseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

Chapter 5

Atlanta Braves (ATL)

Since 1966, the Atlanta Braves competed in Atlanta in the National League, but it has only been since 1994 that they compete in the National League East Division. Since 1994 they have won 4 NL pennants, the last in 2021, and two World Series titles, the last also in 2021.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Atlanta against the Braves. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (DER or DefEff) is one measure of the fielding performance of a team (see §3.2 for more details). The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 ATL team is .685 a bit lower than the MLB average of 0.691 that year [BR]. In other words, the Atlanta Braves have a DER lower than average.

As we will see in the next section, there were 75 errors in the 81 ATL 2023 home games, 42 by the Atlanta Braves and the rest (33) by visiting teams. Over the regular season, home and away, the Braves had 82 errors, close to the MLB average that year of 83.9.

5.1 Season totals

Throughout, `vt` denotes the visiting team and `ht` denotes the home team.

5.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	33
Runs by ht because of errors by vt:	11
Errors by ht:	42
Runs by vt because of errors by ht:	12

This is a good error-to-run conversion rate: $11/33 = 33.3\ldots\%$ on the part of the Braves (compared to $12/42 = 28.5\ldots\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 ATL season (home and away) has about $R/PA = 947/6249$ (about 15.1%) change of resulting in a run. Therefore, a randomly selected error is about twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	3	5
to catcher:	4	7
to 1st baseman:	2	5
to 2nd baseman:	4	4
to 3rd baseman:	7	5
to shortstop:	7	10
to left fielder:	2	2
to center fielder:	1	0
to right fielder:	3	4

5.1.2 Error total by state

This section tabulates the totals by game state.

Errors by state	vt	ht	totals
bases empty	11	17	28
(lone) runner on 1st	11	10	21
(lone) runner on 2nd	1	2	3
(lone) runner on 3rd	0	2	2
two runners on base	9	9	18
bases loaded	1	2	3
total	33	42	75

A table of frequencies of most common errors by the Braves in ATL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	10	23.8%
(1, 0, 0, 0)	3	7.1%
(2, 0, 0, 0)	4	9.5%
(0, 1, 0, 0)	6	14.3%
(1, 1, 0, 0)	2	4.8%
(2, 1, 0, 0)	2	4.8%
(1, 0, 1, 0)	2	4.8%
(0, 1, 1, 0)	4	9.5%
(1, 0, 1, 1)	2	4.8%

A table of frequencies of most common errors by a visiting team in ATL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	18.2%
(1, 0, 0, 0)	5	15.1%
(2, 0, 0, 0)	0	0%
(0, 1, 0, 0)	4	12.1%
(1, 1, 0, 0)	3	9.1%
(2, 1, 0, 0)	4	12.1%
(2, 1, 1, 0)	3	9.1%
(1, 1, 0, 1)	3	9.1%
(2, 0, 1, 1)	1	3%
(0, 1, 1, 1)	1	3%

Recall Figure 4.1 gave a frequency table of game states in all plays. From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Braves homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `ATL2023.evn`):

state	ATL2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	16.8%
(2, 0, 0, 0)	13.3%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	6.7%
(0, 0, 1, 0)	2.4%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	3.3%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	1.1%
(2, 0, 0, 1)	1.8%
(0, 1, 1, 0)	1.1%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.9%

5.1.3 Some interesting errors

Some of the most interesting error plays in the ATL 2023 home games:

- The game SDN vs ATL on 2023-04-08 had only 2 errors but one of them was for a fascinating appeal play.
- The game PHI vs ATL on 2023-05-28 had 3 errors but 2 were from a double-error play.
- The game MIN vs ATL on 2023-06-27 had 34 errors, including one play with 2 errors by 2 different fielders.
- The game WAS vs ATL on 2023-09-29 had 5 errors, including one play with 2 errors by 2 different fielders.

5.1.4 Relative team rankings

We have 81 home games between ATL and some visiting MLB team.

ATL, the home team (ht), *lost* more games than won against¹:

ARI (2/3), CHA (2/3), LAN (2/3), SLN (2/3), HOU (3/3), SDN (3/4), PHI (4/7).

The teams that *split* their series were:

BOS (1/2), WAS (3/6),

and *won* more than lost against

CHN (0/3), CIN (0/3), MIL (0/3), MIN (0/3), NYA (0/3), COL (0/4),
ANA (1/3), BAL (1/3), PIT (1/3), SEA (1/3), SFN (1/3), NYN (1/6), MIA (1/7).

The teams that did not play in any of the 81 home games in ATL 2023 were:

CLE, DET, KCA, OAK, TBA, TEX, TOR.

¹Visiting teams (vt) are listed with that team's W/T record in parentheses (W = numbers of games vt won against ht at ATL, T = Total home games vt played vs ht at ATL).

5.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for ATL 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Sean Murphy	102	.365
3 (1B)	Matt Olson	164	.389
4 (2B)	Ozzie Albies	148	.336
5 (3B)	Austin Riley	160	.345
6 (SS)	Orlando Arcia	140	.321
7 (LF)	Eddie Rosario	305	.305
8 (CF)	Michael Harris II	331	.331
9 (RF)	Ronald Acuña J	416	.416

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 ATL home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

5.2.1 April

	game	SDN vs ATL on 20230406
	state before	(2, 0, 0, 1, 8, 5, 4, 0)
1.	error	play,8,0,azocj001,11,CB.X,S1/BG25-.3-H;B-3(E1/TH)
	state after	(2, 0, 0, 1, 8, 5, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 60 *In the top of the 8th inning, with a runner on 3rd, SDN batter Jose Azocar hit a soft bunt single to the shallow infield. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the batter to reach 3rd.*

	game	SDN vs ATL on 20230407
	state before	(0, 0, 0, 0, 8, 5, 4, 1)
2.	error	play,8,1,arcio002,01,CX,E6/TH/G6+.B-1
	state after	(0, 1, 0, 0, 8, 5, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 61 *In the bottom of the 8th inning, with bases empty, ATL batter Orlando Arcia hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	SDN vs ATL on 20230408
	state before	(1, 1, 0, 0, 4, 1, 0, 0)
3.	error	play,4,0,odorr001,12,FFB>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 4, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 62 This game had 2 errors. The final score was 4-1. *In the top of the 4th inning, with a runner on 1st, SDN batter Rougned Odor had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Odor is still at bat, now with a 2-2 count.*

	game	SDN vs ATL on 20230408
	state before	(1, 0, 1, 1, 4, 2, 0, 0)
4.	error	play,4,0,grist001,20,BBX,FC3/G34.3XH(12);2-H(E2)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 3, 0, 0)
	fielder	2 (catcher on the home team)

Summary 63 This play is one of the most interesting ones of the year! *In the top of the 4th inning, with runners on 2nd and 3rd, SDN batter Trent Grisham hit a ground ball into a fielder's choice at the 1st/2nd base gap. At home plate, the catcher (with an assist from the first*

baseman)² tried to tag out the runner from 3rd. An error by the catcher³ allowed the runner on 2nd to score and the batter to reach 2nd.

	game	CIN vs ATL on 20230411
	state before	(2, 1, 1, 1, 3, 1, 1, 0)
5.	error	play,3,0,stees001,12,BCFX,S4/G34.3-H;2-H;1-3(E2)
	state after	(2, 1, 0, 1, 3, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 64 *In the top of the 3rd inning, with bases loaded, CIN batter Spencer Steer hit a ground ball single to the 1st/2nd base gap. The runners on 2nd and 3rd scored (2 RBIs). An error by the catcher allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	HOU vs ATL on 20230421
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
6.	error	play,1,0,dubom001,21,BFBX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 65 This game had 2 errors. The final score was 6-4. *In the top of the 1st inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

²Video shows that the runner from 3rd (Odor) was out at home plate on an appeal. The catcher thought he had the ball (relayed from the first baseman) and so blocked the plate hoping to tag Odor. (Was this blocking the plate interference, so MLB Rule 6.01(i)(2) applies? Possibly the Official Scorer ruled that this Rule does not apply because Odor could have avoided the collision by sliding into home plate?) However, the ball never made it into the catcher's glove. The runner from 2nd (Kim) scored. *After that*, Odor came back to touch home plate (this violates MLB Rule 5.09(c)). For the appeal: The pitcher got the ball, tossed it to the catcher, who touched home plate. Then the Umpire ruled Odor out.

³Probably this is due to the Official Scorer ruling that the catcher should have caught to throw from the first baseman.

	game	HOU vs ATL on 20230421
	state before	(0, 1, 0, 0, 7, 1, 4, 0)
7.	error	play,7,0,diazy004,00,X,D9/F89XD+.1-H(E9)(NR);B-3
	state after	(0, 0, 0, 1, 7, 2, 4, 0)
	fielder	9 (right field on the home team)

Summary 66 *In the top of the 7th inning, with a runner on 1st, HOU batter Yainer Diaz hit a hard fly ball double to extra deep right center field. An error by the right fielder allowed the runner from 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	HOU vs ATL on 20230422
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
8.	error	play,3,0,dubom001,10,BX,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 67 This game had 3 errors. The final score was 6-3. *In the top of the 3rd inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.*

	game	HOU vs ATL on 20230422
	state before	(0, 0, 0, 0, 6, 2, 1, 0)
9.	error	play,6,0,abrej003,11,BFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 68 *In the top of the 6th inning, with bases empty, HOU batter Jose Abreu hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.*

	game	HOU vs ATL on 20230422
	state before	(2, 1, 0, 0, 9, 5, 3, 0)
10.	error	play,9,0,alvay001,12,SCBX,E4/G34.1-3;B-1
	state after	(2, 1, 0, 1, 9, 5, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 69 *In the top of the 9th inning, with a runner on 1st, HOU batter Yordan Alvarez hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	HOU vs ATL on 20230423
	state before	(0, 1, 1, 0, 6, 0, 1, 1)
11.	error	play,6,1,rilea001,01,CX,54(1)/F0/G56.2-H(E4/TH)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 6, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 70 *In the bottom of the 6th inning, with runners on 1st and 2nd, ATL batter Austin Riley hit a ground ball to short. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for MIA vs ATL on 2023-04-24

position	innings	player	position	innings	player
1	1	Edward Cabrera	1	1	Spencer Strider
1	5	Steven Okert	1	9	Joe Jimenez
1	6	Sean Nolin	2	1	Sean Murphy
2	1	Jacob Stallings	3	1	Matt Olson
3	1	Garrett Cooper	4	1	Ozzie Albies
4	1	Jon Berti	5	1	Austin Riley
5	1	Jean Segura	6	1	Ehire Adrianza
6	1	Garrett Hampson	7	1	Eddie Rosario
7	1	Bryan De La Cruz	8	1	Sam Hilliard
7	8	Avisail Garcia	9	1	Ronald Acuna Jr.
8	1	Jazz Chisholm Jr.			
8	8	Bryan De La Cruz			
9	1	Avisail Garcia			
9	8	Jesus Sanchez			

12.	game	MIA vs ATL on 20230424
	state before	(0, 1, 0, 0, 2, 0, 0, 1)
	error	play,2,1,albio001,22,BS1FBX,FC4/G34.1-3(E4/TH);B-2
	state after	(0, 0, 1, 1, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 71 This game had 4 errors. The final score was 0-11.

In the bottom of the 2nd inning, with a runner on 1st, ATL batter Ozzie Albies hit a ground ball into a fielder's choice at 1st. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	MIA vs ATL on 20230424
	state before	(0, 0, 0, 0, 7, 0, 8, 0)
13.	error	play,7,0,chisj001,20,BBX,E3/G3.B-1
	state after	(0, 1, 0, 0, 7, 0, 8, 0)
	fielder	3 (1st baseman on the home team)

Summary 72 *In the top of the 7th inning, with bases empty, MIA batter Jazz Chisholm hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	MIA vs ATL on 20230424
	state before	(0, 0, 0, 0, 7, 0, 8, 1)
14.	error	play,7,1,albio001,22,BBFFX,S5/G5.B-2(E5/TH)
	state after	(0, 0, 1, 0, 7, 0, 8, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 73 *In the bottom of the 7th inning, with bases empty, ATL batter Ozzie Albies hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	MIA vs ATL on 20230424
	state before	(2, 0, 0, 0, 9, 0, 11, 0)
15.	error	play,9,0,gurry001,02,.FFX,E5/G56.B-1
	state after	(2, 1, 0, 0, 9, 0, 11, 0)
	fielder	5 (3rd baseman on the home team)

Summary 74 *In the top of the 9th inning, with bases empty, MIA batter Yulieski Gurriel hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	MIA vs ATL on 20230426
	state before	(1, 1, 0, 1, 8, 4, 5, 1)
16.	error	play,8,1,ozunm001,11,*BC>X,FC1/G1.3-H(E1/TH)(NR)(UR);1-3;B-2
	state after	(1, 0, 1, 1, 8, 4, 6, 1)
	fielder	1 (pitcher on the visiting team)

Summary 75 *In the bottom of the 8th inning, with runners on 1st and 3rd, ATL batter Marcell Ozuna hit a ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	MIA vs ATL on 20230427
	state before	(0, 1, 1, 1, 5, 0, 2, 1)
17.	error	play,5,1,acunr001,11,F*BX,FC4/G4M.3-H;2-3;1X2(4E6);B-1
	state after	(0, 1, 1, 1, 5, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 76 *In the bottom of the 5th inning, with bases loaded, ATL batter Ronald Acuna hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to 3rd. An error by the shortstop (on a relay from the second baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

5.2.2 May

	game	BAL vs ATL on 20230505
	state before	(0, 1, 0, 0, 3, 0, 1, 0)
18.	error	play,3,0,hendg002,11,CB1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 3, 0, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 77 This game had 2 errors. The final score was 9-4.

In the top of the 3rd inning, with a runner on 1st, BAL batter Gunnar Henderson has a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to advance to 3rd.

	game	BAL vs ATL on 20230505
	state before	(0, 1, 1, 0, 7, 3, 1, 0)
19.	error	play,7,0,matej003,11,C*BX,FC/SH/BG1S-.2-H(NR);1-3;B-2(E1/TH)
	state after	(0, 1, 1, 0, 7, 3, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 78 *In the top of the 7th inning, with runners on 1st and 2nd, BAL batter Jorge Mateo hit a soft bunt to shallow infield. The runner on 2nd scored (no RBI) and the runner on 1st to advance to 3rd. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	BAL vs ATL on 20230506
	state before	(1, 0, 1, 0, 6, 3, 3, 0)
20.	error	play,6,0,hendg002,12,CBCX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 6, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 79 This game had 2 errors. The final score was 4-5.

In the top of the 6th inning, with a runner on 2nd, BAL batter Gunnar Henderson hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	BAL vs ATL on 20230506
	state before	(2, 1, 0, 0, 8, 4, 5, 1)
21.	error	play,8,1,harrm004,11,BS1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 4, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 80 *In the bottom of the 8th inning, with a runner on 1st, ATL batter Michael Harris hit a single had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Harris is still at bat, now with a 2-1 count.*

	game	BAL vs ATL on 20230507
	state before	(1, 0, 0, 0, 2, 0, 1, 1)
22.	error	play,2,1,ozunm001,21,.BFBX,E5/TH/G5.B-1
	state after	(1, 1, 0, 0, 2, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 81 *This game had 2 errors. The final score was 2-3. In the bottom of the 2nd inning, with bases empty, ATL batter Marcell Ozuna hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	BAL vs ATL on 20230507
	state before	(1, 0, 0, 0, 3, 0, 1, 0)
23.	error	play,3,0,mccaj001,12,FCBFX,E5/TH/G5.B-1
	state after	(1, 1, 0, 0, 3, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 82 *In the top of the 3rd inning, with bases empty, BAL batter James McCann hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SEA vs ATL on 2023-05-20

position	innings	player	position	innings	player
1	1	Logan Gilbert	1	1	Jesse Chavez
1	7	Matt Brash	1	3	Michael Tonkin
1	8	Justin Topa	1	5	Joe Jimenez
1	9	Paul Sewald	1	7	Kirby Yates
2	1	Tom Murphy	1	8	Lucas Luetge
3	1	Ty France	1	9	A.J. Minter
4	1	Jose Caballero	2	1	Sean Murphy
5	1	Eugenio Suarez	3	1	Matt Olson
6	1	J.P. Crawford	4	1	Ozzie Albies
7	1	Taylor Trammell	5	1	Austin Riley
8	1	Julio Rodriguez	6	1	Orlando Arcia
9	1	Jarred Kelenic	7	1	Eddie Rosario
			8	1	Michael Harris II
			9	1	Ronald Acuna Jr.

	game	SEA vs ATL on 20230520
	state before	(0, 1, 1, 0, 4, 0, 2, 0)
24.	error	play,4,0,hernt002,12,CS*BX,FC6/G6.2-3;1-2(E6);B-1
	state after	(0, 1, 1, 1, 4, 0, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 83 This game had 3 errors. The final score was 7-3.

In the top of the 4th inning, with runners on 1st and 2nd, SEA batter Teoscar Hernandez hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	SEA vs ATL on 20230520
	state before	(2, 1, 0, 1, 4, 2, 2, 0)
25.	error	play,4,0,crawj002,01,FX,S9/L89.3-H(UR);1-3(E9)
	state after	(2, 1, 0, 1, 4, 3, 2, 0)
	fielder	9 (right field on the home team)

Summary 84 *In the top of the 4th inning, with runners on 1st and 3rd, SEA batter J.P. Crawford hit a line drive single to right center field. The runner on 3rd score (1 RBI). An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	SEA vs ATL on 20230520
	state before	(0, 1, 0, 0, 5, 3, 2, 0)
26.	error	play,5,0,kelej001,10,1BX,S9/L9.1-3;B-2(E9/TH)
	state after	(0, 0, 1, 1, 5, 3, 2, 0)
	fielder	9 (right field on the home team)

Summary 85 *In the top of the 5th inning, with a runner on 1st, SEA batter Jarred Kelenic hit a line drive single to right field. The runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	LAN vs ATL on 20230523
	state before	(2, 1, 0, 0, 1, 1, 0, 1)
27.	error	play,1,1,rilea001,02,CCX,D8/L78D+.1-H;B-3(E6/TH)
	state after	(2, 0, 0, 1, 1, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 86 This game had 3 errors. The final score was 8-1. *In the bottom of the 1st inning, with a runner on 1st, ATL batter Austin Riley hit a hard line drive double to left center field. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 3rd.*

	game	LAN vs ATL on 20230523
	state before	(2, 0, 0, 0, 2, 2, 1, 0)
28.	error	play,2,0,rojam002,01,TX,E3/G3.B-1
	state after	(2, 1, 0, 0, 2, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 87 *In the top of the 2nd inning, with bases empty, LAN batter Miguel Rojas hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	LAN vs ATL on 20230523
	state before	(0, 0, 0, 0, 8, 5, 1, 0)
29.	error	play,8,0,perad001,02,.CFX,E3/G23-.B-1
	state after	(0, 1, 0, 0, 8, 5, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 88 *In the top of the 8th inning, with bases empty, LAN batter David Peralta hit a soft ground ball to shallow left infield. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for PHI vs ATL on 2023-05-28

position	innings	player	position	innings	player
1	1	Dylan Covey	1	1	Spencer Strider
1	1	Jeff Hoffman	1	7	Collin McHugh
1	3	Yunior Marte	1	8	Dereck Rodriguez
1	5	Connor Brogdon	2	1	Sean Murphy
1	6	Matt Strahm	3	1	Matt Olson
1	7	Andrew Vasquez	4	1	Ozzie Albies
2	1	J.T. Realmuto	5	1	Austin Riley
3	1	Kody Clemens	6	1	Orlando Arcia
4	1	Bryson Stott	7	1	Eddie Rosario
5	1	Edmundo Sosa	7	9	Kevin Pillar
6	1	Trea Turner	8	1	Michael Harris II
7	1	Kyle Schwarber	9	1	Ronald Acuna Jr.
8	1	Brandon Marsh			
9	1	Nick Castellanos			

30.	game	PHI vs ATL on 20230528
	state before	(1, 1, 0, 0, 1, 0, 5, 1)
	error	play,1,1,arcio002,32,CB*BBF1>X,FC6/G6.1-3(E6)(E7);B-2
	state after	(1, 0, 1, 1, 1, 0, 5, 1)
	fielder	6, 7 (shortstop, left field on the visiting team)

Summary 89 This game had 3 errors. This particular play had 2 errors, credited to 2 different players. The final score was 4-11.

In the bottom of the 1st inning, with a runner on 1st, ATL batter Orlando Arcia hit a ground ball into a fielder's choice at short. Error by both the shortstop and the right fielder allowed the runner on 1st to reach 3rd and the batter to reach 2nd.

31.	game	PHI vs ATL on 20230528
	state before	(2, 1, 1, 0, 4, 2, 7, 1)
	error	play,4,1,ozunm001,01,CX,E5/G1S.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 4, 2, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 90 *In the bottom of the 4th inning, with runners on 1st and 2nd, ATL batter Marcell Ozuna hit a ground ball to shallow infield. An error by the third baseman allowed the runners on 1st and 2nd to each advance a base and the batter to reach 1st. The bases are loaded.*

5.2.3 June

32.	game	NYN vs ATL on 20230606
	state before	(0, 0, 0, 0, 7, 4, 5, 0)
	error	play,7,0,marts002,22,.CBBTFFX,E5/G56.B-1
	state after	(0, 1, 0, 0, 7, 4, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 91 This game had 2 errors. The final score was 4-6. *In the top of the 7th inning, with bases empty, NYN batter Starling Marte hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

33.	game	NYN vs ATL on 20230606
	state before	(1, 0, 0, 0, 8, 4, 5, 1)
	error	play,8,1,rosae001,12,FFFBX,D7/F7LD.B-3(E7)
	state after	(1, 0, 0, 1, 8, 4, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 92 *In the bottom of the 8th inning, with bases empty, ATL batter Eddie Rosario hit a fly ball double to deep left field. An error by the left fielder allowed the batter to reach 3rd.*

	game	NYN vs ATL on 20230607
	state before	(1, 0, 0, 0, 3, 2, 0, 1)
34.	error	play,3,1,harrm004,21,BBCX,S2/BG2.B-2(E2/TH)
	state after	(1, 0, 1, 0, 3, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 93 *In the bottom of the 3rd inning, with bases empty, ATL batter Michael Harris hit a bunt single to very shallow infield. A throwing error by the catcher allowed the batter to reach 2nd.*

	game	NYN vs ATL on 20230608
	state before	(1, 1, 0, 0, 1, 0, 2, 1)
35.	error	play,1,1,rosae001,21,B*BFX,S9/L9.1-H(E9)(NR)(UR);B-3
	state after	(1, 0, 0, 1, 1, 0, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 94 *In the bottom of the 1st inning, with a runner on 1st, ATL batter Eddie Rosario hit a line drive single to right field. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

Visiting/Home team rosters for WAS vs ATL on 2023-06-09

position	innings	player	position	innings	player
1	1	Josiah Gray	1	1	AJ Smith-Shawver
1	6	Mason Thompson	1	6	Collin McHugh
1	7	Carl Edwards Jr.	1	8	Kirby Yates
1	8	Kyle Finnegan	1	9	Raisel Iglesias
2	1	Keibert Ruiz	2	1	Sean Murphy
3	1	Dominic Smith	2	9	Travis d'Arnaud
4	1	Luis Garcia	3	1	Matt Olson
5	1	Jeimer Candelario	4	1	Ozzie Albies
6	1	CJ Abrams	5	1	Austin Riley
7	1	Corey Dickerson	6	1	Orlando Arcia
7	7	Stone Garrett	7	1	Eddie Rosario
8	1	Alex Call	8	1	Michael Harris II
9	1	Lane Thomas	9	1	Ronald Acuna Jr.

	game	WAS vs ATL on 20230609
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
36.	error	play,1,0,garc1006,22,SB*BF>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 95 This game had 3 errors. The final score was 2-3.

In the top of the 1st inning, with a runner on 1st, WAS batter Luis Garcia had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Garcia is still at bat, now with a full count.

	game	WAS vs ATL on 20230609
	state before	(0, 0, 0, 0, 6, 1, 1, 0)
37.	error	play,6,0,thoml002,32,BBCF.BX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 96 *In the top of the 6th inning, with bases empty, WAS batter Lane Thomas hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	WAS vs ATL on 20230609
	state before	(0, 1, 0, 0, 8, 2, 1, 1)
38.	error	play,8,1,rosae001,02,.FFX,S9/G3.1-3;B-2(E9)
	state after	(0, 0, 1, 1, 8, 2, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 97 *In the bottom of the 8th inning, with a runner on 1st, ATL batter Eddie Rosario hit a ground ball single to 1st. The runner on 1st advanced to 3rd. An error by the right fielder allowed the batter to reach 2nd.*

	game	COL vs ATL on 20230618
	state before	(0, 1, 1, 0, 2, 2, 0, 0)
39.	error	play,2,0,casth001,02,FCX,E6/G8.2-H(NR);1-3;B-2
	state after	(0, 0, 1, 1, 2, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 98 *In the top of the 2nd inning, with runners on 1st and 2nd, COL batter Harold Castro hit a ground ball to center field. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

40.	game	MIN vs ATL on 20230626
	state before	(1, 1, 0, 1, 4, 1, 0, 1)
	error	play,4,1,darnt001,00,X,64(1)/F0/G6.3-H;B-2(E3)
	state after	(2, 0, 0, 1, 4, 1, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 99 *In the bottom of the 4th inning, with runners on 1st and 3rd, ATL batter Travis d’Arnaud hit a ground ball to short. The second baseman (with an assist from the shortstop) forced out the runner from 1st. The runner on 3rd scored (1 RBI). An error by the first baseman allowed the batter to reach 2nd.*

Visiting/Home team rosters for MIN vs ATL on 2023-06-27

position	innings	player	position	innings	player
1	1	Joe Ryan	1	1	Bryce Elder
1	4	Oliver Ortega	1	7	Michael Tonkin
1	7	Brent Headrick	1	9	Ben Heller
2	1	Christian Vazquez	2	1	Sean Murphy
3	1	Alex Kirilloff	3	1	Matt Olson
4	1	Edouard Julien	4	1	Ozzie Albies
5	1	Royce Lewis	5	1	Austin Riley
6	1	Carlos Correa	6	1	Orlando Arcia
7	1	Joey Gallo	7	1	Eddie Rosario
8	1	Michael A. Taylor	7	9	Kevin Pillar
9	1	Max Kepler	8	1	Michael Harris II
			9	1	Ronald Acuna Jr.

41.	game	MIN vs ATL on 20230627
	state before	(1, 0, 1, 1, 1, 0, 0, 0)
	error	play,1,0,buxtb001,22,CSBBX,E6/G6.3-H;B-1
	state after	(1, 1, 1, 0, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 100 This game had 4 errors, including a double-error play. The final score was 2-6.

In the top of the 1st inning, with runners on 2nd and 3rd, MIN batter Byron Buxton hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st. The runner on 3rd scored (1 RBI).

	game	MIN vs ATL on 20230627
	state before	(1, 0, 0, 0, 2, 1, 4, 0)
42.	error	play,2,0,gallj002,00,X,S7/L78D.B-3(E7/TH)(E2/TH)
	state after	(1, 0, 0, 1, 2, 1, 4, 0)
	fielder	7, 2 (left field, catcher on the home team)

Summary 101 This play had 2 errors, by 2 different fielders.

In the top of the 2nd inning, with bases empty, MIN batter Joey Gallo hit a line drive single to deep left center field. A throwing error by the left fielder and a throwing error by the catcher allowed the batter to reach 3rd.

	game	MIN vs ATL on 20230627
	state before	(1, 1, 1, 1, 4, 1, 6, 0)
43.	error	play,4,0,julie001,12,SCBX,7/SF/F7L.3-H;2-3;1-2(E7/TH)
	state after	(2, 0, 1, 1, 4, 2, 6, 0)
	fielder	7 (left field on the home team)

Summary 102 *In the top of the 4th inning, with bases loaded, MIN batter Edouard Julien hit a fly ball to left field. The left fielder put out the batter. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to 3rd. A throwing error by the left fielder allowed the runner from 1st to reach 2nd.*

	game	MIA vs ATL on 20230630
	state before	(2, 0, 0, 0, 5, 3, 7, 0)
44.	error	play,5,0,arral001,21,BBFX,E6/G6S.B-1
	state after	(2, 1, 0, 0, 5, 3, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 103 This game had 3 errors. The final score was 4-16.
In the top of the 5th inning, with bases empty, MIA batter Luis Arraez hit a ground ball to shallow short. An error by the shortstop allowed the batter to reach 1st base.

	game	MIA vs ATL on 20230630
	state before	(0, 1, 0, 0, 5, 3, 10, 1)
45.	error	play,5,1,ozunm001,00,X,FC6/G89D.1-3(E6);B-2
	state after	(0, 0, 1, 1, 5, 3, 10, 1)
	fielder	6 (shortstop on the visiting team)

Summary 104 *In the bottom of the 5th inning, with a runner on 1st, ATL batter Marcell Ozuna hit a ground ball into a fielder's choice to deep right center field. An error by the shortstop allowed the runner on 1st to reach 3rd and the batter to reach 2nd.*

	game	MIA vs ATL on 20230630
	state before	(0, 0, 0, 0, 7, 3, 14, 0)
46.	error	play,7,0,bertj001,22,.BFF.BX,S9/L9D+.B-3(E9)
	state after	(0, 0, 0, 1, 7, 3, 14, 0)
	fielder	9 (right field on the home team)

Summary 105 *In the top of the 7th inning, with bases empty, MIA batter Jon Berti hit a hard line drive single to deep right field. An error by the right fielder allowed the batter to reach 3rd.*

5.2.4 July

	game	MIA vs ATL on 20230702
	state before	(1, 1, 0, 0, 5, 2, 1, 1)
47.	error	play,5,1,acunr001,10,1*B>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 2, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 106 This game had 2 errors. The final score was 3-6.

In the bottom of the 5th inning, with a runner on 1st, ATL batter Ronald Acuna had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Acuna is still at bat, now with a 2-0 count.

	game	MIA vs ATL on 20230702
	state before	(2, 0, 0, 0, 7, 2, 4, 0)
48.	error	play,7,0,davij007,31,BBBCX,E4/G4.B-1
	state after	(2, 1, 0, 0, 7, 2, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 107 *In the top of the 7th inning, with bases empty, MIA batter Jonathan Davis hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	CHA vs ATL on 20230714
	state before	(1, 0, 0, 0, 3, 0, 4, 1)
49.	error	play,3,1,rosae001,11,BFX,E3/G3.B-1
	state after	(1, 1, 0, 0, 3, 0, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 108 *In the bottom of the 3rd inning, with bases empty, ATL batter Eddie Rosario hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for ARI vs ATL on 2023-07-18

position	innings	player	position	innings	player
1	1	Zach Davies	1	1	Bryce Elder
1	4	Jose Ruiz	1	3	Michael Tonkin
1	6	Austin Adams	1	6	Ben Heller
1	7	Tyler Gilbert	1	7	Joe Jimenez
1	8	Miguel Castro	1	8	Kirby Yates
1	9	Kevin Ginkel	1	9	Dereck Rodriguez
2	1	Carson Kelly	1	9	Raisel Iglesias
2	9	Gabriel Moreno	2	1	Sean Murphy
3	1	Christian Walker	3	1	Matt Olson
4	1	Ketel Marte	4	1	Ozzie Albies
5	1	Emmanuel Rivera	5	1	Austin Riley
6	1	Geraldo Perdomo	6	1	Orlando Arcia
7	1	Lourdes Gurriel Jr.	7	1	Sam Hilliard
7	9	Corbin Carroll	7	4	Kevin Pillar
8	1	Alek Thomas	8	1	Michael Harris II
9	1	Corbin Carroll	9	1	Ronald Acuna Jr.
9	9	Jake McCarthy			

	game	ARI vs ATL on 20230718
	state before	(2, 0, 0, 1, 1, 1, 0, 0)
50.	error	play,1,0,gurrl001,31,BBCBX,E6/G6.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 1, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 109 This game had 3 errors. The final score was 16-13. *In the top of the 1st inning, with a runner on 3rd, ARI batter Lourdes Gurriel hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	ARI vs ATL on 20230718
	state before	(1, 0, 1, 0, 2, 5, 5, 0)
51.	error	play,2,0,carrc005,22,FSBFBS,K+WP.2-H(E2/TH)(NR);B-3
	state after	(1, 0, 1, 0, 2, 5, 5, 0)
	fielder	2 (catcher on the home team)

Summary 110 *In the top of the 2nd inning, with a runner on 2nd, ARI batter Corbin Carroll had a 2-2 count. The next pitch was a swinging strike and a wild pitch. A throwing error by the catcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 3rd. The batter strike out but reached on a dropped strike 3.*

	game	ARI vs ATL on 20230718
	state before	(0, 1, 0, 0, 9, 12, 13, 0)
52.	error	play,9,0,thoma004,02,1CSX,FC3/G9.1-3(E3);B-2
	state after	(0, 0, 1, 1, 9, 12, 13, 0)
	fielder	3 (1st baseman on the home team)

Summary 111 *In the top of the 9th inning, with a runner on 1st, ARI batter Alek Thomas hit a ground ball into a fielder's choice at right field. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	ARI vs ATL on 20230719
	state before	(2, 1, 0, 0, 9, 5, 2, 0)
53.	error	play,9,0,walkc002,00,1,SB2.1-3(E3)
	state after	(2, 0, 0, 1, 9, 5, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 112 *In the top of the 9th inning, with a runner on 1st, ARI batter Christian Walker had a 0-0 count. During a pickoff throw to 1st by the pitcher, the runner on 1st stole 2nd. An error by the first baseman allowed that runner to advance to 3rd. Walker is still at bat, with a 0-0 count.*

	game	MIL vs ATL on 20230730
	state before	(2, 1, 0, 1, 1, 0, 0, 0)
54.	error	play,1,0,adamw002,01,CB,WP.3-H(NR);1-2(E2/TH)
	state after	(2, 0, 1, 0, 1, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 113 *In the top of the 1st inning, with runners on 1st and 3rd, MIL batter Willy Adames had a 0-1 count. The next pitch was wild allowing the runner on 3rd to score (no RBI). A throwing error by the catcher allowed the runner on 1st to reach 2nd. Adames is still at bat, now with a 1-1 count.*

5.2.5 August

	game	ANA vs ATL on 20230802
	state before	(0, 0, 0, 0, 9, 4, 12, 0)
55.	error	play,9,0,escoc001,02,.CSX,E4/G34.B-2
	state after	(0, 0, 1, 0, 9, 4, 12, 0)
	fielder	4 (2nd baseman on the home team)

Summary 114 *In the top of the 9th inning, with bases empty, ANA batter Eduardo Escobar hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 2nd base.*

	game	NYA vs ATL on 20230815
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
56.	error	play,1,1,harrm004,02,1CS1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 115 This game had 2 errors. The final score was 0-5. *In the bottom of the 1st inning, with a runner on 1st, ATL batter Michael Harris has a 0-2 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to reach 3rd. Harris is still at bat with a 0-2 count.*

	game	NYA vs ATL on 20230815
	state before	(0, 0, 0, 0, 4, 0, 3, 1)
57.	error	play,4,1,murps001,12,BSCX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 116 *In the bottom of the 4th inning, with bases empty, ATL batter Sean Murphy hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SFN vs ATL on 20230820
	state before	(1, 0, 0, 0, 2, 1, 0, 1)
58.	error	play,2,1,murps001,22,CSBFFF.FBX,E5/G56.B-1
	state after	(1, 1, 0, 0, 2, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 117 This game had 2 errors. The final score was 4-3.
In the bottom of the 2nd inning, with bases empty, ATL batter Sean Murphy hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	SFN vs ATL on 20230820
	state before	(2, 1, 1, 0, 8, 3, 2, 1)
59.	error	play,8,1,olsom001,00,>C,SB3.2-H(E2/TH) (NR) (UR) ;1-2
	state after	(2, 1, 1, 0, 8, 3, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 118 *In the bottom of the 8th inning, with runners on 1st and 2nd, ATL batter Matt Olson had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI) and the runner on 1st to reach 2nd. Olson is still at bat, now with a 0-1 count.*

	game	NYN vs ATL on 20230822
	state before	(2, 1, 0, 0, 2, 0, 2, 1)
60.	error	play,2,1,acunr001,01,FX,FC6/G6D.1-2(E6);B-1
	state after	(2, 1, 1, 0, 2, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 119 *In the bottom of the 2nd inning, with a runner on 1st, ATL batter Ronald Acuna hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	NYN vs ATL on 20230823
	state before	(2, 1, 1, 0, 1, 0, 0, 1)
61.	error	play,1,1,ozunm001,12,CF.F*BX,S8/G6M.2-H;1-3;B-2(E8/TH)
	state after	(2, 0, 1, 1, 1, 0, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 120 *In the bottom of the 1st inning, with runners on 1st and 2nd, ATL batter Marcell Ozuna hit a ground ball single to deep short. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 2nd. A throwing error by the center fielder allowed the batter to reach 2nd.*

5.2.6 September

	game	SLN vs ATL on 20230906
	state before	(1, 0, 0, 0, 8, 8, 6, 0)
62.	error	play,8,0,nootl001,21,CBBX,E4/G34.B-1
	state after	(1, 1, 0, 0, 8, 8, 6, 0)
	fielder	4 (2nd baseman on the home team)

Summary 121 *In the top of the 8th inning, with bases empty, SLN batter Lars Nootbaar hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	PIT vs ATL on 20230908
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
63.	error	play,1,1,acunr001,22,CBFBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 122 *In the bottom of the 1st inning, with bases empty, ATL batter Ronald Acuna hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs ATL on 20230909
	state before	(0, 0, 0, 0, 7, 7, 3, 0)
64.	error	play,7,0,suwij001,22,BCBFX,3E1/G34S.B-1
	state after	(0, 1, 0, 0, 7, 7, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 123 *In the top of the 7th inning, with bases empty, PIT batter Jack Suwinski hit a ground ball to shallow infield. An error by the pitcher (off an assist from the first baseman) allowed the batter to reach 1st base.*

	game	PHI vs ATL on 20230920
	state before	(1, 1, 0, 0, 6, 4, 2, 0)
65.	error	play,6,0,cavej001,20,BB>C,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 4, 2, 0)
	fielder	2 (catcher on the home team)

Summary 124 *In the top of the 6th inning, with a runner on 1st, PHI batter Jake Cave hit a single had a 2-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Cave is still at bat, now with a 2-1 count.*

66.	game	CHN vs ATL on 20230926
	state before	(2, 0, 1, 1, 8, 6, 5, 1)
	error	play,8,1,murps001,32,BBSF.*BX,E9/F89D.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 8, 6, 7, 1)
	fielder	9 (right field on the visiting team)

Summary 125 *In the bottom of the 8th inning, with runners on 2nd and 3rd, ATL batter Sean Murphy hit a fly ball to deep right center field. An error by the right fielder allowed the runners on 2nd and 3rd to score (unearned, no RBIs) and the batter to reach 1st.*

67.	game	CHN vs ATL on 20230927
	state before	(0, 1, 0, 1, 7, 3, 1, 1)
	error	play,7,1,ozunm001,32,FBBS.*BX,FC4/G34.3-H;1-2(E4/TH);B-1
	state after	(0, 1, 1, 0, 7, 3, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 126 *In the bottom of the 7th inning, with runners on 1st and 3rd, ATL batter Marcell Ozuna hit a ground ball into a fielder's choice at the 1st/2nd base gap. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

68.	game	CHN vs ATL on 20230928
	state before	(0, 0, 0, 0, 2, 1, 2, 1)
	error	play,2,1,rosae001,00,X,E7/L78D.B-2
	state after	(0, 0, 1, 0, 2, 1, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 127 This game had 2 errors. The final score was 3-5. *In the bottom of the 2nd inning, with bases empty, ATL batter Eddie Rosario hit a line drive to deep left center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	CHN vs ATL on 20230928
	state before	(1, 1, 0, 1, 3, 1, 4, 1)
69.	error	play,3,1,rosae001,01,FX,E1/G1S-.3-H;1-2;B-1
	state after	(1, 1, 1, 0, 3, 1, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 128 *In the bottom of the 3rd inning, with runners on 1st and 3rd, ATL batter Eddie Rosario hit a soft ground ball to shallow infield. An error by the pitcher allowed the runner on 3rd to score (1 RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.*

Visiting/Home team rosters for WAS vs ATL on 2023-09-29

position	innings	player	position	innings	player
1	1	Trevor Williams	1	1	Allan Winans
1	4	Andres Machado	1	6	Brad Hand
1	5	Jordan Weems	1	7	Jesse Chavez
1	5	Jose A. Ferrer	1	8	Jackson Stephens
1	6	Joe La Sorsa	2	1	Travis d'Arnaud
1	7	Hunter Harvey	3	1	Matt Olson
1	9	Kyle Finnegan	3	4	Luke Williams
2	1	Keibert Ruiz	4	1	Ozzie Albies
3	1	Dominic Smith	5	1	Austin Riley
4	1	Luis Garcia	6	1	Nicky Lopez
5	1	Carter Kieboom	7	1	Eddie Rosario
6	1	CJ Abrams	8	1	Forrest Wall
7	1	Jake Alu	9	1	Ronald Acuna Jr.
8	1	Jacob Young	9	3	Kevin Pillar
9	1	Lane Thomas			

	game	WAS vs ATL on 20230929
	state before	(2, 1, 0, 0, 5, 4, 3, 1)
70.	error	play,5,1,wallf001,11,CB1X,S4/G4.1-H(E4/TH)(E1/TH)(NR);B-3
	state after	(2, 0, 0, 1, 5, 4, 4, 1)
	fielder	4,1 (2nd baseman, pitcher on the visiting team)

Summary 129 This game had 5 errors. This particular play had 2 errors by 2 different fielders. The final score was 10-6.

In the bottom of the 5th inning, with a runner on 1st, ATL batter Forrest Wall hit a ground ball single to 2nd base. A throwing error by the second baseman and a throwing error by the pitcher allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.

	game	WAS vs ATL on 20230929
	state before	(0, 1, 0, 0, 7, 8, 6, 0)
71.	error	play,7,0,ruizk001,10,BX,4E1/G34.1-2;B-1
	state after	(0, 1, 1, 0, 7, 8, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 130 *In the top of the 7th inning, with a runner on 1st, WAS batter Keibert Ruiz hit a ground ball to the 1st/2nd base gap. An error by the pitcher (on an assist from the 2nd baseman) allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	WAS vs ATL on 20230929
	state before	(0, 1, 1, 0, 7, 8, 6, 0)
72.	error	play,7,0,garc1006,10,BX,FC6/G8.2-H(NR)(UR);1-2(E6);B-1
	state after	(0, 1, 1, 0, 7, 9, 6, 0)
	fielder	6 (shortstop on the home team)

Summary 131 *In the top of the 7th inning, with runners on 1st and 2nd, WAS batter Luis Garcia hit a ground ball into a fielder's choice at center field. The runner on 2nd scored (unearned, no RBI). An error by the shortstop allowed the runner on 1st to reach 2nd and the batter to reach 1st.*

	game	WAS vs ATL on 20230929
	state before	(2, 1, 1, 0, 7, 9, 6, 0)
73.	error	play,7,0,alu-j001,12,BCCX,E5/G7S.2-H(NR) (UR) ;1-2;B-1
	state after	(2, 1, 1, 0, 7, 10, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 132 *In the top of the 7th inning, with runners on 1st and 2nd, WAS batter Jake Alu hit a ground ball to shallow left field. An error by the third baseman allowed te runner on 2nd to score (unearned, no RBI), the runner on 1st to reach 2nd, and the batter to reach 1st.*

	game	WAS vs ATL on 20230930
	state before	(2, 0, 1, 0, 1, 3, 0, 1)
74.	error	play,1,1,olsom001,10,BX,E5/G6.2-3;B-1
	state after	(2, 1, 0, 1, 1, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 133 This game had 2 errors. The final score was 3-5.
In the bottom of the 1st inning, with a runner on 2nd, ATL batter Matt Olson hit a ground ball to short. An error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	WAS vs ATL on 20230930
	state before	(0, 0, 0, 0, 8, 3, 5, 1)
75.	error	play,8,1,rosae001,00,..X,S5/G56S-.B-2(E5/TH)
	state after	(0, 0, 1, 0, 8, 3, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 134 *In the bottom of the 8th inning, with bases empty, ATL batter Eddie Rosario hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

Chapter 6

Baltimore Orioles (BAL)

The Baltimore Orioles have won six American League championship pennants since their Baltimore origin in 1954 (in 1966, 1969, 1970, 1971, 1979, 1983). Since 1983, there have been some lesser post-season successes. For example, 2023 was Baltimore's first postseason appearance since 2016, with a regular season win-loss record of 101-61.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Baltimore against the Orioles. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

That year, there were 63 errors in the 81 home games, 31 by the Orioles and the rest by visiting teams. Over all games (home and away), the Orioles committed 71 errors, well below¹ the MLB average of 84. The defensive efficiency record (usually denoted DER or DefEff) is another measure of team fielding performance. The DER of the 2023 BAL team is above average: 0.696 compared with the MLB average of 0.691 that year [BR].

6.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

¹Since the standard deviation of the list of 30 error totals is 13.8, that's about one standard deviation below average.

6.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt :	32
Runs by ht because of errors by vt :	11
Errors by ht :	31
Runs by vt because of errors by ht :	13

From this we can see that the expected number of errors committed by the 2023 Orioles in a *home* game is $31/81 \approx 0.38$. This quantity can also be regarded as their home game error rate. The expected number of errors committed by the 2023 Orioles (home or away) game is $71/81 \approx 0.87$. As mentioned, the MLB average error rate that year was $84/81 \approx 1.04$.

The Orioles scored 11 runs during the 32 plays where a visiting team made a fielding error. Visiting teams scored 13 runs during the 31 plays where the Orioles made a fielding error. Therefore, in 2023 the Orioles had an error-to-run conversion rate of $11/32 = 34.37\ldots\%$, while had a $13/31 = 41.93\%$ conversion rate.

Errors by position	vt	ht
errors due to pitcher:	1	4
to catcher:	8	5
to 1st baseman:	2	2
to 2nd baseman:	8	2
to 3rd baseman:	5	4
to shortstop:	4	9
to left fielder:	2	0
to center fielder:	1	2
to right fielder:	1	2
subtotals	32	30

Twenty-one errors were committed with bases empty (the *empty state*) and 17 were with a (lone) runner on 1st. This is suprising since, according

to the *Uniformity Hypothesis* the frequency the empty state occurs in a play is at least twice that of the frequency of a lone runner on 1st. However, the error rate for the empty state is not doubled, suggesting that the distribution of errors, for a given position, is not uniform.

6.1.2 Error total by state

This section tabulates the totals by game state.

Errors by state	vt	ht	totals
bases empty	8	13	21
(lone) runner on 1st	11	6	17
(lone) runner on 2nd	1	3	4
(lone) runner on 3rd	0	2	2
two runners on base	10	5	15
bases loaded	2	2	4
total	32	31	63

Here is a table of frequencies of most common errors by the Orioles in BAL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	6/31 = 19.35%
(1, 0, 0, 0)	3	9.7%
(2, 0, 0, 0)	4	12.9%
(0, 1, 0, 0)	2	6.4%
(1, 1, 0, 0)	2	6.4%
(2, 1, 0, 0)	2	6.4%
(2, 0, 1, 0)	3	9.7%
(1, 1, 1, 0)	2	6.4%

Here is a table of frequencies of most common errors by a visiting team in BAL 2023 home games:

state	frequency, vt	percentage
(0, 0, 0, 0)	4	$4/32 = 12.5\%$
(1, 0, 0, 0)	2	$2/32 = 6.25\%$
(2, 0, 0, 0)	2	6.25%
(0, 1, 0, 0)	3	9.4%
(1, 1, 0, 0)	4	12.5%
(2, 1, 0, 0)	4	12.5%
(2, 0, 1, 0)	1	3.1%
(0, 1, 1, 0)	1	3.1%
(1, 1, 1, 0)	5	15.6%
(2, 1, 1, 1)	1	3.1%

Recall Figure 4.1 gave a frequency table of game states in all plays. From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Baltimore homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `BAL2023.eva`):

state	BAL2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	16.8%
(2, 0, 0, 0)	13.3%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	6.7%
(0, 0, 1, 0)	2.4%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	3.3%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	1.1%
(2, 0, 0, 1)	1.8%
(0, 1, 1, 0)	1.1%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.9%

6.1.3 Some interesting errors

Some of the most interesting error plays in the BAL 2023 home games:

- The out play, 3,1,mullc002,01,FX,FC/G4.2-3;1X2(4E6)(6);B-1 at 2nd base was *not* negated by the error by the shortstop. For more details, see the White Sox vs Orioles game on 2023-08-28 below².
- One game had 5 errors, four of them by the Orioles. For more details, check out the 2023-07-18 game against the Dodgers below.
- Another game, on 2023-10-01, had only 3 errors but 2 of them were during one player's at-bat³.

6.1.4 Relative team rankings

We have 81 home games between BAL and some visiting MLB team.

BAL, the home team (ht), *lost* more games than won against⁴:

CIN (2/3), CLE (2/3), HOU (2/3), LAN (2/3),
MIN (2/3), SLN (2/3), TEX (2/3),

The teams that *split* their series were:

ANA (2/4), NYA (3/6),

and *won* more than lost against

WAS (0/2), DET (0/3), KCA (0/3), MIA (0/3), NYN (0/3), CHA (1/3),
COL (1/3), PIT (1/3), SEA (1/3), OAK (1/4), TOR (2/6), BOS (3/7), TBA
(3/7)

The teams that did not play in any of the 81 home games in BAL 2023 were:

ARI, ATL, CHN, MIL, PHI, SDN, SFN

²Unfortunately, I was unable to locate video of this play online to explain things further. However, a similar error occurred in the game between the Padres and the Orioles on 2019-06-25. See the 3rd inning play with Dwight Smith at bat in [Jo24].

³Rafael Devers of the Boston Red Sox was that lucky guy.

⁴Visiting teams (vt) are listed with that team's W/T record in parentheses (W = numbers of games vt won against ht at BAL, T = Total home games vt played vs ht at BAL).

6.2 Game-by-game totals

To save space, we only give the team rosters in special cases.

6.2.1 April

	game	NYA vs BAL on 20230407
	state before	(1, 1, 0, 1, 2, 0, 1, 1)
1.	error	play,2,1,haysa001,00,X,FC4/G34.3-H;1-2(E4);B-1
	state after	(1, 1, 1, 0, 2, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 2 *In the bottom of the 2nd inning, with runners on 1st and 3rd, BAL batter Austin Hays hit a ground ball to a fielder's choice at the 1st/2nd base gap. The runner on 3rd scored (1 RBI), An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	NYA vs BAL on 20230408
	state before	(2, 1, 0, 0, 4, 1, 1, 1)
2.	error	play,4,1,fraza001,11,1BF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Remark 3 *In the bottom of the 4th inning, with a runner on 1st, BAL batter Adam Frazier had a 1-1 count. During the next pitch, which was a ball, the runner on 1st base stole 2nd. A throwing error by the catcher allowed the runner to reach 3rd base. Frazier is at bat, now with a 2-1 count.*

	game	OAK vs BAL on 20230410
	state before	(1, 1, 0, 0, 1, 1, 0, 1)
3.	error	play,1,1,mounr001,00,1>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Remark 4 *In the bottom of the 1st inning, with a runner on 1st, BAL batter Ryan Mountcastle had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd base. A throwing error by the catcher allowed the runner to advance to 3rd base. Mountcastle is at bat, now with a 1-0 count.*

	game	OAK vs BAL on 20230412
	state before	(1, 0, 1, 1, 2, 3, 1, 1)
4.	error	play,2,1,mcker001,11,BSX,E9/F9D/SF.3-H;2-3;B-1
	state after	(1, 1, 0, 1, 2, 3, 2, 1)
	fielder	9 (right field on the visiting team)

Remark 5 *In the bottom of the 2nd inning with runners on 2nd and 3rd, BAL batter Ryan McKenna hit a sacrifice fly to deep right field. An error by the right fielder allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	BOS vs BAL on 20230425
	state before	(1, 1, 1, 0, 9, 8, 2, 1)
5.	error	play,9,1,matej003,12,CBSX,FC1/G1.2-3;1X2(1E4);B-1
	state after	(1, 1, 1, 1, 9, 8, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 6 *In the bottom of the 9th inning, with runners on 1st and 2nd, BAL batter Jorge Mateo hit a ground ball to a fielder's choice at the pitcher. The runner on 2nd advanced to 3rd, An error by the second baseman (assisted by the pitcher) allowed the runner from 1st to reach 2nd safely and the batter to advance to 1st.*

	game	BOS vs BAL on 20230426
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
6.	error	play,1,1,rutsa001,02,CFF>X,E4/L89S.1-3;B-1
	state after	(0, 1, 0, 1, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 7 *In the bottom of the 1st inning with a runner on 1st, BAL batter Adley Rutschman hit a line drive to shallow left center field. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

6.2.2 May

	game	TBA vs BAL on 20230509
	state before	(0, 0, 0, 0, 4, 1, 3, 0)
7.	error	play,4,0,franw002,12,FCBX,E3/G3.B-1
	state after	(0, 1, 0, 0, 4, 1, 3, 0)
	fielder	3 (1st baseman on the home team)

Remark 8 *In the top of the 4th inning with bases empty, TBA batter Wander Franco hit a ground ball to 1st. An error by the first baseman allowed the batter to reach 1st base.*

	game	TBA vs BAL on 20230509
	state before	(2, 0, 0, 0, 8, 2, 4, 0)
8.	error	play,8,0,ralel001,00,X,S9/G34.B-2(E9)
	state after	(2, 0, 1, 0, 8, 2, 4, 0)
	fielder	9 (right field on the home team)

Remark 9 *In the top of the 8th inning with bases empty, TBA batter Luke Raley hit a ground ball single to right field. The batter reached second base due to an error by the right fielder.*

	game	PIT vs BAL on 20230512
	state before	(1, 0, 0, 1, 3, 0, 0, 0)
9.	error	play,3,0,hayek001,22,BCBSX,FC6/G6S.3-H(E6/TH) (NR) (UR) ;B-1
	state after	(1, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Remark 10 *In the top of the 3rd inning with a runner on 3rd, PIT batter Ke'Bryan Hayes hit a ground ball to a fielder's choice at short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st base.*

	game	PIT vs BAL on 20230514
	state before	(0, 0, 0, 0, 8, 4, 0, 0)
10.	error	play,8,0,castr006,11,..BS.X,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 8, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Remark 11 *In the top of the 8th inning with bases empty, PIT batter Rodolfo Castro hit a ground ball to 3rd A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs BAL on 20230516
	state before	(2, 1, 1, 0, 1, 1, 0, 1)
11.	error	play,1,1,hendg002,20,BB+2,P02(E2/TH).2-3;1-2
	state after	(2, 0, 1, 1, 1, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Remark 12 *In the bottom of the 1st inning with runners on 1st and 2nd, BAL batter Gunnar Henderson has a 2-0 count. During an attempted pick off at 2nd base a throwing error by the catcher allowed the runners on 1st and 2nd to advance one base each. Henderson is still at bat with a 2-0 count.*

	game	ANA vs BAL on 20230517
	state before	(0, 1, 0, 1, 2, 0, 0, 1)
12.	error	play,2,1,fraza001,00,X,FC/SH/BG5S-.1-2;B-1(E5)
	state after	(0, 1, 1, 1, 2, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Remark 13 *In the bottom of the 2nd inning with runners on 1st and 3rd, BAL batter Adam Frazier bunted into a fielder's choice short of 3rd base. The runner on 1st advanced to 2nd. An error by the third baseman allowed the batter to reach 1st The bases are loaded.*

	game	TEX vs BAL on 20230526
	state before	(1, 1, 1, 0, 4, 5, 1, 0)
13.	error	play,4,0,semim001,10,BX,FC6/G6.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 4, 5, 1, 0)
	fielder	6 (shortstop on the home team)

Remark 14 *In the top of the 4th inning with runners on 1st and 2nd, TEX batter Marcus Semien hit a ground ball to a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are now loaded.*

	game	TEX vs BAL on 20230526
	state before	(2, 0, 0, 0, 7, 10, 1, 1)
14.	error	play,7,1,haysa001,11,BFX,S8/L78.B-2(E7)
	state after	(2, 0, 1, 0, 7, 10, 1, 1)
	fielder	7 (left field on the visiting team)

Remark 15 *In the bottom of the 7th inning with bases empty, BAL batter Austin Hayes hit a line drive single to left-center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	TEX vs BAL on 20230527
	state before	(2, 0, 1, 0, 8, 4, 1, 0)
15.	error	play,8,0,smitj007,01,.FX,4E1/G34.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 8, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Remark 16 *In the top of the 8th inning with a runner on 2nd, TEX batter Josh Smith hit a ground ball to the 1st/2nd base gap. An error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

Visiting/Home team rosters for CLE vs BAL on 2023-05-29

position	innings	player	position	innings	player
1	1	Logan Allen	1	1	Tyler Wells
1	8	Enyel De Los Santos	1	7	Cionel Perez
1	9	Trevor Stephan	1	7	Mike Baumann
2	1	Cam Gallagher	1	8	Mychal Givens
3	1	Josh Naylor	1	9	Keegan Akin
4	1	Andres Gimenez	2	1	James McCann
5	1	Gabriel Arias	3	1	Ryan Mountcastle
6	1	Amed Rosario	4	1	Adam Frazier
7	1	Steven Kwan	5	1	Ramon Urias
8	1	Myles Straw	6	1	Jorge Mateo
9	1	Will Brennan	7	1	Austin Hays
			8	1	Cedric Mullins
			8	9	Ryan McKenna
			9	1	Ryan McKenna
			9	9	Terrin Vavra

	game	CLE vs BAL on 20230529
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
16.	error	play,1,0,rosaa003,12,SFB1X,FC1/G1S.1-2(E1/TH);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Remark 17 This game had 3 errors. The Guardians shut out the Orioles 5-0.

In the top of the 1st inning with a runner on 1st, CLE batter Amed Rosario⁵ hit a ground ball to a fielder's choice at shallow infield. A throwing error on the pitcher allowed the runner on 1st to advance to 2nd base and the batter to reach 1st.

	game	CLE vs BAL on 20230529
	state before	(1, 0, 0, 0, 4, 0, 0, 1)
17.	error	play,4,1,mounr001,22,CS.BBX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 4, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Remark 18 *In the bottom of the 4th inning with bases empty, BAL batter Ryan Mountcastle hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base safely.*

	game	CLE vs BAL on 20230529
	state before	(1, 0, 1, 1, 7, 3, 0, 0)
18.	error	play,7,0,rosaa003,22,.FC*BBX,FC4/G4.3-H(NR) (UR) ;2-H(E4/TH) (NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 7, 5, 0, 0)
	fielder	4 (2nd baseman on the home team)

Remark 19 *In the top of the 7th inning with runners on 2nd and 3rd, CLE batter Amed Rosario hit a ground ball to a fielder's choice at 2nd. The runner on 3rd scored (unearned, no RBI). A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd base.*

	game	CLE vs BAL on 20230531
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
19.	error	play,1,0,ramij003,11,BSX,D7/L7LD+.1-H;B-3(E6/TH)
	state after	(1, 0, 0, 1, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

⁵Rosario was also the batter in another fielder's choice error in this game. See below.

Remark 20 *In the top of the 1st inning with a runner on 1st, CLE batter Jose Ramirez hit a line drive double to deep left field. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to advance to 3rd base.*

6.2.3 June

	game	KCA vs BAL on 20230611
	state before	(0, 0, 0, 0, 4, 3, 6, 0)
20.	error	play,4,0,olive001,01,CX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 4, 3, 6, 0)
	fielder	5 (3rd baseman on the home team)

Remark 21 *In the top of the 4th inning, with bases empty, KCA batter Edward Olivares hit a ground ball to 3rd. A throwing error by the third baseman allowed the batter to advance to 1st base.*

	game	TOR vs BAL on 20230613
	state before	(0, 0, 0, 0, 8, 4, 11, 1)
21.	error	play,8,1,ohear001,10,....BX,E4/G34D.B-1
	state after	(0, 1, 0, 0, 8, 4, 11, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 22 *In the bottom of the 8th inning with bases empty, BAL batter Ryan O'Hearn hit a ground ball to 2nd base. An error by the second baseman allowed the batter to advance to 1st base.*

	game	TOR vs BAL on 20230614
	state before	(1, 1, 0, 0, 7, 1, 0, 1)
22.	error	play,7,1,ohear001,20,B*BX,46(1)/F0/G34.B-2(E6/TH)
	state after	(2, 0, 1, 0, 7, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Remark 23 *In the bottom of the 7th inning, with a runner on 1st, BAL batter Ryan O’Hearn hit a ground ball to the 1st/2nd base gap. The runner advancing from 1st was forced out at 2nd. A throwing error by the shortstop allowed the batter to advance to 2nd base.*

	game	TOR vs BAL on 20230615
	state before	(2, 0, 0, 0, 5, 2, 2, 1)
23.	error	play,5,1,uriar001,10,BX,E3/G1S-.B-1
	state after	(2, 1, 0, 0, 5, 2, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Remark 24 *In the bottom of the 5th inning, with bases empty, BAL batter Ramon Urias hit a soft ground ball to shallow infield. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SEA vs BAL on 2023-06-23

position	innings	player	position	innings	player
1	1	Logan Gilbert	1	1	Kyle Gibson
1	8	Justin Topa	1	4	Logan Gillaspie
1	9	Tayler Saucedo	1	6	Cionel Perez
2	1	Tom Murphy	1	8	Bryan Baker
3	1	Ty France	1	8	Keegan Akin
3	8	Mike Ford	1	9	Josh Lester
4	1	Kolten Wong	2	1	Adley Rutschman
5	1	Eugenio Suarez	2	9	Anthony Bemboom
6	1	Jose Caballero	3	1	Ryan O'Hearn
7	1	Jarred Kelenic	4	1	Adam Frazier
8	1	Julio Rodriguez	4	9	Joey Ortiz
9	1	Teoscar Hernandez	5	1	Ramon Urias
			6	1	Jorge Mateo
			7	1	Austin Hays
			8	1	Aaron Hicks
			8	9	Ryan McKenna
			9	1	Anthony Santander

	game	SEA vs BAL on 20230623
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
24.	error	play,1,1,hendg002,31,BBCBF,FLE7
	state after	(0, 0, 0, 0, 1, 0, 0, 1)
	fielder	7 (left field on the visiting team)

Remark 25 This is an ∞ *play*, a play where the state does not change⁶.
One more thing: this game had 3 errors.

⁶Hypothetically, if this FLE play was repeated infinitely many times, the game would never end!

In the bottom of the 1st inning, with bases empty and a 3-1 count, BAL batter Gunner Henderson hit a foul fly ball outside the left field line. An error by the left fielder allowed the batter to remain safe at bat. Henderson is still at bat, now with a full count.

25.	game	SEA vs BAL on 20230623
	state before	(2, 1, 0, 1, 8, 10, 1, 0)
	error	play,8,0,ralec001,32,BBBCF>X,S8/F8S.3-H;1-3;B-2(E8/TH)
	state after	(2, 0, 1, 1, 8, 11, 1, 0)
	fielder	8 (center field on the home team)

Remark 26 *In the top of the 8th inning with runners on 1st and 3rd, SEA batter Cal Raleigh hit a single to shallow center field. The runner on 3rd scored (1 RBI) and the runner on 1st advanced to 3rd, A throwing error by the center fielder allowed the batter advanced to 2nd.*

26.	game	SEA vs BAL on 20230623
	state before	(1, 0, 0, 0, 9, 13, 1, 1)
	error	play,9,1,lestj002,12,BCSX,E5/G15.B-1
	state after	(1, 1, 0, 0, 9, 13, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Remark 27 *In the bottom of the 9th inning with bases empty, BAL batter Josh Lester hit a ground ball to shallow short. An error by the third baseman allowed the batter to reach 1st base.*

27.	game	SEA vs BAL on 20230625
	state before	(1, 1, 0, 0, 5, 2, 2, 1)
	error	play,5,1,bemba001,01,1FX,S4/F9LS.1-H(E4)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 5, 2, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 28 *In the bottom of the 5th inning with a runner on 1st, BAL batter Anthony Bemboom hit a single to shallow right field. An error by the second baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd base.*

	game	CIN vs BAL on 20230626
	state before	(1, 1, 1, 1, 3, 1, 2, 1)
28.	error	play,3,1,westj002,10,BX,54(1)/F0/G56.3-H;2-H(E4/TH)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 3, 1, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 29 *In the bottom of the 3rd inning with bases loaded, BAL batter Jordan Westburg hit a ground ball to 3rd. The second baseman (assisted by the third baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner from 2nd to score (unearned, no RBI) and the batter to reach 1st base.*

	game	CIN vs BAL on 20230627
	state before	(2, 1, 0, 0, 9, 3, 1, 1)
29.	error	play,9,1,fraza001,10,B>S,CS2(E1/TH)
	state after	(2, 0, 1, 0, 9, 3, 1, 1)
	fielder	1 (pitcher on the visiting team)

Remark 30 *In the bottom of the 9th inning with a runner on 1st, BAL batter Adam Frazier had a 1-0 count. During the next pitch, which was a swinging strike, the runner on 1st attempted to steal 2nd base. The pitcher made a throwing error allowing the runner to reach 2nd base safely⁷. Frazier is still at bat, now with a 1-1 count.*

⁷By Rule 10 from the MLB rule book, this is not a stolen base since it is after the pitch.

	game	CIN vs BAL on 20230628
	state before	(2, 1, 0, 0, 6, 6, 4, 0)
30.	error	play,6,0,indij001,20,BB>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 6, 4, 0)
	fielder	2 (catcher on the home team)

Remark 31 This game had 2 errors. The Reds beat Baltimore 11-7. *In the top of the 6th inning with a runner on 1st, CIN batter Jonathan India had a 2-0 count. At the next pitch, which was a called strike, the runner on 1st base stole 2nd base. A throwing error by the catcher allowed the runner to advance to 3rd base. India is at bat, now with a 2-1 count.*

	game	CIN vs BAL on 20230628
	state before	(1, 0, 0, 0, 10, 10, 7, 0)
31.	error	play,10,0,mail1001,12,CFBX,D8/F89XD.B-3(E8)
	state after	(1, 0, 0, 1, 10, 10, 7, 0)
	fielder	8 (catcher on the home team)

Remark 32 *In the top of the 10th inning with bases empty, CIN batter Luke Maile hits a double to deep right-center field. Due to an error by the center fielder, the batter reached 3rd base.*

6.2.4 July

	game	MIA vs BAL on 20230714
	state before	(0, 1, 0, 0, 2, 0, 0, 1)
32.	error	play,2,1,hicka001,10,B1>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 2, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Remark 33 This game had 2 errors. The Marlins lost to the Orioles 2-5.

In the bottom of the 2nd inning with a runner on 1st, BAL batter Aaron Hicks had a 1-0 count. During the next pitch, which was a ball, the runner on 1st base stole 2nd base. A throwing error by the catcher allowed the runner to reach 3rd base. Hicks is at bat, now with a 2-0 count.

	game	MIA vs BAL on 20230714
	state before	(2, 0, 0, 0, 6, 1, 3, 0)
33.	error	play,6,0,sancj003,00,X,E6/TH/G6.B-2
	state after	(2, 0, 1, 0, 6, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Remark 34 *In the top of the 6th inning, with bases empty, MIA batter Jesus Sanchez hit a ground ball to the shortstop. A throwing error by the shortstop allowed the batter to reach 2nd base.*

	game	MIA vs BAL on 20230716
	state before	(1, 1, 1, 1, 4, 0, 3, 1)
34.	error	play,4,1,mccaj001,22,SBCBX,FC6/G6.3-H(E6)(NR)(UR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 4, 0, 4, 1)
	fielder	6 (shortstop on the visiting team)

Remark 35 *In the bottom of the 4th inning with bases loaded, BAL batter James McCann hit a ground ball to a fielder's choice at short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st base. The bases are still loaded.*

Visiting/Home team rosters for LAN vs BAL on 2023-07-18

position	innings	player	position	innings	player
1	1	Michael Grove	1	1	Tyler Wells
1	6	Yency Almonte	1	3	Cole Irvin
1	7	Phil Bickford	1	7	Cionel Perez
1	8	Nick Robertson	1	8	Mike Baumann
2	1	Will Smith	2	1	Adley Rutschman
3	1	Freddie Freeman	3	1	Ryan O'Hearn
4	1	Mookie Betts	4	1	Adam Frazier
4	8	Yonny Hernandez	5	1	Jordan Westburg
5	1	Max Muncy	6	1	Gunnar Henderson
6	1	Miguel Rojas	7	1	Austin Hays
7	1	David Peralta	8	1	Aaron Hicks
7	4	Chris Taylor	9	1	Colton Cowser
8	1	James Outman			
8	3	Jake Marisnick			
8	4	James Outman			
9	1	Jason Heyward			
9	3	James Outman			
9	4	David Peralta			

	game	LAN vs BAL on 20230718
	state before	(1, 1, 0, 0, 2, 5, 0, 1)
35.	error	play,2,1,hicka001,00,X,D8/L78XD+.1-H;B-3(E8)
	state after	(1, 0, 0, 1, 2, 5, 1, 1)
	fielder	8 (center field on the visiting team)

Remark 36 This game had **5 errors**. The Dodgers beat the Orioles 10-3.

In the bottom of the 2nd inning with a runner on 1st, BAL batter Aaron

Hicks hit a hard line drive double to deep left center field. The runner on 1st scored (1 RBI). An error by the center fielder allowed the batter to reach 3rd base.

	game	LAN vs BAL on 20230718
	state before	(2, 0, 0, 0, 5, 5, 1, 0)
36.	error	play,5,0,rojam002,02,CFX,6E3/G6.B-1
	state after	(2, 1, 0, 0, 5, 5, 1, 0)
	fielder	3 (1st baseman on the home team)

Remark 37 *In the top of the 5th inning with bases empty, LAN batter Miguel Rojas hit a ground ball to the shortstop. An error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st base safely.*

	game	LAN vs BAL on 20230718
	state before	(0, 1, 1, 0, 6, 5, 1, 0)
37.	error	play,6,0,muncm001,01,CX,FC4/G4.2-3;1-2(E4);B-1
	state after	(0, 1, 1, 1, 6, 5, 1, 0)
	fielder	4 (2nd baseman on the home team)

Remark 38 *In the top of the 6th inning with runners on 1st and 2nd, LAN batter Max Muncy hit a ground ball to a fielder's choice at 2nd. The runner on 2nd advanced to 3rd base. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

	game	LAN vs BAL on 20230718
	state before	(0, 1, 1, 0, 7, 5, 1, 0)
38.	error	play,7,0,bettm001,00,.>B,SB3;SB2.2-H(E2/TH)(NR);1-3
	state after	(0, 0, 0, 1, 7, 6, 1, 0)
	fielder	2 (catcher on the home team)

Remark 39 *In the top of the 7th inning with runners on 1st and 2nd, LAN batter Mookie Betts had a 0-0 count. On the next pitch, whcih was a ball, the runner on 2nd stole 3rd and the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (no RBI) and the runner from 1st (now on 2nd after the steal) to advance to 3rd. Betts is still at bat, now with a 1-0 count.*

39.	game	LAN vs BAL on 20230718
	state before	(2, 0, 1, 0, 8, 9, 1, 0)
	error	play,8,0,smitw003,00,X,S6/G56.2-H(E6/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 8, 10, 1, 0)
	fielder	6 (shortstop on the home team)

Remark 40 *In the top of the 8th inning with a runner on 2nd, LAN batter Will Smith hit a ground ball single to short. A throwing error by the shortstop allowed the runner to score (unearned, no RBI) and the batter to advance to 2nd base.*

40.	game	NYA vs BAL on 20230729
	state before	(1, 1, 0, 0, 2, 1, 0, 0)
	error	play,2,0,volpa001,00,1>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 2, 1, 0, 0)
	fielder	2 (catcher on the home team)

Remark 41 *In the top of the 2nd inning with a runner on 1st, NYA batter Anthony Volpe had a 0-0 count. During the first pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner to reach 3rd base. Volpe is still at bat, now with a 1-0 count.*

41.	game	NYA vs BAL on 20230730
	state before	(1, 0, 0, 0, 8, 3, 9, 0)
	error	play,8,0,volpa001,10,BX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 8, 3, 9, 0)
	fielder	6 (shortstop on the home team)

Remark 42 *In the top of the 8th inning with bases empty, NYA batter Anthony Volpe hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

6.2.5 August

	game	NYN vs BAL on 20230804
	state before	(1, 0, 0, 0, 8, 2, 10, 0)
42.	error	play,8,0,orter001,11,CBX,D9/G3.B-3(E9/TH)
	state after	(1, 0, 0, 1, 8, 2, 10, 0)
	fielder	9 (right field on the home team)

Remark 43 *In the top of the 8th inning with bases empty, NYN batter Rafael Ortega hit a ground ball double to 1st. A throwing error by the right fielder allowed the batter to advance to 3rd.*

	game	HOU vs BAL on 20230810
	state before	(2, 1, 0, 0, 8, 2, 5, 0)
43.	error	play,8,0,singj001,12,SBSX,3E1/G1S.1-2;B-1
	state after	(2, 1, 1, 0, 8, 2, 5, 0)
	fielder	1 (pitcher on the home team)

Remark 44 *In the top of the 8th inning with a runner on 1st, HOU batter Jon Singleton hit a ground ball to shallow infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st base safely.*

	game	TOR vs BAL on 20230822
	state before	(1, 1, 1, 0, 5, 3, 1, 1)
44.	error	play,5,1,hendg002,12,CBF>S,K+SB3;SB2.2-H(E2/TH) (NR)
	state after	(2, 0, 1, 0, 5, 3, 2, 1)
	fielder	2 (catcher on the visiting team)

Remark 45 *In the bottom of the 5th inning with runners on 1st and 2nd, BAL batter Gunner Henderson had a 1-2 count. During the next pitch, which was a swinging strike, the runners on 1st and 2nd stole 2nd and 3rd base, respectively. A throwing error by the catcher allowed the runner from 2nd base (now on 3rd after the steal) to score (no RBI). Henderson has struck out.*

Visiting/Home team rosters for TOR vs BAL on 2023-08-23

position	innings	player	position	innings	player
1	1	Kevin Gausman	1	1	Dean Kremer
1	7	Genesis Cabrera	1	7	Jacob Webb
1	8	Bowden Francis	1	8	Yennier Cano
1	8	Trevor Richards	1	9	Shintaro Fujinami
2	1	Alejandro Kirk	2	1	Adley Rutschman
2	8	Danny Jansen	3	1	Ryan O'Hearn
3	1	Vladimir Guerrero Jr.	4	1	Jordan Westburg
4	1	Whit Merrifield	5	1	Ramon Urias
5	1	Matt Chapman	6	1	Gunnar Henderson
6	1	Bo Bichette	7	1	Austin Hays
7	1	Daulton Varsho	8	1	Cedric Mullins
8	1	Kevin Kiermaier	9	1	Anthony Santander
9	1	George Springer	9	9	Ryan McKenna

	game	TOR vs BAL on 20230823
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
45.	error	play,1,1,rutsa001,21,.BFBX,E3/G3.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Remark 46 This game had 3 errors. The Blue Jays we shut out by the Orioles 0-7.

In the the bottom of 1st inning with bases empty, BAL batter Adley Rutschman hit a ground ball to 1st. An error on the first baseman allowed the batter to reach 1st base safely.

	game	TOR vs BAL on 20230823
	state before	(1, 1, 1, 0, 2, 0, 0, 1)
46.	error	play,2,1,westj002,21,*BS*BX,FC4/G4M.2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 47 *In the bottom of the 2nd inning, with runners on 1st and 2nd, BAL batter Jordan Westburg hit a ground ball to a fielder's choice at deep 2nd. The runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner from 1st to advance to 2nd and the batter to reach 1st base. The bases are now loaded.*

	game	TOR vs BAL on 20230823
	state before	(0, 1, 1, 0, 8, 0, 3, 1)
47.	error	play,8,1,haysa001,02,CSB,WP.2-H(E2/TH) (NR) ;1-3
	state after	(0, 0, 0, 1, 8, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Remark 48 *In the bottom of the 8th inning, with runners on 1st and 2nd, BAL batter Austin Hays had a 0-2 count. The next pitch was wild. A throwing error by the catcher allowed the runner on 2nd to score (no RBI) and the runner on 1st to advance to 3rd. Hays was still at bat, now with a 1-2 count.*

	game	COL vs BAL on 20230825
	state before	(2, 1, 0, 0, 8, 4, 5, 1)
48.	error	play,8,1,mullc002,02,LC.>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 4, 5, 1)
	fielder	2 (catcher on the visiting team)

Remark 49 *In the bottom of the 8th inning, with a runner on 1st, BAL batter Curtis Mullins has a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd base. A throwing error by the catcher allowed the runner to advance to 3rd. Mullins is still at bat, now with a 1-2 count.*

49.	game	COL vs BAL on 20230826
	state before	(0, 1, 0, 0, 5, 2, 0, 1)
	error	play,5,1,mullc002,32,BC1BF.FF1B>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Remark 50 *In the bottom of the 5th inning, with a runner on 1st, BAL batter Curtis Mullins with a full count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd base. The catcher made a throwing error allowing the runner to reach 3rd. Mullins has struck out.*

50.	game	COL vs BAL on 20230827
	state before	(0, 0, 0, 0, 9, 3, 3, 0)
	error	play,9,0,goodh001,00,...X,S6/G6.B-2(E6/TH)
	state after	(0, 0, 1, 0, 9, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Remark 51 *In the top of the 9th inning, with bases empty, COL batter Hunter Goodman hit a single to the shortstop. The shortstop made a throwing error, which allowed the batter to reach 2nd base safely.*

51.	game	CHA vs BAL on 20230828
	state before	(1, 1, 1, 0, 3, 0, 2, 1)
	error	play,3,1,mullc002,01,FX,FC/G4.2-3;1X2(4E6)(6);B-1
	state after	(2, 1, 0, 1, 3, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Remark 52 *In the bottom of the 3rd inning, with runners on 1st and 2nd, BAL batter Curtis Mullins hit a ground ball to a fielder's choice at 2nd. The runner on 2nd base advanced to 3rd. In spite of an error by the shortstop, the runner advancing from 1st was tagged out at 2nd. However, the batter reached 1st base safely.*

	game	CHA vs BAL on 20230830
	state before	(2, 0, 1, 0, 3, 6, 4, 0)
52.	error	play,3,0,andre001,00,X,S8/L89D+.2-H;B-2(E8)
	state after	(2, 0, 1, 0, 3, 7, 4, 0)
	fielder	8 (center field on the home team)

Remark 53 *In the top of the 3rd inning, with a batter on 2nd, CHA batter Elvis Andrus hit a line drive single to deep right center field. The runner on 2nd scored (1 RBI). A throwing error by the center fielder allowed the batter to advance to 2nd base.*

6.2.6 September

	game	SLN vs BAL on 20230911
	state before	(2, 1, 0, 0, 2, 1, 0, 1)
53.	error	play,2,1,hicka001,20,.B*BX,S8/G4M.1-H(E5/TH)(NR);B-3
	state after	(2, 0, 0, 1, 2, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Remark 54 This game had 2 errors. The Cardinals lost to the Orioles 5-11.

In the bottom of the 2nd inning, with a runner on 1st, BAL batter Aaron Hicks hit a ground ball single to deep 2nd. A throwing error by the third baseman allowed the runner on 1st to score (no RBI) and the batter to advance to 3rd.

54.	game	SLN vs BAL on 20230911
	state before	(0, 0, 0, 0, 7, 5, 9, 0)
	error	play,7,0,gormn001,12,..CBFX,E1/TH/G23-.B-1
	state after	(0, 1, 0, 0, 7, 5, 9, 0)
	fielder	1 (pitcher on the home team)

Remark 55 *In the bottom of the 7th inning, with bases empty, SLN batter Nolan Gorman hit a soft ground ball to 1st. A throwing error by the pitcher allowed the batter to reach 1st base.*

55.	game	SLN vs BAL on 20230912
	state before	(1, 1, 1, 0, 4, 3, 0, 1)
	error	play,4,1,westj002,02,CFX,E5/G56.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 4, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Remark 56 *In the bottom of the 4th inning, with runners on 1st and 2nd, BAL batter Jordan Westburg hit a ground ball to 3rd base. The third baseman made an error which allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st base. The bases are now loaded.*

56.	game	TBA vs BAL on 20230915
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
	error	play,1,1,hendg002,12,CF.BX,E5/G2.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Remark 57 *In the bottom of the 1st inning with bases empty, BAL batter Gunnar Henderson hit a ground ball to shallow infield. An error by the third baseman allowed the batter to reach 1st base safely.*

	game	TBA vs BAL on 20230916
	state before	(2, 0, 0, 0, 6, 0, 8, 0)
57.	error	play,6,0,arozr001,22,BCBCX,E5/G5.B-1
	state after	(2, 1, 0, 0, 6, 0, 8, 0)
	fielder	5 (3rd baseman on the home team)

Remark 58 *In the top of the 6th inning with bases empty, TBA batter Randy Arozarena hit a ground ball to 3rd. An error by the third baseman allowed the batter to reach 1st base safely.*

	game	BOS vs BAL on 20230928
	state before	(0, 0, 0, 0, 6, 0, 1, 0)
58.	error	play,6,0,wongc001,11,BFX,E6/G78S.B-1
	state after	(0, 1, 0, 0, 6, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Remark 59 *In the top of the 6th inning, with bases empty, BOS batter Conner Wong hit a ground ball to shallow left center field. An error by the shortstop allowed the batter to reach 1st base safely.*

	game	BOS vs BAL on 20230929
	state before	(0, 1, 0, 0, 9, 2, 0, 0)
59.	error	play,9,0,dever001,12,FFFB>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 2, 0, 0)
	fielder	2 (catcher on the home team)

Remark 60 *In the top of the 9th inning with a runner on 1st, BOS batter Rafael Devers was at bat with a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd base. A throwing error by the catcher allowed the runner to reach 3rd. Devers was still at bat, now with a 2-2 count.*

	game	BOS vs BAL on 20230930
	state before	(2, 0, 1, 0, 7, 0, 1, 1)
60.	error	play,7,1,kjerh001,21,BBSX,S4/G4.2-H(E4)(NR)(UR)
	state after	(2, 1, 0, 0, 7, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Remark 61 *In the bottom of the 7th inning with a runner on 2nd, BAL batter Heston Kjerstad hit a ground ball single to 2nd. An error on the second baseman allowed the runner on 2nd to score (unearned, no RBI). The batter is safe at 1st.*

6.2.7 October

Visiting/Home team rosters for BOS vs BAL on 2023-10-01

position	innings	player	position	innings	player
1	1	Tanner Houck	1	1	Kyle Bradish
1	7	Brennan Bernardino	1	3	Danny Coulombe
1	8	John Schreiber	1	4	Tyler Wells
1	9	Nick Robertson	1	5	Shintaro Fujinami
2	1	Connor Wong	1	6	DL Hall
3	1	Bobby Dalbec	1	7	Cionel Perez
4	1	Enmanuel Valdez	1	7	Yennier Cano
5	1	Rafael Devers	1	8	Cole Irvin
5	4	Pablo Reyes	2	1	James McCann
6	1	Trevor Story	3	1	Ryan O'Hearn
7	1	Masataka Yoshida	3	7	Ryan Mountcastle
7	6	Wilyer Abreu	4	1	Adam Frazier
8	1	Wilyer Abreu	4	6	Jordan Westburg
8	6	Ceddanne Rafaela	5	1	Jordan Westburg
9	1	Alex Verdugo	5	6	Gunnar Henderson
9	7	Rob Refsnyder	5	8	Ramon Urias
			6	1	Gunnar Henderson
			6	6	Jorge Mateo
			6	8	Gunnar Henderson
			7	1	Heston Kjerstad
			8	1	Cedric Mullins
			8	8	Jorge Mateo
			9	1	Anthony Santander

61.	game	BOS vs BAL on 20231001
	state before	(1, 1, 1, 0, 3, 0, 0, 0)
	error	play,3,0,dever001,10,B>B,SB3;SB2.2-H(E2/TH) (NR) ;1-3
	state after	(1, 0, 0, 1, 3, 1, 0, 0)
	fielder	2 (catcher on the home team)

Remark 62 This game had 3 errors, two of them in the same inning involving the same batter. The Orioles lost to the Red Sox, 6-1.

In the top of the 3rd inning, with runners on 1st and 2nd, BOS batter Rafael Devers was at bat with a 1-0 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd base and the runner on 1st stole 2nd base. An error on the catcher's throw allowed the runner now on 3rd base to score (no RBI) and the runner now on 2nd to advance to 3rd. Devers is still at bat, now with a 2-0 count.

62.	game	BOS vs BAL on 20231001
	state before	(1, 0, 0, 1, 3, 1, 0, 0)
	error	play,3,0,dever001,20,B>B.X,E6/G6.3-H(UR) ;B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Remark 63 *Still in the same at bat in the top of the 3rd inning, with a runner on 3rd, BOS batter Rafael Devers hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (1 RBI) and the batter to reach 1st.*

63.	game	BOS vs BAL on 20231001
	state before	(1, 1, 1, 1, 7, 2, 0, 0)
	error	play,7,0,mcgur002,12,.CSBX,FC5/G56.3-H(E5) (NR) (UR) ;2-H(NR) (UR) ;1-2;B-1
	state after	(1, 1, 1, 0, 7, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Remark 64 *In the top of the 7th inning, with bases loaded, BOS batter Reese McGuire hit a ground ball to a fielder's choice at 3rd. An error by the third baseman allowed the runners on 2nd and 3rd to score (both unearned, no RBIs), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

Chapter 7

Boston Red Sox (BOS)

The Boston Red Sox compete in the American League East Division of the MLB. Founded in 1901 as one of the American League's eight charter franchises, the team's name "Red Sox" name was chosen by the team owner, John Taylor, around 1908. The Red Sox have won 9 World Series titles and 10 ALED titles, the latest in 2018.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Boston against the Red Sox. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (usually denoted DER or DefEff) is one measure of the fielding performance of a team. High DER is ideal. The DER of the 2023 BOS team is 0.677 compared with the MLB average¹ of 0.691 that year [BR]. This is over one standard deviation below average.

The regular season *error total* E is another team fielding performance metric. Low E is the goal. In 2023, there were 124 errors in the 81 Red Sox home games, 58 by the Red Sox and the rest (66) by visiting teams. Home and away, the Boston Red Sox had 102 errors, well over the 2023 MLB average of 83.9. That is a "high" error rate.

7.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

¹The standard deviation of these DER values is $\sigma = 0.011$.

7.1.1 Errors by position

This section tabulates the error-to-run conversion rate and error totals by position.

The following table only counts the runs made during the same play that the error occurred in.

Runs and errors	totals
Errors by vt:	66
Runs by ht because of errors by vt:	20
Errors by ht:	58
Runs by vt because of errors by ht:	15

This is a very good error-to-run conversion rate: $20/66 = 30.3\%$ on the part of the Red Sox (compared to $15/57 = 25.9\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 BOS season (home and away) has about $R/PA = 772/6174$?? (about 12.5%) change of resulting in a run Therefore, a randomly selected error is over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 102, so the expected number of errors (by either team) in a randomly selected BOS 2023 home game is about 1.26.

Errors by position	vt	ht
errors due to pitcher:	3	7
to catcher:	20	9
to 1st baseman:	4	4
to 2nd baseman:	7	6
to 3rd baseman:	11	11
to shortstop:	13	15
to left fielder:	3	3
to center fielder:	2	0
to right fielder:	3	2

7.1.2 Errors by state

This section tabulates the totals by game state.

Errors by state	vt	ht	totals
bases empty	19	24	43
(lone) runner on 1st	24	11	45
(lone) runner on 2nd	7	7	14
(lone) runner on 3rd	0	0	0
two runners on base	13	11	24
bases loaded	3	5	8
total	66	58	124

A table of frequencies of most common errors by the Red Sox in BOS 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	13.8%
(1, 0, 0, 0)	7	12.1%
(2, 0, 0, 0)	9	15.5%
(0, 1, 0, 0)	4	6.9%
(1, 1, 0, 0)	2	3.45%
(2, 1, 0, 0)	5	8.6%
(1, 0, 1, 0)	4	6.9%
(2, 1, 1, 0)	6	10.3%
(1, 1, 1, 1)	3	5.2%

A table of frequencies of most common errors by a visiting team in BOS 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	9.1%
(1, 0, 0, 0)	9	13.6%
(2, 0, 0, 0)	4	6.1%
(0, 1, 0, 0)	8	12.1%
(1, 1, 0, 0)	8	12.1%
(2, 1, 0, 0)	8	12.1%
(1, 0, 1, 0)	4	6.1%
(1, 1, 1, 0)	3	4.6%
(1, 1, 0, 1)	2	3.03%
(1, 1, 1, 1)	2	3.03%

Recall Figure 4.1 gave a frequency table of game states in all plays. From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Red Sox homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `BOS2023.eva`):

state	BOS2023 percentages
(0, 0, 0, 0)	23.8%
(1, 0, 0, 0)	15.9%
(2, 0, 0, 0)	11.7%
(0, 1, 0, 0)	5.6%
(1, 1, 0, 0)	7.8%
(2, 1, 0, 0)	7.0%
(0, 0, 1, 0)	2.3%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	4.2%
(0, 0, 0, 1)	0.4%
(1, 0, 0, 1)	1.1%
(2, 0, 0, 1)	1.7%
(0, 1, 1, 0)	1.2%
(1, 1, 1, 0)	2.4%
(2, 1, 1, 0)	3.2%

7.1.3 Some interesting errors

Some of the most interesting error plays in the BOS 2023 home games:

- The game TOR vs BOS on 2023-05-03 had 5 errors. An earlier game in the same series had 3 errors.
- The game TBA vs BOS on 2023-06-03 (1/2) had 3 errors. The next game in the doubleheader had 2 more errors.

7.1.4 Relative team rankings

We have 81 home games between BOS and some visiting MLB team.

BOS, the home team (ht), *lost* more games than won against²:

CHA (2/3), CIN (2/3), COL (2/3), LAN (2/3), (3, 3, 'HOU (3/3),
MIA (3/3), PIT (3/3), SLN (3/3), TBA (5/6)

The teams that *split* their series were:

BAL (3/6),

and *won* more than lost against

ATL (0/2), OAK (0/3), CLE (1/3), DET (1/3), MIN (1/3), NYN (1/3),
SEA (1/3), TEX (1/3), ANA (1/4), KCA (1/4), NYA (3/7), TOR (3/7)

The teams that did not play in any of the 81 home games in BOS 2023 were:

ARI, CHN, MIL, PHI, SDN, SFN, WAS

²Visiting teams (vt) are listed with that team's W/T record in parentheses (W = numbers of games vt won against ht at BOS, T = Total home games vt played vs ht at BOS).

7.2 Game-by-game totals

The errors are listed chronologically in the order of the game the error occurred in. To save space, we only give the team rosters in special cases.

A typical roster for BOS 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Connor Wong	122	.288
3 (1B)	Triston Casas	126	.367
4 (2B)	Christian Arroyo	62	.268
5 (3B)	Rafael Devers	151	.351
6 (SS)	Kiké Hernández	64	.279
7 (LF)	Masataka Yoshida	88	.338
8 (CF)	Jarren Duran	76	.346
9 (RF)	Alex Verdugo	140	.324

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 BOS home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

7.2.1 March

	game	BAL vs BOS on 20230330
	state before	(1, 0, 1, 0, 8, 10, 4, 0)
1.	error	play,8,0,uriar001,12,BFCX,FC6/G6.2-3(E6/TH);B-2
	state after	(1, 0, 1, 0, 8, 10, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 135 This home opener had 3 errors. The final score was 10-9.

In the top of the 8th inning, with a runner on 2nd, BAL batter Ramon Urias hit a ground ball into a fielder's choice at short. A throwing error

by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd.

	game	BAL vs BOS on 20230330
	state before	(0, 1, 0, 0, 9, 10, 7, 1)
2.	error	play,9,1,verda001,20,BBX,S8/L78D.1-3;B-2(E8)
	state after	(0, 0, 1, 1, 9, 10, 7, 1)
	fielder	8 (center field on the visiting team)

Summary 136 *In the bottom of the 9th inning, with a runner on 1st, BOS batter Alex Verdugo hit a line drive single to left center field. The runner on 1st advanced to 3rd. An error by the center fielder allowed the batter to reach 2nd.*

	game	BAL vs BOS on 20230330
	state before	(1, 1, 1, 0, 9, 10, 8, 1)
3.	error	play,9,1,yoshm002,32,.CBBBSX,6(1)/F0/G6M.2-H(E6)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 9, 10, 9, 1)
	fielder	6 (shortstop on the visiting team)

Summary 137 *In the bottom of the 9th inning, with runners on 1st and 2nd, BOS batter Masataka Yoshida hit a ground ball to deep short. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

7.2.2 April

	game	BAL vs BOS on 20230401
	state before	(2, 0, 0, 0, 9, 8, 6, 1)
4.	error	play,9,1,yoshm002,10,BX,E7/F7LS.B-1
	state after	(2, 1, 0, 0, 9, 8, 6, 1)
	fielder	7 (left field on the visiting team)

Summary 138 *In the bottom of the 9th inning, with bases empty, BOS batter Masataka Yoshida hit a fly ball to shallow left field. An error by the left fielder allowed the batter to reach 1st base.*

	game	BAL vs BOS on 20230402
	state before	(0, 1, 0, 0, 5, 3, 6, 1)
5.	error	play,5,1,verda001,11,C*BX,E6/F6D.1-2;B-1
	state after	(0, 1, 1, 0, 5, 3, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 139 This game had 2 errors. The final score was .
In the bottom of the 5th inning, with a runner on 1st, BOS batter Alex Verdugo hit a fly ball to deep short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	BAL vs BOS on 20230402
	state before	(1, 0, 1, 0, 8, 5, 8, 1)
6.	error	play,8,1,duvaa001,32,C1SFB>F1>B.BFX,5/L56S.2-3(E4)
	state after	(2, 0, 0, 1, 8, 5, 8, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 140 *In the bottom of the 8th inning, with a runner on 2nd, BOS batter Adam Duvall hit a line drive to shallow short. The third baseman putout the batter. An error by the second baseman allowed the runner on 2nd to advance to 3rd.*

	game	PIT vs BOS on 20230403
	state before	(1, 0, 0, 0, 1, 3, 1, 1)
7.	error	play,1,1,turnj001,00,X,E8/F89D.B-2
	state after	(1, 0, 1, 0, 1, 3, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 141 This game had 2 errors. The final score was 7-6.

In the bottom of the 1st inning, with bases empty, BOS batter Justin Turner hit a fly ball to deep right center field. An error by the center fielder allowed the batter to reach 2nd base.

	game	PIT vs BOS on 20230403
	state before	(1, 0, 0, 0, 7, 7, 6, 0)
8.	error	play,7,0,santc002,02,CFX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 7, 7, 6, 0)
	fielder	6 (shortstop on the home team)

Summary 142 *In the top of the 7th inning, with bases empty, PIT batter Carlos Santana hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs BOS on 20230404
	state before	(2, 1, 0, 0, 6, 3, 1, 1)
9.	error	play,6,1,yoshm002,12,*SFF*BX,E6/TH/G6.1-3;B-1
	state after	(2, 1, 0, 1, 6, 3, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 143 This game had 2 errors. The final score was 4-1.

In the bottom of the 6th inning, with a runner on 1st, BOS batter Masataka Yoshida hit a ground ball to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.

	game	PIT vs BOS on 20230404
	state before	(0, 0, 0, 0, 9, 4, 1, 1)
10.	error	play,9,1,yoshm002,00,.X,E3/G3.B-1
	state after	(0, 1, 0, 0, 9, 4, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 144 *In the bottom of the 9th inning, with bases empty, BOS batter Masataka Yoshida hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

11.	game	PIT vs BOS on 20230405
	state before	(0, 1, 0, 1, 7, 2, 0, 0)
	error	play,7,0,reynb001,00,X,7/SF/F78.3-H;1-3(E7/TH)
	state after	(1, 0, 0, 1, 7, 3, 0, 0)
	fielder	7 (left field on the home team)

Summary 145 This game had 2 errors. The final score was 4-1 (coincidentally, the same score as in the game the day before).

In the top of the 7th inning, with runners on 1st and 3rd, PIT batter Bryan Reynolds hit a fly ball to left center field. The left fielder putout the batter. The runner on 3rd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd.

12.	game	PIT vs BOS on 20230405
	state before	(2, 0, 0, 0, 8, 4, 1, 0)
	error	play,8,0,bae-j001,22,MBFBFFX,S1/L13.B-2(E1/TH)
	state after	(2, 0, 1, 0, 8, 4, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 146 *In the top of the 8th inning, with bases empty, PIT batter Ji Hwan Bae hit a line drive single to shallow right infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

Visiting/Home team rosters for ANA vs BOS on 2023-04-14

position	innings	player	position	innings	player
1	1	Patrick Sandoval	1	1	Tanner Houck
1	4	Jimmy Herget	1	5	Josh Winckowski
1	5	Aaron Loup	1	8	John Schreiber
1	7	Jaime Barria	1	9	Kenley Jansen
2	1	Logan O'Hoppe	2	1	Connor Wong
3	1	Jake Lamb	3	1	Triston Casas
4	1	Brandon Drury	4	1	Christian Arroyo
5	1	Anthony Rendon	5	1	Rafael Devers
6	1	Luis Rengifo	6	1	Yu Chang
7	1	Taylor Ward	7	1	Rob Refsnyder
8	1	Mike Trout	8	1	Enrique Hernandez
9	1	Hunter Renfroe	9	1	Alex Verdugo

13.	game	ANA vs BOS on 20230414
	state before	(0, 0, 0, 0, 4, 2, 0, 1)
	error	play,4,1,dever001,32,CBFFBBFX,E6/G4M.B-1
	state after	(0, 1, 0, 0, 4, 2, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 147 This game had 3 errors. The final score was 3-5.

In the bottom of the 4th inning, with bases empty, BOS batter Rafael Devers hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.

14.	game	ANA vs BOS on 20230414
	state before	(2, 1, 1, 0, 4, 2, 1, 1)
	error	play,4,1,chany001,12,CS*BX,E5/TH/G5S.2-H(NR)(UR);1-3;B-2
	state after	(2, 0, 1, 1, 4, 2, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 148 *In the bottom of the 4th inning, with runners on 1st and 2nd, BOS batter Yu Chang hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	ANA vs BOS on 20230414
	state before	(2, 0, 1, 1, 6, 2, 3, 1)
15.	error	play,6,1,refsr001,20,BBX,E5/TH/G5.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 6, 2, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 149 *In the bottom of the 6th inning, with runners on 2nd and 3rd, BOS batter Rob Refsnyder hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	ANA vs BOS on 20230415
	state before	(2, 1, 1, 0, 2, 4, 2, 0)
16.	error	play,2,0,ohtas001,12,SFBX,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 2, 4, 2, 0)
	fielder	2 (catcher on the home team)

Summary 150 This game had 3 errors, *all* by catcher's interference! The final score was 3-5.

In the top of the 2nd inning, with runners on 1st and 2nd, ANA batter Shohei Ohtani had a 1-2 count. Catcher's interference allowed the batter to take 1st, and (necessarily) the runners on 1st and 2nd each advance one base.

	game	ANA vs BOS on 20230415
	state before	(0, 1, 0, 0, 8, 7, 6, 1)
17.	error	play,8,1,tapir001,02,.LFX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 8, 7, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 151 *In the bottom of the 8th inning, with a runner on 1st, BOS batter Raimel Tapia had a 0-2 count. Catcher's interference allowed the batter to take 1st, and (necessarily) the runner on 1st advanced one base.*

	game	ANA vs BOS on 20230415
	state before	(1, 1, 1, 0, 8, 7, 6, 1)
18.	error	play,8,1,mcgur002,10,.BX,C/E2.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 8, 7, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 152 *In the bottom of the 8th inning, with runners on 1st and 2nd, BOS batter Reese McGuire had a 1-0 count. Catcher's interference allowed the batter to take 1st, and (necessarily) the runner on 1st advanced one base.*

Visiting/Home team rosters for ANA vs BOS on 2023-04-17

position	innings	player	position	innings	player
1	1	Shohei Ohtani	1	1	Brayan Bello
1	3	Tucker Davidson	1	3	Kutter Crawford
1	6	Aaron Loup	2	1	Reese McGuire
1	7	Matt Moore	2	9	Connor Wong
1	8	Jose Quijada	3	1	Triston Casas
1	9	Carlos Estevez	4	1	Christian Arroyo
2	1	Logan O'Hoppe	5	1	Rafael Devers
3	1	Jake Lamb	6	1	Enrique Hernandez
3	7	Brandon Drury	7	1	Rob Refsnyder
4	1	Brandon Drury	8	1	Jarren Duran
4	7	Luis Rengifo	9	1	Raimel Tapia
5	1	Gio Urshela			
6	1	Zach Neto			
7	1	Taylor Ward			
8	1	Brett Phillips			
9	1	Hunter Renfroe			

	game	ANA vs BOS on 20230417
	state before	(1, 0, 0, 0, 6, 5, 1, 0)
19.	error	play,6,0,netoz001,12,CFBX,E6/TH/G6-.B-1
	state after	(1, 1, 0, 0, 6, 5, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 153 This game had 3 errors. The final score was 5-4.
In the top of the 6th inning, with bases empty, ANA batter Zach Neto hit a soft ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	ANA vs BOS on 20230417
	state before	(1, 1, 0, 0, 6, 5, 1, 0)
20.	error	play,6,0,ohtas001,10,B1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 154 *In the top of the 6th inning, with a runner on 1st, ANA batter Shohei Ohtani has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to advance to 2nd.*

	game	ANA vs BOS on 20230417
	state before	(1, 0, 1, 0, 6, 5, 1, 0)
21.	error	play,6,0,ohtas001,10,B1.X,E1/TH/G23-.2-3;B-1
	state after	(1, 1, 0, 1, 6, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 155 *In the top of the 6th inning, with a runner on 2nd, ANA batter Shohei Ohtani hit a soft ground ball to shallow left infield. A throwing error by the pitcher allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	MIN vs BOS on 20230418
	state before	(0, 1, 0, 0, 1, 0, 1, 1)
22.	error	play,1,1,turnj001,22,BSBCFFX,S5/G56S.1-3(E5/TH);B-2
	state after	(0, 0, 1, 1, 1, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 156 This game had 2 errors. The final score was 4-5.

In the bottom of the 1st inning, with a runner on 1st, BOS batter Justin Turner hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	MIN vs BOS on 20230418
	state before	(1, 1, 0, 0, 8, 2, 1, 1)
23.	error	play,8,1,mcgur002,22,.BBCSX,5E3/G6S.1-3;B-1
	state after	(1, 1, 0, 1, 8, 2, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 157 *In the bottom of the 8th inning, with a runner on 1st, BOS batter Reese McGuire hit a ground ball to shallow short. An error by the first baseman (on an assist from the shortstop) allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	MIN vs BOS on 20230419
	state before	(2, 1, 0, 0, 5, 7, 1, 0)
24.	error	play,5,0,gallj002,00,X,E4/P9S.1-3;B-2
	state after	(2, 0, 1, 1, 5, 7, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 158 *In the top of the 5th inning, with a runner on 1st, MIN batter Joey Gallo hit a pop fly to shallow right field. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	CLE vs BOS on 20230429
	state before	(1, 1, 0, 1, 6, 1, 5, 1)
25.	error	play,6,1,yoshm002,00,>B,SB2.3-H(E2/TH) (NR) (UR);1-3
	state after	(1, 0, 0, 1, 6, 1, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 159 *In the bottom of the 6th inning, with runners on 1st and 3rd, BOS batter Masataka Yoshida had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI) and the runner from 1st (now on 2nd) to further advance to 3rd. Yoshida is still at bat, with a 1-0 count.*

	game	CLE vs BOS on 20230430
	state before	(2, 0, 0, 0, 3, 0, 0, 1)
26.	error	play,3,1,turnj001,22,CBFBX,E6/TH/G6D.B-1
	state after	(2, 1, 0, 0, 3, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 160 *This game had 2 errors. The final score was 1-7. In the bottom of the 3rd inning, with bases empty, BOS batter Justin Turner hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs BOS on 20230430
	state before	(0, 0, 0, 0, 5, 0, 0, 0)
27.	error	play,5,0,gonzo001,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 161 *In the top of the 5th inning, with bases empty, CLE batter Oscar Gonzalez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

7.2.3 May

Visiting/Home team rosters for TOR vs BOS on 2023-05-01

position	innings	player	position	innings	player
1	1	Jose Berrios	1	1	Corey Kluber
1	6	Tim Mayza	1	6	Brennan Bernardino
1	7	Nate Pearson	1	7	Chris Martin
1	9	Jordan Romano	1	8	Josh Winckowski
2	1	Danny Jansen	2	1	Reese McGuire
3	1	Brandon Belt	3	1	Triston Casas
3	7	Cavan Biggio	4	1	Enmanuel Valdez
4	1	Whit Merrifield	5	1	Rafael Devers
5	1	Matt Chapman	6	1	Enrique Hernandez
6	1	Bo Bichette	6	9	Christian Arroyo
7	1	Daulton Varsho	7	1	Masataka Yoshida
8	1	Kevin Kiermaier	8	1	Jarren Duran
9	1	Cavan Biggio	9	1	Alex Verdugo
9	7	Nathan Lukes			

	game	TOR vs BOS on 20230501
	state before	(1, 0, 1, 0, 8, 3, 5, 0)
28.	error	play,8,0,kierk001,01,SX,E6/TH/G4.2-H(NR);B-1
	state after	(1, 1, 0, 0, 8, 4, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 162 This game had 3 errors. The final score was 5-1.

In the top of the 8th inning, with a runner on 2nd, TOR batter Kevin Kiermaier hit a ground ball to 2nd. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st base.

	game	TOR vs BOS on 20230501
	state before	(1, 1, 1, 0, 8, 4, 5, 0)
29.	error	play,8,0,guerv002,12,FBFX,6(1)/F0/G6M.2-H(E6/TH)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 8, 5, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 163 *In the top of the 8th inning, with runners on 1st and 2nd, TOR batter Vladimir Guerrero hit a fly ball to short. The shortstop forced out the runner from 1st at 2nd base. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	TOR vs BOS on 20230501
	state before	(1, 1, 0, 0, 8, 5, 5, 1)
30.	error	play,8,1,valde002,10,.B>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 8, 5, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 164 *In the bottom of the 8th inning, with a runner on 1st, BOS batter Enmanuel Valdez had a 1-0 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Valdez is still at bat, now with a 1-1 count.*

Visiting/Home team rosters for TOR vs BOS on 2023-05-03

position	innings	player	position	innings	player
1	1	Alek Manoah	1	1	Nick Pivetta
1	6	Anthony Bass	1	7	Chris Martin
1	6	Trevor Richards	1	8	Kutter Crawford
1	7	Nate Pearson	1	9	Ryan Brasier
2	1	Alejandro Kirk	2	1	Connor Wong
3	1	Vladimir Guerrero Jr.	3	1	Triston Casas
4	1	Whit Merrifield	4	1	Enmanuel Valdez
5	1	Santiago Espinal	5	1	Rafael Devers
6	1	Bo Bichette	6	1	Enrique Hernandez
7	1	Daulton Varsho	7	1	Masataka Yoshida
8	1	Kevin Kiermaier	8	1	Jarren Duran
9	1	George Springer	9	1	Alex Verdugo

	game	TOR vs BOS on 20230503
	state before	(0, 1, 0, 0, 4, 2, 0, 0)
31.	error	play,4,0,kirka001,32,FBF.B*B>B,W.1-3(E2/TH)
	state after	(0, 1, 0, 1, 4, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 165 This game had 5 errors. The final score was 3-8.

In the top of the 4th inning, with a runner on 1st, TOR batter Alejandro Kirk had a full count. During the next pitch, which was a ball, the runner on 1st attempted to advance beyond 2nd. A throwing error by the catcher allowed the runner on 1st to advance to 3rd. The batter walked and took 1st.

	game	TOR vs BOS on 20230503
	state before	(1, 0, 0, 0, 4, 2, 0, 1)
32.	error	play,4,1,dever001,11,FBX,S12/L23.B-2(E2/TH)
	state after	(1, 0, 1, 0, 4, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 166 *In the bottom of the 4th inning, with bases empty, BOS batter Rafael Devers hit a line drive single to left shallow infield. A throwing error by the catcher allowed the batter to reach 2nd.*

	game	TOR vs BOS on 20230503
	state before	(1, 0, 1, 0, 4, 2, 1, 1)
33.	error	play,4,1,casat001,00,2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 4, 2, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 167 *In the bottom of the 4th inning, with a runner on 2nd, BOS batter Triston Casas has a 0-0 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to safely advance to 3rd. Casas is still at bat with a 0-0 count.*

	game	TOR vs BOS on 20230503
	state before	(1, 0, 0, 0, 5, 3, 2, 1)
34.	error	play,5,1,yoshm002,32,CBBBFX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 5, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 168 *In the bottom of the 5th inning, with bases empty, BOS batter Masataka Yoshida hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TOR vs BOS on 20230503
	state before	(1, 0, 0, 0, 8, 3, 7, 1)
35.	error	play,8,1,casat001,02,CSFX,S7/F78S.B-2(E7/TH)
	state after	(1, 0, 1, 0, 8, 3, 7, 1)
	fielder	7 (left field on the visiting team)

Summary 169 *In the bottom of the 8th inning, with bases empty, BOS batter Triston Casas hit a fly ball single to shallow left center field. A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	TOR vs BOS on 20230504
	state before	(1, 1, 0, 0, 5, 1, 8, 0)
36.	error	play,5,0,sprig001,32,BCBBFX,E5/G6.1-2;B-1
	state after	(1, 1, 1, 0, 5, 1, 8, 0)
	fielder	5 (3rd baseman on the home team)

Summary 170 This game had 2 errors. The final score was 5-11. *In the top of the 5th inning, with a runner on 1st, TOR batter George Springer hit a ground ball to short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TOR vs BOS on 20230504
	state before	(2, 1, 0, 0, 6, 4, 8, 1)
37.	error	play,6,1,casat001,00,>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 4, 8, 1)
	fielder	2 (catcher on the visiting team)

Summary 171 *In the bottom of the 6th inning, with a runner on 1st, BOS batter Triston Casas had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Casas is still at bat, now with a 1-0 count.*

	game	SLN vs BOS on 20230512
	state before	(0, 1, 0, 0, 9, 6, 6, 0)
38.	error	play,9,0,gormn001,00,.>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 6, 6, 0)
	fielder	2 (catcher on the home team)

Summary 172 *In the top of the 9th inning, with a runner on 1st, SLN batter Nolan Gorman had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Gorman is still at bat, now with a 0-1 count.*

	game	SLN vs BOS on 20230513
	state before	(1, 1, 1, 1, 9, 2, 3, 0)
39.	error	play,9,0,burla001,10,.BX,46(1)/F0/G4.3-H;2-H(E6/TH)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 9, 4, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 173 *In the top of the 9th inning, with bases loaded, SLN batter Alec Burleson hit a ground ball to 2nd.. The shortstop (with an assist from the second baseman) putout the runner from 1st. The runner from 3rd scored (1 RBI) A throwing error by the shortstop allowed the runner on 2nd to scpre (unearned, no RBI) and the batter to advance to 2nd base.*

	game	SLN vs BOS on 20230514
	state before	(0, 0, 0, 0, 7, 6, 1, 0)
40.	error	play,7,0,donob001,00,X,E3/TH/G34.B-2
	state after	(0, 0, 1, 0, 7, 6, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 174 *In the top of the 7th inning, with bases empty, SLN batter Brendan Donovan hit a ground ball to the 1st/2nd base gap. A throwing error by the first baseman allowed the batter to reach 2nd base.*

	game	SEA vs BOS on 20230515
	state before	(0, 0, 0, 0, 6, 2, 0, 0)
41.	error	play,6,0,frant002,00,X,D9/L9L.B-3(E9/TH)
	state after	(0, 0, 0, 1, 6, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 175 This game had 2 errors. The final score was 10-1.
In the top of the 6th inning, with bases empty, SEA batter Ty France hit a line drive double to right field. A throwing error by the right fielder allowed the batter to reach 3rd.

	game	SEA vs BOS on 20230515
	state before	(0, 1, 0, 0, 8, 7, 1, 0)
42.	error	play,8,0,hernt002,12,CCFBX,FC4/G4.1-2(E4);B-1
	state after	(0, 1, 1, 0, 8, 7, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 176 *In the top of the 8th inning, with a runner on 1st, SEA batter Teoscar Hernandez hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SEA vs BOS on 20230516
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
43.	error	play,1,1,verda001,02,CCFFX,E4/G4.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 177 *In the bottom of the 1st inning, with bases empty, BOS batter Alex Verdugo hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SEA vs BOS on 2023-05-17

position	innings	player	position	innings	player
1	1	Marco Gonzales	1	1	Brayan Bello
1	2	Trevor Gott	1	6	Joely Rodriguez
1	4	Matt Brash	1	7	Chris Martin
1	5	Gabe Speier	1	8	Ryan Sherriff
1	6	Chris Flexen	1	9	Justin Garza
2	1	Cal Raleigh	2	1	Connor Wong
3	1	Ty France	3	1	Justin Turner
4	1	Jose Caballero	3	7	Triston Casas
4	6	Sam Haggerty	4	1	Pablo Reyes
5	1	Eugenio Suarez	4	8	Enmanuel Valdez
6	1	J.P. Crawford	5	1	Rafael Devers
6	6	Jose Caballero	5	8	Pablo Reyes
7	1	Jarred Kelenic	6	1	Enrique Hernandez
8	1	Julio Rodriguez	7	1	Rob Refsnyder
9	1	Teoscar Hernandez	8	1	Jarren Duran
			9	1	Alex Verdugo
			9	3	Raimel Tapia

	game	SEA vs BOS on 20230517
	state before	(2, 1, 0, 0, 2, 0, 5, 1)
44.	error	play,2,1,yoshm002,31,BBBCX,S16/G4M.1-3(E6/TH)
	state after	(2, 1, 0, 1, 2, 0, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 178 This game had 3 errors. The final score was 3-12.

In the bottom of the 2nd inning, with a runner on 1st, BOS batter Masataka Yoshida hit a ground single to 2nd. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	SEA vs BOS on 20230517
	state before	(1, 0, 0, 0, 5, 1, 9, 0)
45.	error	play,5,0,rodrj007,12,FBSX,E5/TH/G56S.B-1
	state after	(1, 1, 0, 0, 5, 1, 9, 0)
	fielder	5 (3rd baseman on the home team)

Summary 179 *In the top of the 5th inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to shallow 3rd. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	SEA vs BOS on 20230517
	state before	(1, 1, 1, 1, 6, 1, 12, 0)
46.	error	play,6,0,rodrj007,01,CX,64(1)/F0/G6.3-H;2-H(E4/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 1, 6, 1, 12, 0)
	fielder	4 (2nd baseman on the home team)

Summary 180 *In the top of the 6th inning, with bases loaded, SEA batter Julio Rodriguez hit a ground ball to short. The second baseman (assisted by the shortstop) forced out the runner from 1st and the runner from 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter reached 2nd.*

	game	CIN vs BOS on 20230530
	state before	(2, 0, 0, 0, 3, 1, 0, 0)
47.	error	play,3,0,stept001,10,BX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 181 This game had 2 errors. The final score was 9-8. *In the top of the 3rd inning, with bases empty, CIN batter Tyler Stephenson hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CIN vs BOS on 20230530
	state before	(1, 1, 1, 0, 6, 2, 0, 0)
48.	error	play,6,0,mclam001,11,BCX,FC6/G6.2-H(NR)(UR);1-3(E6/TH);B-2
	state after	(1, 1, 1, 0, 6, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 182 *In the top of the 6th inning, with runners on 1st and 2nd, CIN batter Matt McLain hit a ground ball into a fielder's choice at short. The runner from 2nd scored (unearned, no RBI). A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	CIN vs BOS on 20230531
	state before	(0, 0, 0, 0, 7, 2, 3, 0)
49.	error	play,7,0,mail1001,21,CBBX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 7, 2, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 183 This game had 2 errors. The final score was 5-4. *In the top of the 7th inning, with bases empty, CIN batter Luke Maile hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CIN vs BOS on 20230531
	state before	(1, 0, 0, 0, 7, 5, 3, 1)
50.	error	play,7,1,valde002,01,CX,C/E2.B-1
	state after	(1, 1, 0, 0, 7, 5, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 184 *In the bottom of the 7th inning, with bases empty, BOS batter Enmanuel Valdez had a 0-1 count. Catcher's interference allowed the batter to take 1st.*

7.2.4 June

Visiting/Home team rosters for TBA vs BOS on 2023-06-03 (1/2)

position	innings	player	position	innings	player
1	1	Trevor Kelley	1	1	Garrett Whitlock
1	3	Joe La Sorsa	1	5	Ryan Sherriff
1	5	Kevin Kelly	1	6	Corey Kluber
1	6	Cooper Criswell	1	7	Josh Winckowski
1	6	Jalen Beeks	1	8	Chris Martin
2	1	Christian Bethancourt	1	9	Kenley Jansen
3	1	Yandy Diaz	2	1	Reese McGuire
4	1	Brandon Lowe	3	1	Justin Turner
5	1	Taylor Walls	4	1	Pablo Reyes
6	1	Wander Franco	5	1	Rafael Devers
7	1	Randy Arozarena	6	1	Enrique Hernandez
8	1	Jose Siri	7	1	Masataka Yoshida
9	1	Josh Lowe	8	1	Rob Refsnyder
			8	6	Jarren Duran
			9	1	Alex Verdugo

	game	TBA vs BOS on 20230603 (1/2)
	state before	(2, 1, 1, 0, 2, 0, 0, 0)
51.	error	play,2,0,bethc001,02,CCFX,S6/G6.2-H(E6/TH) (NR) ;1-3
	state after	(2, 1, 0, 1, 2, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 185 This game (game 1 of a doubleheader) had 3 errors. The final score was 5-8.

In the top of the 2nd inning, with runners on 1st and 2nd, TBA batter Christian Bethancourt hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 3rd. The batter took 1st.

52.	game	TBA vs BOS on 20230603 (1/2)
	state before	(2, 0, 0, 0, 3, 3, 0, 0)
	error	play,3,0,ramih004,01,CX,E6/TH/G6M.B-2
	state after	(2, 0, 1, 0, 3, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 186 *In the top of the 3rd inning, with bases empty, TBA batter Harold Ramirez hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 2nd base.*

53.	game	TBA vs BOS on 20230603 (1/2)
	state before	(2, 1, 0, 0, 7, 5, 7, 1)
	error	play,7,1,verda001,01,C>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 5, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 187 *In the bottom of the 7th inning, with a runner on 1st, BOS batter Alex Verdugo had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Verdugo is still at bat, now with a 1-1 count.*

54.	game	TBA vs BOS on 20230603 (2/2)
	state before	(1, 1, 0, 0, 6, 2, 1, 1)
	error	play,6,1,casat001,12,.F*BS>*B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 2, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 188 This game (game 2 of a doubleheader) had 2 errors. The final score was 4-2.

In the bottom of the 6th inning, with a runner on 1st, BOS batter Triston Casas had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Casas is still at bat, now with a 2-2 count.

	game	TBA vs BOS on 20230603 (2/2)
	state before	(2, 1, 0, 0, 7, 2, 1, 1)
55.	error	play,7,1,dever001,00,X,E4/P3D.1-H(NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 7, 2, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 189 *In the bottom of the 7th inning, with a runner on 1st, BOS batter Rafael Devers hit a pop up to deep 1st base. An error by the second baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	TBA vs BOS on 20230604
	state before	(0, 1, 1, 1, 4, 1, 1, 0)
56.	error	play,4,0,margm001,11,BSX,S7/L78S.3-H;2-H(E7) (NR) ;1-2
	state after	(0, 1, 1, 0, 4, 3, 1, 0)
	fielder	7 (left field on the home team)

Summary 190 This game had 2 errors. The final score was 6-2.
In the top of the 4th inning, with bases loaded, TBA batter Manuel Margot hit a line drive single to shallow left center field. The runner on 3rd scored (1 RBI). An error by the left fielder allowed the runner from 2nd to also score (unearned, no RBI) and the runner on 1st to advance to 2nd. The batter took 1st.

	game	TBA vs BOS on 20230604
	state before	(2, 1, 0, 0, 6, 4, 2, 0)
57.	error	play,6,0,diazy001,12,BFF>X,S9/G4.1-H;B-H(E2/TH) (NR)
	state after	(2, 0, 0, 0, 6, 6, 2, 0)
	fielder	2 (catcher on the home team)

Summary 191 *In the top of the 6th inning, with a runner on 1st, TBA batter Yandy Diaz hit a ground ball single to 2nd. The runner on 1st scored (1 RBI). A throwing error by the catcher allowed the batter to also score (unearned, no RBI).*

	game	TBA vs BOS on 20230605
	state before	(2, 0, 0, 0, 6, 3, 0, 0)
58.	error	play,6,0,margm001,11,FBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 6, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 192 *In the top of the 6th inning, with bases empty, TBA batter Manuel Margot hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	COL vs BOS on 20230612
	state before	(2, 0, 1, 0, 4, 0, 0, 0)
59.	error	play,4,0,gricr001,00,X,E6/TH/G6.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 4, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 193 This game had 2 errors. The final score was 4-3. *In the top of the 4th inning, with a runner on 2nd, COL batter Randal Grichuk hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	COL vs BOS on 20230612
	state before	(2, 1, 1, 1, 10, 3, 2, 0)
60.	error	play,10,0,jonen002,32,.VBC.BS>X,E3/G3.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 10, 4, 2, 0)
	fielder	3 (shortstop on the home team)

Summary 194 *In the top of the 10th inning, with bases loaded, COL batter Nolan Jones hit a ground ball to 1st. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	COL vs BOS on 20230613
	state before	(1, 0, 0, 0, 9, 4, 4, 0)
61.	error	play,9,0,tovae001,12,FFFBFX,6E3/G6.B-1
	state after	(1, 1, 0, 0, 9, 4, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 195 *In the top of the 9th inning, with bases empty, COL batter Ezequiel Tovar hit a ground ball to short. An error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st base.*

	game	COL vs BOS on 20230614
	state before	(0, 0, 0, 0, 2, 0, 1, 0)
62.	error	play,2,0,gricr001,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 196 *In the top of the 2nd inning, with bases empty, COL batter Randal Grichuk hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for NYA vs BOS on 2023-06-16

position	innings	player	position	innings	player
1	1	Domingo German	1	1	Tanner Houck
1	3	Matt Krook	1	5	Joe Jacques
1	4	Albert Abreu	1	7	Corey Kluber
1	6	Nick Ramirez	1	9	Brennan Bernardino
1	8	Isiah Kiner-Falefa	2	1	Connor Wong
2	1	Kyle Higashioka	3	1	Triston Casas
3	1	Anthony Rizzo	4	1	Christian Arroyo
4	1	Gleyber Torres	5	1	Rafael Devers
5	1	Josh Donaldson	6	1	Pablo Reyes
6	1	Anthony Volpe	7	1	Masataka Yoshida
7	1	Willie Calhoun	7	9	Rob Refsnyder
8	1	Billy McKinney	8	1	Adam Duvall
9	1	Jake Bauers	8	9	Jarren Duran
			9	1	Alex Verdugo

63.	game	NYA vs BOS on 20230616
	state before	(2, 1, 1, 0, 4, 1, 10, 1)
	error	play,4,1,wongc001,22,.C*BBSX,S9/L9L.2-H;1-H(NR);B-2(E9/TH)
	state after	(2, 0, 1, 0, 4, 1, 12, 1)
	fielder	9 (right field on the visiting team)

Summary 197 This game had 3 errors. The final score was 5-15.
In the bottom of the 4th inning, with runners on 1st and 2nd, BOS batter Connor Wong hit a line drive single to right field. The runner on 1st and 2nd scored (2 RBI). A throwing error by the right fielder allowed the batter to reach 2nd.

	game	NYA vs BOS on 20230616
	state before	(0, 1, 0, 0, 6, 1, 13, 0)
64.	error	play,6,0,torrg001,00,X,FC6/G6.1-2(E6);B-1
	state after	(0, 1, 1, 0, 6, 1, 13, 0)
	fielder	6 (shortstop on the home team)

Summary 198 *In the top of the 6th inning, with a runner on 1st, NYA batter Gleyber Torres hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	NYA vs BOS on 20230616
	state before	(1, 0, 1, 0, 6, 4, 13, 1)
65.	error	play,6,1,casat001,32,.C.BBFBX,S4/G34.2-H(E4/TH) (NR) (UR)
	state after	(1, 1, 0, 0, 6, 4, 14, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 199 *In the bottom of the 6th inning, with a runner on 2nd, BOS batter Triston Casas hit a ground ball single to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	NYA vs BOS on 20230618 (1/2)
	state before	(2, 0, 0, 0, 5, 2, 0, 0)
66.	error	play,5,0,trevj001,10,BX,E5/G56.B-1
	state after	(2, 1, 0, 0, 5, 2, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 200 This game 1 of a doubleheader had 1 error. The final score was 2-6.

In the top of the 5th inning, with bases empty, NYA batter Jose Trevino hit a ground ball to 3rd. An error by the third baseman allowed the batter to reach 1st base.

	game	NYA vs BOS on 20230618 (2/2)
	state before	(2, 1, 1, 1, 4, 1, 1, 1)
67.	error	play,4,1,mcgur002,00,X,C/E2.3-H(RBI)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 4, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 201 This game 2 of a doubleheader also had 1 error. The final score was 1-4.

In the bottom of the 4th inning, with bases loaded, BOS batter Reese McGuire had a 0-0 count. Catcher's interference allowed the batter to take 1st. Consequently:

(a) The runners on 1st and 2nd each advanced one base.

(b) The runner on 3rd base scored (unearned, 1 RBI).

	game	MIA vs BOS on 20230629
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
68.	error	play,2,0,chisj001,32,BBBCF.X,E3/G9LS.B-2
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 202 *In the top of the 2nd inning, with bases empty, MIA batter Jazz Chisholm hit a ground ball to shallow right field down the 1st base line. An error by the first baseman allowed the batter to reach 2nd base.*

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	game	TEX vs BOS on 20230704
	state before	(2, 0, 0, 0, 8, 6, 1, 0)
69.	error	play,8,0,seagc001,00,X,E6/G4.B-1
	state after	(2, 1, 0, 0, 8, 6, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 203 This game had 2 errors. The final score was 6-2.

In the top of the 8th inning, with bases empty, TEX batter Corey Seager hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.

	game	TEX vs BOS on 20230704
	state before	(2, 0, 1, 0, 9, 6, 2, 1)
70.	error	play,9,1,refsr001,22,B>C.SBX,C/E2.B-1
	state after	(2, 1, 1, 0, 9, 6, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 204 *In the bottom of the 9th inning, with a runner on 2nd, BOS batter Rob Refsnyder had a 2-2 count. Catcher's interference allowed the batter to take 1st. (No other runner advanced.)*

	game	TEX vs BOS on 20230705
	state before	(1, 1, 0, 0, 6, 2, 3, 1)
71.	error	play,6,1,hamid002,00,X,D9/G3.1-H(E9) (NR) (UR) ;B-3
	state after	(1, 0, 0, 1, 6, 2, 4, 1)
	fielder	9 (right field on the visiting team)

Summary 205 *In the bottom of the 6th inning, with a runner on 1st, BOS batter David Hamilton hit a ground ball double to 1st. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

Visiting/Home team rosters for TEX vs BOS on 2023-07-06

position	innings	player	position	innings	player
1	1	Nathan Eovaldi	1	1	Kutter Crawford
1	6	Brock Burke	1	5	Joe Jacques
1	6	Josh Sborz	1	6	Brandon Walter
1	7	Grant Anderson	1	7	Josh Winckowski
1	7	John King	1	9	Chris Martin
1	7	Jose Leclerc	1	9	Justin Garza
2	1	Jonah Heim	2	1	Connor Wong
3	1	Nathaniel Lowe	3	1	Triston Casas
4	1	Marcus Semien	4	1	Enrique Hernandez
5	1	Josh Jung	4	7	Christian Arroyo
6	1	Corey Seager	5	1	Rafael Devers
7	1	Robbie Grossman	6	1	David Hamilton
8	1	Leody Taveras	6	7	Enrique Hernandez
9	1	Adolis Garcia	7	1	Masataka Yoshida
			8	1	Adam Duvall
			9	1	Alex Verdugo

	game	TEX vs BOS on 20230706
	state before	(2, 1, 0, 0, 5, 3, 3, 0)
72.	error	play,5,0,jungj001,01,CX,D7/G56.1-H(E7/TH) (NR)
	state after	(2, 0, 1, 0, 5, 4, 3, 0)
	fielder	7 (left field on the home team)

Summary 206 This game had 3 errors. The final score was 6-10.

In the top of the 5th inning, with a runner on 1st, TEX batter Josh Jung hit a ground ball double to 3rd. A throwing error by the left fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 2nd.

73.	game	TEX vs BOS on 20230706
	state before	(0, 1, 0, 0, 6, 5, 3, 1)
	error	play,6,1,herne001,32,C*BBFBX,54(1)/FO/G56S.B-2(E4/TH)
	state after	(1, 0, 1, 0, 6, 5, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 207 *In the bottom of the 6th inning, with a runner on 1st, BOS batter Enrique Hernandez hit a ground ball to shallow 3rd. The second baseman (with an assist from the third baseman) forced out the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.*

74.	game	TEX vs BOS on 20230706
	state before	(1, 1, 1, 1, 7, 5, 4, 1)
	error	play,7,1,herne001,12,.S.FFBX,S7/L7S.3-H;2-H;1-3(E7/TH);B-2
	state after	(1, 0, 1, 1, 7, 5, 6, 1)
	fielder	7 (left field on the visiting team)

Summary 208 *In the bottom of the 7th inning, with bases loaded, BOS batter Enrique Hernandez (the same batter in the previous error in the 6th inning of this game) hit a line drive single to left center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

75.	game	OAK vs BOS on 20230707
	state before	(2, 1, 1, 0, 5, 0, 5, 0)
	error	play,5,0,kempt001,02,CFFX,S5/G56S.2-H(E5/TH) (NR) (UR);1-3;B-2
	state after	(2, 0, 1, 1, 5, 1, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 209 *In the top of the 5th inning, with runners on 1st and 2nd, OAK batter Tony Kemp hit a ground ball single to shallow 3rd*

base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.

	game	OAK vs BOS on 20230708
	state before	(2, 0, 0, 0, 2, 0, 5, 1)
76.	error	play,2,1,yoshm002,01,CX,E5/G56.B-1
	state after	(2, 1, 0, 0, 2, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 210 This game had 2 errors. The final score was 3-10. *In the bottom of the 2nd inning, with bases empty, BOS batter Masataka Yoshida hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs BOS on 20230708
	state before	(2, 1, 0, 0, 8, 2, 10, 0)
77.	error	play,8,0,diaza003,01,CX,S4/G4.1-3(E4/TH)
	state after	(2, 1, 0, 1, 8, 2, 10, 0)
	fielder	4 (2nd baseman on the home team)

Summary 211 *In the top of the 8th inning, with a runner on 1st, OAK batter Aledmys Diaz hit a ground ball single to 2nd base.. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for OAK vs BOS on 2023-07-09

position	innings	player	position	innings	player
1	1	JP Sears	1	1	Tayler Scott
1	6	Paul Blackburn	1	2	Chris Murphy
1	7	Ken Waldichuk	1	5	Josh Winckowski
1	8	Shintaro Fujinami	1	7	Joely Rodriguez
2	1	Manny Pina	1	8	Chris Martin
2	7	Shea Langeliers	1	9	Kenley Jansen
3	1	Ryan Noda	2	1	Jorge Alfaro
4	1	Tony Kemp	2	9	Connor Wong
4	7	Jordan Diaz	3	1	Justin Turner
5	1	Jace Peterson	4	1	Christian Arroyo
6	1	Tyler Wade	5	1	Rafael Devers
7	1	Seth Brown	6	1	Yu Chang
7	7	Tony Kemp	7	1	Rob Refsnyder
8	1	JJ Bleday	7	9	Jarren Duran
9	1	Cody Thomas	8	1	Enrique Hernandez
9	7	Seth Brown	8	7	Jarren Duran
			8	9	Adam Duvall
			9	1	Adam Duvall
			9	9	Alex Verdugo

	game	OAK vs BOS on 20230709
	state before	(1, 0, 1, 0, 1, 0, 0, 0)
78.	error	play,1,0,brows003,02,SS.B,WP.2-H(E2/TH) (NR) (UR)
	state after	(1, 0, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 212 This game had 3 errors. The final score was 3-4.

In the top of the 1st inning, with a runner on 2nd, OAK batter Seth Brown had a 0-2 count. During the next pitch, which was wild, the runner from 2nd tried to advance. A throwing error by the catcher

allowed this runner from 2nd to score (unearned, no RBI). Brown is still at bat, now with a 1-2 count.

79.	game	OAK vs BOS on 20230709
	state before	(2, 1, 0, 0, 6, 2, 2, 1)
	error	play,6,1,arroc001,01,F>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 2, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 213 *In the bottom of the 6th inning, with a runner on 1st, BOS batter Christian Arroyo had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Arroyo is still at bat, now with a 1-1 count.*

80.	game	OAK vs BOS on 20230709
	state before	(1, 0, 0, 0, 8, 2, 4, 1)
	error	play,8,1,duraj001,32,BCBS.BX,S6/G6.B-2(E6)
	state after	(1, 0, 1, 0, 8, 2, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 214 *In the bottom of the 8th inning, with bases empty, BOS batter Jarren Duran hit a ground ball single to short. An error by the shortstop allowed the batter to reach 2nd.*

81.	game	NYN vs BOS on 20230721
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
	error	play,1,1,turnj001,01,C>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 215 *In the bottom of the 1st inning, with a runner on 1st, BOS batter Justin Turner had a 0-1 count. During the next pitch, which*

was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Turner is still at bat, now with a 1-1 count.

	game	NYN vs BOS on 20230722
	state before	(1, 0, 0, 0, 3, 0, 2, 1)
82.	error	play,3,1,yoshm002,11,CBX,S1/G1.B-2(E1/TH)
	state after	(1, 0, 1, 0, 3, 0, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 216 This game had 2 errors. The final score was 6-8.

In the bottom of the 3rd inning, with bases empty, BOS batter Masataka Yoshida hit a ground ball single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.

	game	NYN vs BOS on 20230722
	state before	(2, 0, 1, 1, 4, 0, 2, 0)
83.	error	play,4,0,mcnej002,01,FX,S8/L8.3-H;2-H;B-H(E2/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 4, 3, 2, 0)
	fielder	2 (catcher on the home team)

Summary 217 *In the top of the 4th inning, with runners on 2nd and 3rd, NYN batter Jeff McNeil hit a line drive single to center field. The runners on 2nd and 3rd each scored (2 RBIs). A throwing error by the catcher allowed the batter to also score (unearned, no RBI).*

	game	NYN vs BOS on 20230723
	state before	(2, 1, 1, 0, 8, 1, 6, 0)
84.	error	play,8,0,voged001,22,..BCC.FFFBF2,P02(E1/TH).2-3;1-2
	state after	(2, 0, 1, 1, 8, 1, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 218 *In the top of the 8th inning, with runners on 1st and 2nd, NYN batter Daniel Vogelbach has a 2-2 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base. Vogelbach is still at bat with a 2-2 count.*

85.	game	ATL vs BOS on 20230725
	state before	(2, 1, 0, 0, 4, 1, 2, 1)
	error	play,4,1,duraj001,10,B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 219 This game had 2 errors, both from a stolen base. The final score was 1-7.

In the bottom of the 4th inning, with a runner on 1st, BOS batter Jarren Duran had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Duran is still at bat, now with a 2-0 count.

86.	game	ATL vs BOS on 20230725
	state before	(2, 1, 0, 0, 7, 1, 5, 0)
	error	play,7,0,acunr001,20,BB>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 1, 5, 0)
	fielder	2 (catcher on the home team)

Summary 220 *In the top of the 7th inning, with a runner on 1st, ATL batter Ronald Acuña had a 2-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Acuña is still at bat, now with a 3-0 count.*

87.	game	ATL vs BOS on 20230726
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
	error	play,2,0,ozunm001,21,CBBX,E5/L5+.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 221 This game had 2 errors. The final score was 3-5.

In the top of the 2nd inning, with bases empty, ATL batter Marcell Ozuna hit a hard line drive to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	ATL vs BOS on 20230726
	state before	(1, 1, 0, 0, 7, 3, 2, 1)
88.	error	play,7,1,chany001,12,.FFB>X,FC/G4.1-2(E4);B-1
	state after	(1, 1, 1, 0, 7, 3, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 222 *In the bottom of the 7th inning, with a runner on 1st, BOS batter Yu Chang hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

7.2.6 August

	game	TOR vs BOS on 20230804
	state before	(2, 1, 1, 0, 2, 3, 0, 0)
89.	error	play,2,0,sprig001,22,CFBF*BX,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 2, 3, 0, 0)
	fielder	2 (catcher on the home team)

Summary 223 *In the top of the 2nd inning, with runners on 1st and 2nd, TOR batter George Springer had a 2-2 count. Catcher's interference allowed the batter to take 1st. Consequently, the runners on 1st and 2nd each advanced one base.*

	game	TOR vs BOS on 20230805
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
90.	error	play,1,0,belbt001,01,FX,E1/BP25.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 224 This game had 2 errors. The final score was 5-4.
In the top of the 1st inning, with bases empty, TOR batter Brandon Belt hit a bunt pop up to very shallow infield. An error by the pitcher allowed the batter to reach 1st base.

91.	game	TOR vs BOS on 20230805
	state before	(2, 0, 0, 0, 3, 1, 0, 0)
	error	play,3,0,sprig001,02,SFX,S5/G5S.B-2(E5/TH)
	state after	(2, 0, 1, 0, 3, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 225 *In the top of the 3rd inning, with bases empty, TOR batter George Springer hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

92.	game	KCA vs BOS on 20230807
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
	error	play,3,1,wongc001,11,CB>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 3, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 226 This game had 2 errors. The final score was 2-6.
In the bottom of the 3rd inning, with a runner on 1st, BOS batter Connor Wong had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Wong is still at bat, now with a 2-1 count.

93.	game	KCA vs BOS on 20230807
	state before	(0, 0, 0, 0, 4, 0, 1, 0)
	error	play,4,0,massm001,01,CX,E4/G4.B-1
	state after	(0, 1, 0, 0, 4, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 227 *In the top of the 4th inning, with bases empty, KCA batter Michael Massey hit a ground ball to 2nd. An error by the second baseman allowed the batter to reach 1st base.*

	game	KCA vs BOS on 20230808
	state before	(2, 1, 1, 0, 4, 3, 0, 0)
94.	error	play,4,0,peres002,00,2,P02(E1/TH).2-3
	state after	(2, 1, 0, 1, 4, 3, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 228 *In the top of the 4th inning, with runners on 1st and 2nd, KCA batter Salvador Perez has a 0-0 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to safely reach 3rd base.*

	game	KCA vs BOS on 20230809
	state before	(2, 0, 0, 0, 2, 1, 0, 0)
95.	error	play,2,0,duffm002,12,BCSX,E3/G34D.B-1
	state after	(2, 1, 0, 0, 2, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 229 This game had 2 errors. The final score was 3-4. *In the top of the 2nd inning, with bases empty, KCA batter Matt Duffy hit a ground ball to deep 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	KCA vs BOS on 20230809
	state before	(2, 0, 1, 0, 4, 2, 3, 1)
96.	error	play,4,1,urial001,32,B*BCBCX,C/E2.B-1
	state after	(2, 1, 1, 0, 4, 2, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 230 *In the bottom of the 4th inning, with a runner on 2nd, BOS batter Luis Urias had a full count. Catcher's interference allowed the batter to take 1st. (No advance for the runner on 2nd base.)*

	game	DET vs BOS on 20230811
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
97.	error	play,1,1,refsr001,01,FX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 231 *In the bottom of the 1st inning, with bases empty, BOS batter Rob Refsnyder hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	DET vs BOS on 20230812
	state before	(2, 0, 1, 0, 4, 2, 0, 1)
98.	error	play,4,1,reyep001,11,BCX,E5/G56.2-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 2, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 232 *In the bottom of the 4th inning, with a runner on 2nd, BOS batter Pablo Reyes hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	DET vs BOS on 20230813
	state before	(0, 0, 0, 0, 6, 2, 6, 0)
99.	error	play,6,0,vierm001,02,.CC.X,E4/G4.B-2(E4/TH)
	state after	(0, 0, 1, 0, 6, 2, 6, 0)
	fielder	4, 4 (2nd baseman on the home team)

Summary 233 This play had 2 errors by the same fielder. *In the top of the 6th inning, with bases empty, DET batter Matt Vierling hit a ground ball to 2nd. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	LAN vs BOS on 20230825
	state before	(2, 0, 0, 0, 3, 0, 3, 1)
100.	error	play,3,1,duvaa001,32,SBBBBF.FX,S6/G6M.B-2(E6/TH)
	state after	(2, 0, 1, 0, 3, 0, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 234 This game had 2 errors. The final score was 7-4.
In the bottom of the 3rd inning, with bases empty, BOS batter Adam Duvall hit a ground single to short. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	LAN vs BOS on 20230825
	state before	(0, 1, 1, 0, 7, 6, 3, 1)
101.	error	play,7,1,turnj001,30,.BBBX,5(2)/F0/G56S.1-H(E5/TH) (NR) (UR);B-2
	state after	(1, 0, 1, 0, 7, 6, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 235 *In the bottom of the 7th inning, with runners on 1st and 2nd, BOS batter Justin Turner hit a fly ball to shallow 3rd. The third baseman tagged out the runner from 2nd. A throwing error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	LAN vs BOS on 20230826
	state before	(0, 1, 0, 0, 7, 4, 6, 1)
102.	error	play,7,1,verda001,01,FN,SB2.1-3(E1/TH)
	state after	(0, 0, 0, 1, 7, 4, 6, 1)
	fielder	1 (pitcher on the visiting team)

Summary 236 This game had 2 errors. The final score was 5-8.
In the bottom of the 7th inning, with a runner on 1st, BOS batter Alex Verdugo had a 0-1 count. During the next pitch, which was a “no pitch”, the runner on 1st stole 2nd. A throwing error by the pitcher allowed that runner to advance to 3rd. Verdugo is still at bat, with a 0-1 count.

103.	game	LAN vs BOS on 20230826
	state before	(1, 1, 1, 1, 8, 5, 7, 1)
	error	play,8,1,urial001,10,BX,FC4/G4.3-H;2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 8, 5, 8, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 237 *In the bottom of the 8th inning, with bases loaded, BOS batter Luis Urias hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

104.	game	HOU vs BOS on 20230828
	state before	(2, 0, 0, 0, 4, 2, 2, 0)
	error	play,4,0,penaj004,10,BX,E1/TH/G15-.B-1
	state after	(2, 1, 0, 0, 4, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 238 *In the top of the 4th inning, with bases empty, HOU batter Jeremy Pena hit a soft ground ball to shallow left infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	J.P. France	1	1	Brayan Bello
1	6	Rafael Montero	1	5	Joe Jacques
1	8	Hector Neris	1	6	John Schreiber
1	9	Bryan Abreu	1	7	Josh Winckowski
2	1	Yainer Diaz	1	9	Mauricio Llovera
3	1	Jose Abreu	2	1	Reese McGuire
4	1	Jose Altuve	3	1	Triston Casas
5	1	Alex Bregman	4	1	David Hamilton
6	1	Jeremy Pena	5	1	Rafael Devers
7	1	Michael Brantley	6	1	Trevor Story
8	1	Mauricio Dubon	7	1	Masataka Yoshida
9	1	Kyle Tucker	8	1	Adam Duvall
			9	1	Alex Verdugo

	game	HOU vs BOS on 20230829
	state before	(1, 1, 0, 1, 3, 2, 0, 1)
105.	error	play,3,1,turnj001,12,S*BFX,FC5/G6.3-H;1-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 3, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 239 This game had 3 errors. The final score was 6-2.

In the bottom of the 3rd inning, with runners on 1st and 3rd, BOS batter Justin Turner hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	HOU vs BOS on 20230829
	state before	(1, 0, 0, 0, 5, 2, 0, 0)
106.	error	play,5,0,dubom001,32,BCFBBFFX,E5/G56.B-1
	state after	(1, 1, 0, 0, 5, 2, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 240 *In the top of the 5th inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	HOU vs BOS on 20230829
	state before	(0, 0, 0, 0, 8, 6, 1, 1)
107.	error	play,8,1,turnj001,32,.BCBBFFF,FLE2
	state after	(0, 0, 0, 0, 8, 6, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 241 This play is an *infinity play*, and could be repeated *ad infinitum*. There is no change at all to the game state.
In the bottom of the 8th inning, with bases empty and a full count, BOS batter Justin Turner hit a pop fly foul behind home plate. An error by the catcher allowed the batter to remain safe at-bat, with the same full count.

	game	HOU vs BOS on 20230830
	state before	(0, 0, 1, 1, 6, 7, 0, 1)
108.	error	play,6,1,wongc001,01,FX,E6/TH/G6.3-H;B-1
	state after	(0, 1, 1, 0, 6, 7, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 242 *In the bottom of the 6th inning, with runners on 2nd and 3rd, BOS batter Connor Wong hit a ground ball to short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 1st.*

7.2.7 September

	game	BAL vs BOS on 20230908
	state before	(1, 1, 1, 0, 6, 1, 0, 0)
109.	error	play,6,0,mullc002,12,FFBX,D9/F9D.2-H;1-H;B-3(E9)
	state after	(1, 0, 0, 1, 6, 3, 0, 0)
	fielder	9 (right field on the home team)

Summary 243 *In the top of the 6th inning, with runners on 1st and 2nd, BAL batter Cedric Mullins hit a fly ball double to deep right field. The runners on 1st and 2nd both scored (2 RBIs). A throwing error by the right fielder allowed the batter to reach 3rd.*

Visiting/Home team rosters for BAL vs BOS on 2023-09-09

position	innings	player	position	innings	player
1	1	Jack Flaherty	1	1	Chris Sale
1	4	Cionel Perez	1	5	Zack Weiss
1	5	Shintaro Fujinami	1	7	Mauricio Llovera
1	7	Danny Coulombe	1	8	John Schreiber
1	7	Jacob Webb	1	9	Joe Jacques
1	8	Jorge Lopez	2	1	Connor Wong
1	9	Yennier Cano	3	1	Triston Casas
2	1	James McCann	4	1	Enmanuel Valdez
3	1	Ryan Mountcastle	5	1	Rafael Devers
4	1	Jordan Westburg	6	1	Trevor Story
4	7	Adam Frazier	7	1	Masataka Yoshida
4	9	Ramon Urias	8	1	Wilyer Abreu
5	1	Gunnar Henderson	9	1	Alex Verdugo
6	1	Jorge Mateo			
7	1	Austin Hays			
8	1	Aaron Hicks			
8	9	Cedric Mullins			
9	1	Anthony Santander			
9	9	Aaron Hicks			

	game	BAL vs BOS on 20230909
	state before	(2, 0, 1, 0, 2, 0, 2, 0)
110.	error	play,2,0,westj002,11,S*BX,S6/G6M.2-H(E6/TH)(NR)(UR)
	state after	(2, 1, 0, 0, 2, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 244 This game had 3 errors. The final score was 13-12. *In the top of the 2nd inning, with a runner on 2nd, BAL batter Jordan Westburg hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI). The*

batter took 1st.

	game	BAL vs BOS on 20230909
	state before	(1, 0, 0, 0, 3, 5, 2, 1)
111.	error	play,3,1,turnj001,02,CSX,E6/G6D.B-1
	state after	(1, 1, 0, 0, 3, 5, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 245 *In the bottom of the 3rd inning, with bases empty, BOS batter Justin Turner hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	BAL vs BOS on 20230909
	state before	(1, 1, 1, 0, 4, 7, 4, 1)
112.	error	play,4,1,turnj001,00,X,E5/G5.2-H(NR) (UR) ;1-3;B-2
	state after	(1, 0, 1, 1, 4, 7, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 246 *In the bottom of the 4th inning, with runners on 1st and 2nd, BOS batter Justin Turner hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (un-earned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

Visiting/Home team rosters for BAL vs BOS on 20230910

position	innings	player	position	innings	player
1	1	Grayson Rodriguez	1	1	Brayan Bello
1	5	Cole Irvin	1	6	Josh Winckowski
1	8	Joey Krehbiel	1	7	Garrett Whitlock
2	1	Adley Rutschman	1	9	Nick Robertson
3	1	Ryan O'Hearn	2	1	Reese McGuire
4	1	Adam Frazier	3	1	Triston Casas
5	1	Ramon Urias	4	1	Luis Urias
6	1	Gunnar Henderson	5	1	Rafael Devers
7	1	Austin Hays	6	1	Trevor Story
8	1	Cedric Mullins	7	1	Wilyer Abreu
9	1	Aaron Hicks	8	1	Ceddanne Rafaela
			9	1	Alex Verdugo

113.	game	BAL vs BOS on 20230910
	state before	(2, 1, 0, 1, 3, 3, 0, 1)
	error	play,3,1,stort001,02,CF.FX,E5/G7L.3-H(NR)(UR);1-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 3, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 247 This game had 3 errors. The final score was 3-7.

In the bottom of the 3rd inning, with runners on 1st and 3rd, BOS batter Trevor Story hit a ground ball to left field. An error by the third baseman allowed the runners on 1st and 3rd to score (unearned, no RBIs) and the batter to reach 2nd.

114.	game	BAL vs BOS on 20230910
	state before	(1, 1, 0, 0, 6, 3, 4, 1)
	error	play,6,1,dever001,10,BX,E3/G3.1-3;B-2
	state after	(1, 0, 1, 1, 6, 3, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 248 *In the bottom of the 6th inning, with a runner on 1st, BOS batter Rafael Devers hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	BAL vs BOS on 20230910
	state before	(2, 1, 0, 0, 8, 3, 7, 1)
115.	error	play,8,1,turnj001,12,1BFF1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 3, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 249 *In the bottom of the 8th inning, with a runner on 1st, BOS batter Justin Turner had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 3rd. A throwing error by the catcher allowed that runner to advance to 3rd. Turner is still at bat, now with a 2-2 count.*

	game	NYA vs BOS on 20230912
	state before	(1, 1, 1, 1, 9, 3, 1, 0)
116.	error	play,9,0,wellla002,12,SFFF*BFFFX,C/E2.3-H(UR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 9, 4, 1, 0)
	fielder	2 (catcher on the home team)

Summary 250 *In the top of the 9th inning, with bases loaded, NYA batter Austin Wells had a 1-2 count. Catcher's interference allowed the batter to take 1st. Consequently:*

- (a) *The runners on 1st and 2nd each advanced one base.*
- (b) *The runner on 3rd base scored (unearned, 1 RBI).*

	game	NYA vs BOS on 20230914 (1/2)
	state before	(0, 0, 0, 0, 2, 0, 1, 1)
117.	error	play,2,1,stort001,00,X,E5/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 251 This game is the first in a doubleheader. The final score was 0-5.

In the bottom of the 2nd inning, with bases empty, BOS batter Trevor Story hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	NYA vs BOS on 20230914 (2/2)
	state before	(0, 1, 1, 0, 1, 0, 0, 1)
118.	error	play,1,1,verda001,01,FX,36(1)/FO/G3.2-H(E6/TH)(NR);B-1
	state after	(1, 1, 0, 0, 1, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 252 This game (the second in a doubleheader) had 2 errors. The final score was 8-5.

In the bottom of the 1st inning, with runners on 1st and 2nd, BOS batter Alex Verdugo hit a ground ball to 1st. At 2nd, the shortstop (with an assist from the first baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	NYA vs BOS on 20230914 (2/2)
	state before	(2, 1, 0, 1, 5, 5, 3, 1)
119.	error	play,5,1,casat001,01,CX,E6/TH/G4MS-.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 5, 5, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 253 *In the bottom of the 5th inning, with runners on 1st and 3rd, BOS batter Triston Casas hit a soft ground ball to shallow 2nd base. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	CHA vs BOS on 20230923
	state before	(1, 0, 1, 0, 5, 0, 0, 1)
120.	error	play,5,1,mcgur002,00,X,C/E2.B-1
	state after	(1, 1, 1, 0, 5, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 254 *In the bottom of the 5th inning, with a runner on 2nd, BOS batter Reese McGuire had a 0-0 count. Catcher's interference allowed the batter to take 1st. (No advance for eht runner on 2nd.)*

	game	CHA vs BOS on 20230924
	state before	(1, 1, 0, 0, 3, 0, 1, 1)
121.	error	play,3,1,rafac001,01,S>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 255 *In the bottom of the 3rd inning, with a runner on 1st, BOS batter Ceddanne Rafaela had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rafaela is still at bat, now with a 2-2 count.*

Visiting/Home team rosters for TBA vs BOS on 2023-09-26

position	innings	player	position	innings	player
1	1	Zach Eflin	1	1	Tanner Houck
1	6	Kevin Kelly	1	4	Zack Kelly
1	7	Colin Poche	1	5	Brennan Bernardino
1	8	Shawn Armstrong	1	6	Mauricio Llovera
1	9	Pete Fairbanks	1	7	John Schreiber
2	1	Rene Pinto	1	8	Chris Murphy
2	8	Christian Bethancourt	1	8	Josh Winckowski
3	1	Jonathan Aranda	2	1	Reese McGuire
4	1	Curtis Mead	2	3	Connor Wong
5	1	Junior Caminero	3	1	Bobby Dalbec
6	1	Taylor Walls	4	1	Enmanuel Valdez
7	1	Raimel Tapia	5	1	Rafael Devers
8	1	Manuel Margot	6	1	Trevor Story
9	1	Josh Lowe	7	1	Masataka Yoshida
			8	1	Wilyer Abreu
			9	1	Alex Verdugo
			9	8	Adam Duvall

	game	TBA vs BOS on 20230926
	state before	(0, 0, 1, 0, 1, 1, 0, 0)
122.	error	play,1,0,parei001,11,CBX,E5/L5D+.2-3;B-1
	state after	(0, 1, 0, 1, 1, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 256 This game had 3 errors. The final score was 9-7.

In the top of the 1st inning, with a runner on 2nd, TBA batter Isaac Paredes hit a hard line drive to deep 3rd base. An error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	TBA vs BOS on 20230926
	state before	(1, 0, 0, 0, 2, 3, 0, 1)
123.	error	play,2,1,stort001,10,BX,E9/F9LS.B-2
	state after	(1, 0, 1, 0, 2, 3, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 257 *In the bottom of the 2nd inning, with bases empty, BOS batter Trevor Story hit a fly ball to shallow right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	TBA vs BOS on 20230926
	state before	(1, 1, 0, 0, 8, 9, 6, 1)
124.	error	play,8,1,dalbb001,21,SBBX,FC3/G3S.1-2(E3/TH);B-1
	state after	(1, 1, 1, 0, 8, 9, 6, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 258 *In the bottom of the 8th inning, with a runner on 1st, BOS batter Bobby Dalbec hit a ground ball into a fielder's choice at shallow 1st. A throwing error by the first baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

Chapter 8

Chicago Cubs (CHN)

The Chicago Cubs compete in the National League Central Division. The Cubs, under a different name, were a founding member of the NL in 1876, but were renamed as the Chicago Cubs in 1903. They have won 3 World Series titles, the last in 2016, and numerous NLCD titles, the last in 2020.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Chicago against the Cubs at Wrigley Field. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 CHN team is .694 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the Cubs is slightly above the MLB average.

In CHN 2023, there were 86 errors in the 81 home games, 44 by the Cubs and the rest (42) by visiting teams. The total number of errors by the Cubs (home and away) over the regular season in 2023 was 92. Since 83.9 was the MLB average, the Cubs were well above average for errors (which is bad).

8.1 Season totals

The batter shall keep at least one foot in the *batter's box* throughout the batter's time at bat, unless one of the following exceptions applies, in which case the batter may leave the batter's box but not the dirt area surrounding home plate:

- (i) The batter swings at a pitch; (ii) An attempted check swing is appealed to a base umpire;
- (iii) The batter is forced off balance or out of the batter's box by a pitch;
- (iv) A member of either team requests and is granted "Time";
- (v) A defensive player attempts a play on a runner at any base;
- (vi) The batter feints a bunt;
- (vii) A wild pitch or passed ball occurs;
- (viii) The pitcher leaves the dirt area of the pitching mound after receiving the ball; or
- (ix) The catcher leaves the catcher's box to give defensive signals.

– **MLB Rule Book** 5.04(b)(4)(A)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

8.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run conversion rate.

Runs and errors	totals
Errors by vt:	41
Runs by ht because of errors by vt:	16
Errors by ht:	45
Runs by vt because of errors by ht:	4

This is a very good error-to-run conversion rate by the Cubs: $16/41 = 39\%$ on the part of the Cubs (compared to the relatively poor $4/44 = 9.1\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 CHN season (home and away) has about $R/PA = 819/6220$ (about 13.2%) change of resulting in a run. Therefore, a randomly selected error is (for CHN) nearly 3 times as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is

85, so the expected number of errors (by either team) in a randomly selected CHN 2023 home game is about 1.

Errors by position	vt	ht
errors due to pitcher:	1	6
to catcher:	9	6
to 1st baseman:	3	6
to 2nd baseman:	3	5
to 3rd baseman:	12	9
to shortstop:	7	7
to left fielder:	2	1
to center fielder:	3	3
to right fielder:	1	1

8.1.2 Error total by state

Any runner is **out** when he runs more than three feet away from his base path to avoid being tagged unless his action is to avoid interference with a fielder fielding a batted ball. A runner's base path is established when the tag attempt occurs and is a straight line from the runner to the base he is attempting to reach safely.

– **MLB Rule Book** 5.09(b)(1)

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	17	18	35
(lone) runner on 1st	11	15	26
(lone) runner on 2nd	3	6	9
(lone) runner on 3rd	0	0	0
two runners on base	6	6	12
bases loaded	4	0	4
total	41	45	86

A table of frequencies of most common errors by the CHN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	13.3%
(1, 0, 0, 0)	6	13.3%
(2, 0, 0, 0)	6	13.3%
(0, 1, 0, 0)	5	11.1%
(1, 1, 0, 0)	5	11.1%
(2, 1, 0, 0)	5	11.1%
(0, 0, 1, 0)	2	4.4%
(2, 0, 1, 0)	3	6.7%
(1, 0, 1, 1)	1	2.2%

A table of frequencies of most common errors by a visiting team in CHN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	14.6%
(1, 0, 0, 0)	5	12.2%
(2, 0, 0, 0)	6	14.6%
(0, 1, 0, 0)	3	7.3%
(1, 1, 0, 0)	2	4.9%
(2, 1, 0, 0)	6	14.6%
(2, 0, 1, 0)	2	4.9%
(1, 1, 1, 0)	1	2.4%
(1, 0, 1, 1)	2	4.9%
(2, 1, 1, 1)	2	4.9%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Cubs homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `CHN2023.evn`):

state	CHN2023 percentages
(0, 0, 0, 0)	23.9%
(1, 0, 0, 0)	16.3%
(2, 0, 0, 0)	12.6%
(0, 1, 0, 0)	5.5%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	6.7%
(0, 0, 1, 0)	1.9%
(1, 0, 1, 0)	3.5%
(2, 0, 1, 0)	3.7%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.9%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.7%
(1, 1, 1, 0)	2.7%
(2, 1, 1, 0)	3.3%

8.1.3 Some interesting errors

There were a number of games with 4 or 5 errors. Surprisingly, the game with 5 errors was with the Texas Rangers who, ironically, had one of the *lowest* error rates in the MLB that year.

Some of the most interesting error plays in the CHN 2023 home games:

- The game TEX vs CHN on 2023-04-08 had 5 errors.
- The game CIN vs CHN on 2023-08-01 had 4 errors.
- The **next day**, the game CIN vs CHN on 2023-08-02 had 4 errors and a double consecutive error.
- The game SFN vs CHN on 2023-09-05 had 4 errors.

8.1.4 Relative team rankings

We have 81 home games between CHN and some visiting MLB team (vt). A number of teams did not play CHN in the regular season. However, of those that did, CHN *lost* more games than won against¹:

BOS (2/3), CLE (2/3), PHI (3/3), ARI (3/4), LAN (3/4), CIN (4/7)

and *won* more than lost against

COL (0/3), SFN (0/3), ATL (1/3), BAL (1/3), KCA (1/3), MIA (1/3),
 NYN (1/3), SDN (1/3), SEA (1/3), TBA (1/3), TEX (1/3), WAS (1/3),
 PIT (2/6), SLN (3/7) .

The teams that *split* their series were:

MIL (3/6), CHA (1/2) .

The teams that did not play in any of the 81 home games in CHN 2023 were:

ANA, DET, HOU, MIN, NYA, OAK, TOR.

8.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster² for CHN 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Yan Gomes	103	.315
3 (1B)	Trey Mancini		.299
4 (2B)	Nico Hoerner	135	.346
5 (3B)	Nick Madrigal	72	.311
6 (SS)	Dansby Swanson	147	.328
7 (LF)	Ian Happ	154	.360
8 (CF)	Cody Bellinger	84	.356
9 (RF)	Seiya Suzuki	132	.357

¹The vt's W/T record in parentheses (W = number of home games vt won against the Cubs, T = Total games vt vs Cubs played at CHN).

²It appears they did not have a regular first baseman, but utility player Trey Mancini was frequently at that position.

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 CHN home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 86 errors in the 81 home games, 45 by the Cubs and the rest (41) by visiting teams.

8.2.1 April

	game	MIL vs CHN on 20230330
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
1.	error	play,3,0,turab002,00,X,S4/G34.B-2(E4/TH)
	state after	(0, 0, 1, 0, 3, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 259 This game has 2 errors. The final score was 0-4.
In the top of the 3rd inning, with bases empty, MIL batter Brice Turang hit a ground ball single to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 2nd.

	game	MIL vs CHN on 20230330
	state before	(1, 1, 0, 1, 3, 0, 0, 1)
2.	error	play,3,1,swand001,01,CX,S9/G3.3-H;1-H(E6/TH)(NR)
	state after	(1, 1, 0, 0, 3, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 260 *In the bottom of the 3rd inning, with runners on 1st and 3rd, CHN batter Dansby Swanson hit a ground ball single to 1st. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 1st to score (unearned, no RBI). The batter took 1st.*

8.2.2 April

	game	MIL vs CHN on 20230402
	state before	(2, 0, 0, 0, 8, 9, 3, 0)
3.	error	play,8,0,mitcg001,00,X,E2/TH/G2-.B-1
	state after	(2, 1, 0, 0, 8, 9, 3, 0)
	fielder	2 (catcher on the home team)

Summary 261 *In the top of the 8th inning, with bases empty, MIL batter Garrett Mitchell hit a soft ground ball to very shallow infield. A throwing error by the catcher allowed the batter to reach 1st base.*

Visiting/Home team rosters for TEX vs CHN on 2023-04-08

position	innings	player	position	innings	player
1	1	Martin Perez	1	1	Justin Steele
1	6	Brock Burke	1	7	Keegan Thompson
1	6	Ian Kennedy	1	8	Brad Boxberger
1	7	Taylor Hearn	1	9	Michael Rucker
1	8	Dane Dunning	2	1	Yan Gomes
2	1	Jonah Heim	3	1	Trey Mancini
3	1	Nathaniel Lowe	4	1	Nico Hoerner
4	1	Marcus Semien	5	1	Nick Madrigal
5	1	Josh Jung	6	1	Dansby Swanson
6	1	Corey Seager	7	1	Ian Happ
7	1	Ezequiel Duran	8	1	Cody Bellinger
7	7	Josh H. Smith	9	1	Patrick Wisdom
8	1	Adolis Garcia			
9	1	Robbie Grossman			

	game	TEX vs CHN on 20230408
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
4.	error	play,1,1,swand001,10,BX,S6/G6.B-2(E6/TH)
	state after	(1, 0, 1, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 262 This game has 5 errors. The final score was 3-10.
In the bottom of the 1st inning, with bases empty, CHN batter Dansby Swanson hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	TEX vs CHN on 20230408
	state before	(2, 0, 0, 0, 2, 0, 1, 1)
5.	error	play,2,1,bellc002,22,CBBCF.FX,E4/G34.B-1
	state after	(2, 1, 0, 0, 2, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 263 *In the bottom of the 2nd inning, with bases empty, CHN batter Cody Bellinger hit a ground ball to 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	TEX vs CHN on 20230408
	state before	(2, 1, 0, 0, 2, 0, 1, 1)
6.	error	play,2,1,gomey001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 264 *In the bottom of the 2nd inning, with a runner on 1st, CHN batter Yan Gomes had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Gomes is still at bat, now with a 0-1 count.*

	game	TEX vs CHN on 20230408
	state before	(1, 1, 0, 0, 6, 1, 3, 1)
7.	error	play,6,1,wisdp001,22,BSBTX,S6/G6.1-3(E6/TH);B-2
	state after	(1, 0, 1, 1, 6, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 265 *In the bottom of the 6th inning, with a runner on 1st, CHN batter Patrick Wisdom hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	TEX vs CHN on 20230408
	state before	(2, 1, 1, 1, 8, 2, 7, 1)
8.	error	play,8,1,happi001,00,X,S8/G4.3-H;2-H;1-H(E8/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 8, 2, 10, 1)
	fielder	8 (center field on the visiting team)

Summary 266 *In the bottom of the 8th inning, with bases loaded, CHN batter Ian Happ hit a ground ball single to 2nd. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the center fielder allowed the runner on 1st to also score (unearned, no RBI). The batter took 1st.*

	game	TEX vs CHN on 20230409
	state before	(1, 0, 1, 1, 2, 0, 0, 0)
9.	error	play,2,0,thomb003,01,FX,FC5/G56S.3-H(E5/TH) (NR);2-3;B-2
	state after	(1, 0, 1, 1, 2, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 267 This game has 2 errors. The final score was 8-2. *In the top of the 2nd inning, with runners on 2nd and 3rd, TEX batter Bubba Thompson hit a ground ball into a fielder's choice at shallow 3rd. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 2nd.*

	game	TEX vs CHN on 20230409
	state before	(1, 0, 0, 0, 4, 2, 2, 0)
10.	error	play,4,0,jankt001,01,CX,E3/G3.B-1
	state after	(1, 1, 0, 0, 4, 2, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 268 *In the top of the 4th inning, with bases empty, TEX batter Travis Jankowski hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	SEA vs CHN on 20230410
	state before	(0, 1, 0, 0, 4, 1, 0, 1)
11.	error	play,4,1,bellc002,01,S1>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 4, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 269 *In the bottom of the 4th inning, with a runner on 1st, CHN batter Cody Bellinger had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Bellinger is still at bat, now with a 1-1 count.*

	game	SEA vs CHN on 20230411
	state before	(0, 1, 1, 0, 2, 2, 0, 0)
12.	error	play,2,0,rodrj007,20,BBX,FC6/G6M.2-3;1-2(E6/TH);B-1
	state after	(0, 1, 1, 1, 2, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 270 *In the top of the 2nd inning, with runners on 1st and 2nd, SEA batter Julio Rodriguez hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

13.	game	LAN vs CHN on 20230420
	state before	(2, 1, 0, 0, 8, 2, 2, 0)
	error	play,8,0,muncm001,21,BC*B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 2, 2, 0)
	fielder	2 (catcher on the home team)

Summary 271 *In the top of the 8th inning, with a runner on 1st, LAN batter Max Muncy had a 2-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Muncy is still at bat, now with a 3-1 count.*

14.	game	LAN vs CHN on 20230422
	state before	(1, 0, 0, 0, 5, 3, 2, 0)
	error	play,5,0,bettm001,11,BCX,E5/G7S.B-1
	state after	(1, 1, 0, 0, 5, 3, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 272 *In the top of the 5th inning, with bases empty, LAN batter Mookie Betts hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.*

15.	game	LAN vs CHN on 20230423
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
	error	play,1,1,hoern001,12,CBSFFX,E5/G7LS.B-2
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 273 This game has 2 errors. The final score was 7-3. *In the bottom of the 1st inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 2nd base.*

	game	LAN vs CHN on 20230423
	state before	(0, 1, 0, 0, 9, 7, 3, 1)
16.	error	play,9,1,manct001,01,CX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 9, 7, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 274 *In the bottom of the 9th inning, with a runner on 1st, CHN batter Trey Mancini had a 0-1 count. Catcher's interference allowed the batter to take 1st. The runner on 1st advanced one base, as required.*

	game	SDN vs CHN on 20230427
	state before	(2, 1, 1, 0, 4, 1, 3, 0)
17.	error	play,4,0,grist001,32,FF.BBFB>F>X,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 4, 1, 3, 0)
	fielder	2 (catcher on the home team)

Summary 275 *In the top of the 4th inning, with runners on 1st and 2nd, SDN batter Trent Grisham had a full count. Catcher's interference allowed the batter to take 1st. Runners on 1st and 2nd base each advanced one base, as required.*

8.2.3 May

	game	MIA vs CHN on 20230506
	state before	(0, 0, 0, 0, 8, 2, 1, 1)
18.	error	play,8,1,bellc002,12,..SBF.X,1E3/G1S+.B-1
	state after	(0, 1, 0, 0, 8, 2, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 276 This game has 2 errors. The final score was 2-4. *In the bottom of the 8th inning, with bases empty, CHN batter Cody Bellinger hit a hard ground ball to shallow 1st. An error by the the first baseman (with an assist from the pitcher) allowed the batter to reach 1st base.*

	game	MIA vs CHN on 20230506
	state before	(2, 1, 0, 0, 8, 2, 4, 1)
19.	error	play,8,1,swand001,20,1BB>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 2, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 277 *In the bottom of the 8th inning, with a runner on 1st, CHN batter Dansby Swanson had a 2-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Swanson is still at bat, now with a 3-0 count.*

	game	NYN vs CHN on 20230523
	state before	(0, 0, 0, 0, 4, 1, 4, 1)
20.	error	play,4,1,taucm001,22,.FFBBFX,E7/L7L.B-2
	state after	(0, 0, 1, 0, 4, 1, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 278 This game has 2 errors. The final score was 2-7. *In the bottom of the 4th inning, with bases empty, CHN batter Mike Tauchman hit a line drive to left field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	NYN vs CHN on 20230523
	state before	(2, 0, 0, 0, 8, 2, 7, 1)
21.	error	play,8,1,swand001,32,FF.BFBFBX,E5/TH/G56.B-2
	state after	(2, 0, 1, 0, 8, 2, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 279 *In the bottom of the 8th inning, with bases empty, CHN batter Dansby Swanson hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	NYN vs CHN on 20230525
	state before	(2, 1, 1, 0, 3, 1, 1, 0)
22.	error	play,3,0,batyb001,12,CBFX,S9/G34.2-H;1-3;B-2(E9/TH)
	state after	(2, 0, 1, 1, 3, 2, 1, 0)
	fielder	9 (right field on the home team)

Summary 280 This game has 2 errors. The final score was 10-1.
In the top of the 3rd inning, with runners on 1st and 2nd, NYN batter Brett Baty hit a ground ball single to the 1st/2nd base gap. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.

	game	NYN vs CHN on 20230525
	state before	(1, 1, 1, 0, 5, 4, 1, 0)
23.	error	play,5,0,batyb001,20,.B*B>C,SB3;SB2.2-H(E2/TH) (NR) (UR)
	state after	(1, 0, 1, 0, 5, 5, 1, 0)
	fielder	2 (catcher on the home team)

Summary 281 *In the top of the 5th inning, with runners on 1st and 2nd, NYN batter Brett Baty (same as in the previous error in this game) had a 2-0 count. During the next pitch, which was a called strike, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Baty is still at bat, now with a 2-1 count.*

	game	CIN vs CHN on 20230526
	state before	(2, 0, 1, 0, 3, 2, 0, 0)
24.	error	play,3,0,senzn001,21,BF.BX,S8/G4M.2-H;B-2(E8)
	state after	(2, 0, 1, 0, 3, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 282 This game has 2 errors. The final score was 9-0.
In the top of the 3rd inning, with a runner on 2nd, CIN batter Nick

Senzel hit a ground ball single to 2nd. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the batter to reach 2nd.

	game	CIN vs CHN on 20230526
	state before	(0, 0, 1, 0, 4, 4, 0, 0)
25.	error	play,4,0,casac001,10,BX,3E1/SH/BG23-.2-3
	state after	(0, 1, 0, 1, 4, 4, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 283 *In the top of the 4th inning, with a runner on 2nd, CIN batter Curt Casali hit a soft sacrifice bunt to very shallow right infield. An error by the pitcher (with an assist from the first baseman³) allowed the runner on 2nd to advance to 3rd and the batter to reach 1st.*

	game	CIN vs CHN on 20230528
	state before	(1, 0, 1, 0, 5, 3, 3, 0)
26.	error	play,5,0,stees001,12,1.SFB2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 5, 3, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 284 *In the top of the 5th inning, with a runner on 2nd, CIN batter Spencer Steer has a 1-2 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to advance to 3rd.*

	game	TBA vs CHN on 20230529
	state before	(0, 0, 0, 0, 4, 0, 0, 1)
27.	error	play,4,1,suzus001,22,BBFF.FFX,S5/G5.B-2(E5/TH)
	state after	(0, 0, 1, 0, 4, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

³As you might imagine, video replay shows the pitcher missed catching the ball tossed to him from the first baseman who'd fielded the bunt.

Summary 285 *In the bottom of the 4th inning, with bases empty, CHN batter Seiya Suzuki hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	TBA vs CHN on 20230531
	state before	(1, 0, 0, 0, 2, 0, 2, 1)
28.	error	play,2,1,wisdp001,21,BBCX,E5/G56.B-1
	state after	(1, 1, 0, 0, 2, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 286 This game has 2 errors. The final score was 4-3.
In the bottom of the 2nd inning, with bases empty, CHN batter Patrick Wisdom hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	TBA vs CHN on 20230531
	state before	(1, 1, 1, 0, 9, 4, 3, 1)
29.	error	play,9,1,wisdp001,12,CCFBX,FC5/G56.2-3;1-2(E5/TH);B-1
	state after	(1, 1, 1, 1, 9, 4, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 287 *In the bottom of the 9th inning, with runners on 1st and 2nd, CHN batter Patrick Wisdom hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. A throwing error by the third baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

8.2.4 June

	game	PIT vs CHN on 20230614
	state before	(1, 1, 0, 0, 7, 5, 7, 0)
30.	error	play,7,0,joe-c001,22,CFFBBFX,S4/G34.1-3(E4/TH);B-2
	state after	(1, 0, 1, 1, 7, 5, 7, 0)
	fielder	4 (2nd baseman on the home team)

Summary 288 *In the top of the 7th inning, with a runner on 1st, PIT batter Connor Joe hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	PIT vs CHN on 20230615
	state before	(2, 1, 1, 1, 5, 2, 5, 1)
31.	error	play,5,1,hoern001,32,.BC*BFB>X,E6/G6.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 5, 2, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 289 *In the bottom of the 5th inning, with bases loaded, CHN batter Nico Hoerner hit a ground ball to shortstop. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	BAL vs CHN on 20230617
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
32.	error	play,2,0,mccaj001,12,BFCFX,S4/G6M.1-3(E4/TH)
	state after	(2, 1, 0, 1, 2, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 290 *In the top of the 2nd inning, with a runner on 1st, BAL batter James McCann hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	BAL vs CHN on 20230618
	state before	(2, 0, 0, 0, 4, 2, 1, 1)
33.	error	play,4,1,bellc002,12,F.FFBX,E9/L89D.B-3
	state after	(2, 0, 0, 1, 4, 2, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 291 *In the bottom of the 4th inning, with bases empty, CHN batter Cody Bellinger hit a line drive to deep right center field. An error by the right fielder allowed the batter to reach 3rd base.*

	game	PHI vs CHN on 20230627
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
34.	error	play,1,1,suzus001,31,BBBFX,E6/G6.B-1
	state after	(1, 1, 0, 0, 1, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 292 *This game has 2 errors. The final score was 5-1. In the bottom of the 1st inning, with bases empty, CHN batter Seiya Suzuki hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PHI vs CHN on 20230627
	state before	(2, 1, 0, 0, 3, 2, 0, 1)
35.	error	play,3,1,hoern001,10,*B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 3, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 293 *In the bottom of the 3rd inning, with a runner on 1st, CHN batter Nico Hoerner had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Hoerner is still at bat, now with a 2-0 count.*

	game	PHI vs CHN on 20230628
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
36.	error	play,1,0,castn001,12,BSCX,E1/TH/G1.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 294 *In the top of the 1st inning, with bases empty, PHI batter Nick Castellanos hit a ground ball to shallow center infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	PHI vs CHN on 20230629
	state before	(0, 0, 0, 0, 5, 3, 1, 1)
37.	error	play,5,1,taucm001,31,SBBBX,E7/F78XD.B-2
	state after	(0, 0, 1, 0, 5, 3, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 295 This game has 2 errors. The final score was 3-1.
In the bottom of the 5th inning, with bases empty, CHN batter Mike Tauchman hit a fly ball to extra deep left center field. An error by the left fielder allowed the batter to reach 2nd base.

	game	PHI vs CHN on 20230629
	state before	(2, 0, 0, 0, 6, 3, 1, 1)
38.	error	play,6,1,younj003,21,FBBX,E8/F89.B-2
	state after	(2, 0, 1, 0, 6, 3, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 296 *In the bottom of the 6th inning, with bases empty, CHN batter Jared Young hit a fly ball to right center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	CLE vs CHN on 20230630
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
39.	error	play,1,0,rosaa003,22,FBFBFX,FC1/G1S.1-2(E1/TH);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 297 *In the top of the 1st inning, with a runner on 1st, CLE batter Amed Rosario hit a ground ball into a fielder's choice at shallow center infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

8.2.5 July

	game	CLE vs CHN on 20230701
	state before	(2, 1, 0, 1, 6, 4, 0, 0)
40.	error	play,6,0,stram002,21,BCBX,S8/G4M.3-H;1-H(E8) (NR) (UR) ;B-3
	state after	(2, 0, 0, 1, 6, 6, 0, 0)
	fielder	8 (center field on the home team)

Summary 298 *In the top of the 6th inning, with runners on 1st and 3rd, CLE batter Myles Straw hit a ground ball single to 2nd. The runner on 3rd scored (1 RBI). An error by the center fielder allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 3rd.*

Visiting/Home team rosters for CLE vs CHN on 2023-07-02

position	innings	player	position	innings	player
1	1	Aaron Civale	1	1	Jameson Taillon
1	10	Sam Hentges	1	10	Adbert Alzolay
1	7	Enyel De Los Santos	1	6	Michael Rucker
1	8	Trevor Stephan	1	7	Anthony Kay
1	9	Emmanuel Clase	1	8	Hayden Wesneski
2	1	Cam Gallagher	2	1	Yan Gomes
3	1	Josh Naylor	3	1	Jared Young
3	10	Gabriel Arias	4	1	Nico Hoerner
4	1	Andres Gimenez	5	1	Christopher Morel
5	1	Jose Ramirez	6	1	Dansby Swanson
6	1	Amed Rosario	7	1	Mike Tauchman
7	1	David Fry	8	1	Cody Bellinger
7	10	Steven Kwan	9	1	Seiya Suzuki
8	1	Myles Straw			
9	1	Will Brennan			

	game	CLE vs CHN on 20230702
	state before	(2, 0, 0, 0, 3, 2, 0, 1)
41.	error	play,3,1,morec001,21,BBSX,S5/G5.B-2(E5/TH)
	state after	(2, 0, 1, 0, 3, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 299 This game has 3 errors. The final score was 8-6.

In the bottom of the 3rd inning, with bases empty, CHN batter Christopher Morel hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	CLE vs CHN on 20230702
	state before	(0, 0, 0, 0, 4, 2, 0, 0)
42.	error	play,4,0,naylj001,22,FCB.BX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 4, 2, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 300 *In the top of the 4th inning, with bases empty, CLE batter Josh Naylor hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CLE vs CHN on 20230702
	state before	(1, 1, 0, 0, 5, 5, 0, 0)
43.	error	play,5,0,naylj001,10,*BX,FC3/G3S-.1-3(E3/TH);B-1
	state after	(1, 1, 0, 1, 5, 5, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 301 *In the top of the 5th inning, with a runner on 1st, CLE batter Josh Naylor hit a ground ball into a fielder's choice at shallow 1st. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	BOS vs CHN on 20230714
	state before	(1, 1, 0, 0, 9, 8, 3, 1)
44.	error	play,9,1,younj003,10,B>T,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 9, 8, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 302 *In the bottom of the 9th inning, with a runner on 1st, CHN batter Jared Young had a 1-0 count. During the next pitch, which was a foul tip, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Young is still at bat, now with a 1-1 count.*

	game	BOS vs CHN on 20230715
	state before	(0, 1, 1, 1, 3, 0, 1, 1)
45.	error	play,3,1,suzus001,10,BX,5(2)/FO/G5S.3-H(E5/TH) (NR);1-2;B-1
	state after	(1, 1, 1, 0, 3, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 303 *In the bottom of the 3rd inning, with bases loaded, CHN batter Seiya Suzuki hit a ground ball to shallow 3rd. The third baseman forced out the runner from 2nd. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	BOS vs CHN on 20230716
	state before	(0, 0, 1, 0, 4, 1, 0, 0)
46.	error	play,4,0,duvaa001,11,BTX,E6/TH/G6+.B-1
	state after	(0, 1, 1, 0, 4, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 304 This game has 2 errors. The final score was 11-5. *In the top of the 4th inning, with a runner on 2nd, BOS batter Adam Duvall hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs CHN on 20230716
	state before	(0, 0, 0, 0, 5, 1, 0, 0)
47.	error	play,5,0,wongc001,01,FX,E6/G78S+.B-2
	state after	(0, 0, 1, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 305 *In the top of the 5th inning, with bases empty, BOS batter Connor Wong hit a hard ground ball to shallow left right field. An error by the shortstop allowed the batter to reach 2nd base.*

	game	WAS vs CHN on 20230718
	state before	(1, 0, 1, 0, 6, 3, 2, 1)
48.	error	play,6,1,bellc002,32,*B>B.FC*BX,S4/G4.2-H(E4/TH) (NR)
	state after	(1, 1, 0, 0, 6, 3, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 306 *In the bottom of the 6th inning, with a runner on 2nd, CHN batter Cody Bellinger hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	SLN vs CHN on 20230720
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
49.	error	play,1,0,arenn001,32,BF1BF1B.>F>X,E5/G56.1-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 307 *This game has 2 errors. The final score was 7-2. In the top of the 1st inning, with a runner on 1st, SLN batter Nolan Arenado hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

50.	game	SLN vs CHN on 20230720
	state before	(1, 1, 0, 0, 2, 1, 0, 0)
	error	play,2,0,dejop001,00,X,FC5/G56.1-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 2, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 308 *In the top of the 2nd inning, with a runner on 1st, SLN batter Paul DeJong hit a ground ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

51.	game	SLN vs CHN on 20230721
	state before	(2, 0, 0, 0, 9, 3, 4, 0)
	error	play,9,0,arenn001,22,BCSFBX,E4/G34.B-1
	state after	(2, 1, 0, 0, 9, 3, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 309 *In the top of the 9th inning, with bases empty, SLN batter Nolan Arenado hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

52.	game	SLN vs CHN on 20230722
	state before	(0, 1, 0, 0, 3, 2, 1, 0)
	error	play,3,0,gormn001,12,CBSFFX,FC1/G15.1-2(E1);B-1
	state after	(0, 1, 1, 0, 3, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 310 *In the top of the 3rd inning, with a runner on 1st, SLN batter Nolan Gorman hit a ground ball into a fielder's choice at shallow center infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SLN vs CHN on 20230723
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
53.	error	play,1,1,happi001,21,CBBX,E1/G1S-.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 311 *In the bottom of the 1st inning, with bases empty, CHN batter Ian Happ hit a soft ground ball to shallow center infield. An error by the pitcher allowed the batter to reach 1st base.*

8.2.6 August

Visiting/Home team rosters for CIN vs CHN on 2023-08-01

position	innings	player	position	innings	player
1	1	Ben Lively	1	1	Justin Steele
1	5	Derek Law	1	7	Hayden Wesneski
1	6	Sam Moll	1	8	Hayden Wesneski
1	7	Fernando Cruz	1	9	Anthony Kay
1	7	Luke Maile	2	1	Tucker Barnhart
2	1	Tyler Stephenson	3	1	Jeimer Candelario
3	1	Spencer Steer	4	1	Nico Hoerner
4	1	Kevin Newman	4	8	Nick Madrigal
5	1	Elly De La Cruz	5	1	Nick Madrigal
6	1	Matt McLain	5	8	Patrick Wisdom
7	1	Nick Senzel	6	1	Dansby Swanson
8	1	TJ Friedl	6	8	Nico Hoerner
9	1	Stuart Fairchild	7	1	Ian Happ
			7	8	Christopher Morel
			8	1	Cody Bellinger
			9	1	Mike Tauchman

Comment: In the 8th inning the Cubs made a number of substitutions. Hayden Wesneski pitched in the 7th and 8th innings but in the 8th inning he was placed 3rd in the batting line-up, while Morel was taken out of the DH position and replaced Happ as a fielder at LF. (While Wesneski substituted for Happ in the batters lineup, he never actually took an at-bat.)

	game	CIN vs CHN on 20230801
	state before	(0, 1, 0, 1, 1, 0, 0, 1)
54.	error	play,1,1,happi001,00,>B,SB2.3-H(E2/TH) (NR)
	state after	(0, 0, 1, 0, 1, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 312 This game has 4 errors. The final score was 9-20.

In the bottom of the 1st inning, with runners on 1st and 3rd, CHN batter Ian Happ had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Happ is still at bat, now with a 1-0 count.

	game	CIN vs CHN on 20230801
	state before	(2, 0, 1, 0, 2, 1, 5, 0)
55.	error	play,2,0,friet001,22,BFCB.X,E3/G34.2-H(NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 2, 2, 5, 0)
	fielder	3 (1st baseman on the home team)

Summary 313 *In the top of the 2nd inning, with a runner on 2nd, CIN batter TJ Friedl hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CIN vs CHN on 20230801
	state before	(0, 1, 0, 0, 2, 2, 5, 1)
56.	error	play,2,1,hoern001,00,>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 2, 2, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 314 *In the bottom of the 2nd inning, with a runner on 1st, CHN batter Nico Hoerner had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Hoerner is still at bat, now with a 1-0 count.*

	game	CIN vs CHN on 20230801
	state before	(2, 0, 0, 0, 5, 2, 13, 0)
57.	error	play,5,0,senzn001,00,X,E5/G56.B-1
	state after	(2, 1, 0, 0, 5, 2, 13, 0)
	fielder	5 (3rd baseman on the home team)

Summary 315 *In the top of the 5th inning, with bases empty, CIN batter Nick Senzel hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for CIN vs CHN on 2023-08-02

position	innings	player	position	innings	player
1	1	Brandon Williamson	1	1	Drew Smyly
1	5	Buck Farmer	1	5	Michael Fulmer
1	6	Alex Young	1	7	Mark Leiter Jr.
1	6	Lucas Sims	1	8	Julian Merryweather
1	7	Derek Law	1	9	Jose Cuas
1	8	Luke Maile	2	1	Yan Gomes
2	1	Tyler Stephenson	3	1	Patrick Wisdom
3	1	Joey Votto	3	7	Cody Bellinger
4	1	Kevin Newman	4	1	Nico Hoerner
4	7	Spencer Steer	5	1	Jeimer Candelario
5	1	Nick Senzel	6	1	Dansby Swanson
6	1	Matt McLain	7	1	Ian Happ
7	1	Spencer Steer	8	1	Cody Bellinger
7	7	Will Benson	8	7	Mike Tauchman
8	1	Stuart Fairchild	9	1	Seiya Suzuki
8	7	TJ Friedl			
9	1	Will Benson			
9	7	Jake Fraley			

	game	CIN vs CHN on 20230802
	state before	(2, 0, 1, 1, 6, 5, 8, 1)
58.	error	play,6,1,bellc002,31,.BBBCX,E5/TH/G56.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 5, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 316 This game had 4 errors, including three errors by the

same fielder and a (rare) double consecutive error. The final score was 6-16.

In the bottom of the 6th inning, with runners on 2nd and 3rd, CHN batter Cody Bellinger hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runners on 2nd and 3rd to both score (unearned, no RBIs) and the batter to reach 1st.

	game	CIN vs CHN on 20230802
	state before	(2, 0, 1, 0, 7, 6, 12, 1)
59.	error	play,7,1,hoern001,00,X,E5/TH/G56.2-H(NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 7, 6, 13, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 317 *In the bottom of the 7th inning, with a runner on 2nd, CHN batter Nico Hoerner hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CIN vs CHN on 20230802
	state before	(2, 0, 1, 0, 7, 6, 13, 1)
60.	error	play,7,1,morec001,00,X,E5/G56.2-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 7, 6, 14, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 318 This play is an example of a *consecutive error*, since Morel is right after Hoerner in the batter's lineup.

In the bottom of the 7th inning, with a runner on 2nd, CHN batter Christopher Morel hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	CIN vs CHN on 20230802
	state before	(2, 1, 0, 0, 7, 6, 14, 1)
61.	error	play,7,1,bellc002,22,*BSBFX,S3/G34.1-3(E3/TH)
	state after	(2, 1, 0, 1, 7, 6, 14, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 319 This play is **another** example of a *consecutive error*, since Bellinger is right after Morel in the batter's lineup.

In the bottom of the 7th inning, with a runner on 1st, CHN batter Cody Bellinger hit a ground ball single to the 1st/2nd base gap. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	ATL vs CHN on 20230805
	state before	(1, 0, 1, 1, 1, 0, 0, 1)
62.	error	play,1,1,bellc002,12,S*B.CX,E3/G34D.3-H;2-H(NR);B-1
	state after	(1, 1, 0, 0, 1, 0, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 320 *In the bottom of the 1st inning, with runners on 2nd and 3rd, CHN batter Cody Bellinger hit a ground ball to 1st base. An error by the first baseman allowed the runners on 2nd and 3rd to score (no RBI, 1 RBI, resp.) and the batter to reach 1st.*

	game	ATL vs CHN on 20230806
	state before	(2, 1, 0, 0, 5, 2, 2, 0)
63.	error	play,5,0,rilea001,10,BX,E5/TH/G5.1-2;B-1
	state after	(2, 1, 1, 0, 5, 2, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 321 *In the top of the 5th inning, with bases empty, ATL batter Austin Riley hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CHA vs CHN on 20230816
	state before	(1, 1, 0, 0, 4, 0, 0, 0)
64.	error	play,4,0,rob1002,00,X,S7/G56+.1-3(E7);B-2
	state after	(1, 0, 1, 1, 4, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 322 *In the top of the 4th inning, with a runner on 1st, CHA batter Luis Robert hit a hard ground ball single to 3rd. An error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

Visiting/Home team rosters for KCA vs CHN on 2023-08-18

position	innings	player	position	innings	player
1	1	Cole Ragans	1	1	Jameson Taillon
1	7	Dylan Coleman	1	7	Michael Fulmer
1	8	Austin Cox	1	8	Jose Cuas
1	8	Taylor Clarke	1	9	Drew Smyly
1	9	Carlos Hernandez	2	1	Yan Gomes
2	1	Freddy Fermin	3	1	Patrick Wisdom
3	1	Salvador Perez	3	9	Jeimer Candelario
4	1	Michael Massey	4	1	Nico Hoerner
5	1	Maikel Garcia	5	1	Jeimer Candelario
5	4	Matt Duffy	5	9	Nick Madrigal
6	1	Bobby Witt Jr.	6	1	Dansby Swanson
7	1	Nelson Velazquez	7	1	Ian Happ
7	9	Dairon Blanco	8	1	Cody Bellinger
8	1	Kyle Isbel	9	1	Seiya Suzuki
9	1	Drew Waters			

	game	KCA vs CHN on 20230818
	state before	(0, 0, 0, 0, 4, 1, 0, 0)
65.	error	play,4,0,wittb002,32,CBSBF.BX,S5/G56.B-2(E5/TH)
	state after	(0, 0, 1, 0, 4, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 323 This game has 3 errors. The final score is 4-3.
In the top of the 4th inning, with bases empty, KCA batter Bobby Witt hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	KCA vs CHN on 20230818
	state before	(1, 1, 1, 1, 4, 2, 1, 1)
66.	error	play,4,1,morec001,11,FBX,5(2)/F0/G56.3-H;1-3(E5/TH);B-1
	state after	(2, 1, 0, 1, 4, 2, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 324 *In the bottom of the 4th inning, with bases loaded, CHN batter Christopher Morel hit a ground ball to 3rd. The runner on 3rd scored (1 RBI) while the third baseman forced out the runner from 2nd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	KCA vs CHN on 20230818
	state before	(1, 0, 0, 0, 6, 3, 1, 0)
67.	error	play,6,0,duffm002,02,SC.X,E3/G4.B-1
	state after	(1, 1, 0, 0, 6, 3, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 325 *In the top of the 6th inning, with bases empty, KCA batter Matt Duffy hit a ground ball to 2nd base. An error by the first baseman allowed the batter to reach 1st base.*

	game	KCA vs CHN on 20230819
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
68.	error	play,1,1,hoern001,12,CBCX,E6/G6.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 326 *In the bottom of the 1st inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs CHN on 20230828
	state before	(1, 1, 0, 0, 1, 1, 0, 0)
69.	error	play,1,0,adamw002,11,1SBX,FC6/G56.1-3(E6/TH);B-2
	state after	(1, 0, 1, 1, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 327 This game has 2 errors. The final score is 6-2.
In the top of the 1st inning, with a runner on 1st, MIL batter Willy Adames hit a ground ball into a fielder's choice at 3rd. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	MIL vs CHN on 20230828
	state before	(2, 0, 0, 0, 2, 4, 1, 0)
70.	error	play,2,0,yelic001,01,CX,S4/G4.B-2(E4/TH)
	state after	(2, 0, 1, 0, 2, 4, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 328 *In the top of the 2nd inning, with bases empty, MIL batter Christian Yelich hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd.*

Visiting/Home team rosters for MIL vs CHN on 2023-08-30

position	innings	player	position	innings	player
1	1	Brandon Woodruff	1	1	Kyle Hendricks
1	7	Elvis Peguero	1	7	Julian Merryweather
1	8	Joel Payamps	1	8	Adbert Alzolay
2	1	William Contreras	1	8	Mark Leiter Jr.
3	1	Rowdy Tellez	2	1	Miguel Amaya
4	1	Brice Turang	2	9	Yan Gomes
5	1	Andruw Monasterio	3	1	Cody Bellinger
6	1	Willy Adames	4	1	Nico Hoerner
7	1	Christian Yelich	5	1	Nick Madrigal
8	1	Sal Frelick	6	1	Dansby Swanson
8	8	Joey Wiemer	7	1	Ian Happ
9	1	Mark Canha	8	1	Mike Tauchman
9	8	Sal Frelick	9	1	Seiya Suzuki

	game	MIL vs CHN on 20230830
	state before	(0, 0, 0, 0, 3, 0, 2, 0)
71.	error	play,3,0,yelic001,32,CBBFBX,E1/TH/G2-.B-2
	state after	(0, 0, 1, 0, 3, 0, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 329 This game has 3 errors. The final score is 2-3.

In the top of the 3rd inning, with bases empty, MIL batter Christian Yelich hit a soft ground ball to very shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd base.

	game	MIL vs CHN on 20230830
	state before	(0, 0, 0, 0, 8, 2, 2, 1)
72.	error	play,8,1,morec001,10,....BX,E6/G56.B-1
	state after	(0, 1, 0, 0, 8, 2, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 330 *In the bottom of the 8th inning, with bases empty, CHN batter Christopher Morel hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs CHN on 20230830
	state before	(0, 0, 0, 0, 9, 2, 3, 0)
73.	error	play,9,0,monaa001,02,.CFX,E3/TH/G4.B-1
	state after	(0, 1, 0, 0, 9, 2, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 331 *In the top of the 9th inning, with bases empty, MIL batter Andruw Monasterio hit a ground ball to 2nd base. A throwing error by the first baseman allowed the batter to reach 1st base.*

8.2.7 September

	game	SFN vs CHN on 20230904
	state before	(2, 1, 0, 0, 8, 0, 3, 1)
74.	error	play,8,1,bellc002,01,.CX,FC4/G4.1-2(E4);B-1
	state after	(2, 1, 1, 0, 8, 0, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 332 *In the bottom of the 8th inning, with a runner on 1st, CHN batter Cody Bellinger hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for SFN vs CHN on 2023-09-05

position	innings	player	position	innings	player
1	1	Ryan Walker	1	1	Kyle Hendricks
1	2	Scott Alexander	1	6	Hayden Wesneski
1	3	Jakob Junis	1	7	Drew Smyly
1	6	Tyler Rogers	1	8	Jose Cuas
1	7	Luke Jackson	1	8	Mark Leiter Jr.
1	8	Taylor Rogers	1	9	Julian Merryweather
2	1	Blake Sabol	2	1	Yan Gomes
2	7	Patrick Bailey	3	1	Jeimer Candelario
3	1	LaMonte Wade Jr.	4	1	Nico Hoerner
3	7	J.D. Davis	5	1	Nick Madrigal
4	1	Thairo Estrada	6	1	Dansby Swanson
5	1	J.D. Davis	7	1	Ian Happ
5	7	Casey Schmitt	8	1	Cody Bellinger
6	1	Brandon Crawford	9	1	Seiya Suzuki
7	1	Joc Pederson			
8	1	Wade Meckler			
8	8	Austin Slater			
9	1	Mike Yastrzemski			
9	7	Austin Slater			
9	8	Mitch Haniger			

	game	SFN vs CHN on 20230905
	state before	(1, 0, 0, 0, 2, 1, 0, 0)
75.	error	play,2,0,davij006,02,CFFFX,S8/G4M.B-2(E8)
	state after	(1, 0, 1, 0, 2, 1, 0, 0)
	fielder	8 (center field on the home team)

Summary 333 This game has 4 errors. The final score is 8-11.

In the top of the 2nd inning, with bases empty, SFN batter J.D. Davis hit a ground ball single to 2nd. An error by the center fielder allowed

the batter to reach 2nd.

	game	SFN vs CHN on 20230905
	state before	(1, 1, 0, 1, 7, 6, 6, 1)
76.	error	play,7,1,madrn001,32,B*B1BC>FX,FC5/G5.3-H;1-3(E5/TH);B-2
	state after	(1, 0, 1, 1, 7, 6, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 334 *In the bottom of the 7th inning, with runners on 1st and 3rd, CHN batter Nick Madrigal hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SFN vs CHN on 20230905
	state before	(2, 0, 1, 0, 8, 7, 10, 0)
77.	error	play,8,0,slata001,22,FBSBX,E6/G6.2-3;B-1
	state after	(2, 1, 0, 1, 8, 7, 10, 0)
	fielder	6 (shortstop on the home team)

Summary 335 *In the top of the 8th inning, with a runner on 2nd, SFN batter Austin Slater hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SFN vs CHN on 20230905
	state before	(1, 0, 0, 0, 8, 7, 10, 1)
78.	error	play,8,1,swand001,10,BX,S8/L89D.B-2(E8)
	state after	(1, 0, 1, 0, 8, 7, 10, 1)
	fielder	8 (center field on the visiting team)

Summary 336 *In the bottom of the 8th inning, with bases empty, CHN batter Dansby Swanson hit a line drive single to deep right center field. An error by the center fielder allowed the batter to reach 2nd.*

79.	game	ARI vs CHN on 20230907
	state before	(1, 0, 0, 0, 8, 5, 1, 0)
	error	play,8,0,thoma004,00,X,E3/G3.B-1
	state after	(1, 1, 0, 0, 8, 5, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 337 *In the top of the 8th inning, with bases empty, ARI batter Alek Thomas hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

80.	game	ARI vs CHN on 20230908
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
	error	play,1,0,martk001,22,FF1B1B>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 338 This game has 2 errors. The final score is 1-0.
In the top of the 1st inning, with a runner on 1st, ARI batter Ketel Marte had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Marte is still at bat, now with a full count.

81.	game	ARI vs CHN on 20230908
	state before	(2, 1, 0, 0, 6, 0, 0, 0)
	error	play,6,0,martk001,01,F1>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 339 *In the top of the 6th inning, with a runner on 1st, ARI batter Ketel Marte (the same batter as in the last error, which was also a stolen base!) had a 0-1 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Marte is still at bat, now with a 0-2 count.*

	game	ARI vs CHN on 20230909
	state before	(2, 1, 0, 0, 10, 3, 1, 0)
82.	error	play,10,0,gurrl001,11,BC1,P01(E3).1-2
	state after	(2, 0, 1, 0, 10, 3, 1, 0)
	fielder	3 (catcher on the home team)

Summary 340 *In the top of the 10th inning, with a runner on 1st, ARI batter Lourdes Gurriel has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the first baseman allowed that runner to reach 2nd. Gurriel is still at bat with a 1-1 count.*

	game	ARI vs CHN on 20230910
	state before	(2, 1, 0, 0, 4, 0, 4, 1)
83.	error	play,4,1,morec001,12,S1S+11P+1.>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 341 *In the bottom of the 4th inning, with a runner on 1st, CHN batter Christopher Morel had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Morel is still at bat, now with a 2-2 count.*

	game	PIT vs CHN on 20230921
	state before	(0, 1, 0, 0, 3, 0, 0, 0)
84.	error	play,3,0,reynb001,12,CBFX,FC6/G6M.1-2(E6);B-1
	state after	(0, 1, 1, 0, 3, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 342 This game has 2 errors. The final score is 8-6. *In the top of the 3rd inning, with a runner on 1st, PIT batter Bryan Reynolds hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	PIT vs CHN on 20230921
	state before	(0, 1, 0, 0, 9, 5, 4, 0)
85.	error	play,9,0,hayek001,02,CFX,D7/L7D+.1XH(742);B-3(E5/TH)
	state after	(1, 0, 0, 1, 9, 5, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 343 *In the top of the 9th inning, with a runner on 1st, PIT batter Ke'Bryan Hayes hit a hard line drive double to deep left field. The runner on 1st (attempting to score) was tagged out at home. A throwing error by the third baseman allowed the batter to reach 3rd.*

	game	COL vs CHN on 20230923
	state before	(1, 0, 0, 0, 7, 3, 3, 0)
86.	error	play,7,0,blacc001,01,FX,E6/F56D.B-2
	state after	(1, 0, 1, 0, 7, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 344 *In the top of the 7th inning, with bases empty, COL batter Charlie Blackmon hit a fly ball to deep short. An error by the shortstop allowed the batter to reach 2nd base.*

Chapter 9

Chicago White Sox (CHA)

The Chicago White Sox compete in the American League Central Division of the MLB. They have been based in Chicago since 1901 and have had their current name since 1904. They have won 3 World Series titles, first in 1906 and the last in 2005. They have won numerous division titles, their last ALCD title was in 2021.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Chicago against the White Sox at their home stadium. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. *High* DER and *low* E is ideal. The DER of the 2023 CHA team is .685 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the White Sox is below the MLB average.

In CHA 2023, there were 97 errors in the 81 home games, 51 by the White Sox and the rest (46) by visiting teams. The total number of errors by the team (home and away) over the regular season of 2023 was 95. As 83.9 was the MLB average with standard deviation about 14. the White Sox were almost one standard deviation over the MLB average (which is bad).

9.1 Season totals

The pitcher, while off the rubber, may throw to any base. If he makes a wild throw, such throw is the throw of an infielder and what follows is governed by the rules covering a ball thrown by a fielder.

– **MLB Rule Book** 5.07(e) Comment

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

9.1.1 Run conversion and error total by position

This section tabulates the totals by position and by game state.

Runs and errors	totals
Errors by vt:	46
Runs by ht because of errors by vt:	9
Errors by ht:	51
Runs by vt because of errors by ht:	23

This is a good¹ error-to-run conversion rate: $9/46 = 19.6\%$ on the part of the White Sox (compared to the much better $23/51 = 45.1\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 CHA season (home and away) has about $R/PA = 641/5980$ (about 10.7%) change of resulting in a run. Therefore, a randomly selected error is (for CHA) nearly 2 times as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 97, so the expected number of errors (by either team) in a randomly selected CHA 2023 home game is about 1.2.

¹But not “very good,” as many other MLB teams that year had rates of 30%.

Errors by position	vt	ht
errors due to pitcher:	6	7
to catcher:	7	8
to 1st baseman:	3	5
to 2nd baseman:	6	3
to 3rd baseman:	11	5
to shortstop:	8	11
to left fielder:	0	4
to center fielder:	2	1
to right fielder:	2	6

9.1.2 Error total by state

With no runners on base, the pitcher is not required to come to a complete stop when using the Set Position. If, however, in the umpire's judgment, a pitcher delivers the ball in a deliberate effort to catch the batter off guard, this delivery shall be deemed a quick pitch, for which the penalty is a ball.

– **MLB Rule Book** 5.07(a)(2) Comment

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	18	15	33
(lone) runner on 1st	12	12	24
(lone) runner on 2nd	8	6	14
(lone) runner on 3rd	0	2	2
two runners on base	6	13	19
bases loaded	2	3	5
total	46	51	97

A table of frequencies of most common errors by the CHA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	13.7%
(1, 0, 0, 0)	4	7.8%
(2, 0, 0, 0)	4	%
(0, 1, 0, 0)	3	5.9%
(1, 1, 0, 0)	3	%
(2, 1, 0, 0)	6	11.8%
(1, 0, 1, 0)	3	%
(1, 1, 1, 0)	4	%
(0, 1, 1, 1)	2	3.9%

A table of frequencies of most common errors by a visiting team in CHA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	10	21.7%
(1, 0, 0, 0)	4	8.7%
(2, 0, 0, 0)	4	%
(0, 1, 0, 0)	2	4.3%
(1, 1, 0, 0)	4	%
(2, 1, 0, 0)	6	13.0%
(1, 0, 1, 0)	4	%
(0, 1, 1, 0)	2	%
(1, 1, 1, 0)	2	%
(1, 1, 1, 1)	1	2.2%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 White Sox homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `CHA2023.eva`):

state	CHA2023 percentages
(0, 0, 0, 0)	24.6%
(1, 0, 0, 0)	18.0%
(2, 0, 0, 0)	14.6%
(0, 1, 0, 0)	5.6%
(1, 1, 0, 0)	7.1%
(2, 1, 0, 0)	7.4%
(0, 0, 1, 0)	2.2%
(1, 0, 1, 0)	2.6%
(2, 0, 1, 0)	3.3%
(0, 0, 0, 1)	0.1%
(1, 0, 0, 1)	1.0%
(2, 0, 0, 1)	1.4%
(0, 1, 1, 0)	1.1%
(1, 1, 1, 0)	2.3%
(2, 1, 1, 0)	2.6%

9.1.3 Some interesting errors

Some of the most interesting error plays in the CHA 2023 home games:

- The game BOS vs CHA on 2023-06-23 has 4 errors.
- The game OAK vs CHA on 2023-08-25 had 4 errors.
- The game KCA vs CHA on 2023-09-12 had 4 errors, including a play with 3 RBIs.

9.1.4 Relative team rankings

In the Windup Position, a pitcher is permitted to have his “free” foot on the rubber, in front of the rubber, behind the rubber or off the side of the rubber. From the Windup Position, the pitcher may:

- (A) deliver the ball to the batter, or
- (B) step and throw to a base in an attempt to pick-off a runner, or
- (C) disengage the rubber (if he does he must drop his hand to his sides).

In disengaging the rubber the pitcher must step off with his pivot foot and not his free foot first.

– **MLB Rule Book** 5.07(a)(1) Comment

We have 81 home games between CHA and some visiting MLB team. A number of teams did not play CHA in the 2023 regular season. However, of those that did, CHA *lost* more games than won against

CHN (2/2), KCA (2/6), ANA (2/3), ARI (2/3), BAL (2/3), HOU (2/3),
MIA (2/3), PHI (2/3), SEA (2/3), SFN (2/3), SLN (2/3), TEX (2/3),
CLE (3/7), TBA (3/4), MIN (4/7), MIL (3/3), SDN (3/3), TOR (3/3),

Here, the team are ordered from “worst” to “best”, with that team’s W/T record in parentheses (W = numbers of games won against CHA, T = Total home games played at CHA). In 2023, the White Sox *won* more than lost against

BOS (1/3), NYA (1/3).

The teams that *split* their series were:

OAK (2/4), DET (3/6).

The teams that did not play in any of the 81 home games in CHA 2023 were:

ATL, CIN, COL, LAN, NYN, PIT, WAS .

9.2 Game-by-game totals

The Official Scorer shall credit a **stolen base** to a runner whenever the runner advances one base when a runner is attempting to steal, and the catcher, after receiving the pitch, makes a wild throw trying to prevent the stolen base, the Official Scorer shall credit the runner with a stolen base. The Official Scorer shall not charge an error unless the wild throw permits the stealing runner to advance one or more extra bases, or permits another runner to advance, in which case the Official Scorer shall credit the runner with the stolen base and charge one error to the catcher.

– **MLB Rule Book** 9.07(b)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for CHA 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Yasmani Grandal	92	.309
3 (1B)	Andrew Vaughn	143	.314
4 (2B)	Elvis Andrus	63	.304
5 (3B)	Yoán Moncada	90	.305
6 (SS)	Tim Anderson	119	.286
7 (LF)	Andrew Benintendi	147	.326
8 (CF)	Luis Robert Jr.	143	.315
9 (RF)	Oscar Colás	69	.257

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 CHA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

As we will see below, there were 95 errors in the 81 home games, 45 by CHA and the rest (50) by visiting teams.

9.2.1 April

	game	SFN vs CHA on 20230405
	state before	(2, 1, 1, 0, 3, 1, 2, 0)
1.	error	play,3,0,pedej001,10,*BX,E3/L3.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 3, 1, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 345 This game had 2 errors. The final score was 3-7.

In the top of the 3rd inning, with runners on 1st and 2nd, SFN batter Joc Pederson hit a line drive to 1st base. An error by the first baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

	game	SFN vs CHA on 20230405
	state before	(0, 0, 0, 0, 7, 1, 4, 1)
2.	error	play,7,1,andre001,01,...CX,E6/G56.B-1
	state after	(0, 1, 0, 0, 7, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 346 *In the bottom of the 7th inning, with bases empty, CHA batter Elvis Andrus hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	SFN vs CHA on 20230406
	state before	(1, 1, 0, 0, 2, 4, 0, 1)
3.	error	play,2,1,zavas001,22,BCBFX,FC6/G6.1-2(E6);B-1
	state after	(1, 1, 1, 0, 2, 4, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 347 This game had 2 errors. The final score was 16-6. *In the bottom of the 2nd inning, with a runner on 1st, CHA batter Seby Zavala hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SFN vs CHA on 20230406
	state before	(1, 1, 0, 0, 5, 8, 3, 1)
4.	error	play,5,1,zavas001,12,SSB+1,P01(E2/TH).1-2
	state after	(1, 0, 1, 0, 5, 8, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 348 *In the bottom of the 5th inning, with a runner on 1st, CHA batter Seby Zavala has a 1-2 count. During an attempt to pickoff the runner on 1st, a throwing error by the catcher allowed that runner to safely advance to 2nd.*

	game	BAL vs CHA on 20230414
	state before	(2, 0, 1, 0, 8, 1, 7, 0)
5.	error	play,8,0,ohear001,10,B2,P02(E1/TH).2-3
	state after	(2, 0, 0, 1, 8, 1, 7, 0)
	fielder	1 (pitcher on the home team)

Summary 349 *In the top of the 8th inning, with a runner on 2nd, BAL batter Ryan O’Hearn has a 1-0 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed that runner to safely advance to 3rd.*

Visiting/Home team rosters for BAL vs CHA on 2023-04-15

position	innings	player	position	innings	player
1	1	Kyle Gibson	1	1	Michael Kopech
1	10	Logan Gillaspie	1	10	Jimmy Lambert
1	6	Danny Coulombe	1	6	Aaron Bummer
1	7	Bryan Baker	1	6	Keynan Middleton
1	7	Danny Coulombe	1	7	Kendall Graveman
1	8	Cionel Perez	1	8	Tanner Banks
1	9	Yennier Cano	1	9	Reynaldo Lopez
2	1	James McCann	2	1	Yasmani Grandal
2	7	Adley Rutschman	3	1	Andrew Vaughn
3	1	Ryan Mountcastle	4	1	Lenyn Sosa
4	1	Adam Frazier	5	1	Jake Burger
5	1	Ramon Urias	6	1	Elvis Andrus
5	7	Gunnar Henderson	7	1	Andrew Benintendi
6	1	Jorge Mateo	8	1	Luis Robert Jr.
7	1	Austin Hays	9	1	Oscar Colas
8	1	Cedric Mullins			
9	1	Anthony Santander			

	game	BAL vs CHA on 20230415
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
6.	error	play,1,1,benia002,22,BCBFX,S1/G56S-.B-3(E1/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 350 This game had 3 errors. The final score was 6-7.

In the bottom of the 1st inning, with bases empty, CHA batter Andrew Benintendi hit a soft ground ball single to shallow 3rd. A throwing error by the pitcher allowed the batter to reach 3rd.

	game	BAL vs CHA on 20230415
	state before	(1, 0, 1, 0, 5, 1, 2, 1)
7.	error	play,5,1,robelt002,11,BCX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 5, 1, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 351 *In the bottom of the 5th inning, with a runner on 2nd, CHA batter Luis Robert hit a ground ball to shortstop. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	BAL vs CHA on 20230415
	state before	(2, 0, 1, 0, 7, 5, 5, 1)
8.	error	play,7,1,jimee001,11,FBX,E5/TH/G56S.2-3;B-1
	state after	(2, 1, 0, 1, 7, 5, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 352 *In the bottom of the 7th inning, with a runner on 2nd, CHA batter Eloy Jimenez hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	BAL vs CHA on 20230416
	state before	(2, 1, 0, 0, 4, 1, 4, 0)
9.	error	play,4,0,vavrt001,12,1FC*B>FX,E5/G56.1-2;B-1
	state after	(2, 1, 1, 0, 4, 1, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 353 *In the top of the 4th inning, with a runner on 1st, BAL batter Terrin Vavra hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TBA vs CHA on 20230427
	state before	(1, 1, 1, 0, 6, 7, 2, 0)
10.	error	play,6,0,franw002,21,BCBX,E3/G3D.2-H(NR);1-2;B-1
	state after	(1, 1, 1, 0, 6, 8, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 354 *In the top of the 6th inning, with runners on 1st and 2nd, TBA batter Wander Franco hit a ground ball to deep 1st base. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

Visiting/Home team rosters for TBA vs CHA on 2023-04-29

position	innings	player	position	innings	player
1	1	Calvin Faucher	1	1	Lance Lynn
1	2	Yonny Chirinos	1	7	Aaron Bummer
1	7	Garrett Cleavinger	1	7	Jimmy Lambert
1	8	Jalen Beeks	1	8	Keynan Middleton
1	9	Ryan Thompson	1	9	Jake Diekman
2	1	Francisco Mejia	2	1	Yasmani Grandal
3	1	Isaac Paredes	3	1	Andrew Vaughn
3	7	Yandy Diaz	4	1	Romy Gonzalez
4	1	Brandon Lowe	5	1	Jake Burger
4	7	Taylor Walls	6	1	Elvis Andrus
5	1	Taylor Walls	7	1	Adam Haseley
5	7	Isaac Paredes	7	3	Gavin Sheets
6	1	Wander Franco	8	1	Luis Robert Jr.
7	1	Randy Arozarena	8	3	Adam Haseley
8	1	Jose Siri	9	1	Gavin Sheets
9	1	Luke Raley	9	3	Oscar Colas
9	7	Manuel Margot			

	game	TBA vs CHA on 20230429
	state before	(0, 1, 1, 0, 2, 0, 0, 1)
11.	error	play,2,1,grany001,32,BBCBFX,E1/TH/G2-.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 2, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 355 This game had 3 errors. The final score was 12-3.

In the bottom of the 2nd inning, with runners on 1st and 2nd, CHA batter Yasmani Grandal hit a soft ground ball to very shallow infield. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

	game	TBA vs CHA on 20230429
	state before	(1, 1, 0, 0, 7, 1, 3, 0)
12.	error	play,7,0,parei001,22,CFB.BX,D7/L7LD+.1-H;B-3(E1/TH)
	state after	(1, 0, 0, 1, 7, 2, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 356 *In the top of the 7th inning, with a runner on 1st, TBA batter Isaac Paredes hit a hard line drive double to deep left field. The runner on 1st scored (1 RBI). A throwing error by the pitcher allowed the batter to reach 3rd.*

	game	TBA vs CHA on 20230429
	state before	(2, 0, 0, 0, 8, 11, 3, 1)
13.	error	play,8,1,jimee001,01,CX,E5/G56.B-1
	state after	(2, 1, 0, 0, 8, 11, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 357 *In the bottom of the 8th inning, with bases empty, CHA batter Eloy Jimenez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	TBA vs CHA on 20230430
	state before	(0, 0, 1, 0, 8, 2, 4, 0)
14.	error	play,8,0,parei001,32,CB>B.FFBX,S9/L9L.2-H(E9)(NR)
	state after	(0, 1, 0, 0, 8, 3, 4, 0)
	fielder	9 (right field on the home team)

Summary 358 This game had 2 errors. The final score was 9-12. *In the top of the 8th inning, with a runner on 2nd, TBA batter Isaac Paredes hit a line drive single to right field. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	TBA vs CHA on 20230430
	state before	(1, 0, 1, 0, 8, 6, 4, 0)
15.	error	play,8,0,loweb001,01,1>S.X,E7/F78+.2-H(NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 8, 7, 4, 0)
	fielder	7 (left field on the home team)

Summary 359 *In the top of the 8th inning, with a runner on 2nd, TBA batter Brandon Lowe hit a hard fly ball to left center field. An error by the left fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

9.2.2 May

	game	MIN vs CHA on 20230504
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
16.	error	play,2,0,miraj002,00,X,E6/TH/G6+.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 360 This game had 2 errors. The final score was 7-3.
In the top of the 2nd inning, with bases empty, MIN batter Jose Miranda hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	MIN vs CHA on 20230504
	state before	(0, 0, 1, 0, 12, 2, 2, 0)
17.	error	play,12,0,miraj002,00,..X,E6/G6.2-H(NR) (UR) ;B-1
	state after	(0, 1, 0, 0, 12, 3, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 361 *In the top of the 12th inning, with a runner on 2nd, MIN batter Jose Miranda hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	HOU vs CHA on 20230513
	state before	(2, 1, 0, 0, 7, 1, 1, 1)
18.	error	play,7,1,vauga001,12,CF1B>*B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 362 *In the bottom of the 7th inning, with a runner on 1st, CHA batter Andrew Vaughn had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Vaughn is still at bat, now with a 2-2 count.*

	game	HOU vs CHA on 20230514
	state before	(0, 1, 0, 1, 1, 0, 0, 0)
19.	error	play,1,0,alvay001,21,S*BBX,D9/L89D+.3-H;1-H;B-3(E6/TH)
	state after	(0, 0, 0, 1, 1, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 363 *In the top of the 1st inning, with runners on 1st and 3rd, HOU batter Yordan Alvarez hit a hard line drive double to deep right center field. The runners on 1st and 3rd both scored (2 RBIs). A throwing error by the shortstop allowed the batter to reach 3rd.*

Visiting/Home team rosters for CLE vs CHA on 2023-05-16

position	innings	player	position	innings	player
1	1	Shane Bieber	1	1	Lance Lynn
1	5	Xzavion Curry	1	8	Keynan Middleton
1	7	Nick Sandlin	1	9	Joe Kelly
1	8	Enyel De Los Santos	2	1	Yasmani Grandal
2	1	Mike Zunino	3	1	Andrew Vaughn
2	7	David Fry	4	1	Hanser Alberto
3	1	Josh Naylor	5	1	Yoan Moncada
3	6	Gabriel Arias	6	1	Tim Anderson
4	1	Andres Gimenez	7	1	Andrew Benintendi
5	1	Gabriel Arias	8	1	Luis Robert Jr.
5	6	Brayan Rocchio	9	1	Gavin Sheets
6	1	Amed Rosario	9	9	Jake Marisnick
7	1	Steven Kwan			
8	1	Myles Straw			
9	1	Will Brennan			

	game	CLE vs CHA on 20230516
	state before	(2, 0, 0, 0, 5, 0, 0, 1)
20.	error	play,5,1,rob1002,32,BBSBSX,C/E2.B-1
	state after	(2, 1, 0, 0, 5, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 364 This game had 3 errors. The final score was 3-8.

In the bottom of the 5th inning, with bases empty, CHA batter Luis Robert had a full count. Catcher's interference allowed the batter to take 1st.

	game	CLE vs CHA on 20230516
	state before	(2, 0, 0, 0, 6, 1, 6, 0)
21.	error	play,6,0,roccb001,00,.X,S2/G2-.B-2(E2/TH)
	state after	(2, 0, 1, 0, 6, 1, 6, 0)
	fielder	2 (catcher on the home team)

Summary 365 *In the top of the 6th inning, with bases empty, CLE batter Brayan Rocchio hit a soft ground ball single to very shallow infield. A throwing error by the the catcher allowed the batter to reach 2nd.*

	game	CLE vs CHA on 20230516
	state before	(0, 0, 0, 0, 8, 1, 7, 0)
22.	error	play,8,0,stram002,02,CCX,E6/TH/G6MS.B-1
	state after	(0, 1, 0, 0, 8, 1, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 366 *In the top of the 8th inning, with bases empty, CLE batter Myles Straw hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs CHA on 20230517
	state before	(1, 0, 1, 0, 6, 1, 5, 1)
23.	error	play,6,1,albeh001,01,FX,E6/TH/G6.2-3;B-1
	state after	(1, 1, 0, 1, 6, 1, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 367 *In the bottom of the 6th inning, with a runner on 2nd, CHA batter Hanser Alberto hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	CLE vs CHA on 20230518
	state before	(2, 1, 0, 0, 4, 0, 1, 1)
24.	error	play,4,1,marij002,11,BSX,E5/G5.1-2;B-1
	state after	(2, 1, 1, 0, 4, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 368 This game had 2 errors. The final score was 3-1.
In the bottom of the 4th inning, with a runner on 1st, CHA batter Jake Marisnick hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CLE vs CHA on 20230518
	state before	(1, 0, 1, 1, 7, 1, 1, 0)
25.	error	play,7,0,roccb001,01,CX,E6/G6MS.3-H;2-3;B-1
	state after	(1, 1, 0, 1, 7, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 369 *In the top of the 7th inning, with runners on 2nd and 3rd, CLE batter Brayan Rocchio hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	KCA vs CHA on 20230519
	state before	(0, 0, 0, 0, 3, 0, 2, 1)
26.	error	play,3,1,andet001,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 370 *In the bottom of the 3rd inning, with bases empty, CHA batter Tim Anderson hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	KCA vs CHA on 20230520
	state before	(0, 1, 0, 0, 6, 1, 2, 1)
27.	error	play,6,1,rob1002,10,BX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 6, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 371 *In the bottom of the 6th inning, with a runner on 1st, CHA batter Luis Robert had a 1-0 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd.*

	game	ANA vs CHA on 20230529
	state before	(2, 1, 0, 0, 3, 4, 1, 0)
28.	error	play,3,0,urshg001,11,BCX,E5/TH/G56S.1-2;B-1
	state after	(2, 1, 1, 0, 3, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 372 *In the top of the 3rd inning, with a runner on 1st, ANA batter Giovanny Urshela hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	ANA vs CHA on 20230530
	state before	(1, 0, 1, 0, 4, 3, 5, 1)
29.	error	play,4,1,zavas001,00,>B,SB3.2-H(E5) (NR)
	state after	(1, 0, 0, 0, 4, 3, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 373 *In the bottom of the 4th inning, with a runner on 2nd, CHA batter Seby Zavala had a ?-? count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. An error by the third baseman allowed that runner to score (unearned, no RBI). Zavala is still at bat, now with a 1-0 count.*

9.2.3 June

	game	DET vs CHA on 20230602
	state before	(2, 0, 0, 0, 5, 0, 0, 1)
30.	error	play,5,1,grany001,02,CFX,E3/G3.B-1
	state after	(2, 1, 0, 0, 5, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 374 This game had 2 errors. The final score was 0-3.

In the bottom of the 5th inning, with bases empty, CHA batter Yasmani Grandal hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.

	game	DET vs CHA on 20230602
	state before	(1, 1, 1, 0, 6, 0, 1, 1)
31.	error	play,6,1,jimee001,11,*BFX,S8/L8D.2-H;1-3;B-2(E8/TH)
	state after	(1, 0, 1, 1, 6, 0, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 375 *In the bottom of the 6th inning, with runners on 1st and 2nd, CHA batter Eloy Jimenez hit a line drive single to deep center field. The runner on 2nd score (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

	game	MIA vs CHA on 20230609
	state before	(0, 0, 0, 0, 3, 0, 1, 1)
32.	error	play,3,1,andre001,10,BX,S5/G56.B-2(E5/TH)
	state after	(0, 0, 1, 0, 3, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 376 This game had 2 errors. The final score was 1-2.

In the bottom of the 3rd inning, with bases empty, CHA batter Elvis Andrus hit a ground ball single to 3rd base. A throwing error by the third basemann allowed the batter to reach 2nd.

	game	MIA vs CHA on 20230609
	state before	(1, 1, 0, 0, 7, 1, 1, 1)
33.	error	play,7,1,rob1002,21,B1BFX,C/E2.1-2;B-1
	state after	(1, 1, 1, 0, 7, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 377 *In the bottom of the 7th inning, with a runner on 1st, CHA batter Luis Robert had a 2-1 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd.*

	game	MIA vs CHA on 20230610
	state before	(0, 1, 1, 1, 9, 0, 1, 0)
34.	error	play,9,0,delab001,02,CSX,FC6/G6.3-H(E6)(NR)(UR);2-3;1-2;B-1
	state after	(0, 1, 1, 1, 9, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 378 *In the top of the 9th inning, with bases loaded, MIA batter Bryan De La Cruz hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), he runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	MIA vs CHA on 20230611
	state before	(1, 0, 0, 0, 9, 4, 5, 0)
35.	error	play,9,0,davij007,22,CBBFX,C/E2.B-1
	state after	(1, 1, 0, 0, 9, 4, 5, 0)
	fielder	2 (catcher on the home team)

Summary 379 *In the top of the 9th inning, with bases empty, MIA batter Jonathan Davis had a 2-2 count. Catcher's interference allowed the batter to take 1st.*

	game	TEX vs CHA on 20230620
	state before	(2, 1, 0, 0, 2, 0, 2, 1)
36.	error	play,2,1,remiz001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 380 This game had 2 errors. The final score was 6-7.

In the bottom of the 2nd inning, with a runner on 1st, CHA batter Zach Remillard had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Remillard is still at bat, now with a 0-1 count.

	game	TEX vs CHA on 20230620
	state before	(0, 1, 1, 0, 7, 4, 4, 0)
37.	error	play,7,0,jungj001,22,.CBCBFF.X,FC6/G6.2-3;1-2(E6);B-1
	state after	(0, 1, 1, 1, 7, 4, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 381 *In the top of the 7th inning, with runners on 1st and 2nd, TEX batter Josh Jung hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	TEX vs CHA on 20230621
	state before	(1, 0, 1, 0, 7, 6, 2, 1)
38.	error	play,7,1,remiz001,00,X,E6/TH/G56.2-3;B-1
	state after	(1, 1, 0, 1, 7, 6, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 382 *In the bottom of the 7th inning, with a runner on 2nd, CHA batter Zach Remillard hit a ground ball to 3rd base. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs CHA on 2023-06-23

position	innings	player	position	innings	player
1	1	Brayan Bello	1	1	Lucas Giolito
1	7	Nick Pivetta	1	7	Gregory Santos
1	8	Chris Martin	1	8	Reynaldo Lopez
1	9	Kenley Jansen	1	9	Keynan Middleton
2	1	Connor Wong	2	1	Seby Zavala
3	1	Justin Turner	2	8	Yasmani Grandal
4	1	Christian Arroyo	3	1	Andrew Vaughn
5	1	Rafael Devers	3	9	Gavin Sheets
6	1	David Hamilton	4	1	Tim Anderson
7	1	Rob Refsnyder	5	1	Jake Burger
8	1	Enrique Hernandez	6	1	Elvis Andrus
9	1	Adam Duvall	7	1	Andrew Benintendi
			8	1	Luis Robert Jr.
			9	1	Gavin Sheets
			9	9	Clint Frazier

	game	BOS vs CHA on 20230623
	state before	(1, 0, 1, 0, 3, 0, 0, 0)
39.	error	play,3,0,refsr001,32,>C.BB2SB2,P02(E4).2-H(NR)(UR)
	state after	(1, 0, 0, 0, 3, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 383 This game had 4 errors. The final score was 3-1.

In the top of the 3rd inning, with a runner on 2nd, BOS batter Rob Refsnyder has a 3-2 count. The pitcher attempted to pickoff the runner on 2nd. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI).

	game	BOS vs CHA on 20230623
	state before	(0, 0, 0, 0, 4, 1, 1, 0)
40.	error	play,4,0,yoshm002,22,BSTBFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 4, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 384 *In the top of the 4th inning, with bases empty, BOS batter Masataka Yoshida hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs CHA on 20230623
	state before	(0, 0, 0, 0, 7, 3, 1, 1)
41.	error	play,7,1,burgj001,10,BX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 7, 3, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 385 *In the bottom of the 7th inning, with bases empty, CHA batter Jake Burger hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs CHA on 20230623
	state before	(2, 1, 0, 0, 8, 3, 1, 1)
42.	error	play,8,1,vauga001,00,X,S4/G34.1-3(E4/TH)
	state after	(2, 1, 0, 1, 8, 3, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 386 *In the bottom of the 8th inning, with a runner on 1st, CHA batter Andrew Vaughn hit a ground ball single to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	BOS vs CHA on 20230624
	state before	(2, 1, 0, 0, 4, 1, 1, 0)
43.	error	play,4,0,casat001,01,FX,E5/G56D.1-2;B-1
	state after	(2, 1, 1, 0, 4, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 387 This game had 2 errors, including an infinity play. The final score was 4-5.

In the top of the 4th inning, with a runner on 1st, BOS batter Triston Casas hit a ground ball to deep 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	BOS vs CHA on 20230624
	state before	(2, 0, 0, 0, 6, 3, 2, 0)
44.	error	play,6,0,arroc001,00,F,FLE3
	state after	(2, 0, 0, 0, 6, 3, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 388 This play is an *infinity play*, which can be repeated over and over without the game state ever changing.

In the top of the 6th inning, with bases empty and a 0-0 count, BOS batter Christian Arroyo hit a (catchable) pop fly foul of the 1st base line. An error by the first baseman allowed the batter to remain safe at bat. Arroyo is still at bat, now with a 1-0 count.

9.2.4 July

	game	TOR vs CHA on 20230706
	state before	(1, 0, 0, 0, 6, 0, 0, 1)
45.	error	play,6,1,andre001,32,FS.FFBFBFBX,E4/G4.B-1
	state after	(1, 1, 0, 0, 6, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 389 *In the bottom of the 6th inning, with bases empty, CHA batter Elvis Andrus hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SLN vs CHA on 2023-07-07

position	innings	player	position	innings	player
1	1	Jordan Montgomery	1	1	Dylan Cease
1	5	JoJo Romero	1	7	Keynan Middleton
1	6	Andre Pallante	1	8	Gregory Santos
1	7	Chris Stratton	1	9	Kendall Graveman
1	7	Kyle Leahy	2	1	Yasmani Grandal
2	1	Ivan Herrera	3	1	Andrew Vaughn
3	1	Paul Goldschmidt	4	1	Elvis Andrus
4	1	Nolan Gorman	4	7	Zach Remillard
5	1	Nolan Arenado	5	1	Jake Burger
6	1	Paul DeJong	6	1	Tim Anderson
7	1	Lars Nootbaar	7	1	Andrew Benintendi
8	1	Dylan Carlson	8	1	Luis Robert Jr.
9	1	Jordan Walker	9	1	Zach Remillard
			9	7	Oscar Colas

	game	SLN vs CHA on 20230707
	state before	(1, 0, 1, 0, 2, 0, 0, 0)
46.	error	play,2,0,dejop001,02,FCF2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 390 This game had 3 errors. The final score was 7-8.
In the top of the 2nd inning, with a runner on 2nd, SLN batter Paul DeJong has a 0-2 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed that runner to advance to 3rd,

	game	SLN vs CHA on 20230707
	state before	(0, 0, 0, 0, 3, 2, 0, 0)
47.	error	play,3,0,noot1001,21,BBCX,S9/L89+.B-2(E9)
	state after	(0, 0, 1, 0, 3, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 391 *In the top of the 3rd inning, with bases empty, SLN batter Lars Nootbaar hit a hard line drive single to right center field. An error by the right fielder allowed the batter to reach 2nd.*

	game	SLN vs CHA on 20230707
	state before	(0, 1, 0, 0, 6, 5, 1, 1)
48.	error	play,6,1,rob1002,00,X,FC/G56.1X2(5E4);B-1
	state after	(0, 1, 1, 0, 6, 5, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 392 *In the bottom of the 6th inning, with a runner on 1st, CHA batter Luis Robert hit a ground ball into a fielder's choice at 3rd. An error by the second baseman (with an assist from the third baseman) allowed the runner on 1st to safely advance to 2 and the batter to reach 1st.*

	game	SLN vs CHA on 20230709
	state before	(0, 0, 0, 0, 3, 0, 0, 1)
49.	error	play,3,1,colao001,12,CFBFX,E3/G3.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 393 This game had 2 errors. The final score was 4-3. *In the bottom of the 3rd inning, with bases empty, CHA batter Oscar Colas hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	SLN vs CHA on 20230709
	state before	(1, 1, 0, 0, 3, 0, 0, 1)
50.	error	play,3,1,andre001,00,.B,PB.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 394 *In the bottom of the 3rd inning, with a runner on 1st, CHA batter Elvis Andrus had a 0-0 count. During the next pitch, which was a passed ball, the runner on 1st advanced to 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Andrus is till at bat, now with a 1-0 count.*

	game	SLN vs CHA on 20230709
	state before	(2, 1, 1, 1, 7, 2, 1, 1)
51.	error	play,7,1,perec005,02,..CFX,E5/G56.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 7, 2, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 395 *In the bottom of the 7th inning, with bases loaded, CHA batter Carlos Perez hit a ground ball 3rd base. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	CHN vs CHA on 20230725
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
52.	error	play,2,0,suzus001,12,FBCX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 396 *In the top of the 2nd inning, with bases empty, CHN batter Seiya Suzuki hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CHN vs CHA on 20230726
	state before	(1, 1, 0, 0, 2, 0, 2, 0)
53.	error	play,2,0,mastm001,32,BBCBCX,S8/L78.1-3;B-2(E8)
	state after	(1, 0, 1, 1, 2, 0, 2, 0)
	fielder	8 (center field on the home team)

Summary 397 This game had 2 errors. The final score was 10-7. *In the top of the 2nd inning, with a runner on 1st, CHN batter Miles Mastrobuoni hit a line drive single to left center field. The runner on 1st advanced to 3rd. An error by the center fielder allowed the batter to reach 2nd.*

	game	CHN vs CHA on 20230726
	state before	(2, 1, 1, 1, 5, 3, 7, 0)
54.	error	play,5,0,morec001,02,FFX,S7/L7.3-H;2-H;1-3(E7)
	state after	(2, 1, 0, 1, 5, 5, 7, 0)
	fielder	7 (left field on the home team)

Summary 398 *In the top of the 5th inning, with bases loaded, CHN batter Christopher Morel hit a line drive single to left field. The runners on 2nd and 3rd scored (2 RBIs). An error by the left fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Tanner Bibee	1	1	Dylan Cease
1	7	Trevor Stephan	1	6	Aaron Bummer
1	9	Emmanuel Clase	1	7	Gregory Santos
2	1	Bo Naylor	1	9	Sammy Peralta
3	1	Josh Naylor	2	1	Seby Zavala
4	1	Andres Gimenez	3	1	Yasmani Grandal
5	1	Jose Ramirez	4	1	Jake Burger
6	1	Gabriel Arias	5	1	Yoan Moncada
7	1	Steven Kwan	6	1	Tim Anderson
8	1	Myles Straw	7	1	Andrew Benintendi
9	1	Will Brennan	8	1	Luis Robert Jr.
9	9	Oscar Gonzalez	9	1	Oscar Colas

	game	CLE vs CHA on 20230727
	state before	(1, 1, 1, 0, 3, 3, 0, 0)
55.	error	play,3,0,brenw002,21,*BCBX,S9/L9.2-H;1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 3, 4, 0, 0)
	fielder	9 (right field on the home team)

Summary 399 This game had 3 errors. The final score was 6-3.

In the top of the 3rd inning, with runners on 1st and 2nd, CLE batter Will Brennan hit a line drive single to right field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.

	game	CLE vs CHA on 20230727
	state before	(0, 1, 0, 0, 7, 4, 3, 0)
56.	error	play,7,0,ramij003,02,1CT1>*B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 7, 4, 3, 0)
	fielder	2 (catcher on the home team)

Summary 400 *In the top of the 7th inning, with a runner on 1st, CLE batter Jose Ramirez had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Ramirez is still at bat, now with a 1-2 count.*

	game	CLE vs CHA on 20230727
	state before	(0, 0, 0, 0, 8, 5, 3, 0)
57.	error	play,8,0,ariag002,11,CBX,D9/L9LD.B-3(E9)
	state after	(0, 0, 0, 1, 8, 5, 3, 0)
	fielder	9 (right field on the home team)

Summary 401 *In the top of the 8th inning, with bases empty, CLE batter Gabriel Arias hit a line drive double to deep right field. An error by the right fielder allowed the batter to reach 3rd.*

Visiting/Home team rosters for CLE vs CHA on 2023-07-28

position	innings	player	position	innings	player
1	1	Xzavion Curry	1	1	Touki Toussaint
1	4	Daniel Norris	1	6	Tanner Banks
1	6	Nick Sandlin	1	9	Gregory Santos
1	8	Michael Kelly	2	1	Seby Zavala
2	1	Bo Naylor	3	1	Gavin Sheets
2	7	David Fry	4	1	Zach Remillard
3	1	Josh Naylor	5	1	Jake Burger
4	1	Andres Gimenez	6	1	Tim Anderson
5	1	Jose Ramirez	7	1	Andrew Benintendi
6	1	Gabriel Arias	8	1	Luis Robert Jr.
6	8	Tyler Freeman	9	1	Oscar Colas
7	1	Steven Kwan			
8	1	Will Brennan			
9	1	Oscar Gonzalez			

	game	CLE vs CHA on 20230728
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
58.	error	play,1,0,gimea001,31,BF1BBX,E3/G3.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 402 This game had 3 errors. The final score was 0-3.
In the top of the 1st inning, with a runner on 1st, CLE batter Andres Gimenez hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CLE vs CHA on 20230728
	state before	(0, 0, 0, 0, 5, 0, 1, 0)
59.	error	play,5,0,gonzo001,01,CX,S9/L89.B-2(E9)
	state after	(0, 0, 1, 0, 5, 0, 1, 0)
	fielder	9 (right field on the home team)

Summary 403 *In the top of the 5th inning, with bases empty, CLE batter Oscar Gonzalez hit a line drive single to right center field. An error by the right fielder allowed the batter to reach 2nd.*

	game	CLE vs CHA on 20230728
	state before	(1, 0, 0, 0, 5, 0, 1, 1)
60.	error	play,5,1,zavas001,22,CSBBX,E5/G56.B-1
	state after	(1, 1, 0, 0, 5, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 404 *In the bottom of the 5th inning, with bases empty, CHA batter Seby Zavala hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	CLE vs CHA on 20230730
	state before	(1, 0, 0, 0, 6, 3, 0, 0)
61.	error	play,6,0,brenw002,01,CX,E1/G15-.B-1
	state after	(1, 1, 0, 0, 6, 3, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 405 This game had 2 errors. The final score was 5-0.
In the top of the 6th inning, with bases empty, CLE batter Will Brennan hit a soft ground ball to very shallow left infield. An error by the pitcher allowed the batter to reach 1st base.

	game	CLE vs CHA on 20230730
	state before	(1, 1, 1, 0, 6, 3, 0, 1)
62.	error	play,6,1,robel002,00,X,9/F89S.2-3(E9/TH);1-2
	state after	(2, 0, 1, 1, 6, 3, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 406 *In the bottom of the 6th inning, with runners on 1st and 2nd, CHA batter Luis Robert hit a fly ball to shallow right center. The batter was put out by the right fielder. A throwing error by the right fielder allowed the runners on 1st and 2nd to each advance one base.*

9.2.5 August

	game	NYA vs CHA on 20230807
	state before	(0, 1, 1, 0, 8, 1, 2, 1)
63.	error	play,8,1,remiz001,00,...X,FC/SH/BG1S.2-3;1-2;B-1(E1)
	state after	(0, 1, 1, 1, 8, 1, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 407 *In the bottom of the 8th inning, with runners on 1st and 2nd, CHA batter Zach Remillard hit a bunt to a fielder's choice at very shallow infield. The runners on 1st and 2nd each advanced one base. An error by the pitcher allowed the batter to reach 1st. The bases are loaded.*

64.	game	NYA vs CHA on 20230809
	state before	(0, 0, 1, 0, 5, 1, 4, 1)
	error	play,5,1,jimee001,22,F1BC.>B.FX,E4/G34.2-H(NR);B-1
	state after	(0, 1, 0, 0, 5, 1, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 408 *In the bottom of the 5th inning, with a runner on 2nd, CHA batter Eloy Jimenez hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

65.	game	MIL vs CHA on 20230811
	state before	(0, 1, 1, 1, 6, 4, 5, 0)
	error	play,6,0,contw002,01,FX,S1/G15-.3-H;2-H(E1/TH)(NR);1-3;B-2
	state after	(0, 0, 1, 1, 6, 6, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 409 *In the top of the 6th inning, with bases loaded, MIL batter William Contreras hit a sft ground ball single to shallow left infield. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

Visiting/Home team rosters for SEA vs CHA on 2023-08-21

position	innings	player	position	innings	player
1	1	Luis Castillo	1	1	Touki Toussaint
1	8	Eduard Bazarido	1	5	Tanner Banks
1	9	Darren McCaughan	1	7	Brent Honeywell
2	1	Cal Raleigh	1	9	Lane Ramsey
2	9	Brian O’Keefe	2	1	Yasmani Grandal
3	1	Ty France	3	1	Gavin Sheets
4	1	Josh Rojas	4	1	Lenyn Sosa
4	7	Jose Caballero	5	1	Yoan Moncada
5	1	Eugenio Suarez	5	8	Zach Remillard
5	7	Josh Rojas	6	1	Elvis Andrus
6	1	J.P. Crawford	7	1	Andrew Benintendi
7	1	Dominic Canzone	8	1	Luis Robert Jr.
8	1	Cade Marlowe	8	7	Trayce Thompson
9	1	Teoscar Hernandez	9	1	Oscar Colas

	game	SEA vs CHA on 20230821
	state before	(2, 1, 0, 1, 1, 3, 0, 0)
66.	error	play,1,0,fordm002,01,C>B,SB2.3-H(E2/TH) (NR)
	state after	(2, 0, 1, 0, 1, 4, 0, 0)
	fielder	2 (catcher on the home team)

Summary 410 This game had 3 errors. The final score was 14-2.

In the top of the 1st inning, with runners on 1st and 3rd, SEA batter Mike Ford had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). Ford is still at bat, now with a 1-1 count.

	game	SEA vs CHA on 20230821
	state before	(2, 1, 1, 0, 1, 4, 0, 0)
67.	error	play,1,0,rojaj001,00,X,S9/L9.2-H;1-3(E9)
	state after	(2, 1, 0, 1, 1, 5, 0, 0)
	fielder	9 (right field on the home team)

Summary 411 *In the top of the 1st inning, with runners on 1st and 2nd, SEA batter Josh Rojas hit a line drive single to right field. The runner on 2nd scored (1 RBI). A throwing error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	SEA vs CHA on 20230821
	state before	(2, 0, 1, 0, 9, 14, 1, 1)
68.	error	play,9,1,sosal001,31,BBFBX,E5/TH/G5.2-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 9, 14, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 412 *In the bottom of the 9th inning, with a runner on 2nd, CHA batter Lenyn Sosa hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	SEA vs CHA on 20230823
	state before	(0, 0, 1, 0, 10, 4, 4, 1)
69.	error	play,10,1,andre001,11,...L>B+2,SB3.2-H(E6/TH)(NR)(UR)
	state after	(0, 0, 0, 0, 10, 4, 5, 1)
	fielder	6 (3rd baseman on the visiting team)

Summary 413 *In the bottom of the 10th inning, with a runner on 2nd, CHA batter Elvis Andrus had a 1-1 count. When the catcher attempted a pickoff on 2nd, the runner on 2nd stole 3rd. A throwing error by the shortstop allowed that runner to score (unearned, no RBI). Andrus is still at bat, with a 1-1 count.*

	game	OAK vs CHA on 20230824
	state before	(1, 0, 0, 0, 5, 3, 2, 1)
70.	error	play,5,1,lee-k002,11,BFX,E1/TH/G1S.B-2
	state after	(1, 0, 1, 0, 5, 3, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 414 This game had 2 errors. The final score was 8-5. *In the bottom of the 5th inning, with bases empty, CHA batter Korey Lee hit a ground ball to very shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd base.*

	game	OAK vs CHA on 20230824
	state before	(2, 1, 0, 0, 7, 5, 5, 0)
71.	error	play,7,0,geloz001,12,S1C*B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 5, 5, 0)
	fielder	2 (catcher on the home team)

Summary 415 *In the top of the 7th inning, with a runner on 1st, OAK batter Zack Gelof had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Gelof is still at bat, now with a 2-2 count.*

Visiting/Home team rosters for OAK vs CHA on 2023-08-25

position	innings	player	position	innings	player
1	1	Zach Neal	1	1	Dylan Cease
1	6	Sean Newcomb	1	5	Tanner Banks
1	8	Francisco Perez	1	8	Bryan Shaw
2	1	Shea Langeliers	2	1	Korey Lee
3	1	Ryan Noda	3	1	Andrew Vaughn
4	1	Zack Gelof	4	1	Lenyn Sosa
5	1	Jonah Bride	5	1	Yoan Moncada
6	1	Nick Allen	6	1	Tim Anderson
7	1	Tony Kemp	7	1	Andrew Benintendi
8	1	Lawrence Butler	8	1	Luis Robert Jr.
9	1	Seth Brown	9	1	Oscar Colas
9	7	Esteury Ruiz			

	game	OAK vs CHA on 20230825
	state before	(2, 1, 1, 0, 2, 1, 0, 0)
72.	error	play,2,0,allen002,01,CX,S7/L78.2-H;1-3;B-2(E7)
	state after	(2, 0, 1, 1, 2, 2, 0, 0)
	fielder	7 (left field on the home team)

Summary 416 This game had 4 errors. The final score was 12-4.
In the top of the 2nd inning, with runners on 1st and 2nd, OAK batter Nick Allen hit a line drive single to left center field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. An error by the left fielder allowed the batter to reach 2nd.

	game	OAK vs CHA on 20230825
	state before	(2, 1, 0, 0, 2, 4, 0, 0)
73.	error	play,2,0,geloz001,00,X,D7/G5.1-H;B-3(E7/TH)
	state after	(2, 0, 0, 1, 2, 5, 0, 0)
	fielder	7 (left field on the home team)

Summary 417 *In the top of the 2nd inning, with a runner on 1st, OAK batter Zack Gelof hit a ground ball double to 3rd. The runner on 1st scored (1 RBI). A throwing error by the left fielder allowed the batter to reach 3rd.*

74.	game	OAK vs CHA on 20230825
	state before	(1, 0, 0, 0, 2, 5, 0, 1)
	error	play,2,1,vauga001,02,CC.FX,E4/G6MS.B-1
	state after	(1, 1, 0, 0, 2, 5, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 418 *In the bottom of the 2nd inning, with bases empty, CHA batter Andrew Vaughn hit a ground ball to shallow short. An error by the second baseman allowed the batter to reach 1st base.*

75.	game	OAK vs CHA on 20230825
	state before	(1, 1, 1, 0, 5, 8, 3, 0)
	error	play,5,0,nodar001,00,.X,6(1)/F0/G6M.2-H(E6/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 5, 9, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 419 *In the top of the 5th inning, with runners on 1st and 2nd, OAK batter Ryan Noda hit a ground ball to short. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

76.	game	OAK vs CHA on 20230826
	state before	(0, 1, 1, 0, 8, 0, 6, 0)
	error	play,8,0,perec003,22,.BCCBX,FC/G56+.2-H(NR) (UR) ;1-3(E5) ;B-1
	state after	(0, 1, 0, 1, 8, 1, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 420 *In the top of the 8th inning, with runners on 1st and 2nd, OAK batter Carlos Perez hit a hard ground ball into a fielder's choice at 3rd. The runner on 2nd scored (unearned, no RBI) while an error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

Visiting/Home team rosters for OAK vs CHA on 2023-08-27

position	innings	player	position	innings	player
1	1	Paul Blackburn	1	1	Mike Clevinger
1	7	Francisco Perez	1	8	Lane Ramsey
1	7	Spencer Patton	1	9	Tanner Banks
2	1	Carlos Perez	2	1	Yasmani Grandal
3	1	Ryan Noda	3	1	Andrew Vaughn
4	1	Zack Gelof	4	1	Lenyn Sosa
5	1	Jordan Diaz	5	1	Yoan Moncada
6	1	Aledmys Diaz	6	1	Tim Anderson
7	1	Tony Kemp	7	1	Andrew Benintendi
8	1	Esteury Ruiz	8	1	Oscar Colas
8	6	Lawrence Butler	8	8	Trayce Thompson
9	1	Seth Brown	9	1	Gavin Sheets
9	6	Esteury Ruiz	9	8	Oscar Colas

	game	OAK vs CHA on 20230827
	state before	(2, 1, 0, 0, 3, 0, 1, 1)
77.	error	play,3,1,vauga001,30,B1BBX,E5/TH/G5.1-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 3, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 421 This game had 3 errors. The final score was 1-6.
In the bottom of the 3rd inning, with a runner on 1st, CHA batter Andrew Vaughn hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 1st.

	game	OAK vs CHA on 20230827
	state before	(2, 1, 0, 1, 3, 0, 2, 1)
78.	error	play,3,1,sheeg001,32,C*B*BFB>B,W+WP.3-H(NR)(UR);1-3(E1)
	state after	(2, 1, 0, 1, 3, 0, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 422 *In the bottom of the 3rd inning, with runners on 1st and 3rd, CHA batter Gavin Sheets had a full count. During the next pitch, which was a wild pitch, the runner on 3rd scored (unearned, no RBI). An error by the pitcher allowed the runner on 1st to advance to 3rd. Sheets took his walk to 1st.*

	game	OAK vs CHA on 20230827
	state before	(0, 0, 0, 0, 6, 1, 2, 1)
79.	error	play,6,1,moncy001,00,...X,D9/L9LS+.B-3(E9)
	state after	(0, 0, 0, 1, 6, 1, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 423 *In the bottom of the 6th inning, with bases empty, CHA batter Yoan Moncada hit a hard line drive double to shallow right field. An error by the right fielder allowed the batter to reach 3rd.*

9.2.6 September

	game	DET vs CHA on 20230901
	state before	(1, 1, 0, 1, 6, 2, 1, 0)
80.	error	play,6,0,meadp001,10,.BX,4/F89S.3-H(E4/TH)(NR)
	state after	(2, 1, 0, 0, 6, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 424 *In the top of the 6th inning, with runners on 1st and 3rd, DET batter Parker Meadows hit a fly ball to shallow right center*

field. The batter was put out by the second baseman. A throwing error by the second baseman allowed the runner on 3rd to score (unearned, no RBI).

	game	DET vs CHA on 20230902
	state before	(2, 0, 0, 0, 1, 4, 0, 1)
81.	error	play,1,1,vauga001,11,BC.X,E5/G56.B-1
	state after	(2, 1, 0, 0, 1, 4, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 425 This game had 2 errors. The final score was 10-0.

In the bottom of the 1st inning, with bases empty, CHA batter Andrew Vaughn hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	DET vs CHA on 20230902
	state before	(1, 1, 0, 0, 6, 8, 0, 0)
82.	error	play,6,0,torks001,11,BSX,FC1/G13.1-3(E1/TH);B-1
	state after	(1, 1, 0, 1, 6, 8, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 426 *In the top of the 6th inning, with a runner on 1st, DET batter Spencer Torkelson hit a ground ball into a fielder's choice at shallow right infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

Visiting/Home team rosters for KCA vs CHA on 2023-09-12

position	innings	player	position	innings	player
1	1	Jordan Lyles	1	1	Touki Toussaint
1	6	Taylor Clarke	1	2	Luis Patino
1	7	Collin Snider	1	5	Tanner Banks
1	8	Angel Zerpa	1	7	Deivi Garcia
1	9	Carlos Hernandez	1	8	Aaron Bummer
2	1	Logan Porter	1	9	Michael Kopech
3	1	Matt Duffy	2	1	Yasmani Grandal
3	8	Nick Pratto	2	6	Carlos Perez
4	1	Michael Massey	3	1	Andrew Vaughn
5	1	Maikel Garcia	4	1	Lenyn Sosa
6	1	Bobby Witt Jr.	5	1	Yoan Moncada
7	1	Edward Olivares	6	1	Zach Remillard
7	8	Dairon Blanco	6	9	Elvis Andrus
8	1	Kyle Isbel	7	1	Trayce Thompson
9	1	Dairon Blanco	8	1	Luis Robert Jr.
9	8	Drew Waters	9	1	Gavin Sheets
			9	9	Zach Remillard

	game	KCA vs CHA on 20230912 (2/2)
	state before	(1, 1, 1, 1, 6, 9, 5, 1)
83.	error	play,6,1,sheeg001,32,B*BC*BSX,D8/F89XD.3-H;2-H;1-H;B-3(E8)
	state after	(1, 0, 0, 1, 6, 9, 8, 1)
	fielder	8 (center field on the visiting team)

Summary 427 This game 2 of a doubleheader had 4 errors. The final score was 11-10 (while 2-6 was the score of game 1).

In the bottom of the 6th inning, with bases loaded, CHA batter Gavin Sheets hit a fly ball double to extra deep right center field. The runners on 1st, 2nd, and 3rd all scored (3 RBIs). An error by the center fielder allowed the batter to reach 3rd. (By the way, the next batter scores

Sheets on a sacrifice fly, but it is an unearned run due to the E8 in this play.)

	game	KCA vs CHA on 20230912 (2/2)
	state before	(0, 0, 0, 0, 7, 10, 9, 1)
84.	error	play,7,1,moncy001,10,.BX,E3/G3.B-1
	state after	(0, 1, 0, 0, 7, 10, 9, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 428 *In the bottom of the 7th inning, with bases empty, CHA batter Yoan Moncada hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	KCA vs CHA on 20230912 (2/2)
	state before	(2, 1, 0, 0, 9, 11, 9, 0)
85.	error	play,9,0,peres002,00,1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 9, 11, 9, 0)
	fielder	2 (catcher on the home team)

Summary 429 *In the top of the 9th inning, with a runner on 1st, KCA batter Salvador Perez had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Perez is still at bat, now with a 1-0 count.*

	game	KCA vs CHA on 20230912 (2/2)
	state before	(2, 0, 0, 1, 9, 11, 9, 0)
86.	error	play,9,0,peres002,10,1>B.X,C/E2.B-1
	state after	(2, 1, 0, 1, 9, 11, 9, 0)
	fielder	2 (catcher on the home team)

Summary 430 *This play is a consecutive error with the same batter. In the top of the 9th inning, with a runner on 3rd, KCA batter Salvador Perez had a 1-0 count. Catcher's interference allowed the batter to take 1st.*

	game	KCA vs CHA on 20230913
	state before	(0, 1, 0, 1, 7, 5, 0, 1)
87.	error	play,7,1,grany001,10,*BX,FC6/G6.3-H;1-3(E6/TH);B-1
	state after	(0, 1, 0, 1, 7, 5, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 431 *In the bottom of the 7th inning, with runners on 1st and 3rd, CHA batter Yasmani Grandal hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	MIN vs CHA on 20230914
	state before	(1, 0, 0, 0, 9, 6, 2, 0)
88.	error	play,9,0,farmk001,02,CCF*S,K.B-1(E2/TH)
	state after	(1, 1, 0, 0, 9, 6, 2, 0)
	fielder	2 (catcher on the home team)

Summary 432 *In the top of the 9th inning, with bases empty, MIN batter Kyle Farmer has a 0-2 count. The next pitch was an uncaught swinging strike. A throwing error by the catcher allowed the batter to reach 1st.*

	game	MIN vs CHA on 20230915
	state before	(1, 0, 0, 0, 3, 4, 0, 0)
89.	error	play,3,0,corrc001,01,CX,E6/G6S.B-1
	state after	(1, 1, 0, 0, 3, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 433 *In the top of the 3rd inning, with bases empty, MIN batter Carlos Correa hit a ground ball to shallow short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIN vs CHA on 20230916
	state before	(0, 0, 0, 0, 1, 1, 0, 1)
90.	error	play,1,1,andet001,01,CX,S5/G5S.B-2(E5/TH)
	state after	(0, 0, 1, 0, 1, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 434 This game had 2 errors. The final score was 6-7.

In the bottom of the 1st inning, with bases empty, CHA batter Tim Anderson hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	MIN vs CHA on 20230916
	state before	(2, 0, 0, 1, 8, 4, 7, 0)
91.	error	play,8,0,wallm001,31,BBCBX,E3/G3.3-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 8, 5, 7, 0)
	fielder	3 (1st baseman on the home team)

Summary 435 *In the top of the 8th inning, with a runner on 3rd, MIN batter Matt Wallner hit a ground ball to 1st base. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	MIN vs CHA on 20230917
	state before	(0, 0, 0, 0, 5, 3, 0, 1)
92.	error	play,5,1,thomt002,20,BBX,E4/TH/G6M.B-1
	state after	(0, 1, 0, 0, 5, 3, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 436 This game had 2 errors. The final score was 4-0.

In the bottom of the 5th inning, with bases empty, CHA batter Trayce Thompson hit a ground ball to short. A throwing error by the second baseman allowed the batter to reach 1st base.

	game	MIN vs CHA on 20230917
	state before	(2, 0, 0, 0, 9, 4, 0, 0)
93.	error	play,9,0,jeffr002,22,B1>C.F.BX,E5/TH/G5.B-1
	state after	(2, 1, 0, 0, 9, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 437 *In the top of the 9th inning, with bases empty, MIN batter Ryan Jeffers hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	ARI vs CHA on 20230926
	state before	(1, 1, 1, 0, 3, 1, 4, 0)
94.	error	play,3,0,markk001,00,X,FC1/G13.2-H(NR)(UR);1-3(E1/TH);B-1
	state after	(1, 1, 0, 1, 3, 2, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 438 This game had 2 errors. The final score was 15-4. *In the top of the 3rd inning, with runners on 1st and 2nd, ARI batter Ketel Marte hit a ground ball into a fielder's choice at shallow right in-field. The runner from 2nd scored (unearned, no RBI) while a throwing error by the pitcher allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

	game	ARI vs CHA on 20230926
	state before	(0, 1, 0, 0, 5, 3, 4, 0)
95.	error	play,5,0,carrc005,01,FX,FC/G3.1X2(4E6);B-1
	state after	(0, 1, 1, 0, 5, 3, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 439 *In the top of the 5th inning, with a runner on 1st, ARI batter Corbin Carroll hit a ground ball into a fielder's choice at 1st. An error by the shortstop (with an assist from the second baseman) allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SDN vs CHA on 20230929
	state before	(2, 0, 0, 0, 8, 3, 0, 0)
96.	error	play,8,0,grist001,12,CF.FBX,E4/G34.B-1
	state after	(2, 1, 0, 0, 8, 3, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 440 *In the top of the 8th inning, with bases empty, SDN batter Trent Grisham hit a ground ball to 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	SDN vs CHA on 20231001
	state before	(2, 1, 0, 0, 1, 0, 0, 1)
97.	error	play,1,1,sheeg001,00,X,4E1/G34.1-2;B-1
	state after	(2, 1, 1, 0, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 441 *In the bottom of the 1st inning, with a runner on 1st, CHA batter Gavin Sheets hit a ground ball to the 1st/2nd base gap. A throwing error by the first baseman (with an assist from the second baseman) allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Chapter 10

Cincinnati Reds (CIN)

The Cincinnati Reds have competed in the National League since 1890, and have been in the NL Central Division since 1994. They have won 5 World Series titles, the last in 1990, and numerous NLCD titles, the last in 2012.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Cincinnati against the Reds. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 CIN team is 0.685 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the Reds is a little lower than the MLB average.

In CIN 2023, there were 91 errors in the 81 home games, 46 by the Reds and the rest (45) by visiting teams. The total number of errors by the Reds only (home and away) over the 2023 regular season was 91 (coincidentally the same as the total number of errors in the Reds' homegames by both sides). Since 83.9 was the MLB average, the Reds were a little above average for errors.

10.1 Season totals

The Official Scorer shall credit a **stolen base** to a runner whenever the runner advances one base unaided by a hit, a putout, an error, a force-out, a fielder's choice, a passed ball, a wild pitch or a balk, subject to the following:

(c) When a runner, attempting to steal, or after being picked off base, evades being put out in a run-down play and advances to the next base without the aid of an error, the Official Scorer shall credit the runner with a stolen base. If another runner also advances on the play, the Official Scorer shall credit both runners with stolen bases. If a runner advances while another runner, attempting to steal, evades being put out in a run-down play and returns safely, without the aid of an error, to the base he originally occupied, the Official Scorer shall credit a stolen base to the runner who advances.

– **MLB Rule Book 9.07(c)**

Throughout, *vt* denotes the visiting team and *ht* denotes the home team.

10.1.1 Run conversion and error total by position

This section tabulates the totals by position and by game state.

Runs and errors	totals
Errors by vt:	46
Runs by ht because of errors by vt:	9
Errors by ht:	45
Runs by vt because of errors by ht:	6

This is a good¹ error-to-run conversion rate: $9/46 = 19.5\%$ on the part of the Reds (compared to $6/45 = 13.3\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 CIN season (home and away) has about $R/PA = 783/6195$ (about 12.6%) change of resulting in a run. Therefore, a randomly selected error is (for the 2023 Reds) about as likely to result in a run than a randomly selected at-bat, in this case.

¹But not great - more common in 2023 MLB teams is a percentage in the 30-40% range.

Note that the total number of errors (by either team) in a home game is 91, so the expected number of errors (by either team) in a randomly selected CIN 2023 home game is about 1.12.

Errors by position	vt	ht
errors due to pitcher:	7	3
to catcher:	6	8
to 1st baseman:	2	3
to 2nd baseman:	6	7
to 3rd baseman:	11	7
to shortstop:	9	12
to left fielder:	1	0
to center fielder:	2	2
to right fielder:	2	3

10.1.2 Error total by state

The Official Scorer shall credit a **stolen base** to a runner whenever the runner advances one base unaided by a hit, a putout, an error, a force-out, a fielder's choice, a passed ball, a wild pitch or a balk, subject to the following:

(d) When a double- or triple-steal is attempted and one runner is thrown out before reaching and holding the base such runner is attempting to steal, no other runner shall be credited with a stolen base.

– **MLB Rule Book 9.07(d)**

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	20	12	32
(lone) runner on 1st	13	24	37
(lone) runner on 2nd	3	3	6
(lone) runner on 3rd	3	1	4
two runners on base	6	5	11
bases loaded	1	0	1
total	46	45	91

A table of frequencies of most common errors by the CIN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	4	8.9%
(1, 0, 0, 0)	2	4.4%
(2, 0, 0, 0)	6	13.3%
(0, 1, 0, 0)	8	17.8%
(1, 1, 0, 0)	9	20%
(2, 1, 0, 0)	7	15.6%
(1, 0, 1, 0)	3	6.7%
(0, 1, 1, 0)	2	4.4%
(1, 1, 1, 0)	2	4.4%

A table of frequencies of most common errors by a visiting team in CIN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	17.4%
(1, 0, 0, 0)	9	19.6%
(2, 0, 0, 0)	3	6.5%
(0, 1, 0, 0)	4	8.7%
(1, 1, 0, 0)	4	8.7%
(2, 1, 0, 0)	5	10.9%
(2, 0, 0, 1)	3	6.5%
(2, 1, 1, 0)	3	6.5%
(1, 1, 0, 1)	1	2.2%
(1, 1, 1, 1)	1	2.2%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Reds homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `CIN2023.evn`):

state	CIN2023 percentages
(0, 0, 0, 0)	24.1%
(1, 0, 0, 0)	16.3%
(2, 0, 0, 0)	13.4%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.8%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	4.1%
(0, 0, 0, 1)	0.3%
(1, 0, 0, 1)	1.0%
(2, 0, 0, 1)	1.5%
(0, 1, 1, 0)	1.0%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.6%

10.1.3 Some interesting errors

Some of the most interesting error plays in the CIN 2023 home games:

- The game WAS vs CIN on 2023-08-05 has 4 errors.
- The game CLE vs CIN on 2023-08-16 had 4 errors.
- The game TOR vs CIN on 2023-08-20 has 6 errors, which is the largest number in this team error analysis study of 2023 homegames.

10.1.4 Relative team rankings

We have 81 home games between CIN and some visiting MLB team (vt). A number of teams did not play CIN in the regular season. However, of those that did, CIN *lost* more games than won against²:

ATL (2/3), CHA (2/3), MIA (2/3), MIN (2/3), TBA (2/3), TOR (2/3)
 NYA (3/3), WAS (3/3), SLN (4/7), MIL (6/7)

and *won* more than lost against

ARI (0/3), COL (0/3), TEX (0/3)
 LAN (1/3), NYN (1/3), SDN (1/3), SEA (1/3),

The teams that *split* their series were:

CLE (1/2), PHI (2/4), SFN (2/4), CHN (3/6), PIT (3/6)

The teams that did not play in any of the 81 home games in CIN 2023 were:

ANA, BAL, BOS, DET, HOU, KCA, OAK.

10.2 Game-by-game totals

The Official Scorer shall credit a **stolen base** to a runner whenever the runner advances one base unaided by a hit, a putout, an error, a force-out, a fielder's choice, a passed ball, a wild pitch or a balk, subject to the following:

(g) The Official Scorer shall not score a stolen base when a runner advances solely because of the defensive team's indifference to the runner's advance. The Official Scorer shall score such a play as a fielder's choice.

– **MLB Rule Book 9.07(g)**

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for CIN 2023 is as follows [BA].

²The vt's W/T record is given in parentheses: (W = numbers of homegames vt won against the Reds, T = Total home games vt vs Reds at CIN).

position	player	games played	OBP
1 (C)	Tyler Stephenson	92	.317
3 (1B)	Spencer Steer	73	.356
4 (2B)	Jonathan India	104	.338
5 (3B)	Nick Senzel	57	.297
6 (SS)	Elly De La Cruz	69	.300
7 (LF)	Will Benson	46	.321
8 (CF)	TJ Friedl	126	.352
9 (RF)	Jake Fraley	55	.330

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 CIN home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

As we will see below, there were 91 errors in the 81 home games, 45 by CIN and the rest (46) by visiting teams.

10.2.1 April

	game	PIT vs CIN on 20230330
	state before	(2, 1, 0, 0, 3, 1, 1, 1)
1.	error	play,3,1,stept001,02,CC>X,E4/G6S.1-2;B-1
	state after	(2, 1, 1, 0, 3, 1, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 442 *In the bottom of the 3rd inning, with a runner on 1st, CIN batter Tyler Stephenson hit a ground ball to shallow short. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

10.2.2 April

	game	PIT vs CIN on 20230401
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
2.	error	play,1,0,santc002,31,C1BBBX,S9/G4.1-3(E9)
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 443 *In the top of the 1st inning, with a runner on 1st, PIT batter Carlos Santana hit a ground ball single to 2nd. An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	CHN vs CIN on 20230403
	state before	(0, 1, 0, 0, 2, 3, 3, 1)
3.	error	play,2,1,friet001,11,CB.X,S1/BG13.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 2, 3, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 444 This game had 2 errors. The final score was 6-7. *In the bottom of the 2nd inning, with a runner on 1st, CIN batter TJ Friedl hit a bunt single to shallow right infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	CHN vs CIN on 20230403
	state before	(0, 1, 0, 0, 7, 6, 7, 0)
4.	error	play,7,0,manct001,32,B*BCS*B+11>F>X,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 7, 6, 7, 0)
	fielder	2 (catcher on the home team)

Summary 445 *In the top of the 7th inning, with a runner on 1st, CHN batter Trey Mancini had a full count. Catcher's interference allowed the batter to take 1st. The runner on 1st advanced to 2nd.*

	game	CHN vs CIN on 20230404
	state before	(2, 0, 0, 0, 5, 1, 3, 0)
5.	error	play,5,0,hoern001,10,BX,E6/G6.B-1
	state after	(2, 1, 0, 0, 5, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 446 This game had 2 errors. The final score was 12-5.
In the top of the 5th inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	CHN vs CIN on 20230404
	state before	(0, 1, 1, 0, 7, 6, 3, 0)
6.	error	play,7,0,hosme001,21,BC*BX,FC4/G34.2-3;1-2(E4);B-1
	state after	(0, 1, 1, 1, 7, 6, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 447 *In the top of the 7th inning, with runners on 1st and 2nd, CHN batter Eric Hosmer hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	PHI vs CIN on 20230413
	state before	(0, 0, 1, 0, 1, 1, 0, 1)
7.	error	play,1,1,stees001,20,BB>B,SB3.2-H(E5)(NR)
	state after	(0, 0, 0, 0, 1, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 448 *In the bottom of the 1st inning, with a runner on 2nd, CIN batter Spencer Steer had a 2-0 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. An error by the third baseman allowed that runner to score (unearned, no RBI). Steer is still at bat, now with a 3-0 count.*

	game	PHI vs CIN on 20230414
	state before	(1, 0, 0, 0, 9, 8, 2, 1)
8.	error	play,9,1,senzn001,12,CFBX,E5/TH/G6.B-2
	state after	(1, 0, 1, 0, 9, 8, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 449 *In the bottom of the 9th inning, with bases empty, CIN batter Nick Senzel hit a ground ball to short. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	PHI vs CIN on 20230415
	state before	(2, 1, 0, 0, 5, 0, 5, 0)
9.	error	play,5,0,turtn001,20,BBX,FC/G6.1-3(E6);BX2(34)
	state after	(0, 0, 0, 0, 5, 0, 5, 1)
	fielder	6 (shortstop on the home team)

Summary 450 *In the top of the 5th inning, with a runner on 1st, PHI batter Trea Turner hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 3rd. The batter attempted to reach 2nd but was tagged or by the second baseman (with an assist from the first baseman). As this is the 3rd out, the half-inning is over.*

	game	PHI vs CIN on 20230416
	state before	(2, 0, 0, 1, 3, 10, 0, 1)
10.	error	play,3,1,fralj001,21,BFBX,E3/G9LD.3-H(NR)(UR);B-3
	state after	(2, 0, 0, 1, 3, 10, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 451 *This game had 2 errors. The final score was 14-3. In the bottom of the 3rd inning, with a runner on 3rd, CIN batter Jake Fraley hit a ground ball to deep right field. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 3rd.*

11.	game	PHI vs CIN on 20230416
	state before	(1, 0, 1, 0, 5, 11, 3, 0)
	error	play,5,0,bohama001,22,BFBFX,S9/L9S.2-H(E9)(NR)
	state after	(1, 1, 0, 0, 5, 12, 3, 0)
	fielder	9 (right field on the home team)

Summary 452 *In the top of the 5th inning, with a runner on 2nd, PHI batter Alec Bohm hit a line drive single to shallow right field. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

12.	game	TBA vs CIN on 20230417
	state before	(1, 0, 0, 0, 2, 0, 1, 1)
	error	play,2,1,senzn001,12,FFFBX,E6/G6.B-1
	state after	(1, 1, 0, 0, 2, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 453 This game had 2 errors. The final score was 1-8. *In the bottom of the 2nd inning, with bases empty, CIN batter Nick Senzel hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

13.	game	TBA vs CIN on 20230417
	state before	(1, 1, 1, 0, 6, 0, 4, 0)
	error	play,6,0,parei001,10,BX,FC4/G4.2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 6, 0, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 454 *In the top of the 6th inning, with runners on 1st and 2nd, TBA batter Isaac Paredes hit a ground ball into a fielder's choice at 2nd. The runner on 2nd to advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

14.	game	TEX vs CIN on 20230424
	state before	(2, 0, 0, 1, 1, 0, 0, 1)
	error	play,1,1,stept001,10,BX,E5/TH/G56S.3-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 1, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 455 *In the bottom of the 1st inning, with a runner on 3rd, CIN batter Tyler Stephenson hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

15.	game	TEX vs CIN on 20230425
	state before	(0, 0, 0, 0, 7, 6, 0, 1)
	error	play,7,1,ramoh001,11,CBX,S6/G56.B-2(E6/TH)
	state after	(0, 0, 1, 0, 7, 6, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 456 *In the bottom of the 7th inning, with bases empty, CIN batter Henry Ramos hit a ground ball single to left short. A throwing error by the shortstop allowed the batter to reach 2nd.*

16.	game	TEX vs CIN on 20230426
	state before	(1, 0, 0, 0, 7, 2, 3, 1)
	error	play,7,1,senzn001,11,SBX,E5/G7LS.B-2
	state after	(1, 0, 1, 0, 7, 2, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 457 *This game had 2 errors. The final score was 3-5. In the bottom of the 7th inning, with bases empty, CIN batter Nick Senzel hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 2nd base.*

	game	TEX vs CIN on 20230426
	state before	(0, 0, 0, 0, 9, 3, 3, 1)
17.	error	play,9,1,newmk001,00,...X,S6/G6.B-2(E6/TH)
	state after	(0, 0, 1, 0, 9, 3, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 458 *In the bottom of the 9th inning, with bases empty, CIN batter Kevin Newman hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

10.2.3 May

	game	CHA vs CIN on 20230505
	state before	(2, 0, 0, 0, 7, 5, 4, 1)
18.	error	play,7,1,maill001,20,BBX,E2/P23.B-1
	state after	(2, 1, 0, 0, 7, 5, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 459 *In the bottom of the 7th inning, with bases empty, CIN batter Luke Maile hit a pop up to shallow right infield. An error by the catcher allowed the batter to reach 1st base.*

	game	CHA vs CIN on 20230507
	state before	(2, 0, 0, 0, 3, 11, 1, 0)
19.	error	play,3,0,benia002,22,CBBSX,E3/G3.B-1
	state after	(2, 1, 0, 0, 3, 11, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 460 This game had 2 errors. The final score was 17-4. *In the top of the 3rd inning, with bases empty, CHA batter Andrew Benintendi hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	CHA vs CIN on 20230507
	state before	(0, 0, 0, 0, 9, 17, 4, 1)
20.	error	play,9,1,fairs001,01,.FX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 9, 17, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 461 *In the bottom of the 9th inning, with bases empty, CIN batter Stuart Fairchild hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	NYN vs CIN on 20230510
	state before	(1, 1, 0, 0, 9, 2, 1, 0)
21.	error	play,9,0,mcnej002,12,F1BFFX,FC4/G34.1-2(E4/TH);B-1
	state after	(1, 1, 1, 0, 9, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 462 *In the top of the 9th inning, with a runner on 1st, NYN batter Jeff McNeil hit a ground ball into a fielder's choice at 1st/2nd. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	NYN vs CIN on 20230511
	state before	(2, 0, 0, 0, 8, 0, 5, 0)
22.	error	play,8,0,batyb001,11,CBX,E3/G3.B-1
	state after	(2, 1, 0, 0, 8, 0, 5, 0)
	fielder	3 (1st baseman on the home team)

Summary 463 *In the top of the 8th inning, with bases empty, NYN batter Brett Baty hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

23.	game	NYA vs CIN on 20230519
	state before	(1, 1, 1, 0, 7, 3, 2, 0)
	error	play,7,0,higak001,01,S2,P02(E1/TH).2-3;1-2
	state after	(1, 0, 1, 1, 7, 3, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 464 *In the top of the 7th inning, with runners on 1st and 2nd, NYA batter Kyle Higashioka has a 0-1 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base. Higashioka is still at bat with a 0-1 count.*

24.	game	SLN vs CIN on 20230522
	state before	(0, 1, 0, 0, 5, 3, 4, 0)
	error	play,5,0,contw001,00,1,P01(E1/TH).1-2
	state after	(0, 0, 1, 0, 5, 3, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 465 *In the top of the 5th inning, with a runner on 1st, SLN batter Willson Contreras has a 0-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to safely advance to 2nd. Contreras is still at bat with a 0-0 count.*

25.	game	SLN vs CIN on 20230524
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
	error	play,1,1,stees001,12,BFCFX,S9/G34.1-3;B-2(E9)
	state after	(1, 0, 1, 1, 1, 0, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 466 *In the bottom of the 1st inning, with a runner on 1st, CIN batter Spencer Steer hit a ground ball single to 1st/2nd. The runner on 1st advanced to 3rd. An error by the right fielder allowed the batter to reach 2nd.*

	game	SLN vs CIN on 20230525
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
26.	error	play,1,1,friet001,02,CCX,E4/G4.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 467 *In the bottom of the 1st inning, with bases empty, CIN batter TJ Friedl hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

10.2.4 June

Visiting/Home team rosters for MIL vs CIN on 2023-06-02

position	innings	player	position	innings	player
1	1	Corbin Burnes	1	1	Brandon Williamson
1	10	Elvis Peguero	1	10	Alex Young
1	11	Bryse Wilson	1	10	Fernando Cruz
1	7	Joel Payamps	1	7	Buck Farmer
1	8	Peter Strzelecki	1	8	Lucas Sims
1	9	Devin Williams	1	9	Alexis Diaz
2	1	Victor Caratini	2	1	Luke Maile
3	1	Darin Ruf	2	10	Curt Casali
3	3	Rowdy Tellez	3	1	Kevin Newman
4	1	Owen Miller	3	10	Spencer Steer
4	7	Andruw Monasterio	4	1	Jonathan India
5	1	Mike Brosseau	5	1	Nick Senzel
5	7	Owen Miller	5	10	Kevin Newman
6	1	Andruw Monasterio	6	1	Matt McLain
6	7	Brice Turang	7	1	Will Benson
7	1	Christian Yelich	7	10	Nick Senzel
8	1	Joey Wiemer	8	1	Stuart Fairchild
9	1	Brian Anderson	9	1	Jake Fraley

	game	MIL vs CIN on 20230602
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
27.	error	play,2,0,anded006,11,T*BN,OA.1-2(E2)
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 468 This game had 3 errors. The final score was 5-4. *In the top of the 2nd inning, with a runner on 1st, MIL batter Brian Anderson had a 1-1 count. An error by the catcher allowed the runner on 1st to advance to 2nd. Anderson is still at bat with a 1-1 count.*

	game	MIL vs CIN on 20230602
	state before	(1, 1, 0, 0, 4, 2, 3, 0)
28.	error	play,4,0,monaa001,11,*BC1X,FC6/G6M.1-3(E6/TH);B-2
	state after	(1, 0, 1, 1, 4, 2, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 469 *In the top of the 4th inning, with a runner on 1st, MIL batter Andruw Monasterio hit a ground ball into a fielder's choice at short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	MIL vs CIN on 20230602
	state before	(1, 0, 0, 0, 9, 3, 3, 1)
29.	error	play,9,1,bensw001,00,X,E5/G56S.B-1
	state after	(1, 1, 0, 0, 9, 3, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 470 *In the bottom of the 9th inning, with bases empty, CIN batter Will Benson hit a ground ball to shallow 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	LAN vs CIN on 20230606
	state before	(2, 1, 1, 0, 5, 8, 3, 1)
30.	error	play,5,1,stept001,11,C.*BX,E5/TH/G5.2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 5, 8, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 471 This game had 2 errors. The final score was 8-9.

In the bottom of the 5th inning, with runners on 1st and 2nd, CIN batter Tyler Stephenson hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	LAN vs CIN on 20230606
	state before	(2, 1, 0, 0, 6, 8, 4, 1)
31.	error	play,6,1,mclam001,02,CS1X,S8/F8S.1-3;B-2(E8/TH)
	state after	(2, 0, 1, 1, 6, 8, 4, 1)
	fielder	8 (center field on the visiting team)

Summary 472 *In the bottom of the 6th inning, with a runner on 1st, CIN batter Matt McLain hit a fly ball single to shallow center field. The runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

	game	LAN vs CIN on 20230607
	state before	(0, 0, 0, 0, 3, 3, 2, 0)
32.	error	play,3,0,freef001,12,CCBX,S8/L8D.B-2(E8)
	state after	(0, 0, 1, 0, 3, 3, 2, 0)
	fielder	8 (center field on the home team)

Summary 473 *In the top of the 3rd inning, with bases empty, LAN batter Freddie Freeman hit a line drive single to deep center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	COL vs CIN on 20230619
	state before	(1, 0, 0, 0, 6, 4, 3, 1)
33.	error	play,6,1,indij001,22,BCBC.X,E5/G56.B-1
	state after	(1, 1, 0, 0, 6, 4, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 474 *In the bottom of the 6th inning, with bases empty, CIN batter Jonathan India hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for COL vs CIN on 2023-06-20

position	innings	player	position	innings	player
1	1	Noah Davis	1	1	Ben Lively
1	5	Peter Lambert	1	5	Alex Young
1	8	Brent Suter	1	6	Fernando Cruz
2	1	Elias Diaz	1	7	Daniel Duarte
3	1	Mike Moustakas	1	8	Casey Legumina
4	1	Coco Montes	1	9	Alexis Diaz
5	1	Ryan McMahon	2	1	Curt Casali
6	1	Harold Castro	2	9	Luke Maile
7	1	Jurickson Profar	3	1	Spencer Steer
8	1	Brenton Doyle	4	1	Jonathan India
9	1	Nolan Jones	5	1	Elly De La Cruz
			6	1	Matt McLain
			7	1	Will Benson
			7	9	Nick Senzel
			8	1	TJ Friedl
			9	1	Jake Fraley

	game	COL vs CIN on 20230620
	state before	(1, 0, 0, 0, 5, 2, 5, 1)
34.	error	play,5,1,dela003,12,CBCX,S1/G1S+.B-2(E1/TH)
	state after	(1, 0, 1, 0, 5, 2, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 475 This game had 3 errors, 2 from pickoff attempts. The final score was 6-8.

In the bottom of the 5th inning, with bases empty, CIN batter Elly De La Cruz hit a hard ground ball single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.

	game	COL vs CIN on 20230620
	state before	(1, 0, 1, 0, 5, 2, 5, 1)
35.	error	play,5,1,fralj001,00,2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 5, 2, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 476 *In the bottom of the 5th inning, with a runner on 2nd, CIN batter Jake Fraley has a 0-0 count. The pitcher attempted to pickoff the runner on 2nd. An error by the pitcher allowed the runner on 2nd to advance to 3rd. Fraley is still at bat with a 0-0 count.*

	game	COL vs CIN on 20230620
	state before	(2, 0, 1, 0, 5, 2, 6, 1)
36.	error	play,5,1,stees001,10,B2,P02(E1/TH).2-H(NR)(UR)
	state after	(2, 0, 0, 0, 5, 2, 7, 1)
	fielder	1 (pitcher on the visiting team)

Summary 477 *In the bottom of the 5th inning, with a runner on 2nd, CIN batter Spencer Steer has a 1-0 count. The pitcher attempted (for the 2nd time this inning) to pickoff the runner on 2nd. An error by the pitcher allowed the runner on 2nd to score (unearned, no RBI). Steer is still at bat with a 1-0 count.*

	game	COL vs CIN on 20230621
	state before	(2, 1, 0, 0, 8, 3, 5, 1)
37.	error	play,8,1,bensw001,20,.11BB1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 8, 3, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 478 *In the bottom of the 8th inning, with a runner on 1st, CIN batter Will Benson has a 2-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed the runner on 1st to advance to 2nd. Benson is still at bat with a 2-0 count.*

	game	ATL vs CIN on 20230623
	state before	(2, 1, 0, 0, 4, 5, 4, 0)
38.	error	play,4,0,albio001,00,1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 5, 4, 0)
	fielder	2 (catcher on the home team)

Summary 479 *In the top of the 4th inning, with a runner on 1st, ATL batter Ozzie Albies had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Albies is still at bat, now with a 1-0 count.*

	game	ATL vs CIN on 20230624
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
39.	error	play,1,0,albio001,32,BSFF1BB>X,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 480 This game had 2 errors. The final score was 7-6. *In the top of the 1st inning, with a runner on 1st, ATL batter Ozzie Albies had a full count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced one base.*

	game	ATL vs CIN on 20230624
	state before	(2, 1, 0, 0, 5, 6, 4, 1)
40.	error	play,5,1,stees001,12,CB1S>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 6, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 481 *In the bottom of the 5th inning, with a runner on 1st, CIN batter Spencer Steer had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Steer is still at bat, now with a 2-2 count.*

	game	ATL vs CIN on 20230625
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
41.	error	play,1,0,rilea001,32,BBCCX,E5/TH/G56.B-2
	state after	(2, 0, 1, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 482 *In the top of the 1st inning, with bases empty, ATL batter Austin Riley hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	SDN vs CIN on 20230630
	state before	(1, 0, 0, 0, 3, 0, 0, 1)
42.	error	play,3,1,stept001,00,X,E4/L8.B-1
	state after	(1, 1, 0, 0, 3, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 483 This game had 2 errors. The final score was 5-7 in extra innings.

In the bottom of the 3rd inning, with bases empty, CIN batter Tyler Stephenson hit a line drive to center field. An error by the second baseman allowed the batter to reach 1st base.

	game	SDN vs CIN on 20230630
	state before	(2, 0, 0, 0, 6, 1, 0, 1)
43.	error	play,6,1,delae003,11,BCX,E8/F78D+.B-2
	state after	(2, 0, 1, 0, 6, 1, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 484 *In the bottom of the 6th inning, with bases empty, CIN batter Elly De La Cruz hit a hard fly ball to deep left center field. An error by the center fielder allowed the batter to reach 2nd base.*

10.2.5 July

	game	SDN vs CIN on 20230701
	state before	(0, 1, 0, 0, 6, 1, 1, 0)
44.	error	play,6,0,kim-h002,20,.BBX,FC6/G6.1-2(E6);B-1
	state after	(0, 1, 1, 0, 6, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 485 This game had 2 errors. The final score was 12-5.
In the top of the 6th inning, with a runner on 1st, SDN batter Ha-Seong Kim hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	SDN vs CIN on 20230701
	state before	(1, 0, 0, 0, 7, 7, 1, 0)
45.	error	play,7,0,kim-h002,10,BX,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 7, 7, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 486 *In the top of the 7th inning, with bases empty, SDN batter Ha-Seong Kim hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	SDN vs CIN on 20230702
	state before	(0, 0, 0, 0, 3, 0, 0, 1)
46.	error	play,3,1,maill001,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 487 *In the bottom of the 3rd inning, with bases empty, CIN batter Luke Maile hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs CIN on 20230715
	state before	(1, 1, 0, 0, 8, 3, 0, 0)
47.	error	play,8,0,adamw002,11,SBX,E5/G6S.1-2;B-1
	state after	(1, 1, 1, 0, 8, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 488 *In the top of the 8th inning, with a runner on 1st, MIL batter Willy Adames hit a ground ball to short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SFN vs CIN on 20230718
	state before	(1, 1, 1, 0, 3, 4, 4, 1)
48.	error	play,3,1,indij001,00,N,CS3(E1/TH).2-H(NR)(UR);1-3
	state after	(1, 0, 0, 1, 3, 4, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 489 This game had 2 errors. The final score was 11-10. *In the bottom of the 3rd inning, with runners on 1st and 2nd, CIN batter Jonathan India had a 0-0 count when the runner on 2nd attempted to steal 3rd. A throwing error by the pitcher allowed that runner to safely make 3rd base then score (unearned, no RBI), and the runner on 1st to advance to 3rd. India is still at bat with a 0-0 count.*

	game	SFN vs CIN on 20230718
	state before	(1, 1, 0, 0, 6, 7, 7, 0)
49.	error	play,6,0,matol002,11,CBX,FC6/G6.1X2(6E4);B-1
	state after	(1, 1, 1, 0, 6, 7, 7, 0)
	fielder	4 (2nd baseman on the home team)

Summary 490 *In the top of the 6th inning, with a runner on 1st, SFN batter Luis Matos hit a ground ball into a fielder's choice at short. An error by the second baseman (with an assist from the shortstop) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	ARI vs CIN on 20230721
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
50.	error	play,1,1,senzn001,11,BCX,E6/G6.B-1
	state after	(1, 1, 0, 0, 1, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 491 *In the bottom of the 1st inning, with bases empty, CIN batter Nick Senzel hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	ARI vs CIN on 20230722
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
51.	error	play,1,1,friet001,22,*BBC1SFFX,FC4/G4.1-2(E4);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 492 *In the bottom of the 1st inning, with a runner on 1st, CIN batter TJ Friedl hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	ARI vs CIN on 20230723
	state before	(0, 1, 0, 0, 5, 0, 3, 0)
52.	error	play,5,0,martk001,32,BBBC.CX,S3/G23.1-H(E3/TH)(NR);B-2
	state after	(0, 0, 1, 0, 5, 1, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 493 *In the top of the 5th inning, with a runner on 1st, ARI batter Ketel Marte hit a ground ball single to shallow 1st base. A throwing error by the first baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

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	game	WAS vs CIN on 20230804
	state before	(0, 1, 0, 0, 4, 0, 3, 1)
53.	error	play,4,1,newmk001,21,B1BSX,E4/G34.1-2;B-1
	state after	(0, 1, 1, 0, 4, 0, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 494 *In the bottom of the 4th inning, with a runner on 1st, CIN batter Kevin Newman hit a ground ball to 1st/2nd base. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for WAS vs CIN on 2023-08-05

position	innings	player	position	innings	player
1	1	Joan Adon	1	1	Andrew Abbott
1	7	Andres Machado	1	6	Derek Law
1	8	Jordan Weems	1	9	Daniel Duarte
1	9	Kyle Finnegan	2	1	Luke Maile
2	1	Riley Adams	2	9	Tyler Stephenson
3	1	Joey Meneses	3	1	Joey Votto
3	8	Dominic Smith	4	1	Matt McLain
4	1	Michael Chavis	5	1	Nick Senzel
5	1	Ildemaro Vargas	6	1	Elly De La Cruz
6	1	CJ Abrams	7	1	Spencer Steer
7	1	Stone Garrett	8	1	TJ Friedl
8	1	Alex Call	9	1	Will Benson
9	1	Lane Thomas			

	game	WAS vs CIN on 20230805
	state before	(0, 1, 0, 0, 4, 2, 0, 0)
54.	error	play,4,0,chavm001,11,*SB1X,S8/G6M.1-H(E8)(NR);B-2
	state after	(0, 0, 1, 0, 4, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 495 This game had 4 errors, including a double error. The final score was 7-3.

In the top of the 4th inning, with a runner on 1st, WAS batter Michael Chavis hit a ground ball single to deep short. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.

	game	WAS vs CIN on 20230805
	state before	(1, 0, 1, 0, 4, 4, 0, 0)
55.	error	play,4,0,menej001,00,>C,SB3.2-H(E2/TH) (NR)
	state after	(1, 0, 0, 0, 4, 5, 0, 0)
	fielder	2 (catcher on the home team)

Summary 496 *In the top of the 4th inning, with a runner on 2nd, WAS batter Joey Meneses had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI). Meneses is still at bat, now with a 0-1 count.*

	game	WAS vs CIN on 20230805
	state before	(2, 1, 0, 0, 8, 6, 3, 0)
56.	error	play,8,0,menej001,21,1FB*BX,FC6/G6.1-3(E6) (E6/TH) ;B-1
	state after	(2, 1, 0, 1, 8, 6, 3, 0)
	fielder	6,6 (shortstop on the home team)

Summary 497 This play contains 2 errors by the *same* fielder. *In the top of the 8th inning, with a runner on 1st, WAS batter Joey Meneses (the same batter in the previous error-play) hit a ground ball into a fielder's choice at short. An error by the shortstop, as well as a throwing error by the shortstop, allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	WAS vs CIN on 20230806
	state before	(1, 1, 0, 0, 6, 5, 3, 0)
57.	error	play,6,0,thoml002,00,>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 5, 3, 0)
	fielder	2 (catcher on the home team)

Summary 498 *In the top of the 6th inning, with a runner on 1st, WAS batter Lane Thomas had a 0-0 count. During the next pitch,*

which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Thomas is still at bat, now with a 1-0 count.

	game	MIA vs CIN on 20230807
	state before	(2, 0, 0, 0, 3, 1, 0, 0)
58.	error	play,3,0,bertj001,10,BX,E6/G6.B-1
	state after	(2, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 499 This game had 3 errors. The final score was 2-5.

In the top of the 3rd inning, with bases empty, MIA batter Jon Berti hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	MIA vs CIN on 20230807
	state before	(2, 0, 0, 1, 3, 1, 2, 1)
59.	error	play,3,1,stees001,02,FSFFF,FLE2
	state after	(2, 0, 0, 1, 3, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 500 This play is an *infinity play*, one that can be repeated over and over without change to the game state.

In the bottom of the 3rd inning, with a 0-2 count and a runner on 3rd, CIN batter Spencer Steer hit a pop fly foul behind home plate. An error by the catcher allowed the batter to remain safe. Steer is still at bat with a 0-2 count.

	game	MIA vs CIN on 20230807
	state before	(1, 1, 0, 0, 7, 1, 4, 1)
60.	error	play,7,1,stees001,01,C1>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 501 *In the bottom of the 7th inning, with a runner on 1st, CIN batter Spencer Steer had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Steer is still at bat, now with a 1-1 count.*

	game	MIA vs CIN on 20230808
	state before	(0, 0, 0, 0, 5, 1, 1, 1)
61.	error	play,5,1,fairs001,22,CBBS.X,T9/F9LS.B-H(E4/TH) (NR)
	state after	(0, 0, 0, 0, 5, 1, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 502 *In the bottom of the 5th inning, with bases empty, CIN batter Stuart Fairchild hit a fly ball to shallow right field. A throwing error by the second baseman allowed the batter to score (unearned, no RBI).*

	game	MIA vs CIN on 20230809
	state before	(0, 0, 0, 0, 8, 4, 4, 1)
62.	error	play,8,1,mclam001,00,.X,E6/G2.B-1
	state after	(0, 1, 0, 0, 8, 4, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 503 *In the bottom of the 8th inning, with bases empty, CIN batter Matt McLain hit a ground ball to the infield. An error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs CIN on 20230815
	state before	(1, 0, 1, 0, 7, 2, 0, 0)
63.	error	play,7,0,kwans001,12,1BC>S.FFFX,S9/L89.2-H;B-2(E9)
	state after	(1, 0, 1, 0, 7, 3, 0, 0)
	fielder	9 (right field on the home team)

Summary 504 *In the top of the 7th inning, with a runner on 2nd, CLE batter Steven Kwan hit a line drive single to right center field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for CLE vs CIN on 2023-08-16

position	innings	player	position	innings	player
1	1	Noah Syndergaard	1	1	Andrew Abbott
1	5	Daniel Norris	1	6	Fernando Cruz
1	6	Eli Morgan	1	7	Sam Moll
1	8	Sam Hentges	1	8	Derek Law
2	1	Cam Gallagher	1	9	Alan Busenitz
2	3	Bo Naylor	2	1	Luke Maile
3	1	Gabriel Arias	3	1	Joey Votto
4	1	Tyler Freeman	4	1	Matt McLain
5	1	Jose Ramirez	5	1	Spencer Steer
6	1	Brayan Rocchio	6	1	Elly De La Cruz
7	1	Steven Kwan	7	1	Stuart Fairchild
8	1	Myles Straw	8	1	TJ Friedl
9	1	Ramon Laureano	9	1	Will Benson

	game	CLE vs CIN on 20230816
	state before	(0, 0, 0, 0, 3, 1, 1, 1)
64.	error	play,3,1,mclam001,22,.CBFBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 505 This game had 4 errors. The final score was 2-7.
In the bottom of the 3rd inning, with bases empty, CIN batter Matt McLain hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	CLE vs CIN on 20230816
	state before	(1, 1, 1, 1, 5, 2, 5, 1)
65.	error	play,5,1,bensw001,01,.FX,3/G3.3XH(3E2)(NR)(UR);2-3;1-2
	state after	(2, 0, 1, 1, 5, 2, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 506 *In the bottom of the 5th inning, with bases loaded, CIN batter Will Benson hit a ground ball to 1st and was put out by the first baseman. An error by the catcher allowed the runner on 3rd to safely score (unearned, no RBI), and the runners on 1st and 2nd to each advance to one base.*

	game	CLE vs CIN on 20230816
	state before	(2, 1, 0, 0, 6, 2, 5, 1)
66.	error	play,6,1,stees001,00,X,FC5/G56.1-2(E5);B-1
	state after	(2, 1, 1, 0, 6, 2, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 507 *In the bottom of the 6th inning, with a runner on 1st, CIN batter Spencer Steer hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	CLE vs CIN on 20230816
	state before	(2, 1, 1, 0, 6, 2, 5, 1)
67.	error	play,6,1,vottj001,10,BX,S7/G6.2-H(UR);1-3(E7/TH)
	state after	(2, 1, 0, 1, 6, 2, 6, 1)
	fielder	7 (left field on the visiting team)

Summary 508 *In the bottom of the 6th inning, with runners on 1st and 2nd, CIN batter Joey Votto hit a ground ball single to left field. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

68.	game	TOR vs CIN on 20230819
	state before	(0, 1, 0, 0, 4, 3, 1, 1)
	error	play,4,1,delae003,12,*BCCFX,T9/L89D.1-H;B-H(E9/TH) (NR) (UR)
	state after	(0, 0, 0, 0, 4, 3, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 509 This game had 2 errors. The final score was 4-3.

In the bottom of the 4th inning, with a runner on 1st, CIN batter Elly De La Cruz hit a line drive triple to deep right center field. The runner on 1st scored (1 RBI). A throwing error by the right fielder allowed the batter to also score (unearned, no RBI).

69.	game	TOR vs CIN on 20230819
	state before	(1, 1, 0, 0, 9, 4, 3, 0)
	error	play,9,0,bichb001,32,BBBFF>S,K+SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 9, 4, 3, 0)
	fielder	2 (catcher on the home team)

Summary 510 *In the top of the 9th inning, with a runner on 1st, TOR batter Bo Bichette had a full count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

Visiting/Home team rosters for TOR vs CIN on 2023-08-20

position	innings	player	position	innings	player
1	1	Hyun Jin Ryu	1	1	Hunter Greene
1	6	Genesis Cabrera	1	4	Buck Farmer
1	7	Bowden Francis	1	6	Sam Moll
2	1	Danny Jansen	1	7	Derek Law
3	1	Vladimir Guerrero Jr.	1	9	Alex Young
3	4	Cavan Biggio	2	1	Luke Maile
4	1	Whit Merrifield	2	8	Tyler Stephenson
5	1	Matt Chapman	3	1	Christian Encarnacion-Strand
6	1	Bo Bichette	4	1	Matt McLain
6	8	Santiago Espinal	5	1	Noelvi Marte
7	1	Daulton Varsho	6	1	Elly De La Cruz
8	1	Kevin Kiermaier	7	1	Spencer Steer
9	1	George Springer	8	1	TJ Friedl
			9	1	Stuart Fairchild
			9	3	TJ Hopkins

	game	TOR vs CIN on 20230820
	state before	(1, 0, 0, 1, 1, 0, 0, 0)
70.	error	play,1,0,bel tb 001,32,SSBBBX,FC4/G4.3-H(E4)(NR)(UR);B-1
	state after	(1, 1, 0, 0, 1, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 511 This game had 6 errors, including a consecutive error. The final score was 10-3.

In the top of the 1st inning, with a runner on 3rd, TOR batter Brandon Belt hit a ground ball into a fielder's choice at 2nd. An error by the secon baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.

71.	game	TOR vs CIN on 20230820
	state before	(2, 1, 0, 0, 1, 1, 0, 0)
	error	play,1,0,sprig001,01,CX,FC5/G56.1-3(E5/TH);B-1
	state after	(2, 1, 0, 1, 1, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 512 *In the top of the 1st inning, with a runner on 1st, TOR batter George Springer hit a round ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

72.	game	TOR vs CIN on 20230820
	state before	(1, 1, 0, 1, 2, 5, 0, 1)
	error	play,2,1,martn004,32,*BBFS.BX,7/F78S.3-H(E5/TH)(NR)(UR);1-H(NR)(UR)
	state after	(2, 0, 0, 0, 2, 5, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 513 *In the bottom of the 2nd inning, with runners on 1st and 3rd, CIN batter Noelvi Marte hit a fly ball to shallow left center field. The left fielder put out the batter. A throwing error by the third baseman allowed the runners on 1st and 3rd to score (unearned, no RBI).*

73.	game	TOR vs CIN on 20230820
	state before	(2, 0, 0, 0, 2, 5, 2, 1)
	error	play,2,1,friet001,12,CCBX,E3/TH/G3.B-1
	state after	(2, 1, 0, 0, 2, 5, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 514 This is an example of a *consecutive error*, since Friedl bats right after Marte in the CIN lineup.

In the bottom of the 2nd inning, with runners on 1st and 3rd, CIN batter TJ Friedl hit a ground ball to 1st base. A throwing error by the first baseman allowed the batter to reach 1st base.

	game	TOR vs CIN on 20230820
	state before	(1, 1, 0, 0, 6, 9, 0, 0)
74.	error	play,6,0,varsd001,21,BBTX,56(1)/F0/G6.B-2(E6/TH)
	state after	(2, 0, 1, 0, 6, 9, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 515 *In the top of the 6th inning, with a runner on 1st, TOR batter Daulton Varsho hit a ground ball ball to short. The runner from 1st was forced out at 2nd by the shortstop (with an assist from the third baseman). A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	TOR vs CIN on 20230820
	state before	(0, 0, 0, 0, 8, 9, 1, 0)
75.	error	play,8,0,bichb001,11,.FBX,E6/G2.B-1
	state after	(0, 1, 0, 0, 8, 9, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 516 *In the top of the 8th inning, with bases empty, TOR batter Bo Bichette hit a ground ball to the infield. An error by the shortstop allowed the batter to reach 1st base.*

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	game	CHN vs CIN on 20230901
	state before	(1, 1, 0, 0, 9, 2, 2, 1)
76.	error	play,9,1,bensw001,32,.F1BF>FBB>S,K+SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 9, 2, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 517 *In the bottom of the 9th inning, with a runner on 1st, CIN batter Will Benson had a full count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	CHN vs CIN on 20230902
	state before	(2, 0, 0, 0, 5, 0, 0, 0)
77.	error	play,5,0,hoern001,31,FBBBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 5, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 518 This game had 2 errors. The final score was 1-2.

In the top of the 5th inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	CHN vs CIN on 20230902
	state before	(1, 1, 0, 0, 9, 1, 0, 0)
78.	error	play,9,0,gomey001,21,BBFX,FC4/G4.1X2(4E6);B-1
	state after	(1, 1, 1, 0, 9, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 519 *In the top of the 9th inning, with a runner on 1st, CHN batter Yan Gomes hit a ground ball into a fielder's choice at 2nd. An error by the shortstop (with an assist from the second baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	CHN vs CIN on 20230903
	state before	(0, 1, 1, 0, 8, 7, 5, 0)
79.	error	play,8,0,mastm001,12,.L*BLX,FC/SH/BG1S.2-3;1-2;B-1(E1/TH)
	state after	(0, 1, 1, 1, 8, 7, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 520 *In the top of the 8th inning, with runners on 1st and 2nd, CHN batter Miles Mastrobuoni hit a bunt to a fielder's choice³ at*

³Since the throw is made to 1st, it's not clear to me why this is scored as a fielder's choice. The MLB Rule Book definition has this phrase: "...The term is also used by scorers ...to account for the advance of a runner (other than by stolen base or error) while a fielder is attempting to put out another runner..." Seems to vaguely apply to this situation. Maybe that is why?

very shallow infield. The runners on 1st and 2nd each advanced one base. A throwing error by the pitcher allowed the batter to reach 1st. The bases are loaded.

	game	SEA vs CIN on 20230904
	state before	(1, 0, 0, 0, 3, 0, 5, 1)
80.	error	play,3,1,stept001,00,X,E5/TH/G56S.B-1
	state after	(1, 1, 0, 0, 3, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 521 This game had 2 errors. The final score was 3-6.
In the bottom of the 3rd inning, with bases empty, CIN batter Tyler Stephenson hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	SEA vs CIN on 20230904
	state before	(1, 1, 0, 0, 6, 2, 5, 1)
81.	error	play,6,1,friet001,00,X,FC/SH/BG1S.1-2;B-1(E1/TH)
	state after	(1, 1, 1, 0, 6, 2, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 522 This error is similar to that on 2023-09-03 against the Cubs. Similar questions apply.
In the bottom of the 6th inning, with a runner on 1st, CIN batter TJ Friedl hit a hit a bunt to a fielder's choice at very shallow infield. The runner on 1st advanced to 2nd. A throwing error by the pitcher allowed the batter to reach 1st.

	game	SEA vs CIN on 20230906
	state before	(1, 1, 0, 0, 3, 2, 1, 0)
82.	error	play,3,0,hernt002,00,X,FC5/G5.1-2(E5);B-1
	state after	(1, 1, 1, 0, 3, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 523 *In the top of the 3rd inning, with a runner on 1st, SEA batter Teoscar Hernandez hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SLN vs CIN on 20230908
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
83.	error	play,1,0,arenn001,21,SBBX,S5/G5.1-3(E5/TH)
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 524 *In the top of the 1st inning, with a runner on 1st, SLN batter Nolan Arenado hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	SLN vs CIN on 20230910
	state before	(0, 0, 0, 0, 4, 0, 2, 0)
84.	error	play,4,0,arenn001,00,X,E5/L56+.B-1
	state after	(0, 1, 0, 0, 4, 0, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 525 This game had 2 errors. The final score was 1-7. *In the top of the 4th inning, with bases empty, SLN batter Nolan Arenado hit a hard line drive to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SLN vs CIN on 20230910
	state before	(2, 0, 0, 0, 6, 1, 5, 1)
85.	error	play,6,1,indij001,32,CBBBCX,E6/G6.B-1
	state after	(2, 1, 0, 0, 6, 1, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 526 *In the bottom of the 6th inning, with bases empty, CIN batter Jonathan India hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIN vs CIN on 20230918
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
86.	error	play,2,0,farmk001,00,.X,E6/G6.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 527 *In the top of the 2nd inning, with bases empty, MIN batter Kyle Farmer hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIN vs CIN on 20230919
	state before	(2, 1, 0, 0, 5, 2, 0, 0)
87.	error	play,5,0,castw003,12,CBF1FX,S4/G4.1-3(E4/TH)
	state after	(2, 1, 0, 1, 5, 2, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 528 *In the top of the 5th inning, with a runner on 1st, MIN batter Willi Castro hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	MIN vs CIN on 20230920
	state before	(0, 1, 0, 0, 9, 2, 3, 0)
88.	error	play,9,0,farmk001,00,>S,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 2, 3, 0)
	fielder	2 (catcher on the home team)

Summary 529 *In the top of the 9th inning, with a runner on 1st, MIN batter Kyle Farmer had a 0-0 count. During the next pitch, which*

was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Farmer is still at bat, now with a 0-1 count.

	game	PIT vs CIN on 20230922
	state before	(0, 0, 0, 0, 7, 4, 5, 0)
89.	error	play,7,0,bae-j001,01,..CX,E4/G4.B-1
	state after	(0, 1, 0, 0, 7, 4, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 530 This game had 2 errors. The final score was 7-5.

In the top of the 7th inning, with bases empty, PIT batter Ji Hwan Bae hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.

	game	PIT vs CIN on 20230922
	state before	(1, 1, 0, 1, 7, 5, 5, 0)
90.	error	play,7,0,andum001,10,BX,46(1)/FO/G4.3-H(UR);B-2(E6/TH)
	state after	(2, 0, 1, 0, 7, 6, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 531 *In the top of the 7th inning, with runners on 1st and 3rd, PIT batter Miguel Andujar hit a ground ball to 2nd. The runner fro 1st was forced out at 2nd by the shortstop (with an assist from the second baseman). The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	PIT vs CIN on 20230924
	state before	(2, 1, 1, 0, 6, 2, 2, 1)
91.	error	play,6,1,stept001,10,BX,FC/G6M.2XH(62);1-2(E6);B-1
	state after	(0, 0, 0, 0, 7, 2, 2, 0)
	fielder	6 (shortstop on the visiting team)

Summary 532 *In the bottom of the 6th inning, with runners on 1st and 2nd, CIN batter Tyler Stephenson hit a ground ball into a fielder's choice at short. The runner from 2nd was tagged out at home by the catcher (with an assist from the shortstop). An error by the shortstop allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

Chapter 11

Cleveland Guardians (CLE)

Since 1994, the Cleveland Guardians compete in the American League Central Division of the MLB. Formerly the “Indians”, the team adopted the current name following the 2021 season. The team’s name references the Guardians of Traffic, eight Art Deco sculptures by Henry Hering on the city’s Hope Memorial Bridge.

While the Guardians won the ALCD title as recently as 2022 and 2024, the team’s last World Series championship was in 1948.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Cleveland against the Guardians. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder’s choice or an attempted pickoff.

The DER of the 2023 CLE team is 0.698 compared with the MLB average of 0.691 that year [BR], about a half a standard deviation better than average.

There were 78 errors in the 81 home games, 44 by the Guardians and the rest (34) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 83, almost identical to the MLB error total average of 83.9.

11.1 Season totals

The Official Scorer shall credit the batter with a run batted in for every run that scores by reason of the batter becoming a runner with the bases full (because of a base on balls, an award

of first base for being touched by a pitched ball or for interference or obstruction).

– **MLB Rule Book** 9.04(a)(2)

Throughout, `vt` denotes the visiting team and `ht` denotes the home team.

11.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	34
Runs by ht because of errors by vt:	13
Errors by ht:	44
Runs by vt because of errors by ht:	18

This is a very good error-to-run conversion rate: $13/34 = 38.2\ldots\%$ on the part of the Guardians (compared to $18/44 = 40.9\ldots\%$ on the part of the visiting teams, which is also very good). A typical *play* (not necessarily with an error) in the 2023 CLE season (home and away) has about a R/PA $662/6096 = 10.8\ldots\%$ chance of resulting in a run¹. Therefore, a randomly selected error is over *three times* as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	2	4
to catcher:	4	8
to 1st baseman:	1	2
to 2nd baseman:	4	6
to 3rd baseman:	6	9
to shortstop:	5	7
to left fielder:	2	2
to center fielder:	4	1
to right fielder:	4	4

¹Just compute Runs over Plate Appearances, R/PA, from [BR].

11.1.2 Error total by state

This section tabulates error data by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	9	14	23
(lone) runner on 1st	9	10	19
(lone) runner on 2nd	3	4	7
(lone) runner on 3rd	1	1	2
two runners on base	11	11	22
bases loaded	1	4	5
total	34	44	78

A table of frequencies of most common errors by the Guardians in CLE 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	9	20.5%
(1, 0, 0, 0)	3	6.8%
(2, 0, 0, 0)	2	4.6%
(0, 1, 0, 0)	4	9.1%
(1, 1, 0, 0)	2	4.6%
(2, 1, 0, 0)	4	9.1%
(2, 0, 1, 0)	2	4.6%
(1, 1, 1, 0)	4	9.1%
(1, 1, 1, 1)	3	6.8%

A table of frequencies of most common errors by a visiting team in CLE 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	17.6%
(1, 0, 0, 0)	3	8.8%
(2, 0, 0, 0)	0	0.0%
(0, 1, 0, 0)	7	20.6%
(1, 1, 0, 0)	0	0.0%
(2, 1, 0, 0)	2	5.9%
(0, 0, 1, 0)	2	5.9%
(2, 1, 1, 0)	2	5.9%
(0, 1, 0, 1)	4	11.8%
(1, 1, 0, 1)	3	6.9%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Guardians homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `CLE2023.eva`):

state	CLE2023 percentages
(0, 0, 0, 0)	24.0%
(1, 0, 0, 0)	16.0%
(2, 0, 0, 0)	13.0%
(0, 1, 0, 0)	5.6%
(1, 1, 0, 0)	6.6%
(2, 1, 0, 0)	6.6%
(0, 0, 1, 0)	2.7%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	4.0%
(0, 0, 0, 1)	0.4%
(1, 0, 0, 1)	1.1%
(2, 0, 0, 1)	1.5%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.1%
(2, 1, 1, 0)	2.9%

11.1.3 Some interesting errors

Some of the most interesting error plays in the CLE 2023 home games:

- The game BOS vs CLE on 2023-06-07 had 4 errors.
- The game BAL vs CLE on 2023-09-22 had 5 errors.

11.1.4 Relative team rankings

We have 81 home games between TBA and some visiting MLB team (vt). A number of teams did not play TBA in the regular season. However, of those

that did, CLE *lost* more games than won against²:

ATL (2/3), COL (2/3), LAN (2/3), MIA (2/3), MIL (2/3),
 NYA (2/3), SEA (2/3), CHA (4/6), DET (5/7) .

The teams that *split* their series were:

CIN (1/2), BAL (2/4), TOR (2/4), MIN (3/6) .

The teams that TBA *won* more than lost against

TEX (0/3), ANA (1/3), BOS (1/3), HOU (1/3),
 OAK (1/3), PHI (1/3), SLN (1/3), TBA (1/3), KCA (2/7),

The teams that did not play in any of the 81 home games in CLE 2023 were:

ARI, CHN, NYN, PIT SDN, SFN, WAS .

11.2 Game-by-game totals

The batter becomes a runner and is entitled to first base without liability to be put out (provided he advances to and touches first base) when the catcher or any fielder interferes with him.

– **MLB Rule Book** 6.01(c)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for CLE 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Bo Naylor	68	.339
3 (1B)	Josh Naylor	91	.354
4 (2B)	Andrés Gimenez	151	.314
5 (3B)	José Ramírez	125	.306
6 (SS)	Amed Rosario	88	.356
7 (LF)	Steven Kwan	154	.340
8 (CF)	Myles Straw	145	.301
9 (RF)	Will Brennan	111	.299

²We recorded that vt's W/T record in parentheses (W = numbers of games vt won against TBA, T = Total home games vt played vs the Rays at Tampa Bay.

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 Guardians home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Rel].

11.2.1 April

	game	SEA vs CLE on 20230409
	state before	(1, 0, 0, 0, 3, 2, 1, 0)
1.	error	play,3,0,frant002,11,BSX,E6/G6.B-1
	state after	(1, 1, 0, 0, 3, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 533 This game had 2 errors. The final score was 6-7.
In the top of the 3rd inning, with bases empty, SEA batter Ty France hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.

	game	SEA vs CLE on 20230409
	state before	(0, 1, 1, 0, 12, 6, 5, 1)
2.	error	play,12,1,naylj001,12,SBF2,P02(E1/TH).2-3;1-2
	state after	(0, 0, 1, 1, 12, 6, 5, 1)
	fielder	1 (shortstop on the home team)

Summary 534 *In the bottom of the 12th inning, with runners on 1st and 2nd, CLE batter Josh Naylor has a 1-2 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base. Naylor is still at bat with a 1-2 count.*

	game	NYA vs CLE on 20230410
	state before	(0, 0, 0, 0, 2, 2, 0, 1)
3.	error	play,2,1,gimea001,10,BX,E1/BP34S-.B-1
	state after	(0, 1, 0, 0, 2, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 535 *In the bottom of the 2nd inning, with bases empty, CLE batter Andres Gimenez hit a soft bunt pop up to shallow right infield. An error by the pitcher allowed the batter to reach 1st base.*

	game	NYA vs CLE on 20230411
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
4.	error	play,1,0,judga001,00,X,FC5/G5D.1-2(E5);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 536 *In the top of the 1st inning, with a runner on 1st, NYA batter Aaron Judge hit a ground ball into a fielder's choice at deep 3rd. . An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for NYA vs CLE on 2023-04-12

position	innings	player	position	innings	player
1	1	Clarke Schmidt	1	1	Peyton Battenfield
1	5	Ron Marinaccio	1	5	Nick Sandlin
1	6	Michael King	1	7	Trevor Stephan
1	8	Wandy Peralta	1	8	James Karinchak
1	9	Clay Holmes	1	9	Emmanuel Clase
2	1	Kyle Higashioka	2	1	Cam Gallagher
3	1	Anthony Rizzo	3	1	Josh Bell
4	1	Oswaldo Cabrera	4	1	Andres Gimenez
4	9	Gleyber Torres	5	1	Jose Ramirez
5	1	Isiah Kiner-Falefa	6	1	Amed Rosario
6	1	Anthony Volpe	7	1	Steven Kwan
7	1	Franchy Cordero	8	1	Myles Straw
8	1	Aaron Hicks	9	1	Will Brennan
9	1	Giancarlo Stanton			
9	9	Oswaldo Cabrera			

	game	NYA vs CLE on 20230412
	state before	(1, 1, 1, 0, 5, 0, 3, 0)
5.	error	play,5,0,higak001,11,B*SX,S8/F8XD+.2-H;1-H(E4/TH) (NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 5, 2, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 537 This game had 3 errors. The final score was 4-3.

In the top of the 5th inning, with runners on 1st and 2nd, NYA batter Kyle Higashioka hit a hard fly ball single to extra deep center field. The runner on 2nd scored (1 RBI). A throwing error by the center field allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.

	game	NYA vs CLE on 20230412
	state before	(0, 0, 0, 0, 8, 3, 3, 1)
6.	error	play,8,1,ramij003,01,.CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 8, 3, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 538 *In the bottom of the 8th inning, with bases empty, CLE batter Jose Ramirez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	NYA vs CLE on 20230412
	state before	(1, 0, 0, 0, 9, 3, 3, 0)
7.	error	play,9,0,stanm004,02,CSX,S6/G6.B-2(E6/TH)
	state after	(1, 0, 1, 0, 9, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 539 *In the top of the 9th inning, with bases empty, NYA batter Giancarlo Stanton hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	MIA vs CLE on 20230422
	state before	(1, 1, 0, 1, 2, 1, 0, 0)
8.	error	play,2,0,stalj001,22,CB>FB1>X,E6/G6.3-H;1-2;B-1
	state after	(1, 1, 1, 0, 2, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 540 *In the top of the 2nd inning, with runners on 1st and 3rd, MIA batter Jacob Stallings hit a ground ball to short. The runner from 3rd scored (1 RBI). An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	MIA vs CLE on 20230423
	state before	(1, 1, 0, 1, 7, 1, 5, 1)
9.	error	play,7,1,brenw002,21,.FBBX,FC3/G4S.3XH(3E2/TH) (NR) (UR);1-3;B-2
	state after	(1, 0, 1, 1, 7, 1, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 541 *In the bottom of the 7th inning, with runners on 1st and 3rd, CLE batter Will Brennan hit a ground ball into a fielder's choice at shallow 2nd. A throwing error by the catcher (with an assist from the first baseman) allowed the runner on 3rd to safely score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	COL vs CLE on 20230424
	state before	(1, 1, 1, 1, 4, 1, 0, 0)
10.	error	play,4,0,tovae001,12,BFSFX,S7/L78.3-H;2-H;1-3(E7/TH);B-2
	state after	(1, 0, 1, 1, 4, 3, 0, 0)
	fielder	7 (left field on the home team)

Summary 542 *In the top of the 4th inning, with bases loaded, COL batter Ezequiel Tovar hit a line drive single to left center field. The runners from 2nd and 3rd both scored (2 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter reached 2nd.*

	game	COL vs CLE on 20230425
	state before	(0, 1, 0, 0, 6, 5, 0, 1)
11.	error	play,6,1,rosaa003,01,F>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 6, 5, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 543 *In the bottom of the 6th inning, with a runner on 1st, CLE batter Amed Rosario had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rosario is still at bat, with a 1-1 count.*

	game	COL vs CLE on 20230426
	state before	(1, 1, 1, 1, 7, 1, 3, 1)
12.	error	play,7,1,naylj001,10,BF,FLE2
	state after	(1, 1, 1, 1, 7, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 544 This game had an *infinity play*, one that does not change the game state. The final score was 1-4.
In the bottom of the 7th inning, with a 1-0 count and bases loaded, CLE batter Josh Naylor hit a foul fly ball behind home plate. An error by the catcher allowed the batter to remain at-bat. The bases are still loaded but Naylor now has a 1-1 count.

11.2.2 May

	game	MIN vs CLE on 20230505
	state before	(2, 1, 0, 0, 3, 0, 0, 1)
13.	error	play,3,1,kwans001,11,BC1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 3, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 545 *In the bottom of the 3rd inning, with a runner on 1st, CLE batter Steven Kwan has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed the runner on 1st to safely advance to 2nd. Kwan is still at bat with a 1-1 count.*

	game	MIN vs CLE on 20230506
	state before	(0, 1, 0, 1, 4, 0, 0, 1)
14.	error	play,4,1,naylj001,01,CX,S5/G6.3-H;1-3(E5/TH)
	state after	(0, 1, 0, 1, 4, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 546 *In the bottom of the 4th inning, with runners on 1st and 3rd, CLE batter Josh Naylor hit a ground ball single to 3rd base. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	DET vs CLE on 20230509
	state before	(1, 1, 1, 0, 2, 0, 1, 0)
15.	error	play,2,0,ibana001,22,CB*BFX,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 2, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 547 This game had 2 errors. The final score was 0-2. *In the top of the 2nd inning, with runners on 1st and 2nd, DET batter Andy Ibanez hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	DET vs CLE on 20230509
	state before	(0, 0, 0, 0, 5, 0, 1, 0)
16.	error	play,5,0,mckiz001,11,BCX,E4/G4.B-1
	state after	(0, 1, 0, 0, 5, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 548 *In the top of the 5th inning, with bases empty, DET batter Zach McKinstry hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	DET vs CLE on 20230510
	state before	(2, 1, 1, 0, 3, 3, 0, 0)
17.	error	play,3,0,badda001,10,BX,D9/G3.2-H;1-H(E9/TH)(NR)(UR);B-3
	state after	(2, 0, 0, 1, 3, 5, 0, 0)
	fielder	9 (right field on the home team)

Summary 549 *In the top of the 3rd inning, with runners on 1st and 2nd, DET batter Akil Baddoo hit a ground ball double to 1st. The runner on 2nd scored (1 RBI). A throwing error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	ANA vs CLE on 20230512
	state before	(0, 0, 0, 0, 5, 3, 2, 0)
18.	error	play,5,0,ohas001,32,CBFBBX,C/E2.B-1
	state after	(0, 1, 0, 0, 5, 3, 2, 0)
	fielder	2 (catcher on the home team)

Summary 550 This game had 2 errors. The final score was 5-4.
In the top of the 5th inning, with bases empty, ANA batter Shohei Ohtani had a full count. Catcher's interference allowed the batter to take 1st.

	game	ANA vs CLE on 20230512
	state before	(1, 1, 0, 1, 9, 3, 4, 0)
19.	error	play,9,0,renfh001,22,BBSSF.FF.>F>X,FC4/G4.3-H;1-3(E4/TH);B-1
	state after	(1, 1, 0, 1, 9, 4, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 551 *In the top of the 9th inning, with runners on 1st and 3rd, ANA batter Hunter Renfroe hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	ANA vs CLE on 20230513
	state before	(2, 0, 1, 0, 7, 3, 2, 0)
20.	error	play,7,0,renf1001,10,>B.X,S9/G34.2-H;B-2(E9/TH)
	state after	(2, 0, 1, 0, 7, 4, 2, 0)
	fielder	9 (right field on the home team)

Summary 552 *This game had 2 errors. The final score was 6-8. In the top of the 7th inning, with a runner on 2nd, ANA batter Luis Rengifo hit a ground ball single to the 1st/2nd base gap. The runner on 2nd scored (1 RBI). A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	ANA vs CLE on 20230513
	state before	(2, 0, 0, 0, 8, 4, 2, 0)
21.	error	play,8,0,thaim001,00,X,E6/P5D.B-1
	state after	(2, 1, 0, 0, 8, 4, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 553 *In the top of the 8th inning, with bases empty, ANA batter Matt Thaiss hit a pop up to deep 3rd. An error by the shortstop allowed the batter to reach 1st base.*

	game	ANA vs CLE on 20230514
	state before	(2, 1, 1, 0, 2, 0, 0, 1)
22.	error	play,2,1,brenw002,00,X,E6/G4.2-H(NR)(UR);1-3;BX2(46)
	state after	(0, 0, 0, 0, 3, 0, 1, 0)
	fielder	6 (shortstop on the visiting team)

Summary 554 *In the bottom of the 2nd inning, with runners on 1st and 2nd, CLE batter Will Brennan hit a ground ball to 2nd base. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd. However, the batter was tagged out attempting to reach 2nd.*

	game	CHA vs CLE on 20230523
	state before	(0, 0, 1, 1, 7, 1, 2, 0)
23.	error	play,7,0,grany001,22,BSFBX,E5/G56S.3-H;2-3;B-1
	state after	(0, 1, 0, 1, 7, 2, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 555 *In the top of the 7th inning, with runners on 2nd and 3rd, CHA batter Yasmani Grandal hit a ground ball to shallow 3rd base. An error by the third baseman allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	CHA vs CLE on 20230524
	state before	(2, 1, 0, 0, 3, 1, 0, 0)
24.	error	play,3,0,benia002,21,BCB1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 3, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 556 *In the top of the 3rd inning, with a runner on 1st, CHA batter Andrew Benintendi has a 2-1 count. The pitcher attempted to pick off the runner on 1st. An error by the pitcher allowed that runner to safely reach 2nd.*

	game	SLN vs CLE on 20230527
	state before	(0, 0, 0, 0, 8, 1, 1, 1)
25.	error	play,8,1,ramij003,22,.BCBFX,S9/G34.B-2(E9)
	state after	(0, 0, 1, 0, 8, 1, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 557 *In the bottom of the 8th inning, with bases empty, CLE batter Jose Ramirez hit a ground ball single to the 1st/2nd base gap. An error by the right fielder allowed the batter to reach 2nd.*

11.2.3 June

	game	BOS vs CLE on 20230606
	state before	(0, 0, 0, 0, 7, 1, 2, 0)
26.	error	play,7,0,reyep001,01,.CX,E6/G6.B-1
	state after	(0, 1, 0, 0, 7, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 558 *In the top of the 7th inning, with bases empty, BOS batter Pablo Reyes hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs CLE on 2023-06-07

position	innings	player	position	innings	player
1	1	Kutter Crawford	1	1	Tanner Bibee
1	4	Brennan Bernardino	1	6	Eli Morgan
1	5	Chris Murphy	1	7	Sam Hentges
1	5	Justin Garza	1	8	Trevor Stephan
2	1	Connor Wong	1	9	Emmanuel Clase
3	1	Triston Casas	2	1	Mike Zunino
4	1	Enmanuel Valdez	3	1	Josh Naylor
4	7	Pablo Reyes	4	1	Andres Gimenez
4	8	Christian Arroyo	5	1	Jose Ramirez
5	1	Rafael Devers	6	1	Amed Rosario
6	1	Enrique Hernandez	7	1	Steven Kwan
6	8	Pablo Reyes	8	1	Myles Straw
7	1	Masataka Yoshida	9	1	Will Brennan
8	1	Jarren Duran			
8	8	Enrique Hernandez			
9	1	Alex Verdugo			
9	8	Jarren Duran			

	game	BOS vs CLE on 20230607
	state before	(0, 0, 0, 0, 4, 1, 1, 0)
27.	error	play,4,0,herne001,31,BBBCX,S4/G4S.B-2(E4/TH)
	state after	(0, 0, 1, 0, 4, 1, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 559 This game had 4 errors. The final score was 2-5.

In the top of the 4th inning, with bases empty, BOS batter Enrique Hernandez hit a ground ball single to shallow 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd.

	game	BOS vs CLE on 20230607
	state before	(2, 0, 0, 1, 4, 1, 1, 0)
28.	error	play,4,0,valde002,12,BCFX,E3/G3.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 4, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 560 *In the top of the 4th inning, with a runner on 3rd, BOS batter Enmanuel Valdez hit a ground ball to 1st base. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	BOS vs CLE on 20230607
	state before	(0, 1, 0, 0, 4, 2, 1, 1)
29.	error	play,4,1,ramij003,12,C*BCFX,S34/G3.1-3(E4/TH);B-2
	state after	(0, 0, 1, 1, 4, 2, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 561 *In the bottom of the 4th inning, with a runner on 1st, CLE batter Jose Ramirez hit a ground ball single to 1st base. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	BOS vs CLE on 20230607
	state before	(0, 1, 0, 1, 4, 2, 2, 1)
30.	error	play,4,1,bellj005,00,.X,FC/G5.3-H(UR);1-2(E5);B-1
	state after	(0, 1, 1, 0, 4, 2, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 562 *In the bottom of the 4th inning, with runners on 1st and 3rd, CLE batter Josh Bell hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	HOU vs CLE on 20230609
	state before	(1, 1, 1, 0, 12, 7, 6, 1)
31.	error	play,12,1,brenw002,11,FBX,6(1)/F0/G6.2-H(E6/TH) (NR) (UR);B-1
	state after	(2, 1, 0, 0, 12, 7, 7, 1)
	fielder	6 (3rd baseman on the visiting team)

Summary 563 This game had 2 errors. The final score was 9-10.

In the bottom of the 12th inning, with runners on 1st and 2nd, CLE batter Will Brennan hit a ground ball to short. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	HOU vs CLE on 20230609
	state before	(1, 0, 1, 0, 13, 8, 7, 1)
32.	error	play,13,1,kwans001,11,BFX,S7/L78.2-H(UR);B-2(E7/TH)
	state after	(1, 0, 1, 0, 13, 8, 8, 1)
	fielder	7 (3rd baseman on the visiting team)

Summary 564 *In the bottom of the 13th inning, with a runner on 2nd, CLE batter Steven Kwan hit a line drive single to left center field. A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	HOU vs CLE on 20230610
	state before	(0, 1, 0, 0, 4, 3, 1, 0)
33.	error	play,4,0,meyej002,11,F1B>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 4, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 565 This game had 2 errors. The final score was 6-4.

In the top of the 4th inning, with a runner on 1st, HOU batter Jake Meyers had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Meyers is still at bat, now with a 2-1 count.

	game	HOU vs CLE on 20230610
	state before	(0, 1, 0, 0, 9, 6, 3, 1)
34.	error	play,9,1,rosaa003,12,CF*BX,FC5/G56.1-3(E5/TH);B-1
	state after	(0, 1, 0, 1, 9, 6, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 566 *In the bottom of the 9th inning, with a runner on 1st, CLE batter Amed Rosario hit a ground ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	HOU vs CLE on 20230611
	state before	(0, 0, 0, 0, 4, 0, 2, 0)
35.	error	play,4,0,julkc001,01,CX,E5/TH/G6.B-1
	state after	(0, 1, 0, 0, 4, 0, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 567 *In the top of the 4th inning, with bases empty, HOU batter Corey Julks hit a ground ball to short. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	HOU vs CLE on 20230611
	state before	(1, 0, 0, 0, 7, 0, 5, 0)
36.	error	play,7,0,diazy004,10,BX,E5/TH/G5.B-1
	state after	(1, 1, 0, 0, 7, 0, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 568 *In the top of the 7th inning, with bases empty, HOU batter Yainer Diaz hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs CLE on 20230620
	state before	(1, 1, 1, 0, 5, 1, 0, 0)
37.	error	play,5,0,ruize001,22,BF*BFX,C/E2.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 5, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 569 *In the top of the 5th inning, with runners on 1st and 2nd, OAK batter Esteury Ruiz had a 2-2 count. Catcher's interference allowed the batter to take 1st. The runners on 1st and 2nd each advanced one base, loading the bases.*

	game	OAK vs CLE on 20230621
	state before	(0, 1, 0, 1, 3, 4, 0, 1)
38.	error	play,3,1,rosaa003,22,SF1BB>X,S9/G34.3-H;1-H(E9) (NR)
	state after	(0, 1, 0, 0, 3, 4, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 570 This game had 2 errors. The final score was 6-7.
In the bottom of the 3rd inning, with runners on 1st and 3rd, CLE batter Amed Rosario hit a ground ball single to the 1st/2nd base gap. The runner on 3rd scored (1 RBI). An error by the right fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 1st.

	game	OAK vs CLE on 20230621
	state before	(1, 1, 0, 1, 8, 6, 5, 1)
39.	error	play,8,1,kwans001,12,*BCFX,D9/L9LS.3-H;1-H(E9/TH) (NR) (UR) ;B-3
	state after	(1, 0, 0, 1, 8, 6, 7, 1)
	fielder	9 (right field on the visiting team)

Summary 571 *In the bottom of the 8th inning, with runners on 1st and 3rd, CLE batter Steven Kwan hit a line drive double to shallow right field. A throwing error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	MIL vs CLE on 20230624
	state before	(1, 0, 1, 0, 1, 0, 0, 0)
40.	error	play,1,0,tellr001,11,BCX,E9/L89D+.2-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 1, 1, 0, 0)
	fielder	9 (right field on the home team)

Summary 572 This game had 2 errors. The final score was 2-4.

In the top of the 1st inning, with a runner on 2nd, MIL batter Rowdy Tellez hit a hard line drive to deep right center field. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.

	game	MIL vs CLE on 20230624
	state before	(0, 0, 1, 0, 8, 2, 3, 1)
41.	error	play,8,1,gimea001,12,.BCLX,S87/L78.2-H;B-2(E7)
	state after	(0, 0, 1, 0, 8, 2, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 573 *In the bottom of the 8th inning, with a runner on 2nd, CLE batter Andres Gimenez hit a line drive single to left center field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the batter to reach 2nd.*

	game	MIL vs CLE on 20230625
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
42.	error	play,1,1,rosaa003,01,SX,E4/F3D.BX2(46)
	state after	(2, 0, 0, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 574 *In the bottom of the 1st inning, with bases empty, CLE batter Amed Rosario hit a fly ball to deep 1st base. An error by the second baseman allowed the batter to reach 1st. The batter attempted to advance to 2nd, but was tagged out by the shortstop (with an assist from the second baseman).*

11.2.4 July

	game	ATL vs CLE on 20230704
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
43.	error	play,1,0,albio001,01,F1X,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 575 *In the top of the 1st inning, with a runner on 1st, ATL batter Ozzie Albies had a 0-1 count. Catcher's interference allowed the batter to take 1st and the runners on 1st base to take 2nd.*

	game	KCA vs CLE on 20230706
	state before	(2, 0, 0, 0, 5, 1, 1, 0)
44.	error	play,5,0,olive001,00,X,E6/G56.B-1
	state after	(2, 1, 0, 0, 5, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 576 This game had 2 errors. The final score was 1-6.
In the top of the 5th inning, with bases empty, KCA batter Edward Olivares hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.

	game	KCA vs CLE on 20230706
	state before	(1, 0, 0, 1, 6, 1, 3, 1)
45.	error	play,6,1,brenw002,12,>C.S*BFX,E4/G34.3-H(NR) (UR) ;B-1
	state after	(1, 1, 0, 0, 6, 1, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 577 *In the bottom of the 6th inning, with a runner on 3rd, CLE batter Will Brennan hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	KCA vs CLE on 20230708
	state before	(1, 1, 0, 0, 2, 0, 1, 0)
46.	error	play,2,0,massm001,22,CBFBX,6(1)/F0/G6.B-2(E6/TH)
	state after	(2, 0, 1, 0, 2, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 578 This game had 2 errors. The final score was 6-10.
In the top of the 2nd inning, with a runner on 1st, KCA batter Michael Massey hit a ground ball to short. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	KCA vs CLE on 20230708
	state before	(0, 1, 0, 1, 4, 0, 4, 1)
47.	error	play,4,1,rosaa003,01,1FX,FC5/G56.3-H;1-2(E5);B-1
	state after	(0, 1, 1, 0, 4, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 579 *In the bottom of the 4th inning, with runners on 1st and 3rd, CLE batter Amed Rosario hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the third baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	KCA vs CLE on 20230709
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
48.	error	play,1,1,kwans001,11,CBX,E4/TH/G4.B-2
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 580 *In the bottom of the 1st inning, with bases empty, CLE batter Steven Kwan hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd base.*

Visiting/Home team rosters for PHI vs CLE on 2023-07-23

position	innings	player	position	innings	player
1	1	Aaron Nola	1	1	Xzavion Curry
1	10	Jeff Hoffman	1	10	Trevor Stephan
1	10	Yunior Marte	1	4	Michael Kelly
1	8	Gregory Soto	1	5	Sam Hentges
1	9	Craig Kimbrel	1	6	Nick Sandlin
2	1	J.T. Realmuto	1	7	Eli Morgan
3	1	Bryce Harper	1	8	Enyel De Los Santos
4	1	Bryson Stott	1	9	Tim Herrin
5	1	Alec Bohm	2	1	Cam Gallagher
6	1	Trea Turner	2	8	Bo Naylor
7	1	Jake Cave	3	1	Josh Bell
8	1	Brandon Marsh	3	10	David Fry
9	1	Nick Castellanos	4	1	Andres Gimenez
			5	1	Jose Ramirez
			6	1	Tyler Freeman
			7	1	Steven Kwan
			8	1	Will Brennan
			8	10	Myles Straw
			9	1	David Fry
			9	10	Will Brennan

	game	PHI vs CLE on 20230723
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
49.	error	play,1,0,castn001,01,CX,S5/G56.1-3(E5/TH)
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 581 This game had 3 errors. The final score was 8-5.

In the top of the 1st inning, with a runner on 1st, PHI batter Nick Castellanos hit a ground ball single to 3rd base. A throwing error by

the third baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	PHI vs CLE on 20230723
	state before	(2, 0, 1, 0, 6, 3, 3, 0)
50.	error	play,6,0,marsb002,21,BFBX,E5/P5D.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 4, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 582 *In the top of the 6th inning, with a runner on 2nd, PHI batter Brandon Marsh hit a pop up to deep 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	PHI vs CLE on 20230723
	state before	(1, 0, 0, 0, 7, 4, 3, 1)
51.	error	play,7,1,fry-d001,01,CX,E5/TH/G56D.B-1
	state after	(1, 1, 0, 0, 7, 4, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 583 *In the bottom of the 7th inning, with bases empty, CLE batter David Fry hit a ground ball to deep 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	KCA vs CLE on 20230724
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
52.	error	play,1,1,kwans001,12,FFBX,6E3/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 584 This game had 2 errors. The final score was 5-3. *In the bottom of the 1st inning, with bases empty, CLE batter Steven Kwan hit a ground ball to short. An error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st base.*

	game	KCA vs CLE on 20230724
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
53.	error	play,2,0,duffm002,32,CBC.BBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 585 *In the top of the 2nd inning, with bases empty, KCA batter Matt Duffy hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	KCA vs CLE on 20230725
	state before	(1, 1, 0, 0, 5, 0, 1, 0)
54.	error	play,5,0,wated001,32,F1BBSB>S,K+SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 586 *In the top of the 5th inning, with a runner on 1st, KCA batter Drew Waters had a full count. During the next pitch, which was a swinging strike, the runner on 1st stole 3rd. A throwing error by the catcher allowed that runner to advance to 3rd. Waters, the batter, has struck out.*

11.2.5 August

	game	CHA vs CLE on 20230804
	state before	(2, 1, 0, 0, 2, 0, 1, 1)
55.	error	play,2,1,kwans001,01,C>*B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 587 *In the bottom of the 2nd inning, with a runner on 1st, CLE batter Steven Kwan had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Kwan is still at bat, now with a 1-1 count.*

	game	CHA vs CLE on 20230805
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
56.	error	play,1,1,gimea001,22,FBFBX,E7/L7+.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 588 This game had 2 errors. The final score was 5-3.

In the bottom of the 1st inning, with bases empty, CLE batter Andres Gimenez hit a hard line drive to left field. An error by the left fielder allowed the batter to reach 1st base.

	game	CHA vs CLE on 20230805
	state before	(2, 1, 1, 0, 8, 7, 2, 1)
57.	error	play,8,1,brenw002,00,X,S8/L78S.3-H;2-H;1-3(E8)
	state after	(2, 1, 0, 1, 8, 7, 4, 1)
	fielder	8 (center field on the visiting team)

Summary 589 *In the bottom of the 8th inning, with runners on 1st and 2nd, CLE batter Will Brennan hit a line drive single to left center field. The runners on 2nd and 3rd scored (2 RBIs). An error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	CHA vs CLE on 20230806
	state before	(0, 0, 1, 0, 5, 2, 2, 1)
58.	error	play,5,1,stram002,02,FFX,S16/G6.2-H(E6/TH)(NR)(UR)
	state after	(0, 1, 0, 0, 5, 2, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 590 This game had 2 errors. The final score was 5-3.

In the bottom of the 5th inning, with a runner on 2nd, CLE batter Myles Straw hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.

	game	CHA vs CLE on 20230806
	state before	(2, 1, 1, 1, 9, 2, 3, 0)
59.	error	play,9,0,remiz001,02,SCX,E5/TH/G56.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 9, 3, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 591 *In the top of the 9th inning, with bases loaded, CHA batter Zach Remillard hit a ground ball. A throwing error by the third baseman allowed the runners to each advance one base and the batter to reach 1st. In particular, the runner on 3rd scored (unearned, no RBI). The bases are still loaded.*

Visiting/Home team rosters for DET vs CLE on 2023-08-19

position	innings	player	position	innings	player
1	1	Matt Manning	1	1	Tanner Bibee
1	7	Alex Lange	1	6	Nick Sandlin
1	7	Will Vest	1	7	Tim Herrin
1	8	Andrew Vasquez	1	8	Trevor Stephan
1	9	Jason Foley	1	9	James Karinchak
2	1	Jake Rogers	2	1	Bo Naylor
3	1	Spencer Torkelson	3	1	Kole Calhoun
4	1	Zach McKinstry	4	1	Andres Gimenez
4	7	Zack Short	5	1	Jose Ramirez
5	1	Matt Vierling	6	1	Gabriel Arias
5	7	Zach McKinstry	7	1	Steven Kwan
6	1	Javier Baez	8	1	Myles Straw
7	1	Akil Baddoo	9	1	Will Brennan
7	7	Matt Vierling	9	9	Ramon Laureano
8	1	Riley Greene			
9	1	Kerry Carpenter			

	game	DET vs CLE on 20230819
	state before	(0, 1, 0, 0, 5, 3, 1, 1)
60.	error	play,5,1,naylb001,11,BFX,D8/L8XD+.1-H(E8/TH) (NR) (UR) ;B-3(E2)
	state after	(0, 0, 0, 1, 5, 3, 2, 1)
	fielder	8,2 (center field,catcher on the visiting team)

Summary 592 This game had 3 errors, including this *double error* play. The final score was 4-3.

In the bottom of the 5th inning, with a runner on 1st, CLE batter Bo Naylor hit a hard line drive double to extra deep center field. A throwing error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and an error by the catcher allowed the batter to reach 3rd.

	game	DET vs CLE on 20230819
	state before	(2, 1, 0, 0, 9, 4, 3, 0)
61.	error	play,9,0,rogej004,00,1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 9, 4, 3, 0)
	fielder	2 (catcher on the home team)

Summary 593 *In the top of the 9th inning, with a runner on 1st, DET batter Jake Rogers had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rogers is still at bat, now with a 1-0 count.*

	game	LAN vs CLE on 20230823
	state before	(0, 1, 0, 0, 8, 4, 1, 0)
62.	error	play,8,0,rosaa003,11,BF1,P01(E1/TH) .1-2
	state after	(0, 0, 1, 0, 8, 4, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 594 *In the top of the 8th inning, with a runner on 1st, LAN batter Amed Rosario has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 2nd. Rosario is still at bat with a 1-1 count.*

	game	LAN vs CLE on 20230824
	state before	(1, 1, 1, 1, 5, 3, 2, 0)
63.	error	play,5,0,heywj001,10,BX,E8/F4D.3-H(NR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 5, 4, 2, 0)
	fielder	8 (center field on the home team)

Summary 595 This game had 2 errors, both with bases loaded. The final score was 9-3.

In the top of the 5th inning, with bases loaded, LAN batter Jason Heyward hit a fly ball to deep 2nd. An error by the center fielder allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.

	game	LAN vs CLE on 20230824
	state before	(1, 1, 1, 1, 5, 4, 2, 0)
64.	error	play,5,0,outmj002,22,*B*BFSFFX,S9/L89+.3-H;2-H(UR);1-3(E9/TH);B-2
	state after	(1, 0, 1, 1, 5, 6, 2, 0)
	fielder	9 (right field on the home team)

Summary 596 *In the top of the 5th inning, with bases loaded, LAN batter James Outman hit a hard line drive single to right center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

11.2.6 September

Visiting/Home team rosters for TBA vs CLE on 2023-09-03

position	innings	player	position	innings	player
1	1	Taj Bradley	1	1	Xzavion Curry
1	6	Kevin Kelly	1	6	Sam Hentges
1	7	Colin Poche	1	7	Nick Sandlin
1	8	Robert Stephenson	1	8	Trevor Stephan
1	9	Andrew Kittredge	1	9	Eli Morgan
1	9	Pete Fairbanks	2	1	Cam Gallagher
2	1	Rene Pinto	2	7	Bo Naylor
2	7	Christian Bethancourt	3	1	Kole Calhoun
3	1	Yandy Diaz	4	1	Tyler Freeman
4	1	Brandon Lowe	4	7	Jose Tena
5	1	Isaac Paredes	5	1	Jose Ramirez
6	1	Taylor Walls	6	1	Gabriel Arias
7	1	Luke Raley	7	1	Steven Kwan
8	1	Jose Siri	8	1	Myles Straw
9	1	Josh Lowe	9	1	Ramon Laureano

	game	TBA vs CLE on 20230903
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
65.	error	play,3,0,wallt003,31,.BBBCX,E4/TH/G34.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 597 This game had **5 errors**, 3 of them in the 9th inning. The final score was 6-2.
In the top of the 3rd inning, with bases empty, TBA batter Davis Taylor Walls hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 1st base.

	game	TBA vs CLE on 20230903
	state before	(0, 0, 0, 0, 3, 2, 0, 1)
66.	error	play,3,1,stram002,12,CBFFX,E6/G6D.B-1
	state after	(0, 1, 0, 0, 3, 2, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 598 *In the bottom of the 3rd inning, with bases empty, CLE batter Myles Straw hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs CLE on 20230903
	state before	(0, 1, 1, 0, 9, 3, 2, 0)
67.	error	play,9,0,loweb001,01,CX,S1/G25-.2-H(E1/TH) (NR) ;1-3
	state after	(0, 1, 0, 1, 9, 4, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 599 *In the top of the 9th inning, with runners on 1st and 2nd, TBA batter Brandon Lowe hit a ground ball single to shallow left infield. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 3rd. The batter took 1st.*

	game	TBA vs CLE on 20230903
	state before	(0, 1, 0, 1, 9, 5, 2, 0)
68.	error	play,9,0,arozr001,32,*BF*BTBFX,FC5/G56.3-H(UR) ;1X2(5E4) ;B-1
	state after	(0, 1, 1, 0, 9, 1, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 600 *In the top of the 9th inning, with runners on 1st and 3rd, TBA batter Randy Arozarena hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the second baseman (with an assist from the third baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

69.	game	TBA vs CLE on 20230903
	state before	(0, 1, 0, 0, 9, 5, 2, 1)
	error	play,9,1,naylb001,32,BCS*BFFBX,FC/G4.1X2(4E6);B-1
	state after	(0, 1, 1, 0, 9, 5, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 601 *In the bottom of the 9th inning, with a runner on 1st, CLE batter Bo Naylor hit a ground ball into a fielder's choice at 2nd. An error by the shortstop (with an assist from the second baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

70.	game	MIN vs CLE on 20230904
	state before	(0, 1, 0, 0, 6, 13, 1, 1)
	error	play,6,1,gimea001,22,BFBFX,S8/L78D.1-3(E8);B-2
	state after	(0, 0, 1, 1, 6, 13, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 602 This game had 2 errors. The final score was 20-6. *In the bottom of the 6th inning, with a runner on 1st, CLE batter Andres Gimenez hit a line drive single to deep left center field. An error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

71.	game	MIN vs CLE on 20230904
	state before	(2, 1, 0, 1, 7, 18, 2, 1)
	error	play,7,1,brenw002,12,BFFFFFFX,S8/L8D+.3-H;1-3(E8/TH);B-2
	state after	(2, 0, 1, 1, 7, 18, 3, 1)
	fielder	8 (center field on the visiting team)

Summary 603 *In the bottom of the 7th inning, with runners on 1st and 3rd, CLE batter Will Brennan hit a hard line drive single to deep center field. The runner on 3rd scored (1 RBI). A throwing error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

Visiting/Home team rosters for BAL vs CLE on 2023-09-22

position	innings	player	position	innings	player
1	1	Dean Kremer	1	1	Shane Bieber
1	4	Tyler Wells	1	6	James Karinchak
1	6	Shintaro Fujinami	1	7	Enyel De Los Santos
1	7	Danny Coulombe	1	8	Reynaldo Lopez
1	7	Jorge Lopez	1	9	Emmanuel Clase
1	8	DL Hall	2	1	Bo Naylor
1	9	Yennier Cano	2	5	David Fry
2	1	Adley Rutschman	3	1	Josh Naylor
3	1	Ryan O'Hearn	4	1	Andres Gimenez
4	1	Jordan Westburg	5	1	Jose Ramirez
4	8	Adam Frazier	6	1	Gabriel Arias
5	1	Ramon Urias	7	1	Steven Kwan
6	1	Gunnar Henderson	8	1	Myles Straw
7	1	Austin Hays	9	1	Will Brennan
7	7	Aaron Hicks			
7	9	Ryan McKenna			
8	1	Aaron Hicks			
8	7	Cedric Mullins			
9	1	Anthony Santander			
9	9	Aaron Hicks			

	game	BAL vs CLE on 20230922
	state before	(0, 0, 1, 0, 3, 3, 3, 0)
72.	error	play,3,0,rutsa001,01,CX,FC1/G4MS.2-H(E1/TH) (NR) (UR) ;B-2
	state after	(0, 0, 1, 0, 3, 4, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 604 This game had **5 errors**. The final score was 8-9.

In the top of the 3rd inning, with a runner on 2nd, BAL batter Adley Rutschman hit a ground ball into a fielder's choice at 2nd. A throwing

error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.

	game	BAL vs CLE on 20230922
	state before	(0, 1, 0, 0, 4, 4, 3, 1)
73.	error	play,4,1,stram002,32,BBFFFBX,E9/L9LD.1-3;B-2
	state after	(0, 0, 1, 1, 4, 4, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 605 *In the bottom of the 4th inning, with a runner on 1st, CLE batter Myles Straw hit a line drive to deep right field. An error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	BAL vs CLE on 20230922
	state before	(1, 1, 0, 1, 4, 4, 4, 1)
74.	error	play,4,1,naylj001,10,BX,FC/G6M.3-H(UR);1-3(E6);B-1
	state after	(1, 1, 0, 1, 4, 4, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 606 *In the bottom of the 4th inning, with runners on 1st and 3rd, CLE batter Josh Naylor hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	BAL vs CLE on 20230922
	state before	(1, 1, 1, 0, 7, 4, 5, 0)
75.	error	play,7,0,santa003,22,FF*BBFX,S7/L78S.2-H;1-3(E7/TH);B-2
	state after	(1, 0, 1, 1, 7, 5, 5, 0)
	fielder	7 (left field on the home team)

Summary 607 *In the top of the 7th inning, with runners on 1st and 2nd, BAL batter Anthony Santander hit a line drive single to shallow left center field. A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	BAL vs CLE on 20230922
	state before	(0, 0, 0, 0, 8, 5, 6, 0)
76.	error	play,8,0,kjerh001,00,.X,C/E2.B-1
	state after	(0, 1, 0, 0, 8, 5, 6, 0)
	fielder	2 (catcher on the home team)

Summary 608 *In the top of the 8th inning, with bases empty, BAL batter Heston Kjerstad had a 0-0 count. Catcher's interference allowed the batter to take 1st.*

	game	CIN vs CLE on 20230926
	state before	(0, 0, 0, 0, 5, 5, 7, 0)
77.	error	play,5,0,encac001,22,BFF.BX,1E3/G23-.B-1
	state after	(0, 1, 0, 0, 5, 5, 7, 0)
	fielder	3 (1st baseman on the home team)

Summary 609 *In the top of the 5th inning, with bases empty, CIN batter Christian Encarnacion-Strand hit a soft ground ball to shallow right infield. An error by the the first baseman allowed the batter to reach 1st base.*

	game	CIN vs CLE on 20230927
	state before	(2, 1, 0, 0, 7, 1, 4, 0)
78.	error	play,7,0,bensw001,22,FBF>FB>*B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 1, 4, 0)
	fielder	2 (catcher on the home team)

Summary 610 *In the top of the 7th inning, with a runner on 1st, CIN batter Will Benson had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Benson is still at bat, with a full count.*

Chapter 12

Colorado Rockies (COL)

Since their first season in 1993, the Denver-based Colorado Rockies have competed in the National League West Division. They have not won a NLWD title, nor have they won a World Series title. However, they did win the NL championship in 2007 based on a wild-card berth in the post-season.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Denver against the Rockies. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 COL team is .670 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the Colorado Rockies is almost 2 standard deviations below the MLB average.

In COL2023 home games, there were 92 errors in the 81 home games, 45 by the Rockies and the rest (47) by visiting teams. The total number of errors by the team (home and away) over the 2023 regular season was 81, slightly lower than the MLB average of 83.9 (which is good).

12.1 Season totals

The Official Scorer shall not charge an error when a runner or runners advance as the result of a passed ball, a wild pitch or a balk.

– **MLB Rule Book** 9.12(f)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

12.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	47
Runs by ht because of errors by vt:	11
Errors by ht:	45
Runs by vt because of errors by ht:	11

The Rockies have a good error-to-run conversion rate, $11/47 = 23.4\%$, almost the same rate as the visiting teams: $11/45 = 24.4\%$. A typical *play* (not necessarily with an error) in the 2023 COL season (home and away) has about $R/PA = 721/6055$ (about 11.9%) chance of resulting in a run. Therefore, a randomly selected error is (for COL) about twice as likely to result in a run than a randomly selected at-bat, at least in this case. Note that the total number of errors (by either team) in a home game is 92, so the expected number of errors (by either team) in a randomly selected COL 2023 home game is about 1.14?.

Errors by position	vt	ht
errors due to pitcher:	7	5
to catcher:	2	5
to 1st baseman:	1	5
to 2nd baseman:	7	8
to 3rd baseman:	10	8
to shortstop:	14	3
to left fielder:	2	3
to center fielder:	3	2
to right fielder:	0	6

12.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	25	13	38
(lone) runner on 1st	10	18	28
(lone) runner on 2nd	3	4	7
(lone) runner on 3rd	1	0	1
two runners on base	7	9	16
bases loaded	1	1	2
total	47	45	92

A table of frequencies of most common errors by the COL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	17.8%
(1, 0, 0, 0)	4	8.9%
(2, 0, 0, 0)	1	2.2%
(0, 1, 0, 0)	11	24.4%
(1, 1, 0, 0)	4	8.9%
(2, 1, 0, 0)	3	6.7%
(2, 0, 1, 0)	3	6.7%
(1, 1, 1, 0)	3	6.7%
(1, 1, 1, 1)	1	2.2%

A table of frequencies of most common errors by a visiting team in a COL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	17.0%
(1, 0, 0, 0)	8	17.0%
(2, 0, 0, 0)	9	19.1%
(0, 1, 0, 0)	5	10.6%
(1, 1, 0, 0)	2	4.3%
(2, 1, 0, 0)	3	6.4%
(1, 1, 1, 0)	4	8.5%
(2, 1, 1, 0)	2	4.3%
(2, 0, 1, 1)	1	2.1%
(0, 1, 1, 1)	1	2.1%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Rockies homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file COL2023.evn):

state	COL2023 percentages
(0, 0, 0, 0)	24.1%
(1, 0, 0, 0)	16.3%
(2, 0, 0, 0)	13.4%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.8%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	4.1%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	1.0%
(2, 0, 0, 1)	1.5%
(0, 1, 1, 0)	1.0%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.7%

12.1.3 Some interesting errors

Some of the most interesting error plays in the COL 2023 home games:

- The game ANA vs COL on 2023-06-23 had 4 errors and on double error.
- The game LAN vs COL on 2023-09-28 had 4 errors.

12.1.4 Relative team rankings

We have 81 home games between the Rockies and some visiting MLB team (vt). A number of teams did not play COL (at their home in Denver) in the regular season. However, of those that did, COL *lost* more games than won against¹:

¹The vt's W/T record is in parentheses (W = numbers of (home) games vt won against COL, T = Total home games vt vs Rockies played at COL).

DET (2/3), MIN (2/3), OAK (2/3), PHI (2/3),
 SLN (2/3), TOR (2/3), ATL (3/3), PIT (3/3),
 ARI (4/6), SDN (4/6), LAN (4/7), SFN (4/7)

and *won* more than lost against

MIL (0/3), MIA (1/4), ANA (1/3), CHA (1/3),
 CHN (1/3), CIN (1/3), NYA (1/3), NYN (1/3) .

The teams that *split* their series were:

HOU (1/2), WAS (2/4).

The teams that did not play in any of the 81 Denver home games against the Rockies were:

BAL, BOS, CLE, KCA, SEA, TBA, TEX.

12.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for COL 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Elias Díaz	126	.316
3 (1B)	Elehuris Montero	55	.290
4 (2B)	Harold Castro	68	.275
5 (3B)	Ryan McMahon	130	.287
6 (SS)	Ezequiel Tovar	153	.322
7 (LF)	Jurickson Profar	89	.316
8 (CF)	Brenton Doyle	125	.250
9 (RF)	Kris Bryant	46	.313

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 COL home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

12.2.1 April

	game	WAS vs COL on 20230407
	state before	(2, 1, 0, 0, 3, 4, 1, 0)
1.	error	play,3,0,thom1002,11,BCX,E5/G6.1-2;B-1
	state after	(2, 1, 1, 0, 3, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 611 This game had 2 errors. The final score was 10-5.
In the top of the 3rd inning, with a runner on 1st, WAS batter Lane Thomas hit a ground ball to short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	WAS vs COL on 20230407
	state before	(1, 0, 1, 0, 8, 9, 5, 0)
2.	error	play,8,0,ruizk001,11,B.FX,E3/G9LS.2-H(NR)(UR);B-1
	state after	(1, 1, 0, 0, 8, 10, 5, 0)
	fielder	3 (1st baseman on the home team)

Summary 612 *In the top of the 8th inning, with a runner on 2nd, WAS batter Keibert Ruiz hit a ground ball to short right field down the 1st base line. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	WAS vs COL on 20230409
	state before	(2, 1, 0, 0, 5, 3, 2, 0)
3.	error	play,5,0,garrs003,22,F*SBBX,S6/G6.1-3(E6/TH)
	state after	(2, 1, 0, 1, 5, 3, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 613 *In the top of the 5th inning, with a runner on 1st, WAS batter Stone Garrett hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for SLN vs COL on 2023-04-10

position	innings	player	position	innings	player
1	1	Steven Matz	1	1	German Marquez
1	6	Zack Thompson	1	6	Brent Suter
1	7	Drew VerHagen	1	8	Justin Lawrence
1	8	Chris Stratton	2	1	Elias Diaz
2	1	Willson Contreras	3	1	C.J. Cron
3	1	Brendan Donovan	4	1	Alan Trejo
4	1	Nolan Gorman	4	8	Ryan McMahon
5	1	Nolan Arenado	5	1	Elehuris Montero
6	1	Tommy Edman	5	8	Alan Trejo
7	1	Alec Burleson	6	1	Ezequiel Tovar
8	1	Dylan Carlson	7	1	Jurickson Profar
8	8	Tyler O'Neill	8	1	Yonathan Daza
9	1	Jordan Walker	9	1	Kris Bryant

	game	SLN vs COL on 20230410
	state before	(2, 0, 0, 0, 2, 0, 0, 1)
4.	error	play,2,1,monte001,01,CX,E4/G6M.B-1
	state after	(2, 1, 0, 0, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 614 This game had 3 errors. Furthermore, the next game in this series against the Cardinals also had 3 errors. The final score was 4-6.

In the bottom of the 2nd inning, with bases empty, COL batter Elehuris Montero hit a ground ball to short. An error by the second baseman allowed the batter to reach 1st base.

	game	SLN vs COL on 20230410
	state before	(0, 0, 0, 0, 6, 2, 3, 0)
5.	error	play,6,0,donob001,32,.B.FBBFX,E7/F7LD.B-2
	state after	(0, 0, 1, 0, 6, 2, 3, 0)
	fielder	7 (left field on the home team)

Summary 615 *In the top of the 6th inning, with bases empty, SLN batter Brendan Donovan hit a fly ball to deep left field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	SLN vs COL on 20230410
	state before	(2, 1, 1, 0, 7, 3, 5, 1)
6.	error	play,7,1,monte001,01,CX,E6/G6S.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 7, 3, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 616 *In the bottom of the 7th inning, with runners on 1st and 2nd, COL batter Elehuris Montero hit a ground ball to short. An error by the shortstop allowed the runners on 1t and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for SLN vs COL on 2023-04-11

position	innings	player	position	innings	player
1	1	Miles Mikolas	1	1	Kyle Freeland
1	6	Genesis Cabrera	1	7	Dinelson Lamet
1	7	Giovanny Gallegos	1	7	Jake Bird
1	9	Ryan Helsley	1	9	Connor Seabold
2	1	Willson Contreras	1	9	Pierce Johnson
3	1	Paul Goldschmidt	2	1	Elias Diaz
4	1	Taylor Motter	3	1	C.J. Cron
4	7	Nolan Gorman	4	1	Ryan McMahon
5	1	Nolan Arenado	5	1	Harold Castro
6	1	Tommy Edman	6	1	Ezequiel Tovar
7	1	Brendan Donovan	7	1	Jurickson Profar
8	1	Tyler O'Neill	8	1	Yonathan Daza
9	1	Jordan Walker	9	1	Kris Bryant

	game	SLN vs COL on 20230411
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
7.	error	play,1,1,bryak001,10,BX,E5/G56.B-1
	state after	(1, 1, 0, 0, 1, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 617 This game had 3 errors. The final score was 9-6.
In the bottom of the 1st inning, with bases empty, COL batter Kris Bryant hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	SLN vs COL on 20230411
	state before	(0, 1, 0, 0, 3, 2, 0, 1)
8.	error	play,3,1,profj001,11,BCX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 3, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 618 *In the bottom of the 3rd inning, with a runner on 1st, COL batter Jurickson Profar had a 1-1 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd, as required.*

	game	SLN vs COL on 20230411
	state before	(2, 1, 1, 0, 7, 2, 6, 0)
9.	error	play,7,0,edmat001,11,CBX,E4/G34+.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 7, 2, 6, 0)
	fielder	4 (2nd baseman on the home team)

Summary 619 *In the top of the 7th inning, with runners on 1st and 2nd, SLN batter Tommy Edman hit a hard ground ball to 1st/2nd. An error by the second baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for PIT vs COL on 2023-04-17

position	innings	player	position	innings	player
1	1	Rich Hill	1	1	Kyle Freeland
1	7	Wil Crowe	1	3	Ty Blach
1	8	Jose Hernandez	1	5	Connor Seabold
2	1	Austin Hedges	1	8	Jake Bird
3	1	Carlos Santana	1	9	Alan Trejo
3	7	Connor Joe	2	1	Elias Diaz
4	1	Mark Mathias	3	1	C.J. Cron
5	1	Ke'Bryan Hayes	3	8	Mike Moustakas
5	9	Tucupita Marcano	4	1	Ryan McMahon
6	1	Rodolfo Castro	4	8	Harold Castro
7	1	Bryan Reynolds	5	1	Elehuris Montero
7	7	Jack Suwinski	6	1	Ezequiel Tovar
8	1	Ji Hwan Bae	7	1	Jurickson Profar
9	1	Connor Joe	8	1	Yonathan Daza
9	7	Canaan Smith-Njigba	9	1	Kris Bryant

	game	PIT vs COL on 20230417
	state before	(0, 1, 0, 0, 2, 1, 0, 0)
10.	error	play,2,0,castr006,00,X,E5/G56D.1-3;B-1
	state after	(0, 1, 0, 1, 2, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 620 This game had 2 errors. Furthermore, each game in this 3 game Pirates-Rockies series had 2 errors (for a total of 6). The final score was 14-3.

In the top of the 2nd inning, with a runner on 1st, PIT batter Rodolfo Castro hit a ground ball to deep 3rd base. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.

	game	PIT vs COL on 20230417
	state before	(0, 1, 0, 0, 5, 9, 1, 0)
11.	error	play,5,0,bae-j001,11,BCX,E5/TH/G6S-.1-2;B-1
	state after	(0, 1, 1, 0, 5, 9, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 621 *In the top of the 5th inning, with a runner on 1st, PIT batter Ji Hwan Bae hit a soft ground ball to shallow short. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	PIT vs COL on 20230418
	state before	(1, 0, 0, 0, 2, 1, 3, 0)
12.	error	play,2,0,smitc008,12,CFBFX,E3/G34.B-1
	state after	(1, 1, 0, 0, 2, 1, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 622 This game had 2 errors. The final score was 5-3.

In the top of the 2nd inning, with bases empty, PIT batter Canaan Smith-Njigba hit a ground ball to 1st/2nd. An error by the first baseman allowed the batter to reach 1st base.

	game	PIT vs COL on 20230418
	state before	(0, 0, 0, 0, 6, 4, 3, 1)
13.	error	play,6,1,diaze005,11,BFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 4, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 623 *In the bottom of the 6th inning, with bases empty, COL batter Elias Diaz hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs COL on 20230419
	state before	(1, 1, 0, 0, 4, 9, 1, 1)
14.	error	play,4,1,treja001,12,CFFBX,FC6/G6.1X2(6E4);B-1
	state after	(1, 1, 1, 0, 4, 9, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 624 This game had 2 errors. The final score was 14-3. *In the bottom of the 4th inning, with a runner on 1st, COL batter Alan Trejo hit a ground ball into a fielder's choice at short. An error by the second baseman (with an assist from the shortstop) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	PIT vs COL on 20230419
	state before	(2, 1, 0, 0, 8, 12, 2, 1)
15.	error	play,8,1,tovae001,11,CBX,E6/G6M.1-2;B-1
	state after	(2, 1, 1, 0, 8, 12, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 625 *In the bottom of the 8th inning, with a runner on 1st, COL batter Ezequiel Tovar hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	ARI vs COL on 20230428
	state before	(1, 1, 0, 1, 7, 2, 1, 0)
16.	error	play,7,0,rojaj001,01,.CX,FC5/G25.3-H;1X3(5E4);B-1
	state after	(1, 1, 0, 1, 7, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 626 *In the top of the 7th inning, with runners on 1st and 3rd, ARI batter Josh Rojas hit a ground ball into a fielder's choice at shallow left infield. The runner on 3rd scored (1 RBI). An error by the second baseman (with an assist from the third baseman) allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

	game	ARI vs COL on 20230429
	state before	(1, 0, 0, 0, 3, 7, 1, 0)
17.	error	play,3,0,rivee001,00,X,S8/L89D+.B-2(E8)
	state after	(1, 0, 1, 0, 3, 7, 1, 0)
	fielder	8 (center field on the home team)

Summary 627 *In the top of the 3rd inning, with bases empty, ARI batter Emmanuel Rivera hit a hard line drive single to deep right center field. An error by the center field?? allowed the runner on ?? to advance to ??*

12.2.2 May

	game	MIL vs COL on 20230503
	state before	(1, 0, 0, 0, 4, 0, 0, 1)
18.	error	play,4,1,bryak001,32,CBBS.BX,E5/G56.B-1
	state after	(1, 1, 0, 0, 4, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 628 *In the bottom of the 4th inning, with bases empty, COL batter Kris Bryant hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

19.	game	PHI vs COL on 20230512
	state before	(0, 1, 0, 0, 7, 3, 1, 1)
	error	play,7,1,gricr001,11,*BSX,S4/G6M.1-3(E4/TH)
	state after	(0, 1, 0, 1, 7, 3, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 629 This game had 2 errors. In fact, there was at least one error in each of the games in this Phillies-Rockies series. The final score was 6-3.

In the bottom of the 7th inning, with a runner on 1st, COL batter Randal Grichuk hit a ground ball single to short. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

20.	game	PHI vs COL on 20230512
	state before	(2, 0, 1, 0, 8, 5, 3, 0)
	error	play,8,0,castn001,00,>C,SB3.2-H(E2/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 8, 6, 3, 0)
	fielder	2 (catcher on the home team)

Summary 630 *In the top of the 8th inning, with a runner on 2nd, PHI batter Nick Castellanos had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI). Castellanos is still at bat, now with a 0-1 count.*

21.	game	PHI vs COL on 20230513
	state before	(0, 0, 0, 0, 8, 5, 3, 1)
	error	play,8,1,cronc002,00,.X,E6/L78D+.B-1
	state after	(0, 1, 0, 0, 8, 5, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 631 *In the bottom of the 8th inning, with bases empty, COL batter C.J. Cron hit a hard line drive to deep left center. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for PHI vs COL on 2023-05-14

position	innings	player	position	innings	player
1	1	Aaron Nola	1	1	Kyle Freeland
1	8	Andrew Vasquez	1	7	Jake Bird
2	1	J.T. Realmuto	1	8	Justin Lawrence
3	1	Alec Bohm	2	1	Elias Diaz
4	1	Bryson Stott	3	1	C.J. Cron
5	1	Edmundo Sosa	3	4	Mike Moustakas
6	1	Trea Turner	4	1	Harold Castro
7	1	Kyle Schwarber	5	1	Ryan McMahon
8	1	Brandon Marsh	6	1	Ezequiel Tovar
9	1	Nick Castellanos	7	1	Jurickson Profar
			8	1	Brenton Doyle
			9	1	Kris Bryant

	game	PHI vs COL on 20230514
	state before	(0, 1, 1, 1, 1, 0, 0, 1)
22.	error	play,1,1,cronc002,01,CX,46(1)/F0/G4.3-H;2-H(E6/TH)(NR);B-1
	state after	(1, 1, 0, 0, 1, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 632 This game had 3 errors. The final score was 0-4. *In the bottom of the 1st inning, with bases loaded, COL batter C.J. Cron hit a ground ball to 2nd. The shortstop (with an assist from the second baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	PHI vs COL on 20230514
	state before	(2, 0, 0, 0, 2, 0, 2, 1)
23.	error	play,2,1,doylb001,22,BCBSX,E5/G5.B-1
	state after	(2, 1, 0, 0, 2, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 633 *In the bottom of the 2nd inning, with bases empty, COL batter Brenton Doyle hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	PHI vs COL on 20230514
	state before	(1, 0, 0, 0, 6, 0, 4, 0)
24.	error	play,6,0,castn001,00,X,E5/TH/G5.B-2
	state after	(1, 0, 1, 0, 6, 0, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 634 *In the top of the 6th inning, with bases empty, PHI batter Nick Castellanos hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	CIN vs COL on 20230515
	state before	(0, 0, 0, 0, 4, 2, 4, 0)
25.	error	play,4,0,senzn001,00,X,E9/L9D+.B-2
	state after	(0, 0, 1, 0, 4, 2, 4, 0)
	fielder	9 (right field on the home team)

Summary 635 *In the top of the 4th inning, with bases empty, CIN batter Nick Senzel hit a hard line drive to deep right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	CIN vs COL on 20230517
	state before	(2, 0, 0, 0, 4, 2, 0, 1)
26.	error	play,4,1,diaze005,12,CS.FBX,3E1/G3.B-1
	state after	(2, 1, 0, 0, 4, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 636 *In the bottom of the 4th inning, with bases empty, COL batter Elias Diaz hit a ground ball to 1st. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	MIA vs COL on 20230522
	state before	(0, 0, 0, 0, 7, 2, 3, 0)
27.	error	play,7,0,hampg001,21,FBBX,E4/G4.B-2
	state after	(0, 0, 1, 0, 7, 2, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 637 *In the top of the 7th inning, with bases empty, MIA batter Garrett Hampson hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 2nd base.*

	game	MIA vs COL on 20230523
	state before	(1, 0, 0, 0, 4, 1, 0, 1)
28.	error	play,4,1,gricr001,22,CBSBFX,E8/L78XD+.B-2
	state after	(1, 0, 1, 0, 4, 1, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 638 *In the bottom of the 4th inning, with bases empty, COL batter Randal Grichuk hit a hard line drive to extra deep left center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	NYN vs COL on 20230526
	state before	(1, 1, 1, 0, 3, 2, 1, 0)
29.	error	play,3,0,mcnej002,11,FBX,FC3/G3.2-3;1-2(E3/TH);B-1
	state after	(1, 1, 1, 1, 3, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 639 This game had 2 errors. The final score was 5-2.

In the top of the 3rd inning, with runners on 1st and 2nd, NYN batter Jeff McNeil hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

30.	game	NYN vs COL on 20230526
	state before	(1, 1, 0, 0, 5, 3, 1, 0)
	error	play,5,0,mcnej002,22,BBCFFX,FC5/G56.1X2(5E4);B-1
	state after	(1, 1, 1, 0, 5, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 640 *In the top of the 5th inning, with a runner on 1st, NYN batter Jeff McNeil hit a ground ball into a fielder's choice at 3rd. An error by the second baseman (with an assist from the third baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

31.	game	NYN vs COL on 20230527
	state before	(0, 1, 0, 0, 5, 1, 6, 0)
	error	play,5,0,vienm001,12,CBFX,S8/L8D+.1-3(E8)
	state after	(0, 1, 0, 1, 5, 1, 6, 0)
	fielder	8 (center field on the home team)

Summary 641 *In the top of the 5th inning, with a runner on 1st, NYN batter Mark Vientos hit a hard line drive single to deep center field. An error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

32.	game	NYN vs COL on 20230528
	state before	(0, 0, 0, 0, 3, 0, 0, 1)
	error	play,3,1,blacc001,00,X,3E1/G3.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 642 *In the bottom of the 3rd inning, with bases empty, COL batter Charlie Blackmon hit a ground ball to 1st. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

12.2.3 June

	game	SFN vs COL on 20230608
	state before	(2, 0, 0, 0, 2, 1, 4, 0)
33.	error	play,2,0,sabob001,10,BX,S1/BG2-.B-2(E1/TH)
	state after	(2, 0, 1, 0, 2, 1, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 643 This game had 2 errors. The final score was 6-4.
In the top of the 2nd inning, with bases empty, SFN batter Blake Sabol hit a soft bunt single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.

	game	SFN vs COL on 20230608
	state before	(2, 1, 0, 0, 2, 1, 4, 1)
34.	error	play,2,1,gricr001,10,B1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 2, 1, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 644 *In the bottom of the 2nd inning, with a runner on 1st, COL batter Randal Grichuk has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to safely advance to 2nd. Grichuk is still at bat with a 1-0 count.*

	game	SDN vs COL on 20230610
	state before	(1, 0, 0, 0, 5, 1, 1, 1)
35.	error	play,5,1,tovae001,12,FSBX,S8/L8D+.B-2(E8)
	state after	(1, 0, 1, 0, 5, 1, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 645 *In the bottom of the 5th inning, with bases empty, COL batter Ezequiel Tovar hit a hard line drive single to deep center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	SDN vs COL on 20230611
	state before	(0, 1, 0, 0, 3, 0, 1, 0)
36.	error	play,3,0,sotoj001,10,B1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 3, 0, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 646 This game had 2 errors. The final score was 4-5. *In the top of the 3rd inning, with a runner on 1st, SDN batter Juan Soto has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to advance to 3rd. Soto is still at bat with a 1-0 count.*

	game	SDN vs COL on 20230611
	state before	(1, 1, 0, 0, 9, 3, 3, 0)
37.	error	play,9,0,carpm002,00,X,S9/G34+.1-3(E9)
	state after	(1, 1, 0, 1, 9, 3, 3, 0)
	fielder	9 (right field on the home team)

Summary 647 *In the top of the 9th inning, with a runner on 1st, SDN batter Matt Carpenter hit a hard ground ball single to 1st/2nd. An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for ANA vs COL on 2023-06-23

position	innings	player	position	innings	player
1	1	Michael Lorenzen	1	1	Austin Gomber
1	6	Chasen Shreve	1	8	Jake Bird
1	6	Mason Englert	1	9	Pierce Johnson
2	1	Jake Rogers	2	1	Elias Diaz
2	8	Eric Haase	3	1	C.J. Cron
3	1	Spencer Torkelson	4	1	Harold Castro
4	1	Zack Short	5	1	Ryan McMahon
5	1	Andy Ibanez	6	1	Ezequiel Tovar
6	1	Javier Baez	7	1	Nolan Jones
7	1	Zach McKinstry	8	1	Brenton Doyle
7	7	Eric Haase	9	1	Kris Bryant
7	8	Zach McKinstry			
8	1	Jake Marisnick			
8	7	Matt Vierling			
9	1	Matt Vierling			
9	7	Zach McKinstry			
9	8	Kerry Carpenter			

	game	ANA vs COL on 20230623
	state before	(0, 1, 0, 0, 2, 0, 1, 0)
38.	error	play,2,0,padlk001,10,BX,D7/F78XD.1-H(E7)(NR);B-3(E7/TH)
	state after	(0, 0, 0, 1, 2, 1, 1, 0)
	fielder	7,7 (left field on the home team)

Summary 648 This game had 4 errors, including a double error. This one play had 2 errors by the same fielder. The final score was 4-7.

In the top of the 2nd inning, with a runner on 1st, ANA batter Kevin Padlo hit a fly ball double to extra deep left center field. An error by the left fielder allowed the runner on 1st to score (unearned, no RBI). A throwing error by the left fielder (his second error in this one play)

allowed the batter to reach 3rd.

	game	ANA vs COL on 20230623
	state before	(2, 1, 1, 0, 8, 4, 3, 0)
39.	error	play,8,0,ohtas001,31,FBBB+2,OA.2-3(E2/TH);1-2
	state after	(2, 1, 1, 0, 8, 4, 3, 0)
	fielder	2 (catcher on the home team)

Summary 649 *In the top of the 8th inning, with runners on 1st and 2nd, ANA batter Shohei Ohtani had a 3-1 count. A throwing error by the catcher (attempting to pickoff the runner on 2nd) allowed that runner to advance to 3rd. Ohtani is still at bat with a 3-1 count.*

	game	ANA vs COL on 20230623
	state before	(1, 1, 1, 0, 8, 4, 3, 1)
40.	error	play,8,1,mcmar001,11,.BSX,FC3/G3.2-3;1X2(3E6);B-1
	state after	(1, 1, 1, 1, 8, 4, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 650 *In the bottom of the 8th inning, with runners on 1st and 2nd, COL batter Ryan McMahon hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. An error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	ANA vs COL on 20230625
	state before	(0, 0, 0, 0, 2, 0, 1, 0)
41.	error	play,2,0,drurb001,32,FBB.BSX,E4/G34.B-1
	state after	(0, 1, 0, 0, 2, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 651 This game had 2 errors. The final score was 3-4.
In the top of the 2nd inning, with bases empty, ANA batter Brandon Drury hit a ground ball to 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.

	game	ANA vs COL on 20230625
	state before	(2, 1, 0, 0, 7, 2, 3, 1)
42.	error	play,7,1,tovae001,10,B1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 7, 2, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 652 *In the bottom of the 7th inning, with a runner on 1st, COL batter Ezequiel Tovar has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to reach 3rd. Tovar is still at-bat with a 1-0 count.*

	game	LAN vs COL on 20230627
	state before	(0, 0, 0, 0, 7, 5, 0, 0)
43.	error	play,7,0,bettm001,11,.BSX,E5/G5.B-1
	state after	(0, 1, 0, 0, 7, 5, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 653 *In the top of the 7th inning, with bases empty, LAN batter Mookie Betts hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	LAN vs COL on 20230628
	state before	(2, 0, 1, 0, 6, 6, 8, 1)
44.	error	play,6,1,diaze005,10,BX,S8/L89D+.2-H;B-2(E4)
	state after	(2, 0, 1, 0, 6, 6, 9, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 654 *In the bottom of the 6th inning, with a runner on 2nd, COL batter Elias Diaz hit a hard line drive single to right center field. The runner on 2nd scored (1 RBI). An error by the second baseman allowed the batter to reach 2nd.*

Visiting/Home team rosters for DET vs COL on 2023-06-30

position	innings	player	position	innings	player
1	1	Michael Lorenzen	1	1	Austin Gomber
1	6	Chasen Shreve	1	8	Jake Bird
1	6	Mason Englert	1	9	Pierce Johnson
2	1	Jake Rogers	2	1	Elias Diaz
2	8	Eric Haase	3	1	C.J. Cron
3	1	Spencer Torkelson	4	1	Harold Castro
4	1	Zack Short	5	1	Ryan McMahon
5	1	Andy Ibanez	6	1	Ezequiel Tovar
6	1	Javier Baez	7	1	Nolan Jones
7	1	Zach McKinstry	8	1	Brenton Doyle
7	7	Eric Haase	9	1	Kris Bryant
7	8	Zach McKinstry			
8	1	Jake Marisnick			
8	7	Matt Vierling			
9	1	Matt Vierling			
9	7	Zach McKinstry			
9	8	Kerry Carpenter			

	game	DET vs COL on 20230630
	state before	(0, 1, 0, 0, 3, 0, 5, 0)
45.	error	play,3,0,vierm001,12,CSBC,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 0, 5, 0)
	fielder	2 (catcher on the home team)

Summary 655 This game had 3 errors. The final score was 5-8.

In the top of the 3rd inning, with a runner on 1st, DET batter Matt Vierling had a 1-2 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.

	game	DET vs COL on 20230630
	state before	(1, 0, 0, 0, 7, 3, 8, 0)
46.	error	play,7,0,haase001,12,.SBSX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 7, 3, 8, 0)
	fielder	6 (shortstop on the home team)

Summary 656 *In the top of the 7th inning, with bases empty, DET batter Eric Haase hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	DET vs COL on 20230630
	state before	(2, 0, 0, 0, 8, 5, 8, 1)
47.	error	play,8,1,doylb001,22,BCFBFX,E6/G6.B-1
	state after	(2, 1, 0, 0, 8, 5, 8, 1)
	fielder	6 (shortstop on the visiting team)

Summary 657 *In the bottom of the 8th inning, with bases empty, COL batter Brenton Doyle hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

12.2.4 July

	game	DET vs COL on 20230701
	state before	(2, 0, 0, 0, 3, 0, 0, 1)
48.	error	play,3,1,bryak001,11,FBX,E5/G5.B-2(E3)
	state after	(2, 0, 1, 0, 3, 0, 0, 1)
	fielder	5,3 (3rd baseman, 1st baseman on the visiting team)

Summary 658 This play had 2 errors by 2 different fielders. The final score was 4-2.

In the bottom of the 3rd inning, with bases empty, COL batter Kris Bryant hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 2nd.

49.	game	DET vs COL on 20230702
	state before	(2, 0, 0, 0, 5, 8, 4, 1)
	error	play,5,1,diaze005,01,CX,E5/TH/G56.B-2
	state after	(2, 0, 1, 0, 5, 8, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 659 *In the bottom of the 5th inning, with bases empty, COL batter Elias Diaz hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

50.	game	NYA vs COL on 20230715
	state before	(0, 0, 0, 0, 8, 6, 2, 0)
	error	play,8,0,lemad001,32,.BBCBC.X,S9/L89D+.B-2(E9)
	state after	(0, 0, 1, 0, 8, 6, 2, 0)
	fielder	9 (right field on the home team)

Summary 660 *In the top of the 8th inning, with bases empty, NYA batter DJ LeMahieu hit a hard line drive single to deep right center field. An error by the right fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for NYA vs COL on 2023-07-16

position	innings	player	position	innings	player
1	1	Gerrit Cole	1	1	Chase Anderson
1	11	Nick Ramirez	1	10	Matt Koch
1	11	Ron Marinaccio	1	11	Gavin Hollowell
1	7	Wandy Peralta	1	6	Jake Bird
1	8	Clay Holmes	1	6	Justin Lawrence
1	8	Tommy Kahnle	1	8	Pierce Johnson
1	9	Ian Hamilton	1	9	Daniel Bard
2	1	Jose Trevino	2	1	Austin Wynns
2	9	Kyle Higashioka	2	9	Elias Diaz
3	1	Anthony Rizzo	3	1	Michael Toglia
3	9	DJ LeMahieu	4	1	Harold Castro
4	1	Gleyber Torres	4	9	Alan Trejo
5	1	DJ LeMahieu	5	1	Ryan McMahon
5	9	Oswald Peraza	6	1	Ezequiel Tovar
6	1	Anthony Volpe	7	1	Nolan Jones
7	1	Isiah Kiner-Falefa	8	1	Brenton Doyle
8	1	Harrison Bader	8	9	Harold Castro
9	1	Oswaldo Cabrera	9	1	Kris Bryant

	game	NYA vs COL on 20230716
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
51.	error	play,1,1,mcmar001,32,CSBFBBX,E6/G6M.B-2
	state after	(2, 0, 1, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 661 This game had 3 errors. The final score was 7-8.

In the bottom of the 1st inning, with bases empty, COL batter Ryan McMahon hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 2nd base.

	game	NYA vs COL on 20230716
	state before	(1, 1, 1, 0, 6, 0, 1, 0)
52.	error	play,6,0,kinei001,12,CS*BX,S8/G6M.2-H;1-H(E5/TH) (NR);B-3
	state after	(1, 0, 0, 1, 6, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 662 *In the top of the 6th inning, with runners on 1st and 2nd, NYA batter Isiah Kiner-Falefa hit a ground ball single to deep short. The runner on 2nd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 3rd.*

	game	NYA vs COL on 20230716
	state before	(0, 1, 0, 0, 9, 3, 5, 0)
53.	error	play,9,0,torrg001,21,BCBX,S5/G56S-.1-H(E5/TH) (NR);B-2
	state after	(0, 0, 1, 0, 9, 4, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 663 *In the top of the 9th inning, with a runner on 1st, NYA batter Gleyber Torres hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	HOU vs COL on 20230718
	state before	(0, 1, 0, 0, 4, 3, 3, 0)
54.	error	play,4,0,diazy004,11,C*BX,FC1/G1.1-2(E1);B-1
	state after	(0, 1, 1, 0, 4, 3, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 664 *In the top of the 4th inning, with a runner on 1st, HOU batter Yainer Diaz hit a ground ball into a fielder's choice at very shallow infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for OAK vs COL on 2023-07-28

position	innings	player	position	innings	player
1	1	JP Sears	1	1	Kyle Freeland
1	6	Austin Pruitt	1	5	Gavin Hollowell
1	6	Sam Moll	1	6	Connor Seabold
1	7	Tayler Scott	1	9	Matt Koch
1	8	Lucas Erceg	2	1	Elias Diaz
1	8	Sam Long	3	1	Elehuris Montero
1	9	Trevor May	3	9	Michael Toglia
2	1	Shea Langeliers	4	1	Alan Trejo
3	1	Jordan Diaz	4	7	Harold Castro
4	1	Zack Gelof	4	9	Ryan McMahon
5	1	Aledmys Diaz	5	1	Ryan McMahon
5	9	Jace Peterson	5	9	Elehuris Montero
6	1	Nick Allen	6	1	Ezequiel Tovar
7	1	Tony Kemp	7	1	Jurickson Profar
8	1	JJ Bleday	8	1	Brenton Doyle
9	1	Ramon Laureano	8	7	Randal Grichuk
			9	1	Randal Grichuk
			9	7	Nolan Jones

	game	OAK vs COL on 20230728
	state before	(0, 0, 0, 0, 2, 1, 0, 0)
55.	error	play,2,0,diaza003,20,BBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 2, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 665 This game had 3 errors. The final score was 8-5.

In the top of the 2nd inning, with bases empty, OAK batter Aledmys Diaz hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	OAK vs COL on 20230728
	state before	(1, 1, 1, 1, 2, 3, 0, 0)
56.	error	play,2,0,diazj008,02,CFX,FC/G4.3-H(UR);2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 2, 4, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 666 *In the top of the 2nd inning, with bases loaded, OAK batter Jordan Diaz hit a ground ball into a fielder's choice at 2nd. The runner on 3rd score (1 RBI) and the runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	OAK vs COL on 20230728
	state before	(1, 1, 1, 0, 4, 5, 0, 0)
57.	error	play,4,0,laurr001,12,*SS*BFX,FC4/G4.2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 4, 5, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 667 *In the top of the 4th inning, with runners on 1st and 2nd, OAK batter Ramon Laureano hit a ground ball into a fielder's choice at 2nd. The runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	OAK vs COL on 20230730
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
58.	error	play,1,1,tovae001,22,FBBFX,E3/P3.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 668 *In the bottom of the 1st inning, with bases empty, COL batter Ezequiel Tovar hit a pop up to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SDN vs COL on 2023-07-31

position	innings	player	position	innings	player
1	1	Seth Lugo	1	1	Austin Gomber
1	8	Tom Cosgrove	1	10	Brad Hand
1	9	Nick Martinez	1	10	Matt Koch
2	1	Gary Sanchez	1	7	Jake Bird
3	1	Matthew Batten	1	8	Daniel Bard
4	1	Jake Cronenworth	1	8	Justin Lawrence
5	1	Manny Machado	2	1	Elias Diaz
6	1	Xander Bogaerts	3	1	Elehuris Montero
7	1	Juan Soto	4	1	Harold Castro
8	1	Trent Grisham	4	10	Alan Trejo
9	1	Fernando Tatis Jr.	5	1	Ryan McMahon
			6	1	Ezequiel Tovar
			7	1	Jurickson Profar
			8	1	Brenton Doyle
			9	1	Michael Toggia

	game	SDN vs COL on 20230731
	state before	(0, 1, 0, 0, 5, 1, 0, 1)
59.	error	play,5,1,toglm001,00,X,FC1/G4MS.1X2(1E6);B-1
	state after	(0, 1, 1, 0, 5, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 669 This game had 3 errors. The final score was 3-4 (in extra innings).

In the bottom of the 5th inning, with a runner on 1st, COL batter Michael Toggia hit a ground ball into a fielder's choice at 2nd. An error by the shortstop (with an assist from the pitcher) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.

	game	SDN vs COL on 20230731
	state before	(1, 1, 1, 0, 5, 1, 0, 1)
60.	error	play,5,1,monte001,12,SBSFX,D8/L78D+.2-H;1-H(E8) (NR) (UR)
	state after	(1, 0, 1, 0, 5, 1, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 670 *In the bottom of the 5th inning, with runners on 1st and 2nd, COL batter Elehuris Montero hit a hard line drive double to deep left center field.. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the runner on 1st to aslo score (unearned, no RBI). The batter took 2nd.*

	game	SDN vs COL on 20230731
	state before	(0, 0, 1, 0, 10, 3, 3, 1)
61.	error	play,10,1,doylb001,00,.X,FC2/BG2.2X3(2E5);B-1
	state after	(0, 1, 0, 1, 10, 3, 3, 1)
	fielder	5 (center field on the visiting team)

Summary 671 *In the bottom of the 10th inning, with a runner on 2nd, COL batter Brenton Doyle hit a ground ball into a fielder's choice at very shallow infield. An error by the third baseman (with an assist from the catcher) allowed the runner on 2nd to safely advance to 3rd and the batter to reach 1st.*

12.2.5 August

	game	SDN vs COL on 20230802
	state before	(0, 1, 0, 0, 2, 1, 0, 0)
62.	error	play,2,0,cronj001,22,BBCF1,SB2.1-3(E3/TH)
	state after	(0, 0, 0, 1, 2, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 672 *In the top of the 2nd inning, with a runner on 1st, SDN batter Jake Cronenworth had a 2-2 count. The pitcher attempted to pickoff the runner on 1st but that runner safely stole 2nd. A throwing error by the first baseman allowed that runner to advance to 3rd. Cronenworth is still at bat, with a 2-2 count.*

	game	ARI vs COL on 20230815
	state before	(0, 0, 0, 0, 7, 2, 2, 0)
63.	error	play,7,0,perdg001,00,.X,S1/BG2-.B-2(E1/TH)
	state after	(0, 0, 1, 0, 7, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 673 This game had 2 errors. The final score was 8-5.
In the top of the 7th inning, with bases empty, ARI batter Gerardo Perdomo hit a soft bunt single to very shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.

	game	ARI vs COL on 20230815
	state before	(2, 0, 1, 1, 7, 2, 2, 0)
64.	error	play,7,0,walkc002,01,C+3,P03(E2/TH).3-H(NR)(UR);2XH(72)
	state after	(0, 0, 0, 0, 7, 3, 2, 1)
	fielder	2 (catcher on the home team)

Summary 674 *In the top of the 7th inning, with 2 outs and runners on 2nd and 3rd, ARI batter Christian Walker has a 0-1 count. During an attempted pickoff of the runner on 3rd, a throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). The runner on 2nd advanced to third but was tagged out by the catcher (with an assist from the left fielder) when he attempted to score. This is the 3rd out, marking the end of the half-inning.*

	game	CHA vs COL on 20230819
	state before	(1, 1, 1, 0, 1, 1, 0, 1)
65.	error	play,1,1,rodgb002,22,FBBFX,S6/G56.2-H(E6)(NR);1-2
	state after	(1, 1, 1, 0, 1, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 675 This game had 2 errors. The final score was 5-11.
In the bottom of the 1st inning, with runners on 1st and 2nd, COL batter Brendan Rodgers hit a ground ball single to 3rd. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 2nd. The batter took 1st.

	game	CHA vs COL on 20230819
	state before	(0, 0, 0, 0, 3, 2, 5, 1)
66.	error	play,3,1,monte001,21,BBFX,E5/G7S.B-1
	state after	(0, 1, 0, 0, 3, 2, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 676 *In the bottom of the 3rd inning, with bases empty, COL batter Elehuris Montero hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.*

	game	CHA vs COL on 20230820
	state before	(2, 1, 1, 0, 3, 1, 0, 0)
67.	error	play,3,0,moncy001,01,F>*B,SB3.2-H(E2/TH)(NR)(UR);1-2
	state after	(2, 0, 1, 0, 3, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 677 This game had 2 errors. The final score was 10-5.
In the top of the 3rd inning, with runners on 1st and 2nd, CHA batter Yoan Moncada had a 0-1 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI). Moncada is still at bat, now with a 1-1 count.

	game	CHA vs COL on 20230820
	state before	(2, 1, 1, 0, 4, 2, 1, 1)
68.	error	play,4,1,toglm001,12,SS.*BX,E4/G34.2-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 4, 2, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 678 *In the bottom of the 4th inning, with runners on 1st and 2nd, COL batter Michael Toggia hit a ground ball to 1st/2nd. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	ATL vs COL on 20230828
	state before	(2, 0, 1, 0, 9, 13, 4, 0)
69.	error	play,9,0,rilea001,01,CX,E9/F89S.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 9, 14, 4, 0)
	fielder	9 (right field on the home team)

Summary 679 *In the top of the 9th inning, with a runner on 2nd, ATL batter Austin Riley hit a fly ball to shallow right center field. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	ATL vs COL on 20230829
	state before	(1, 1, 0, 0, 3, 2, 1, 0)
70.	error	play,3,0,olsom001,22,FSBBX,FC4/G34.1-3(E4/TH);B-2
	state after	(1, 0, 1, 1, 3, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 680 *In the top of the 3rd inning, with a runner on 1st, ATL batter Matt Olson hit a ground ball into a fielder's choice at 1st/2nd. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

71.	game	ATL vs COL on 20230830
	state before	(0, 1, 1, 0, 4, 1, 1, 0)
	error	play,4,0,ozunm001,32,CS*BB*BFX,FC6/G6.2-H(NR);1-3(E6/TH);B-2
	state after	(0, 0, 1, 1, 4, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 681 This game had 2 errors. The final score was 7-3.

In the top of the 4th inning, with runners on 1st and 2nd, ATL batter Marcell Ozuna hit a ground ball into a fielder's choice at short. The runner on 1st scored (unearned, no RBI). A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

72.	game	ATL vs COL on 20230830
	state before	(1, 0, 0, 0, 5, 2, 1, 1)
	error	play,5,1,doylb001,22,SCBBX,E6/G6.B-1
	state after	(1, 1, 0, 0, 5, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 682 *In the bottom of the 5th inning, with bases empty, COL batter Brenton Doyle hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

12.2.6 September

73.	game	TOR vs COL on 20230901
	state before	(0, 0, 0, 0, 6, 4, 2, 1)
	error	play,6,1,diaze005,22,.BCFBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 6, 4, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 683 This game had 2 errors. The final score was 1-9.

In the bottom of the 6th inning, with bases empty, COL batter Elias Diaz

hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	TOR vs COL on 20230901
	state before	(0, 0, 0, 1, 9, 13, 5, 1)
74.	error	play,9,1,monte001,00,X,E5/G7LS.3-H(NR);B-1
	state after	(0, 1, 0, 0, 9, 13, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 684 *In the bottom of the 9th inning, with a runner on 3rd, COL batter Elehuris Montero hit a ground ball to shallow left field. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	TOR vs COL on 20230902
	state before	(0, 0, 0, 0, 2, 5, 0, 1)
75.	error	play,2,1,jonen002,01,CX,E6/G4.B-1
	state after	(0, 1, 0, 0, 2, 5, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 685 This game had 2 errors. The final score was 7-8. *In the bottom of the 2nd inning, with bases empty, COL batter Nolan Jones hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	TOR vs COL on 20230902
	state before	(0, 0, 0, 0, 4, 5, 2, 1)
76.	error	play,4,1,boucs001,22,BBCFX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 4, 5, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 686 *In the bottom of the 4th inning, with bases empty, COL batter Sean Bouchard hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	TOR vs COL on 20230903
	state before	(2, 0, 1, 0, 5, 2, 3, 0)
77.	error	play,5,0,guerv002,22,B.CF*BX,S7/L7D+.2-H;B-2(E7/TH)
	state after	(2, 0, 1, 0, 5, 3, 3, 0)
	fielder	7 (left field on the home team)

Summary 687 This game had 2 errors. The final score was 7-5.

In the top of the 5th inning, with a runner on 2nd, TOR batter Vladimir Guerrero hit a hard line drive single to deep left field. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the batter to reach 2nd.

	game	TOR vs COL on 20230903
	state before	(0, 1, 0, 0, 6, 5, 5, 0)
78.	error	play,6,0,varsd001,32,BBCC1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 6, 5, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 688 *In the top of the 6th inning, with a runner on 1st, TOR batter Daulton Varsho has a full count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 3rd. Varsho is still at-bat with a full count.*

	game	CHN vs COL on 20230913
	state before	(2, 0, 0, 0, 5, 3, 3, 1)
79.	error	play,5,1,bryak001,02,CFX,E7/F78XD.B-2
	state after	(2, 0, 1, 0, 5, 3, 3, 1)
	fielder	7 (left field on the visiting team)

Summary 689 *In the bottom of the 5th inning, with bases empty, COL batter Kris Bryant hit a fly ball to extra deep left center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	SFN vs COL on 20230915
	state before	(1, 1, 1, 0, 9, 2, 1, 1)
80.	error	play,9,1,monte001,10,BX,S7/G56.2-H;1-H(E7/TH)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 9, 2, 3, 1)
	fielder	7 (left field on the visiting team)

Summary 690 *In the bottom of the 9th inning, with runners on 1st and 2nd, COL batter Elehuris Montero hit a ground ball single to 3rd. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 2nd.*

	game	SFN vs COL on 20230916 (2/2)
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
81.	error	play,1,1,tovae001,12,CSBFX,E6/G6.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 691 This game 2 of a doubleheader had 2 errors. The final score was 2-5.

In the bottom of the 1st inning, with bases empty, COL batter Ezequiel Tovar hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	SFN vs COL on 20230916 (2/2)
	state before	(0, 1, 0, 0, 3, 1, 3, 1)
82.	error	play,3,1,bryak001,02,CSB,WP.1-3(E2/TH)
	state after	(0, 0, 0, 1, 3, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 692 *In the bottom of the 3rd inning, with a runner on 1st, COL batter Kris Bryant had a 0-2 count. The next pitch was a wild pitch. The runner on 1st advanced to 2nd but a throwing error by the catcher allowed the runner to advance to 3rd. Bryant is still at bat, now with a 1-2 count.*

Visiting/Home team rosters for SFN vs COL on 2023-09-17

position	innings	player	position	innings	player
1	1	Sean Manaia	1	1	Chris Flexen
1	6	John Brebbia	1	6	Evan Justice
1	7	Tyler Rogers	1	6	Matt Koch
1	8	Luke Jackson	1	7	Matt Carasiti
1	9	Camilo Doval	1	8	Noah Davis
1	9	Taylor Rogers	2	1	Austin Wynns
2	1	Patrick Bailey	3	1	Elehuris Montero
3	1	LaMonte Wade Jr.	4	1	Brendan Rodgers
4	1	Thairo Estrada	5	1	Alan Trejo
5	1	J.D. Davis	6	1	Ezequiel Tovar
6	1	Brandon Crawford	7	1	Nolan Jones
7	1	Mitch Haniger	8	1	Brenton Doyle
8	1	Mike Yastrzemski	9	1	Hunter Goodman
8	6	Austin Slater			
9	1	Michael Conforto			
9	6	Mike Yastrzemski			

	game	SFN vs COL on 20230917
	state before	(1, 0, 0, 0, 5, 1, 0, 1)
83.	error	play,5,1,doylb001,00,X,E1/BG1S-.B-1
	state after	(1, 1, 0, 0, 5, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 693 This game had 3 errors. The final score was 11-10.
In the bottom of the 5th inning, with bases empty, COL batter Brenton Doyle hit a soft bunt to shallow center infield. An error by the pitcher allowed the batter to reach 1st base.

	game	SFN vs COL on 20230917
	state before	(1, 1, 0, 0, 6, 9, 2, 1)
84.	error	play,6,1,treja001,02,.FFX,FC1/G2.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 6, 9, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 694 *In the bottom of the 6th inning, with a runner on 1st, COL batter Alan Trejo hit a ground ball into a fielder's choice at very shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SFN vs COL on 20230917
	state before	(2, 0, 1, 1, 9, 11, 7, 1)
85.	error	play,9,1,bryak001,32,.B.BBCFFX,E1/G23-.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 9, 11, 8, 1)
	fielder	1 (pitcher on the visiting team)

Summary 695 *In the bottom of the 9th inning, with runners on 2nd and 3rd, COL batter Kris Bryant hit a soft ground ball to shallow right infield. An error by the pitcher allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd and the batter to reach 1st.*

	game	LAN vs COL on 20230926
	state before	(0, 1, 0, 0, 5, 5, 0, 0)
86.	error	play,5,0,outmj002,02,SFFX,S3/G34.1-3(E3)
	state after	(0, 1, 0, 1, 5, 5, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 696 *In the top of the 5th inning, with a runner on 1st, LAN batter James Outman hit a ground ball single to 1st/2nd. An error by the first baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for LAN vs COL on 2023-09-28

position	innings	player	position	innings	player
1	1	Ryan Yarbrough	1	1	Chris Flexen
1	5	Shelby Miller	1	7	Jake Bird
1	7	Caleb Ferguson	1	8	Brent Suter
1	7	Evan Phillips	1	9	Connor Seabold
1	8	Miguel Rojas	2	1	Austin Wynns
2	1	Austin Barnes	3	1	Elehuris Montero
3	1	Freddie Freeman	4	1	Brendan Rodgers
3	5	Enrique Hernandez	5	1	Alan Trejo
4	1	Mookie Betts	6	1	Ezequiel Tovar
4	5	Kolten Wong	7	1	Nolan Jones
5	1	Max Muncy	8	1	Brenton Doyle
6	1	Chris Taylor	9	1	Sean Bouchard
6	5	Amed Rosario			
7	1	David Peralta			
8	1	Enrique Hernandez			
8	5	Chris Taylor			
9	1	Jason Heyward			

	game	LAN vs COL on 20230928
	state before	(2, 0, 0, 0, 3, 3, 7, 1)
87.	error	play,3,1,wynna001,12,CFBX,E4/G4.B-1
	state after	(2, 1, 0, 0, 3, 3, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 697 This game had 4 errors. The final score was 5-14.
In the bottom of the 3rd inning, with bases empty, COL batter Austin Wynns hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.

	game	LAN vs COL on 20230928
	state before	(1, 1, 0, 0, 6, 3, 9, 0)
88.	error	play,6,0,heywj001,12,CSBX,S9/L9L.1-3;B-2(E9)
	state after	(1, 0, 1, 1, 6, 3, 9, 0)
	fielder	9 (right field on the home team)

Summary 698 *In the top of the 6th inning, with a runner on 1st, LAN batter Jason Heyward hit a line drive single to right field. The runner on 1st advanced to 3rd. An error by the right fielder allowed the batter to reach 2nd.*

	game	LAN vs COL on 20230928
	state before	(1, 0, 1, 0, 7, 4, 13, 1)
89.	error	play,7,1,tovae001,22,*BCBFX,E4/G4M.2-3;B-1
	state after	(1, 1, 0, 1, 7, 4, 13, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 699 *In the bottom of the 7th inning, with a runner on 2nd, COL batter Ezequiel Tovar hit a ground ball to 2nd base. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	LAN vs COL on 20230928
	state before	(0, 1, 0, 0, 8, 4, 13, 1)
90.	error	play,8,1,boucs001,01,CX,E5/G5.1-2;B-1
	state after	(0, 1, 1, 0, 8, 4, 13, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 700 *In the bottom of the 8th inning, with a runner on 1st, COL batter Sean Bouchard hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

12.2.7 October

	game	MIN vs COL on 20231001
	state before	(2, 1, 0, 0, 4, 2, 0, 0)
91.	error	play,4,0,kiria001,11,B+1SX,S9/L9+.1-3(E9)
	state after	(2, 1, 0, 1, 4, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 701 This game had 2 errors. The final score was 2-3 (in extra innings).

In the top of the 4th inning, with a runner on 1st, MIN batter Alex Kirilloff hit a hard line drive single to right field. An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	MIN vs COL on 20231001
	state before	(1, 0, 0, 0, 6, 2, 1, 1)
92.	error	play,6,1,doylb001,00,X,S5/BG25-.B-2(E5/TH)
	state after	(1, 0, 1, 0, 6, 2, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 702 *In the bottom of the 6th inning, with bases empty, COL batter Brenton Doyle hit a soft bunt single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

Chapter 13

Detroit Tigers (DET)

Since 1998, the Detroit Tigers compete in the American League Central Division. One of the AL's eight charter franchises, the club is the only Western League team still in its original city. They are also the oldest continuous one name, one city franchise in the AL. Since they were established in 1901, the Tigers have won 4 World Series titles, the latest in 1984. They have won four ALCD titles, the latest in 2014.

Baseball trivia: The Tigers and Chicago White Sox, one of the oldest active rivalries in the league today, have yet to meet in the postseason. However, in 1935, the Tigers played the Chicago Cubs in the World Series. The Tigers 1935 lineup featured four future Hall of Famers – Mickey Cochrane, Goose Goslin, Charlie Gehringer, and the next AL MVP, Hank Greenberg. In fact, Greenberg led the league in home runs, extra-base hits and RBIs (48 more than the next closest player, Lou Gehrig, with 120 RBIs). The Tigers beat the Chicago Cubs and won the 1935 World Series.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Detroit against the Tigers. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 DET team is 0.699 higher than the MLB average

DER of 0.691 that year [BR] (which is good).

That year, there were 104 errors in the 81 home games, 62 by the Tigers and the rest (42) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 100, well above the MLB average of 83.9 (which is bad).

These statistics are unusual in that the Tigers have a “good” DER and a “bad” E.

13.1 Season totals

Throughout, *vt* denotes the visiting team and *ht* denotes the home team.

13.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position.

Runs and errors	totals
Errors by vt:	42
Runs by ht because of errors by vt:	14
Errors by ht:	62
Runs by vt because of errors by ht:	15

This is a very good error-to-run conversion rate: $14/42 = 33.3\%$ on the part of the Tigers (compared to $15/62 = 24.2\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 DET season (home and away) has $R/PA = 661/6080$ (about 10.9%) chance of resulting in a run.

Note that the total number of errors is 104, so the expected number of errors in a randomly selected DET2023 home game is about 1.28.

Errors by position	vt	ht
errors due to pitcher:	6	2
to catcher:	4	6
to 1st baseman:	0	8
to 2nd baseman:	8	5
to 3rd baseman:	3	18
to shortstop:	13	11
to left fielder:	3	5
to center fielder:	2	4
to right fielder:	2	1

13.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	13	24	37
(lone) runner on 1st	13	11	24
(lone) runner on 2nd	5	7	12
(lone) runner on 3rd	1	0	1
two runners on base	9	19	28
bases loaded	1	1	2
total	42	62	104

A table of frequencies of most common errors by the Tigers in DET 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	13	21%
(1, 0, 0, 0)	8	12.9%
(2, 0, 0, 0)	3	4.84%
(0, 1, 0, 0)	5	8.07%
(1, 1, 0, 0)	3	4.84%
(2, 1, 0, 0)	3	4.84%
(1, 0, 1, 0)	3	4.84%
(1, 1, 0, 1)	2	3.23%
(2, 1, 1, 0)	10	16.1%

A table of frequencies of most common errors by a visiting team in DET 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	14.3%
(1, 0, 0, 0)	5	11.9%
(2, 0, 0, 0)	2	4.76%
(0, 1, 0, 0)	5	11.9%
(1, 1, 0, 0)	5	11.9%
(2, 1, 0, 0)	3	7.14%
(1, 0, 1, 0)	2	4.76%
(1, 1, 0, 1)	3	7.14%
(2, 1, 1, 1)	1	2.38%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Tigers homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `DET2023.eva`):

state	DET2023 percentages
(0, 0, 0, 0)	24.5%
(1, 0, 0, 0)	16.9%
(2, 0, 0, 0)	12.8%
(0, 1, 0, 0)	6.0%
(1, 1, 0, 0)	7.0%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.4%
(1, 0, 1, 0)	2.8%
(2, 0, 1, 0)	3.6%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.9%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	3.0%
(2, 1, 1, 0)	2.9%

13.1.3 Some interesting errors

Some of the most interesting error plays in the DET 2023 home games:

- In the game against the Boston Red Sox on 2023-04-08, there were 3 errors but only one batter was at bat during each error.
- The game ARI vs DET on 20230609 had 3 errors, two by catcher's interference.
- The game HOU vs DET on 2023-08-25 had 4 errors, including two with Peña at bat. One of those with Peña at bat was a consecutive error.

13.1.4 Relative team rankings

We have 81 home games between DET and some visiting MLB team (vt). A number of teams did not play DET in the regular season. However, of those

that did, DET *lost* more games than won against¹:

ATL (2/3), CHN (2/3), CIN (2/3), HOU (2/3), OAK (2/3), SDN (2/3), SEA
(2/3), TBA (2/3),
TEX (2/3), TOR (2/3), ANA (3/3), ARI (3/3), BOS (3/3), BAL (3/4), NYA
(3/4) .

The teams that *split* their series were:

PIT (1/2) .

The teams that DET *won* more than lost against

NYN (0/3), SFN (0/3), KCA (1/6), CLE (2/6), CHA (2/7), MIN (3/7) .

The teams that did not play in any of the 81 home games in DET 2023 were:

COL, LAN, MIA, MIL, PHI, SLN, WAS .

13.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for DET 2023 is as follows [BA].

position	player	games played
1 (C)	Jake Rogers	101
3 (1B)	Spencer Torkelson	156
4 (2B)	Andy Ibáñez	78
5 (3B)	Nick Maton	54
6 (SS)	Javier Beáz	131
7 (LF)	Akil Baddoo	97
8 (CF)	Riley Greene	81
9 (RF)	Kerry Carpenter	81

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2019 Tigers home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

¹We recorded that vt's W/T record in parentheses (W = numbers of games vt won against DET, T = Total home games vt played vs the Tigers at Detroit.

13.2.1 April

	game	BOS vs DET on 20230406
	state before	(2, 0, 0, 0, 5, 2, 3, 1)
1.	error	play,5,1,torks001,00,X,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 5, 2, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 703 *In the bottom of the 5th inning, with bases empty, DET batter Spencer Torkelson hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs DET on 2023-04-08

position	innings	player	position	innings	player
1	1	Tanner Houck	1	1	Joey Wentz
1	6	Zack Kelly	1	2	Garrett Hill
1	8	Kaleb Ort	1	6	Tyler Alexander
1	9	Ryan Brasier	1	9	Zach McKinstry
2	1	Connor Wong	2	1	Jake Rogers
3	1	Justin Turner	3	1	Spencer Torkelson
3	8	Triston Casas	4	1	Zach McKinstry
4	1	Christian Arroyo	4	9	Nick Maton
5	1	Rafael Devers	5	1	Nick Maton
5	8	Yu Chang	5	9	Ryan Kreidler
6	1	Enrique Hernandez	6	1	Javier Baez
7	1	Rob Refsnyder	7	1	Akil Baddoo
8	1	Adam Duvall	8	1	Riley Greene
9	1	Alex Verdugo	9	1	Matt Vierling
9	9	Raimel Tapia			

	game	BOS vs DET on 20230408
	state before	(1, 0, 0, 0, 6, 8, 2, 1)
2.	error	play,6,1,baezj001,00,X,C/E2.B-1
	state after	(1, 1, 0, 0, 6, 8, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 704 This game had 3 errors, each with the same batter. The Boston Red Sox beat the Tigers 14-5.

In the bottom of the 6th inning, with bases empty, DET batter Javier Baez had a 0-0 count. Catcher's interference allowed the batter to take 1st base.

	game	BOS vs DET on 20230408
	state before	(2, 0, 0, 0, 8, 12, 2, 1)
3.	error	play,8,1,baezj001,02,CFX,E6/G6.B-1
	state after	(2, 1, 0, 0, 8, 12, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 705 *In the bottom of the 8th inning, with bases empty, DET batter Javier Baez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs DET on 20230408
	state before	(2, 1, 0, 1, 9, 14, 4, 1)
4.	error	play,9,1,baezj001,02,C*SX,E6/TH/G6.3-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 9, 14, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 706 *In the bottom of the 9th inning, with runners on 1st and 3rd, DET batter Javier Baez hit a ground ball to short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st base.*

	game	BOS vs DET on 20230409
	state before	(1, 1, 0, 0, 5, 2, 1, 0)
5.	error	play,5,0,turnj001,01,1CX,FC6/G6.1-2(E6/TH);B-1
	state after	(1, 1, 1, 0, 5, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 707 *In the top of the 5th inning, with a runner on 1st, BOS batter Justin Turner hit to fielder's choice at short. A throwing error by the shortstop allowed runner on 1st to safely reach 2nd and the batter to reach 1st base.*

	game	SFN vs DET on 20230414
	state before	(2, 1, 0, 0, 9, 4, 4, 0)
6.	error	play,9,0,estr001,12,.1CBFFX,E7/F7S.1-3;B-2
	state after	(2, 0, 1, 1, 9, 4, 4, 0)
	fielder	7 (left field on the home team)

Summary 708 *In the top of the 9th inning, with a runner on 1st, SFN batter Thairo Estrada hit a fly ball to shallow left field. An error by the left fielder allowed the runner on 1st to reach 3rd and the batter to reach 2nd base.*

	game	SFN vs DET on 20230415
	state before	(1, 0, 1, 0, 5, 6, 2, 1)
7.	error	play,5,1,haase001,01,SX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 5, 6, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 709 This game had 2 errors. This game had a *consecutive error* because the batter after Haase in the game was Vierling (see the error list below). The Giants lost to the Tigers 6-7.

In the bottom of the 5th inning, with a runner on 2nd, DET batter Eric Haase hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	SFN vs DET on 20230415
	state before	(1, 1, 0, 1, 5, 6, 2, 1)
8.	error	play,5,1,vierm001,21,BBSX,FC6/G78.3-H(UR);1-2(E6);B-1
	state after	(1, 1, 1, 0, 5, 6, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 710 *In the bottom of the 5th inning, with runners on 1st and 3rd, DET batter Matt Vierling hit to a fielder's choice at short. The runner on 3rd scored (1 RBI). An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CLE vs DET on 20230418
	state before	(1, 0, 0, 0, 4, 0, 1, 0)
9.	error	play,4,0,ramij003,10,BX,S7/L7LS.B-2(E7)
	state after	(1, 0, 1, 0, 4, 0, 1, 0)
	fielder	7 (left field on the home team)

Summary 711 *In the top of the 4th inning, with bases empty, CLE batter Jose Ramirez hit a line drive single to shallow left field. An error by the left fielder allowed the batter to reach 2nd.*

	game	BAL vs DET on 20230429
	state before	(0, 1, 0, 1, 6, 6, 0, 1)
10.	error	play,6,1,baezj001,01,CX,FC/G6.3-H;1-3(E6);B-1
	state after	(0, 1, 0, 1, 6, 6, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 712 *In the bottom of the 6th inning, with runners on 1st and 3rd, DET batter Javier Baez hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	BAL vs DET on 20230430
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
11.	error	play,2,0,hendg002,32,CBCB1.*B>X,E4/G4.1-2;B-1
	state after	(0, 1, 1, 0, 2, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 713 *In the top of the 2nd inning, with a runner on 1st, BAL batter Gunnar Henderson hit a ground ball to 2nd base. An error by the second baseman allowed the runner on 1st to advance to 2nd and batter to reach 1st base.*

13.2.2 May

Visiting/Home team rosters for NYN vs DET on 2023-05-03 (1/2)

position	innings	player	position	innings	player
1	1	Joey Lucchesi	1	1	Joey Wentz
1	5	Jimmy Yacabonis	1	7	Will Vest
1	8	Adam Ottavino	1	8	Tyler Alexander
2	1	Tomas Nido	1	9	Alex Lange
3	1	Pete Alonso	2	1	Jake Rogers
4	1	Jeff McNeil	3	1	Spencer Torkelson
5	1	Brett Baty	4	1	Jonathan Schoop
6	1	Francisco Lindor	4	8	Nick Maton
7	1	Tommy Pham	4	9	Zach McKinstry
8	1	Brandon Nimmo	5	1	Zack Short
9	1	Starling Marte	5	9	Nick Maton
			6	1	Javier Baez
			7	1	Andy Ibanez
			7	8	Zach McKinstry
			7	9	Akil Baddoo
			8	1	Riley Greene
			9	1	Matt Vierling

	game	NYN vs DET on 20230503 (1/2)
	state before	(0, 0, 0, 0, 5, 2, 4, 0)
12.	error	play,5,0,batyb001,00,X,S89/L8S+.B-2(E8)
	state after	(0, 0, 1, 0, 5, 2, 4, 0)
	fielder	8 (center field on the home team)

Summary 714 This first game of a doubleheader had 2 errors. The Tigers beat the Mets 5-6.

In the top of the 5th inning, with bases empty, NYN batter Brett Baty hit a hard line drive single to shallow center field. An error by the center fielder allowed the batter to reach 1st base.

	game	NYN vs DET on 20230503 (1/2)
	state before	(0, 0, 1, 0, 5, 2, 4, 0)
13.	error	play,5,0,nidot001,00,X,S7/G6.2-H(E7/TH)(NR);B-2
	state after	(0, 0, 1, 0, 5, 3, 4, 0)
	fielder	7 (left field on the home team)

Summary 715 This is a *consecutive* error, as Baty is listed just before Nido in the batting lineup.

In the top of the 5th inning, with a runner on 2nd, NYN batter Tomas Nido hit a ground ball single to short. A throwing error by the left fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd base.

	game	NYN vs DET on 20230503 (2/2)
	state before	(0, 0, 0, 0, 3, 0, 3, 0)
14.	error	play,3,0,alvaf001,00,X,E5/G7S.B-1
	state after	(0, 1, 0, 0, 3, 0, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 716 This second game of a doubleheader. The Tigers beat the Mets 1-8.

In the top of the 3rd inning, with bases empty, NYN batter Francisco Alvarez hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.

Visiting/Home team rosters for SEA vs DET on 2023-05-12

position	innings	player	position	innings	player
1	1	Marco Gonzales	1	1	Matthew Boyd
1	7	Trevor Gott	1	2	Mason Englert
1	8	Justin Topa	1	5	Will Vest
1	9	Tayler Saucedo	1	6	Tyler Holton
2	1	Tom Murphy	1	7	Jose Cisnero
3	1	Ty France	1	8	Tyler Alexander
4	1	Jose Caballero	2	1	Jake Rogers
5	1	Eugenio Suarez	3	1	Spencer Torkelson
6	1	J.P. Crawford	4	1	Jonathan Schoop
7	1	Jarred Kelenic	4	8	Nick Maton
8	1	Julio Rodriguez	5	1	Andy Ibanez
9	1	Teoscar Hernandez	5	9	Zach McKinstry
			6	1	Javier Baez
			7	1	Eric Haase
			8	1	Riley Greene
			9	1	Matt Vierling
			9	8	Zach McKinstry
			9	9	Akil Baddoo

	game	SEA vs DET on 20230512
	state before	(1, 1, 0, 1, 2, 5, 0, 0)
15.	error	play,2,0,kelej001,01,.C.X,36(1)/F0/G3S.3-H(E6/TH)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 2, 6, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 717 This game had 3 errors. The Mariners beat the Tigers

9-2.

In the top of the 2nd inning, with runners on 1st and 3rd, SEA batter Jarred Kelenic hit a ground ball to shallow right infield. The shortstop (with assist by first baseman) put out the runner from 1st at 2nd base. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), and the batter to reach 1st base.

	game	SEA vs DET on 20230512
	state before	(1, 0, 0, 0, 4, 6, 0, 0)
16.	error	play,4,0,rodrj007,00,X,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 4, 6, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 718 *In the top of the 4th inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.*

	game	SEA vs DET on 20230512
	state before	(0, 1, 0, 0, 8, 7, 2, 1)
17.	error	play,8,1,torks001,02,CCX,FC1/G1.1-3(E1/TH);B-1
	state after	(0, 1, 0, 1, 8, 7, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 719 *In the bottom of the 8th inning, with a runner on 1st, DET batter Spencer Torkelson hit a into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	SEA vs DET on 20230514
	state before	(2, 1, 0, 0, 3, 3, 2, 1)
18.	error	play,3,1,torks001,01,CS,WP.1-3(E2/TH)
	state after	(2, 0, 0, 1, 3, 3, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 720 *In the bottom of the 3rd inning, with a runner on 1st, DET batter Spencer Torkelson had a 0-1 count. During the next pitch, which was a wild pitch², the runner on 1st advanced to 2nd. A throwing error by the catcher allowed the runner to further advance to 3rd. Torkelson is still at bat, now with a 0-2 count.*

	game	PIT vs DET on 20230516
	state before	(1, 1, 0, 1, 3, 0, 2, 1)
19.	error	play,3,1,cabrm001,21,BBF1,P03(E1/TH).3-H(NR);1-3
	state after	(1, 0, 0, 1, 3, 0, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 721 *In the bottom of the 3rd inning, with runners on 1st and 3rd, DET batter Miguel Cabrera has a 2-1 count. The pitcher attempted to pickoff the runner on 3rd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI) and the runner from 1st to reach 3rd. Cabrera is still at bat with a 2-1 count.*

	game	PIT vs DET on 20230517
	state before	(0, 1, 0, 0, 7, 4, 0, 0)
20.	error	play,7,0,mccua001,01,FX,FC6/G6.1X2(6E4);B-1
	state after	(0, 1, 1, 0, 7, 4, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 722 *In the top of the 7th inning, with a runner on 1st, PIT batter Andrew McCutchen hit a ground ball to a fielder's choice at short. An error by the second baseman allowed the runner from 1st to safely reach 2nd and the batter to reach 1st base.*

²I assume that the batter swung at the wild pitch, itself possibly an intentional pitch out, so that the count before the swing was 0-1. For the curious, the next play was play,3,1,torks001,22,CS.BBT,K.

	game	CHA vs DET on 20230525
	state before	(2, 1, 0, 0, 7, 2, 5, 1)
21.	error	play,7,1,mckiz001,22,BBCS1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 7, 2, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 723 This game had 2 errors. The final score was 2-7.

In the bottom of the 7th inning, with a runner on 1st, DET batter Zach McKinstry has a 2-2 count. The pitcher attempted to pick off the runner on 1st. An error by the pitcher allowed the runner on 1st to advance to 3rd. McKinstry is still at bat with a 2-2 count.

	game	CHA vs DET on 20230525
	state before	(0, 0, 0, 0, 9, 2, 7, 0)
22.	error	play,9,0,andet001,21,.CBBF,FLE3
	state after	(0, 0, 0, 0, 9, 2, 7, 0)
	fielder	3 (1st baseman on the home team)

Summary 724 This game has an infinity play!

In the top of the 9th inning, with bases empty and a 2-1 count, CHA batter Tim Anderson hit a pop fly foul of the 1st base line. An error by the first baseman allowed the battter to remain at bat, now with a 2-2 count.

	game	CHA vs DET on 20230526
	state before	(0, 1, 0, 0, 6, 8, 1, 1)
23.	error	play,6,1,maton001,11,BCX,FC4/G4M.1-2(E4);B-1
	state after	(0, 1, 1, 0, 6, 8, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 725 This game had 2 errors. The final score was 12-3.

In the bottom of the 6th inning, with a runner on 1st, DET batter Nick Maton hit to a fielder's choice at deep 2nd base. An error by the seond baseman allowed the runner on 1st to safely reach 2nd and the batter to reach 1st base.

	game	CHA vs DET on 20230526
	state before	(2, 1, 1, 0, 8, 9, 3, 0)
24.	error	play,8,0,robel002,02,FSX,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 8, 9, 3, 0)
	fielder	2 (catcher on the home team)

Summary 726 *In the top of the 8th inning, with runners on 1st and 2nd, CHA batter Luis Robert had a 0-2 count. Catcher's interference allowed the batter to reach 1st base. The bases are loaded.*

	game	CHA vs DET on 20230527
	state before	(2, 1, 1, 0, 7, 0, 2, 0)
25.	error	play,7,0,burgj001,12,.FBC.FX,E4/TH/G34+.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 7, 0, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 727 *In the top of the 7th inning, with runners on 1st and 2nd, CHA batter Jake Burger hit a hard ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	TEX vs DET on 20230529
	state before	(1, 1, 1, 0, 7, 5, 0, 0)
26.	error	play,7,0,lowen001,00,>C,SB3.1-2(E2/TH)
	state after	(1, 0, 1, 1, 7, 5, 0, 0)
	fielder	2 (catcher on the home team)

Summary 728 *In the top of the 7th inning, with runners on 1st and 2nd, TEX batter Nathaniel Lowe had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd. A throwing error by the catcher allowed the runner on 1st to advance to 2nd. Lowe is still at bat, now with a 0-1 count.*

	game	TEX vs DET on 20230530
	state before	(1, 0, 0, 0, 3, 1, 1, 0)
27.	error	play,3,0,semim001,11,SBX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 3, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 729 This game had 2 errors. The Texas Rangers topped the Tigers 10-6.

In the top of the 3rd inning, with bases empty, TEX batter Marcus Semien hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.

	game	TEX vs DET on 20230530
	state before	(2, 1, 1, 1, 9, 9, 6, 0)
28.	error	play,9,0,jankt001,22,BCC.BFX,E6/TH/G4M.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 9, 10, 6, 0)
	fielder	6 (shortstop on the home team)

Summary 730 *In the top of the 9th inning, with bases loaded, TEX batter Travis Jankowski hit a ground ball to deep 2nd base. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st base. The bases are still loaded.*

	game	TEX vs DET on 20230531
	state before	(1, 0, 0, 0, 2, 0, 0, 1)
29.	error	play,2,1,nevit001,01,CX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 731 *In the bottom of the 2nd inning, with bases empty, DET batter Tyler Nevin hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

13.2.3 June

Visiting/Home team rosters for ARI vs DET on 2023-06-09

position	innings	player	position	innings	player
1	1	Merrill Kelly	1	1	Michael Lorenzen
1	7	Drey Jameson	1	7	Will Vest
1	9	Miguel Castro	1	8	Garrett Hill
2	1	Jose Herrera	2	1	Jake Rogers
3	1	Christian Walker	3	1	Spencer Torkelson
4	1	Ketel Marte	4	1	Zack Short
5	1	Josh Rojas	5	1	Nick Maton
6	1	Geraldo Perdomo	6	1	Javier Baez
7	1	Corbin Carroll	7	1	Akil Baddoo
8	1	Jake McCarthy	7	5	Andy Ibanez
9	1	Pavin Smith	8	1	Jake Marisnick
			9	1	Zach McKinstry

	game	ARI vs DET on 20230609
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
30.	error	play,1,1,baezj001,02,.SFX,S5/G56S.BX3(25)(E5/TH)
	state after	(2, 0, 0, 0, 1, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 732 This game had 3 errors, 2 by catcher's interference. The final score was 11-6.

In the bottom of the 1st inning, with bases empty, DET batter Javier Baez hit a ground ball single to shallow 3rd base. Due to a throwing error by the third baseman, the batter reached 2nd. However, the batter was put out at 3rd by the third baseman (assisted by the shortstop).

	game	ARI vs DET on 20230609
	state before	(2, 1, 0, 1, 7, 4, 2, 0)
31.	error	play,7,0,smitp002,21,.BBC1>X,C/E2.1-2;B-1
	state after	(2, 1, 1, 1, 7, 4, 2, 0)
	fielder	2 (catcher on the home team)

Summary 733 *In the top of the 7th inning, with runners on 1st and 3rd, ARI batter Pavin Smith had a 2-1 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

	game	ARI vs DET on 20230609
	state before	(0, 1, 0, 0, 8, 9, 3, 0)
32.	error	play,8,0,rojaj001,11,.C*BX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 8, 9, 3, 0)
	fielder	2 (catcher on the home team)

Summary 734 *In the top of the 8th inning, with a runner on 1st, ARI batter Josh Rojas had a 1-1 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	ARI vs DET on 20230611
	state before	(2, 1, 1, 0, 9, 4, 4, 0)
33.	error	play,9,0,walkc002,12,CF.BX,D7/L7LS.2-H;1-H(E7) (NR)
	state after	(2, 0, 1, 0, 9, 6, 4, 0)
	fielder	7 (left field on the home team)

Summary 735 *In the top of the 9th inning, with runners on 1st and 2nd, ARI batter Christian Walker hit a line drive double to shallow left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the runner on 1st to also score (unearned, no RBI). The batter is on 2nd.*

34.	game	ATL vs DET on 20230612
	state before	(1, 0, 1, 0, 6, 3, 0, 0)
	error	play,6,0,harrm004,12,.1>C.BCX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 6, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 736 This game had 3 errors. The final score was 5-6.
In the top of the 6th inning, with a runner on 2nd, ATL batter Michael Harris hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

35.	game	ATL vs DET on 20230612
	state before	(2, 1, 1, 0, 9, 5, 4, 1)
	error	play,9,1,shorz001,12,CBFX,S7/L7D.2-H;1-3;B-2(E7/TH)
	state after	(2, 0, 1, 1, 9, 5, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 737 *In the bottom of the 9th inning, with runners on 1st and 2nd, DET batter Zack Short hit a line drive single to deep left field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to take 2nd.*

36.	game	ATL vs DET on 20230612
	state before	(0, 0, 1, 0, 10, 5, 5, 1)
	error	play,10,1,rogej004,01,...LX,FC/SH/BG56S-.2-3;B-1(E5)
	state after	(0, 1, 0, 1, 10, 5, 5, 1)
	fielder	5 (left field on the visiting team)

Summary 738 *In the bottom of the 10th inning, with a runner on 2nd, DET batter Jake Rogers hit a soft sacrafice bunt into a fielder's choice at shalllow 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the batter to reach 1st.*

	game	ATL vs DET on 20230614
	state before	(1, 1, 1, 0, 4, 2, 4, 0)
37.	error	play,4,0,arcio002,20,BBX,S7/L78.2-H;1-3(E7);B-2
	state after	(1, 0, 1, 1, 4, 3, 4, 0)
	fielder	7 (left field on the home team)

Summary 739 This game had 2 errors. The final score was 10-7.

In the top of the 4th inning, with runners on 1st and 2nd, ATL batter Orlando Arcia hit a line drive single to left center field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the runner on 1s3 to advance to 3rd and the batter to reach 2nd base.

	game	ATL vs DET on 20230614
	state before	(1, 0, 1, 1, 4, 3, 4, 0)
38.	error	play,4,0,pillk001,32,BBCFFFFBF.FX,E5/G56.3-H;2-3;B-1
	state after	(1, 1, 0, 1, 4, 4, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 740 *In the top of the 4th inning, with runners on 2nd and 3rd, ATL batter Kevin Pillar hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st base.*

	game	KCA vs DET on 20230619
	state before	(2, 0, 1, 0, 5, 3, 0, 0)
39.	error	play,5,0,wittb002,32,BCC.BBX,S6/G6.2-H(E6/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 5, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 741 *In the top of the 5th inning, with a runner on 2nd, KCA batter Bobby Witt hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	KCA vs DET on 20230620
	state before	(1, 1, 0, 0, 4, 0, 0, 0)
40.	error	play,4,0,bland001,00,X,E5/G56.1-2;B-1
	state after	(1, 1, 1, 0, 4, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 742 *In the top of the 4th inning, with a runner on 1st, KCA batter Dairon Blanco hit a ground ball to 3rd base. An error by the 3rd baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	KCA vs DET on 20230621
	state before	(0, 0, 0, 0, 8, 3, 5, 1)
41.	error	play,8,1,torks001,10,...BX,E4/F4.B-3
	state after	(0, 0, 0, 1, 8, 3, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 743 *In the bottom of the 8th inning, with bases empty, DET batter Spencer Torkelson hit a fly ball to 2nd base. An error by the second baseman allowed the batter to reach 3rd.*

	game	MIN vs DET on 20230623
	state before	(2, 1, 0, 0, 3, 3, 0, 1)
42.	error	play,3,1,torks001,02,CS+1,P01(E2/TH).1-2
	state after	(2, 0, 1, 0, 3, 3, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 744 This game had 3 errors. The final score was 4-1. *In the bottom of the 3rd inning, with a runner on 1st, DET batter Spencer Torkelson has a 0-2 count. The catcher attempted a pickoff at 1st. A throwing error by the catcher allowed the runner on 1st to advance to 2nd. Torkelson is still at bat with a 0-2.*

	game	MIN vs DET on 20230623
	state before	(0, 1, 1, 0, 6, 3, 0, 0)
43.	error	play,6,0,keplm001,21,*BBSX,E8/F8S.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 6, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 745 *In the top of the 6th inning, with runners on 1st and 2nd, MIN batter Max Kepler hit a fly ball to shallow center field. An error by the center fielder allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	MIN vs DET on 20230623
	state before	(2, 0, 0, 0, 9, 4, 1, 0)
44.	error	play,9,0,corrc001,11,FBX,E5/G56.B-1
	state after	(2, 1, 0, 0, 9, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 746 *In the top of the 9th inning, with bases empty, MIN batter Carlos Correa hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	MIN vs DET on 20230624
	state before	(2, 1, 1, 0, 8, 2, 3, 0)
45.	error	play,8,0,solad001,11,..CBX,FC/G5D.2-3;1-2(E5);B-1
	state after	(2, 1, 1, 1, 8, 2, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 747 *In the top of the 8th inning, with runners on 1st and 2nd, MIN batter Donovan Solano hit a ground ball into a fielder's choice at deep 3rd base. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

46.	game	MIN vs DET on 20230625
	state before	(2, 1, 1, 0, 8, 2, 3, 0)
	error	play,8,0,castw003,12,*BCSFFX,E5/TH/G56.2-H(NR) (UR);1-3;B-2
	state after	(2, 0, 1, 1, 8, 3, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 748 *In the top of the 8th inning, with runners on 1st and 2nd, MIN batter Willi Castro hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the second baseman to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd base.*

47.	game	OAK vs DET on 20230705
	state before	(1, 0, 0, 0, 6, 5, 0, 0)
	error	play,6,0,langs001,12,BSF.X,S5/G56.B-2(E5/TH)
	state after	(1, 0, 1, 0, 6, 5, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 749 *In the top of the 6th inning, with bases empty, OAK batter Shea Langeliers hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

48.	game	OAK vs DET on 20230706
	state before	(1, 1, 1, 0, 3, 0, 2, 1)
	error	play,3,1,carpk001,00,X,S7/L7+.2-H;1-3(E7)
	state after	(1, 1, 0, 1, 3, 0, 3, 1)
	fielder	7 (left fielder on the visiting team)

Summary 750 This game had 2 errors. The final score was 0-9. *In the bottom of the 3rd inning, with runners on 1st and 2nd, DET batter Kerry Carpenter hit a hard line drive single to left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the runner on 1st to advance to 3rd. The batter is on 1st.*

	game	OAK vs DET on 20230706
	state before	(0, 0, 0, 0, 8, 0, 9, 0)
49.	error	play,8,0,diazj008,00,....X,S5/G56.B-2(E5/TH)
	state after	(0, 0, 1, 0, 8, 0, 9, 0)
	fielder	5 (3rd baseman on the home team)

Summary 751 *In the top of the 8th inning, with bases empty, OAK batter Jordan Diaz hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	TOR vs DET on 20230707
	state before	(1, 0, 0, 0, 7, 7, 1, 0)
50.	error	play,7,0,chapm001,12,BCC.X,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 7, 7, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 752 *In the top of the 7th inning, with bases empty, TOR batter Matt Chapman hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.*

	game	SDN vs DET on 20230721
	state before	(0, 1, 0, 0, 2, 3, 0, 1)
51.	error	play,2,1,baejj001,00,X,S8/G4M.1-3;B-2(E8)
	state after	(0, 0, 1, 1, 2, 3, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 753 *In the bottom of the 2nd inning, with a runner on 1st, DET batter Javier Baez hit a ground ball single to shallow center field. The runner on 1st advanced to third. An error by the center fielder allowed the batter to reach 2nd base.*

	game	SDN vs DET on 20230722
	state before	(1, 1, 0, 0, 6, 10, 3, 1)
52.	error	play,6,1,cabrm001,32,CBFBBX,E6/G6.1-3;B-1
	state after	(1, 1, 0, 1, 6, 10, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 754 *In the bottom of the 6th inning, with a runner on 1st, DET batter Miguel Cabrera hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	SDN vs DET on 20230723
	state before	(2, 0, 0, 0, 3, 0, 1, 0)
53.	error	play,3,0,kim-h002,12,CBCX,E5/G56.B-1
	state after	(2, 1, 0, 0, 3, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 755 This game had 2 errors. The final score was 1-3. *In the top of the 3rd inning, with bases empty, SDN batter Ha-Seong Kim hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	SDN vs DET on 20230723
	state before	(2, 0, 1, 0, 9, 1, 3, 0)
54.	error	play,9,0,cronj001,12,CF>*B.X,E6/G6.2-3;B-1
	state after	(2, 1, 0, 1, 9, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 756 *In the top of the 9th inning, with a runner on 2nd, SDN batter Jake Cronenworth hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	ANA vs DET on 20230725
	state before	(2, 1, 1, 0, 5, 4, 1, 0)
55.	error	play,5,0,monim001,00,X,E3/G3.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 5, 4, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 757 This game had 2 errors. The final score was 7-6.
In the top of the 5th inning, with runners on 1st and 2nd, ANA batter Mickey Moniak hit a ground ball to 1st base. An error by the first baseman allowed the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are now loaded.

	game	ANA vs DET on 20230725
	state before	(0, 0, 0, 0, 9, 6, 2, 1)
56.	error	play,9,1,baezj001,12,.BCSX,E4/TH/G4.B-1
	state after	(0, 1, 0, 0, 9, 6, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 758 *In the bottom of the 9th inning, with bases empty, DET batter Javier Baez hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	ANA vs DET on 20230727 (1/2)
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
57.	error	play,1,0,rengl001,00,X,E4/G34.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 759 This game 1 of a doubleheader game had 1 error. The Angels shut out the Tigers 6-0.
In the top of the 1st inning, with bases empty, ANA batter Luis Rengifo hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.

Visiting/Home team rosters for ANA vs DET on 2023-07-27 (2/2)

position	innings	player	position	innings	player
1	1	Patrick Sandoval	1	1	Matt Manning
1	6	Jacob Webb	1	6	Trey Wingenter
1	7	Gerardo Reyes	1	8	Chasen Shreve
1	8	Aaron Loup	1	9	Brendan White
1	8	Zack Weiss	2	1	Jake Rogers
1	9	Jaime Barria	3	1	Spencer Torkelson
2	1	Matt Thaiss	4	1	Andy Ibanez
3	1	Eduardo Escobar	5	1	Zack Short
4	1	Luis Rengifo	5	8	Nick Maton
4	7	Andrew Velazquez	6	1	Javier Baez
5	1	Mike Moustakas	7	1	Zach McKinstry
5	7	Luis Rengifo	8	1	Matt Vierling
6	1	Zach Neto	9	1	Kerry Carpenter
7	1	Taylor Ward			
8	1	Mickey Moniak			
9	1	Hunter Renfroe			

	game	ANA vs DET on 20230727 (2/2)
	state before	(0, 0, 0, 0, 3, 5, 0, 0)
58.	error	play,3,0,wardt002,02,SS.X,E6/TH/G6.B-2
	state after	(0, 0, 1, 0, 3, 5, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 760 This game 2 of a doubleheader had 3 errors, two with Vierling at bat. The final score was 11-4.

In the top of the 3rd inning, with bases empty, ANA batter Taylor Ward hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 2nd base.

	game	ANA vs DET on 20230727 (2/2)
	state before	(0, 1, 0, 0, 3, 5, 0, 1)
59.	error	play,3,1,vierm001,11,CBX,E1/TH/G1.1-3;B-1
	state after	(0, 1, 0, 1, 3, 5, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 761 *In the bottom of the 3rd inning, with a runner on 1st, DET batter Matt Vierling hit a ground ball to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	ANA vs DET on 20230727 (2/2)
	state before	(2, 0, 0, 1, 7, 8, 2, 1)
60.	error	play,7,1,vierm001,22,CFB*BX,E6/G6M.3-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 7, 8, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 762 *In the bottom of the 7th inning, with a runner on 3rd, (the same) DET batter Matt Vierling hit a ground ball to shallow left field. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st base.*

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	game	TBA vs DET on 20230804
	state before	(2, 1, 1, 0, 1, 0, 0, 0)
61.	error	play,1,0,parei001,11,CBX,D7/L7L.2-H;1-H;B-3(E6/TH)
	state after	(2, 0, 0, 1, 1, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 763 This game had 2 errors. The final score was 8-0. *In the top of the 1st inning, with runners on 1st and 2nd, TBA batter Isaac Paredes hit a line drive double to left field. The runners on 1st and 2nd both scored (2 RBI). A throwing error by the shrtstop allowed the batter to reach 3rd.*

	game	TBA vs DET on 20230804
	state before	(0, 0, 0, 0, 6, 4, 0, 0)
62.	error	play,6,0,lowej002,32,.BBFSBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 6, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 764 *In the top of the 6th inning, with bases empty, TBA batter Josh Lowe hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for TBA vs DET on 2023-08-05

position	innings	player	position	innings	player
1	1	Aaron Civale	1	1	Tarik Skubal
1	5	Kevin Kelly	1	6	Jose Cisnero
1	6	Robert Stephenson	1	7	Jason Foley
1	7	Jake Diekman	1	8	Tyler Holton
1	8	Jason Adam	1	9	Alex Lange
2	1	Rene Pinto	1	9	Beau Brieske
2	7	Christian Bethancourt	2	1	Jake Rogers
3	1	Yandy Diaz	3	1	Spencer Torkelson
4	1	Isaac Paredes	4	1	Andy Ibanez
5	1	Curtis Mead	5	1	Matt Vierling
6	1	Wander Franco	6	1	Zack Short
7	1	Randy Arozarena	7	1	Akil Baddoo
8	1	Jose Siri	8	1	Riley Greene
8	7	Manuel Margot	9	1	Kerry Carpenter
9	1	Manuel Margot			
9	7	Josh Lowe			

	game	TBA vs DET on 20230805
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
63.	error	play,2,0,ramih004,00,X,S8/G4M.B-2(E8)
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 765 This game had 3 errors. The final score was 2-4.
In the top of the 2nd inning, with bases empty, TBA batter Harold Ramirez hit a ground ball single to shallow center field. An error by the center fielder allowed the batter to reach 2nd base.

	game	TBA vs DET on 20230805
	state before	(1, 0, 1, 0, 2, 0, 0, 0)
64.	error	play,2,0,margm001,22,*BFBFX,S8/L89.2-H(UR);B-2(E8)
	state after	(1, 0, 1, 0, 2, 1, 0, 0)
	fielder	8 (center field on the home team)

Summary 766 *In the top of the 2nd inning, with a runner on 2nd, TBA batter Manuel Margot hit a line drive single to right center field. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the batter to reach 2nd.*

	game	TBA vs DET on 20230805
	state before	(0, 0, 0, 0, 7, 2, 4, 1)
65.	error	play,7,1,greer003,11,....CBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 7, 2, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 767 *In the bottom of the 7th inning, with bases empty, DET batter Riley Greene hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs DET on 20230806
	state before	(1, 1, 0, 1, 1, 0, 0, 0)
66.	error	play,1,0,arozr001,01,SX,FC4/G4M.3-H;1-2(E4);B-1
	state after	(1, 1, 1, 0, 1, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 768 *In the top of the 1st inning, with runners on 1st and 3rd, TBA batter Randy Arozarena hit a ground ball into a fielder's choice at shallow center field. The runner on 3rd scored (1 RBI). An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	MIN vs DET on 20230807
	state before	(0, 0, 0, 0, 3, 8, 0, 1)
67.	error	play,3,1,maton001,12,CFBX,E6/G4MD.B-1
	state after	(0, 1, 0, 0, 3, 8, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 769 *In the bottom of the 3rd inning, with bases empty, DET batter Nick Maton hit a ground ball to shallow center field. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIN vs DET on 20230808
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
68.	error	play,1,1,cabrm001,00,X,E4/TH/G4S.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 1, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 770 *In the bottom of the 1st inning, with a runner on 2nd, DET batter Miguel Cabrera hit a ground ball to shallow 2nd base. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st base.*

	game	MIN vs DET on 20230809
	state before	(2, 1, 1, 0, 1, 0, 0, 1)
69.	error	play,1,1,carpk001,11,FBX,S9/G34+.2-H;1-H(E9) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 1, 0, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 771 This game had 2 errors, with the same player at bat for both errors. The final score was 5-9.

In the bottom of the 1st inning, with runners on 1st and 2nd, DET batter Kerry Carpenter hit a hard ground ball single to the 1st/2nd base gap. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the runner on 1st to also score (unearned, no RBI) and the the batter to reach 2nd base.

	game	MIN vs DET on 20230809
	state before	(1, 0, 1, 0, 8, 4, 8, 1)
70.	error	play,8,1,carpk001,10,B.X,S4/G34.2-H(E4/TH) (NR) (UR)
	state after	(1, 1, 0, 0, 8, 4, 9, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 772 *In the bottom of the 8th inning, with a runner on 2nd, (the same) DET batter Kerry Carpenter hit a ground ball single to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI). The batter is on 1st.*

Visiting/Home team rosters for CHN vs DET on 2023-08-21

position	innings	player	position	innings	player
1	1	Javier Assad	1	1	Alex Faedo
1	6	Hayden Wesneski	1	7	Brendan White
1	7	Jose Cuas	1	8	Andrew Vasquez
1	8	Daniel Palencia	1	8	Will Vest
1	8	Michael Fulmer	1	9	Beau Brieske
1	9	Mark Leiter Jr.	2	1	Carson Kelly
2	1	Yan Gomes	3	1	Spencer Torkelson
2	9	Miguel Amaya	4	1	Zach McKinstry
3	1	Jeimer Candelario	5	1	Matt Vierling
4	1	Nico Hoerner	6	1	Javier Baez
5	1	Nick Madrigal	7	1	Akil Baddoo
6	1	Dansby Swanson	8	1	Parker Meadows
7	1	Mike Tauchman	9	1	Riley Greene
8	1	Cody Bellinger			
9	1	Seiya Suzuki			

71.	game	CHN vs DET on 20230821
	state before	(1, 1, 1, 0, 2, 0, 0, 0)
	error	play,2,0,candj002,12,SBFX,FC3/G9LS.2-H(NR);1-3(E3);B-1
	state after	(1, 1, 0, 1, 2, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 773 This game had 3 errors. The final score was 7-6.

In the top of the 2nd inning, with runners on 1st and 2nd, CHN batter Jeimer Candelario hit a ground ball into a fielder's choice at shallow right field. The runner from 2nd scored (no RBI). An error by the 1st baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.

	game	CHN vs DET on 20230821
	state before	(0, 0, 0, 0, 5, 4, 0, 1)
72.	error	play,5,1,baezj001,00,X,E6/G6MS.B-1
	state after	(0, 1, 0, 0, 5, 4, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 774 *In the bottom of the 5th inning, with bases empty, DET batter Javier Baez hit a ground ball to shallow left field. An error by the shortstop allowed the batter to reach 1st base.*

	game	CHN vs DET on 20230821
	state before	(2, 1, 1, 0, 8, 5, 2, 0)
73.	error	play,8,0,suzus001,12,CCBX,6E3/G6.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 8, 5, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 775 *In the top of the 8th inning, with runners on 1st and 2nd, CHN batter Seiya Suzuki hit a ground ball to short. An error by the first baseman (assist from shortstop) allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st base. The bases are loaded.*

	game	CHN vs DET on 20230822
	state before	(2, 1, 0, 0, 2, 0, 2, 0)
74.	error	play,2,0,candj002,01,TX,E3/G3.1-3;B-1
	state after	(2, 1, 0, 1, 2, 0, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 776 This game had 3 errors. The final score was 6-8. *In the top of the 2nd inning, with a runner on 1st, CHN batter Jeimer Candelario hit a ground ball to 1st base. An error by the first baseman allowed the runner from 1st to advance to 3rd and the batter to reach 1st base.*

	game	CHN vs DET on 20230822
	state before	(1, 1, 1, 0, 4, 4, 3, 1)
75.	error	play,4,1,vierm001,11,F.BX,S5/G5.2-H(E5/TH)(NR);1-3
	state after	(1, 1, 0, 1, 4, 4, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 777 *In the bottom of the 4th inning, with runners on 1st and 2nd, DET batter Matt Vierling hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (no RBI) and the runner on 1st to advance to 3rd. The batter took 1st.*

	game	CHN vs DET on 20230822
	state before	(1, 1, 0, 0, 6, 6, 7, 1)
76.	error	play,6,1,torks001,02,CSB,WP.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 6, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 778 *In the bottom of the 6th inning, with a runner on 1st, DET batter Spencer Torkelson had a 0-2 count. The next pitch was a wild pitch which allowed the runner on 1st to advance to 2nd. A throwing error by the catcher allowed the runner to reach 3rd. Torkelson is still at bat, now with a 1-2 count.*

Visiting/Home team rosters for HOU vs DET on 2023-08-25

position	innings	player	position	innings	player
1	1	Framber Valdez	1	1	Matt Manning
1	8	Bryan Abreu	1	7	Brendan White
1	9	Ryan Pressly	1	9	Alex Lange
2	1	Martin Maldonado	2	1	Jake Rogers
3	1	Jose Abreu	3	1	Spencer Torkelson
4	1	Jose Altuve	4	1	Andy Ibanez
5	1	Alex Bregman	4	9	Zack Short
6	1	Jeremy Peña	5	1	Zack Short
7	1	Yordan Alvarez	5	9	Matt Vierling
7	9	Chas McCormick	6	1	Javier Baez
8	1	Chas McCormick	7	1	Matt Vierling
8	9	Mauricio Dubon	7	9	Riley Greene
9	1	Kyle Tucker	8	1	Parker Meadows
			9	1	Riley Greene
			9	9	Kerry Carpenter

	game	HOU vs DET on 20230825
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
77.	error	play,3,0,abrej003,12,BS.CX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 779 This game had 4 errors. The final score was 1-4.

In the top of the 3rd inning, with bases empty, HOU batter Jose Abreu hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.

	game	HOU vs DET on 20230825
	state before	(0, 1, 0, 0, 3, 0, 0, 0)
78.	error	play,3,0,penaj004,12,CSBFX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 3, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 780 This is a *consecutive error* in the sense that Peña is listed immediately after Abreu in the Astros batting line up.

In the top of the 3rd inning, with a runner on 1st, HOU batter Jeremy Peña has a 1-2 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	HOU vs DET on 20230825
	state before	(2, 0, 0, 0, 5, 1, 0, 0)
79.	error	play,5,0,penaj004,00,X,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 5, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 781 *In the top of the 5th inning, with bases empty, HOU batter Jeremy Peña hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.*

	game	HOU vs DET on 20230825
	state before	(1, 0, 0, 0, 9, 1, 0, 0)
80.	error	play,9,0,tuckk001,12,CBFX,C/E2.B-1
	state after	(1, 1, 0, 0, 9, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 782 *In the top of the 9th inning, with bases empty, HOU batter Kyle Tucker had a 1-2 count. Catcher's interference allowed the batter to reach 1st base.*

	game	HOU vs DET on 20230826
	state before	(2, 1, 1, 0, 6, 6, 2, 0)
81.	error	play,6,0,diazy004,01,CX,FC6/G6.2-3;1-2(E6);B-1
	state after	(2, 1, 1, 1, 6, 6, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 783 *In the top of the 6th inning, with runners on 1st and 2nd, HOU batter Yainer Diaz hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

	game	HOU vs DET on 20230827
	state before	(1, 1, 0, 0, 8, 7, 0, 0)
82.	error	play,8,0,tuckk001,00,X,FC/G3.1-2(E3);B-1
	state after	(1, 1, 1, 0, 8, 7, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 784 *In the top of the 8th inning, with a runner on 1st, HOU batter Kyle Tucker hit a ground ball into a fielder's choice at 1st. An error by the first baseman allowed the runner on 1st to advance to 2nd base and the batter to reach 1st.*

	game	NYA vs DET on 20230828
	state before	(2, 1, 1, 0, 1, 0, 0, 0)
83.	error	play,1,0,kinei001,11,BCX,E3/G34.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 785 *In the top of the 1st inning, with runners on 1st and 2nd, NYA batter Isiah Kiner-Falefa hit a ground ball to 1st/2nd base gap. An error by the first baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st base. The bases are loaded.*

	game	NYA vs DET on 20230829
	state before	(1, 1, 0, 0, 3, 1, 0, 1)
84.	error	play,3,1,kellc002,01,CX,S8/L8S.1-3(E8);B-2
	state after	(1, 0, 1, 1, 3, 1, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 786 This game had 2 errors. The final score was 4-2.

In the bottom of the 3rd inning, with a runner on 1st, DET batter Carson Kelly hit a line drive single to shallow center field. An error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.

	game	NYA vs DET on 20230829
	state before	(1, 0, 0, 0, 6, 2, 0, 0)
85.	error	play,6,0,judga001,00,X,S5/G5.B-2(E5/TH)
	state after	(1, 0, 1, 0, 6, 2, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 787 *In the top of the 6th inning, with bases empty, NYA batter Aaron Judge hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	NYA vs DET on 20230831
	state before	(1, 1, 1, 0, 10, 3, 3, 1)
86.	error	play,10,1,shorz001,02,CCX,14(1)/FO/G1-.2-H(E4/TH)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 10, 3, 4, 1)
	fielder	4 (3rd baseman on the home team)

Summary 788 *In the bottom of the 10th inning, with runners on 1st and 2nd, DET batter Zack Short hit a soft ground ball to shallow infield. At 2nd, the second baseman (assisted by the pitcher) forced out the runner from 1st. A throwing error by the second baseman allowed the runner from 2nd to score (unearned, no RBI) and the batter to reach 1st base.*

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	game	CHA vs DET on 20230909
	state before	(0, 1, 0, 0, 4, 0, 1, 1)
87.	error	play,4,1,meadp001,00,X,FC4/G34.1-2(E4);B-1
	state after	(0, 1, 1, 0, 4, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 789 This game had 2 errors. The final score was 1-3.

In the bottom of the 4th inning, with a runner on 1st, DET batter Parker Meadows hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CHA vs DET on 20230909
	state before	(2, 1, 0, 0, 8, 1, 3, 0)
88.	error	play,8,0,jimée001,10,1BX,FC6/G56.1-2(E6);B-1
	state after	(2, 1, 1, 0, 8, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 790 *In the top of the 8th inning, with a runner on 1st, CHA batter Eloy Jimenez hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CHA vs DET on 20230910
	state before	(1, 1, 0, 1, 3, 0, 2, 1)
89.	error	play,3,1,lipca001,11,BCX,S9/G34.3-H;1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 3, 0, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 791 This game had 3 errors. The final score was 2-3.

In the bottom of the 3rd inning, with runners on 1st and 3rd, DET

batter Andre Lipcius hit a ground ball single to the 1st/2nd base gap. The runner on 3rd scored (1 RBI) and the runner from 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.

90.	game	CHA vs DET on 20230910
	state before	(1, 0, 0, 0, 5, 1, 3, 0)
	error	play,5,0,sheeg001,01,FX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 5, 1, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 792 In the top of the 5th inning, with bases empty, CHA batter Gavin Sheets hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.

91.	game	CHA vs DET on 20230910
	state before	(1, 0, 0, 0, 7, 2, 3, 1)
	error	play,7,1,torks001,20,BBX,E4/F34D.B-2
	state after	(1, 0, 1, 0, 7, 2, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 793 In the bottom of the 7th inning, with bases empty, DET batter Spencer Torkelson hit a fly ball to shallow right field. An error by the second baseman allowed the batter to reach 2nd base.

92.	game	CIN vs DET on 20230912
	state before	(0, 0, 0, 0, 2, 0, 3, 0)
	error	play,2,0,stept001,11,CBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 794 This game had 2 errors. The final score was 6-5. In the top of the 2nd inning, with bases empty, CIN batter Tyler Stephenson hit a ground ball to 3rd base. A throwing error by the 3rd baseman allowed the batter to reach 1st base.

	game	CIN vs DET on 20230912
	state before	(0, 1, 0, 0, 10, 6, 5, 0)
93.	error	play,10,0,vottj001,22,FFBBX,E4/G34.1-2;B-1
	state after	(0, 1, 1, 0, 10, 6, 5, 0)
	fielder	4 (3rd baseman on the home team)

Summary 795 *In the top of the 10th inning, with a runner on 1st, CIN batter Joey Votto hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner from 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for CIN vs DET on 2023-09-14

position	innings	player	position	innings	player
1	1	Derek Law	1	1	Reese Olson
1	2	Sam Moll	1	7	Will Vest
1	3	Ben Lively	1	8	Jason Foley
1	7	Buck Farmer	1	8	Tyler Holton
2	1	Tyler Stephenson	1	9	Jose Cisnero
3	1	Joey Votto	2	1	Jake Rogers
4	1	Jonathan India	3	1	Tyler Nevin
5	1	Spencer Steer	4	1	Zach McKinstry
5	8	Christian Encarnacion-Strand	5	1	Matt Vierling
6	1	Elly De La Cruz	6	1	Javier Baez
6	8	Noelvi Marte	7	1	Akil Baddoo
7	1	Will Benson	8	1	Parker Meadows
7	8	Spencer Steer	9	1	Kerry Carpenter
8	1	TJ Friedl			
9	1	Nick Martini			
9	8	Nick Senzel			

	game	CIN vs DET on 20230914
	state before	(1, 0, 0, 0, 2, 0, 1, 1)
94.	error	play,2,1,baezj001,00,X,E6/G6+.B-2
	state after	(1, 0, 1, 0, 2, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 796 This game had 3 errors. The final score was 2-8.
In the bottom of the 2nd inning, with bases empty, DET batter Javier Baez hit a hard ground ball to short. An error by the shortstop allowed the batter to reach 2nd base.

	game	CIN vs DET on 20230914
	state before	(0, 0, 0, 0, 7, 1, 3, 0)
95.	error	play,7,0,indij001,12,.CC.BFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 7, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 797 *In the top of the 7th inning, with bases empty, CIN batter Jonathan India hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CIN vs DET on 20230914
	state before	(1, 1, 0, 0, 7, 1, 3, 1)
96.	error	play,7,1,mckiz001,12,C>LB>X,E1/TH/G1.1-2;B-1
	state after	(1, 1, 1, 0, 7, 1, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 798 *In the bottom of the 7th inning, with a runner on 1st, DET batter Zach McKinstry hit a ground ball to shallow infield. A throwing error by the pitcher allowed the runner from 1st to advance to 2nd and the batter to reach 1st base.*

	game	KCA vs DET on 20230926
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
97.	error	play,2,0,velan001,01,CX,E1/TH/G2-.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 799 This game had 2 errors. The final score was 3-6.
In the top of the 2nd inning, with bases empty, KCA batter Nelson Velazquez hit a soft ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.

	game	KCA vs DET on 20230926
	state before	(1, 1, 0, 0, 6, 2, 0, 1)
98.	error	play,6,1,meadp001,01,CX,FC4/G34.1-2(E4);B-1
	state after	(1, 1, 1, 0, 6, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 800 *In the bottom of the 6th inning, with a runner on 1st, DET batter Parker Meadows hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the second fielder allowed the runner from 1st to advance to 2nd and the batter to reach 1st base.*

	game	KCA vs DET on 20230928
	state before	(2, 0, 1, 0, 1, 0, 0, 0)
99.	error	play,1,0,massm001,32,C1>FF>B.FBBFX,E3/G3.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 801 *In the top of the 1st inning, with a runner on 2nd, KCA batter Michael Massey hit a ground ball to 1st base. An error by the first baseman allowed the runner from 2nd to advance to 3rd and the batter to reach 1st base.*

100.	game	CLE vs DET on 20230929
	state before	(1, 0, 1, 0, 3, 6, 2, 0)
	error	play,3,0,stram002,10,BX,S9/L9S.2-H(E9)(NR);B-2
	state after	(1, 0, 1, 0, 3, 7, 2, 0)
	fielder	9 (right field on the home team)

Summary 802 *In the top of the 3rd inning, with a runner on 2nd, CLE batter Myles Straw hit a line drive single to shallow right field. An error by the right fielder allowed the runner from 2nd to score (no RBI) and the batter to reach 2nd base.*

101.	game	CLE vs DET on 20230930
	state before	(0, 0, 0, 0, 7, 0, 3, 1)
	error	play,7,1,meadp001,22,...BCBSX,E1/P2.B-1
	state after	(0, 1, 0, 0, 7, 0, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 803 This game had 2 errors. The Tigers shut out the Guardians 0-8.

In the bottom of the 7th inning, with bases empty, DET batter Parker Meadows hit a pop up to shallow infield. An error by the pitcher allowed the batter to reach 1st base.

102.	game	CLE vs DET on 20230930
	state before	(2, 1, 1, 1, 7, 0, 6, 1)
	error	play,7,1,kellc002,00,X,S7/L78.3-H(UR);2-H(TUR);1-3(E7/TH);B-2
	state after	(2, 0, 1, 1, 7, 0, 8, 1)
	fielder	7 (left field on the visiting team)

Summary 804 *In the bottom of the 7th inning, with bases loaded, DET batter Carson Kelly hit a line drive single to left center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

103.	game	CLE vs DET on 20231001
	state before	(0, 0, 0, 0, 4, 0, 3, 0)
	error	play,4,0,laurr001,21,C.BBX,3E1/G3.B-1
	state after	(0, 1, 0, 0, 4, 0, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 805 This game had 2 errors. The final score was 2-5.

In the top of the 4th inning, with bases empty, CLE batter Ramon Laureano hit a ground ball to 1st. An error by the pitcher (assist from the first baseman) allowed the batter to reach 1st base.

104.	game	CLE vs DET on 20231001
	state before	(2, 0, 1, 0, 7, 2, 5, 1)
	error	play,7,1,ibana001,02,*SCX,E5/G56.2XH(52);B-1
	state after	(0, 0, 0, 0, 8, 2, 5, 0)
	fielder	5 (3rd baseman on the visiting team)

Summary 806 *In the bottom of the 7th inning, with 2 outs and a runner on 2nd, DET batter Andy Ibanez hit a ground ball to 3rd base. The catcher (assist from the third baseman) tagged out the runner from 2nd at home³. An error by the third baseman allowed the batter to reach 1st base. With the out at home, the half-inning is over.*

³Note the overall error designation does not negate the out at home. Indeed, the score at the bottom of the 7th inning is already the final 9th inning score.

Chapter 14

Houston Astros (HOU)

The Houston Astros, founded in 1962, now compete in the American League West Division, having moved to the division in 2013 after spending their first 51 seasons in the National League. They have won their division in 7 of the last 8 years and have won 2 World Series titles, once in 2017 and again in 2022.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Houston against the Astros. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 HOU team is .697 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the Astros is above the MLB average (which is good).

In HOU 2023, there were 82 errors in the 81 home games, 37 by the Astros and the rest (45) by visiting teams. The total number of errors by the team (home and away) over the regular season of 2023 was 81. Since 83.9 was the MLB average, the Astros were slightly below average for errors (which is good).

14.1 Season totals

Effect of Removing Pivot Foot From Plate If the pitcher removes his pivot foot from contact with the pitcher's plate by stepping backward with that foot, he thereby becomes an infielder and if he makes a wild throw from that position, it shall be considered the same as a wild throw by any other infielder.

– **MLB Rule Book** 5.07(c)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

14.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	45
Runs by ht because of errors by vt:	8
Errors by ht:	37
Runs by vt because of errors by ht:	10

This is a relatively poor error-to-run conversion rate: $8/45 = 17.78\%$ on the part of the Astros (compared to $10/37 = 27.03\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 HOU season (home and away) has about $R/PA = 827/6253$ (about 13.2%) change of resulting in a run. Therefore, a randomly selected home game error is (for HOU) about 35% more likely to result in a run than a randomly selected at-bat, at least in this case (for many teams, it is over 200% more likely).

Note that the total number of errors (by either team) in a home game is 82, so the expected number of errors (by either team) in a randomly selected HOU 2023 home game is about 1.

Errors by position	vt	ht
errors due to pitcher:	4	4
to catcher:	9	7
to 1st baseman:	2	4
to 2nd baseman:	6	6
to 3rd baseman:	11	5
to shortstop:	10	7
to left fielder:	0	0
to center fielder:	0	3
to right fielder:	3	1

14.1.2 Error total by state

One **run** shall be scored each time a runner legally advances to and touches first, second, third and home base before three men are put out to end the inning.

Exception: A run is not scored if the runner advances to home base during a play in which the third out is made (1) by the batter-runner before he touches first base; (2) by any runner being forced out; or (3) by a preceding runner who is declared out because he failed to touch one of the bases.

– **MLB Rule Book** 5.08(a)

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	20	11	31
(lone) runner on 1st	9	15	24
(lone) runner on 2nd	6	2	7
(lone) runner on 3rd	1	2	3
two runners on base	6	5	11
bases loaded	3	2	5
total	45	37	82

A table of frequencies of most common errors by the Astros in the HOU 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	21.6%
(1, 0, 0, 0)	2	4.3%
(2, 0, 0, 0)	1	2.7%
(0, 1, 0, 0)	5	13.5%
(1, 1, 0, 0)	6	16.2%
(2, 1, 0, 0)	4	10.8%
(1, 1, 1, 0)	2	5.4%
(2, 1, 1, 0)	2	5.4%
(1, 1, 1, 1)	1	2.7%

A table of frequencies of most common errors by a visiting team in HOU 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	15.6%
(1, 0, 0, 0)	7	15.6%
(2, 0, 0, 0)	6	13.3%
(0, 1, 0, 0)	4	8.9%
(1, 1, 0, 0)	5	11.1%
(2, 1, 0, 0)	0	0%
(2, 0, 1, 0)	3	6.7%
(0, 1, 1, 0)	2	4.4%
(1, 1, 1, 0)	4	8.9%
(1, 1, 1, 1)	2	4.4%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Astros homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `HOU2023.eva`):

state	HOU2023 percentages
(0, 0, 0, 0)	24.0%
(1, 0, 0, 0)	16.9%
(2, 0, 0, 0)	13.1%
(0, 1, 0, 0)	5.8%
(1, 1, 0, 0)	7.2%
(2, 1, 0, 0)	7.3%
(0, 0, 1, 0)	1.8%
(1, 0, 1, 0)	3.1%
(2, 0, 1, 0)	4.1%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.6%
(2, 0, 0, 1)	1.0%
(0, 1, 1, 0)	1.2%
(1, 1, 1, 0)	2.8%
(2, 1, 1, 0)	3.3%

14.1.3 Some interesting errors

Some of the most interesting error plays in the HOU 2023 home games:

- The game SEA vs HOU on 2023-08-18 had 4 errors.
- The game BOS vs HOU on 2023-08-22 had 4 errors.
- The game SDN vs HOU on 2023-09-10 had 4 errors.

14.1.4 Relative team rankings

We have 81 home games between HOU and some visiting MLB team (vt). A number of teams did not play HOU in the regular season. However, of those that did, HOU lost more games than won against¹:

¹That team's W/T record in parentheses (W = numbers of games vt won against HOU, T = Total home games vt played vs Astros at HOU).

BAL (2/3), DET (2/3), MIN (2/3), PHI (2/3), SFN (2/3),
TBA (2/3), CIN (3/3), KCA (3/3), NYA (3/3), SEA (6/7)

and won more than lost against

NYN (1/3), SDN (1/3), TOR (1/3), WAS (1/3), OAK (2/6), ANA (2/7),
COL (0/2), CHN (0/3), CLE (0/3), OAK (2/6), ANA (2/7),

The teams that split their series were:

BOS (2/4), CHA (2/4), TEX (3/6) .

The teams that did not play in any of the 81 home games in HOU 2023 were:

ARI, ATL, LAN, MIA, MIL, PIT, SLN.

14.2 Game-by-game totals

It is interference by a batter or a runner when, after a third strike that is not caught by the catcher, the batter-runner clearly hinders the catcher in his attempt to field the ball. Such batter-runner is out, the ball is dead, and all other runners return to the bases they occupied at the time of the pitch. If a pitch that is not caught remains in the vicinity of home plate and it is inadvertently deflected by the batter or umpire, that ball is a dead ball and the runners should return to the bases they occupied at the time of the pitch (but if the pitch was strike three, the batter is out).

– **MLB Rule Book** 6.01(a)(1)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for HOU 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Martín Maldonado	116	.258
3 (1B)	José Abreu	134	.296
4 (2B)	Jose Altuve	87	.393
5 (3B)	Alex Bregman	160	.363
6 (SS)	Jeremy Peña	150	.324
7 (LF)	Corey Julks	69	.297
8 (CF)	Jake Meyers	105	.296
9 (RF)	Kyle Tucker	153	.369

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 HOU home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 82 errors in the 81 home games, 37 by HOU and the rest (45) by visiting teams.

Visiting/Home team rosters for CHA vs HOU on 2023-03-30

position	innings	player	position	innings	player
1	1	Dylan Cease	1	1	Framber Valdez
1	7	Aaron Bummer	1	6	Bryan Abreu
1	8	Kendall Graveman	1	7	Hector Neris
1	9	Reynaldo Lopez	1	8	Rafael Montero
2	1	Yasmani Grandal	1	9	Ryan Pressly
3	1	Andrew Vaughn	2	1	Martin Maldonado
4	1	Elvis Andrus	3	1	Jose Abreu
5	1	Yoan Moncada	4	1	Mauricio Dubon
6	1	Tim Anderson	5	1	Alex Bregman
7	1	Andrew Benintendi	6	1	Jeremy Pena
8	1	Luis Robert Jr.	7	1	Yordan Alvarez
9	1	Romy Gonzalez	8	1	Jake Meyers
9	7	Oscar Colas	9	1	Kyle Tucker

	game	CHA vs HOU on 20230330
	state before	(1, 1, 1, 0, 7, 0, 0, 0)
1.	error	play,7,0,rob1002,00,X,C/E2.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 7, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 807 This game had 2 errors. The final score was 3-2.

In the top of the 7th inning, with runners on 1st and 2nd, CHA batter Luis Robert had a 0-0 count. Catcher's interference allowed the batter to take 1st. Runners on 1st and 2nd base, each advanced one base. The bases are loaded.

	game	CHA vs HOU on 20230330
	state before	(0, 0, 0, 0, 8, 0, 1, 0)
2.	error	play,8,0,moncy001,00,.X,E3/G9LD.BX3(945)
	state after	(1, 0, 0, 0, 8, 0, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 808 *In the top of the 8th inning, with bases empty, CHA batter Yoan Moncada hit a ground ball to deep right field. An error by the first baseman allowed the batter to advance to 2nd base. The batter attempted to reach 3rd but was tagged out by the third baseman (with assists from the right fielder and the second baseman). The bases are empty.*

14.2.1 April

	game	CHA vs HOU on 20230401
	state before	(0, 0, 0, 0, 4, 3, 2, 1)
3.	error	play,4,1,hensd002,32,CBFB.BX,E5/TH/G5.B-2
	state after	(0, 0, 1, 0, 4, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 809 *In the bottom of the 4th inning, with bases empty, HOU batter David Hensley hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	DET vs HOU on 20230403
	state before	(2, 0, 0, 0, 3, 2, 0, 1)
4.	error	play,3,1,alvay001,11,CBX,E3/G3.B-1
	state after	(2, 1, 0, 0, 3, 2, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 810 This game had 2 errors. The final score was 7-6. *In the bottom of the 3rd inning, with bases empty, HOU batter Yordan Alvarez hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	DET vs HOU on 20230403
	state before	(1, 1, 0, 0, 5, 4, 4, 1)
5.	error	play,5,1,tuckk001,00,X,C/E2.1-2;B-1
	state after	(1, 1, 1, 0, 5, 4, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 811 *In the bottom of the 5th inning, with a runner on 1st, HOU batter Kyle Tucker had a 0-0 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd base.*

	game	DET vs HOU on 20230405
	state before	(0, 0, 1, 0, 7, 1, 4, 1)
6.	error	play,7,1,brega001,00,X,E6/TH/G6.2-3;B-1
	state after	(0, 1, 0, 1, 7, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 812 *In the bottom of the 7th inning, with a runner on 2nd, HOU batter Alex Bregman hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

Visiting/Home team rosters for TEX vs HOU on 2023-04-15

position	innings	player	position	innings	player
1	1	Jon Gray	1	1	Hunter Brown
1	3	Cole Ragans	1	8	Bryan Abreu
1	7	Ian Kennedy	1	9	Rafael Montero
1	7	Josh Sborz	2	1	Yainer Diaz
2	1	Jonah Heim	3	1	Jose Abreu
3	1	Nathaniel Lowe	4	1	Mauricio Dubon
4	1	Marcus Semien	5	1	Alex Bregman
5	1	Josh Jung	6	1	Jeremy Pena
6	1	Josh H. Smith	7	1	Corey Julks
7	1	Travis Jankowski	8	1	Jake Meyers
8	1	Leody Taveras	9	1	Kyle Tucker
9	1	Adolis Garcia			

	game	TEX vs HOU on 20230415
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
7.	error	play,3,0,grosr001,12,CFBX,E4/TH/G34.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 813 This game had 3 errors. The final score was 2-8.

In the top of the 3rd inning, with bases empty, TEX batter Robbie Grossman hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 1st base.

	game	TEX vs HOU on 20230415
	state before	(2, 1, 0, 0, 4, 2, 2, 0)
8.	error	play,4,0,tavel001,00,1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 4, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 814 *In the top of the 4th inning, with a runner on 1st, TEX batter Leody Taveras has a 0-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 2nd. Taveras is still at bat with a 0-0 count.*

	game	TEX vs HOU on 20230415
	state before	(1, 1, 1, 1, 7, 2, 4, 1)
9.	error	play,7,1,abrej003,20,BBX,FC6/G6.3-H;2-H(NR);1-2(E6);B-1
	state after	(1, 1, 1, 0, 7, 2, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 815 *In the bottom of the 7th inning, with bases loaded, HOU batter Jose Abreu hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). The runner on 2nd score (unearned, no RBI) while an error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	TEX vs HOU on 20230416
	state before	(0, 0, 1, 0, 7, 0, 0, 0)
10.	error	play,7,0,garca005,31,BCBBX,E6/TH/G6.B-1
	state after	(0, 1, 1, 0, 7, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 816 This game had 2 errors. The final score was 9-1. *In the top of the 7th inning, with a runner on 2nd, TEX batter Adolis Garcia hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	TEX vs HOU on 20230416
	state before	(0, 0, 0, 0, 8, 6, 1, 0)
11.	error	play,8,0,lowen001,32,.CF.BBFBFX,E3/G3.B-1
	state after	(0, 1, 0, 0, 8, 6, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 817 *In the top of the 8th inning, with bases empty, TEX batter Nathaniel Lowe hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	TOR vs HOU on 20230417
	state before	(1, 1, 1, 0, 1, 0, 2, 1)
12.	error	play,1,1,penaj004,22,*BF2TBX,C/E2.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 1, 0, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 818 *In the bottom of the 1st inning, with runners on 1st and 2nd, HOU batter Jeremy Pena had a 2-2 count. Catcher's interference allowed the batter to take 1st. The runners on 1st and 2nd base each advanced one base, as required. The bases are loaded.*

	game	TOR vs HOU on 20230418
	state before	(1, 1, 0, 0, 7, 4, 0, 1)
13.	error	play,7,1,penaj004,00,X,E4/G34.1-2;B-1
	state after	(1, 1, 1, 0, 7, 4, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 819 *In the bottom of the 7th inning, with a runner on 1st, HOU batter Jeremy Pena hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TOR vs HOU on 20230419
	state before	(0, 0, 0, 0, 6, 0, 2, 1)
14.	error	play,6,1,dubom001,00,X,E4/TH/G4.B-1
	state after	(0, 1, 0, 0, 6, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 820 *In the bottom of the 6th inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	PHI vs HOU on 20230428
	state before	(1, 0, 0, 0, 4, 1, 1, 0)
15.	error	play,4,0,castn001,22,BFBFX,E5/TH/G5S.B-1
	state after	(1, 1, 0, 0, 4, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 821 *This game had 2 errors. The final score was 3-1. In the top of the 4th inning, with bases empty, PHI batter Nick Castellanos hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	PHI vs HOU on 20230428
	state before	(0, 1, 0, 0, 5, 2, 1, 1)
16.	error	play,5,1,julkc001,12,CS*BFX,S6/G6S.1-3(E6/TH)
	state after	(0, 1, 0, 1, 5, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 822 *In the bottom of the 5th inning, with a runner on 1st, HOU batter Corey Julks hit a ground ball single to shallow short. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	PHI vs HOU on 20230430
	state before	(2, 0, 1, 0, 8, 2, 4, 0)
17.	error	play,8,0,realj001,21,BCBX,E5/G56.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 8, 3, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 823 *In the top of the 8th inning, with a runner on 2nd, PHI batter J.T. Realmuto hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

14.2.2 May

	game	SFN vs HOU on 20230502
	state before	(2, 0, 0, 0, 8, 2, 0, 1)
18.	error	play,8,1,diazy004,22,CBBFX,S1/G13.B-2(E1/TH)
	state after	(2, 0, 1, 0, 8, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 824 *In the bottom of the 8th inning, with bases empty, HOU batter Yainer Diaz hit a ground ball single to right infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	SFN vs HOU on 20230503
	state before	(1, 0, 0, 0, 2, 0, 0, 1)
19.	error	play,2,1,meyej002,32,CBFF.BBX,E3/G3.B-2
	state after	(1, 0, 1, 0, 2, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 825 This game had 2 errors. The final score was 4-2. *In the bottom of the 2nd inning, with bases empty, HOU batter Jake Meyers hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 2nd base.*

	game	SFN vs HOU on 20230503
	state before	(0, 0, 0, 0, 5, 0, 0, 1)
20.	error	play,5,1,hensd002,10,BX,E4/TH/G4.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 826 *In the bottom of the 5th inning, with bases empty, HOU batter David Hensley hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	CHN vs HOU on 20230517
	state before	(2, 0, 0, 0, 5, 6, 1, 1)
21.	error	play,5,1,meyej002,11,*SBX,E5/G56.B-1
	state after	(2, 1, 0, 0, 5, 6, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 827 *In the bottom of the 5th inning, with bases empty, HOU batter Jake Meyers hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs HOU on 20230519
	state before	(0, 1, 1, 0, 2, 0, 0, 1)
22.	error	play,2,1,penaj004,10,BX,46(1)/F0/G34.2-H(E6/TH) (NR) (UR) ;B-1
	state after	(1, 1, 0, 0, 2, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 828 *In the bottom of the 2nd inning, with runners on 1st and 2nd, HOU batter Jeremy Pena hit a ground ball to the 1st/2nd base gap. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	OAK vs HOU on 20230520
	state before	(0, 1, 0, 0, 6, 1, 2, 0)
23.	error	play,6,0,nodar001,10,B>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 6, 1, 2, 0)
	fielder	2 (catcher on the home team)

Summary 829 This game had 2 errors, both on a play with stolen bases and with the same player at bat. The final score was 2-3.

In the top of the 6th inning, with a runner on 1st, OAK batter Ryan Noda had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at bat, now with a 2-0 count.

	game	OAK vs HOU on 20230520
	state before	(1, 1, 0, 0, 8, 2, 2, 0)
24.	error	play,8,0,nodar001,10,B1>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 8, 2, 2, 0)
	fielder	2 (catcher on the home team)

Summary 830 *In the top of the 8th inning, with a runner on 1st, OAK batter Ryan Noda had a 1-0 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at bat, now with a 1-1 count.*

14.2.3 June

Visiting/Home team rosters for ANA vs HOU on 2023-06-02

position	innings	player	position	innings	player
1	1	Shohei Ohtani	1	1	Framber Valdez
1	7	Chase Silseth	1	8	Phil Maton
2	1	Chad Wallach	1	9	Bryan Abreu
3	1	Jared Walsh	2	1	Martin Maldonado
4	1	Brandon Drury	3	1	Jose Abreu
5	1	Gio Urshela	4	1	Jose Altuve
6	1	Zach Neto	5	1	Alex Bregman
6	6	Luis Rengifo	6	1	Jeremy Pena
7	1	Taylor Ward	7	1	Yordan Alvarez
8	1	Mike Trout	8	1	Chas McCormick
9	1	Hunter Renfroe	9	1	Kyle Tucker

	game	ANA vs HOU on 20230602
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
25.	error	play,1,1,penaj004,02,CCX,E5/L56+.B-2(E5/TH)
	state after	(1, 0, 1, 0, 1, 0, 0, 1)
	fielder	5, (3rd baseman on the visiting team)

Summary 831 This game had 3 errors, including this play which had 2 errors by the same fielder. The final score was 2-6.

In the bottom of the 1st inning, with bases empty, HOU batter Jeremy Pena hit a hard line drive to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	ANA vs HOU on 20230602
	state before	(2, 0, 0, 1, 8, 0, 6, 0)
26.	error	play,8,0,drurb001,21,BSBX,S5/G56.3-H;B-2(E5/TH)
	state after	(2, 0, 1, 0, 8, 1, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 832 *In the top of the 8th inning, with a runner on 3rd, ANA batter Brandon Drury hit a ground ball single to 3rd base. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	ANA vs HOU on 20230603
	state before	(1, 1, 1, 0, 4, 1, 2, 1)
27.	error	play,4,1,penaj004,22,BCFBX,FC6/G6.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 4, 1, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 833 *In the bottom of the 4th inning, with runners on 1st and 2nd, HOU batter Jeremy Pena hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for WAS vs HOU on 2023-06-14

position	innings	player	position	innings	player
1	1	Josiah Gray	1	1	Framber Valdez
1	8	Jordan Weems	1	8	Hector Neris
1	9	Hunter Harvey	1	9	Ryan Pressly
2	1	Keibert Ruiz	2	1	Martin Maldonado
3	1	Dominic Smith	3	1	Jose Abreu
3	9	Michael Chavis	4	1	Jose Altuve
4	1	Ildemaro Vargas	5	1	Alex Bregman
4	9	Luis Garcia	6	1	Mauricio Dubon
5	1	Jeimer Candelario	7	1	Corey Julks
6	1	CJ Abrams	8	1	Jake Meyers
7	1	Stone Garrett	9	1	Kyle Tucker
7	9	Corey Dickerson			
8	1	Alex Call			
8	9	Lane Thomas			
9	1	Lane Thomas			
9	9	Ildemaro Vargas			

	game	WAS vs HOU on 20230614
	state before	(0, 0, 0, 0, 9, 1, 4, 0)
28.	error	play,9,0,menej001,01,.FX,E5/TH/G56.B-2
	state after	(0, 0, 1, 0, 9, 1, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 834 This game had 3 errors, all in the 9th inning. The final score was 4-5.

In the top of the 9th inning, with bases empty, WAS batter Joey Meneses hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.

	game	WAS vs HOU on 20230614
	state before	(1, 0, 0, 1, 9, 2, 4, 0)
29.	error	play,9,0,vargi001,01,SX,FC1/G1S.3-H(E1/TH) (NR) (UR);B-1
	state after	(1, 1, 0, 0, 9, 3, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 835 *In the top of the 9th inning, with a runner on 3rd, WAS batter Ildemaro Vargas hit a ground ball into a fielder's choice at very shallow infield. A throwing error by the pitcher allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	WAS vs HOU on 20230614
	state before	(1, 1, 1, 1, 9, 4, 4, 1)
30.	error	play,9,1,meyej002,22,C*BTFBX,62(3)/F0/G6.2-H(E2/TH) (NR) (UR);1-2;B-1
	state after	(2, 1, 1, 0, 9, 4, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 836 *In the bottom of the 9th inning, with bases loaded, HOU batter Jake Meyers hit a ground ball to short. At home plate, the catcher (with an assist from the shortstop) forced out the runner from 3rd. A throwing error by the catcher allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	CIN vs HOU on 20230616
	state before	(2, 1, 0, 0, 5, 0, 0, 0)
31.	error	play,5,0,newmk001,31,BBBC>X,E6/G6.1-2;B-1
	state after	(2, 1, 1, 0, 5, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 837 *In the top of the 5th inning, with a runner on 1st, CIN batter Kevin Newman hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CIN vs HOU on 20230617
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
32.	error	play,1,0,mclam001,12,FSBX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 838 This game had 2 errors. The final score was 10-3.

In the top of the 1st inning, with bases empty, CIN batter Matt McLain hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	CIN vs HOU on 20230617
	state before	(1, 1, 0, 0, 1, 2, 0, 0)
33.	error	play,1,0,stees001,10,1B>X,FC6/G6.1-2(E6);B-1
	state after	(1, 1, 1, 0, 1, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 839 *In the top of the 1st inning, with a runner on 1st, CIN batter Spencer Steer hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for CIN vs HOU on 2023-06-18

position	innings	player	position	innings	player
1	1	Luke Weaver	1	1	Ronel Blanco
1	10	Alex Young	1	10	Seth Martinez
1	6	Daniel Duarte	1	6	Phil Maton
1	8	Lucas Sims	1	8	Bryan Abreu
1	9	Ian Gibaut	1	9	Ryan Pressly
2	1	Tyler Stephenson	2	1	Martin Maldonado
3	1	Spencer Steer	2	9	Yainer Diaz
4	1	Jonathan India	3	1	Jose Abreu
4	10	Matt McLain	4	1	Mauricio Dubon
5	1	Kevin Newman	5	1	Alex Bregman
5	8	Nick Senzel	6	1	Jeremy Pena
6	1	Elly De La Cruz	7	1	Corey Julks
7	1	Jake Fraley	7	10	Chas McCormick
7	10	Will Benson	8	1	Chas McCormick
8	1	TJ Friedl	8	10	Jake Meyers
9	1	Will Benson	9	1	Kyle Tucker
9	10	Stuart Fairchild			

	game	CIN vs HOU on 20230618
	state before	(0, 0, 0, 0, 3, 2, 3, 1)
34.	error	play,3,1,diazy004,01,FX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 3, 2, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 840 This game had 3 errors. The final score was 9-7.

In the bottom of the 3rd inning, with bases empty, HOU batter Yainer Diaz hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	CIN vs HOU on 20230618
	state before	(1, 0, 0, 0, 4, 2, 3, 1)
35.	error	play,4,1,brega001,00,X,T9/L9LD.B-H(E4/TH) (NR)
	state after	(1, 0, 0, 0, 4, 2, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 841 *In the bottom of the 4th inning, with bases empty, HOU batter Alex Bregman hit a line drive triple to deep right field. A throwing error by the second baseman allowed the batter to score (unearned, no RBI).*

	game	CIN vs HOU on 20230618
	state before	(0, 0, 1, 0, 10, 6, 6, 0)
36.	error	play,10,0,friet001,20,...BBX,S5/BG25.2-H(E5/TH) (NR) (UR)
	state after	(0, 1, 0, 0, 10, 7, 6, 0)
	fielder	5 (2nd baseman on the visiting team)

Summary 842 *In the top of the 10th inning, with a runner on 2nd, CIN batter TJ Friedl hit a bunt single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	NYN vs HOU on 20230619
	state before	(1, 1, 0, 0, 6, 6, 0, 0)
37.	error	play,6,0,batyb001,12,C*B*S>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 6, 0, 0)
	fielder	2 (catcher on the home team)

Summary 843 This game had 2 errors. The final score was 11-1. *In the top of the 6th inning, with a runner on 1st, NYN batter Brett Baty had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Baty is still at bat, now with a 2-2 count.*

	game	NYN vs HOU on 20230619
	state before	(2, 1, 1, 1, 9, 9, 1, 0)
38.	error	play,9,0,voged001,22,F*BF*BX,S8/L8D+.3-H;2-H;1-3(E8)
	state after	(2, 1, 0, 1, 9, 11, 1, 0)
	fielder	8 (center field on the home team)

Summary 844 *In the top of the 9th inning, with bases loaded, NYN batter Daniel Vogelbach hit a hard line drive single to deep center field. The runners on 2nd and 3rd scored (2 RBIs). A error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for NYN vs HOU on 2023-06-21

position	innings	player	position	innings	player
1	1	Tylor Megill	1	1	Cristian Javier
1	3	Dominic Leone	1	3	Seth Martinez
1	4	Grant Hartwig	1	4	Phil Maton
1	6	Josh Walker	1	6	Rafael Montero
1	7	Adam Ottavino	1	7	Hector Neris
2	1	Omar Narvaez	1	8	Bryan Abreu
3	1	Pete Alonso	1	9	Ryan Pressly
4	1	Jeff McNeil	2	1	Martin Maldonado
5	1	Brett Baty	3	1	Mauricio Dubon
6	1	Francisco Lindor	4	1	Jose Altuve
7	1	Tommy Pham	5	1	Alex Bregman
8	1	Brandon Nimmo	6	1	Jeremy Pena
9	1	Starling Marte	7	1	Corey Julks
			8	1	Chas McCormick
			9	1	Kyle Tucker

	game	NYN vs HOU on 20230621
	state before	(1, 0, 1, 0, 1, 0, 0, 1)
39.	error	play,1,1,tuckk001,22,BSBCX,C/E2.B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 845 This game had 3 errors. The final score was 8-10.
In the bottom of the 1st inning, with a runner on 2nd, HOU batter Kyle Tucker had a 2-2 count. Catcher's interference allowed the batter to take 1st.

	game	NYN vs HOU on 20230621
	state before	(0, 1, 0, 0, 2, 0, 2, 0)
40.	error	play,2,0,batyb001,12,CFBX,S9/G3.1-3;B-2(E4/TH)
	state after	(0, 0, 1, 1, 2, 0, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 846 *In the top of the 2nd inning, with a runner on 1st, NYN batter Brett Baty hit a ground ball single to 1st. The runner on 1st advanced to 3rd. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	NYN vs HOU on 20230621
	state before	(0, 1, 0, 0, 7, 8, 9, 1)
41.	error	play,7,1,penaj004,21,BFBX,S1/G1.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 7, 8, 9, 1)
	fielder	1 (pitcher on the visiting team)

Summary 847 *In the bottom of the 7th inning, with a runner on 1st, HOU batter Jeremy Pena hit a ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

14.2.4 July

	game	COL vs HOU on 20230704
	state before	(1, 1, 1, 0, 7, 0, 3, 1)
42.	error	play,7,1,abrej003,11,ABX,5(2)/FO/G5.1-3(E5/TH);B-1
	state after	(2, 1, 0, 1, 7, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 848 *In the bottom of the 7th inning, with runners on 1st and 2nd, HOU batter Jose Abreu hit a ground ball to 3rd. The third baseman forced out the runner from 2nd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	SEA vs HOU on 20230707
	state before	(2, 0, 1, 0, 3, 0, 0, 1)
43.	error	play,3,1,dubom001,20,BBX,E4/TH/G4M.2-3;B-1
	state after	(2, 1, 0, 1, 3, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 849 *In the bottom of the 3rd inning, with a runner on 2nd, HOU batter Mauricio Dubon hit a ground ball to 2nd base. A throwing error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SEA vs HOU on 20230708
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
44.	error	play,1,1,tuckk001,31,FBBB>X,D9/G3.1-H;B-3(E9/TH)
	state after	(1, 0, 0, 1, 1, 0, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 850 *In the bottom of the 1st inning, with a runner on 1st, HOU batter Kyle Tucker hit a ground ball double to 1st. The runner on 1st scored while a throwing error by the right fielder allowed the batter to reach 3rd.*

	game	SEA vs HOU on 20230709
	state before	(0, 0, 0, 0, 4, 0, 0, 0)
45.	error	play,4,0,rodrj007,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 4, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 851 *In the top of the 4th inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for TEX vs HOU on 2023-07-24

position	innings	player	position	innings	player
1	1	Jon Gray	1	1	Brandon Bielak
1	6	Jose Leclerc	1	5	Parker Mushinski
1	7	Aroldis Chapman	1	7	Phil Maton
1	8	Brock Burke	1	7	Ryne Stanek
1	9	Alex Speas	1	8	Bryan Abreu
2	1	Jonah Heim	1	9	Ryan Pressly
3	1	Nathaniel Lowe	2	1	Martin Maldonado
4	1	Marcus Semien	3	1	Jose Abreu
5	1	Josh Jung	4	1	Mauricio Dubon
6	1	Ezequiel Duran	5	1	Alex Bregman
7	1	Josh H. Smith	6	1	Jeremy Pena
8	1	Leody Taveras	7	1	Chas McCormick
9	1	Travis Jankowski	7	8	Corey Julks
			8	1	Jake Meyers
			8	8	Chas McCormick
			9	1	Kyle Tucker

	game	TEX vs HOU on 20230724
	state before	(1, 1, 1, 0, 2, 2, 4, 0)
46.	error	play,2,0,smitj007,32,*BBCBCFX,E3/G9L.2-H(NR)(UR);1-3;B-1
	state after	(1, 1, 0, 1, 2, 3, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 852 This game had 3 errors. The final score was 9-10.
In the top of the 2nd inning, with runners on 1st and 2nd, TEX batter Josh Smith hit a ground ball to right field. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.

	game	TEX vs HOU on 20230724
	state before	(0, 0, 0, 0, 7, 6, 6, 0)
47.	error	play,7,0,jankt001,10,.BX,S8/L8D.B-2(E8)
	state after	(0, 0, 1, 0, 7, 6, 6, 0)
	fielder	8 (center field on the home team)

Summary 853 *In the top of the 7th inning, with bases empty, TEX batter Travis Jankowski hit a line drive single to deep center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	TEX vs HOU on 20230724
	state before	(1, 0, 0, 0, 7, 9, 9, 1)
48.	error	play,7,1,diazy004,21,BBSX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 7, 9, 9, 1)
	fielder	6 (shortstop on the visiting team)

Summary 854 *In the bottom of the 7th inning, with bases empty, HOU batter Yainer Diaz hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	TEX vs HOU on 20230725
	state before	(2, 1, 0, 1, 6, 0, 2, 0)
49.	error	play,6,0,garca005,01,CX,FC6/G6D.3-H(NR) (UR) ; 1-2(E6) ; B-1
	state after	(2, 1, 1, 0, 6, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 855 *In the top of the 6th inning, with runners on 1st and 3rd, TEX batter Adolis Garcia hit a ground ball into a fielder's choice at short. The runner fom 3rd scored (unearned, no RBI) while an error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	TBA vs HOU on 20230729
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
50.	error	play,1,0,loweb001,00,>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 856 This game had 2 errors. The final score was 4-17. *In the top of the 1st inning, with a runner on 1st, TBA batter Brandon Lowe had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Lowe is still at bat, now with a 1-0 count.*

	game	TBA vs HOU on 20230729
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
51.	error	play,1,1,penaj004,11,BFX,E5/G56S.B-2
	state after	(1, 0, 1, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 857 *In the bottom of the 1st inning, with bases empty, HOU batter Jeremy Pena hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 2nd base.*

	game	TBA vs HOU on 20230730
	state before	(1, 1, 1, 1, 1, 0, 0, 0)
52.	error	play,1,0,ralel001,11,SBX,E3/G3.3-H;2-3;1-2;B-1
	state after	(1, 1, 1, 1, 1, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 858 This game had 2 errors. The final score was 8-2.

In the top of the 1st inning, with bases loaded, TBA batter Luke Raley hit a ground ball to 1st base. An error by the first baseman allowed the runners to each advance one base and the batter to reach 1st. In particular, the runner on 3rd score (1 RBI). The bases are loaded.

	game	TBA vs HOU on 20230730
	state before	(1, 0, 0, 0, 2, 4, 0, 1)
53.	error	play,2,1,abrej003,32,BBBSX.X,E1/G6MS-.B-1
	state after	(1, 1, 0, 0, 2, 4, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 859 *In the bottom of the 2nd inning, with bases empty, HOU batter Jose Abreu hit a soft ground ball to short. An error by the pitcher allowed the batter to reach 1st base.*

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	game	CLE vs HOU on 20230802
	state before	(2, 0, 1, 0, 6, 2, 3, 1)
54.	error	play,6,1,penaj004,22,CBBSX,E6/G6.2-3;B-1
	state after	(2, 1, 0, 1, 6, 2, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 860 This game had 2 errors. The final score was 2-3.

In the bottom of the 6th inning, with a runner on 2nd, HOU batter Jeremy Pena hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	CLE vs HOU on 20230802
	state before	(0, 0, 0, 0, 9, 2, 3, 0)
55.	error	play,9,0,gonzo001,11,.SBX,E4/G4.B-1
	state after	(0, 1, 0, 0, 9, 2, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 861 *In the top of the 9th inning, with bases empty, CLE batter Oscar Gonzalez hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	ANA vs HOU on 20230812
	state before	(2, 0, 0, 0, 4, 0, 0, 0)
56.	error	play,4,0,mousm001,12,SBFX,E5/G6S.B-1
	state after	(2, 1, 0, 0, 4, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 862 *In the top of the 4th inning, with bases empty, ANA batter Mike Moustakas hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs HOU on 20230813
	state before	(0, 1, 0, 0, 9, 2, 1, 0)
57.	error	play,9,0,drurb001,32,FBBF1B>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 9, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 863 *In the top of the 9th inning, with a runner on 1st, ANA batter Brandon Drury had a full count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

Visiting/Home team rosters for SEA vs HOU on 2023-08-18

position	innings	player	position	innings	player
1	1	Bryce Miller	1	1	J.P. France
1	7	Justin Topa	1	7	Kendall Graveman
1	9	Andres Munoz	1	8	Jose Urquidy
2	1	Cal Raleigh	2	1	Martin Maldonado
3	1	Ty France	3	1	Yainer Diaz
4	1	Josh Rojas	4	1	Jose Altuve
5	1	Eugenio Suarez	5	1	Alex Bregman
6	1	Jose Caballero	6	1	Jeremy Pena
7	1	Cade Marlowe	7	1	Yordan Alvarez
8	1	Julio Rodriguez	8	1	Mauricio Dubon
9	1	Teoscar Hernandez	9	1	Chas McCormick

	game	SEA vs HOU on 20230818
	state before	(1, 1, 0, 0, 2, 0, 0, 0)
58.	error	play,2,0,fordm002,22,BFBF1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 864 This game had 4 errors. The final score was 2-0.

In the top of the 2nd inning, with a runner on 1st, SEA batter Mike Ford has a 2-2 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to advance to 2nd. Ford is still at bat with a 2-2 count.

	game	SEA vs HOU on 20230818
	state before	(0, 0, 0, 0, 7, 2, 0, 0)
59.	error	play,7,0,rodrj007,10,.BX,E6/L6D.B-1
	state after	(0, 1, 0, 0, 7, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 865 *In the top of the 7th inning, with bases empty, SEA batter Julio Rodriguez hit a line drive to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	SEA vs HOU on 20230818
	state before	(1, 1, 0, 0, 7, 2, 0, 0)
60.	error	play,7,0,frant002,00,1,P01(E1/TH).1-3
	state after	(1, 0, 0, 1, 7, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 866 *In the top of the 7th inning, with a runner on 1st, SEA batter Ty France has a 0-0 count. The pitcher attempted to pick off the runner on 1st. An error by the pitcher allowed that runner to advance to 3rd. France is still at bat with a 0-0 count.*

	game	SEA vs HOU on 20230818
	state before	(1, 0, 0, 0, 7, 2, 0, 1)
61.	error	play,7,1,diazy004,11,BCX,E6/G6.B-1
	state after	(1, 1, 0, 0, 7, 2, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 867 *In the bottom of the 7th inning, with bases empty, HOU batter Yainer Diaz hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	SEA vs HOU on 20230819
	state before	(1, 1, 0, 0, 2, 2, 0, 1)
62.	error	play,2,1,penaj004,31,*BBBC1X,FC5/G5.1-2(E5);B-1
	state after	(1, 1, 1, 0, 2, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 868 This game had 2 errors. The final score was 10-3.
In the bottom of the 2nd inning, with a runner on 1st, HOU batter

Jeremy Pena hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.

	game	SEA vs HOU on 20230819
	state before	(2, 1, 0, 0, 3, 2, 0, 0)
63.	error	play,3,0,hernt002,30,BBBX,D78/L78D+.1-H(E8)(NR)
	state after	(2, 0, 1, 0, 3, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 869 *In the top of the 3rd inning, with a runner on 1st, SEA batter Teoscar Hernandez hit a hard line drive double to deep left center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 2nd.*

	game	BOS vs HOU on 20230821
	state before	(0, 1, 1, 0, 1, 3, 0, 1)
64.	error	play,1,1,tuckk001,32,BCBBSFX,C/E2.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 1, 3, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 870 *This game had 2 errors. The final score was 4-9. In the bottom of the 1st inning, with runners on 1st and 2nd, HOU batter Kyle Tucker had a full count. Catcher's interference allowed the batter to take 1st. The runners on 1st and 2nd base each advance one base, as required. The bases are loaded.*

	game	BOS vs HOU on 20230821
	state before	(2, 0, 0, 0, 6, 4, 7, 1)
65.	error	play,6,1,diazy004,32,BBCFBX,E9/F3D.B-1
	state after	(2, 1, 0, 0, 6, 4, 7, 1)
	fielder	9 (right field on the visiting team)

Summary 871 *In the bottom of the 6th inning, with bases empty, HOU batter Yainer Diaz hit a fly ball to deep 1st. An error by the right fielder allowed the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs HOU on 2023-08-22

position	innings	player	position	innings	player
1	1	Tanner Houck	1	1	Justin Verlander
1	6	Josh Winckowski	1	7	Kendall Graveman
1	7	Mauricio Llovera	1	8	Ryne Stanek
1	8	Brennan Bernardino	1	9	Rafael Montero
2	1	Connor Wong	2	1	Martin Maldonado
3	1	Triston Casas	3	1	Yainer Diaz
4	1	Luis Urias	4	1	Jose Altuve
5	1	Rafael Devers	4	8	Mauricio Dubon
6	1	Trevor Story	5	1	Alex Bregman
7	1	Masataka Yoshida	6	1	Jeremy Pena
8	1	Adam Duvall	7	1	Chas McCormick
9	1	Alex Verdugo	8	1	Mauricio Dubon
9	4	Wilyer Abreu	8	8	Jake Meyers
			9	1	Kyle Tucker

	game	BOS vs HOU on 20230822
	state before	(2, 0, 0, 0, 1, 0, 2, 1)
66.	error	play,1,1,mccoc001,01,MX,E5/TH/G56S.B-1
	state after	(2, 1, 0, 0, 1, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 872 This game had 4 errors. The final score was 3-7.
In the bottom of the 1st inning, with bases empty, HOU batter Chas McCormick hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	BOS vs HOU on 20230822
	state before	(2, 1, 1, 0, 6, 0, 2, 0)
67.	error	play,6,0,stort001,31,B*BBCX,FC6/G56.2-3;1X2(6E4);B-1
	state after	(2, 1, 1, 1, 6, 0, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 873 *In the top of the 6th inning, with runners on 1st and 2nd, BOS batter Trevor Story hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the second baseman (with an assist from the shortstop) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	BOS vs HOU on 20230822
	state before	(1, 1, 0, 0, 7, 0, 2, 1)
68.	error	play,7,1,tuckk001,32,.CBBBFX,FC1/G1S.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 7, 0, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 874 *In the bottom of the 7th inning, with a runner on 1st, HOU batter Kyle Tucker hit a ground ball into a fielder's choice at very shallow infield. A throwing error by the pitcher allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	BOS vs HOU on 20230822
	state before	(1, 1, 1, 0, 7, 0, 2, 1)
69.	error	play,7,1,diazy004,11,BFX,S9/G34.2-H(UR);1-3(E9/TH);B-2
	state after	(1, 0, 1, 1, 7, 0, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 875 This play is a *consecutive error*, as Diaz bats right after Tucker in the lineup.

In the bottom of the 7th inning, with runners on 1st and 2nd, HOU batter Yainer Diaz hit a ground ball single to the 1st/2nd base gap. A throwing error by the right fielder allowed the runner on 1st to safely advance to 3rd and the batter to reach 2nd.

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	game	NYA vs HOU on 20230902
	state before	(0, 1, 0, 0, 5, 3, 4, 0)
70.	error	play,5,0,domij003,22,BC*BCX,FC4/G9S.1-3(E4);B-1
	state after	(0, 1, 0, 1, 5, 3, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 876 *In the top of the 5th inning, with a runner on 1st, NYA batter Jasson Dominguez hit a ground ball into a fielder's choice at shallow right field. An error by the second baseman allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

Visiting/Home team rosters for SDN vs HOU on 2023-09-10

position	innings	player	position	innings	player
1	1	Matt Waldron	1	1	J.P. France
1	6	Rich Hill	1	7	Parker Mushinski
1	6	Tim Hill	1	8	Joel Kuhnel
2	1	Luis Campusano	2	1	Yainer Diaz
3	1	Garrett Cooper	3	1	Jose Abreu
4	1	Ha-Seong Kim	3	7	Jon Singleton
4	7	Matthew Batten	4	1	Jose Altuve
4	8	Eguy Rosario	5	1	Alex Bregman
5	1	Eguy Rosario	6	1	Jeremy Pena
5	8	Brett Sullivan	6	7	Grae Kessinger
6	1	Xander Bogaerts	7	1	Yordan Alvarez
6	7	Ha-Seong Kim	7	7	Chas McCormick
6	8	Matthew Batten	8	1	Chas McCormick
7	1	Juan Soto	8	7	Jake Meyers
7	7	Jose Azocar	9	1	Kyle Tucker
8	1	Trent Grisham	9	7	Mauricio Dubon
9	1	Jurickson Profar			

	game	SDN vs HOU on 20230910
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
71.	error	play,3,1,diazy004,00,>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 3, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 877 This game had 4 errors. The final score was 2-12.

In the bottom of the 3rd inning, with a runner on 1st, HOU batter Yainer Diaz had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Diaz is still at bat, now with a 1-0 count.

	game	SDN vs HOU on 20230910
	state before	(0, 0, 0, 1, 4, 0, 3, 1)
72.	error	play,4,1,abrej003,31,>B.>C..BBX,E5/G56.3-H(NR);B-1
	state after	(0, 1, 0, 0, 4, 0, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 878 *In the bottom of the 4th inning, with a runner on 3rd, HOU batter Jose Abreu hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	SDN vs HOU on 20230910
	state before	(0, 1, 1, 1, 6, 1, 8, 1)
73.	error	play,6,1,penaj004,22,FBFFBX,FC/G6.3XH(5E2)(NR)(UR);2-3;1-2;B-1
	state after	(0, 1, 1, 1, 6, 1, 9, 1)
	fielder	2 (catcher on the visiting team)

Summary 879 *In the bottom of the 6th inning, with bases loaded, HOU batter Jeremy Pena hit a ground ball into a fielder's choice at short. An error by the catcher (with an assist from the third baseman) allowed the runner on 3rd to score (unearned, no RBI), runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	SDN vs HOU on 20230910
	state before	(2, 0, 0, 0, 8, 2, 12, 1)
74.	error	play,8,1,brega001,11,CBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 8, 2, 12, 1)
	fielder	6 (shortstop on the visiting team)

Summary 880 *In the bottom of the 8th inning, with bases empty, HOU batter Alex Bregman hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	BAL vs HOU on 20230918
	state before	(0, 0, 0, 0, 9, 8, 7, 1)
75.	error	play,9,1,alvay001,00,...X,E5/TH/G6.B-2
	state after	(0, 0, 1, 0, 9, 8, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 881 *In the bottom of the 9th inning, with bases empty, HOU batter Yordan Alvarez hit a ground ball to short. A throwing error by the third baseman allowed the batter to reach 2nd base.*

Visiting/Home team rosters for BAL vs HOU on 2023-09-19

position	innings	player	position	innings	player
1	1	Kyle Gibson	1	1	Hunter Brown
1	5	DL Hall	1	6	Phil Maton
1	6	Jorge Lopez	1	7	Jose Urquidy
1	7	Jack Flaherty	1	8	Bennett Sousa
1	9	Cionel Perez	2	1	Yainer Diaz
1	9	Yennier Cano	3	1	Jose Abreu
2	1	James McCann	4	1	Jose Altuve
3	1	Ryan O'Hearn	5	1	Alex Bregman
4	1	Adam Frazier	6	1	Jeremy Pena
5	1	Gunnar Henderson	7	1	Chas McCormick
6	1	Jorge Mateo	8	1	Mauricio Dubon
7	1	Austin Hays	9	1	Kyle Tucker
8	1	Cedric Mullins			
9	1	Anthony Santander			

	game	BAL vs HOU on 20230919
	state before	(0, 0, 0, 0, 5, 5, 3, 1)
76.	error	play,5,1,dubom001,01,FX,S6/G56.B-2(E6/TH)
	state after	(0, 0, 1, 0, 5, 5, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 882 This game had 3 errors. The final score was 9-5.

In the bottom of the 5th inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball single to 5rd. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	BAL vs HOU on 20230919
	state before	(2, 1, 1, 0, 6, 6, 3, 0)
77.	error	play,6,0,hendg002,12,BFSX,S9/L9S.2-H;1-3;B-2(E9/TH)
	state after	(2, 0, 1, 1, 6, 7, 3, 0)
	fielder	9 (right field on the home team)

Summary 883 *In the top of the 6th inning, with runners on 1st and 2nd, BAL batter Gunnar Henderson hit a line drive single to shallow right field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	BAL vs HOU on 20230919
	state before	(0, 0, 0, 0, 6, 7, 3, 1)
78.	error	play,6,1,tuckk001,02,SSX,E4/G34.B-1
	state after	(0, 1, 0, 0, 6, 7, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 884 *In the bottom of the 6th inning, with bases empty, HOU batter Kyle Tucker hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	BAL vs HOU on 20230920
	state before	(2, 0, 1, 0, 6, 1, 0, 1)
79.	error	play,6,1,penaj004,32,BSCBFFBFX,C/E2.B-1
	state after	(2, 1, 1, 0, 6, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 885 *In the bottom of the 6th inning, with a runner on 2nd, HOU batter Jeremy Pena had a full count. Catcher's interference allowed the batter to take 1st.*

	game	KCA vs HOU on 20230922
	state before	(2, 1, 0, 0, 8, 7, 2, 0)
80.	error	play,8,0,isbek001,21,*B1FBX,E4/TH/G4.1-2;B-1
	state after	(2, 1, 1, 0, 8, 7, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 886 *In the top of the 8th inning, with a runner on 1st, KCA batter Kyle Isbel hit a ground ball to 2nd base. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	KCA vs HOU on 20230923
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
81.	error	play,1,0,wittb002,01,C>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 887 *In the top of the 1st inning, with a runner on 1st, KCA batter Bobby Witt had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Witt is still at bat, now with a 1-1 count.*

	game	KCA vs HOU on 20230924
	state before	(0, 1, 0, 0, 3, 6, 2, 1)
82.	error	play,3,1,tuckk001,10,BX,C/E2.1-2;B-1
	state after	(0, 1, 1, 0, 3, 6, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 888 *In the bottom of the 3rd inning, with a runner on 1st, HOU batter Kyle Tucker had a 1-0 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd, as required.*

Chapter 15

Kansas City Royals (KCA)

The Kansas City Royals was founded as an expansion franchise in 1969 in the American League Central Division. The team has made four World Series trips, winning in 1985 and 2015.

The name “Royals” pays homage to the *American Royal*, a live-stock show held annually in Kansas City since 1899, as well as identically-named baseball teams from the former Negro League in the area.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Kansas City against the Royals. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder’s choice or an attempted pick off.

The defensive efficiency ratio (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 KCA team is 0.685 compared with the MLB average¹ of 0.691 that year [BR]. As they are slightly below average, so give the Royals a C or C- on their DER test.

As we will see in the next section, there were 81 errors in the 81 home games in 2023, 36 by the Royals and the rest (45) by visiting teams. The total number of errors by the Royals team (home and away) over the regular season in 2023 was 80. This is slightly beliw average (which is good).

¹The standard deviation of these DER values is $\sigma = 0.011$.

15.1 Season totals

The Official Scorer shall credit an automatic putout when the batter is called out on an Infield Fly that is not caught.

– **MLB Rule Book** 9.09(c)(1)

Throughout, *vt* denotes the visiting team and *ht* denotes the home team.

15.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Before errors by position are tabulated, we tabulate some error-to-runs conversion data.

Runs and errors	totals
Errors by vt:	45
Runs by ht because of errors by vt:	9
Errors by ht:	36
Runs by vt because of errors by ht:	4

The Royals have a good error-to-run conversion rate of $9/45 = 20\%$ (on the part of the Royals), compared to $4/36 = 11.1\ldots\%$ on the part of the visiting teams. For comparison, a typical *play* (not necessarily with an error) in the 2023 KCA season (home and away) has about $R/PA = 676/5985 = 11.3\ldots\%$ chance of resulting in a run. Therefore, a randomly selected error is about twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	12	5
to catcher:	5	2
to 1st baseman:	3	3
to 2nd baseman:	5	1
to 3rd baseman:	5	9
to shortstop:	9	6
to left fielder:	1	4
to center fielder:	2	3
to right fielder:	3	3

15.1.2 Error total by state

This section tabulates the totals by game state.

A table of top frequencies of game states for errors by the Royals in KCA 2023 home games:

Errors by state	ht	vt	totals
bases empty	15	17	32
(lone) runner on 1st	12	12	24
(lone) runner on 2nd	4	4	8
(lone) runner on 3rd	1	2	3
two runners on base	4	10	14
bases loaded	0	0	0
total	36	45	81

state	frequency	percentage
(0, 0, 0, 0)	9	25%
(1, 0, 0, 0)	2	5.5%
(2, 0, 0, 0)	4	11.1%
(0, 1, 0, 0)	4	11.1%
(1, 1, 0, 0)	6	16.7%
(2, 1, 0, 0)	2	5.5%
(0, 0, 1, 0)	2	5.5%
(1, 1, 1, 0)	2	5.5%
(2, 1, 1, 0)	1	2.78%

A table of top frequencies of game states for errors by a visiting team in KCA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	10	22.2%
(1, 0, 0, 0)	2	4.4%
(2, 0, 0, 0)	5	11.1%
(0, 1, 0, 0)	1	2.2%
(1, 1, 0, 0)	5	11.1%
(2, 1, 0, 0)	6	13.3%
(1, 0, 1, 0)	2	4.4%
(0, 1, 1, 0)	3	6.6%
(1, 1, 1, 0)	5	11.1%
(1, 1, 0, 1)	1	2.2%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Royals homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `KCA2023.eva`):

state	KCA2023 percentages
(0, 0, 0, 0)	23.9%
(1, 0, 0, 0)	16.6%
(2, 0, 0, 0)	13.4%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	7.0%
(2, 1, 0, 0)	6.4%
(0, 0, 1, 0)	2.2%
(1, 0, 1, 0)	2.8%
(2, 0, 1, 0)	3.5%
(0, 0, 0, 1)	0.5%
(1, 0, 0, 1)	1.2%
(2, 0, 0, 1)	1.6%
(0, 1, 1, 0)	1.7%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.4%

15.1.3 Some interesting errors

Some of the most interesting error plays in the KCA 2023 home games:

- The game between the Reds and the Royals on 2023-06-14 had 4 errors.
- The game CLE vs KCA on 2023-09-20 had 3 errors, including a consecutive error.

15.1.4 Relative team rankings

We have 81 home games between KCA and some visiting MLB team (vt). A number of teams did not play KCA in the regular season. However, of those that did, KCA *lost* more games than won against²:

²We recorded that vt's W/T record in parentheses (W = numbers of games vt won against KCA, T = Total home games vt played vs the Royals at Kansas City.

ANA (2/3), BAL (2/3), BOS (2/3), COL (2/3), OAK (2/3), TBA (2/3), WAS
(2/3),
ATL (3/3), CIN (3/3), PIT (3/3), TEX (3/3), SEA (3/4), TOR (3/4), DET
(5/7) .

The teams that *split* their series were:

SLN (1/2), MIN (3/6),

The teams that KCA *won* more than lost against

NYN (0/3), HOU (1/3), LAN (1/3), NYA (1/3), CLE (2/6), CHA (2/7) .

The teams that did not play in any of the 81 home games in KCA 2023 were:

ARI, CHN, MIA, MIL, PHI, SDN, SFN .

15.2 Game-by-game totals

The Official Scorer shall score a sacrifice bunt when, before two are out, the batter advances one or more runners with a bunt and is put out at first base, or would have been put out except for a fielding error, unless, in the judgment of the Official Scorer, the batter was bunting exclusively for a base hit and not sacrificing his own chance of reaching first base for the purpose of advancing a runner or runners, in which case the Official Scorer shall charge the batter with a time at bat.

– **MLB Rule Book 9.08(a)**

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for KCA 2023, listing the player with the most games in that position³, as well as their on-base percentage, is as follows [BA].

³Actually, MJ Melendez had 59 starts as a left fielder, more than Olivares. To avoid repetition of players, we list Olivares, who was the usual starter at LF when MJ played RF.

position	player	games played	OBP
1 (C)	Salvador Perez	91	.292
3 (1B)	Nick Pratto	78	.307
4 (2B)	Michael Massey	119	.274
5 (3B)	Maikel Garcia	104	.323
6 (SS)	Bobby Witt Jr.	149	.319
7 (LF)	Edward Olivares	45	.317
8 (CF)	Kyle Isbel	90	.282
9 (RF)	MJ Melendez	72	.316

In general, the rosters were found using my Python function `team_roster` and sometimes `how_many_calls_with_playerID_atbat`. Moreover, the tabulation of all plays with an error in 2023 KCA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

15.2.1 April

	game	MIN vs KCA on 20230401
	state before	(0, 0, 0, 0, 4, 1, 0, 1)
1.	error	play,4,1,dozih001,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 889 *In the bottom of the 4th inning, with bases empty, KCA batter Hunter Dozier hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	ATL vs KCA on 20230414
	state before	(0, 1, 1, 0, 1, 4, 0, 1)
2.	error	play,1,1,pasqv001,22,CBFFFFBX,36(1)/F0/G3.2-H(NR)(UR);BX2(6E1)
	state after	(1, 0, 1, 0, 1, 4, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 890 This game had 2 errors. The final score was 10-3.

In the bottom of the 1st inning, with runners on 1st and 2nd, KCA batter Vinnie Pasquantino hit a ground ball to 1st. The shortstop forced out the runner from 1st. The runner from 2nd scored (unearned, no RBI). An error by the pitcher allowed the batter to safely reach 2nd.

	game	ATL vs KCA on 20230414
	state before	(2, 0, 0, 0, 3, 7, 1, 1)
3.	error	play,3,1,melem001,11,SBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 3, 7, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 891 *In the bottom of the 3rd inning, with bases empty, KCA batter MJ Melendez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	ATL vs KCA on 20230416
	state before	(0, 1, 1, 0, 3, 3, 0, 1)
4.	error	play,3,1,lopen001,11,CBX,FC/SH/BG2-.2-3;1-2;B-1(E1/TH)
	state after	(0, 1, 1, 1, 3, 3, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 892 This game had 2 errors. The final score was 5-4.

In the bottom of the 3rd inning, with runners on 1st and 2nd, KCA batter Nicky Lopez hit a soft bunt to shallow infield. The runners on 1st and 2nd each advanced one base. A throwing error by the pitcher allowed the batter to reach 1st. The bases are loaded.

	game	ATL vs KCA on 20230416
	state before	(1, 0, 0, 0, 5, 4, 1, 0)
5.	error	play,5,0,acunr001,20,BBX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 5, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 893 *In the top of the 5th inning, with bases empty, ATL batter Ronald Acuna hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TEX vs KCA on 20230417
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
6.	error	play,1,0,semim001,10,BX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 894 *This game had 2 errors. The final score was 4-0. In the top of the 1st inning, with bases empty, TEX batter Marcus Semien hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TEX vs KCA on 20230417
	state before	(2, 0, 0, 0, 3, 4, 0, 1)
7.	error	play,3,1,eaton001,01,CX,6E3/G6.B-1
	state after	(2, 1, 0, 0, 3, 4, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 895 *In the bottom of the 3rd inning, with bases empty, KCA batter Nate Eaton hit a ground ball to short. An error by the first baseman (assisted by the shortstop) allowed the batter to reach 1st base.*

	game	TEX vs KCA on 20230419
	state before	(0, 0, 0, 0, 7, 5, 3, 0)
8.	error	play,7,0,semim001,21,...BSBX,C/E2.B-1
	state after	(0, 1, 0, 0, 7, 5, 3, 0)
	fielder	2 (catcher on the home team)

Summary 896 *In the top of the 7th inning, with bases empty, TEX batter Marcus Semien had a 2-1 count. Catcher's interference allowed the batter to reach 1st base.*

15.2.2 May

	game	BAL vs KCA on 20230502
	state before	(2, 1, 1, 0, 8, 10, 7, 0)
9.	error	play,8,0,uriar001,00,X,S7/L7.2-H;1-3(E7)
	state after	(2, 1, 0, 1, 8, 11, 7, 0)
	fielder	7 (left field on the home team)

Summary 897 *In the top of the 8th inning, with runners on 1st and 2nd, BAL batter Ramon Urias hit a line drive single to left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the runner on 1st to reach 3rd.*

	game	BAL vs KCA on 20230504
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
10.	error	play,1,0,mounr001,00,X,E9/F9LS.B-2
	state after	(2, 0, 1, 0, 1, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 898 *In the top of the 1st inning, with bases empty, BAL batter Ryan Mountcastle hit a fly ball to right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	OAK vs KCA on 20230505
	state before	(0, 1, 1, 0, 1, 0, 0, 0)
11.	error	play,1,0,rookb001,31,BBFB2,SB3.2-H(E1/TH) (NR) (UR) ;1-2
	state after	(0, 1, 1, 0, 1, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 899 This game had 2 errors. The final score was 12-8. *In the top of the 1st inning, with runners on 1st and 2nd, OAK batter Brent Rooker had a 3-1 count. During the next pitch, which was a ??, the runner on 2nd stole 3rd. A throwing error by the pitcher allowed that runner (now on 3rd) to score (unearned, no RBI) and the runner on 1st to reach 2nd. Rooker is still at bat, with a 3-1 count.*

	game	OAK vs KCA on 20230505
	state before	(1, 0, 0, 1, 9, 11, 8, 0)
12.	error	play,9,0,nodar001,22,CS.>FB>B.X,FC3/G3.3-H;B-2(E3/TH)
	state after	(1, 0, 0, 1, 9, 11, 8, 0)
	fielder	3 (1st baseman on the home team)

Summary 900 *In the top of the 9th inning, with a runner on 3rd, OAK batter Ryan Noda hit a ground ball into a fielder's choice at 1st. The runner on 3rd scored (1 RBI). A throwing error by the first baseman allowed the batter to reach 2nd.*

	game	OAK vs KCA on 20230506
	state before	(1, 1, 0, 0, 2, 2, 1, 1)
13.	error	play,2,1,bradj001,01,CX,E1/G2-.1-2;B-1
	state after	(1, 1, 1, 0, 2, 2, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 901 *In the bottom of the 2nd inning, with a runner on 1st, KCA batter Jackie Bradley hit a soft ground ball to the shallow infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CHA vs KCA on 20230508
	state before	(0, 1, 0, 0, 2, 0, 1, 0)
14.	error	play,2,0,sheeg001,00,X,E7/L78.1-2;B-1
	state after	(0, 1, 1, 0, 2, 0, 1, 0)
	fielder	7 (left field on the home team)

Summary 902 This game had 2 errors. The final score was 5-12. *In the top of the 2nd inning, with a runner on 1st, CHA batter Gavin Sheets hit a line drive to left center field. An error by the left fielder allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CHA vs KCA on 20230508
	state before	(2, 0, 0, 0, 8, 5, 12, 1)
15.	error	play,8,1,olive001,21,BFBX,E8/L89XD.B-3
	state after	(2, 0, 0, 1, 8, 5, 12, 1)
	fielder	8 (center field on the visiting team)

Summary 903 *In the bottom of the 8th inning, with bases empty, KCA batter Edward Olivares hit a line drive to extra deep right center field. An error by the center fielder allowed the batter to reach 3rd base.*

	game	CHA vs KCA on 20230510
	state before	(2, 0, 0, 0, 7, 1, 9, 1)
16.	error	play,7,1,massm001,10,BX,D9/L9L.B-3(E9)
	state after	(2, 0, 0, 1, 7, 1, 9, 1)
	fielder	9 (right field on the visiting team)

Summary 904 *In the bottom of the 7th inning, with bases empty, KCA batter Michael Massey hit a line drive double to right field. An error by the right fielder allowed the batter to reach 3rd.*

	game	DET vs KCA on 20230522
	state before	(1, 1, 0, 0, 7, 5, 5, 0)
17.	error	play,7,0,haase001,01,C1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 7, 5, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 905 *In the top of the 7th inning, with a runner on 1st, DET batter Eric Haase has a 0-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 2nd. Haase is still at bat with a 0-1 count.*

	game	DET vs KCA on 20230523
	state before	(1, 0, 0, 0, 4, 0, 0, 1)
18.	error	play,4,1,olive001,00,X,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 4, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 906 This game had 2 errors. The final score was 1-4.

In the bottom of the 4th inning, with bases empty, KCA batter Edward Olivares hit a ground ball to shorts. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	DET vs KCA on 20230523
	state before	(1, 1, 1, 0, 6, 1, 4, 0)
19.	error	play,6,0,badda001,21,CBBX,FC6/G6.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 6, 1, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 907 *In the top of the 6th inning, with runners on 1st and 2nd, DET batter Akil Baddoo hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

	game	WAS vs KCA on 20230526
	state before	(0, 1, 1, 0, 4, 1, 0, 1)
20.	error	play,4,1,olive001,01,C>C,SB3;SB2.2-H(E2/TH)(NR);1-3
	state after	(0, 0, 0, 1, 4, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 908 *In the bottom of the 4th inning, with runners on 1st and 2nd, KCA batter Edward Olivares had a 0-1 count. During the next pitch, which was a called strike, the runners on 2nd and 3rd each stole their next base. A throwing error by the catcher allowed the runner*

from 2nd who is now on 3rd) to score (unearned, no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Olivares is still at bat, now with a 0-2 count.

	game	WAS vs KCA on 20230527
	state before	(0, 1, 0, 0, 6, 2, 2, 0)
21.	error	play,6,0,smitd008,11,BFX,S9/G34.1-3(E9)
	state after	(0, 1, 0, 1, 6, 2, 2, 0)
	fielder	9 (right field on the home team)

Summary 909 *In the top of the 6th inning, with a runner on 1st, WAS batter Dominic Smith hit a ground ball single to the 1st/2nd base gap. An error by the right fielder allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	WAS vs KCA on 20230528
	state before	(1, 0, 0, 0, 9, 2, 2, 1)
22.	error	play,9,1,melem001,20,BBX,E3/G3.B-2
	state after	(1, 0, 1, 0, 9, 2, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 910 *In the bottom of the 9th inning, with bases empty, KCA batter MJ Melendez hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 2nd base.*

15.2.3 June

	game	COL vs KCA on 20230603
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
23.	error	play,1,0,profj001,32,BFBFFFBX,FC6/G6.1-2(E6);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 911 This game had 2 errors. The final score was 6-4.

In the top of the 1st inning, with a runner on 1st, COL batter Jurickson Profar hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	COL vs KCA on 20230603
	state before	(1, 0, 0, 0, 4, 5, 2, 0)
24.	error	play,4,0,tovae001,22,BCBFFX,S8/G6M.B-2(E8)
	state after	(1, 0, 1, 0, 4, 5, 2, 0)
	fielder	8 (center field on the home team)

Summary 912 *In the top of the 4th inning, with bases empty, COL batter Ezequiel Tovar hit a ground ball single to deep short. An error by the center fielder allowed the batter to reach 2nd.*

	game	CIN vs KCA on 20230612
	state before	(2, 1, 0, 0, 6, 3, 3, 1)
25.	error	play,6,1,bland001,12,SS.*BF1,POCS2(134E1)
	state after	(2, 0, 1, 0, 6, 3, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 913 *In the bottom of the 6th inning, with a runner on 1st, KCA batter Dairon Blanco has a 1-2 count. The pitcher attempted to pickoff the runner at 1st but an error by the pitcher allowed the runner to safely reach 2nd. Blanco is still at bat with a 1-2 count.*

	game	CIN vs KCA on 20230613
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
26.	error	play,1,1,wittb002,12,BTCX,E1/TH/G15-.1-3;B-1
	state after	(1, 1, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 914 This game had 2 errors. The final score was 5-4.

In the bottom of the 1st inning, with a runner on 1st, KCA batter Bobby Witt hit a soft ground ball to the shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.

27.	game	CIN vs KCA on 20230613
	state before	(0, 1, 0, 0, 2, 0, 1, 0)
	error	play,2,0,stees001,02,CS1>B,SB2.1-3(E4)
	state after	(0, 0, 0, 1, 2, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 915 *In the top of the 2nd inning, with a runner on 1st, CIN batter Spencer Steer had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A error by the second baseman allowed that runner to advance to 3rd. Steer is still at bat, now with a 1-2 count.*

Visiting/Home team rosters for CIN vs KCA on 2023-06-14

position	innings	player	position	innings	player
1	1	Ben Lively	1	1	Daniel Lynch
1	6	Alex Young	1	8	Nick Wittgren
1	7	Buck Farmer	1	9	Jose Cuas
1	8	Fernando Cruz	2	1	Freddy Fermin
1	9	Alexis Diaz	3	1	Nick Pratto
1	9	Daniel Duarte	4	1	Michael Massey
2	1	Tyler Stephenson	5	1	Nicky Lopez
3	1	Spencer Steer	6	1	Bobby Witt Jr.
4	1	Matt McLain	7	1	Edward Olivares
5	1	Kevin Newman	8	1	Dairon Blanco
6	1	Elly De La Cruz	9	1	MJ Melendez
7	1	Stuart Fairchild			
8	1	Jose Barrero			
9	1	TJ Hopkins			

	game	CIN vs KCA on 20230614
	state before	(2, 1, 0, 0, 1, 0, 0, 1)
28.	error	play,1,1,wittb002,10,BX,S9/L89S.1-3;B-2(E9/TH)
	state after	(2, 0, 1, 1, 1, 0, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 916 This game had 4 errors. The final score was 7-4.

In the bottom of the 1st inning, with a runner on 1st, KCA batter Bobby Witt hit a line drive single to right center field. The runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.

	game	CIN vs KCA on 20230614
	state before	(1, 1, 1, 0, 3, 1, 0, 1)
29.	error	play,3,1,wittb002,00,X,S9/L9LS.2-H;1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 3, 1, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 917 *In the bottom of the 3rd inning, with runners on 1st and 2nd, KCA batter Bobby Witt hit a line drive single to shallow right field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	CIN vs KCA on 20230614
	state before	(2, 1, 0, 0, 7, 7, 3, 1)
30.	error	play,7,1,olive001,01,.CX,S8/G6M.1-3;B-2(E8)
	state after	(2, 0, 1, 1, 7, 7, 3, 1)
	fielder	8 (center field on the visiting team)

Summary 918 *In the bottom of the 7th inning, with a runner on 1st, KCA batter Edward Olivares hit a ground ball single to deep short. The runner on 1st advanced to 3rd. An error by the center fielder allowed the batter to reach 2nd.*

	game	CIN vs KCA on 20230614
	state before	(1, 1, 1, 0, 9, 7, 3, 1)
31.	error	play,9,1,olive001,21,BBCX,S5/G56S.2-H(NR);1-3(E5/TH);B-2
	state after	(1, 0, 1, 1, 9, 7, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 919 *In the bottom of the 9th inning, with runners on 1st and 2nd, KCA batter Edward Olivares hit a ground ball single to shallow 3rd base. The runner on 2nd scored (no RBI) A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	ANA vs KCA on 20230617
	state before	(0, 1, 0, 0, 9, 9, 8, 1)
32.	error	play,9,1,garc003,01,.1C>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 9, 8, 1)
	fielder	2 (catcher on the visiting team)

Summary 920 *In the bottom of the 9th inning, with a runner on 1st, KCA batter Maikel Garcia hit a single had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 3rd. A throwing error by the catcher allowed that runner to advance to 3rd. Garcia is still at bat, now with a 1-1 count.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Gavin Williams	1	1	Brady Singer
1	8	Trevor Stephan	1	7	Carlos Hernandez
1	9	Emmanuel Clase	1	8	Aroldis Chapman
2	1	Bo Naylor	1	9	Scott Barlow
3	1	Josh Naylor	2	1	Freddy Fermin
3	9	Gabriel Arias	3	1	Nick Pratto
4	1	Andres Gimenez	4	1	Nicky Lopez
5	1	Jose Ramirez	5	1	Maikel Garcia
6	1	Amed Rosario	6	1	Bobby Witt Jr.
7	1	Steven Kwan	7	1	MJ Melendez
8	1	Myles Straw	8	1	Kyle Isbel
9	1	Will Brennan	9	1	Drew Waters

	game	CLE vs KCA on 20230627
	state before	(2, 1, 0, 0, 6, 0, 0, 0)
33.	error	play,6,0,naylj001,22,VFF.BFX,E6/G6M.1-2;B-1
	state after	(2, 1, 1, 0, 6, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 921 This game had 3 errors. The final score was 2-1.

In the top of the 6th inning, with a runner on 1st, CLE batter Josh Naylor hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CLE vs KCA on 20230627
	state before	(1, 0, 0, 1, 8, 0, 0, 1)
34.	error	play,8,1,lopen001,12,>B.FSX,FC1/G1.3-H(E1/TH) (NR) (UR) ;B-1
	state after	(1, 1, 0, 0, 8, 0, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 922 *In the bottom of the 8th inning, with a runner on 3rd, KCA batter Nicky Lopez hit a ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st base.*

	game	CLE vs KCA on 20230627
	state before	(2, 0, 0, 0, 9, 2, 1, 1)
35.	error	play,9,1,melem001,22,BFSFBX,E4/G34.B-1
	state after	(2, 1, 0, 0, 9, 2, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 923 *In the bottom of the 9th inning, with bases empty, KCA batter MJ Melendez hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	LAN vs KCA on 20230630
	state before	(0, 0, 0, 0, 7, 5, 3, 0)
36.	error	play,7,0,martj006,22,....FBF.BX,E5/G5.B-1
	state after	(0, 1, 0, 0, 7, 5, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 924 *In the top of the 7th inning, with bases empty, LAN batter J.D. Martinez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

15.2.4 July

	game	LAN vs KCA on 20230701
	state before	(2, 0, 0, 0, 5, 3, 5, 0)
37.	error	play,5,0,vargm001,01,FX,E5/G5D.B-2
	state after	(2, 0, 1, 0, 5, 3, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 925 This game had 2 errors. The final score was 4-6.

In the top of the 5th inning, with bases empty, LAN batter Miguel Vargas hit a ground ball to deep 3rd base. An error by the third baseman allowed the batter to reach 2nd base.

	game	LAN vs KCA on 20230701
	state before	(0, 0, 0, 0, 7, 3, 5, 1)
38.	error	play,7,1,bland001,00,X,3E1/G3.B-1
	state after	(0, 1, 0, 0, 7, 3, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 926 *In the bottom of the 7th inning, with bases empty, KCA batter Dairon Blanco hit a ground ball to 1st. An error by the pitcher (assisted by the first baseman) allowed the batter to reach 1st base.*

	game	LAN vs KCA on 20230702
	state before	(0, 0, 0, 0, 6, 1, 7, 1)
39.	error	play,6,1,fermf002,22,..CBCFBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 1, 7, 1)
	fielder	6 (shortstop on the visiting team)

Summary 927 *In the bottom of the 6th inning, with bases empty, KCA batter Freddy Fermin hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs KCA on 20230716
	state before	(2, 1, 0, 0, 8, 4, 8, 1)
40.	error	play,8,1,wated001,12,.SBF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 4, 8, 1)
	fielder	2 (catcher on the visiting team)

Summary 928 *In the bottom of the 8th inning, with a runner on 1st, KCA batter Drew Waters had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Waters is still at bat, now with a 2-2 count.*

	game	DET vs KCA on 20230717
	state before	(1, 1, 0, 0, 4, 0, 0, 1)
41.	error	play,4,1,fermf002,21,B*BS1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 4, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 929 This game had 2 errors. The final score was 3-2.
In the bottom of the 4th inning, with a runner on 1st, KCA batter Freddy Fermin has a 2-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner from 1st advance to 2nd. Fermin is still at bat with a 2-1 count.

	game	DET vs KCA on 20230717
	state before	(0, 0, 0, 0, 7, 0, 2, 1)
42.	error	play,7,1,fermf002,22,CFFF.BBX,S6/G6M.B-2(E6/TH)
	state after	(0, 0, 1, 0, 7, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 930 *In the bottom of the 7th inning, with bases empty, KCA batter Freddy Fermin hit a ground ball single to deep short. A throwing error by the shortstop allowed the batter to advance to 2nd.*

	game	DET vs KCA on 20230718
	state before	(0, 0, 0, 0, 6, 5, 7, 0)
43.	error	play,6,0,baezj001,00,.X,D7/G5.B-3(E7)
	state after	(0, 0, 0, 1, 6, 5, 7, 0)
	fielder	7 (left field on the home team)

Summary 931 *In the top of the 6th inning, with bases empty, DET batter Javier Baez hit a ground ball double to 3rd. An error by the left fielder allowed the batter to reach 2nd.*

	game	MIN vs KCA on 20230728
	state before	(2, 0, 1, 0, 5, 2, 1, 0)
44.	error	play,5,0,keplm001,12,BFFFX,3E1/G3.2-3;B-1
	state after	(2, 1, 0, 1, 5, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 932 *In the top of the 5th inning, with a runner on 2nd, MIN batter Max Kepler hit a ground ball to 1st. An error by the pitcher (with an assist from the first baseman) allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

15.2.5 August

	game	NYN vs KCA on 20230801
	state before	(1, 1, 1, 0, 7, 1, 1, 1)
45.	error	play,7,1,isbek001,22,CBCFF.B>B,SB3;SB2.2-H(E2/TH)(NR);1-3
	state after	(1, 0, 0, 1, 7, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 933 This game had 2 errors. The final score was 6-7. *In the bottom of the 7th inning, with runners on 1st and 2nd, KCA batter Kyle Isbel had a 2-2 count. During the next pitch, which was a ball, the runners on 1st and 2nd each stole their next base. A throwing error by the catcher allowed the runner now on 3rd to score (unearned, no RBI) and the runner from 1st to advance to 3rd. Isbel is still at bat, with a full count.*

	game	NYN vs KCA on 20230801
	state before	(2, 1, 0, 0, 10, 6, 6, 1)
46.	error	play,10,1,bland001,00,X,E5/TH/G5.1-2;B-1
	state after	(2, 1, 1, 0, 10, 6, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 934 *In the bottom of the 10th inning, with a runner on 1st, KCA batter Dairon Blanco hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	NYN vs KCA on 20230803
	state before	(1, 1, 0, 1, 8, 0, 7, 1)
47.	error	play,8,1,tayls003,12,FBSX,FC5/G56.3-H;1X2(5E4);B-1
	state after	(1, 1, 0, 1, 8, 0, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 935 *In the bottom of the 8th inning, with runners on 1st and 3rd, KCA batter Samad Taylor hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the second baseman allowed the runner on 1st to safely reach 2nd and the batter to reach 1st.*

	game	SLN vs KCA on 20230811
	state before	(1, 1, 0, 0, 3, 0, 9, 0)
48.	error	play,3,0,arenn001,10,BX,E5/G56.1-2;B-1
	state after	(1, 1, 1, 0, 3, 0, 9, 0)
	fielder	5 (3rd baseman on the home team)

Summary 936 This game had 2 errors. The final score was 8-12. *In the top of the 3rd inning, with a runner on 1st, SLN batter Nolan Arenado hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SLN vs KCA on 20230811
	state before	(0, 0, 0, 0, 7, 6, 9, 1)
49.	error	play,7,1,massm001,20,.BBX,E4/G89.B-2
	state after	(0, 0, 1, 0, 7, 6, 9, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 937 *In the bottom of the 7th inning, with bases empty, KCA batter Michael Massey hit a ground ball to right center field. An error by the second baseman allowed the batter to reach 2nd base.*

	game	SLN vs KCA on 20230812
	state before	(1, 1, 0, 0, 5, 3, 2, 0)
50.	error	play,5,0,arenn001,10,BX,E6/G6D.1-2;B-1
	state after	(1, 1, 1, 0, 5, 3, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 938 *In the top of the 5th inning, with a runner on 1st, SLN batter Nolan Arenado hit a ground ball to deep short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SEA vs KCA on 20230815
	state before	(0, 0, 0, 0, 4, 4, 0, 0)
51.	error	play,4,0,canzd001,12,FBCFX,4E1/G3.B-1
	state after	(0, 1, 0, 0, 4, 4, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 939 This game had 2 errors. The final score was 10-8. *In the top of the 4th inning, with bases empty, SEA batter Dominic Canzone hit a ground ball to 1st. An error by the pitcher allowed the batter to reach 1st base.*

	game	SEA vs KCA on 20230815
	state before	(0, 0, 0, 0, 9, 8, 5, 1)
52.	error	play,9,1,velan001,01,.CX,E4/TH/G4.B-1
	state after	(0, 1, 0, 0, 9, 8, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 940 *In the bottom of the 9th inning, with bases empty, KCA batter Nelson Velazquez hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	SEA vs KCA on 20230816
	state before	(0, 0, 0, 1, 2, 4, 1, 1)
53.	error	play,2,1,velan001,00,F,FLE2
	state after	(0, 0, 0, 1, 2, 4, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 941 This game had 2 errors and an ∞ play. The final score was 6-5.

In the bottom of the 2nd inning, with no outs and a runner on 3rd, KCA batter Nelson Velazquez hit a foul fly ball behind home plate. An error by the catcher allowed the batter to remain safely at bat.

	game	SEA vs KCA on 20230816
	state before	(0, 0, 0, 0, 8, 4, 4, 0)
54.	error	play,8,0,moord002,22,.CCBBX,E5/G5S.B-1
	state after	(0, 1, 0, 0, 8, 4, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 942 *In the top of the 8th inning, with bases empty, SEA batter Dylan Moore hit a ground ball to shallow 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SEA vs KCA on 2023-08-17

position	innings	player	position	innings	player
1	1	George Kirby	1	1	Angel Zerpa
1	7	Isaiah Campbell	1	5	Max Castillo
1	8	Trent Thornton	1	6	Jackson Kowar
1	9	Matt Brash	1	8	Carlos Hernandez
2	1	Brian O’Keefe	1	8	Taylor Hearn
2	9	Cal Raleigh	1	9	John McMillon
3	1	Ty France	2	1	Freddy Fermin
4	1	Dylan Moore	3	1	Matt Duffy
5	1	Eugenio Suarez	4	1	Michael Massey
6	1	Jose Caballero	5	1	Maikel Garcia
7	1	Cade Marlowe	6	1	Bobby Witt Jr.
8	1	Julio Rodriguez	7	1	MJ Melendez
9	1	Sam Haggerty	8	1	Kyle Isbel
9	6	Dominic Canzone	9	1	Nelson Velazquez

	game	SEA vs KCA on 20230817
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
55.	error	play,1,0,rodrj007,21,BFBX,S5/G5.B-2(E5/TH)
	state after	(0, 0, 1, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 943 This game had 3 errors. The final score was 6-4.

In the top of the 1st inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball single to 3rd base. A throwing error by the third basemann allowed the batter to reach 2nd.

	game	SEA vs KCA on 20230817
	state before	(1, 1, 0, 0, 6, 1, 1, 0)
56.	error	play,6,0,canzd001,00,.>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 1, 1, 0)
	fielder	2 (catcher on the home team)

Summary 944 *In the top of the 6th inning, with a runner on 1st, SEA batter Dominic Canzone had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Canzone is still at bat, now with a 1-0 count.*

	game	SEA vs KCA on 20230817
	state before	(0, 0, 0, 0, 6, 2, 1, 1)
57.	error	play,6,1,garcm003,11,.CBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 945 *In the bottom of the 6th inning, with bases empty, KCA batter Maikel Garcia hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs KCA on 20230830
	state before	(1, 1, 0, 0, 2, 3, 0, 0)
58.	error	play,2,0,rivaa001,01,C1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 2, 3, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 946 *In the top of the 2nd inning, with a runner on 1st, PIT batter Alfonso Rivas has a 0-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 2nd. Rivas is still at bat with a 0-1 count.*

15.2.6 September

	game	BOS vs KCA on 20230902
	state before	(2, 0, 0, 0, 2, 3, 0, 0)
59.	error	play,2,0,verda001,32,CC.BBBX,D8/L78XD+.B-3(E8)
	state after	(2, 0, 0, 1, 2, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 947 This game had 2 errors. The final score was 9-5.

In the top of the 2nd inning, with bases empty, BOS batter Alex Verdugo hit a hard line drive double to extra deep left center field. An error by the center fielder allowed the batter to reach 3rd.

	game	BOS vs KCA on 20230902
	state before	(2, 1, 0, 0, 5, 8, 1, 0)
60.	error	play,5,0,dever001,00,X,E3/G9LS.1-3;B-2
	state after	(2, 0, 1, 1, 5, 8, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 948 *In the top of the 5th inning, with a runner on 1st, BOS batter Rafael Devers hit a ground ball to shallow right field down the 1st base line. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	BOS vs KCA on 20230903
	state before	(2, 0, 0, 0, 3, 0, 0, 0)
61.	error	play,3,0,verda001,01,CX,E5/G56.B-1
	state after	(2, 1, 0, 0, 3, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 949 This game had 2 errors. The final score was 7-3.

In the top of the 3rd inning, with bases empty, BOS batter Alex Verdugo hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	BOS vs KCA on 20230903
	state before	(0, 0, 1, 0, 7, 4, 1, 0)
62.	error	play,7,0,refsr001,21,.FBBX,E6/G6.2-H(NR) (UR) ;B-2
	state after	(0, 0, 1, 0, 7, 5, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 950 *In the top of the 7th inning, with a runner on 2nd, BOS batter Rob Refsnyder hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CHA vs KCA on 20230904
	state before	(2, 0, 1, 0, 8, 0, 12, 0)
63.	error	play,8,0,thomt002,32,BBBCFX,S3/G34.2-H(E3/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 8, 1, 12, 0)
	fielder	3 (1st baseman on the home team)

Summary 951 *In the top of the 8th inning, with a runner on 2nd, CHA batter Trayce Thompson hit a ground ball single to the 1st/2nd base gap. A throwing error by the first baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	CHA vs KCA on 20230905
	state before	(1, 1, 1, 0, 6, 6, 4, 1)
64.	error	play,6,1,massm001,31,.CBBB2,P02(E1/TH) .2-3;1-2
	state after	(1, 0, 1, 1, 6, 6, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 952 This game had 2 errors. The final score was 6-7. *In the bottom of the 6th inning, with runners on 1st and 2nd, KCA batter Michael Massey has a 3-1 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance a base. Massey is still at bat with a 3-1 count.*

	game	CHA vs KCA on 20230905
	state before	(0, 0, 0, 0, 9, 6, 5, 1)
65.	error	play,9,1,loftn001,00,.X,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 9, 6, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 953 *In the bottom of the 9th inning, with bases empty, KCA batter Nick Loftin hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs KCA on 20230906
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
66.	error	play,1,1,beatm001,11,BCX,FC1/G1.1-3(E1/TH);B-1
	state after	(1, 1, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 954 *In the bottom of the 1st inning, with a runner on 1st, KCA batter Matt Beaty hit a ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	HOU vs KCA on 20230915
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
67.	error	play,3,0,penaj004,10,.BX,E8/L8D+.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 955 *In the top of the 3rd inning, with bases empty, HOU batter Jeremy Pena hit a hard line drive to deep center field. An error by the center fielder allowed the batter to reach 1st base.*

Visiting/Home team rosters for HOU vs KCA on 2023-09-16

position	innings	player	position	innings	player
1	1	J.P. France	1	1	Cole Ragans
1	5	Rafael Montero	1	7	Collin Snider
1	6	Phil Maton	1	7	James McArthur
1	7	Hector Neris	1	9	Carlos Hernandez
1	8	Joel Kuhnel	1	9	Taylor Clarke
2	1	Yainer Diaz	2	1	Salvador Perez
3	1	Jose Abreu	2	5	Logan Porter
4	1	Jose Altuve	3	1	Nick Pratto
5	1	Alex Bregman	4	1	Michael Massey
6	1	Jeremy Pena	5	1	Maikel Garcia
7	1	Chas McCormick	6	1	Bobby Witt Jr.
8	1	Mauricio Dubon	7	1	MJ Melendez
9	1	Kyle Tucker	8	1	Kyle Isbel
			9	1	Dairon Blanco

	game	HOU vs KCA on 20230916
	state before	(1, 1, 0, 0, 3, 0, 3, 0)
68.	error	play,3,0,dubom001,10,BX,FC5/G56.1-3(E5);B-1
	state after	(1, 1, 0, 1, 3, 0, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 956 This game had 3 errors. The final score was 8-10.

In the top of the 3rd inning, with a runner on 1st, HOU batter Mauricio Dubon hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	HOU vs KCA on 20230916
	state before	(1, 0, 1, 0, 6, 3, 5, 1)
69.	error	play,6,1,garc003,10,BX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 6, 3, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 957 *In the bottom of the 6th inning, with a runner on 2nd, KCA batter Maikel Garcia hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	HOU vs KCA on 20230916
	state before	(1, 1, 1, 0, 7, 6, 7, 1)
70.	error	play,7,1,blan001,10,BX,FC5/G56.2-H(NR)(UR);1-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 7, 6, 8, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 958 *In the bottom of the 7th inning, with runners on 1st and 2nd, KCA batter Dairon Blanco hit a ground ball into a fielder's choice at 3rd. The runner on 2nd scored (unearned, no RBI). A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	HOU vs KCA on 20230917
	state before	(1, 1, 0, 0, 2, 0, 0, 1)
71.	error	play,2,1,port1001,22,BBC1FFX,E1/TH/G25.1-3;B-2
	state after	(1, 0, 1, 1, 2, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 959 This game had 2 errors. The final score was 7-1. *In the bottom of the 2nd inning, with a runner on 1st, KCA batter Logan Porter hit a ground ball to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	HOU vs KCA on 20230917
	state before	(0, 0, 0, 0, 9, 4, 1, 0)
72.	error	play,9,0,meyej002,10,...BX,E7/F7D.B-3
	state after	(0, 0, 0, 1, 9, 4, 1, 0)
	fielder	7 (left field on the home team)

Summary 960 *In the top of the 9th inning, with bases empty, HOU batter Jake Meyers hit a fly ball to deep left field. An error by the left fielder allowed the batter to reach 3rd base.*

Visiting/Home team rosters for CLE vs KCA on 2023-09-19

position	innings	player	position	innings	player
1	1	Logan Allen	1	1	Steven Cruz
1	4	Michael Kelly	1	2	Alec Marsh
1	5	James Karinchak	1	7	Carlos Hernandez
1	6	Eli Morgan	1	7	Taylor Clarke
1	7	Xzavion Curry	1	8	James McArthur
1	8	Nick Sandlin	1	9	Collin Snider
2	1	Bo Naylor	2	1	Tyler Cropley
3	1	Josh Naylor	2	9	Logan Porter
4	1	Andres Gimenez	3	1	Nick Pratto
5	1	Jose Ramirez	4	1	Nick Loftin
6	1	Gabriel Arias	4	8	Michael Massey
7	1	Steven Kwan	5	1	Maikel Garcia
8	1	Ramon Laureano	6	1	Bobby Witt Jr.
9	1	Will Brennan	7	1	MJ Melendez
			8	1	Kyle Isbel
			9	1	Nelson Velazquez

	game	CLE vs KCA on 20230919
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
73.	error	play,1,1,garc003,12,FBCX,S5/G5S-.B-2(E5/TH)
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 961 This game had 3 errors. The final score was 6-7.
In the bottom of the 1st inning, with bases empty, KCA batter Maikel Garcia hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	CLE vs KCA on 20230919
	state before	(0, 0, 1, 0, 5, 0, 4, 0)
74.	error	play,5,0,ramij003,20,BBX,E9/L9+.2-3;B-2
	state after	(0, 0, 1, 1, 5, 0, 4, 0)
	fielder	9 (right field on the home team)

Summary 962 *In the top of the 5th inning, with a runner on 2nd, CLE batter Jose Ramirez hit a hard line drive to right field. An error by the right fielder allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd base.*

	game	CLE vs KCA on 20230919
	state before	(0, 1, 0, 1, 5, 2, 4, 1)
75.	error	play,5,1,velan001,00,X,E3/G9L.3-H;1-3;B-1
	state after	(0, 1, 0, 1, 5, 2, 5, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 963 *In the bottom of the 5th inning, with runners on 1st and 3rd, KCA batter Nelson Velazquez hit a ground ball to right field. An error by the first baseman allowed the runner on 3rd to score (1 RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.*

Visiting/Home team rosters for CLE vs KCA on 2023-09-20

position	innings	player	position	innings	player
1	1	Lucas Giolito	1	1	Zack Greinke
1	6	James Karinchak	1	4	Angel Zerpa
1	7	Michael Kelly	2	1	Logan Porter
1	8	Enyel De Los Santos	3	1	Nick Loftin
2	1	Cam Gallagher	3	8	Nick Pratto
2	7	David Fry	4	1	Michael Massey
3	1	Kole Calhoun	5	1	Maikel Garcia
4	1	Andres Gimenez	6	1	Bobby Witt Jr.
5	1	Jose Ramirez	7	1	Edward Olivares
6	1	Jose Tena	7	8	Dairon Blanco
6	7	Tyler Freeman	8	1	Drew Waters
7	1	Will Brennan	9	1	Nelson Velazquez
8	1	Myles Straw			
9	1	Ramon Laureano			

	game	CLE vs KCA on 20230920
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
76.	error	play,1,1,garc003,00,X,S7/L78+.B-2(E7)
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 964 This game had 3 errors, including a consecutive error. The same fielder was involved in 2 of these 3 errors. The final score was 2-6.

In the bottom of the 1st inning, with bases empty, KCA batter Maikel Garcia hit a hard line drive single to left center field. An error by the left fielder allowed the batter to reach 2nd.

	game	CLE vs KCA on 20230920
	state before	(0, 0, 1, 0, 1, 0, 0, 1)
77.	error	play,1,1,wittb002,01,CX,E6/TH/G6D.2-3;B-1
	state after	(0, 1, 0, 1, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 965 This is a *consecutive error* since Witt is listed immediately after Garcia in the batter's lineup.

In the bottom of the 1st inning, with a runner on 2nd, KCA batter Bobby Witt hit a ground ball to deep short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	CLE vs KCA on 20230920
	state before	(2, 0, 1, 0, 8, 2, 5, 1)
78.	error	play,8,1,melem001,00,X,E6/L8.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 8, 2, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 966 *In the bottom of the 8th inning, with a runner on 2nd, KCA batter MJ Melendez hit a line drive to center field. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	NYA vs KCA on 20230930
	state before	(2, 1, 0, 0, 5, 2, 2, 1)
79.	error	play,5,1,velan001,01,S1>C,CS2(2E4).1-3
	state after	(2, 0, 1, 0, 5, 2, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 967 This game had 2 errors. The final score was 5-2.

In the bottom of the 5th inning, with a runner on 1st, KCA batter Nelson Velazquez had a 0-1 count. During the next pitch, which was a

called strike, the runner on 1st stole 2nd due to an error by the second baseman. That runner advanced to 3rd. Velezquez is still at bat, now with a 0-2 count.

	game	NYA vs KCA on 20230930
	state before	(1, 1, 1, 0, 9, 5, 2, 0)
80.	error	play,9,0,higak001,00,X,FC6/G6.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 9, 5, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 968 *In the top of the 9th inning, with runners on 1st and 2nd, NYA batter Kyle Higashioka hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	NYA vs KCA on 20231001
	state before	(1, 0, 1, 0, 6, 2, 5, 1)
81.	error	play,6,1,duffm002,02,CF2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 6, 2, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 969 *In the bottom of the 6th inning, with a runner on 2nd, KCA batter Matt Duffy has a 0-2 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed that runner to safely reach 3rd. Duffy is still at bat with a 0-2 count.*

Chapter 16

Los Angeles Angels (ANA)

The Los Angeles Angels¹ compete in the American League West division. Founded in Los Angeles in 1961 by Gene Autry, the team didn't move to nearby Anaheim until 1966. The first MLB team to originate in California, they moved to Anaheim in 1966. They won the World Series in 2002 and last won an ALW division title in 2014.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Anaheim against the Angels. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 ANA team is .682 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the ANA is almost one standard deviation below the MLB average.

In ANA 2023, there were 82 errors in the 81 home games, 41 by the ANA and the rest (41) by visiting teams. The total number of errors by the team (home and away) over the 2023 regular season was 95, almost one standard deviation above the MLB average (which is bad).

¹This team was formerly known as the Anaheim Angels. They didn't move or change stadiums but they did legally change their name to the Los Angeles Angels in 2015.

16.1 Season totals

At any time during the pitcher's preliminary movements and until his natural pitching motion commits him to the pitch, he may throw to any base provided he steps directly toward such base before making the throw.

– **MLB Rule Book** 5.07(d)

Throughout, `vt` denotes the visiting team and `ht` denotes the home team.

16.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	44
Runs by ht because of errors by vt:	11
Errors by ht:	41
Runs by vt because of errors by ht:	16

This is a good error-to-run conversion rate: $11/44 = 26.8\%$ on the part of the Angels (compared to the much better $16/41 = 39.0\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 ANA season (home and away) has about $R/PA = 739/6145$ (about 12.0%) change of resulting in a run. Therefore, a randomly selected error is over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 82, so the expected number of errors (by either team) in a randomly selected ANA 2023 home game is about 1.

Errors by position	vt	ht
errors due to pitcher:	3	3
to catcher:	8	4
to 1st baseman:	6	5
to 2nd baseman:	3	3
to 3rd baseman:	8	8
to shortstop:	7	9
to left fielder:	4	1
to center fielder:	1	2
to right fielder:	1	4

16.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	18	15	33
(lone) runner on 1st	11	12	23
(lone) runner on 2nd	1	3	4
(lone) runner on 3rd	1	1	2
two runners on base	10	8	18
bases loaded	0	1	1
total	41	40	81

A table of frequencies of most common errors by the Angels in the ANA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	19.5%
(1, 0, 0, 0)	4	9.8%
(2, 0, 0, 0)	3	7.3%
(0, 1, 0, 0)	7	17.1%
(1, 1, 0, 0)	2	4.9%
(2, 1, 0, 0)	3	7.3%
(1, 0, 1, 0)	2	4.9%
(1, 1, 1, 0)	4	9.8%
(2, 0, 1, 1)	2	4.9%

A table of frequencies of most common errors by a visiting team in ANA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	14.6%
(1, 0, 0, 0)	6	14.6%
(2, 0, 0, 0)	6	14.6%
(0, 1, 0, 0)	3	7.3%
(1, 1, 0, 0)	5	12.2%
(2, 1, 0, 0)	3	7.3%
(0, 1, 1, 0)	2	4.9%
(1, 1, 1, 0)	3	7.3%
(2, 1, 1, 1)	4	9.8%
(1, 1, 0, 1)	1	2.4%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Angels homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `ANA2023.eva`):

state	ANA2023 percentages
(0, 0, 0, 0)	25.2%
(1, 0, 0, 0)	17.7%
(2, 0, 0, 0)	14.3%
(0, 1, 0, 0)	5.6%
(1, 1, 0, 0)	6.8%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.7%
(1, 0, 1, 0)	2.5%
(2, 0, 1, 0)	3.5%
(0, 0, 0, 1)	0.1%
(1, 0, 0, 1)	0.9%
(2, 0, 0, 1)	1.1%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.0%
(2, 1, 1, 0)	3.1%

16.1.3 Some interesting errors

Some of the most interesting error plays in the ANA 2023 home games:

- The game OAK vs ANA on 2023-04-26 had 4 errors.
- The game TBA vs ANA on 2023-08-19 (game 2 of the doubleheader that day) had 4 errors.

16.1.4 Relative team rankings

We have 81 home games between the Los Angeles Angels and some visiting MLB team (vt). A number of teams did not play the Angels in the regular season. However, of those that did, ANA *lost* more games than won against²:

²We record the W/L numbers in parentheses (W = numbers of home games vt won against ANA, T = Total number of home games vt vs Angels played at ANA).

ARI (2/3), TBA (2/3), TOR (2/3), MIA (2/3), LAN (2/2),
BAL (3/3), CIN (3/3), MIA (3/3), HOU (4/6), TEX (4/6), SEA (5/7),

and *won* more than lost against

BOS (0/3), CHN (0/3), NYA (0/3), OAK (1/7), CLE (1/4),
KCA (1/3), MIL (1/3), PIT (1/3), SFN (1/3), WAS (1/3).

The teams that *split* their series were:

CHA (2/4).

The teams that did not play in any of the 81 home games in ANA 2023 were:

ATL, COL, MIL, NYN, PHI, SDN, SLN.

16.2 Game-by-game totals

The Official Scorer shall charge an error to a fielder who causes another fielder to misplay a ball—for example, by knocking the ball out of the other fielder’s glove. On such a play, when the Official Scorer charges an error to the **interfering fielder**, the Official Scorer shall not charge an error to the fielder with whom the other fielder interfered.

– **MLB Rule Book** 9.12(a)(1) Comment

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for ANA 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Matt Thaiss	82	.319
3 (1B)	Nolan Schanuel		.402
4 (2B)	Brandon Drury	93	.306
5 (3B)	Anthony Rendon	43	.361
6 (SS)	Zach Neto	85	.308
7 (LF)	Taylor Ward	94	.335
8 (CF)	Mike Trout	80	.367
9 (RF)	Hunter Renfroe	122	.304

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 ANA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 82 errors in the 81 home games, 41 by ANA and the rest (41) by visiting teams.

16.2.1 April

Visiting/Home team rosters for TOR vs ANA on 2023-04-07

position	innings	player	position	innings	player
1	1	Chris Bassitt	1	1	Patrick Sandoval
1	7	Yimi Garcia	1	7	Jimmy Herget
1	8	Erik Swanson	1	7	Matt Moore
1	9	Jordan Romano	1	8	Andrew Wantz
2	1	Alejandro Kirk	2	1	Logan O’Hoppe
3	1	Vladimir Guerrero Jr.	3	1	Jake Lamb
4	1	Santiago Espinal	4	1	Brandon Drury
5	1	Matt Chapman	5	1	Gio Urshela
6	1	Bo Bichette	6	1	David Fletcher
7	1	Whit Merrifield	7	1	Taylor Ward
7	8	Daulton Varsho	8	1	Mike Trout
8	1	Kevin Kiermaier	9	1	Hunter Renfroe
9	1	George Springer			

		game
		TOR vs ANA on 20230407
		state before
		(1, 0, 0, 0, 1, 0, 2, 1)
1.	error	play,1,1,renfh001,11,BCX,E6/TH/G6+.B-1
		state after
		(1, 1, 0, 0, 1, 0, 2, 1)
		fielder
		6 (shortstop on the visiting team)

Summary 970 This game had 2 errors. The final score was 4-3.

In the bottom of the 1st inning, with bases empty, ANA batter Hunter Renfroe hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	TOR vs ANA on 20230407
	state before	(1, 1, 0, 0, 4, 1, 2, 1)
2.	error	play,4,1,ohopl001,22,CBBF*B>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 4, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 971 *In the bottom of the 4th inning, with a runner on 1st, ANA batter Logan O’Hoppe had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. O’Hoppe is still at bat, now with a full count.*

	game	TOR vs ANA on 20230408
	state before	(0, 0, 0, 0, 3, 4, 0, 1)
3.	error	play,3,1,urshg001,21,BFBX,S6/G4.B-3(E6)
	state after	(0, 0, 0, 1, 3, 4, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 972 This game had 2 errors. The final score was 5-9.

In the bottom of the 3rd inning, with bases empty, ANA batter Giovanny Urshela hit a ground ball single to short. An error by the shortstop allowed the batter to advance to 3rd.

	game	TOR vs ANA on 20230408
	state before	(0, 0, 0, 0, 5, 5, 4, 1)
4.	error	play,5,1,thaim001,11,BSX,E1/TH/G1.B-1
	state after	(0, 1, 0, 0, 5, 5, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 973 *In the bottom of the 5th inning, with bases empty, ANA batter Matt Thaiss hit a ground ball to the pitcher. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	TOR vs ANA on 20230409
	state before	(0, 1, 0, 0, 6, 4, 6, 0)
5.	error	play,6,0,merrw001,21,.BFB>X,E1/G6MS-.1-2;B-1
	state after	(0, 1, 1, 0, 6, 4, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 974 *In the top of the 6th inning, with a runner on 1st, TOR batter Whit Merrifield hit a soft ground ball to short. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	WAS vs ANA on 20230410
	state before	(2, 1, 0, 1, 4, 1, 4, 0)
6.	error	play,4,0,thoml002,10,BX,S4/G4.3-H;1-3(E4/TH);B-2
	state after	(2, 0, 1, 1, 4, 2, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 975 *In the top of the 4th inning, with a runner on 3rd, WAS batter Lane Thomas hit a ground ball single to 2nd base. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter reached 2nd.*

	game	WAS vs ANA on 20230412
	state before	(1, 0, 1, 0, 3, 0, 0, 1)
7.	error	play,3,1,troum001,10,B>B,SB3.2-H(E2/TH)(NR)
	state after	(1, 0, 0, 0, 3, 0, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 976 This game had 2 errors. The final score was 2-3.

In the bottom of the 3rd inning, with a runner on 2nd, ANA batter Mike Trout had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Trout is still at bat, now with a 2-0 count.

	game	WAS vs ANA on 20230412
	state before	(0, 1, 0, 0, 7, 2, 3, 1)
8.	error	play,7,1,renda001,12,.BCLX,E3/G3.1-3;B-1
	state after	(0, 1, 0, 1, 7, 2, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 977 *In the bottom of the 7th inning, with a runner on 1st, ANA batter Anthony Rendon hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	KCA vs ANA on 20230422
	state before	(1, 1, 1, 1, 6, 4, 4, 0)
9.	error	play,6,0,lopen001,01,.CX,36(1)/F0/G3.3-H;2-H(E3)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 6, 6, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 978 This game had 2 errors. The final score was 11-8.

In the top of the 6th inning, with bases loaded, KCA batter Nicky Lopez hit a ground ball to 1st. At second, the shortstop (with an assist from the first baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.

	game	KCA vs ANA on 20230422
	state before	(0, 1, 0, 0, 9, 6, 8, 0)
10.	error	play,9,0,isbek001,11,SBX,D9/F89XD.1-H;B-3(E9/TH)
	state after	(0, 0, 0, 1, 9, 7, 8, 0)
	fielder	9 (right field on the home team)

Summary 979 *In the top of the 9th inning, with a runner on 1st, KCA batter Kyle Isbel hit a fly ball double to extra deep right center field. The runner on 1st scored (1 RBI). A throwing error by the right fielder allowed the batter to reach 3rd.*

	game	KCA vs ANA on 20230423
	state before	(1, 0, 0, 0, 3, 1, 1, 1)
11.	error	play,3,1,wardt002,00,X,E5/G56.B-1
	state after	(1, 1, 0, 0, 3, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 980 *In the bottom of the 3rd inning, with bases empty, ANA batter Taylor Ward hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs ANA on 20230424
	state before	(2, 0, 0, 0, 1, 2, 0, 1)
12.	error	play,1,1,ohtas001,32,BBCCBX,C/E2.B-1
	state after	(2, 1, 0, 0, 1, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 981 *In the bottom of the 1st inning, with bases empty, ANA batter Shohei Ohtani had a full count. Catcher's interference allowed the batter to take 1st.*

Visiting/Home team rosters for OAK vs ANA on 2023-04-26

position	innings	player	position	innings	player
1	1	Luis Medina	1	1	Patrick Sandoval
1	6	Shintaro Fujinami	1	8	Chase Silseth
1	8	Chad Smith	2	1	Matt Thaiss
2	1	Carlos Perez	3	1	Gio Urshela
3	1	Jesus Aguilar	4	1	Brandon Drury
4	1	Jordan Diaz	5	1	Anthony Rendon
5	1	Aledmys Diaz	6	1	Zach Neto
6	1	Kevin Smith	7	1	Taylor Ward
7	1	Brent Rooker	8	1	Mike Trout
8	1	Esteury Ruiz	8	9	Brett Phillips
9	1	Conner Capel	9	1	Hunter Renfroe

	game	OAK vs ANA on 20230426
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
13.	error	play,1,0,rookb001,01,S>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 982 This game had 4 errors. The final score was 3-11.

In the top of the 1st inning, with a runner on 1st, OAK batter Brent Rooker had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rooker is still at bat, now with a 1-1 count.

	game	OAK vs ANA on 20230426
	state before	(1, 1, 1, 0, 2, 1, 4, 1)
14.	error	play,2,1,troum001,11,*BFX,FC6/G6.2-3(E6/TH);1-2;B-1
	state after	(1, 1, 1, 1, 2, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 983 *In the bottom of the 2nd inning, with runners on 1st and 2nd, ANA batter Mike Trout hit a ground ball into a fielder's choice at short. . A throwing error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	OAK vs ANA on 20230426
	state before	(1, 0, 0, 0, 3, 2, 6, 1)
15.	error	play,3,1,urshg001,02,CSX,S5/G5S-.B-2(E5/TH)
	state after	(1, 0, 1, 0, 3, 2, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 984 *In the bottom of the 3rd inning, with bases empty, ANA batter Giovanny Urshela hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	OAK vs ANA on 20230426
	state before	(1, 1, 0, 0, 6, 3, 8, 1)
16.	error	play,6,1,ohtas001,11,*BTX,S7/L78.1-H(E7)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 6, 3, 9, 1)
	fielder	7 (left field on the visiting team)

Summary 985 *In the bottom of the 6th inning, with a runner on 1st, ANA batter Shohei Ohtani hit a line drive single to left center field. An error by the left fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	OAK vs ANA on 20230427
	state before	(1, 1, 1, 0, 8, 5, 8, 0)
17.	error	play,8,0,diaza003,00,X,FC4/G8S.2-H(NR);1-3(E4);B-1
	state after	(1, 1, 0, 1, 8, 6, 8, 0)
	fielder	4 (2nd baseman on the home team)

Summary 986 *In the top of the 8th inning, with runners on 1st and 2nd, OAK batter Aledmys Diaz hit a ground ball into a fielder’s choice at shallow center field. The runner from 2nd scored (no RBI) while an error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

16.2.2 May

Visiting/Home team rosters for TEX vs ANA on 2023-05-05

position	innings	player	position	innings	player
1	1	Dane Dunning	1	1	Tyler Anderson
1	10	Josh Sborz	1	10	Carlos Estevez
1	6	Brock Burke	1	6	Jaime Barria
1	8	Jonathan Hernandez	2	1	Matt Thaiss
1	9	Will Smith	2	9	Chad Wallach
2	1	Jonah Heim	3	1	Brandon Drury
3	1	Nathaniel Lowe	4	1	Luis Rengifo
4	1	Marcus Semien	5	1	Anthony Rendon
5	1	Josh Jung	6	1	Zach Neto
6	1	Ezequiel Duran	7	1	Taylor Ward
7	1	Bubba Thompson	8	1	Mike Trout
7	8	Travis Jankowski	9	1	Hunter Renfroe
8	1	Leody Taveras			
9	1	Adolis Garcia			

	game	TEX vs ANA on 20230505
	state before	(2, 0, 0, 0, 3, 1, 0, 1)
18.	error	play,3,1,netoz001,12,FS.BX,E1/G4MS-.B-1
	state after	(2, 1, 0, 0, 3, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 987 This game had 3 errors. The final score was 4-5.

In the bottom of the 3rd inning, with bases empty, ANA batter Zach Neto hit a soft ground ball to shallow 2nd base. An error by the pitcher allowed the batter to reach 1st base.

	game	TEX vs ANA on 20230505
	state before	(2, 0, 1, 1, 4, 2, 0, 0)
19.	error	play,4,0,lowen001,01,FX,E6/TH/G6M.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 4, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 988 *In the top of the 4th inning, with runners on 2nd and 3rd, TEX batter Nathaniel Lowe hit a ground ball to short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	TEX vs ANA on 20230505
	state before	(0, 1, 0, 0, 8, 3, 0, 0)
20.	error	play,8,0,jankt001,00,.X,FC5/G6S.1-3(E5/TH);B-1
	state after	(0, 1, 0, 1, 8, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 989 *In the top of the 8th inning, with a runner on 1st, TEX batter Travis Jankowski hit a ground ball into a fielder's choice at shallow short. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	TEX vs ANA on 20230507
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
21.	error	play,1,0,grosr001,12,BCSX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 990 This game had 3 errors. The final score was 16-8.

In the top of the 1st inning, with bases empty, TEX batter Robbie Grossman hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	TEX vs ANA on 20230507
	state before	(1, 0, 1, 1, 4, 7, 5, 0)
22.	error	play,4,0,jungj001,12,FCBX,FC5/G56S.3-H(E5)(NR)(UR);2-H(E2)(NR);B-2
	state after	(1, 0, 1, 0, 4, 9, 5, 0)
	fielder	5,2 (3rd baseman,catcher on the home team)

Summary 991 This one play had 2 errors by 2 different fielders.

In the top of the 4th inning, with runners on 2nd and 3rd, TEX batter Josh Jung hit a ground ball into a fielder's choice at shallow 3rd. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI). An error by the catcher allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 2nd.

	game	HOU vs ANA on 20230508
	state before	(2, 0, 1, 0, 3, 2, 1, 0)
23.	error	play,3,0,abrej003,02,T.FX,D9/G3+.2-H;B-3(E9)
	state after	(2, 0, 0, 1, 3, 3, 1, 0)
	fielder	9 (right field on the home team)

Summary 992 This game had 2 errors. The final score was 4-6.

In the top of the 3rd inning, with a runner on 2nd, HOU batter Jose Abreu hit a hard ground ball double to 1st. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the batter to reach 3rd.

	game	HOU vs ANA on 20230508
	state before	(0, 0, 0, 0, 7, 3, 4, 0)
24.	error	play,7,0,mccoc001,01,CX,E5/G5.B-1
	state after	(0, 1, 0, 0, 7, 3, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 993 *In the top of the 7th inning, with bases empty, HOU batter Chas McCormick hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for HOU vs ANA on 2023-05-09

position	innings	player	position	innings	player
1	1	Framber Valdez	1	1	Shohei Ohtani
1	9	Ryan Pressly	1	8	Andrew Wantz
2	1	Martin Maldonado	2	1	Chris Okey
3	1	Jose Abreu	2	8	Matt Thaiss
4	1	Mauricio Dubon	3	1	Gio Urshela
4	5	David Hensley	4	1	Brandon Drury
5	1	Alex Bregman	5	1	Anthony Rendon
6	1	Jeremy Pena	6	1	Zach Neto
7	1	Corey Julks	7	1	Taylor Ward
8	1	Jake Meyers	8	1	Mike Trout
9	1	Kyle Tucker	9	1	Hunter Renfroe

	game	HOU vs ANA on 20230509
	state before	(1, 0, 0, 0, 4, 0, 1, 0)
25.	error	play,4,0,tuckk001,00,X,C/E2.B-1
	state after	(1, 1, 0, 0, 4, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 994 This game had 3 errors. The final score was 3-1.
In the top of the 4th inning, with bases empty, HOU batter Kyle Tucker had a 0-0 count. Catcher's interference allowed the batter to take 1st.

	game	HOU vs ANA on 20230509
	state before	(2, 0, 0, 0, 5, 3, 1, 1)
26.	error	play,5,1,urshg001,22,BFFBX,E4/G34.B-1
	state after	(2, 1, 0, 0, 5, 3, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 995 *In the bottom of the 5th inning, with bases empty, ANA batter Giovanny Urshela hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	HOU vs ANA on 20230509
	state before	(2, 1, 0, 0, 6, 3, 1, 1)
27.	error	play,6,1,ohsas001,00,X,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 6, 3, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 996 *In the bottom of the 6th inning, with a runner on 1st, ANA batter Shohei Ohtani had a 0-0 count. Catcher's interference allowed the batter to take 1st. The runner on 1st advanced to 2nd base.*

	game	HOU vs ANA on 20230510
	state before	(2, 1, 0, 0, 4, 2, 2, 0)
28.	error	play,4,0,hensd002,01,C1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 4, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 997 *In the top of the 4th inning, with a runner on 1st, HOU batter David Hensley has a 0-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 3rd. Hensley still is at bat with a 0-1 count.*

	game	MIN vs ANA on 20230519
	state before	(1, 1, 0, 1, 4, 0, 2, 1)
29.	error	play,4,1,wallc001,12,L1FB>S,K+SB2.3-H(E2/TH)(NR)(UR);1-3
	state after	(2, 0, 0, 1, 4, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 998 *In the bottom of the 4th inning, with runners on 1st and 3rd, ANA batter Chad Wallach had a 1-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	MIN vs ANA on 20230520
	state before	(0, 1, 0, 0, 7, 4, 2, 0)
30.	error	play,7,0,larnt001,32,CSBBFBX,T9/L9L+.1-H;B-H(E4/TH) (NR)
	state after	(0, 0, 0, 0, 7, 6, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 999 *In the top of the 7th inning, with a runner on 1st, MIN batter Trevor Larnach hit a hard line drive triple to right field. The runner on 1st scored (1 RBI). A throwing error by the second baseman allowed the batter to score as well (unearned, no RBI).*

	game	MIN vs ANA on 20230521
	state before	(1, 1, 0, 0, 8, 2, 3, 1)
31.	error	play,8,1,renl001,10,B>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 8, 2, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1000 This game had 2 errors. The final score was 2-4. *In the bottom of the 8th inning, with a runner on 1st, ANA batter Luis Rengifo had a 1-0 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rengifo is still at bat, with a 1-1 count.*

	game	MIN vs ANA on 20230521
	state before	(2, 0, 0, 1, 8, 2, 3, 1)
32.	error	play,8,1,thaim001,22,BBCCX,D7/G5.3-H;B-3(E7)
	state after	(2, 0, 0, 1, 8, 2, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 1001 *In the bottom of the 8th inning, with a runner on 3rd, ANA batter Matt Thaiss hit a ground ball double to 3rd base. The runner on 3rd scored (1 RBI). An error by the left fielder allowed the batter to reach 3rd.*

	game	MIA vs ANA on 20230527
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
33.	error	play,1,0,davij007,12,BCFX,E1/TH/G13.B-2
	state after	(0, 0, 1, 0, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1002 *This game had 2 errors. The final score was 8-5. In the top of the 1st inning, with bases empty, MIA batter Jonathan Davis hit a ground ball to very shallow right infield. A throwing error by the pitcher allowed the batter to reach 2nd base.*

	game	MIA vs ANA on 20230527
	state before	(1, 1, 1, 0, 10, 4, 4, 0)
34.	error	play,10,0,coopg002,00,.X,E7/F78XD.2-H(NR)(UR);1-3;B-2
	state after	(1, 0, 1, 1, 10, 5, 4, 0)
	fielder	7 (pitcher on the home team)

Summary 1003 *In the top of the 10th inning, with runners on 1st and 2nd, MIA batter Garrett Cooper hit a fly ball to extra deep left center field. An error by the left fielder allowed the runner on 2nd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 2nd.*

16.2.3 June

	game	CHN vs ANA on 20230606
	state before	(1, 1, 1, 0, 5, 4, 1, 1)
35.	error	play,5,1,rengl001,31,BF*BBX,FC3/G34.2-3;1-2(E3);B-1
	state after	(1, 1, 1, 1, 5, 4, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1004 *In the bottom of the 5th inning, with runners on 1st and 2nd, ANA batter Luis Rengifo hit a ground ball into a fielder's choice at the 1st/2nd base gap. The runner on 2nd advanced to 3rd. An error by the first baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	CHN vs ANA on 20230608
	state before	(0, 0, 0, 0, 4, 1, 2, 1)
36.	error	play,4,1,wallc001,01,FX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1005 *This game had 2 errors. The final score was 1-3. In the bottom of the 4th inning, with bases empty, ANA batter Chad Wallach hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	CHN vs ANA on 20230608
	state before	(2, 1, 1, 0, 6, 1, 2, 1)
37.	error	play,6,1,ohtas001,00,>B,SB3;SB2.2-H(E2/TH) (NR) (UR)
	state after	(2, 0, 1, 0, 6, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1006 *In the bottom of the 6th inning, with runners on 1st and 2nd, ANA batter Shohei Ohtani had a 0-0 count. During the next pitch, which was a ball, the runners on 1st and 2nd stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Ohtani is still at bat, now with a 1-0 count.*

	game	SEA vs ANA on 20230609
	state before	(2, 0, 0, 0, 3, 2, 0, 1)
38.	error	play,3,1,wardt002,20,BBX,E6/G6D.B-1
	state after	(2, 1, 0, 0, 3, 2, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1007 *In the bottom of the 3rd inning, with bases empty, ANA batter Taylor Ward hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for SEA vs ANA on 2023-06-11

position	innings	player	position	innings	player
1	1	Logan Gilbert	1	1	Griffin Canning
1	4	Matt Brash	1	6	Chris Devenski
1	5	Tayler Saucedo	1	7	Jose Soriano
1	6	Penn Murfee	1	9	Jimmy Herget
1	7	Chris Flexen	2	1	Matt Thaiss
2	1	Cal Raleigh	2	7	Chad Wallach
3	1	Ty France	3	1	Jared Walsh
4	1	Jose Caballero	4	1	Brandon Drury
5	1	Eugenio Suarez	5	1	Luis Rengifo
6	1	J.P. Crawford	6	1	Zach Neto
7	1	Jarred Kelenic	7	1	Taylor Ward
8	1	Julio Rodriguez	8	1	Mike Trout
9	1	Teoscar Hernandez	9	1	Hunter Renfroe

	game	SEA vs ANA on 20230611
	state before	(0, 0, 0, 0, 4, 2, 6, 1)
39.	error	play,4,1,netoz001,00,X,E6/G6D+.B-1
	state after	(0, 1, 0, 0, 4, 2, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1008 This game had 3 errors. The final score was 4-9.
In the bottom of the 4th inning, with bases empty, ANA batter Zach Neto hit a hard ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.

40.	game	SEA vs ANA on 20230611
	state before	(0, 1, 0, 0, 6, 3, 7, 0)
	error	play,6,0,kelej001,12,CBTF.X,FC3/G34S.1-2(E3/TH);B-1
	state after	(0, 1, 1, 0, 6, 3, 7, 0)
	fielder	3 (1st baseman on the home team)

Summary 1009 *In the top of the 6th inning, with a runner on 1st, SEA batter Jarred Kelenic hit a ground ball into a fielder's choice at shallow right infield. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

41.	game	SEA vs ANA on 20230611
	state before	(0, 0, 0, 0, 6, 4, 7, 1)
	error	play,6,1,netoz001,00,X,E1/TH/BG23.B-1
	state after	(0, 1, 0, 0, 6, 4, 7, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1010 *In the bottom of the 6th inning, with bases empty, ANA batter Zach Neto hit a bunt to very shallow right infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

42.	game	LAN vs ANA on 20230621
	state before	(2, 1, 1, 0, 5, 1, 0, 1)
	error	play,5,1,renfh001,12,.FBSF,FLE5
	state after	(2, 1, 1, 0, 5, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1011 This play is an *infinity play*, which can be repeated infinitely often without ending the game.

In the bottom of the 5th inning, with runners on 1st and 2nd, ANA batter Hunter Renfroe hit a pop fly to foul of the 3rd base line. An error by the third baseman allowed the batter to remain safe at bat. Hunter is still at bat, now with a 2-2 count.

	game	CHA vs ANA on 20230627
	state before	(1, 1, 0, 0, 7, 1, 4, 1)
43.	error	play,7,1,wardt002,20,BB+1,P01(E2/TH).1-3
	state after	(1, 0, 0, 1, 7, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1012 *In the bottom of the 7th inning, with a runner on 1st, ANA batter Taylor Ward has a 2-0 count. A throwing error by the catcher in attempting to pickoff the runner at 1st allowed that runner to safely reach 2nd. Ward is still at bat with a 2-0 count.*

	game	ARI vs ANA on 20230630
	state before	(0, 1, 0, 0, 8, 6, 1, 1)
44.	error	play,8,1,troum001,32,CBSB*BX,FC3/G3.1-3(E3/TH);B-1
	state after	(0, 1, 0, 1, 8, 6, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1013 *In the bottom of the 8th inning, with a runner on 1st, ANA batter Mike Trout hit a ground ball into a fielder's choice at 1st. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

16.2.4 July

	game	ARI vs ANA on 20230701
	state before	(2, 1, 0, 1, 6, 1, 1, 0)
45.	error	play,6,0,martk001,11,BSX,E5/TH/G56+.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 6, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1014 *In the top of the 6th inning, with runners on 1st and 3rd, ARI batter Ketel Marte hit a hard ground ball to 3rd base. A*

throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	HOU vs ANA on 20230714
	state before	(0, 1, 0, 0, 2, 0, 0, 1)
46.	error	play,2,1,renfh001,00,X,FC6/G6MS.1-2(E6);B-1
	state after	(0, 1, 1, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1015 This game had 2 errors. The final score was 7-5.
In the bottom of the 2nd inning, with a runner on 1st, ANA batter Hunter Renfroe hit a ground ball into a fielder's choice at shallow short. An error by the shortstop allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.

	game	HOU vs ANA on 20230714
	state before	(1, 0, 1, 0, 5, 3, 2, 0)
47.	error	play,5,0,abrej003,00,X,E5/G56.2-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 5, 4, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1016 *In the top of the 5th inning, with a runner on 2nd, HOU batter Jose Abreu hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for HOU vs ANA on 2023-07-15

position	innings	player	position	innings	player
1	1	Framber Valdez	1	1	Reid Detmers
1	10	Phil Maton	1	10	Carlos Estevez
1	7	Bryan Abreu	1	7	Chris Devenski
1	7	Ryne Stanek	1	7	Zack Weiss
1	8	Hector Neris	1	8	Matt Moore
1	9	Ryan Pressly	1	8	Tucker Davidson
2	1	Martin Maldonado	2	1	Matt Thaiss
2	10	Yainer Diaz	3	1	Mike Moustakas
3	1	Jose Abreu	4	1	Luis Rengifo
4	1	Mauricio Dubon	4	10	Andrew Velazquez
5	1	Alex Bregman	5	1	Eduardo Escobar
6	1	Jeremy Pena	6	1	Zach Neto
6	10	Grae Kessinger	7	1	Taylor Ward
7	1	Corey Julks	8	1	Mickey Moniak
8	1	Chas McCormick	8	10	Trey Cabbage
9	1	Kyle Tucker	9	1	Hunter Renfroe

	game	HOU vs ANA on 20230715
	state before	(0, 0, 0, 0, 4, 4, 3, 1)
48.	error	play,4,1,escoe001,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 4, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1017 This game had 3 errors. The final score was 12-13. *In the bottom of the 4th inning, with bases empty, ANA batter Eduardo Escobar hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	HOU vs ANA on 20230715
	state before	(1, 0, 0, 0, 9, 11, 9, 0)
49.	error	play,9,0,abrej003,31,BBSBX,S5/G56.B-2(E5/TH)
	state after	(1, 0, 1, 0, 9, 11, 9, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1018 *In the top of the 9th inning, with bases empty, HOU batter Jose Abreu hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	HOU vs ANA on 20230715
	state before	(1, 1, 1, 0, 10, 12, 12, 1)
50.	error	play,10,1,wardt002,22,CBSBFFX,46(1)/FO/G34+.2-H(E6/TH)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 10, 12, 13, 1)
	fielder	6 (3rd baseman on the home team)

Summary 1019 *In the bottom of the 10th inning, with runners on 1st and 2nd, ANA batter Taylor Ward hit a hard ground ball to the 1st/2nd base gap. At 2nd, the shortstop (with an assist from the second baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the runner from 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	NYA vs ANA on 20230717
	state before	(0, 0, 0, 0, 9, 3, 3, 0)
51.	error	play,9,0,lemad001,02,..CSFX,S9/L9LS.B-2(E9)
	state after	(0, 0, 1, 0, 9, 3, 3, 0)
	fielder	9 (right field on the home team)

Summary 1020 *In the top of the 9th inning, with bases empty, NYA batter DJ LeMahieu hit a line drive single to shallow right field. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	NYA vs ANA on 20230718
	state before	(1, 0, 0, 0, 6, 1, 5, 0)
52.	error	play,6,0,stanm004,00,X,E6/TH/G6+.B-1
	state after	(1, 1, 0, 0, 6, 1, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 1021 *In the top of the 6th inning, with bases empty, NYA batter Giancarlo Stanton hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	NYA vs ANA on 20230719
	state before	(1, 0, 0, 0, 6, 2, 6, 1)
53.	error	play,6,1,netoz001,11,CBX,S6/G56.B-2(E6/TH)
	state after	(1, 0, 1, 0, 6, 2, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1022 *In the bottom of the 6th inning, with bases empty, ANA batter Zach Neto hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	PIT vs ANA on 20230723
	state before	(2, 0, 1, 1, 7, 2, 7, 0)
54.	error	play,7,0,mccua001,30,B*BBN,0A.3-H(E2/TH) (NR)
	state after	(2, 0, 1, 0, 7, 3, 7, 0)
	fielder	2 (catcher on the home team)

Summary 1023 *In the top of the 7th inning, with runners on 2nd and 3rd, PIT batter Andrew McCutchen had a 3-0 count. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). McCutchen is still at bat with a 3-0 count.*

16.2.5 August

	game	SEA vs ANA on 20230803
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
55.	error	play,1,0,rodrj007,22,CSBBFFX,FC3/G3.1-3(E3/TH);B-1
	state after	(0, 1, 0, 1, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1024 *In the top of the 1st inning, with a runner on 1st, SEA batter Julio Rodriguez hit a ground ball into a fielder's choice at 1st. A throwing error by the first baseman allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

	game	SEA vs ANA on 20230806
	state before	(2, 0, 0, 0, 1, 1, 0, 0)
56.	error	play,1,0,ralec001,21,B.CBX,E9/F9D.B-2
	state after	(2, 0, 1, 0, 1, 1, 0, 0)
	fielder	9 (right field on the home team)

Summary 1025 *In the top of the 1st inning, with bases empty, SEA batter Cal Raleigh hit a fly ball to deep right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	SFN vs ANA on 20230807
	state before	(2, 1, 0, 0, 6, 1, 1, 1)
57.	error	play,6,1,monim001,02,CFX,S8/L8+.1-H(E8)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 6, 1, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 1026 This game had 2 errors. The final score was 8-3. *In the bottom of the 6th inning, with a runner on 1st, ANA batter Mickey Moniak hit a hard line drive single to center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	SFN vs ANA on 20230807
	state before	(2, 0, 0, 0, 7, 2, 2, 0)
58.	error	play,7,0,matol002,02,CSX,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 7, 2, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1027 *In the top of the 7th inning, with bases empty, SFN batter Luis Matos hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	SFN vs ANA on 20230808
	state before	(2, 1, 1, 0, 1, 0, 2, 1)
59.	error	play,1,1,renfh001,11,BCX,D8/L89XD.2-H;1-H;B-3(E4/TH)
	state after	(2, 0, 0, 1, 1, 0, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1028 This game had 2 errors. The final score was 5-7.
In the bottom of the 1st inning, with runners on 1st and 2nd, ANA batter Hunter Renfroe hit a line drive double to extra deep right center field. The runners on 1st and 2nd both scored (2 RBIs). A throwing error by the second baseman allowed the batter to reach 3rd.

	game	SFN vs ANA on 20230808
	state before	(1, 1, 0, 0, 5, 3, 5, 1)
60.	error	play,5,1,mousm001,02,FFX,E4/TH/G34.1-2;B-1
	state after	(1, 1, 1, 0, 5, 3, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1029 *In the bottom of the 5th inning, with a runner on 1st, ANA batter Mike Moustakas hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SFN vs ANA on 20230809
	state before	(1, 1, 1, 0, 2, 0, 0, 0)
61.	error	play,2,0,crawb001,32,BBBFC+2,P02(E2/TH).2-3;1-2
	state after	(1, 0, 1, 1, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1030 This game had 3 errors. The final score was 1-4.

In the top of the 2nd inning, with runners on 1st and 2nd, SFN batter Brandon Crawford has a full count. During an attempted pickoff at 2nd, a throwing error by the catcher allowed the runners on 1st and 2nd to each advance one base. Crawford is still at bat with a full count.

	game	SFN vs ANA on 20230809
	state before	(0, 1, 1, 0, 6, 1, 0, 1)
62.	error	play,6,1,drurb001,12,FFBX,S7/L7S.2-H;1-3(E7);B-2
	state after	(0, 0, 1, 1, 6, 1, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 1031 *In the bottom of the 6th inning, with runners on 1st and 2nd, ANA batter Brandon Drury hit a line drive single to shallow left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SFN vs ANA on 20230809
	state before	(1, 1, 0, 0, 9, 1, 4, 0)
63.	error	play,9,0,sabob001,12,.CSBX,FC3/G3.1-3(E3/TH);B-1
	state after	(1, 1, 0, 1, 9, 1, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1032 *In the top of the 9th inning, with a runner on 1st, SFN batter Blake Sabol hit a ground ball into a fielder's choice at 1st. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

64.	game	TBA vs ANA on 20230818
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
	error	play,1,0,parei001,10,BX,D7/L7L+.1-H;B-3(E6)
	state after	(2, 0, 0, 1, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1033 *In the top of the 1st inning, with a runner on 1st, TBA batter Isaac Paredes hit a hard line drive double to left field. The runner on 1st scored (1 RBI). An error by the shortstop allowed the batter to reach 3rd.*

65.	game	TBA vs ANA on 20230819 (1/2)
	state before	(1, 1, 1, 0, 3, 2, 2, 1)
	error	play,3,1,mousm001,00,X,E3/G3.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 3, 2, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1034 This is game 1 of a doubleheader. The final score was 6-7.

In the bottom of the 3rd inning, with runners on 1st and 2nd, ANA batter Mike Moustakas hit a ground ball to 1st base. An error by the first baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

Visiting/Home team rosters for TBA vs ANA on 2023-08-19 (2/2)

position	innings	player	position	innings	player
1	1	Zach Eflin	1	1	Patrick Sandoval
1	7	Jacob Lopez	1	5	Jaime Barria
1	8	Cooper Criswell	1	8	Jimmy Herget
2	1	Christian Bethancourt	1	9	Eduardo Escobar
3	1	Yandy Diaz	2	1	Logan O'Hoppe
3	7	Jonathan Aranda	3	1	Nolan Schanuel
4	1	Isaac Paredes	4	1	Brandon Drury
5	1	Curtis Mead	5	1	Mike Moustakas
6	1	Osleivis Basabe	5	7	Luis Rengifo
7	1	Randy Arozarena	6	1	Luis Rengifo
8	1	Jose Siri	6	7	Andrew Velazquez
9	1	Josh Lowe	7	1	Randal Grichuk
			8	1	Jordyn Adams
			9	1	Hunter Renfroe

	game	TBA vs ANA on 20230819 (2/2)
	state before	(2, 0, 0, 0, 5, 2, 0, 0)
66.	error	play,5,0,arozr001,31,CBBBX,E8/F8S.B-1
	state after	(2, 1, 0, 0, 5, 2, 0, 0)
	fielder	8 (center field on the home team)

Summary 1035 This game 2 of a doubleheader had 4 errors. The final score was 18-4.

In the top of the 5th inning, with bases empty, TBA batter Randy Arozarena hit a fly ball to shallow center field. An error by the center fielder allowed the batter to reach 1st base.

	game	TBA vs ANA on 20230819 (2/2)
	state before	(0, 0, 0, 0, 6, 4, 1, 0)
67.	error	play,6,0,bethc001,00,X,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 6, 4, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1036 *In the top of the 6th inning, with bases empty, TBA batter Christian Bethancourt hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs ANA on 20230819 (2/2)
	state before	(0, 0, 0, 0, 7, 11, 1, 0)
68.	error	play,7,0,diazy001,21,...CBBX,S8/L8D.B-2(E8)
	state after	(0, 0, 1, 0, 7, 11, 1, 0)
	fielder	8 (center field on the home team)

Summary 1037 *In the top of the 7th inning, with bases empty, TBA batter Yandy Diaz hit a line drive single to deep center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	TBA vs ANA on 20230819 (2/2)
	state before	(2, 0, 0, 0, 9, 16, 4, 1)
69.	error	play,9,1,adamj003,21,BCBX,E5/G6.B-1
	state after	(2, 1, 0, 0, 9, 16, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1038 *In the bottom of the 9th inning, with bases empty, ANA batter Jordyn Adams hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.*

70.	game	CIN vs ANA on 20230822
	state before	(2, 0, 1, 1, 5, 1, 3, 0)
	error	play,5,0,delae003,12,S.C*BX,E3/G3.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 3, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1039 *In the top of the 5th inning, with runners on 2nd and 3rd, CIN batter Elly De La Cruz hit a ground ball to 1st base. An error by the first baseman allowed the runners on 2nd and 3rd to both score (unearned, no RBIs) and the batter to reach 1st.*

71.	game	CIN vs ANA on 20230823 (1/2)
	state before	(2, 1, 0, 0, 5, 1, 3, 0)
	error	play,5,0,mclam001,11,CBX,E6/TH/G6.1-2;B-1
	state after	(2, 1, 1, 0, 5, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1040 This game 1 of a doubleheader had 2 errors. The final score was 9-4.

In the top of the 5th inning, with a runner on 1st, CIN batter Matt McLain hit a ground ball to short. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

72.	game	CIN vs ANA on 20230823 (1/2)
	state before	(2, 1, 0, 0, 8, 8, 4, 1)
	error	play,8,1,schan001,12,FBCX,S3/G4S.1-3(E3/TH)
	state after	(2, 1, 0, 1, 8, 8, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1041 *In the bottom of the 8th inning, with a runner on 1st, ANA batter Nolan Schanuel hit a ground ball single to shallow 2nd. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

16.2.6 September

	game	BAL vs ANA on 20230905
	state before	(1, 0, 1, 0, 3, 1, 0, 0)
73.	error	play,3,0,mounr001,01,FX,S7/L78.2-H(E7) (NR) (UR)
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	7 (left field on the home team)

Summary 1042 *In the top of the 3rd inning, with a runner on 2nd, BAL batter Ryan Mountcastle hit a line drive single to left center field. An error by the left fielder allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	BAL vs ANA on 20230906
	state before	(0, 0, 0, 0, 3, 1, 0, 0)
74.	error	play,3,0,matej003,11,SBX,E6/G6+.B-1
	state after	(0, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1043 *In the top of the 3rd inning, with bases empty, BAL batter Jorge Mateo hit a hard ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs ANA on 20230907
	state before	(1, 0, 0, 0, 5, 0, 0, 1)
75.	error	play,5,1,parik002,11,S.BX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 5, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1044 *In the bottom of the 5th inning, with bases empty, ANA batter Kyren Paris hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CLE vs ANA on 20230909
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
76.	error	play,1,1,drurb001,11,SBX,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 1, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1045 *In the bottom of the 1st inning, with bases empty, ANA batter Brandon Drury hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	TEX vs ANA on 20230926
	state before	(0, 0, 0, 0, 9, 1, 9, 0)
77.	error	play,9,0,hedga001,00,.X,E5/L5.B-1
	state after	(0, 1, 0, 0, 9, 1, 9, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1046 *In the top of the 9th inning, with bases empty, TEX batter Austin Hedges hit a line drive to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs ANA on 20230929
	state before	(2, 1, 1, 0, 4, 1, 1, 1)
78.	error	play,4,1,schan001,10,BX,S7/L7S.2-H;1-H(E7) (E7/TH) (NR) (UR);B-2
	state after	(2, 0, 1, 0, 4, 1, 3, 1)
	fielder	7,7 (left field on the visiting team)

Summary 1047 *This one play had 2 errors by the same fielder. In the bottom of the 4th inning, with runners on 1st and 2nd, ANA batter Nolan Schanuel hit a line drive single to shallow left field. The runner on 2nd scored (1 RBI). Errors (there were 2) by the left fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	OAK vs ANA on 20230930
	state before	(0, 0, 0, 0, 5, 1, 0, 0)
79.	error	play,5,0,diazj008,10,BX,E6/G6.B-1
	state after	(0, 1, 0, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1048 This game had 2 errors. The final score was 7-3.
In the top of the 5th inning, with bases empty, OAK batter Jordan Diaz hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	OAK vs ANA on 20230930
	state before	(2, 0, 0, 0, 8, 6, 3, 1)
80.	error	play,8,1,fletd002,01,CX,S9/L9LS.B-3(E9)
	state after	(2, 0, 0, 1, 8, 6, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1049 *In the bottom of the 8th inning, with bases empty, ANA batter David Fletcher hit a line drive single to shallow right field. An error by the right fielder allowed the batter to reach 3rd.*

16.2.7 October

	game	OAK vs ANA on 20231001
	state before	(0, 1, 0, 0, 2, 0, 1, 0)
81.	error	play,2,0,bledj001,00,X,6(1)/F0/G6.B-2(E6/TH)
	state after	(1, 0, 1, 0, 2, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1050 This game had 2 errors. The final score was 3-7.
In the top of the 2nd inning, with a runner on 1st, OAK batter JJ Bleday hit a ground ball to short. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	OAK vs ANA on 20231001
	state before	(0, 1, 1, 0, 4, 0, 2, 1)
82.	error	play,4,1,adelj001,00,X,S3/G34.2-H(E3/TH)(NR);1-3
	state after	(0, 1, 0, 1, 4, 0, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1051 *In the bottom of the 4th inning, with runners on 1st and 2nd, ANA batter Jo Adell hit a ground ball single to the 1st/2nd base gap. A throwing error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 3rd. The batter took 1st.*

Chapter 17

Los Angeles Dodgers (LAN)

In 1957 the Dodgers moved to LA from Brooklyn. A few years later, the Dodgers won the World Series, beating the Chicago White Sox in six games in the 1959. Just last year, the Dodgers secured their 10th NL West Division title in 11 years (each of the ten years previous to 2023, except for 2021). Within that, they won the NL Pennant 3 times (2017, 2018, 2020), and within that, they won the World Series once (2020).

This chapter is devoted to listing the errors in each regular season game of 2023 played in Los Angeles against the Dodgers. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 LAN team is 0.709 compared with the MLB average¹ of 0.691 that year [BR], about one standard deviation above normal. This qualifies as “high DER.”

There were 87 errors in the 81 LAN 2023 home games, 40 by the Dodgers and the rest (47) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 is 76. The Dodgers were not quite one standard deviation below the MLB average, but we can call their error total “low E.”

¹The standard deviation of these DER values is $\sigma = 0.011$.

17.1 Season totals

The Official Scorer shall credit the batter with a run batted in for every run that scores by reason of the batter becoming a runner with the bases full (because of a base on balls, an award of first base for being touched by a pitched ball or for interference or obstruction).

– **MLB Rule Book** 9.04(a)(2)

Throughout, *vt* denotes the visiting team and *ht* denotes the home team.

17.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	47
Runs by ht because of errors by vt:	19
Errors by ht:	40
Runs by vt because of errors by ht:	10

This is an excellent error-to-run conversion rate: $19/47 = 40\%$ on the part of the Nationals (compared to $10/40 = 25\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 LAN season (home and away) has about $R/PA = 906/6333$ (14.3%) change of resulting in a run. Therefore, a randomly selected LAN 2023 error is almost 3 times as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors is 87, so the expected number of errors (by either team) in a randomly selected LAN 2023 home game is about 1.07.

17.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	17	18	35
(lone) runner on 1st	12	7	19
(lone) runner on 2nd	4	7	11
(lone) runner on 3rd	0	1	1
two runners on base	13	6	19
bases loaded	1	1	2
total	47	40	87

A table of frequencies of most common errors by the Dodgers in LAN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	$6/40 = 15.0\%$
(1, 0, 0, 0)	7	17.5%
(2, 0, 0, 0)	5	12.5%
(0, 1, 0, 0)	2	5%
(1, 1, 0, 0)	2	5%
(2, 1, 0, 0)	3	7.5%
(1, 0, 1, 0)	2	5%
(2, 0, 1, 0)	2	5%
(2, 1, 1, 0)	3	7.5%

A table of frequencies of most common errors by a visiting team in LAN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	$6/47 = 12.8\%$
(1, 0, 0, 0)	5	10.6%
(2, 0, 0, 0)	6	12.8%
(0, 1, 0, 0)	5	10.6%
(1, 1, 0, 0)	3	6.38%
(2, 1, 0, 0)	4	8.51%
(1, 0, 1, 0)	2	4.26%
(1, 1, 1, 0)	3	6.38%
(2, 1, 1, 0)	2	4.26%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department:

the 2023 Dodgers homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `LAN2023.evn`):

state	LAN2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.1%
(2, 0, 0, 0)	13.4%
(0, 1, 0, 0)	4.9%
(1, 1, 0, 0)	6.6%
(2, 1, 0, 0)	7.0%
(0, 0, 1, 0)	2.2%
(1, 0, 1, 0)	2.7%
(2, 0, 1, 0)	3.6%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.0%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.2%
(2, 1, 1, 0)	3.6%

17.1.3 Some interesting errors

Some of the most interesting error plays in the LAN 2023 home games:

- interesting play, SFN vs LAN on 2023-06-16.

Note: In their series with the Giants (June 16, 17, 18, where they lost every game) there were at least 3 errors in each game.

- interesting play, NYN vs LAN on 2023-04-18

Note: This game had 3 errors, one of which was a double error on one player.

- interesting play, COL vs LAN on 2023-08-10

Note: This was Freddie Freeman's only error all year long!

17.1.4 Relative team rankings

We have 81 home games between LAN and some visiting MLB team (vt). In the lists below, we have ordered the visiting teams with that team's W/T record in parentheses².

A number of teams did not play LAN in the regular season. However, of those that did, LAN *lost* more games than won against:

CHN (2/3), CIN (2/3), NYA (2/3), NYN (2/3), TOR (2/3),
ATL (3/4), SFN (4/7)

and *won* more than lost against

ANA (0/2), MIL (0/3), OAK (0/3), PHI (0/3), SLN (0/3), COL (0/6), CHA
(1/3), DET (1/3), HOU (1/3),
MIA (1/3), MIN (1/3), WAS (1/3), PIT (1/4), SDN (2/6), ARI (2/7).

The teams that *split* their series were:

(none)

The teams that did not play in any of the 81 home games in LAN 2023 were:

BAL, BOS, CLE, KCA, SEA, TBA, TEX .

17.2 Game-by-game totals

The batter becomes a runner and is entitled to first base without liability to be put out (provided he advances to and touches first base) when the catcher or any fielder interferes with him.

– **MLB Rule Book** 6.01(c)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for LAN 2023 is as follows [BA].

²Where W = numbers of home games vt won against LAN, T = Total home games vt played against the Dodgers at Los Angeles.

position	player	games played
1 (C)	Will Smith	112
3 (1B)	Freddie Freeman	163
4 (2B)	Miguel Vargas	76
5 (3B)	Max Muncy	126
6 (SS)	Miguel Rojas	121
7 (LF)	David Peralta	119
8 (CF)	James Outman	135
9 (RF)	Mookie Betts	107

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 Dodgers home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Rel].

17.2.1 March

	game	ARI vs LAN on 20230330
	state before	(1, 0, 0, 0, 3, 2, 0, 0)
1.	error	play,3,0,martk001,00,X,E6/G6D.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1052 This game had 2 errors. The final score was 2-8.

In the top of the 3rd inning, with bases empty, ARI batter Ketel Marte hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.

	game	ARI vs LAN on 20230330
	state before	(0, 0, 0, 0, 8, 2, 7, 1)
2.	error	play,8,1,outmj002,11,FBX,S9/L9L.B-2(E9)
	state after	(0, 0, 1, 0, 8, 2, 7, 1)
	fielder	9 (right field on the visiting team)

Summary 1053 *In the bottom of the 8th inning, with bases empty, LAN batter James Outman hit a line drive single to right field. An error by the right fielder allowed the batter to reach 2nd base.*

17.2.2 April

	game	ARI vs LAN on 20230401
	state before	(1, 0, 0, 0, 7, 1, 8, 0)
3.	error	play,7,0,longe001,32,CSFBBFFBF.X,E5/G56.B-1
	state after	(1, 1, 0, 0, 7, 1, 8, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1054 *In the top of the 7th inning, with bases empty, ARI batter Evan Longoria hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	ARI vs LAN on 20230402
	state before	(2, 1, 0, 0, 7, 1, 1, 0)
4.	error	play,7,0,thoma004,10,B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 1, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1055 *In the top of the 7th inning, with a runner on 1st, ARI batter Alek Thomas had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner (now on 2nd) to reach 3rd. Thomas is still at bat, now with a 2-0 count.*

	game	COL vs LAN on 20230403
	state before	(1, 0, 0, 0, 8, 4, 13, 1)
5.	error	play,8,1,martj006,10,BX,E9/F89XD.B-2
	state after	(1, 0, 1, 0, 8, 4, 13, 1)
	fielder	9 (right field on the visiting team)

Summary 1056 *In the bottom of the 8th inning, with bases empty, LAN batter J.D. Martinez hit a fly ball to very deep right center field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	COL vs LAN on 20230404
	state before	(0, 0, 0, 0, 7, 0, 4, 0)
6.	error	play,7,0,monte001,22,...FBBFX,E5/G56+.B-1
	state after	(0, 1, 0, 0, 7, 0, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1057 *In the top of the 7th inning, with bases empty, COL batter Elehuris Montero hit a hard ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	CHN vs LAN on 20230414
	state before	(2, 0, 0, 0, 2, 0, 0, 1)
7.	error	play,2,1,vargm001,01,FX,E5/G56.B-1
	state after	(2, 1, 0, 0, 2, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1058 *In the bottom of the 2nd inning, with bases empty, LAN batter Miguel Vargas hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	CHN vs LAN on 20230416
	state before	(1, 0, 0, 0, 5, 0, 1, 0)
8.	error	play,5,0,bellc002,02,CSX,E4/G34.B-1
	state after	(1, 1, 0, 0, 5, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1059 *In the top of the 5th inning, with bases empty, CHN batter Cody Bellinger hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for NYN vs LAN on 2023-04-18

position	innings	player	position	innings	player
1	1	Tylor Megill	1	1	Clayton Kershaw
1	6	Denyi Reyes	1	8	Evan Phillips
1	7	Jeff Bringham	1	9	Brusdar Graterol
1	8	John Curtiss	2	1	Austin Barnes
2	1	Tomas Nido	3	1	Freddie Freeman
2	8	Francisco Alvarez	4	1	Miguel Vargas
3	1	Pete Alonso	5	1	Max Muncy
4	1	Jeff McNeil	6	1	Miguel Rojas
5	1	Eduardo Escobar	6	6	Luke Williams
5	8	Brett Baty	7	1	David Peralta
6	1	Francisco Lindor	8	1	James Outman
7	1	Mark Canha	9	1	Jason Heyward
8	1	Tommy Pham			
9	1	Starling Marte			

	game	NYN vs LAN on 20230418
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
9.	error	play,1,0,nimmb001,12,CBFX,E9/L89XD+.B-3
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 1060 This game had 2 or 3 errors, depending on if you count double errors as 1 or 2. The Dodgers shut out the Mets, 0-5.

In the top of the 1st inning, with bases empty, NYN batter Brandon Nimmo hit a hard line drive to extra deep right center field. An error by the right fielder allowed the batter to reach 3rd base.

	game	NYN vs LAN on 20230418
	state before	(2, 1, 0, 0, 3, 0, 3, 1)
10.	error	play,3,1,heywj001,20,VBX,S8/G4+.1-3(E8);B-2(E8/TH)
	state after	(2, 0, 1, 1, 3, 0, 3, 1)
	fielder	8, 8 (center field on the visiting team)

Summary 1061 This is a double error, and both errors by the same fielder.

In the bottom of the 3rd inning, with a runner on 1st, LAN batter Jason Heyward hit a hard ground ball single to 2nd. An error by the center fielder allowed the runner from 1st to advance to 3rd. A (separate) throwing error allowed the batter to reach 2nd.

	game	SLN vs LAN on 20230428
	state before	(1, 1, 1, 0, 1, 2, 1, 1)
11.	error	play,1,1,heywj001,12,BCFX,S9/G34.2-H;1-3(E9)
	state after	(1, 1, 0, 1, 1, 2, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 1062 This game had 2 errors. The final score was 3-7.

In the bottom of the 1st inning, with runners on 1st and 2nd, LAN batter Jason Heyward hit a ground ball single to the 1st/2nd base gap. The runner from 2nd scored (1 RBI). An error by the right fielder allowed the runner on 1st to reach 3rd. The batter took 1st.

	game	SLN vs LAN on 20230428
	state before	(1, 1, 1, 0, 5, 2, 4, 1)
12.	error	play,5,1,heywj001,31,BBFBX,C/E2.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 5, 2, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1063 *In the bottom of the 5th inning, with runners on 1st and 2nd, LAN batter Jason Heyward had a 3-1 count. Catcher's interference allowed the runner on 1st and 2nd to each advance one base and the batter to take 1st. The bases are loaded.*

	game	SLN vs LAN on 20230429
	state before	(1, 1, 0, 0, 2, 0, 0, 1)
13.	error	play,2,1,thomt002,22,C1S*B1BF>S,K+SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1064 This game had 2 errors. The final score was 0-1. *In the bottom of the 2nd inning, with a runner on 1st, LAN batter Trayce Thompson had a 2-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to reach 3rd. Thompson has struck out.*

	game	SLN vs LAN on 20230429
	state before	(2, 1, 0, 0, 8, 0, 1, 0)
14.	error	play,8,0,nootl001,12,CCBX,E6/G6.1-2;B-1
	state after	(2, 1, 1, 0, 8, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1065 *In the top of the 8th inning, with a runner on 1st, SLN batter Lars Nootbaar hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

17.2.3 May

	game	PHI vs LAN on 20230501
	state before	(1, 1, 0, 0, 4, 1, 5, 1)
15.	error	play,4,1,bettm001,00,X,S8/L8D+.1-3(E8/TH);B-2
	state after	(1, 0, 1, 1, 4, 1, 5, 1)
	fielder	8 (center field on the visiting team)

Summary 1066 *In the bottom of the 4th inning, with a runner on 1st, LAN batter Mookie Betts hit a hard line drive single to deep center*

field. A throwing error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	PHI vs LAN on 20230501
	state before	(2, 1, 1, 1, 5, 1, 8, 0)
16.	error	play,5,0,castn001,21,.BSBX,S7/L78D+.3-H;2-H;1-3;B-2(E7)
	state after	(2, 0, 1, 1, 5, 3, 8, 0)
	fielder	7 (left field on the home team)

Summary 1067 *In the top of the 5th inning, with bases loaded, PHI batter Nick Castellanos hit a hard line drive single to deep left center field. The runner from 2nd and 3rd both scored (2 RBIs), the runner from 1st advanced to 3rd. An error by the left fielder allowed the batter to reach 2nd.*

	game	PHI vs LAN on 20230502
	state before	(2, 0, 0, 0, 3, 0, 3, 0)
17.	error	play,3,0,sosae001,20,BBX,C/E2.B-1
	state after	(2, 1, 0, 0, 3, 0, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1068 *In the top of the 3rd inning, with bases empty, PHI batter Edmundo Sosa had a 2-0 count. Catcher's interference allowed the batter to reach 1st base.*

	game	PHI vs LAN on 20230503
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
18.	error	play,1,0,castn001,00,X,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1069 This game had 2 errors. The final score was 6-10.
In the top of the 1st inning, with runners on 1st and 2nd, PHI batter Nick Castellanos hit a ground ball into a fielder's choice at 3rd. The runner at 2nd advanced to 3rd. An error by the third baseman allowed the runner from 1st to reach 2nd and the batter to reach 1st. The bases are loaded.

	game	PHI vs LAN on 20230503
	state before	(2, 0, 1, 1, 8, 5, 4, 1)
19.	error	play,8,1,barna001,12,.FC>B.X,E5/G78S.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 8, 5, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1070 *In the bottom of the 8th inning, with runners on 2nd and 3rd, LAN batter Austin Barnes hit a ground ball to shallow left center field. An error by the 3rd baseman allowed the runners on 2nd and 3rd to score (no RBIs) and the batter to reach 1st.*

	game	SDN vs LAN on 20230512
	state before	(2, 0, 0, 0, 4, 0, 2, 1)
20.	error	play,4,1,rojam002,22,SBS.BX,E6/TH/G6+.B-1
	state after	(2, 1, 0, 0, 4, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1071 *In the bottom of the 4th inning, with bases empty, LAN batter Miguel Rojas hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	SDN vs LAN on 20230513
	state before	(1, 0, 0, 0, 3, 2, 3, 0)
21.	error	play,3,0,tatif002,22,FSBBX,E6/TH/G56.B-1
	state after	(1, 1, 0, 0, 3, 2, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1072 This game had 2 errors. The final score was 2-4.

In the top of the 3rd inning, with bases empty, SDN batter Fernando Tatis hit a ground ball to 3rd base. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	SDN vs LAN on 20230513
	state before	(1, 0, 0, 0, 6, 2, 4, 0)
22.	error	play,6,0,machm001,01,FX,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 6, 2, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1073 *In the top of the 6th inning, with bases empty, SDN batter Manny Machado hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	MIN vs LAN on 20230515
	state before	(2, 0, 0, 0, 1, 0, 3, 1)
23.	error	play,1,1,heywj001,12,CFBX,C/E2.B-1
	state after	(2, 1, 0, 0, 1, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1074 This game had 2 errors. The final score was 8-9.

In the bottom of the 1st inning, with bases empty, LAN batter Jason Heyward had a 1-2 count. Catcher's interference allowed the batter to reach 1st base.

	game	MIN vs LAN on 20230515
	state before	(2, 0, 0, 0, 5, 2, 5, 1)
24.	error	play,5,1,smitw003,12,BSFFX,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 5, 2, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1075 *In the bottom of the 5th inning, with bases empty, LAN batter Will Smith hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	MIN vs LAN on 20230517
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
25.	error	play,1,1,freef001,12,SBFX,E3/G34S.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1076 *In the bottom of the 1st inning, with a runner on 1st, LAN batter Freddie Freeman hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	MIN vs LAN on 20230517
	state before	(1, 1, 1, 0, 7, 2, 2, 0)
26.	error	play,7,0,kiria001,12,CBFFN,SB3.2-H(E1/TH) (NR) (UR);1-3
	state after	(1, 0, 0, 1, 7, 3, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1077 *In the top of the 7th inning, with runners on 1st and 2nd, MIN batter Alex Kirilloff had a 1-2 count. During the next pitch, which was a “no pitch” (N) credited as a ball, the runner on 2nd stole 3rd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI) and the runner on 1st to advance to 3rd. Kirilloff is still at bat with a 2-0 count.*

	game	MIN vs LAN on 20230517
	state before	(0, 0, 0, 0, 9, 3, 7, 0)
27.	error	play,9,0,farmk001,12,.CBFFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 9, 3, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 1078 *In the top of the 9th inning, with bases empty, MIN batter Kyle Farmer hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	WAS vs LAN on 20230529
	state before	(0, 1, 0, 0, 5, 1, 0, 1)
28.	error	play,5,1,vargm001,10,B1X,FC6/G6M.1-3(E6);B-1
	state after	(0, 1, 0, 1, 5, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1079 *In the bottom of the 5th inning, with a runner on 1st, LAN batter Miguel Vargas hit a ground ball into a fielder's choice at deep short. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	WAS vs LAN on 20230530
	state before	(0, 1, 0, 0, 8, 3, 6, 1)
29.	error	play,8,1,rojam002,00,>X,S6/G6.1-3(E6/TH)
	state after	(0, 1, 0, 1, 8, 3, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1080 *In the bottom of the 8th inning, with a runner on 1st, LAN batter Miguel Rojas hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	WAS vs LAN on 20230531
	state before	(0, 0, 0, 0, 3, 3, 3, 1)
30.	error	play,3,1,freef001,32,CBBFBX,E4/G34D.B-1
	state after	(0, 1, 0, 0, 3, 3, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1081 *This game had 2 errors. The final score was 10-6. In the bottom of the 3rd inning, with bases empty, LAN batter Freddie Freeman hit a ground ball to the 1st/2nd base gao. An error by the second baseman allowed the batter to reach 1st base.*

	game	WAS vs LAN on 20230531
	state before	(0, 1, 0, 0, 8, 5, 5, 0)
31.	error	play,8,0,abrac001,01,FX,FC1/G13-.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 8, 5, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1082 *In the top of the 8th inning, with a runner on 1st, WAS batter C.J. Abrams hit a ground ball into a fielder's choice at shallow infield. An error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

17.2.4 June

	game	NYA vs LAN on 20230602
	state before	(2, 1, 0, 0, 4, 2, 7, 1)
32.	error	play,4,1,smitw003,12,B1FCF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 2, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 1083 *In the bottom of the 4th inning, with a runner on 1st, LAN batter Will Smith had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to reach 3rd. Smith is still at bat, now with a 2-2 count.*

	game	NYA vs LAN on 20230604
	state before	(1, 0, 0, 0, 3, 0, 0, 0)
33.	error	play,3,0,torrg001,11,SBX,E5/G6.B-1
	state after	(1, 1, 0, 0, 3, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1084 *This game had 2 errors. The final score was 4-1. In the top of the 3rd inning, with bases empty, NYA batter Gleyber Torres hit a ground ball to short. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	NYA vs LAN on 20230604
	state before	(1, 1, 0, 0, 7, 0, 0, 0)
34.	error	play,7,0,kinei001,00,X,S1/BG25.1-3(E1/TH);B-2
	state after	(1, 0, 1, 1, 7, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1085 *In the top of the 7th inning, with a runner on 1st, NYA batter Isiah Kiner-Falefa hit a bunt single towards 3rd. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	CHA vs LAN on 20230613
	state before	(1, 0, 0, 0, 2, 0, 4, 0)
35.	error	play,2,0,burgj001,10,BX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 2, 0, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1086 This game had 2 errors. The final score was 1-5. *In the top of the 2nd inning, with bases empty, CHA batter Jake Burger hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs LAN on 20230613
	state before	(0, 1, 1, 0, 6, 0, 4, 1)
36.	error	play,6,1,rojam002,12,..CCFBFX,FC5/G5.2-3(E5);1-2;B-1
	state after	(0, 1, 1, 1, 6, 0, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1087 *In the bottom of the 6th inning, with runners on 1st and 2nd, LAN batter Miguel Rojas hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	CHA vs LAN on 20230614
	state before	(0, 0, 0, 0, 6, 2, 0, 0)
37.	error	play,6,0,burgj001,11,CBX,E6/G6+.B-1
	state after	(0, 1, 0, 0, 6, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1088 This game had 2 errors. The final score was 8-4. *In the top of the 6th inning, with bases empty, CHA batter Jake Burger hit a hard ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs LAN on 20230614
	state before	(0, 1, 1, 0, 6, 2, 0, 1)
38.	error	play,6,1,heywj001,11,BTX,S9/L9+.2-H(E9/TH)(NR);1-3;B-2
	state after	(0, 0, 1, 1, 6, 2, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 1089 *In the bottom of the 6th inning, with runners on 1st and 2nd, LAN batter Jason Heyward hit a hard line drive single to right field. A throwing error by the right fielder allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

Visiting/Home team rosters for SFN vs LAN on 2023-06-16

position	innings	player	position	innings	player
1	1	John Brebbia	1	1	Emmet Sheehan
1	10	Taylor Rogers	1	10	Caleb Ferguson
1	11	Jakob Junis	1	11	Alex Vesia
1	2	Ryan Walker	1	7	Brusdar Graterol
1	3	Sean Manaea	1	8	Tayler Scott
1	7	Luke Jackson	1	8	Victor Gonzalez
1	8	Tyler Rogers	1	9	Nick Robertson
1	9	Camilo Doval	2	1	Will Smith
2	1	Patrick Bailey	3	1	Freddie Freeman
3	1	LaMonte Wade Jr.	4	1	Miguel Vargas
3	8	David Villar	5	1	Chris Taylor
4	1	Thairo Estrada	5	5	Michael Busch
5	1	Wilmer Flores	6	1	Miguel Rojas
5	7	Casey Schmitt	7	1	David Peralta
6	1	Brandon Crawford	7	10	Jason Heyward
7	1	Michael Conforto	7	11	J.D. Martinez
7	8	Austin Slater	7	3	Jonny Deluca
8	1	Luis Matos	8	1	Jonny Deluca
9	1	Mike Yastrzemski	8	3	James Outman
			9	1	Mookie Betts

	game	SFN vs LAN on 20230616
	state before	(2, 0, 1, 1, 5, 0, 2, 1)
39.	error	play,5,1,outmj002,10,BX,S7/L78+.3-H;2-H;B-2(E7)
	state after	(2, 0, 1, 0, 5, 0, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 1090 This game had 4 errors. The final score was 7-5 (11 innings).

In the bottom of the 5th inning, with runners on 2nd and 3rd, LAN

batter James Outman hit a hard line drive single to left center field. The runners on 2nd and 3rd both scored. An error by the left fielder allowed the batter to reach 2nd.

	game	SFN vs LAN on 20230616
	state before	(1, 1, 1, 0, 8, 2, 4, 0)
40.	error	play,8,0,pedej001,21,C*BBX,S7/G6.2-H;1-3(E7/TH);B-2
	state after	(1, 0, 1, 1, 8, 3, 4, 0)
	fielder	7 (left field on the home team)

Summary 1091 *In the top of the 8th inning, with runners on 1st and 2nd, SFN batter Joc Pederson hit a ground ball single to left field. The runner on 2nd scored. An error by the left fielder allowed runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SFN vs LAN on 20230616
	state before	(1, 0, 1, 0, 11, 7, 5, 1)
41.	error	play,11,1,bettm001,22,BBCSX,E5/P25.2XH(96421)(E1/TH);B-3
	state after	(2, 0, 0, 1, 11, 7, 5, 1)
	fielder	5, 1 (left fielder, pitcher on the home team)

Summary 1092 This is a double error in one play.

In the bottom of the 11th inning, with a runner on 2nd, LAN batter Mookie Betts hit a pop up to 3rd base. An error by the third baseman allowed the batter to safely advance to 1st. The pitcher tagged out³ the runner on 2nd at home plate. A throwing error by the pitcher allowed the batter to reach 3rd.

	game	SFN vs LAN on 20230617
	state before	(1, 0, 1, 0, 5, 0, 0, 0)
42.	error	play,5,0,crawb001,00,2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 5, 0, 0, 0)
	fielder	1 (pitcher on the home team)

³This out is not negated by the error E5 nor by the error E1/TH because of the assist notation (96421).

Summary 1093 This game had 3 errors. The final score was 15-0.
In the top of the 5th inning, with a runner on 2nd, SFN batter Brandon Crawford has a 0-0 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to advance to 3rd. Crawford is still at bat, with the same 0-0 count.

	game	SFN vs LAN on 20230617
	state before	(1, 1, 0, 0, 7, 9, 0, 0)
43.	error	play,7,0,matol002,01,.CX,E5/TH/G56.1-2;B-1
	state after	(1, 1, 1, 0, 7, 9, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1094 *In the top of the 7th inning, with a runner on 1st, SFN batter Luis Matos hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	SFN vs LAN on 20230617
	state before	(2, 0, 1, 0, 9, 13, 0, 0)
44.	error	play,9,0,wadel001,12,BCSX,S5/G5.2-H(NR);B-2(E5/TH)
	state after	(2, 0, 1, 0, 9, 14, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1095 *In the top of the 9th inning, with a runner on 2nd, SFN batter LaMonte Wade hit a ground ball single to 3rd base. The runner on 2nd scored (no RBI). A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	SFN vs LAN on 20230618
	state before	(2, 0, 1, 0, 4, 1, 0, 0)
45.	error	play,4,0,yastm001,00,X,S9/L9S+.2-H;B-2(E9/TH)
	state after	(2, 0, 1, 0, 4, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 1096 This game had 3 errors. The final score was 7-3.

In the top of the 4th inning, with a runner on 2nd, SFN batter Mike Yastrzemski hit a hard line drive single to shallow right field. The runner on 2nd scored (1 RBI). A throwing error by the right fielder allowed the batter to reach 2nd.

	game	SFN vs LAN on 20230618
	state before	(0, 0, 0, 0, 6, 7, 2, 1)
46.	error	play,6,1,outmj002,10,..BX,S5/G5S.B-2(E5/TH)
	state after	(0, 0, 1, 0, 6, 7, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1097 *In the bottom of the 6th inning, with bases empty, LAN batter James Outman hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	SFN vs LAN on 20230618
	state before	(1, 0, 1, 0, 9, 7, 2, 1)
47.	error	play,9,1,freef001,10,BX,FC3/G3.2-3;1-2(E3);B-1
	state after	(1, 1, 1, 1, 9, 7, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1098 *In the bottom of the 9th inning, with a runner on 2nd, LAN batter Freddie Freeman hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	HOU vs LAN on 20230625
	state before	(0, 0, 1, 0, 10, 4, 4, 0)
48.	error	play,10,0,julkc001,00,..X,S9/L9L.2-H(UR);B-2(E9)
	state after	(0, 0, 1, 0, 10, 5, 4, 0)
	fielder	9 (1st baseman on the visiting team)

Summary 1099 *In the top of the 10th inning, with a runner on 2nd, HOU batter Corey Julks hit a line drive single to right field. The runner on 2nd scores. An error by the right fielder allowed the batter to reach 2nd.*

17.2.5 July

	game	PIT vs LAN on 20230703
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
49.	error	play,2,0,castr006,32,B*BBC1S>X,S1/G23.1-3(E1/TH)
	state after	(2, 1, 0, 1, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1100 This game had 2 errors. The final score was 2-5.
In the top of the 2nd inning, with a runner on 1st, PIT batter Rodolfo Castro hit a ground ball single to shallow 1st. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	PIT vs LAN on 20230703
	state before	(2, 0, 1, 1, 5, 1, 4, 1)
50.	error	play,5,1,heywj001,32,BBSFBX,E3/L3.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 5, 1, 5, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1101 *In the bottom of the 5th inning, with runners on 2nd and 3rd, LAN batter Jason Heyward hit a line drive to 1st base. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	PIT vs LAN on 20230704
	state before	(1, 1, 1, 0, 6, 5, 6, 0)
51.	error	play,6,0,davih001,12,CCBFX,S7/G56.2-H;1-3;B-2(E7/TH)
	state after	(1, 0, 1, 1, 6, 6, 6, 0)
	fielder	7 (left field on the home team)

Summary 1102 This game had 2 errors. The final score was 9-7.

In the top of the 6th inning, with runners on 1st and 2nd, PIT batter Henry Davis hit a ground ball single to short. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd.

	game	PIT vs LAN on 20230704
	state before	(1, 0, 0, 0, 8, 6, 7, 1)
52.	error	play,8,1,outmj002,11,CBX,E4/G4.B-1
	state after	(1, 1, 0, 0, 8, 6, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1103 *In the bottom of the 8th inning, with bases empty, LAN batter James Outman hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	ANA vs LAN on 20230707
	state before	(1, 1, 1, 0, 8, 4, 8, 1)
53.	error	play,8,1,freef001,10,BX,16(1)/F0/G1S.2-H(E6/TH)(NR);B-1
	state after	(2, 1, 0, 0, 8, 4, 9, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1104 *In the bottom of the 8th inning, with runners on 1st and 2nd, LAN batter Freddie Freeman hit a ground ball to shallow in-field. At 2nd, the shortstop forced out the runner from 1st. A throwing error by the shortstop (with assist from the pitcher) allowed the runner from 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	ANA vs LAN on 20230708
	state before	(2, 0, 1, 1, 2, 0, 4, 1)
54.	error	play,2,1,smitw003,00,X,S7/L78D+.3-H;2-H;B-2(E7/TH)
	state after	(2, 0, 1, 0, 2, 0, 6, 1)
	fielder	7 (left field on the visiting team)

Summary 1105 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, LAN batter Will Smith hit a hard line drive single to deep left centerfield. The runners from 2nd and 3rd each scored (2 RBIs). A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	TOR vs LAN on 20230724
	state before	(0, 1, 0, 0, 2, 0, 1, 0)
55.	error	play,2,0,varsd001,02,FS1,P01(E1/TH).1-2
	state after	(0, 0, 1, 0, 2, 0, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1106 *This game had 2 errors. The final score was 6-3. In the top of the 2nd inning, with a runner on 1st, TOR batter Daulton Varsho has a 0-2 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner from 1st to reach 2nd. Varsho is still at bat with a 0-2 count.*

	game	TOR vs LAN on 20230724
	state before	(0, 0, 1, 0, 11, 3, 3, 0)
56.	error	play,11,0,chapm001,12,...FCBX,E5/TH/G56+.B-1
	state after	(0, 1, 1, 0, 11, 3, 3, 0)
	fielder	5 (pitcher on the home team)

Summary 1107 *In the top of the 11th inning, with a runner on 2nd, TOR batter Matt Chapman hit a hard ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CIN vs LAN on 20230729
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
57.	error	play,1,1,perad001,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1108 This game had 2 errors. The final score was 2-3.
In the bottom of the 1st inning, with bases empty, LAN batter David Peralta hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.

	game	CIN vs LAN on 20230729
	state before	(0, 0, 1, 0, 6, 0, 2, 0)
58.	error	play,6,0,friet001,22,BBLCX,E4/G4.2-3;B-1
	state after	(0, 1, 0, 1, 6, 0, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1109 *In the top of the 6th inning, with a runner on 2nd, CIN batter TJ Friedl hit a ground ball to 2nd base. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	CIN vs LAN on 20230730
	state before	(0, 1, 1, 0, 1, 1, 0, 0)
59.	error	play,1,0,fralj001,22,*BF2*BFX,S8/L89S.2-H;1-3;B-2(E8/TH)
	state after	(0, 0, 1, 1, 1, 2, 0, 0)
	fielder	8 (center field on the home team)

Summary 1110 *In the top of the 1st inning, with runners on 1st and 2nd, CIN batter Jake Fraley hit a line drive single to shallow right center field. The runner from 2nd scored (1 RBI) and the runner from 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

17.2.6 August

	game	OAK vs LAN on 20230803
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
60.	error	play,1,1,rosaa003,11,FBX,E6/G6.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1111 This game had 2 errors. The final score was 2-8.
In the bottom of the 1st inning, with a runner on 2nd, LAN batter Amed Rosario hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	OAK vs LAN on 20230803
	state before	(1, 0, 1, 1, 6, 1, 3, 1)
61.	error	play,6,1,rosaa003,01,CX,D9/L9L.3-H;2-H;B-3(E9/TH)
	state after	(1, 0, 0, 1, 6, 1, 5, 1)
	fielder	9 (right field on the visiting team)

Summary 1112 *In the bottom of the 6th inning, with runners on 2nd and 3rd, LAN batter Amed Rosario hit a line drive double to right field. The runners from 2nd and 3rd scored (2 RBIs). A throwing error by the right fielder allowed the batter to reach 3rd.*

	game	COL vs LAN on 20230810
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
62.	error	play,1,0,mcmar001,22,CBBF.X,E3/G3.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1113 *In the top of the 1st inning, with bases empty, COL batter Ryan McMahon hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	COL vs LAN on 20230811
	state before	(0, 1, 1, 0, 4, 0, 1, 0)
63.	error	play,4,0,rodgb002,12,SSBX,S5/G5S.2-H(E5/TH)(NR)(UR);1-3;B-2
	state after	(0, 0, 1, 1, 4, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1114 This game had 2 errors. The final score was 1-6.

In the top of the 4th inning, with runners on 1st and 2nd, COL batter Brendan Rodgers hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd base.

	game	COL vs LAN on 20230811
	state before	(1, 1, 1, 1, 6, 1, 4, 1)
64.	error	play,6,1,freef001,31,BFBBX,E7/F78XD/SF.3-H;2-3;1-2;B-1
	state after	(1, 1, 1, 1, 6, 1, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 1115 *In the bottom of the 6th inning, with bases loaded, LAN batter Freddie Freeman hit a sacrifice fly to deep left center field. An error by the left fielder allowed the runner on 3rd to score (1 RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are still loaded.*

	game	COL vs LAN on 20230813
	state before	(1, 0, 1, 0, 3, 1, 2, 1)
65.	error	play,3,1,rosaa003,00,X,S9/L9.2-H(E9)(NR);B-2
	state after	(1, 0, 1, 0, 3, 1, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1116 *In the bottom of the 3rd inning, with a runner on 2nd, LAN batter Amed Rosario hit a line drive single to right field. An error by the right fielder allowed the runner on 2nd to score (no RBI) and the batter to reach 2nd.*

game	MIL vs LAN on 20230815
state before	(0, 1, 0, 0, 4, 1, 0, 1)
error	play,4,1,freef001,02,FFX,FC4/G4.1-2(E4);B-1
state after	(0, 1, 1, 0, 4, 1, 0, 1)
fielder	4 (2nd baseman on the visiting team)

Summary 1117 This game had 2 errors. The final score was 2-6.
In the bottom of the 4th inning, with a runner on 1st, LAN batter Freddie Freeman hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to safely reach 2nd and the batter to advance to 1st.

game	MIL vs LAN on 20230815
state before	(1, 0, 0, 0, 6, 1, 1, 1)
error	play,6,1,smitw003,11,CBX,E5/TH/G56.B-1
state after	(1, 1, 0, 0, 6, 1, 1, 1)
fielder	5 (3rd baseman on the visiting team)

Summary 1118 *In the bottom of the 6th inning, with bases empty, LAN batter Will Smith hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for MIL vs LAN on 2023-08-16

position	innings	player	position	innings	player
1	1	Wade Miley	1	1	Clayton Kershaw
1	6	Abner Uribe	1	6	Brusdar Graterol
1	7	Andrew Chafin	1	7	Ryan Brasier
1	8	Trevor Megill	1	8	Gus Varland
2	1	William Contreras	2	1	Will Smith
3	1	Carlos Santana	3	1	Freddie Freeman
4	1	Andruw Monasterio	4	1	Amed Rosario
5	1	Brian Anderson	5	1	Max Muncy
6	1	Willy Adames	6	1	Miguel Rojas
7	1	Christian Yelich	7	1	Chris Taylor
8	1	Joey Wiemer	8	1	Enrique Hernandez
8	8	Sal Frelick	9	1	Mookie Betts
9	1	Sal Frelick			
9	8	Tyrone Taylor			

	game	MIL vs LAN on 20230816
	state before	(1, 1, 0, 1, 1, 0, 0, 1)
66.	error	play,1,1,martj006,10,1BX,C/E2.1-2;B-1
	state after	(1, 1, 1, 1, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1119 This game had 3 errors, including 2 by catcher’s interference (both with J.D. Martinez at bat). The final score was 1-7. *In the bottom of the 1st inning, with runners on 1st and 3rd, LAN batter J.D. Martinez had a 1-0 count. Catcher’s interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base. The bases are loaded.*

	game	MIL vs LAN on 20230816
	state before	(2, 1, 1, 1, 2, 1, 2, 1)
67.	error	play,2,1,martj006,22,CBBFX,C/E2.3-H(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 2, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1120 *In the bottom of the 2nd inning, with 2 outs and bases loaded, LAN batter J.D. Martinez had a 2-2 count. Catcher's interference allowed each runner to advance one base and the batter to reach 1st base. As a result, the runner from 3rd scored (1 RBI)⁴. The bases are still loaded.*

	game	MIL vs LAN on 20230816
	state before	(2, 1, 0, 0, 6, 1, 4, 1)
68.	error	play,6,1,freef001,02,FFX,E8/F78XD+.1-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 6, 1, 5, 1)
	fielder	8 (center field on the visiting team)

Summary 1121 *In the bottom of the 6th inning, with a runner on 1st, LAN batter Freddie Freeman hit a hard fly ball to deep left center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	MIA vs LAN on 20230819
	state before	(1, 1, 0, 1, 8, 1, 0, 1)
69.	error	play,8,1,barna001,01,LX,S3/BG23.3-H;1-3(E3/TH);B-2
	state after	(1, 0, 1, 1, 8, 1, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1122 *In the bottom of the 8th inning, with runners on 1st and 3rd, LAN batter Austin Barnes hit a bunt single to 1st base. The runner from 3rd scored. A throwing error by the first baseman allowed the runner on 1st to reach 3rd and the batter to reach 2nd.*

⁴See MLB Rule 9.04(a)(2).

70.	game	ARI vs LAN on 20230829
	state before	(2, 1, 0, 0, 3, 0, 5, 1)
	error	play,3,1,taylc001,20,BBX,D7/L5D.1-H;B-3(E6/TH)
	state after	(2, 0, 0, 1, 3, 0, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1123 *In the bottom of the 3rd inning, with a runner on 1st, LAN batter Chris Taylor hit a line drive double to deep 3rd base. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 3rd.*

71.	game	ARI vs LAN on 20230830
	state before	(0, 0, 0, 0, 3, 0, 2, 1)
	error	play,3,1,perad001,12,BSFFX,E1/G1S-.B-1
	state after	(0, 1, 0, 0, 3, 0, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1124 *In the bottom of the 3rd inning, with bases empty, LAN batter David Peralta hit a soft ground ball to the pitcher. An error by the pitcher allowed the batter to reach 1st base.*

	game	ATL vs LAN on 20230831
	state before	(2, 0, 0, 0, 6, 7, 4, 0)
	error	play,6,0,lopen001,12,BS.FX,E5/TH/G56.B-2
	state after	(2, 0, 1, 0, 6, 7, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1125 *In the top of the 6th inning, with bases empty, ATL batter Nicky Lopez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

17.2.7 September

72.	game	ATL vs LAN on 20230901
	state before	(0, 0, 0, 0, 4, 3, 0, 0)
	error	play,4,0,darnt001,21,CBBX,E5/G5.B-1
	state after	(0, 1, 0, 0, 4, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1126 This game had 2 errors. The final score was 6-3.

In the top of the 4th inning, with bases empty, ATL batter Travis d'Arnaud hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	ATL vs LAN on 20230901
	state before	(1, 1, 0, 0, 8, 6, 0, 1)
	error	play,8,1,rojam002,02,SCFFX,FC/G4.1-2(E4);B-1
	state after	(1, 1, 1, 0, 8, 6, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1127 *In the bottom of the 8th inning, with a runner on 1st, LAN batter Miguel Rojas hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to reach 2nd and the batter to reach 1st.*

	game	ATL vs LAN on 20230903
	state before	(0, 1, 0, 0, 5, 0, 0, 1)
	error	play,5,1,rojam002,02,CF1X,D8/L78XD+.1-H;B-3(E6/TH)
	state after	(0, 0, 0, 1, 5, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1128 This game had 2 errors. The final score was 1-3.

In the bottom of the 5th inning, with a runner on 1st, LAN batter Miguel Rojas hit a hard line drive double to deep left center field. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 3rd.

game	ATL vs LAN on 20230903
state before	(0, 0, 0, 0, 8, 1, 2, 1)
error	play,8,1,herne001,02,CC.X,E1/TH/G2-.B-1
state after	(0, 1, 0, 0, 8, 1, 2, 1)
fielder	1 (pitcher on the visiting team)

Summary 1129 *In the bottom of the 8th inning, with bases empty, LAN batter Enrique Hernandez hit a soft ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	SDN vs LAN on 20230911
	state before	(0, 0, 0, 0, 9, 7, 7, 0)
73.	error	play,9,0,grist001,22,.CBC.BFX,E7/F78XD.B-2
	state after	(0, 0, 1, 0, 9, 7, 7, 0)
	fielder	7 (left field on the home team)

Summary 1130 This game had 2 errors. The final score was 11-8. *In the top of the 9th inning, with bases empty, SDN batter Trent Grisham hit a fly ball to deep left center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	SDN vs LAN on 20230911
	state before	(1, 0, 0, 1, 9, 7, 7, 0)
	error	play,9,0,tatif002,12,BSSX,E1/G1-.B-1
	state after	(1, 1, 0, 1, 9, 7, 7, 0)
	fielder	1 (pitcher on the home team)

Summary 1131 *In the top of the 9th inning, with a runner on 3rd, SDN batter Fernando Tatis hit a soft ground ball to shallow infield. An error by the pitcher allowed the batter to reach 1st base.*

	game	SDN vs LAN on 20230912
	state before	(2, 0, 0, 0, 7, 2, 7, 0)
74.	error	play,7,0,rosae002,10,BX,E5/G56.B-1
	state after	(2, 1, 0, 0, 7, 2, 7, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1132 *In the top of the 7th inning, with bases empty, SDN batter Eguy Rosario hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for DET vs LAN on 2023-09-20

position	innings	player	position	innings	player
1	1	Reese Olson	1	1	Bobby Miller
1	7	Miguel Diaz	1	7	Ryan Brasier
1	8	Will Vest	1	8	Evan Phillips
2	1	Carson Kelly	1	9	Brusdar Graterol
3	1	Spencer Torkelson	2	1	Austin Barnes
4	1	Andy Ibanez	3	1	Freddie Freeman
4	8	Zach McKinstry	4	1	Mookie Betts
5	1	Matt Vierling	5	1	Max Muncy
6	1	Zach McKinstry	6	1	Miguel Rojas
6	8	Zack Short	7	1	David Peralta
7	1	Akil Baddoo	8	1	James Outman
8	1	Parker Meadows	9	1	Jason Heyward
9	1	Kerry Carpenter			

	game	DET vs LAN on 20230920
	state before	(2, 0, 0, 0, 7, 2, 1, 0)
75.	error	play,7,0,meadp001,12,CFBFFX,E5/TH/G6.B-1
	state after	(2, 1, 0, 0, 7, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1133 This game had 3 errors. The final score was 4-2.

In the top of the 7th inning, with bases empty, DET batter Parker Meadows hit a ground ball to short. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	DET vs LAN on 20230920
	state before	(2, 0, 1, 0, 8, 3, 1, 0)
76.	error	play,8,0,ibana001,11,BFX,S9/L9LS.2-H;B-3(E9/TH)
	state after	(2, 0, 0, 1, 8, 4, 1, 0)
	fielder	9 (right field on the home team)

Summary 1134 *In the top of the 8th inning, with a runner on 2nd, DET batter Andy Ibanez hit a line drive single to shallow right field. A throwing error by the right fielder allowed the batter to reach 3rd base.*

	game	DET vs LAN on 20230920
	state before	(2, 0, 0, 0, 9, 4, 2, 1)
77.	error	play,9,1,heywj001,11,BCX,E4/G34.B-1
	state after	(2, 1, 0, 0, 9, 4, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1135 *In the bottom of the 9th inning, with bases empty, LAN batter Jason Heyward hit a ground ball to 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	SFN vs LAN on 20230921
	state before	(2, 0, 0, 0, 4, 0, 2, 1)
78.	error	play,4,1,rosaa003,01,CX,S5/G56S-.B-2(E5/TH)
	state after	(2, 0, 1, 0, 4, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1136 This game had 2 errors. The final score was 2-7.
In the bottom of the 4th inning, with bases empty, LAN batter Amed Rosario hit a soft ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.

game	SFN vs LAN on 20230921
state before	(1, 0, 0, 0, 7, 2, 3, 1)
error	play,7,1,taylc001,02,SF.X,E5/G5.B-1
state after	(1, 1, 0, 0, 7, 2, 3, 1)
fielder	5 (3rd baseman on the visiting team)

Summary 1137 *In the bottom of the 7th inning, with bases empty, LAN batter Chris Taylor hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

game	SFN vs LAN on 20230924
state before	(1, 0, 0, 0, 3, 0, 2, 1)
error	play,3,1,bettm001,21,CBBX,S5/G56.B-2(E5/TH)
state after	(1, 0, 1, 0, 3, 0, 2, 1)
fielder	5 (3rd baseman on the visiting team)

Summary 1138 *In the bottom of the 3rd inning, with bases empty, LAN batter Mookie Betts hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

Chapter 18

Miami Marlins (MIA)

Originating in 1993 as the Florida Marlins, the **Miami Marlins** compete in the National League East Division. The franchise adopted the nickname “Marlins” from previous minor league teams, such as the Miami Marlins of the Triple-A ’s International League from 1956 to 1960, In fact, Hall of Famer Satchel Paige pitched for Miami Marlins (AAA) from 1956 to 1958.

The modern Miami Marlins have won the World Series title twice (1997, 2003). Their last post-season appearance was in 2023 (as a Wild Card).

This chapter is devoted to listing the errors in each regular season game of 2023 played at Miami against the Marlins. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder’s choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 MIA team is 0.683 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the 2023 Miami Marlins is less than one standard deviation away from the MLB average.

In MIA 2023, there were 83 errors in the 81 home games, 46 by the MIA and the rest (37) by visiting teams. The total number of errors by the Marlins (home and away) over the regular season of 2023 was 98, well over 83.9, the

2023 MLB average. The Marlins were well above average for errors (which is bad).

18.1 Season totals

Set Position shall be indicated by the pitcher when he stands facing the batter with his pivot foot in contact with, and his other foot in front of, the pitcher's plate, holding the ball in both hands in front of his body and coming to a complete stop. From such Set Position he may deliver the ball to the batter, throw to a base or step backward off the pitcher's plate with his pivot foot.

– **MLB Rule Book** 5.07(a)(2), part 1/2

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

18.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	37
Runs by ht because of errors by vt:	14
Errors by ht:	46
Runs by vt because of errors by ht:	13

This is an excellent error-to-run conversion rate: $14/37 = 37.8\%$ on the part of the Marlins (compared to $13/46 = 28.2\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 MIA season (home and away) has about $R/PA = 666/6043$ (about 11%) change of resulting in a run. Therefore, a randomly selected error is (for MIA) over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	7	10
to catcher:	7	6
to 1st baseman:	4	3
to 2nd baseman:	5	1
to 3rd baseman:	7	8
to shortstop:	5	9
to left fielder:	1	3
to center fielder:	1	4
to right fielder:	0	2

18.1.2 Error total by state

Before assuming Set Position, the pitcher may elect to make any natural preliminary motion such as that known as the **stretch**. But if he so elects, he shall come to Set Position before delivering the ball to the batter. After assuming Set Position, any natural motion associated with his delivery of the ball to the batter commits him to the pitch without alteration or interruption.

– **MLB Rule Book** 5.07(a)(2), part 2/2

The error totals are collected by game state in this section.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	12	16	28
(lone) runner on 1st	11	16	27
(lone) runner on 2nd	2	8	10
(lone) runner on 3rd	1	0	1
two runners on base	6	5	11
bases loaded	5	1	6
total	37	46	83

A table of frequencies of most common errors by the MIA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	9	19.6%
(1, 0, 0, 0)	3	6.5%
(2, 0, 0, 0)	4	8.6%
(0, 1, 0, 0)	4	8.6%
(1, 1, 0, 0)	7	15.2%
(2, 1, 0, 0)	5	10.9%
(1, 0, 1, 0)	3	6.5%
(2, 0, 1, 0)	4	8.6%
(0, 0, 1, 1)	2	4.3%

A table of frequencies of most common errors by a visiting team in MIA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	2	5.4%
(1, 0, 0, 0)	6	16.2%
(2, 0, 0, 0)	4	10.8%
(0, 1, 0, 0)	4	10.8%
(1, 1, 0, 0)	4	10.8%
(2, 1, 0, 0)	3	8.1%
(1, 1, 1, 0)	2	5.4%
(2, 1, 1, 0)	2	5.4%
(0, 1, 0, 1)	1	2.7%
(1, 1, 1, 1)	4	10.8%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Marlins homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `MIA2023.evn`):

state	MIA2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	16.2%
(2, 0, 0, 0)	13.2%
(0, 1, 0, 0)	6.3%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	7.0%
(0, 0, 1, 0)	2.3%
(1, 0, 1, 0)	3.5%
(2, 0, 1, 0)	4.1%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.6%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.7%
(1, 1, 1, 0)	2.4%
(2, 1, 1, 0)	2.8%

18.1.3 Some interesting errors

Some of the most interesting error plays in the MIA 2023 home games:

- The game SDN vs MIA on 2023-05-30 had 4 errors.
- The game DET vs MIA on 2023-07-30 had 4 errors.
- The game HOU vs MIA on 2023-08-15 had 3 errors including a double error play.

18.1.4 Relative team rankings

If there is a runner, or runners, it is a **balk** when

- (10) The pitcher, after coming to a legal pitching position, removes one hand from the ball other than in an actual pitch, or in throwing to a base;
- (11) The pitcher, while touching his plate, accidentally or intentionally has the ball slip or fall out of his hand or glove;
- (12) The pitcher, while giving an intentional base on balls, pitches when the catcher is not in the catcher's box;

(13) The pitcher delivers the pitch from Set Position without coming to a stop.

– **MLB Rule Book** 6.02(a)(10)-(13)

We have 81 home games between MIA and some visiting MLB team (vt). Each list below is given with that team's W/T record in parentheses¹.

A number of teams did not play the Marlins in the regular season. However, of those that did, MIA *lost* more games than won against:

TBA (2/2), CIN (2/3), COL (2/3), DET (2/3), HOU (2/3),
SDN (2/3), TOR (2/3), PHI (4/7), NYN (5/7)

and *won* more than lost against

CHN (0/3), KCA (0/3), OAK (0/3), ARI (1/3), LAN (1/3),
MIL (1/3), MIN (1/3), NYA (1/3), SFN (1/3), PIT (1/4), WAS (2/6).

The teams that *split* their series were:

SLN (2/4), ATL (3/6).

The teams that did not play in any of the 81 home games in MIA 2023 were:

ANA, BAL, BOS, CHA, CLE, LAN, SEA, TEX.

18.2 Game-by-game totals

The Official Scorer shall charge a catcher with a **passed ball** when the catcher fails to hold or to control a legally pitched ball that should have been held or controlled with ordinary effort, thereby permitting a runner or runners to advance. When the third strike is a passed ball, permitting the batter to reach first base, the Official Scorer shall score a strikeout and a passed ball.

– **MLB Rule Book** 9.13(b)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for MIA 2023 is as follows [BA].

¹Here, W = numbers of home games the vt won against MIA, T = Total home games vt vs MIA played at Miami.

position	player	games played	OBP
1 (C)	Nick Fortes	105	.263
3 (1B)	Yuli Gurriel	93	.304
4 (2B)	Luis Arraez	134	.393
5 (3B)	Jean Segura	84	.248
6 (SS)	Joey Wendle	108	.277
7 (LF)	Bryan De La Cruz	143	.304
8 (CF)	Jazz Chisholm Jr	95	.304
9 (RF)	Jesús Sánchez	107	.327

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 MIA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

As we will see below, there were 83 errors in the 81 home games, 46 by MIA and the rest (37) by visiting teams.

	game	NYN vs MIA on 20230330
	state before	(1, 1, 0, 0, 6, 1, 0, 0)
1.	error	play,6,0,marts002,01,SX,S8/G4M.1-3;B-2(E8/TH)
	state after	(1, 0, 1, 1, 6, 1, 0, 0)
	fielder	8 (center field on the home team)

Summary 1139 *In the top of the 6th inning, with a runner on 1st, NYN batter Starling Marte hit a ground ball single to deep 2nd. The runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

18.2.1 April

	game	NYN vs MIA on 20230402
	state before	(2, 1, 1, 1, 1, 0, 0, 0)
2.	error	play,1,0,mcnej002,12,CBFF.FFX,S1/G23-.3-H;2-H(E1/TH)(NR)(UR);1-3
	state after	(2, 1, 0, 1, 1, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1140 *In the top of the 1st inning, with bases loaded, NYN batter Jeff McNeil hit a soft ground ball single to shallow right infield.. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the runner on 2nd to also score (unearned, no RBI) and the runner on 1st to advance to 3rd. The batter took 1st.*

	game	MIN vs MIA on 20230403
	state before	(1, 1, 0, 0, 4, 6, 0, 0)
3.	error	play,4,0,keplm001,22,CCBFBX,S5/G6MS.1-3(E5/TH)
	state after	(1, 1, 0, 1, 4, 6, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1141 *In the top of the 4th inning, with a runner on 1st, MIN batter Max Kepler hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	MIN vs MIA on 20230405
	state before	(0, 0, 0, 0, 3, 0, 1, 0)
4.	error	play,3,0,castw003,12,FSBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 3, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1142 *In the top of the 3rd inning, with bases empty, MIN batter Willi Castro hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	ARI vs MIA on 20230414
	state before	(1, 1, 1, 0, 1, 0, 0, 1)
5.	error	play,1,1,delab001,12,CBCFX,5(2)/F0/G56.1-3(E5/TH);B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1143 *In the bottom of the 1st inning, with runners on 1st and 2nd, MIA batter Bryan De La Cruz hit a ground ball to 3rd. The runner from 2nd was forced out at 3rd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	ARI vs MIA on 20230415
	state before	(2, 1, 1, 0, 1, 1, 0, 1)
6.	error	play,1,1,garca003,22,BCFBFFX,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 1, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1144 *In the bottom of the 1st inning, with runners on 1st and 2nd, MIA batter Avisail Garcia had a 2-2 count. Catcher's interference allowed the batter to take 1st. The runners on 1st and 2nd base each advanced one base, as required. The bases are loaded.*

	game	ARI vs MIA on 20230415
	state before	(1, 1, 1, 0, 7, 2, 1, 0)
7.	error	play,7,0,martk001,01,CX,3/P3F/FL.2-3(E3/TH);1-2
	state after	(2, 0, 1, 1, 7, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1145 *In the top of the 7th inning, with runners on 1st and 2nd, ARI batter Ketel Marte hit a pop fly foul of the 1st base line. The batter was put out by the first baseman. A throwing error by the first baseman allowed the runners on 1st and 2nd to each advance one base.*

	game	ARI vs MIA on 20230416
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
8.	error	play,2,0,carrc005,10,BX,E7/L78D+.B-2
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1146 This game had 2 errors. The final score was 5-0.
In the top of the 2nd inning, with bases empty, ARI batter Corbin Carroll hit a hard line drive to deep left center field. An error by the left fielder allowed the batter to reach 2nd base.

	game	ARI vs MIA on 20230416
	state before	(0, 0, 0, 0, 6, 1, 0, 0)
9.	error	play,6,0,rojaj001,11,CBX,3E1/G34.B-2
	state after	(0, 0, 1, 0, 6, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1147 *In the top of the 6th inning, with bases empty, ARI batter Josh Rojas hit a ground ball to 1st/2nd. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 2nd base.*

	game	SFN vs MIA on 20230417
	state before	(1, 0, 0, 0, 4, 3, 1, 0)
10.	error	play,4,0,ramoh002,01,CX,E1/TH/BG1S-.B-1
	state after	(1, 1, 0, 0, 4, 3, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1148 *In the top of the 4th inning, with bases empty, SFN batter Heliot Ramos hit a soft bunt to shallow center infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	SFN vs MIA on 20230418
	state before	(1, 0, 0, 0, 4, 2, 4, 1)
11.	error	play,4,1,garca003,10,BX,C/E2.B-1
	state after	(1, 1, 0, 0, 4, 2, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1149 *In the bottom of the 4th inning, with bases empty, MIA batter Avisail Garcia had a 1-0 count. Catcher's interference allowed the batter to take 1st.*

	game	CHN vs MIA on 20230428
	state before	(1, 1, 0, 0, 7, 2, 1, 0)
12.	error	play,7,0,gomey001,02,.CF1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 7, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1150 *In the top of the 7th inning, with a runner on 1st, CHN batter Yan Gomes has a 0-2 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 2nd. Gomes is still at-bat with a 0-2 count.*

	game	CHN vs MIA on 20230430
	state before	(0, 1, 0, 1, 3, 1, 0, 1)
13.	error	play,3,1,coopg002,02,SSFS,K+PB.3-H(NR);1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1151 *In the bottom of the 3rd inning, with runners on 1st and 3rd, MIA batter Garrett Cooper had a 0-2 count. During the next pitch, which was a swinging strike and a passed ball, the runner on 3rd scored (no RBI). A throwing error by the catcher allowed the runner on 1st to advance to 3rd. The batter struck out.*

18.2.2 May

	game	ATL vs MIA on 20230502
	state before	(1, 0, 0, 0, 3, 2, 0, 1)
14.	error	play,3,1,bertj001,10,BX,E5/TH/G5.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1152 *In the bottom of the 3rd inning, with bases empty, MIA batter Jon Berti hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	ATL vs MIA on 20230503
	state before	(2, 0, 1, 0, 2, 7, 1, 1)
15.	error	play,2,1,bertj001,10,BX,E6/TH/G6.2-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 2, 7, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1153 *In the bottom of the 2nd inning, with a runner on 2nd, MIA batter Jon Berti hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

Visiting/Home team rosters for ATL vs MIA on 2023-05-04

position	innings	player	position	innings	player
1	1	Dylan Dodd	1	1	Jesus Luzardo
1	7	Michael Tonkin	1	6	Huascar Brazoban
1	8	Nick Anderson	1	8	Dylan Floro
1	9	A.J. Minter	1	9	A.J. Puk
2	1	Sean Murphy	1	9	Steven Okert
3	1	Matt Olson	2	1	Nick Fortes
4	1	Ozzie Albies	3	1	Yuli Gurriel
5	1	Austin Riley	4	1	Luis Arraez
6	1	Vaughn Grissom	5	1	Jean Segura
7	1	Eddie Rosario	6	1	Jon Berti
8	1	Kevin Pillar	7	1	Bryan De La Cruz
8	7	Sam Hilliard	8	1	Peyton Burdick
9	1	Ronald Acuna Jr.	8	9	Jazz Chisholm Jr.
9	7	Kevin Pillar	9	1	Garrett Hampson
			9	9	Peyton Burdick

	game	ATL vs MIA on 20230504
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
16.	error	play,1,0,acunr001,12,CSBX,E3/P3D.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1154 This game had 3 errors. The final score was 6-3.

In the top of the 1st inning, with bases empty, ATL batter Ronald Acuna hit a pop up to deep 1st base. An error by the first baseman allowed the batter to reach 1st base.

	game	ATL vs MIA on 20230504
	state before	(2, 1, 0, 0, 2, 3, 2, 0)
17.	error	play,2,0,olsom001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 3, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1155 *In the top of the 2nd inning, with a runner on 1st, ATL batter Matt Olson had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Olson is still at bat, now with a 0-1 count.*

	game	ATL vs MIA on 20230504
	state before	(0, 0, 0, 0, 5, 4, 2, 0)
18.	error	play,5,0,rilea001,11,BCX,E1/TH/G6S.B-1
	state after	(0, 1, 0, 0, 5, 4, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1156 *In the top of the 5th inning, with bases empty, ATL batter Austin Riley hit a ground ball to shallow short. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	CIN vs MIA on 20230512
	state before	(0, 0, 0, 0, 4, 1, 0, 1)
19.	error	play,4,1,arral001,21,BBCX,E3/G9LS.B-1
	state after	(0, 1, 0, 0, 4, 1, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1157 *In the bottom of the 4th inning, with bases empty, MIA batter Luis Arraez hit a ground ball to short right field down the 1st base line. An error by the first baseman allowed the batter to reach 1st base.*

20.	game	CIN vs MIA on 20230514
	state before	(2, 1, 1, 0, 4, 0, 0, 1)
	error	play,4,1,fortn001,22,FBCBX,E6/G6.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 4, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1158 *In the bottom of the 4th inning, with runners on 1st and 2nd, MIA batter Nick Fortes hit a ground ball to short. An error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

21.	game	WAS vs MIA on 20230516
	state before	(1, 1, 1, 1, 2, 0, 0, 1)
	error	play,2,1,wendj002,11,CBX,36(1)/F0/G3.3-H;2-H(E1)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 2, 0, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1159 *This game had 2 errors. The final score was 4-5. In the bottom of the 2nd inning, with bases loaded, MIA batter Joey Wendle hit a ground ball to 1st. At 2nd, the shortstop (with an assist from the first baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). An error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

22.	game	WAS vs MIA on 20230516
	state before	(1, 1, 0, 0, 8, 1, 2, 0)
	error	play,8,0,menej001,32,BB1SS>FB>B,W.1-3(E2/TH)
	state after	(1, 1, 0, 1, 8, 1, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1160 *In the top of the 8th inning, with a runner on 1st, WAS batter Joey Meneses had a full count. During the next pitch,*

which was a ball, the runner on 1st ran to 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter took 1st (walk).

	game	WAS vs MIA on 20230517
	state before	(0, 1, 0, 0, 7, 2, 4, 0)
23.	error	play,7,0,vargi001,11,.CBX,S6/G6.1-3(E6/TH);B-2
	state after	(0, 0, 1, 1, 7, 2, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1161 *In the top of the 7th inning, with a runner on 1st, WAS batter Ildemaro Vargas hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	WAS vs MIA on 20230518
	state before	(1, 0, 0, 0, 8, 3, 4, 1)
24.	error	play,8,1,delab001,10,BX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 8, 3, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1162 *In the bottom of the 8th inning, with bases empty, MIA batter Bryan De La Cruz hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SDN vs MIA on 2023-05-30

position	innings	player	position	innings	player
1	1	Ryan Weathers	1	1	Sandy Alcantara
1	5	Domingo Tapia	1	7	Tanner Scott
1	6	Tim Hill	1	8	Andrew Nardi
1	7	Steven Wilson	1	8	JT Chargois
1	8	Nick Martinez	1	9	Bryan Hoeing
1	9	Brent Honeywell	1	9	Dylan Floro
2	1	Gary Sanchez	2	1	Jacob Stallings
2	8	Austin Nola	2	9	Nick Fortes
3	1	Jake Cronenworth	3	1	Yuli Gurriel
4	1	Rougned Odor	4	1	Luis Arraez
5	1	Ha-Seong Kim	5	1	Jean Segura
6	1	Xander Bogaerts	6	1	Jon Berti
7	1	Juan Soto	7	1	Bryan De La Cruz
8	1	Trent Grisham	8	1	Garrett Hampson
9	1	Fernando Tatis Jr.	9	1	Jorge Soler
			9	9	Jesus Sanchez

	game	SDN vs MIA on 20230530
	state before	(2, 1, 0, 0, 2, 0, 0, 1)
25.	error	play,2,1,bertj001,22,*BC1F*BX,S4/G34.1-3(E4/TH)
	state after	(2, 1, 0, 1, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1163 This game had 4 errors. The final score was 9-4.

In the bottom of the 2nd inning, with a runner on 1st, MIA batter Jon Berti hit a ground ball single to 1st/2nd. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	SDN vs MIA on 20230530
	state before	(2, 1, 0, 0, 5, 1, 3, 0)
26.	error	play,5,0,sotoj001,00,>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 1, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1164 *In the top of the 5th inning, with a runner on 1st, SDN batter Juan Soto had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Soto is still at bat, now with a 1-0 count.*

	game	SDN vs MIA on 20230530
	state before	(0, 1, 0, 0, 9, 4, 4, 0)
27.	error	play,9,0,sotoj001,01,C>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 4, 4, 0)
	fielder	2 (catcher on the home team)

Summary 1165 *In the top of the 9th inning, with a runner on 1st, SDN batter Juan Soto (same batter as in the 5th inning error above) had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Soto is still at bat, now with a 1-1 count.*

	game	SDN vs MIA on 20230530
	state before	(0, 1, 0, 1, 9, 5, 4, 0)
28.	error	play,9,0,odorr001,11,BFX,FC6/G6S.3-H(E6/TH) (NR) (UR) ; 1-3;B-2
	state after	(0, 0, 1, 1, 9, 6, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1166 *In the top of the 9th inning, with runners on 1st and 3rd, SDN batter Rougned Odor hit a ground ball into a fielder's choice at short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SDN vs MIA on 20230531
	state before	(2, 0, 0, 0, 3, 1, 0, 0)
29.	error	play,3,0,kim-h002,00,X,E5/G7S.B-1
	state after	(2, 1, 0, 0, 3, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1167 This game had 2 errors. The final score was 1-2.
In the top of the 3rd inning, with bases empty, SDN batter Ha-Seong Kim hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.

	game	SDN vs MIA on 20230531
	state before	(0, 1, 0, 0, 8, 1, 0, 1)
30.	error	play,8,1,davij007,00,X,FC/SH/BG1S-.1-2;B-1(E1)
	state after	(0, 1, 1, 0, 8, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1168 *In the bottom of the 8th inning, with a runner on 1st, MIA batter Jonathan Davis hit a soft bunt to a fielder's choice at shallow center infield. The runner on 1st advanced to second². An error by the pitcher allowed the batter to reach 1st.*

18.2.3 June

	game	SDN vs MIA on 20230601
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
31.	error	play,3,1,wendj002,20,BB1>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 3, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

²Since the play was made by the fielder (pitcher) to 1st, if defensive difference was what allowed this advance, then that could be the reason why this play was scored as a fielder's choice.

Summary 1169 *In the bottom of the 3rd inning, with a runner on 1st, MIA batter Joey Wendle had a 2-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Wendle is still at bat, now with a 2-1 count.*

	game	OAK vs MIA on 20230602
	state before	(0, 0, 0, 0, 6, 0, 4, 0)
32.	error	play,6,0,ruize001,12,BSFX,S1/G25-.BX3(95)(E1/TH)
	state after	(1, 0, 0, 0, 6, 0, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 1170 *In the top of the 6th inning, with bases empty, OAK batter Esteury Ruiz hit a soft ground ball single to left shallow infield. A throwing error by the pitcher allowed the batter to advance to 2nd. However, the batter was tagged out at 3rd when he attempted to reach 3rd.*

	game	OAK vs MIA on 20230603
	state before	(2, 1, 0, 0, 9, 1, 12, 0)
33.	error	play,9,0,laurr001,01,CX,FC6/G6.1-2(E6);B-1
	state after	(2, 1, 1, 0, 9, 1, 12, 0)
	fielder	6 (shortstop on the home team)

Summary 1171 *In the top of the 9th inning, with a runner on 1st, OAK batter Ramon Laureano hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	OAK vs MIA on 20230604
	state before	(1, 0, 1, 0, 3, 4, 1, 0)
34.	error	play,3,0,bledj001,01,>FN,SB3.2-H(E1/TH)(NR)
	state after	(1, 0, 0, 0, 3, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1172 *In the top of the 3rd inning, with a runner on 2nd, OAK batter JJ Bleday had a 0-1 count. During the next pitch, which was a no pitch, the runner on 2nd stole 3rd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI). Bleday is still at bat, with a 0-1 count.*

	game	KCA vs MIA on 20230605
	state before	(2, 1, 0, 0, 4, 4, 2, 1)
35.	error	play,4,1,wendj002,01,CX,S2/BG2-.1-3(E2/TH)
	state after	(2, 1, 0, 1, 4, 4, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1173 *This game had 2 errors. The final score was 6-9. In the bottom of the 4th inning, with a runner on 1st, MIA batter Joey Wendle hit a soft bunt single to very shallow infield. A throwing error by the the catcher allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	KCA vs MIA on 20230605
	state before	(2, 0, 0, 0, 5, 4, 6, 1)
36.	error	play,5,1,gurry001,12,BFCX,E8/F89.B-2
	state after	(2, 0, 1, 0, 5, 4, 6, 1)
	fielder	8 (center field on the visiting team)

Summary 1174 *In the bottom of the 5th inning, with bases empty, MIA batter Yulieski Gurriel hit a fly ball to right center field. An error by the center fielder allowed the batter to reach 2nd base.*

Visiting/Home team rosters for KCA vs MIA on 2023-06-06

position	innings	player	position	innings	player
1	1	Zack Greinke	1	1	Jesus Luzardo
1	5	Jackson Kowar	1	8	Tanner Scott
1	7	Jose Cuas	1	9	Bryan Hoeing
1	8	Nick Wittgren	2	1	Nick Fortes
2	1	Salvador Perez	3	1	Garrett Cooper
3	1	Nick Pratto	4	1	Luis Arraez
4	1	Matt Duffy	5	1	Jean Segura
5	1	Maikel Garcia	6	1	Joey Wendle
6	1	Bobby Witt Jr.	7	1	Bryan De La Cruz
7	1	Edward Olivares	8	1	Jonathan Davis
8	1	Drew Waters	9	1	Jesus Sanchez
9	1	MJ Melendez			

	game	KCA vs MIA on 20230606
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
37.	error	play,1,1,solej001,31,CBBBX,E3/TH/G34.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1175 This game had 3 errors. The final score was 1-6.
In the bottom of the 1st inning, with bases empty, MIA batter Jorge Soler hit a ground ball to 1st/2nd. A throwing error by the first baseman allowed the batter to reach 1st base.

	game	KCA vs MIA on 20230606
	state before	(0, 1, 0, 0, 5, 0, 1, 1)
38.	error	play,5,1,seguj002,22,>FB*BFX,S7/L7S+.1-3(E7);B-2
	state after	(0, 0, 1, 1, 5, 0, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 1176 *In the bottom of the 5th inning, with a runner on 1st, MIA batter Jean Segura hit a hard line drive single to shallow left field. A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	KCA vs MIA on 20230606
	state before	(2, 1, 1, 1, 5, 0, 2, 1)
39.	error	play,5,1,delab001,00,X,D9/L9D+.3-H;2-H;1-H;B-3(E4/TH)
	state after	(2, 0, 0, 1, 5, 0, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1177 *In the bottom of the 5th inning, with bases loaded, MIA batter Bryan De La Cruz hit a hard line drive double to deep right field. The runners on 1st, 2nd, 3rd each scored (3 RBIs). A throwing error by the second baseman allowed the batter to reach 3rd.*

	game	TOR vs MIA on 20230619
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
40.	error	play,1,1,solej001,21,SBBX,FC5/G6S.1-2(E5);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1178 *In the bottom of the 1st inning, with a runner on 1st, MIA batter Jorge Soler hit a ground ball into a fielder's choice at short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	TOR vs MIA on 20230620
	state before	(1, 0, 0, 0, 5, 0, 0, 0)
41.	error	play,5,0,chapm001,10,BX,D8/L89XD+.B-3(E8)
	state after	(1, 0, 0, 1, 5, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1179 *In the top of the 5th inning, with bases empty, TOR batter Matt Chapman hit a hard line drive double to extra deep right center field. An error by the center fielder allowed the batter to reach 3rd.*

	game	PIT vs MIA on 20230622
	state before	(2, 0, 1, 0, 4, 1, 1, 0)
42.	error	play,4,0,castr006,12,BFS.>*B,SB3.2-H(E2/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 4, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1180 *This game had 2 errors. The final score was 4-6. In the top of the 4th inning, with a runner on 2nd, PIT batter Rodolfo Castro had a 1-2 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to ascore (unearned, no RBI). Castro is still at bat, with a 2-2 count.*

	game	PIT vs MIA on 20230622
	state before	(1, 0, 0, 0, 9, 4, 6, 0)
43.	error	play,9,0,castr006,12,CBCX,E6/G6.B-1
	state after	(1, 1, 0, 0, 9, 4, 6, 0)
	fielder	6 (shortstop on the home team)

Summary 1181 *In the top of the 9th inning, with bases empty, PIT batter Rodolfo Castro hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs MIA on 20230623
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
44.	error	play,1,0,joe-c001,00,X,S7/L7S+.B-2(E7)
	state after	(2, 0, 1, 0, 1, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1182 This game had 2 errors. The final score was 3-1.

In the top of the 1st inning, with bases empty, PIT batter Connor Joe hit a hard line drive single to shallow left field. An error by the left fielder allowed the batter to reach 2nd.

	game	PIT vs MIA on 20230623
	state before	(1, 1, 0, 0, 4, 0, 1, 1)
45.	error	play,4,1,wendj002,11,1BFX,FC5/G56.1X2(5E6);B-1
	state after	(1, 1, 1, 0, 4, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1183 *In the bottom of the 4th inning, with a runner on 1st, MIA batter Joey Wendle hit a ground ball into a fielder's choice at 3rd. An error by the shortstop (with an assist from the third baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

18.2.4 July

	game	SLN vs MIA on 20230704
	state before	(1, 1, 1, 1, 4, 0, 4, 1)
46.	error	play,4,1,delab001,31,.BFBBX,FC1/G1S.3-H(E1)(NR)(UR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 4, 0, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1184 *In the bottom of the 4th inning, with bases loaded, MIA batter Bryan De La Cruz hit a ground ball into a fielder's choice at shallow center infield. An error by the pitcher allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	SLN vs MIA on 20230705
	state before	(1, 1, 1, 0, 9, 9, 8, 1)
47.	error	play,9,1,wendj002,12,.SSB>X,FC1/G1S-.2-H;1-H(E1/TH) (NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 9, 9, 10, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1185 *In the bottom of the 9th inning, with runners on 1st and 2nd, MIA batter Joey Wendle hit a ground ball into a fielder's choice at shallow center infield. The runner on 2nd scored (1 RBI). An error by the pitcher allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	PHI vs MIA on 20230707
	state before	(2, 0, 0, 0, 4, 0, 2, 0)
48.	error	play,4,0,realj001,00,X,C/E2.B-1
	state after	(2, 1, 0, 0, 4, 0, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1186 *In the top of the 4th inning, with bases empty, PHI batter J.T. Realmuto had a 0-0 count. Catcher's interference allowed the batter to take 1st.*

	game	PHI vs MIA on 20230708
	state before	(1, 1, 0, 0, 3, 3, 3, 0)
49.	error	play,3,0,realj001,01,*SX,E5/G56S.1-2;B-1
	state after	(1, 1, 1, 0, 3, 3, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1187 *This game had 2 errors. The final score was 3-5. In the top of the 3rd inning, with a runner on 1st, PHI batter J.T. Realmuto hit a ground ball to shallow 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	PHI vs MIA on 20230708
	state before	(1, 1, 0, 0, 4, 3, 3, 1)
50.	error	play,4,1,hampg001,00,X,E1/BP25.1-2;B-1
	state after	(1, 1, 1, 0, 4, 3, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1188 *In the bottom of the 4th inning, with a runner on 1st, MIA batter Garrett Hampson hit a bunt pop-up to shallow left infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	PHI vs MIA on 20230709
	state before	(2, 0, 0, 0, 3, 0, 4, 1)
51.	error	play,3,1,sancj003,11,BSX,E3/G3.B-1
	state after	(2, 1, 0, 0, 3, 0, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1189 *In the bottom of the 3rd inning, with bases empty, MIA batter Jesus Sanchez hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	COL vs MIA on 20230721
	state before	(0, 0, 0, 0, 2, 2, 0, 0)
52.	error	play,2,0,gricr001,10,BX,S9/F9L.B-2(E9)
	state after	(0, 0, 1, 0, 2, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 1190 *In the top of the 2nd inning, with bases empty, COL batter Randal Grichuk hit a fly ball single to right field. An error by the right fielder allowed the batter to reach 2nd.*

	game	DET vs MIA on 20230728
	state before	(0, 0, 1, 0, 2, 0, 2, 1)
53.	error	play,2,1,wendj002,22,CFBFBX,E4/G34.2-3;B-1
	state after	(0, 1, 0, 1, 2, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1191 This game had 2 errors. The final score was 5-6.
In the bottom of the 2nd inning, with a runner on 2nd, MIA batter Joey Wendle hit a ground ball to 1st/2nd. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	DET vs MIA on 20230728
	state before	(2, 1, 0, 0, 4, 0, 4, 0)
54.	error	play,4,0,cabrm001,12,C1C.BX,E5/G56.1-H(NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 4, 1, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1192 *In the top of the 4th inning, with a runner on 1st, DET batter Miguel Cabrera hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	DET vs MIA on 20230729
	state before	(1, 0, 1, 0, 7, 4, 0, 0)
55.	error	play,7,0,vierm001,11,.BCX,S6/G6.2-3(E6/TH)
	state after	(1, 1, 0, 1, 7, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1193 *In the top of the 7th inning, with a runner on 2nd, DET batter Matt Vierling hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to safely advance to 3rd. The batter took 1st.*

Visiting/Home team rosters for DET vs MIA on 2023-07-30

position	innings	player	position	innings	player
1	1	Tarik Skubal	1	1	Jesus Luzardo
1	5	Brendan White	1	5	George Soriano
1	7	Tyler Holton	1	7	A.J. Puk
1	8	Jose Cisnero	1	7	Jorge Lopez
2	1	Jake Rogers	1	8	Tanner Scott
3	1	Spencer Torkelson	1	9	David Robertson
4	1	Andy Ibanez	2	1	Nick Fortes
4	6	Zack Short	3	1	Yuli Gurriel
5	1	Zack Short	4	1	Luis Arraez
5	6	Zach McKinstry	5	1	Jean Segura
6	1	Javier Baez	6	1	Jon Berti
7	1	Matt Vierling	6	8	Joey Wendle
8	1	Riley Greene	7	1	Bryan De La Cruz
9	1	Kerry Carpenter	8	1	Garrett Hampson
			9	1	Avisail Garcia

	game	DET vs MIA on 20230730
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
56.	error	play,2,0,baezj001,11,CBX,FC4/G34.1-2(E4);B-1
	state after	(0, 1, 1, 0, 2, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1194 This game had 4 errors. The final score was 6-8.

In the top of the 2nd inning, with a runner on 1st, DET batter Javier Baez hit a ground ball into a fielder's choice at 1st/2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	DET vs MIA on 20230730
	state before	(0, 0, 1, 1, 3, 0, 0, 0)
57.	error	play,3,0,torks001,12,F*BFX,E6/G6.3-H(NR);2-3;B-2
	state after	(0, 0, 1, 1, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1195 *In the top of the 3rd inning, with runners on 2nd and 3rd, DET batter Spencer Torkelson hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd and the batter to reach 2nd.*

	game	DET vs MIA on 20230730
	state before	(1, 0, 0, 0, 6, 4, 4, 1)
58.	error	play,6,1,bertj001,32,BB.CFBX,E5/TH/G2.B-1
	state after	(1, 1, 0, 0, 6, 4, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1196 *In the bottom of the 6th inning, with bases empty, MIA batter Jon Berti hit a ground ball to very shallow infield. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	DET vs MIA on 20230730
	state before	(1, 1, 0, 0, 6, 4, 4, 1)
59.	error	play,6,1,garca003,12,SF*B>B,SB2.1X3(25E4)
	state after	(1, 0, 0, 1, 6, 4, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1197 This play is an example of a *consecutive error*, since Garcia bats right after Berti (in the previous error) in the MIA line-up for that day.

In the bottom of the 6th inning, with a runner on 1st, MIA batter Avisail Garcia had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. An error by the second baseman (with an assist from the catcher then the third baseman)³ allowed that runner to

³Possibly this was an attempted squeeze play between 2nd and 3rd.

safely advance to 3rd. Garcia is still at bat, with a 2-2 count.

	game	PHI vs MIA on 20230731
	state before	(2, 1, 0, 0, 8, 3, 2, 0)
60.	error	play,8,0,schw001,11,>FB1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 8, 3, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1198 *In the top of the 8th inning, with a runner on 1st, PHI batter Kyle Schwarber has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 2nd. Schwarber is still at-bat with a 1-1 count.*

18.2.5 August

	game	NYA vs MIA on 20230813
	state before	(1, 1, 1, 1, 9, 7, 3, 1)
61.	error	play,9,1,bellj005,20,BBX,E1/TH/G1.3-H;2-H(NR);1-3;B-1
	state after	(1, 1, 0, 1, 9, 7, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1199 *In the bottom of the 9th inning, with bases loaded, MIA batter Josh Bell hit a ground ball to the pitcher. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the runner on 2nd to also score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

Visiting/Home team rosters for HOU vs MIA on 2023-08-15

position	innings	player	position	innings	player
1	1	Cristian Javier	1	1	Johnny Cueto
1	5	Rafael Montero	1	6	George Soriano
1	6	Hunter Brown	1	7	A.J. Puk
1	8	Bryan Abreu	1	8	Steven Okert
1	9	Ryan Pressly	2	1	Jacob Stallings
2	1	Yainer Diaz	3	1	Josh Bell
3	1	Jon Singleton	4	1	Luis Arraez
4	1	Jose Altuve	5	1	Jake Burger
4	1	Mauricio Dubon	6	1	Joey Wendle
5	1	Alex Bregman	7	1	Bryan De La Cruz
6	1	Jeremy Pena	8	1	Jazz Chisholm Jr.
7	1	Chas McCormick	9	1	Jesus Sanchez
8	1	Jake Meyers			
9	1	Kyle Tucker			

	game	HOU vs MIA on 20230815
	state before	(2, 0, 1, 0, 5, 3, 4, 0)
62.	error	play,5,0,tuckk001,00,X,E8/F78XD.2-H(NR)(UR);B-3
	state after	(2, 0, 0, 1, 5, 4, 4, 0)
	fielder	8 (center field on the home team)

Summary 1200 This game had 3 errors, including a 2 errors in 1 play. The final score was 6-5.

In the top of the 5th inning, with a runner on 2nd, HOU batter Kyle Tucker hit a fly ball to extra deep left center field. An error by the center fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 3rd.

	game	HOU vs MIA on 20230815
	state before	(2, 1, 0, 0, 5, 4, 4, 1)
63.	error	play,5,1,bellj005,12,FBS1X,E4/TH/G4.1-H(E1/TH) (NR) (UR);B-2
	state after	(2, 0, 1, 0, 5, 4, 5, 1)
	fielder	4,1 (2nd baseman,pitcher on the visiting team)

Summary 1201 This play had 2 errors by different fielders.

In the bottom of the 5th inning, with a runner on 1st, MIA batter Josh Bell hit a ground ball to 2nd base. A throwing error by the second baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.

	game	HOU vs MIA on 20230816
	state before	(0, 0, 0, 0, 3, 5, 2, 0)
64.	error	play,3,0,alvay001,11,BCX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 3, 5, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1202 This game had 2 errors. The final score was 12-5.

In the top of the 3rd inning, with bases empty, HOU batter Yordan Alvarez hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	HOU vs MIA on 20230816
	state before	(1, 1, 1, 1, 4, 6, 2, 1)
65.	error	play,4,1,bertj001,02,FC.X,46(1)/FO/G4.3-H;2-H(E6/TH) (NR) (UR);B-1
	state after	(2, 1, 0, 0, 4, 6, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1203 *In the bottom of the 4th inning, with bases loaded, MIA batter Jon Berti hit a ground ball to 2nd. At 2nd, the shortstop (with an assist from the second baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 1st.*

66.	game	WAS vs MIA on 20230826
	state before	(2, 0, 1, 0, 4, 0, 0, 0)
	error	play,4,0,smitd008,00,X,S6/G4.2-H(E6/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 4, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1204 *In the top of the 4th inning, with a runner on 2nd, WAS batter Dominic Smith hit a ground ball single to 2nd. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

67.	game	TBA vs MIA on 20230829
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
	error	play,1,0,arozr001,32,CBBSB>F1>FX,8/F8XD.1-3(E8)
	state after	(2, 0, 0, 1, 1, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1205 *In the top of the 1st inning, with a runner on 1st, TBA batter Randy Arozarena hit a fly ball to extra deep center. The batter was put out by the center fielder. An error by the center fielder allowed the runner on 1st to advance to 3rd.*

18.2.6 September

Visiting/Home team rosters for LAN vs MIA on 2023-09-06

position	innings	player	position	innings	player
1	1	Lance Lynn	1	1	JT Chargois
1	5	Victor Gonzalez	1	2	Steven Okert
1	7	Gus Varland	1	4	A.J. Puk
2	1	Will Smith	1	5	Edward Cabrera
3	1	Freddie Freeman	1	9	Bryan Hoeing
3	6	Michael Busch	2	1	Jacob Stallings
4	1	Mookie Betts	3	1	Luis Arraez
4	6	Amed Rosario	3	8	Yuli Gurriel
5	1	Max Muncy	4	1	Xavier Edwards
6	1	Miguel Rojas	5	1	Jake Burger
7	1	David Peralta	5	8	Jon Berti
8	1	James Outman	6	1	Joey Wendle
9	1	Jason Heyward	7	1	Bryan De La Cruz
			7	9	Dane Myers
			8	1	Jazz Chisholm Jr.
			8	8	Garrett Hampson
			9	1	Jesus Sanchez

	game	LAN vs MIA on 20230906
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
68.	error	play,1,1,bellj005,12,BCC.X,C/E2.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1206 This game had 3 errors. The final score was 4-11.
In the bottom of the 1st inning, with bases empty, MIA batter Josh Bell had a 1-2 count. Catcher's interference allowed the batter to take 1st.

	game	LAN vs MIA on 20230906
	state before	(0, 0, 0, 0, 4, 1, 0, 1)
69.	error	play,4,1,burgj001,01,CX,E5/G6S.B-1
	state after	(0, 1, 0, 0, 4, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1207 *In the bottom of the 4th inning, with bases empty, MIA batter Jake Burger hit a ground ball to shallow short. An error by the third baseman allowed the batter to reach 1st base.*

	game	LAN vs MIA on 20230906
	state before	(0, 0, 1, 0, 9, 1, 11, 0)
70.	error	play,9,0,heywj001,01,CX,E6/G6.2-H(NR);B-1
	state after	(0, 1, 0, 0, 9, 2, 11, 0)
	fielder	6 (shortstop on the home team)

Summary 1208 *In the top of the 9th inning, with a runner on 2nd, LAN batter Jason Heyward hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	LAN vs MIA on 20230907
	state before	(1, 1, 1, 0, 4, 1, 0, 0)
71.	error	play,4,0,barna001,22,FBFFBF2,P02(E1/TH).2-3;1-2
	state after	(1, 0, 1, 1, 4, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1209 This game had 2 errors. The final score was 9-0. *In the top of the 4th inning, with runners on 1st and 2nd, LAN batter Austin Barnes has a 2-2 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each safely advance one base. Barnes is still at-bat with a 2-2 count.*

	game	LAN vs MIA on 20230907
	state before	(1, 0, 1, 0, 6, 3, 0, 0)
72.	error	play,6,0,smitw003,22,BFFFBX,S5/G56.2-3(E5/TH);B-2
	state after	(1, 0, 1, 1, 6, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1210 *In the top of the 6th inning, with a runner on 2nd, LAN batter Will Smith hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd.*

	game	ATL vs MIA on 20230915
	state before	(2, 0, 0, 1, 2, 0, 3, 1)
73.	error	play,2,1,bellj005,22,BS*BFX,C/E2.B-1
	state after	(2, 1, 0, 1, 2, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1211 *In the bottom of the 2nd inning, with a runner on 3rd, MIA batter Josh Bell had a 2-2 count. Catcher's interference allowed the batter to take 1st.*

	game	ATL vs MIA on 20230917
	state before	(1, 0, 0, 0, 4, 0, 5, 1)
74.	error	play,4,1,fortn001,11,CBX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 4, 0, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1212 *In the bottom of the 4th inning, with bases empty, MIA batter Nick Fortes hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	NYN vs MIA on 20230918
	state before	(0, 0, 0, 0, 6, 1, 1, 0)
75.	error	play,6,0,nimmb001,22,CFBBX,E5/G7S.B-1
	state after	(0, 1, 0, 0, 6, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1213 This game had 2 errors. The final score was 2-1.

In the top of the 6th inning, with bases empty, NYN batter Brandon Nimmo hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.

	game	NYN vs MIA on 20230918
	state before	(1, 1, 0, 0, 9, 2, 1, 0)
76.	error	play,9,0,vienm001,22,SBBSF.X,E5/G5.1-2;B-1
	state after	(1, 1, 1, 0, 9, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1214 *In the top of the 9th inning, with a runner on 1st, NYN batter Mark Vientos hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Joey Lucchesi	1	1	Braxton Garrett
1	6	Reed Garrett	1	7	A.J. Puk
1	9	Trevor Gott	1	7	Andrew Nardi
2	1	Francisco Alvarez	1	9	Tanner Scott
3	1	Pete Alonso	2	1	Nick Fortes
4	1	Jeff McNeil	3	1	Josh Bell
5	1	Ronny Mauricio	3	8	Yuli Gurriel
6	1	Francisco Lindor	4	1	Xavier Edwards
7	1	Tim Locastro	5	1	Jake Burger
8	1	Brandon Nimmo	6	1	Garrett Hampson
9	1	DJ Stewart	7	1	Jon Berti
			7	9	Bryan De La Cruz
			8	1	Jazz Chisholm Jr.
			9	1	Bryan De La Cruz
			9	9	Jesus Sanchez

	game	NYN vs MIA on 20230919
	state before	(2, 0, 0, 0, 3, 0, 0, 0)
77.	error	play,3,0,alonp001,21,SBBX,S7/L7L+.B-2(E7)
	state after	(2, 0, 1, 0, 3, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1215 This game had 3 errors. The final score was 3-4.
In the top of the 3rd inning, with bases empty, NYN batter Pete Alonso hit a hard line drive single to left field. An error by the left fielder allowed the batter to reach 2nd.

	game	NYN vs MIA on 20230919
	state before	(2, 0, 1, 0, 3, 0, 0, 0)
78.	error	play,3,0,lindf001,10,BX,S1/G25-.2-H(E1/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 3, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1216 *In the top of the 3rd inning, with a runner on 2nd, NYN batter Francisco Lindor hit a soft ground single to shallow left infield. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	NYN vs MIA on 20230919
	state before	(0, 1, 1, 0, 5, 1, 1, 1)
79.	error	play,5,1,edwax001,01,LX,S1/BG25-.2-H(E1/TH) (NR) (UR) ;1X3(75) ;B-2
	state after	(1, 0, 1, 0, 5, 1, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1217 *In the bottom of the 5th inning, with runners on 1st and 2nd, MIA batter Xavier Edwards hit a soft bunt single to shallow left infield. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI). The runner on 1st advanced to 2nd but was tagged out by the third baseman (with an assist from the left fielder) when he attempted to reach 3rd. The batter reached 2nd.*

	game	NYN vs MIA on 20230920
	state before	(0, 1, 0, 0, 3, 1, 0, 0)
80.	error	play,3,0,maurr002,21,B*BFX,S9/L9S.1-H(E9/TH) (NR) ;B-3(E5/TH)
	state after	(0, 0, 0, 1, 3, 2, 0, 0)
	fielder	9,5 (right field,3rd baseman on the home team)

Summary 1218 This play had 2 errors by different fielders.
In the top of the 3rd inning, with a runner on 1st, NYN batter Ronny

Mauricio hit a line drive single to shallow right field. A throwing error by the right fielder allowed the runner on 1st to score (unearned, no RBI). A throwing error by the third baseman allowed the batter to reach 1st.

	game	MIL vs MIA on 20230922
	state before	(2, 0, 0, 0, 8, 16, 1, 1)
81.	error	play,8,1,edwax001,00,X,E3/G34.B-1
	state after	(2, 1, 0, 0, 8, 16, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1219 *In the bottom of the 8th inning, with bases empty, MIA batter Xavier Edwards hit a ground ball to 1st/2nd. An error by the first baseman allowed the batter to reach 1st base.*

	game	MIL vs MIA on 20230923
	state before	(1, 1, 0, 0, 2, 0, 3, 1)
82.	error	play,2,1,fortn001,00,1X,E5/G5.1XH(62);B-1
	state after	(2, 1, 0, 0, 2, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1220 *This game had 2 errors. The final score was 4-5. In the bottom of the 2nd inning, with a runner on 1st, MIA batter Nick Fortes hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st. However, when the runner from 1st (now on 3rd) attempted to score he was tagged out by the catcher (with an assist from the third baseman).*

	game	MIL vs MIA on 20230923
	state before	(0, 0, 1, 1, 6, 0, 4, 0)
83.	error	play,6,0,contw002,12,BSFX,E3/G34.3-H;2-3;B-1
	state after	(0, 1, 0, 1, 6, 1, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1221 *In the top of the 6th inning, with runners on 2nd and 3rd, MIL batter William Contreras hit a ground ball to 1st/2nd. An error by the first baseman allowed the runner on 3rd to score (1 RBI), the runner on 2nd to advance to 3rd and the batter to reach 1st.*

Chapter 19

Milwaukee Brewers (MIL)

The Milwaukee Brewers started playing in the American League East in 1972. However, in 1998 the Brewers joined the *National* League Central¹. While they've never won an NL championship, in each of the last 5 years, the Brewers have won over 50% of their games.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Milwaukee against the Brewers. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pickoff.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 MIL team is 0.719, the highest in the league. If we compare this with the MLB average² of 0.691 that year [BR], we see that it is more than 2 standard deviations above average. So, the Brewers do indeed have a high DER.

As we will see in the next section, that year there were 81 errors (by both teams) in the 81 home games, 39 by the Brewers and the rest (42) by visiting teams. The total number of errors by the team (home and away) over the regular season of 2023 was 77. The Brewers are below average (which is good), but only by a half of a standard deviation.

¹The Houston Astros is another team to play in both leagues.

²The standard deviation of these DER values is $\sigma = 0.011$.

19.1 Season totals

An error is a statistic charged against a fielder whose action has assisted the team on offense, as set forth in this Rule 9.12.

– **MLB Rule Book 9.12**

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

19.1.1 Run conversion and error total by position

This section tabulates error-to-run rates and error totals by position.

Runs and errors	totals
Errors by vt:	42
Runs by ht because of errors by vt:	20
Errors by ht:	39
Runs by vt because of errors by ht:	7

This is an excellent error-to-run conversion rate: $20/42 = 47.6\%$ on the part of the Brewers (compared to $7/39 = 17.9\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 MIL season (home and away) has about $R/PA = 728/6084 = 11.9\%$ chance of resulting in a run. For MIL 2023, a randomly selected error is almost 3 times as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	4	6
to catcher:	9	4
to 1st baseman:	3	6
to 2nd baseman:	3	5
to 3rd baseman:	9	5
to shortstop:	5	6
to left fielder:	3	1
to center fielder:	1	2
to right fielder:	4	1

19.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	13	10	23
(lone) runner on 1st	10	18	28
(lone) runner on 2nd	6	4	10
(lone) runner on 3rd	0	1	1
two runners on base	12	6	18
bases loaded	1	0	1
total	42	39	81

A table of frequencies of most common errors by the Brewers in MIL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	5	12.8 %
(1, 0, 0, 0)	4	10.3 %
(2, 0, 0, 0)	1	2.6 %
(0, 1, 0, 0)	4	10.3 %
(1, 1, 0, 0)	8	20.5 %
(2, 1, 0, 0)	6	15.4 %
(0, 0, 1, 0)	3	7.7 %
(1, 1, 1, 0)	4	10.3 %
(2, 1, 0, 1)	1	2.6 %

A table of frequencies of most common errors by a visiting team in MIL 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	14.3%
(1, 0, 0, 0)	3	7.1%
(2, 0, 0, 0)	4	9.5%
(0, 1, 0, 0)	1	2.4%
(1, 1, 0, 0)	4	9.5%
(2, 1, 0, 0)	5	11.9%
(1, 0, 1, 0)	3	7.1%
(0, 1, 1, 0)	2	4.8%
(1, 1, 1, 0)	2	4.8%
(2, 1, 1, 0)	3	7.1%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Brewers homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `MIL2023.evn`):

state	MIL2023 percentages
(0, 0, 0, 0)	24.2%
(1, 0, 0, 0)	16.0%
(2, 0, 0, 0)	13.1%
(0, 1, 0, 0)	6.2%
(1, 1, 0, 0)	6.8%
(2, 1, 0, 0)	7.4%
(0, 0, 1, 0)	2.1%
(1, 0, 1, 0)	3.0%
(2, 0, 1, 0)	3.6%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	01.0%
(2, 0, 0, 1)	1.6%
(0, 1, 1, 0)	2.0%
(1, 1, 1, 0)	2.4%
(2, 1, 1, 0)	2.4%

19.1.3 Some interesting errors

Some of the most interesting error plays in the MIL 2023 home games:

- The game KCA vs MIL on 2023-05-14 had 3 errors including a *double error* (2 errors in 1 play). The game played the day before had an ∞ play with did not change the game state.
- The game SDN vs MIL on 2023-08-26 had 3 errors including a double steal and a consecutive error.

19.1.4 Relative team rankings

We have 81 home games between MIL and some visiting MLB team (vt). A number of teams did not play the Brewers in the regular season. However, of those that did, MIL *lost* more games than won against³:

ARI (2/3), ATL (2/3), BOS (2/3), DET (2/3), LAN (2/3),
OAK (3/3), SFN (3/4),

The teams that *split* their series were:

(none)

The teams that MIL *won* more than lost against

MIN (0/2), KCA (0/3), NYN (0/3), SDN (0/3), ANA (1/3), BAL (1/3), COL
(1/3), HOU (1/3),
PHI (1/3), WAS (1/3), MIA (1/4), CIN (2/6), SLN (2/6), PIT (2/7), CHN (3/7)

The teams that did not play in any of the 81 home games in MIL 2023 were:

CHA, CLE, NYA, SEA, TBA, TEX, TOR .

19.2 Game-by-game totals

The Official Scorer shall not charge an error if the pitcher fails to cover first base on a play, thereby allowing a batter-runner to reach first base safely.

– **MLB Rule Book** 9.12(a)(1) Comment

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for MIL 2023 is as follows [BA].

³We recorded that vt's W/T record in parentheses (W = numbers of games vt won against MIL, T = Total home games vt played vs the Brewers at Milwaukee.

position	player	games played	OBP
1 (C)	William Contreras	109	.367
3 (1B)	Rowdy Tellez	76	.291
4 (2B)	Brice Turang	120	.285
5 (3B)	Brian Anderson	70	.310
6 (SS)	Willy Adames	140	.310
7 (LF)	Christian Yelich	122	.370
8 (CF)	Joey Wiemer	121	.283
9 (RF)	Tyrone Taylor	61	.267

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 MIL home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

19.2.1 April

	game	SLN vs MIL on 20230408
	state before	(2, 1, 0, 0, 9, 6, 0, 0)
1.	error	play,9,0,oneit001,10,BX,S5/G5.1-3(E5/TH)
	state after	(2, 1, 0, 1, 9, 6, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1222 *In the top of the 9th inning, with a runner on 1st, SLN batter Tyler O'Neill hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	SLN vs MIL on 20230409
	state before	(2, 1, 0, 0, 3, 0, 0, 0)
2.	error	play,3,0,burla001,32,FTB*BFFFFB>F>FX,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 3, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1223 *In the top of the 3rd inning, with a runner on 1st, SLN batter Alec Burleson had a full count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to take 1st base.*

	game	BOS vs MIL on 20230423
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
3.	error	play,1,0,turnj001,32,CC1*BBF*B1X,S9/L89S.1-3(E9)
	state after	(1, 1, 0, 1, 1, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 1224 *In the top of the 1st inning, with a runner on 1st, BOS batter Justin Turner hit a line drive single to shallow right center field. An error by the right window allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	DET vs MIL on 20230426
	state before	(2, 1, 1, 0, 2, 0, 3, 1)
4.	error	play,2,1,adamw002,00,X,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 2, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1225 *In the bottom of the 2nd inning, with runners on 1st and 2nd, MIL batter Willy Adames had a 0-0 count. Catcher's interference allowed each runner to advance one base and the batter to take 1st. The bases are loaded.*

	game	ANA vs MIL on 20230428
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
5.	error	play,1,0,ohtas001,10,BX,36(1)/F0/G3.B-2(E6/TH)
	state after	(2, 0, 1, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1226 This game had 2 errors. The final score was 1-2.

In the top of the 1st inning, with a runner on 1st, ANA batter Shohei Ohtani hit a ground ball to 1st. The shortstop (with an assist from the first baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	ANA vs MIL on 20230428
	state before	(0, 0, 0, 0, 8, 1, 1, 0)
6.	error	play,8,0,wardt002,01,.CX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 8, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1227 *In the top of the 8th inning, with bases empty, ANA batter Taylor Ward hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs MIL on 20230429
	state before	(0, 1, 1, 0, 5, 1, 3, 1)
7.	error	play,5,1,yelic001,21,B2*BCX,FC5/G56.2-3;1-2(E5);B-1
	state after	(0, 1, 1, 1, 5, 1, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1228 *In the bottom of the 5th inning, with runners on 1st and 2nd, MIL batter Christian Yelich hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

19.2.2 May

	game	LAN vs MIL on 20230508
	state before	(0, 0, 0, 0, 5, 0, 0, 1)
8.	error	play,5,1,contw002,01,CX,E5/G6.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1229 *In the bottom of the 5th inning, with bases empty, MIL batter William Contreras hit a ground ball to shortstop. An error by the third baseman allowed the batter to reach 1st base.*

	game	LAN vs MIL on 20230509
	state before	(0, 0, 1, 0, 1, 1, 0, 0)
9.	error	play,1,0,smitw003,30,>B.BB2,P02(E1/TH).2-3
	state after	(0, 0, 0, 1, 1, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1230 *This game had 2 errors. The final score was 6-2. In the top of the 1st inning, with a runner on 2nd, LAN batter Will Smith has a 3-0 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed that runner to safely reach 3rd. Smith is still at bat with a 3-0 count.*

	game	LAN vs MIL on 20230509
	state before	(1, 0, 0, 0, 9, 6, 2, 1)
10.	error	play,9,1,andeb006,12,CBFFX,E4/TH/G4.B-1
	state after	(1, 1, 0, 0, 9, 6, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1231 *In the bottom of the 9th inning, with bases empty, MIL batter Brian Anderson hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	LAN vs MIL on 20230510
	state before	(1, 1, 0, 0, 5, 2, 1, 0)
11.	error	play,5,0,thomt002,22,1BFBCF>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1232 *In the top of the 5th inning, with a runner on 1st, LAN batter Trayce Thompson had a 2-2 count. During the next pitch, which was a balll, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Thompson is still at bat, now with a full count.*

	game	KCA vs MIL on 20230512
	state before	(2, 1, 0, 0, 1, 0, 0, 1)
12.	error	play,1,1,yelic001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1233 *This game had 2 errors. The final score was 1-5. In the bottom of the 1st inning, with a runner on 1st, MIL batter Christian Yelich had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Christian Yelich is still at bat, with a 0-1 count.*

	game	KCA vs MIL on 20230512
	state before	(0, 0, 1, 1, 7, 1, 3, 1)
13.	error	play,7,1,adamw002,11,B.FX,C/E2.B-1
	state after	(0, 1, 1, 1, 7, 1, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1234 *In the bottom of the 7th inning, with runners on 2nd and 3rd, MIL batter Willy Adames had a 1-1 count. Catcher's interference allowed the batter to reach 1st base.. The bases are loaded.*

	game	KCA vs MIL on 20230513
	state before	(2, 1, 0, 0, 4, 2, 2, 0)
14.	error	play,4,0,wittb002,01,CF,FLE3
	state after	(2, 1, 0, 0, 4, 2, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1235 This play is an ∞ play. The game state does not change. Repeat this play infinitely many times and the game will never end!

In the top of the 4th inning, with a runner on 1st, KCA batter Bobby Witt hit a pop fly foul of the 1st base line. An error by the first baseman allowed the batter to remain safe. Witt is still at bat, not with a 0-2 count.

Visiting/Home team rosters for KCA vs MIL on 2023-05-14

position	innings	player	position	innings	player
1	1	Jordan Lyles	1	1	Colin Rea
1	3	Jose Cuas	1	4	Eric Lauer
1	5	Josh Staumont	2	1	Victor Caratini
1	6	Carlos Hernandez	3	1	Rowdy Tellez
1	7	Amir Garrett	4	1	Brice Turang
1	8	Scott Barlow	5	1	Owen Miller
2	1	Salvador Perez	6	1	Willy Adames
3	1	Vinnie Pasquantino	7	1	Christian Yelich
4	1	Michael Massey	8	1	Joey Wiemer
5	1	Hunter Dozier	9	1	Tyrone Taylor
6	1	Maikel Garcia			
7	1	Nick Pratto			
8	1	Nate Eaton			
9	1	MJ Melendez			

	game	KCA vs MIL on 20230514
	state before	(0, 1, 1, 0, 3, 3, 1, 0)
15.	error	play,3,0,garc003,00,X,6/L6.2-3(E6/TH)
	state after	(1, 1, 0, 1, 3, 3, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1236 This game had 3 errors. The final score was 6-9.

In the top of the 3rd inning, with runners on 1st and 2nd, KCA batter

Maikel Garcia hit a line drive to shortstop. The shortstop put out the batter. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd.

16.	game	KCA vs MIL on 20230514
	state before	(0, 1, 0, 1, 3, 4, 2, 1)
	error	play,3,1,winkj002,01,1C>B,SB2.3-H(E6) (NR);1-H(E8/TH) (NR)
	state after	(0, 0, 0, 0, 3, 4, 4, 1)
	fielder	6,8 (shortstop, center field on the visiting team)

Summary 1237 This play had 2 errors by 2 different fielders.

In the bottom of the 3rd inning, with runners on 1st and 3rd, MIL batter Jesse Winker had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI). A throwing error by the center fielder allowed the runner from 1st (now on 2nd) to score (unearned, no RBI). The bases are empty. Winker is still at bat, now with a 1-1 count.

17.	game	HOU vs MIL on 20230522
	state before	(0, 0, 0, 0, 8, 12, 2, 1)
	error	play,8,1,winkj002,32,...CFBBFBF,FLE2
	state after	(0, 0, 0, 0, 8, 12, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1238 This is an ∞ play. There is no change to the game state.

In the bottom of the 8th inning, with no outs bases empty, MIL batter Jesse Winker had a full count when he hit a pop fly behind home plate in foul territory. An error by the catcher allowed the batter to extend his at bat. Winker is still at bat with a full count.

	game	HOU vs MIL on 20230523
	state before	(2, 0, 1, 0, 4, 0, 1, 1)
18.	error	play,4,1,carav001,22,B>B.CC+2,P02(E2/TH).2-H(NR)(UR)
	state after	(2, 1, 0, 0, 4, 0, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1239 *In the bottom of the 4th inning, with a runner on 2nd, MIL batter Victor Caratini has a 2-2 count. Trying to pickoff the runner on 2nd, a throwing error by the catcher allowed that runner to score (unearned, no RBI). Caratini is still at bat with a 2-2 count.*

	game	HOU vs MIL on 20230524
	state before	(1, 0, 0, 0, 6, 0, 2, 0)
19.	error	play,6,0,dubom001,02,CFX,E3/G3.B-1
	state after	(1, 1, 0, 0, 6, 0, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1240 *In the top of the 6th inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for SFN vs MIL on 2023-05-26

position	innings	player	position	innings	player
1	1	Alex Wood	1	1	Freddy Peralta
1	6	Tristan Beck	1	3	Elvis Peguero
2	1	Patrick Bailey	1	6	Ethan Small
3	1	LaMonte Wade Jr.	1	9	Mike Brosseau
4	1	Brett Wisely	2	1	William Contreras
5	1	J.D. Davis	3	1	Mike Brosseau
6	1	Casey Schmitt	3	9	Darin Ruf
7	1	Mitch Haniger	4	1	Owen Miller
8	1	Mike Yastrzemski	5	1	Brian Anderson
8	6	Bryce Johnson	6	1	Willy Adames
9	1	Michael Conforto	6	3	Brice Turang
9	6	Mike Yastrzemski	7	1	Christian Yelich
			7	7	Jesse Winker
			8	1	Joey Wiemer
			9	1	Tyrone Taylor

	game	SFN vs MIL on 20230526
	state before	(0, 0, 0, 0, 3, 3, 1, 0)
20.	error	play,3,0,sabob001,00,.X,E4/G34.B-1
	state after	(0, 1, 0, 0, 3, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1241 This game had 4 errors. The final score was 15-1.
In the top of the 3rd inning, with bases empty, SFN batter Blake Sabol hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.

	game	SFN vs MIL on 20230526
	state before	(0, 0, 1, 0, 3, 4, 1, 0)
21.	error	play,3,0,confm001,22,SF.BBX,FC1/G1-.2-3(E1/TH);B-1
	state after	(0, 1, 0, 1, 3, 4, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1242 *In the top of the 3rd inning, with a runner on 2nd, SFN batter Michael Conforto hit a soft ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 2nd to advance to 3rd and the batter to reach 1st.*

	game	SFN vs MIL on 20230526
	state before	(2, 1, 0, 0, 6, 12, 1, 1)
22.	error	play,6,1,andeb006,00,X,S5/G56S-.1-3(E5/TH)
	state after	(2, 1, 0, 1, 6, 12, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1243 *In the bottom of the 6th inning, with a runner on 1st, MIL batter Brian Anderson hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	SFN vs MIL on 20230526
	state before	(0, 0, 0, 0, 9, 15, 1, 1)
23.	error	play,9,1,andeb006,00,X,E4/TH/G4M.B-1
	state after	(0, 1, 0, 0, 9, 15, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1244 *In the bottom of the 9th inning, with bases empty, MIL batter Brian Anderson hit a ground ball to deep 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	SFN vs MIL on 20230527
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
24.	error	play,1,0,confm001,10,B2,P02(E1/TH).2-3
	state after	(1, 1, 0, 1, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1245 *In the top of the 1st inning, with runners on 1st and 2nd, SFN batter Michael Conforto has a 1-0 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed the runner on 2nd to advance to 3rd. Conforto is still at bat with a 1-0 count.*

19.2.3 June

	game	BAL vs MIL on 20230607
	state before	(0, 0, 0, 0, 8, 0, 10, 1)
25.	error	play,8,1,tellr001,20,..BBX,3E1/G3.B-1
	state after	(0, 1, 0, 0, 8, 0, 10, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1246 *In the bottom of the 8th inning, with bases empty, MIL batter Rowdy Tellez hit a ground ball to 1st. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	BAL vs MIL on 20230608
	state before	(1, 0, 0, 0, 8, 1, 3, 0)
26.	error	play,8,0,rutsa001,01,CX,S5/G56S.B-2(E5/TH)
	state after	(1, 0, 1, 0, 8, 1, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1247 *In the top of the 8th inning, with bases empty, BAL batter Adley Rutschman hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	OAK vs MIL on 20230609
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
27.	error	play,1,0,nodar001,22,C1BF.B>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1248 This game had 2 errors. The final score was 5-2.
In the top of the 1st inning, with a runner on 1st, OAK batter Ryan Noda had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at bat, now with a full count.

	game	OAK vs MIL on 20230609
	state before	(1, 1, 0, 0, 6, 5, 2, 1)
28.	error	play,6,1,wiemj001,11,*BC1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 5, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1249 *In the bottom of the 6th inning, with a runner on 1st, MIL batter Joey Wiemer has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to safely reach 2nd. Wiemer is still at bat withh a 1-1 count.*

	game	OAK vs MIL on 20230610
	state before	(2, 1, 0, 0, 10, 2, 1, 0)
29.	error	play,10,0,ruize001,22,CFBBX,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 10, 2, 1, 0)
	fielder	2 (pitcher on the visiting team)

Summary 1250 *In the top of the 10th inning, with a runner on 1st, OAK batter Esteury Ruiz had a 2-2 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	OAK vs MIL on 20230611
	state before	(0, 0, 0, 0, 4, 0, 2, 0)
30.	error	play,4,0,nodar001,01,CX,S8/L8+.B-2(E8)
	state after	(0, 0, 1, 0, 4, 0, 2, 0)
	fielder	8 (center field on the home team)

Summary 1251 This game had 2 errors. The final score was 8-6.
In the top of the 4th inning, with bases empty, OAK batter Ryan Noda hit a hard line drive single to center field. An error by the center fielder allowed the batter to reach 2nd.

	game	OAK vs MIL on 20230611
	state before	(2, 1, 1, 0, 6, 4, 2, 1)
31.	error	play,6,1,perkb002,01,CX,S9/L89.2-H;1-3(E9)
	state after	(2, 1, 0, 1, 6, 4, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1252 *In the bottom of the 6th inning, with runners on 1st and 2nd, MIL batter Blake Perkins hit a line drive single to right center field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	PIT vs MIL on 20230616
	state before	(2, 1, 1, 0, 1, 0, 0, 1)
32.	error	play,1,1,urial001,02,CC.X,C/E2.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1253 *In the bottom of the 1st inning, with runners on 1st and 2nd, MIL batter Luis Urias had a 0-2 count. Catcher's interference allowed the runners on 1st and 2nd to each advance one base and the batter to take 1st. The bases are loaded.*

33.	game	PIT vs MIL on 20230618
	state before	(1, 1, 0, 0, 3, 2, 0, 1)
	error	play,3,1,winkj002,10,B1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 3, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1254 *In the bottom of the 3rd inning, with a runner on 1st, MIL batter Jesse Winker has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to safely advance to 2nd. Winker is still at bat with a 1-0 count.*

34.	game	ARI vs MIL on 20230619
	state before	(1, 1, 1, 0, 6, 7, 1, 0)
	error	play,6,0,mark001,22,BCSBX,64(1)/FO/G6.2-H(E4/TH)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 8, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1255 *In the top of the 6th inning, with runners on 1st and 2nd, ARI batter Ketel Marte hit a ground ball to short. The second baseman (with an assist from the shortstop) forced out the runner from 1st. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

35.	game	ARI vs MIL on 20230620
	state before	(2, 1, 0, 0, 6, 4, 3, 0)
	error	play,6,0,kellc002,11,FBX,S6/G6S.1-3(E6/TH)
	state after	(2, 1, 0, 1, 6, 4, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1256 *In the top of the 6th inning, with a runner on 1st, ARI batter Carson Kelly hit a ground ball single to shallow short. A throwing error by the shortstop allowed the runner from 1st to advance to 3rd. The batter took 1st.*

	game	ARI vs MIL on 20230621
	state before	(1, 1, 0, 0, 6, 0, 1, 0)
36.	error	play,6,0,rivee001,12,CSFFBX,S8/L78D+.1-H(E8)(NR);B-2
	state after	(1, 0, 1, 0, 6, 1, 1, 0)
	fielder	8 (center field on the home team)

Summary 1257 *In the top of the 6th inning, with a runner on 1st, ARI batter Emmanuel Rivera hit a hard line drive single to deep left center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

19.2.4 July

	game	CHN vs MIL on 20230703
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
37.	error	play,1,0,younj003,01,CX,E3/G3.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1258 This game had 2 errors. The final score was 6-8. *In the top of the 1st inning, with bases empty, CHN batter Jared Young hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	CHN vs MIL on 20230703
	state before	(1, 0, 1, 0, 3, 6, 1, 1)
38.	error	play,3,1,yelic001,12,*BFCX,E3/G9LD.2-H(NR);B-3
	state after	(1, 0, 0, 1, 3, 6, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1259 *In the bottom of the 3rd inning, with a runner on 2nd, MIL batter Christian Yelich hit a ground ball to deep right field. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 3rd.*

	game	CHN vs MIL on 20230704
	state before	(2, 0, 1, 0, 6, 4, 1, 1)
39.	error	play,6,1,contw002,22,.CBBFFX,S2/G2-.2-H(E2/TH) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 6, 4, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1260 *In the bottom of the 6th inning, with a runner on 2nd, MIL batter William Contreras hit a soft ground ball single to shallow infield. A throwing error by the ctacher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CHN vs MIL on 20230705
	state before	(1, 1, 0, 0, 3, 0, 0, 0)
40.	error	play,3,0,suzus001,00,1,P01(E3) .1-2
	state after	(1, 0, 1, 0, 3, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1261 This game had 2 errors. The final score was 4-3. *In the top of the 3rd inning, with a runner on 1st, CHN batter Seiya Suzuki has a 0-0 count. Attempting to pickoff the runner on 1st, an error by the first baseman allowed that runner to safely reach 2nd. Suzuki is still at bat with a 0-0 count.*

	game	CHN vs MIL on 20230705
	state before	(2, 0, 1, 0, 9, 3, 3, 0)
41.	error	play,9,0,hoern001,01,SX,E5/TH/G56.2-H(NR) (UR) ;BX2(324)
	state after	(0, 0, 0, 0, 9, 4, 3, 1)
	fielder	5 (3rd baseman on the home team)

Summary 1262 *In the top of the 9th inning, with 2 outs a runner on 2nd, CHN batter Nico Hoerner hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI). The second baseman (assisted by the first baseman and catcher) tagged out the batter at 2nd base. As this is the 3rd out, it marks the end of the half-inning.*

Visiting/Home team rosters for CHN vs MIL on 2023-07-06

position	innings	player	position	innings	player
1	1	Marcus Stroman	1	1	Freddy Peralta
1	6	Daniel Palencia	1	6	Bryse Wilson
1	7	Mark Leiter Jr.	1	7	Elvis Peguero
1	8	Michael Fulmer	1	8	J.C. Mejia
2	1	Yan Gomes	1	9	Joel Payamps
3	1	Jared Young	2	1	Victor Caratini
4	1	Christopher Morel	3	1	Owen Miller
5	1	Patrick Wisdom	4	1	Brice Turang
6	1	Nico Hoerner	5	1	Brian Anderson
7	1	Ian Happ	6	1	Willy Adames
8	1	Cody Bellinger	7	1	Christian Yelich
9	1	Seiya Suzuki	8	1	Joey Wiemer
			9	1	Raimel Tapia
			9	8	Blake Perkins

	game	CHN vs MIL on 20230706
	state before	(1, 1, 0, 0, 4, 1, 0, 0)
42.	error	play,4,0,younj003,10,BX,3/G3S.1-3(E3/TH)
	state after	(2, 0, 0, 1, 4, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1263 This game had 3 errors. The final score was 5-6.

In the top of the 4th inning, with a runner on 1st, CHN batter Jared Young hit a ground ball to shallow 1st. The batter was put out by the first baseman. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd.

	game	CHN vs MIL on 20230706
	state before	(1, 0, 0, 0, 6, 3, 4, 0)
43.	error	play,6,0,gomey001,00,.X,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 6, 3, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1264 *In the top of the 6th inning, with bases empty, CHN batter Yan Gomes hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	CHN vs MIL on 20230706
	state before	(0, 1, 0, 0, 7, 3, 4, 1)
44.	error	play,7,1,adamw002,10,B1>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 7, 3, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1265 *In the bottom of the 7th inning, with a runner on 1st, MIL batter Willy Adames had a 1-0 count. During the next pitch, which was a called strike the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Adames is still at bat, now with a 1-1 count.*

	game	CIN vs MIL on 20230708
	state before	(0, 0, 0, 0, 4, 1, 4, 0)
45.	error	play,4,0,delae003,11,CBX,S6/G4M.B-2(E6/TH)
	state after	(0, 0, 1, 0, 4, 1, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1266 *In the top of the 4th inning, with bases empty, CIN batter Elly De La Cruz hit a ground ball single to deep 2nd. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	CIN vs MIL on 20230709
	state before	(1, 0, 0, 0, 7, 0, 1, 1)
46.	error	play,7,1,turab002,12,FBFX,E4/G4.B-1
	state after	(1, 1, 0, 0, 7, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1267 *In the bottom of the 7th inning, with bases empty, MIL batter Brice Turang hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	ATL vs MIL on 20230722
	state before	(1, 0, 1, 0, 8, 3, 3, 1)
47.	error	play,8,1,monaa001,11,CBX,FC6/G6.2X3(6E5);B-1
	state after	(2, 1, 0, 0, 8, 3, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1268 This game had 2 errors. The final score was 3-4. *In the bottom of the 8th inning, with a runner on 2nd, MIL batter Andruw Monasterio hit a ground ball into a fielder's choice at short. An error by the third baseman (with an assist from the shortstop) allowed the runner from 2nd to safely take 3rd and the batter to reach 1st.*

	game	ATL vs MIL on 20230722
	state before	(1, 1, 0, 1, 8, 3, 3, 1)
48.	error	play,8,1,frels001,10,B1X,9/SF/L9.3-H(UR);1-2(E9/TH)
	state after	(2, 0, 1, 0, 8, 3, 4, 1)
	fielder	9 (right field on the visiting team)

Summary 1269 *In the bottom of the 8th inning, with a runner on 1st and 3rd, MIL batter Sal Frelick hit a line drive sacrifice fly to right field. The right fielder put out the batter but the runner on 3rd scored (1 RBI). A throwing error by the right fielder allowed the runner on 1st to advance to 2nd.*

	game	CIN vs MIL on 20230724
	state before	(1, 1, 0, 0, 3, 2, 1, 1)
49.	error	play,3,1,adamw002,22,FF*BFBX,FC5/G5S-.1-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 3, 2, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1270 *In the bottom of the 3rd inning, with a runner on 1st, MIL batter Willy Adames hit a soft ground ball into a fielder's choice at 3rd. A throwing error by the third baseman fielder allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	CIN vs MIL on 20230726
	state before	(0, 0, 0, 0, 7, 0, 0, 0)
50.	error	play,7,0,fralj001,01,.FX,E4/G34.B-1
	state after	(0, 1, 0, 0, 7, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1271 *In the top of the 7th inning, with bases empty, CIN batter Jake Fraley hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

19.2.5 August

Visiting/Home team rosters for PIT vs MIL on 2023-08-04

position	innings	player	position	innings	player
1	1	Quinn Priester	1	1	Colin Rea
1	5	Yerry De Los Santos	1	5	Bryse Wilson
1	6	Ryan Borucki	1	7	Abner Uribe
1	7	Carmen Mlodzinski	1	9	Hoby Milner
1	8	Colin Holderman	2	1	William Contreras
1	9	David Bednar	3	1	Carlos Santana
2	1	Endy Rodriguez	4	1	Brice Turang
3	1	Alfonso Rivas	4	7	Andruw Monasterio
4	1	Vinny Capra	5	1	Brian Anderson
5	1	Jared Triolo	6	1	Willy Adames
6	1	Alika Williams	7	1	Christian Yelich
7	1	Joshua Palacios	8	1	Joey Wiemer
8	1	Jack Suwinski	9	1	Sal Frelick
9	1	Henry Davis			

	game	PIT vs MIL on 20230804
	state before	(0, 1, 0, 1, 1, 3, 0, 1)
51.	error	play,1,1,santc002,22,CB*BFX,S8/G4M.3-H;1-3;B-2(E8/TH)
	state after	(0, 0, 1, 1, 1, 3, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 1272 This game had 3 errors. The final score was 8-4. *In the bottom of the 1st inning, with runners on 1st and 3rd, MIL batter Carlos Santana hit a ground ball single to deep 2nd. The runner on 3rd scored (1 RBI), the runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

	game	PIT vs MIL on 20230804
	state before	(2, 0, 0, 0, 2, 6, 2, 1)
52.	error	play,2,1,wiemj001,01,CX,E6/G6.B-1
	state after	(2, 1, 0, 0, 2, 6, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1273 *In the bottom of the 2nd inning, with bases empty, MIL batter Joey Wiemer hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs MIL on 20230804
	state before	(1, 1, 1, 0, 7, 8, 3, 1)
53.	error	play,7,1,canhm001,12,*BFCX,S7/G6.2-H;1-3(E7/TH);B-2
	state after	(1, 0, 1, 1, 7, 8, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 1274 *In the bottom of the 7th inning, with runners on 1st and 2nd, MIL batter Mark Canha hit a ground ball single to short. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	PIT vs MIL on 20230806
	state before	(1, 1, 1, 0, 9, 4, 0, 0)
54.	error	play,9,0,hayek001,01,FX,8/L8XD+.2-3(E6/TH);1-2
	state after	(2, 0, 1, 1, 9, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1275 This game had 2 errors. The final score was 4-1. *In the top of the 9th inning, with runners on 1st and 2nd, PIT batter Ke'Bryan Hayes hit a hard line drive tp extra deep center field. The center fielder put out the batter. A throwing error by the shortstop allowed the runner on 2nd to safely reach 3rd and the runner on 1st to advance to 2nd.*

	game	PIT vs MIL on 20230806
	state before	(2, 1, 0, 1, 9, 4, 0, 1)
55.	error	play,9,1,canhm001,21,BBFX,E5/TH/G5.3-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 9, 4, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1276 *In the bottom of the 9th inning, with runners on 1st and 3rd, MIL batter Mark Canha hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd. and the batter to reach 1st.*

Visiting/Home team rosters for COL vs MIL on 2023-08-09

position	innings	player	position	innings	player
1	1	Chris Flexen	1	1	Adrian Houser
1	10	Justin Lawrence	1	10	Bryse Wilson
1	6	Daniel Bard	1	6	Hoby Milner
1	7	Tyler Kinley	1	7	Abner Uribe
1	8	Brent Suter	1	8	Joel Payamps
1	9	Matt Koch	1	9	Devin Williams
2	1	Austin Wynn	2	1	William Contreras
3	1	Michael Toglia	3	1	Mark Canha
4	1	Harold Castro	3	10	Carlos Santana
4	9	Alan Trejo	4	1	Brice Turang
5	1	Ryan McMahon	5	1	Andruw Monasterio
6	1	Ezequiel Tovar	6	1	Willy Adames
7	1	Nolan Jones	7	1	Tyrone Taylor
8	1	Brenton Doyle	7	10	Mark Canha
8	9	Cole Tucker	8	1	Blake Perkins
9	1	Cole Tucker	8	10	Sal Frelick
9	9	Harold Castro	9	1	Sal Frelick
			9	10	Tyrone Taylor

	game	COL vs MIL on 20230809
	state before	(2, 1, 0, 1, 7, 4, 5, 0)
56.	error	play,7,0,profj001,11,FBX,E4/G4+.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 7, 5, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1277 This game had 3 errors. The final score was 6-7.
In the top of the 7th inning, with runners on 1st and 3rd, COL batter Jurickson Profar hit a hard ground ball to 2nd base. An error by the second baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.

	game	COL vs MIL on 20230809
	state before	(2, 1, 0, 0, 10, 5, 5, 0)
57.	error	play,10,0,tovae001,01,FX,D7/L78XD+.1-H(E7)(NR)(UR);B-3
	state after	(2, 0, 0, 1, 10, 6, 5, 0)
	fielder	7 (2nd baseman on the home team)

Summary 1278 *In the top of the 10th inning, with a runner on 1st, COL batter Ezequiel Tovar hit a hard line drive double to extra deep left center field. An error by the left fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	COL vs MIL on 20230809
	state before	(2, 0, 0, 1, 10, 6, 6, 1)
58.	error	play,10,1,monaa001,00,X,E6/TH/G6.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 10, 6, 7, 1)
	fielder	6 (2nd baseman on the home team)

Summary 1279 *In the bottom of the 10th inning, with a runner on 3rd, MIL batter Andruw Monasterio hit a ground ball to short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	MIN vs MIL on 20230822
	state before	(1, 1, 1, 0, 6, 3, 2, 1)
59.	error	play,6,1,canhm001,12,CFBX,S9/G34.2-H;1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 6, 3, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1280 *In the bottom of the 6th inning, with runners on 1st and 2nd, MIL batter Mark Canha hit a ground ball single to the 1st/2nd base gap. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	MIN vs MIL on 20230823
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
60.	error	play,1,0,solad001,00,X,S7/L78S+.1-3(E7)
	state after	(0, 1, 0, 1, 1, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1281 *This game had 2 errors. The final score was 7-8. In the top of the 1st inning, with a runner on 1st, MIN batter Donovan Solano hit a hard line drive single to left center field. An error by the left fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	MIN vs MIL on 20230823
	state before	(0, 0, 1, 0, 10, 7, 6, 1)
61.	error	play,10,1,adamw002,31,..*BBBCX,S8/G6M.2-H(UR);B-2(E8)
	state after	(0, 0, 1, 0, 10, 7, 7, 1)
	fielder	8 (left field on the home team)

Summary 1282 *In the bottom of the 10th inning, with a runner on 2nd, MIL batter Willy Adames hit a ground ball single to deep short. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the batter to reach 2nd.*

	game	SDN vs MIL on 20230825
	state before	(2, 1, 0, 0, 5, 1, 5, 0)
62.	error	play,5,0,tatif002,01,C>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 1, 5, 0)
	fielder	2 (catcher on the home team)

Summary 1283 This game had 2 errors. The final score was 3-7. *In the top of the 5th inning, with a runner on 1st, SDN batter Fernando Tatis had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Tatis is still at bat, now with a 1-1 count.*

	game	SDN vs MIL on 20230825
	state before	(2, 1, 0, 0, 7, 1, 5, 1)
63.	error	play,7,1,adamw002,12,BCSX,S9/L89S.1-H(E9/TH)(NR);B-2
	state after	(2, 0, 1, 0, 7, 1, 6, 1)
	fielder	9 (right field on the visiting team)

Summary 1284 *In the bottom of the 7th inning, with a runner on 1st, MIL batter Willy Adames hit a line drive single to right center field. A throwing error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for SDN vs MIL on 2023-08-26

position	innings	player	position	innings	player
1	1	Pedro Avila	1	1	Freddy Peralta
1	5	Scott Barlow	1	6	Elvis Peguero
1	6	Tom Cosgrove	1	8	Joel Payamps
1	8	Robert Suarez	1	9	Devin Williams
2	1	Luis Campusano	2	1	William Contreras
3	1	Garrett Cooper	3	1	Carlos Santana
4	1	Ha-Seong Kim	4	1	Brice Turang
5	1	Manny Machado	5	1	Andruw Monasterio
6	1	Xander Bogaerts	6	1	Willy Adames
7	1	Ben Gamel	7	1	Christian Yelich
8	1	Trent Grisham	8	1	Sal Frelick
9	1	Fernando Tatis Jr.	9	1	Tyrone Taylor

64.	game	SDN vs MIL on 20230826
	state before	(1, 1, 1, 0, 5, 1, 1, 1)
	error	play,5,1,yelic001,00,>B,SB3;SB2.2-H(E2/TH) (NR)
	state after	(1, 1, 0, 0, 5, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1285 This game had 3 errors, including a consecutive error and a double steal. The final score was 4-5.

In the bottom of the 5th inning, with a runners on 1st and 2nd, MIL batter Christian Yelich had a 0-0 count. During the next pitch, which was a ball, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Yelich is still at bat, now with a 1-0 count.

	game	SDN vs MIL on 20230826
	state before	(1, 1, 0, 0, 5, 1, 2, 1)
65.	error	play,5,1,contw002,11,FBX,D7/G5.2-H;1-H;B-3(E7)
	state after	(1, 0, 0, 1, 5, 1, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 1286 This play is a *consecutive error* in the sense that Contreras bats consecutively after Yelich.

In the bottom of the 5th inning, with a runner on 1st, MIL batter William Contreras hit a ground ball double to 3rd. The runners on 1st and 2nd scored (2 RBIs). An error by the left fielder allowed the batter to reach 3rd.

	game	SDN vs MIL on 20230826
	state before	(0, 1, 0, 0, 7, 2, 5, 0)
66.	error	play,7,0,gameb001,10,BX,FC3/G34.1X2(3E6);B-1
	state after	(0, 1, 1, 0, 7, 2, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 1287 *In the top of the 7th inning, with a runner on 1st, SDN batter Ben Gamel hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	SDN vs MIL on 20230827
	state before	(2, 0, 0, 0, 5, 4, 2, 1)
67.	error	play,5,1,santc002,10,BX,E3/G34.B-1
	state after	(2, 1, 0, 0, 5, 4, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1288 *In the bottom of the 5th inning, with bases empty, MIL batter Carlos Santana hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

19.2.6 September

68.	game	PHI vs MIL on 20230901
	state before	(2, 1, 1, 1, 8, 5, 4, 1)
	error	play,8,1,millo002,22,BFBFX,E5/G5.3-H(NR) (UR) ; 2-H(NR) (UR) ; 1-H(NR) (UR) ; B-
	state after	(2, 0, 1, 0, 8, 5, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1289 *In the bottom of the 8th inning, with bases loaded, MIL batter Owen Miller hit a ground ball to 3rd base. An error by the third baseman allowed all three runners to score (no RBIs) and the batter to reach 2nd.*

69.	game	PHI vs MIL on 20230902
	state before	(1, 1, 0, 0, 8, 3, 7, 0)
	error	play,8,0,harpb003,12,SS.FBFX,E1/TH/G23.1-3;B-2
	state after	(1, 0, 1, 1, 8, 3, 7, 0)
	fielder	1 (pitcher on the home team)

Summary 1290 *In the top of the 8th inning, with a runner on 1st, PHI batter Bryce Harper hit a ground ball to the shallow right infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

70.	game	PHI vs MIL on 20230903
	state before	(2, 0, 0, 0, 1, 0, 1, 1)
	error	play,1,1,adamw002,32,CBS.BBX,E5/TH/G5-.B-1
	state after	(2, 1, 0, 0, 1, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1291 This game had 2 errors. The final score was 4-2. *In the bottom of the 1st inning, with bases empty, MIL batter Willy Adames hit a soft ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	PHI vs MIL on 20230903
	state before	(1, 0, 0, 0, 8, 4, 2, 1)
71.	error	play,8,1,wiemj001,01,CX,E6/TH/G56.B-1
	state after	(1, 1, 0, 0, 8, 4, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1292 *In the bottom of the 8th inning, with bases empty, MIL batter Joey Wiemer hit a ground ball to 3rd base. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	MIA vs MIL on 20230911
	state before	(2, 1, 0, 1, 6, 0, 10, 1)
72.	error	play,6,1,wiemj001,00,X,E6/TH/G6MS.3-H(NR)(UR);1-3;B-2
	state after	(2, 0, 1, 1, 6, 0, 11, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1293 *In the bottom of the 6th inning, with runners on 1st and 3rd, MIL batter Joey Wiemer hit a ground ball to shallow short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	MIA vs MIL on 20230913
	state before	(0, 0, 0, 0, 5, 1, 0, 1)
73.	error	play,5,1,carav001,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 5, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1294 This game had 2 errors. The final score was 2-0. *In the bottom of the 5th inning, with bases empty, MIL batter Victor Caratini hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.*

74.	game	MIA vs MIL on 20230913
	state before	(0, 0, 0, 0, 9, 2, 0, 1)
	error	play,9,1,taylt002,32,.BBCCBX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 9, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1295 *In the bottom of the 9th inning, with bases empty, MIL batter Tyrone Taylor hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

75.	game	WAS vs MIL on 20230916
	state before	(1, 1, 0, 0, 6, 1, 5, 0)
	error	play,6,0,thoml002,01,1C1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 1, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1296 *In the top of the 6th inning, with a runner on 1st, WAS batter Lane Thomas has a 0-1 count. While the pitcher attempted to pickoff the runner on 1st, a throwing error allowed that runner to safely reach 2nd. Thomas is still at bat with a 0-1 count.*

76.	game	SLN vs MIL on 20230926
	state before	(0, 1, 0, 0, 5, 2, 1, 0)
	error	play,5,0,fermj001,21,BFB>X,FC4/G4M.1-3(E4);B-2
	state after	(0, 0, 1, 1, 5, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1297 *In the top of the 5th inning, with a runner on 1st, SLN batter Jose Fermin hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SLN vs MIL on 20230927
	state before	(0, 0, 1, 0, 6, 2, 0, 1)
77.	error	play,6,1,taylt002,11,C*BX,D7/G56.2-H;B-3(E7/TH)
	state after	(0, 0, 1, 0, 6, 2, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 1298 This game had 2 errors. The final score was 2-3.

In the bottom of the 6th inning, with a runner on 2nd, MIL batter Tyrone Taylor hit a ground ball double to 3rd. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the batter to reach 3rd.

	game	SLN vs MIL on 20230927
	state before	(1, 1, 1, 0, 8, 2, 2, 0)
78.	error	play,8,0,nootl001,22,BCBCX,3E1/G3.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 8, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1299 *In the top of the 8th inning, with runners on 1st and 2nd, SLN batter Lars Nootbaar hit a ground ball to 1st. An error by the pitcher (on an assist by the first baseman) allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	SLN vs MIL on 20230928
	state before	(1, 0, 0, 0, 6, 0, 2, 0)
79.	error	play,6,0,palar002,01,CX,E3/TH/G34.B-1
	state after	(1, 1, 0, 0, 6, 0, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1300 *In the top of the 6th inning, with bases empty, SLN batter Richie Palacios hit a ground ball to the 1st/2nd base gap. A throwing error by the first baseman allowed the batter to reach 1st base.*

	game	CHN vs MIL on 20230929
	state before	(2, 0, 0, 0, 3, 0, 0, 1)
80.	error	play,3,1,yelic001,11,BTX,3E1/G3.B-1
	state after	(2, 1, 0, 0, 3, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1301 *In the bottom of the 3rd inning, with bases empty, MIL batter Christian Yelich ground ball to 1st. An error by the pitcher (on an assist by the first baseman) allowed the batter to reach 1st.*

	game	CHN vs MIL on 20230930
	state before	(0, 1, 1, 0, 1, 6, 0, 1)
81.	error	play,1,1,santc002,31,BF*BB+1,0A.2-3(E3)
	state after	(0, 1, 0, 1, 1, 6, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1302 *In the bottom of the 1st inning, with runners on 1st and 2nd, MIL batter Carlos Santana had a 3-1 count. The catcher attempted a pickoff at 1st. An error by the first baseman allowed the runner on 2nd to advance to 3rd. Santana is still at bat with a 3-1 count.*

Chapter 20

Minnesota Twins (MIN)

Originally founded as the Washington Senators, the team moved to Minnesota and was renamed the Minnesota Twins (after the adjacent “twin cities” Minneapolis and Saint Paul) for the start of the 1961 season. The Twins compete in the American League Central Division. Since 1961, they have won two World Series titles (1987, 1991), and last won the ALCD title in 2023.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Minneapolis against the Twins. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder’s choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 MIN team is 0.697 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the MIN is a little above the MLB average.

In MIN 2023, there were 61 errors in the 81 home games, 36 by the MIN and the rest (25) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 66, low (which is good) compared to the MLB average of 83.9. The Twins in 2023 had one of the lowest team error totals in the MLB.

20.1 Season totals

The intent of this rule is to avoid unnecessary delays. The umpire shall insist that the catcher return the ball promptly to the pitcher, and that the pitcher take his position on the rubber promptly. Obvious delay by the pitcher should instantly be penalized by the umpire.

– **MLB Rule Book** 5.07(c), part 3/3

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

20.1.1 Run conversion and error total by position

This section tabulates error-to-run rates and error totals by fielder position.

Runs and errors	totals
Errors by vt:	25
Runs by ht because of errors by vt:	10
Errors by ht:	36
Runs by vt because of errors by ht:	3

This is an excellent error-to-run conversion rate: $10/25 = 40\%$ on the part of the Twins (compared to $3/36 = 8.3\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 MIN season (home and away) has about $R/PA = 778/6219$ (about 12.5%) change of resulting in a run. Therefore, a randomly selected error is (for MIN) over three times as likely to result in a run than a randomly selected at-bat, at least in this case.

Errors by position	vt	ht
errors due to pitcher:	0	4
to catcher:	2	7
to 1st baseman:	3	3
to 2nd baseman:	5	4
to 3rd baseman:	5	8
to shortstop:	8	3
to left fielder:	1	1
to center fielder:	0	4
to right fielder:	1	0

20.1.2 Error total by state

A fielder may reach into, but not step into, a dugout to make a catch, and if he holds the ball, the catch shall be allowed. A fielder, in order to make a catch on a foul ball nearing a dugout or other out-of-play area (such as the stands), must have one or both feet on or over the playing surface (including the lip of the dugout) and neither foot on the ground inside the dugout or in any other out-of-play area.

– **MLB Rule Book** 5.09(a)(1) Comment

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	14	13	27
(lone) runner on 1st	2	14	16
(lone) runner on 2nd	1	4	5
(lone) runner on 3rd	0	1	1
two runners on base	6	4	10
bases loaded	2	0	2
total	25	36	61

A table of frequencies of most common errors by the MIN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	19.4%
(1, 0, 0, 0)	5	13.9%
(2, 0, 0, 0)	1	2.8%
(0, 1, 0, 0)	5	13.9%
(1, 1, 0, 0)	3	8.3%
(2, 1, 0, 0)	6	16.7%
(0, 0, 1, 0)	2	5.6%
(1, 1, 1, 0)	1	2.8%
(1, 1, 0, 1)	1	2.8%

A table of frequencies of most common errors by a visiting team in MIN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	9	36.0%
(1, 0, 0, 0)	2	8%
(2, 0, 0, 0)	3	12.0%
(0, 1, 0, 0)	1	4%
(1, 1, 0, 0)	0	0%
(2, 1, 0, 0)	1	4%
(0, 1, 1, 0)	2	8%
(2, 1, 1, 0)	2	8%
(1, 0, 1, 1)	1	4%
(2, 1, 1, 1)	2	8%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Twins homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `MIN2023.eva`):

state	MIN2023 percentages
(0, 0, 0, 0)	24.7%
(1, 0, 0, 0)	16.6%
(2, 0, 0, 0)	13.6%
(0, 1, 0, 0)	5.5%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	7.6%
(0, 0, 1, 0)	2.0%
(1, 0, 1, 0)	2.6%
(2, 0, 1, 0)	4.0%
(0, 0, 0, 1)	0.3%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.5%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	3.3%

20.1.3 Some interesting errors

Some of the most interesting error plays in the MIN 2023 home games:

- The game CHA vs MIN on 2023-04-10 had 3 errors, one by the first baseman on the VT, one by the second baseman on the VT, and one by the third baseman on the VT.
- Each of the games in the 3 game series NYA vs MIN on 2023-04-24 through 2023-04-26 had 2 errors, for 6 total.
- The game SFN vs MIN on 2023-05-24 had 6 errors which is the largest number in this team error analysis study of 2023 homegames¹.
- The game SDN vs MIN on 2023-05-09 had a 2 errors in 2 different plays but in the same at-bat.

¹See also TOR vs CIN on 2023-08-20, which also has 6 errors.

20.1.4 Relative team rankings

We have 81 home games between the Twins and some visiting MLB team (vt). A number of teams did not play MIN in the 2023 regular season in Minnesota. However, of those that did, MIN *lost* more games than won against²:

SEA (2/3), SFN (2/3), TBA (2/3), TOR (2/3), WAS (2/3)
BAL (3/3), DET (4/6), CLE (4/7)

and *won* more than lost against

MIL (0/2), ARI (0/3), KCA (1/7), CHA (1/6), TEX (1/4),
ANA (1/3), CHN (1/3), HOU (1/3),
NYA (1/3), NYN (1/3), OAK (1/3), PIT (1/3), SDN (1/3),

The teams that *split* their series were:

BOS (2/4).

The teams that did not play in any of the 81 home games in MIN 2023 were:

ATL, CIN, COL, LAN, MIA, PHI, SLN.

20.2 Game-by-game totals

A **catch** is legal if the ball is finally held by any fielder, even though juggled, or held by another fielder before it touches the ground. Runners may leave their bases the instant the first fielder touches the ball. A fielder may reach over a fence, railing, rope or other line of demarcation to make a catch. He may jump on top of a railing, or canvas that may be in foul ground. No interference should be allowed when a fielder reaches over a fence, railing, rope or into a stand to catch a ball. He does so at his own risk.

– **MLB Rule Book** 5.09(a)(1) Comment

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for MIN 2023 is as follows [BA].

²The visiting team's W/T record is given in parentheses (W = numbers of vt games vs Twins the vt won at MIN, T = Total games vt vs MIN played at Minnesota).

position	player	games played	OBP
1 (C)	Christian Vázquez	94	.280
3 (1B)	Donovan Solano	85	.369
4 (2B)	Edouard Julien	75	.381
5 (3B)	Royce Lewis	49	.372
6 (SS)	Carlos Correa	135	.312
7 (LF)	Willi Castro	54	.339
8 (CF)	Michael Taylor	126	.278
9 (RF)	Max Kepler	124	.332

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 MIN home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were ?? errors in the 81 home games, ?? by XYZ and the rest (??) by visiting teams.

20.2.1 April

	game	HOU vs MIN on 20230408
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
1.	error	play,2,0,tuckk001,00,X,E2/G2-.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1303 *In the top of the 2nd inning, with bases empty, HOU batter Kyle Tucker hit a soft ground ball to very shallow infield. An error by the catcher allowed the batter to reach 1st base.*

	game	HOU vs MIN on 20230409
	state before	(0, 0, 0, 0, 4, 2, 0, 1)
2.	error	play,4,1,buxtb001,10,BX,S5/G6.B-2(E5/TH)
	state after	(0, 0, 1, 0, 4, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1304 This game had 2 errors. The final score was 5-1.
In the bottom of the 4th inning, with bases empty, MIN batter Byron Buxton hit a ground ball single to short. A throwing error by the third baseman allowed the batter to reach 2nd.

	game	HOU vs MIN on 20230409
	state before	(0, 1, 1, 0, 8, 4, 1, 0)
3.	error	play,8,0,abrej003,31,BBCBX,E3/G23-.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 8, 4, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1305 *In the top of the 8th inning, with runners on 1st and 2nd, HOU batter Jose Abreu hit a soft ground ball to shallow right infield. An error by the first baseman allowed the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Dylan Cease	1	1	Kenta Maeda
1	6	Jimmy Lambert	1	7	Caleb Thielbar
1	7	Kendall Graveman	1	8	Jovani Moran
1	8	Aaron Bummer	1	9	Jovani Moran
1	8	Reynaldo Lopez	2	1	Christian Vazquez
2	1	Yasmani Grandal	3	1	Donovan Solano
3	1	Gavin Sheets	4	1	Nick Gordon
3	8	Jake Burger	5	1	Willi Castro
4	1	Elvis Andrus	5	9	Jose Miranda
4	6	Romy Gonzalez	6	1	Kyle Farmer
5	1	Hanser Alberto	7	1	Trevor Larnach
6	1	Tim Anderson	7	9	Willi Castro
6	6	Elvis Andrus	8	1	Michael A. Taylor
7	1	Andrew Benintendi	9	1	Matt Wallner
8	1	Luis Robert Jr.	9	9	Trevor Larnach
9	1	Oscar Colas			

	game	CHA vs MIN on 20230410
	state before	(2, 1, 1, 1, 3, 0, 0, 1)
4.	error	play,3,1,gordn001,12,CFFBX,FC5/G56.3-H(NR)(UR);2-3;1-2(E5);B-1
	state after	(2, 1, 1, 1, 3, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1306 This game had 3 errors. The final score was 4-3.

In the bottom of the 3rd inning, with bases loaded, MIN batter Nick Gordon hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (unearned, no RBI) and the runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	CHA vs MIN on 20230410
	state before	(1, 0, 0, 0, 5, 4, 2, 1)
5.	error	play,5,1,gordn001,21,CBBX,E4/G34.B-1
	state after	(1, 1, 0, 0, 5, 4, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1307 *In the bottom of the 5th inning, with bases empty, MIN batter Nick Gordon hit a ground ball to 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.*

	game	CHA vs MIN on 20230410
	state before	(2, 1, 0, 0, 5, 4, 2, 1)
6.	error	play,5,1,wallm001,21,BFBX,E3/G9LS.1-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 5, 4, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1308 *In the bottom of the 5th inning, with a runner on 1st, MIN batter Matt Wallner hit a ground ball to shallow right field down the 1st base line. An error by the first baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CHA vs MIN on 20230411
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
7.	error	play,1,0,robcl002,00,X,S1/G2-.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1309 This game had 2 errors. The final score was 3-4 (in extra innings).

In the top of the 1st inning, with a runner on 1st, CHA batter Luis Robert hit a soft ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	CHA vs MIN on 20230411
	state before	(0, 0, 1, 0, 10, 3, 3, 1)
8.	error	play,10,1,taylm002,00,...X,S5/BG2.2-H(E5/TH) (NR) (UR)
	state after	(0, 1, 0, 0, 10, 3, 4, 1)
	fielder	5 (pitcher on the home team)

Summary 1310 *In the bottom of the 10th inning, with a runner on 2nd, MIN batter Michael Taylor hit a bunt single to shallow infield. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	WAS vs MIN on 20230421
	state before	(1, 0, 0, 0, 9, 3, 2, 0)
9.	error	play,9,0,garc1006,20,BBX,3E1/G34.B-1
	state after	(1, 1, 0, 0, 9, 3, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1311 *In the top of the 9th inning, with bases empty, WAS batter Luis Garcia hit a ground ball to 1st/2nd. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	WAS vs MIN on 20230422
	state before	(0, 0, 0, 0, 5, 5, 1, 1)
10.	error	play,5,1,gallj002,12,BSFFS,K+PB.B-2(E2/TH)
	state after	(0, 0, 1, 0, 5, 5, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1312 *In the bottom of the 5th inning, with bases empty, MIN batter Joey Gallo had a 1-2 count. During the next pitch which was a swinging strike but a passed ball. The batter runs for 1st (on an uncaught 3rd strike). A throwing error by the catcher allowed the batter to reach 2nd.*

	game	WAS vs MIN on 20230423
	state before	(0, 0, 0, 0, 3, 1, 0, 1)
11.	error	play,3,1,solad001,12,CFFBX,E5/G5.B-1
	state after	(0, 1, 0, 0, 3, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1313 *In the bottom of the 3rd inning, with bases empty, MIN batter Donovan Solano hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for NYA vs MIN on 2023-04-24

position	innings	player	position	innings	player
1	1	Jhony Brito	1	1	Sonny Gray
1	3	Greg Weissert	1	8	Griffin Jax
1	5	Albert Abreu	1	9	Jovani Moran
1	7	Wandy Peralta	2	1	Christian Vazquez
1	8	Jimmy Cordero	3	1	Joey Gallo
2	1	Jose Trevino	4	1	Jorge Polanco
3	1	Anthony Rizzo	5	1	Jose Miranda
4	1	Gleyber Torres	6	1	Carlos Correa
5	1	DJ LeMahieu	7	1	Trevor Larnach
6	1	Anthony Volpe	8	1	Michael A. Taylor
7	1	Oswaldo Cabrera	9	1	Max Kepler
8	1	Aaron Hicks			
9	1	Franchy Cordero			

	game	NYA vs MIN on 20230424
	state before	(0, 1, 0, 0, 6, 0, 4, 1)
12.	error	play,6,1,keplm001,22,CFB11*BX,16(1)/F0/G13.B-2(E6/TH)
	state after	(1, 0, 1, 0, 6, 0, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1314 This game had 2 errors. In fact, each game in this 3 game Yankees-Twins series had 2 errors. The final score was 1-6.

In the bottom of the 6th inning, with a runner on 1st, MIN batter Max Kepler hit a ground ball to shallow right infield. At 2nd, the shortstop (with an assist from the pitcher) forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.

13.	game	NYA vs MIN on 20230424
	state before	(2, 1, 1, 0, 8, 0, 5, 0)
	error	play,8,0,judga001,32,*BF*BBC>F.>F>X,E5/TH/G56S.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 8, 0, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1315 *In the top of the 8th inning, with runners on 1st and 2nd, NYA batter Aaron Judge hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

14.	game	NYA vs MIN on 20230425
	state before	(0, 0, 0, 0, 3, 1, 0, 1)
	error	play,3,1,taylm002,21,CBBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1316 This game had 2 errors. The final score was 1-6 (the same final score as the last game).

In the bottom of the 3rd inning, with bases empty, MIN batter Michael Taylor hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	NYA vs MIN on 20230425
	state before	(1, 1, 1, 0, 5, 1, 2, 0)
15.	error	play,5,0,judga001,00,X,54(1)/F0/G5.2XH(4E3)(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 2, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1317 *In the top of the 5th inning, with runners on 1st and 2nd, NYA batter Aaron Judge hit a ground ball to 3rd. The second baseman (with an assist from the third baseman) forced out the runner from 1st. An error by the first baseman (with an assist from the second baseman) allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	NYA vs MIN on 20230426
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
16.	error	play,2,0,torrg001,01,SX,D8/G4.B-3(E8/TH)
	state after	(0, 0, 0, 1, 2, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1318 This game had 2 errors. The final score was 12-6. *In the top of the 2nd inning, with bases empty, NYA batter Gleyber Torres hit a ground double to center field.. A throwing error by the cnter fielder allowed the batter to reach 3rd.*

	game	NYA vs MIN on 20230426
	state before	(0, 1, 0, 0, 4, 5, 1, 0)
17.	error	play,4,0,cabro002,01,FX,S1/BG2-.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 4, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1319 *In the top of the 4th inning, with a runner on 1st, NYA batter Oswaldo Cabrera hit a soft bunt single to very shallow in-field. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	KCA vs MIN on 20230427
	state before	(0, 0, 0, 0, 3, 0, 3, 0)
18.	error	play,3,0,dozih001,22,CBBFX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 3, 0, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1320 *In the top of the 3rd inning, with bases empty, KCA batter Hunter Dozier hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	KCA vs MIN on 20230429
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
19.	error	play,2,0,pratn001,01,CX,S8/F8S.B-2(E8)
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1321 *In the top of the 2nd inning, with bases empty, KCA batter Nick Pratto hit a fly ball single to shallow center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	KCA vs MIN on 20230430
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
20.	error	play,2,0,pratn001,22,CB1FB>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1322 This game had 2 errors. The final score was 4-8. *In the top of the 2nd inning, with a runner on 1st, KCA batter Nick Pratto had a 2-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	KCA vs MIN on 20230430
	state before	(1, 0, 1, 0, 7, 1, 8, 0)
21.	error	play,7,0,duffm002,12,SC*BX,E4/G34.2-H(NR)(UR);B-1
	state after	(1, 1, 0, 0, 7, 2, 8, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1323 *In the top of the 7th inning, with a runner on 2nd, KCA batter Matt Duffy hit a ground ball to 1st/3rd. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

20.2.2 May

	game	SDN vs MIN on 20230509
	state before	(2, 1, 0, 0, 7, 2, 1, 0)
22.	error	play,7,0,cronj001,01,F1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1324 This game had 2 errors. They were in different plays but in the same inning with the same batter! The final score was 5-1. *In the top of the 7th inning, with a runner on 1st, SDN batter Jake Cronenworth had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Cronenworth is still at bat, now with a 1-1 count.*

	game	SDN vs MIN on 20230509
	state before	(2, 0, 0, 1, 7, 2, 1, 0)
23.	error	play,7,0,cronj001,31,F1>B.B*BB+3,W.3-H(E2/TH)(NR)(UR)
	state after	(2, 1, 0, 0, 7, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1325 *In the top of the 7th inning, with a runner on 3rd, SDN batter Jake Cronenworth had a 3-1 count. During the next pitch, which was a ball, the catcher threw a pickoff to 3rd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). The batter took 1st (walk).*

	game	SDN vs MIN on 20230510
	state before	(1, 1, 0, 0, 8, 1, 2, 0)
24.	error	play,8,0,tatif002,00,X,S7/L78D+.1-3;B-2(E7/TH)
	state after	(1, 0, 1, 1, 8, 1, 2, 0)
	fielder	7 (left field on the home team)

Summary 1326 *In the top of the 8th inning, with a runner on 1st, SDN batter Fernando Tatis hit a hard line drive single to deep left center field. The runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	CHN vs MIN on 20230512
	state before	(1, 0, 1, 1, 2, 0, 0, 1)
25.	error	play,2,1,taylm002,00,X,FC5/G56.3-H(E5/TH) (NR) (UR);2-3;B-1
	state after	(1, 1, 0, 1, 2, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1327 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, MIN batter Michael Taylor hit a ground ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runner on 3rd to safely score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	CHN vs MIN on 20230513
	state before	(2, 1, 1, 1, 8, 1, 9, 1)
26.	error	play,8,1,corrc001,12,CBFFX,S6/G56.3-H;2-H;1-3(E6/TH);B-2
	state after	(2, 0, 1, 1, 8, 1, 11, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1328 *In the bottom of the 8th inning, with bases loaded, MIN batter Carlos Correa hit a ground ball single to 3rd. The runners on 2nd and 3 scored (2 RBIs). A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SFN vs MIN on 20230522
	state before	(0, 0, 0, 0, 3, 4, 0, 0)
27.	error	play,3,0,davij006,00,.X,E5/G56.B-1
	state after	(0, 1, 0, 0, 3, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1329 *This game had 2 errors. The final score was 4-1. In the top of the 3rd inning, with bases empty, SFN batter J.D. Davis hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SFN vs MIN on 20230522
	state before	(2, 0, 1, 0, 8, 4, 0, 0)
28.	error	play,8,0,schmc003,21,.BBCX,E6/G6+.2-3;B-1
	state after	(2, 1, 0, 1, 8, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1330 *In the top of the 8th inning, with a runner on 2nd, SFN batter Casey Schmitt hit a hard ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SFN vs MIN on 20230523
	state before	(0, 0, 0, 0, 2, 0, 2, 1)
29.	error	play,2,1,farmk001,12,CCBX,E3/G34.B-1
	state after	(0, 1, 0, 0, 2, 0, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1331 *In the bottom of the 2nd inning, with bases empty, MIN batter Kyle Farmer hit a ground ball to 1st/2nd. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Anthony DeSclafani	1	1	Joe Ryan
1	6	Ryan Walker	1	6	Brock Stewart
1	8	Taylor Rogers	1	6	Jovani Moran
2	1	Blake Sabol	1	8	Jose De Leon
3	1	LaMonte Wade Jr.	1	9	Jhoan Duran
4	1	Brett Wisely	2	1	Ryan Jeffers
4	7	Casey Schmitt	3	1	Alex Kirilloff
5	1	Casey Schmitt	4	1	Edouard Julien
5	7	Wilmer Flores	5	1	Donovan Solano
6	1	Brandon Crawford	6	1	Kyle Farmer
7	1	Mitch Haniger	7	1	Willi Castro
8	1	Mike Yastrzemski	8	1	Michael A. Taylor
8	7	Brett Wisely	9	1	Matt Wallner
9	1	Michael Conforto			

	game	SFN vs MIN on 20230524
	state before	(0, 1, 1, 0, 2, 1, 2, 1)
30.	error	play,2,1,taylm002,12,LBMS,K.2X3(2E6);1-2
	state after	(1, 0, 1, 1, 2, 1, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1332 This game had 6 errors (tied³. with the largest seen among all games in the 2023 regular season). The final score was 1-7.
In the bottom of the 2nd inning, with runners on 1st and 2nd, MIN

³See also TOR vs CIN on 2023-08-20, which also has 6 errors.

batter Michael Taylor had a 1-2 count. During the next pitch, which was a swinging strike, the runner on 2nd attempted to reach 3rd. An error by the shortstop (with an assist from the catcher) allowed that runner to safely reach 3rd and the runner on on 1st to advance to 2nd. The batter struck out.

	game	SFN vs MIN on 20230524
	state before	(1, 1, 0, 0, 3, 1, 3, 0)
31.	error	play,3,0,sabob001,12,BFFX,FC1/G1-.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 3, 1, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1333 *In the top of the 3rd inning, with a runner on 1st, SFN batter Blake Sabol hit a soft ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SFN vs MIN on 20230524
	state before	(2, 0, 0, 0, 3, 1, 3, 1)
32.	error	play,3,1,farmk001,32,FBFBFFX,E6/G6.B-1
	state after	(2, 1, 0, 0, 3, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1334 *In the bottom of the 3rd inning, with bases empty, MIN batter Kyle Farmer hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	SFN vs MIN on 20230524
	state before	(2, 1, 1, 0, 3, 1, 3, 1)
33.	error	play,3,1,wallm001,11,BFX,E4/G34D.2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 3, 1, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1335 *In the bottom of the 3rd inning, with runners on 1st and 2nd, MIN batter Matt Wallner hit a ground ball to deep 1st/2nd. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	SFN vs MIN on 20230524
	state before	(2, 1, 0, 1, 3, 1, 4, 1)
34.	error	play,3,1,jeffr002,02,CS.>B,SB2;SBH.3-H(NR)(UR);1-3(E4/TH)
	state after	(2, 0, 0, 1, 3, 1, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1336 This play is an example of a *double steal* and a *steal of home*. It is **also** an example of a *consecutive play* since Jeffers bats right after Wallner in the Twins lineup in that game.

In the bottom of the 3rd inning, with runners on 1st and 3rd, MIN batter Ryan Jeffers had a 0-2 count. During the next pitch, which was a bal, the runner on 1st stole 2nd and the runner on 3rd stole home (1 RBI)⁴. The runner from 2nd (now on 3rd) also scored (unearned, no RBI) while a throwing error by the second baseman allowed the runner on 1st to advance to 3rd. Jeffers is still at bat, now with a 1-2 count.

	game	SFN vs MIN on 20230524
	state before	(1, 0, 0, 0, 5, 1, 6, 0)
35.	error	play,5,0,sabob001,12,FSBFX,E4/TH/G3.B-1
	state after	(1, 1, 0, 0, 5, 1, 6, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1337 *In the top of the 5th inning, with bases empty, SFN batter Blake Sabol hit a ground ball to 1st base. A throwing error by the second baseman allowed the batter to reach 1st base.*

⁴The Retrosheet record has both SBH and 3-H but these are not to be double counted.

	game	TOR vs MIN on 20230526
	state before	(1, 0, 0, 0, 4, 3, 0, 0)
36.	error	play,4,0,beltb001,00,X,E5/G6S.B-1
	state after	(1, 1, 0, 0, 4, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1338 *In the top of the 4th inning, with bases empty, TOR batter Brandon Belt hit a ground ball to shallow short. An error by the third baseman allowed the batter to reach 1st base.*

	game	TOR vs MIN on 20230528
	state before	(2, 1, 0, 0, 8, 2, 0, 0)
37.	error	play,8,0,bichb001,11,BFX,S7/L8D.1-H(E8) (NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 8, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 1339 *In the top of the 8th inning, with a runner on 1st, TOR batter Bo Bichette hit a line drive single to deep center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

20.2.3 June

	game	CLE vs MIN on 20230601
	state before	(2, 1, 0, 0, 8, 6, 3, 0)
38.	error	play,8,0,brenw002,00,X,E4/G34.1-2;B-1
	state after	(2, 1, 1, 0, 8, 6, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1340 *In the top of the 8th inning, with a runner on 1st, CLE batter Will Brennan hit a ground ball to 1st/2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

39.	game	MIL vs MIN on 20230613
	state before	(2, 0, 0, 0, 5, 3, 2, 0)
	error	play,5,0,yelic001,32,BSBSBX,E5/G6.B-1
	state after	(2, 1, 0, 0, 5, 3, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1341 *In the top of the 5th inning, with bases empty, MIL batter Christian Yelich hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.*

40.	game	MIL vs MIN on 20230614
	state before	(0, 0, 0, 0, 2, 2, 0, 1)
	error	play,2,1,larnt001,12,CCBFX,E3/G3.B-1
	state after	(0, 1, 0, 0, 2, 2, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1342 *In the bottom of the 2nd inning, with bases empty, MIN batter Trevor Larnach hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

41.	game	DET vs MIN on 20230615
	state before	(0, 0, 0, 0, 9, 8, 4, 1)
	error	play,9,1,buxtb001,22,.CBS.BX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 9, 8, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1343 *In the bottom of the 9th inning, with bases empty, MIN batter Byron Buxton hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

42.	game	BOS vs MIN on 20230619
	state before	(2, 1, 0, 0, 4, 1, 0, 0)
	error	play,4,0,reyep001,02,FC+1X,S4/G4.1-3;B-2(E4/TH)
	state after	(2, 0, 1, 1, 4, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1344 *In the top of the 4th inning, with a runner on 1st, BOS batter Pablo Reyes hit a ground ball single to 2nd base. The runner on 1st advanced to 3rd. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	BOS vs MIN on 20230621
	state before	(0, 0, 0, 0, 4, 2, 3, 0)
43.	error	play,4,0,yoshm002,11,BCX,S8/L78D.B-2(E8)
	state after	(0, 0, 1, 0, 4, 2, 3, 0)
	fielder	8 (center field on the home team)

Summary 1345 This game had 2 errors. The final score was 4-5 (in extra innings).

In the top of the 4th inning, with bases empty, BOS batter Masataka Yoshida hit a line drive single to deep left center field.. An error by the center fielder allowed the batter to reach 2nd.

	game	BOS vs MIN on 20230621
	state before	(2, 1, 0, 0, 5, 3, 3, 0)
44.	error	play,5,0,yoshm002,00,1,P01(E3).1-2
	state after	(2, 0, 1, 0, 5, 3, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1346 *In the top of the 5th inning, with a runner on 1st, BOS batter Masataka Yoshida has a 0-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the first baseman allowed that runner to reach 2nd. Yoshida is still at bat with a 0-0 count.*

20.2.4 July

	game	KCA vs MIN on 20230704
	state before	(0, 0, 0, 0, 4, 0, 4, 0)
45.	error	play,4,0,garcn003,01,CX,E5/G56S.B-1
	state after	(0, 1, 0, 0, 4, 0, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1347 *In the top of the 4th inning, with bases empty, KCA batter Maikel Garcia hit a ground ball to shallow 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

46.	game	KCA vs MIN on 20230705
	state before	(2, 1, 1, 0, 3, 0, 1, 1)
	error	play,3,1,kiria001,22,CFBFFBFX,S7/L7.2-H;1-H(E7/TH)(NR)(UR);B-3
	state after	(2, 0, 0, 1, 3, 0, 3, 1)
	fielder	7 (left field on the visiting team)

Summary 1348 *In the bottom of the 3rd inning, with runners on 1st and 2nd, MIN batter Alex Kirilloff hit a line drive single to left field. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 3rd.*

47.	game	CHA vs MIN on 20230721
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
	error	play,1,1,julie001,00,X,S7/L7D+.BX2(7E4)
	state after	(1, 0, 1, 0, 1, 1, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1349 *This game had 2 errors. The final score was 3-10. In the bottom of the 1st inning, with bases empty, MIN batter Edouard Julien hit a hard line drive single to deep left field. An error by the second baseman (with an assist from the left fielder) allowed the batter to safely reach 2nd.*

48.	game	CHA vs MIN on 20230721
	state before	(2, 0, 0, 0, 7, 3, 8, 1)
	error	play,7,1,corrc001,02,CSX,E9/F9LD.B-2
	state after	(2, 0, 1, 0, 7, 3, 8, 1)
	fielder	9 (right field on the visiting team)

Summary 1350 *In the bottom of the 7th inning, with bases empty, MIN batter Carlos Correa hit a fly ball to deep right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	CHA vs MIN on 20230723
	state before	(1, 1, 0, 0, 5, 2, 0, 0)
49.	error	play,5,0,remiz001,10,B>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1351 *In the top of the 5th inning, with a runner on 1st, CHA batter Zach Remillard had a 1-0 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Remillard is still at bat, now with a 1-1 count.*

20.2.5 August

	game	ARI vs MIN on 20230806
	state before	(0, 1, 0, 0, 8, 2, 2, 0)
50.	error	play,8,0,ahmen001,02,1LL1>C,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 8, 2, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1352 *In the top of the 8th inning, with a runner on 1st, ARI batter Nick Ahmed had a 0-2 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	DET vs MIN on 20230816
	state before	(0, 1, 0, 0, 9, 7, 4, 0)
51.	error	play,9,0,greer003,12,C1BC>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 9, 7, 4, 0)
	fielder	2 (catcher on the home team)

Summary 1353 *In the top of the 9th inning, with a runner on 1st, DET batter Riley Greene had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Greene is still at bat, now with a 2-2 count.*

	game	PIT vs MIN on 20230819
	state before	(1, 0, 0, 0, 6, 0, 2, 0)
52.	error	play,6,0,pegul001,00,X,S5/G25.B-2(E5/TH)
	state after	(1, 0, 1, 0, 6, 0, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1354 *In the top of the 6th inning, with bases empty, PIT batter Liover Peguero hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	TEX vs MIN on 20230827
	state before	(0, 0, 1, 0, 10, 5, 5, 0)
53.	error	play,10,0,garvm001,00,..X,E6/G6.B-1
	state after	(0, 1, 1, 0, 10, 5, 5, 0)
	fielder	6 (3rd baseman on the home team)

Summary 1355 *In the top of the 10th inning, with a runner on 2nd, TEX batter Mitch Garver hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs MIN on 20230828
	state before	(1, 1, 0, 1, 3, 4, 6, 0)
54.	error	play,3,0,gimea001,00,X,FC5/G56S.3-H;1-2(E5/TH);B-1
	state after	(1, 1, 1, 0, 3, 5, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1356 *In the top of the 3rd inning, with runners on 1st and 3rd, CLE batter Andres Gimenez hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st,*

	game	CLE vs MIN on 20230829
	state before	(1, 0, 1, 0, 1, 0, 0, 1)
55.	error	play,1,1,keplm001,00,X,E5/G56.2-3;B-1
	state after	(1, 1, 0, 1, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1357 *In the bottom of the 1st inning, with a runner on 2nd, MIN batter Max Kepler hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

20.2.6 September

	game	NYN vs MIN on 20230908
	state before	(0, 1, 1, 0, 7, 2, 2, 1)
56.	error	play,7,1,julie001,02,CC>B,SB3;SB2.2-H(E2/TH) (NR)
	state after	(0, 1, 0, 0, 7, 2, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1358 *In the bottom of the 7th inning, with runners on 1st and 2nd, MIN batter Edouard Julien had a 0-2 count. During the next pitch, which was a ball the runners on 1st and 2nd stole their next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Julien is still at bat, now with a 1-2 count.*

	game	TBA vs MIN on 20230911
	state before	(0, 0, 0, 0, 3, 1, 0, 0)
57.	error	play,3,0,arozr001,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1359 *In the top of the 3rd inning, with bases empty, TBA batter Randy Arozarena hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TBA vs MIN on 20230913
	state before	(0, 0, 0, 0, 7, 4, 4, 1)
58.	error	play,7,1,farmk001,11,.CBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 7, 4, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1360 This game had 2 errors, in different plays, both by the same fielder. The final score was 5-4.

In the bottom of the 7th inning, with bases empty, MIN batter Kyle Farmer hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	TBA vs MIN on 20230913
	state before	(2, 0, 0, 0, 8, 4, 4, 1)
59.	error	play,8,1,solad001,00,X,E6/G6.B-1
	state after	(2, 1, 0, 0, 8, 4, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1361 *In the bottom of the 8th inning, with bases empty, MIN batter Donovan Solano hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

60.	game	ANA vs MIN on 20230922
	state before	(0, 0, 0, 0, 7, 3, 4, 1)
	error	play,7,1,keplm001,00,.X,E4/G34+.B-1
	state after	(0, 1, 0, 0, 7, 3, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1362 *In the bottom of the 7th inning, with bases empty, MIN batter Max Kepler hit a hard ground ball to 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.*

61.	game	ANA vs MIN on 20230923
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
	error	play,2,0,wallc001,32,BFBBF1>F>F>X,E5/TH/G56S.1-3;B-2
	state after	(2, 0, 1, 1, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1363 *In the top of the 2nd inning, with a runner on 1st, ANA batter Chad Wallach hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

Chapter 21

New York Mets (NYN)

The New York Mets were founded in 1962 to replace New York's departed NL teams, the Brooklyn Dodgers (who moved to Los Angeles in 1957) and the New York Giants (who moved to San Francisco in 1958). The Mets have won 2 World Series titles, the last in 1986, and 6 NLED titles, the last in 2015.

This chapter is devoted to listing the errors in each regular season game of 2023 played in New York City (in Queens) against the Mets. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 NYN team is .689, almost the same as the MLB average of 0.691 that year [BR].

In 2023, there were 83 errors in the 81 NYN home games, 45 by the NYN and the rest (38) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 88, slightly higher than the MLB average of 83.9.

21.1 Season totals

A batted ball not touched by a fielder, which hits the pitcher's rubber and rebounds into foul territory, between home and first,

or between home and third base is a **foul ball**. If a fly ball lands in the infield between home and first base, or home and third base, and then bounces to foul territory without touching a player or umpire and before passing first or third base, it is a foul ball; or if the ball settles on foul territory or is touched by a player on foul territory, it is a foul ball. If a fly ball lands on or beyond first or third base and then bounces to foul territory, it is a fair hit.

– **MLB Rule Book** Definitions (2/2)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

21.1.1 Run conversion and error total by position

This section tabulates the totals by position.

Runs and errors	totals
Errors by vt:	38
Runs by ht because of errors by vt:	13
Errors by ht:	45
Runs by vt because of errors by ht:	11

This is a good error-to-run conversion rate: $13/38 = 0.342$ (about 34.2%) on the part of the Mets (compared to $11/45 = 0.244\dots$, about 24.4%, on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 NYN season (home and away) has about $R/PA = 0.118\dots$ (11.8%) change of resulting in a run. Therefore, a randomly selected error is over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 83, so the expected number of errors (by either team) in a randomly selected NYN 2023 home game is about 0.94.

Errors by position	vt	ht
errors due to pitcher:	5	6
to catcher:	8	10
to 1st baseman:	5	6
to 2nd baseman:	2	3
to 3rd baseman:	6	10
to shortstop:	7	6
to left fielder:	2	8
to center fielder:	1	2
to right fielder:	2	2

21.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	14	15	29
(lone) runner on 1st	13	14	27
(lone) runner on 2nd	1	3	4
(lone) runner on 3rd	0	3	3
two runners on base	6	8	14
bases loaded	4	2	6
total	38	45	83

A table of frequencies of most common errors by the Mets in NYN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	13.3%
(1, 0, 0, 0)	5	11.1%
(2, 0, 0, 0)	4	8.9%
(0, 1, 0, 0)	5	11.1%
(1, 1, 0, 0)	3	6.7%
(2, 1, 0, 0)	6	13.3%
(2, 0, 1, 0)	2	4.4%
(1, 1, 1, 0)	2	4.4%
(0, 1, 1, 1)	1	2.2%

A table of frequencies of most common errors by a visiting team in NYN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	5	13.2%
(1, 0, 0, 0)	6	15.8%
(2, 0, 0, 0)	3	7.9%
(0, 1, 0, 0)	5	13.2%
(1, 1, 0, 0)	3	7.9%
(2, 1, 0, 0)	5	13.2%
(1, 1, 1, 0)	3	7.9%
(1, 0, 1, 1)	2	5.3%
(0, 1, 1, 1)	1	2.6%
(2, 1, 1, 1)	2	5.3%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Mets homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `NYN2023.evn`):

state	NYN2023 percentages
(0, 0, 0, 0)	24.8%
(1, 0, 0, 0)	16.8%
(2, 0, 0, 0)	14.0%
(0, 1, 0, 0)	6.0%
(1, 1, 0, 0)	7.1%
(2, 1, 0, 0)	7.7%
(0, 0, 1, 0)	1.6%
(1, 0, 1, 0)	2.6%
(2, 0, 1, 0)	3.6%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.1%
(2, 1, 1, 0)	3.1%

21.1.3 Some interesting errors

Some of the most interesting error plays in the NYN 2023 home games:

- The game WAS vs NYN on 2023-04-26 had 4 errors.
- The game NYA vs NYN on 2023-06-14 had only 2 errors, but they were consecutive.

21.1.4 Relative team rankings

We have 81 home games between the Mets and some visiting MLB team (vt). A number of teams did not play NYN in the regular season. However, of those that did, the Mets *lost* more games than won against¹:

¹We recorded that vt's W/T record in parentheses (W = numbers of games vt won against NYN, T = Total home games vt played vs the Mets at New York.

ANA (2/3), CIN (2/3), COL (2/3), LAN (2/3), SLN (2/3), TEX (2/3),
TOR (3/3), MIL (3/4), ATL (5/7) .

The teams that *split* their series were:

NYA (1/2) .

The teams that NYN *won* more than lost against

CLE (0/3), CHA (1/3), CHN (1/3), PIT (1/3), SDN (1/3),
SEA (1/3), SFN (1/3), TBA (1/3), ARI (1/4), PHI (1/6), MIA (2/6), WAS
(3/7),

The teams that did not play in any of the 81 home games in NYN 2023 were:

BAL, BOS, DET, HOU, KCA, MIN, OAK .

21.2 Game-by-game totals

The pitcher shall not, while in the 18-foot circle surrounding the pitcher's plate, touch the ball after touching his mouth or lips, or touch his mouth or lips while he is in contact with the pitcher's plate. The pitcher must clearly wipe the fingers of his pitching hand dry before touching the ball or the pitcher's plate.

– **MLB Rule Book** 6.02(c)(1)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for 2023 New York Mets is as follows [BA].

position	player	games played	OBP
1 (C)	Francisco Alvarez	108	.284
3 (1B)	Pete Alonso	144	.318
4 (2B)	Jeff McNeil	109	.333
5 (3B)	Brett Baty	102	.275
6 (SS)	Francisco Lindor	158	.336
7 (LF)	Mark Canha	51	.343
8 (CF)	Brandon Nimmo	138	.363
9 (RF)	Starling Marte	86	.301

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 NYN home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

21.2.1 April

	game	MIA vs NYN on 20230407
	state before	(2, 1, 0, 0, 6, 0, 4, 1)
1.	error	play,6,1,alonp001,00,>*B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1364 *In the bottom of the 6th inning, with a runner on 1st, NYN batter Pete Alonso had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Pete Alonso is still at bat, now with a 1-0 count.*

	game	MIA vs NYN on 20230408
	state before	(1, 0, 0, 0, 4, 0, 1, 1)
2.	error	play,4,1,phamt001,32,BCBBSX,E5/G56S.B-1
	state after	(1, 1, 0, 0, 4, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1365 *In the bottom of the 4th inning, with bases empty, NYN batter Tommy Pham hit a ground ball to shallow 3rd. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for MIA vs NYN on 2023-04-09

position	innings	player	position	innings	player
1	1	Braxton Garrett	1	1	Carlos Carrasco
1	5	Tanner Scott	1	5	Stephen Nogosek
1	7	Dylan Floro	1	9	Dennis Santana
1	7	JT Chargois	2	1	Francisco Alvarez
1	9	A.J. Puk	3	1	Pete Alonso
2	1	Nick Fortes	4	1	Jeff McNeil
3	1	Yuli Gurriel	4	2	Luis Guillorme
4	1	Luis Arraez	5	1	Eduardo Escobar
5	1	Jean Segura	6	1	Francisco Lindor
6	1	Jon Berti	7	1	Tim Lincecum
7	1	Bryan De La Cruz	8	1	Tommy Pham
8	1	Jazz Chisholm Jr.	9	1	Starling Marte
8	7	Garrett Hampson	9	2	Jeff McNeil
9	1	Avisail Garcia			
9	4	Jesus Sanchez			

	game	MIA vs NYN on 20230409
	state before	(2, 1, 0, 0, 8, 7, 2, 0)
3.	error	play,8,0,bertj001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 7, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1366 This game had 3 errors. The final score was 7-2.

In the top of the 8th inning, with a runner on 1st, MIA batter Jon Berti had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Jon Berti is still at bat, now with a 0-1 count.

	game	MIA vs NYN on 20230409
	state before	(1, 0, 0, 0, 8, 7, 2, 1)
4.	error	play,8,1,mcnej002,22,CBBCX,E7/F78.B-2
	state after	(1, 0, 1, 0, 8, 7, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 1367 *In the bottom of the 8th inning, with bases empty, NYN batter Jeff McNeil hit a fly ball to left center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	MIA vs NYN on 20230409
	state before	(0, 0, 0, 0, 9, 7, 2, 1)
5.	error	play,9,1,locat001,11,.CBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 9, 7, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1368 *In the bottom of the 9th inning, with bases empty, NYN batter Tim Locastro hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	SDN vs NYN on 20230410
	state before	(0, 1, 0, 0, 3, 0, 0, 0)
6.	error	play,3,0,nolaa002,11,LB+1,OA.1-2(E2/TH)
	state after	(0, 0, 1, 0, 3, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1369 *In the top of the 3rd inning, with a runner on 1st, SDN batter Austin Nola had a 1-1 count. The catcher attempted to pickoff the runner on 1st. A throwing error by the catcher allowed the runner on 1st to advance to 2nd. Nola is still at-bat with a 1-1 count.*

	game	SDN vs NYN on 20230412
	state before	(0, 1, 1, 0, 6, 2, 3, 1)
7.	error	play,6,1,escoe001,00,.X,FC/SH/BG1S.2-3;1-2;B-1(E1)
	state after	(0, 1, 1, 1, 6, 2, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1370 *In the bottom of the 6th inning, with runners on 1st and 2nd, NYN batter Eduardo Escobar hit a bunt into a fielder's choice at very shallow infield. The runners on 1st and 2nd each advanced one base. An error by the pitcher allowed the batter to reach 1st. The bases are loaded.*

	game	WAS vs NYN on 20230425
	state before	(1, 1, 0, 0, 4, 2, 0, 0)
8.	error	play,4,0,abrac001,00,X,FC6/G4.1-2(E6);B-1
	state after	(1, 1, 1, 0, 4, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1371 *In the top of the 4th inning, with a runner on 1st, WAS batter C. J. Abrams hit a ground ball into a fielder's choice at 2nd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for WAS vs NYN on 2023-04-26

position	innings	player	position	innings	player
1	1	MacKenzie Gore	1	1	Kodai Senga
1	7	Carl Edwards Jr.	1	6	Jeff Brigham
1	7	Hunter Harvey	1	7	Brooks Raley
1	9	Kyle Finnegan	1	8	Adam Ottavino
2	1	Keibert Ruiz	1	9	David Robertson
3	1	Dominic Smith	2	1	Tomas Nido
4	1	Luis Garcia	2	8	Francisco Alvarez
5	1	Jeimer Candelario	3	1	Pete Alonso
6	1	CJ Abrams	4	1	Jeff McNeil
7	1	Alex Call	5	1	Eduardo Escobar
8	1	Victor Robles	5	8	Brett Baty
9	1	Lane Thomas	6	1	Francisco Lindor
			7	1	Mark Canha
			8	1	Brandon Nimmo
			9	1	Starling Marte

	game	WAS vs NYN on 20230426
	state before	(1, 0, 0, 0, 5, 2, 1, 0)
9.	error	play,5,0,menej001,10,BX,E5/G7S.B-1
	state after	(1, 1, 0, 0, 5, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1372 This game had 4 errors. The final score was 4-1.

In the top of the 5th inning, with bases empty, WAS batter Joey Meneses hit a ground ball to shalow left field. An error by the third baseman allowed the batter to reach 1st base.

	game	WAS vs NYN on 20230426
	state before	(0, 0, 0, 0, 5, 2, 1, 1)
10.	error	play,5,1,escoc001,11,S.BX,E6/G6.B-1
	state after	(0, 1, 0, 0, 5, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1373 *In the bottom of the 5th inning, with bases empty, NYN batter Eduardo Escobar hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	WAS vs NYN on 20230426
	state before	(0, 0, 0, 0, 6, 2, 1, 0)
11.	error	play,6,0,smitd008,01,.CX,E3/P23.B-2
	state after	(0, 0, 1, 0, 6, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1374 *In the top of the 6th inning, with bases empty, WAS batter Dominic Smith hit a pop up to shallow 1st base. An error by the first baseman allowed the batter to reach 2nd base.*

	game	WAS vs NYN on 20230426
	state before	(2, 1, 0, 0, 8, 3, 1, 0)
12.	error	play,8,0,calla002,10,1B1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 8, 3, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1375 *In the top of the 8th inning, with a runner on 1st, WAS batter Alex Call has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed the runner on 1st to safely advance to 2nd.*

	game	WAS vs NYN on 20230427
	state before	(2, 0, 0, 0, 3, 1, 1, 1)
13.	error	play,3,1,alonp001,22,BFSBX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 3, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1376 This game had 2 errors. The final score was 8-9.
In the bottom of the 3rd inning, with bases empty, NYN batter Pete Alonso hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	WAS vs NYN on 20230427
	state before	(0, 1, 1, 0, 8, 3, 7, 0)
14.	error	play,8,0,garrs003,01,FX,FC6/G6M.2-3;1-2(E6);B-1
	state after	(0, 1, 1, 1, 8, 3, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 1377 *In the top of the 8th inning, with runners on 1st and 2nd, WAS batter Stone Garrett hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

21.2.2 May

	game	ATL vs NYN on 20230501 (1/2)
	state before	(0, 0, 0, 0, 4, 0, 1, 1)
15.	error	play,4,1,mcnej002,00,X,E6/G4M.B-1
	state after	(0, 1, 0, 0, 4, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1378 This is game 1 of a double header. This final score was 3-5.

In the bottom of the 4th inning, with bases empty, NYN batter Jeff McNeil hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.

	game	ATL vs NYN on 20230501 (2/2)
	state before	(0, 1, 1, 1, 7, 9, 5, 1)
16.	error	play,7,1,voged001,11,C*BX,36(1)/FO/G34.3-H;2-H(E6/TH)(NR);B-1
	state after	(1, 1, 0, 0, 7, 9, 7, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1379 This is game 2 of a double header. This final score was 9-8.

In the bottom of the 7th inning, with bases loaded, NYN batter Daniel Vogelbach hit a fly ball to the 1st/2nd base gap. At 2nd, the shortstop (with an assist from the first baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	COL vs NYN on 20230505
	state before	(2, 1, 0, 0, 8, 0, 1, 0)
17.	error	play,8,0,diaze005,31,BBSB+1,P01(E2/TH).1-2
	state after	(2, 0, 1, 0, 8, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1380 *In the top of the 8th inning, with a runner on 1st, COL batter Elias Diaz has a 3-1 count. During a pickoff attempt at 1st, a throwing error by the catcher allowed the runner on 1st to advance to 2nd. Diaz is still at-bat with a 3-1 count.*

	game	COL vs NYN on 20230507
	state before	(1, 0, 1, 1, 1, 1, 1, 1)
18.	error	play,1,1,batyb001,10,BX,S9/G34.3-H;2-H;B-2(E9/TH)
	state after	(1, 0, 1, 0, 1, 1, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1381 *In the bottom of the 1st inning, with runners on 2nd and 3rd, NYN batter Brett Baty hit a ground ball single to the 1st/2nd base gap. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the right fielder allowed the better to reach 2nd.*

	game	TBA vs NYN on 20230518
	state before	(0, 0, 0, 0, 3, 1, 1, 0)
19.	error	play,3,0,bethc001,02,CSFFFX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 3, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1382 *In the top of the 3rd inning, with bases empty, TBA batter Christian Bethancourt hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CLE vs NYN on 20230519
	state before	(2, 0, 0, 0, 4, 4, 0, 1)
20.	error	play,4,1,batyb001,22,FBFBX,4E3/G4.B-1
	state after	(2, 1, 0, 0, 4, 4, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1383 *In the bottom of the 4th inning, with bases empty, NYN batter Brett Baty hit a ground ball to 2nd base. An error by the first baseman (with an assist from the second baseman) allowed the batter to reach 1st base.*

21.2.3 June

	game	PHI vs NYN on 20230601
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
21.	error	play,1,0,castn001,12,FSBF>B,SB3;SB2.2-H(E2/TH) (NR) ;1-3
	state after	(1, 0, 0, 1, 1, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1384 *In the top of the 1st inning, with runners on 1st and 2nd, PHI batter Nick Castellanos had a 1-2 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd and the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Castellanos is still at bat, now with a 2-2 count.*

	game	TOR vs NYN on 20230603
	state before	(1, 0, 0, 0, 8, 1, 1, 1)
22.	error	play,8,1,alonp001,11,BSX,E5/TH/G56S.B-1
	state after	(1, 1, 0, 0, 8, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1385 *In the bottom of the 8th inning, with bases empty, NYN batter Pete Alonso hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TOR vs NYN on 20230604
	state before	(2, 0, 1, 0, 3, 3, 0, 0)
23.	error	play,3,0,varsd001,31,BC*BBB+2,W.2-H(E2/TH)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 3, 3, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1386 *In the top of the 3rd inning, with a runner on 2nd, TOR batter Daulton Varsho had a 3-1 count. After the next pitch, which was a ball, the catcher tried to pickoff the runner on 2nd. A throwing error by the catcher allowed the runner on 2nd to score (un-earned, no RBI) and the batter to reach 2nd.*

	game	NYA vs NYN on 20230613
	state before	(1, 1, 1, 0, 3, 1, 4, 1)
24.	error	play,3,1,guill001,01,2FX,FC4/G4.2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 3, 1, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1387 *In the bottom of the 3rd inning, with runners on 1st and 2nd, NYN batter Luis Guillorme hit a ground ball into a fielder's choice at 2nd. The runner on 2nd to advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	NYA vs NYN on 20230614
	state before	(1, 1, 1, 0, 7, 1, 1, 0)
25.	error	play,7,0,kinei001,22,BC*BFX,64(1)/F0/G6.2-H(E4/TH)(NR);B-1
	state after	(2, 1, 0, 0, 7, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1388 *This game had 2 errors. The final score was 3-4. In the top of the 7th inning, with runners on 1st and 2nd, NYA batter Isiah Kiner-Falefa hit a ground ball to short. The second baseman (with an assist from the shortstop) forced out the runner from 1st. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	NYA vs NYN on 20230614
	state before	(2, 1, 0, 0, 7, 2, 1, 0)
26.	error	play,7,0,mckib001,11,.SB1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 2, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1389 *This play is a consecutive error since McKinney bats right after Kiner-Falefa.*

In the top of the 7th inning, with a runner on 1st, NYA batter Billy McKinney had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. McKinney is still at bat, now with a 2-1 count.

	game	SLN vs NYN on 20230617
	state before	(2, 0, 0, 0, 5, 4, 1, 0)
27.	error	play,5,0,contw001,01,CX,E6/G6D.B-1
	state after	(2, 1, 0, 0, 5, 4, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1390 *In the top of the 5th inning, with bases empty, SLN batter Willson Contreras hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	SLN vs NYN on 20230618
	state before	(1, 0, 0, 0, 2, 2, 1, 0)
28.	error	play,2,0,kniza001,02,.CFX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 2, 2, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1391 This game had 2 errors. The final score was 8-7. *In the top of the 2nd inning, with bases empty, SLN batter Andrew Knizner hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	SLN vs NYN on 20230618
	state before	(2, 0, 0, 0, 7, 7, 7, 0)
29.	error	play,7,0,carld002,32,BSBBF.X,E3/G34.B-1
	state after	(2, 1, 0, 0, 7, 7, 7, 0)
	fielder	3 (1st baseman on the home team)

Summary 1392 *In the top of the 7th inning, with bases empty, SLN batter Dylan Carlson hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	MIL vs NYN on 20230626
	state before	(2, 0, 0, 0, 2, 0, 0, 0)
30.	error	play,2,0,wiemj001,00,X,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1393 This game had 2 errors. The final score was 2-1.
In the top of the 2nd inning, with bases empty, MIL batter Joey Wiemer hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	MIL vs NYN on 20230626
	state before	(0, 1, 0, 0, 4, 0, 0, 1)
31.	error	play,4,1,lindf001,10,B>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 4, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1394 *In the bottom of the 4th inning, with a runner on 1st, NYN batter Francisco Lindor had a 1-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Lindor is still at bat, now with a 1-1 count.*

	game	MIL vs NYN on 20230629
	state before	(1, 1, 0, 0, 7, 3, 2, 1)
32.	error	play,7,1,mendd002,11,.C1BX,FC5/G6.1-3(E5);B-1
	state after	(1, 1, 0, 1, 7, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1395 This game had 2 errors. The final score was 3-2.
In the bottom of the 7th inning, with a runner on 1st, NYN batter Danny Mendick hit a ground ball into a fielder's choice at short. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	MIL vs NYN on 20230629
	state before	(0, 0, 0, 0, 8, 3, 2, 0)
33.	error	play,8,0,perkb002,00,...L,FLE2
	state after	(0, 0, 0, 0, 8, 3, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1396 This is an *infinity play*, with no change in game state. In the top of the 8th inning, with bases empty and a 0-0 count, MIL batter Blake Perkins hit a fly ball foul behind the plate. An error by the catcher allowed the batter to remain safe. Perkins is still at bat, now with a 1-0 count.

Visiting/Home team rosters for SFN vs NYN on 2023-06-30

position	innings	player	position	innings	player
1	1	Alex Cobb	1	1	Carlos Carrasco
1	6	Taylor Rogers	1	6	Jeff Brigham
1	7	Sean Manaea	1	7	Brooks Raley
1	8	Tyler Rogers	1	8	David Robertson
1	9	Camilo Doval	1	9	Grant Hartwig
2	1	Patrick Bailey	2	1	Omar Narvaez
3	1	LaMonte Wade Jr.	3	1	Pete Alonso
4	1	Thairo Estrada	4	1	Jeff McNeil
5	1	J.D. Davis	5	1	Brett Baty
6	1	Brandon Crawford	6	1	Francisco Lindor
6	7	Casey Schmitt	7	1	Tommy Pham
7	1	Blake Sabol	8	1	Brandon Nimmo
7	8	Luis Matos	9	1	Mark Canha
7	9	Austin Slater			
8	1	Luis Matos			
8	8	Bryce Johnson			
9	1	Joc Pederson			
9	9	Luis Matos			

	game	SFN vs NYN on 20230630
	state before	(0, 1, 0, 0, 2, 1, 1, 1)
34.	error	play,2,1,batyb001,21,BSBX,FC3/G34S.1-3(E3/TH);B-2
	state after	(0, 0, 1, 1, 2, 1, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1397 This game had 3 errors. The final score was 5-4.
In the bottom of the 2nd inning, with a runner on 1st, NYN batter Brett Baty hit a ground ball into a fielder's choice at 1st. A throwing error by the 1st baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	SFN vs NYN on 20230630
	state before	(1, 0, 1, 1, 2, 1, 1, 1)
35.	error	play,2,1,narvo001,10,BX,8/SF/L8.3-H(UR);2-3(E8/TH)
	state after	(2, 0, 0, 1, 2, 1, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 1398 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, NYN batter Omar Narvaez hit a line drive to center field. The batter was put out by the center fielder. The runner on 3rd scored (1 RBI). A throwing error by the center fielder allowed the runner on 2nd to advance to 3rd.*

	game	SFN vs NYN on 20230630
	state before	(1, 0, 0, 0, 8, 2, 4, 0)
36.	error	play,8,0,pedej001,12,CBSFX,E3/TH/G3.B-1
	state after	(1, 1, 0, 0, 8, 2, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1399 *In the top of the 8th inning, with bases empty, SFN batter Joc Pederson hit a ground ball to 1st base. A throwing error by the first baseman allowed the batter to reach 1st base.*

21.2.4 July

	game	SFN vs NYN on 20230701
	state before	(0, 1, 0, 0, 7, 0, 4, 0)
37.	error	play,7,0,bailp001,22,BFSBX,FC3/G34.1-3(E3/TH);B-1
	state after	(0, 1, 0, 1, 7, 0, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1400 *In the top of the 7th inning, with a runner on 1st, SFN batter Patrick Bailey hit a ground ball into a fielder's choice at 1st. A throwing error by the 1st allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	SFN vs NYN on 20230702
	state before	(2, 1, 1, 1, 3, 1, 1, 1)
38.	error	play,3,1,mcnej002,02,CCX,S4/G4M.3-H;2-H(E6)(NR)(UR);1-2
	state after	(2, 1, 1, 0, 3, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1401 This game had 2 errors. The final score was 4-8. *In the bottom of the 3rd inning, with bases loaded, NYN batter Jeff McNeil hit a ground ball single to 2nd base. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 2nd to also score (unearned, no RBI) and the runner on 1st to advance to 2nd. The batter took 1st.*

	game	SFN vs NYN on 20230702
	state before	(2, 1, 0, 0, 7, 4, 6, 1)
39.	error	play,7,1,batyb001,02,CCX,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 7, 4, 6, 1)
	fielder	2 (catcher on the visiting team)

Summary 1402 *In the bottom of the 7th inning, with a runner on 1st, NYN batter Brett Baty had a 0-2 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced to 2nd.*

	game	LAN vs NYN on 20230714
	state before	(1, 0, 0, 0, 5, 3, 0, 1)
40.	error	play,5,1,batyb001,22,BCSBX,C/E2.B-1
	state after	(1, 1, 0, 0, 5, 3, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1403 *In the bottom of the 5th inning, with bases empty, NYN batter Brett Baty had a 2-2 count. Catcher's interference allowed the batter to take 1st.*

	game	LAN vs NYN on 20230715
	state before	(2, 0, 0, 0, 7, 1, 1, 0)
41.	error	play,7,0,rojam002,20,BBX,E9/F9LS.B-2
	state after	(2, 0, 1, 0, 7, 1, 1, 0)
	fielder	9 (right field on the home team)

Summary 1404 *This game had 2 errors. The final score was 5-1. In the top of the 7th inning, with bases empty, LAN batter Miguel Rojas hit a fly ball to shallow right field. An error by the right fielder allowed the batter to reach 2nd base.*

	game	LAN vs NYN on 20230715
	state before	(1, 0, 1, 1, 9, 2, 1, 0)
42.	error	play,9,0,muncm001,01,CX,E5/P56.3-H(NR);B-1
	state after	(1, 1, 1, 0, 9, 3, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1405 *In the top of the 9th inning, with runners on 2nd and 3rd, LAN batter Max Muncy hit a pop up to 3rd base. An error by the third baseman allowed the runner on 3rd to score (no RBI) and the batter to reach 1st.*

	game	CHA vs NYN on 20230720
	state before	(0, 0, 0, 0, 6, 2, 1, 0)
43.	error	play,6,0,rob1002,00,.X,E3/G34.B-1
	state after	(0, 1, 0, 0, 6, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1406 *In the top of the 6th inning, with bases empty, CHA batter Luis Robert hit a ground ball to 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	WAS vs NYN on 20230727
	state before	(2, 1, 0, 0, 4, 0, 0, 1)
44.	error	play,4,1,batyb001,01,C11,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 4, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1407 *This game had 2 errors. The final score was 1-2. In the bottom of the 4th inning, with a runner on 1st, NYN batter Brett Baty has a 0-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Baty is still at bat with a 0-1 count.*

	game	WAS vs NYN on 20230727
	state before	(1, 0, 1, 0, 6, 0, 0, 0)
45.	error	play,6,0,candj002,32,C1B>B.FBX,C/E2.B-1
	state after	(1, 1, 1, 0, 6, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1408 *In the top of the 6th inning, with a runner on 2nd, WAS batter Jeimer Candelario had a full count. Catcher's interference allowed the batter to take 1st.*

	game	WAS vs NYN on 20230728
	state before	(1, 0, 0, 0, 4, 0, 0, 1)
46.	error	play,4,1,mcnej002,11,CBF,FLE2
	state after	(1, 0, 0, 0, 4, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1409 This is an *infinity play*, with no change in game state. The final score was 1-5.

In the bottom of the 4th inning, with bases empty and a 1-1 count, NYN batter Jeff McNeil hit a fly ball foul behind home plate. An error by the catcher allowed the batter to remain safe. McNeil is still at bat, now with a 1-2 count.

Visiting/Home team rosters for WAS vs NYN on 2023-07-29

position	innings	player	position	innings	player
1	1	Patrick Corbin	1	1	Carlos Carrasco
1	6	Jordan Weems	1	3	Reed Garrett
1	8	Kyle Finnegan	1	6	Dominic Leone
1	8	Mason Thompson	1	7	Drew Smith
1	9	Andres Machado	1	8	Trevor Gott
2	1	Riley Adams	1	9	Grant Hartwig
3	1	Dominic Smith	2	1	Francisco Alvarez
4	1	Luis Garcia	3	1	Mark Canha
5	1	Jeimer Candelario	3	8	Mark Vientos
6	1	CJ Abrams	4	1	Danny Mendick
7	1	Stone Garrett	4	8	Jeff McNeil
8	1	Alex Call	5	1	Mark Vientos
9	1	Lane Thomas	5	8	Brett Baty
			6	1	Francisco Lindor
			7	1	Tommy Pham
			8	1	Brandon Nimmo
			9	1	Jeff McNeil
			9	8	Mark Canha
			9	9	DJ Stewart

	game	WAS vs NYN on 20230729
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
47.	error	play,1,0,abrac001,12,SBF.X,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1410 This game had 3 errors. The final score was 11-6. *In the top of the 1st inning, with bases empty, WAS batter C.J. Abrams hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	WAS vs NYN on 20230729
	state before	(1, 1, 0, 0, 8, 8, 4, 0)
48.	error	play,8,0,abrac001,10,B1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 8, 8, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 1411 *In the top of the 8th inning, with a runner on 1st, WAS batter C. J. Abrams has a 1-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Abrahms is still at bat with a 1-0 count.*

	game	WAS vs NYN on 20230729
	state before	(2, 0, 0, 1, 9, 9, 4, 0)
49.	error	play,9,0,garrs003,00,X,E5/G56.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 9, 10, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1412 *In the top of the 9th inning, with a runner on 3rd, WAS batter Stone Garrett hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 3rd to score (no RBI) and the batter to reach 1st.*

	game	WAS vs NYN on 20230730
	state before	(1, 0, 0, 0, 7, 1, 5, 0)
50.	error	play,7,0,garrs003,01,.FX,E8/F89S.B-2
	state after	(1, 0, 1, 0, 7, 1, 5, 0)
	fielder	8 (center field on the home team)

Summary 1413 *In the top of the 7th inning, with bases empty, WAS batter Stone Garrett hit a fly ball to right center field. An error by the center fielder allowed the batter to reach 2nd base.*

21.2.5 August

	game	ATL vs NYN on 20230811
	state before	(2, 1, 1, 1, 4, 0, 0, 0)
51.	error	play,4,0,rosae001,20,BBX,S8/L89D.3-H;2-H;1-H(E8)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 3, 0, 0)
	fielder	8 (center field on the home team)

Summary 1414 *In the top of the 4th inning, with bases loaded, ATL batter Eddie Rosario hit a line drive single to deep right center field. The runners on 2nd and 3rd scored (2 RBIs). An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	ATL vs NYN on 20230812 (1/2)
	state before	(0, 1, 0, 0, 8, 1, 0, 0)
52.	error	play,8,0,rilea001,11,BFX,FC4/G89.1-H(E4)(E8)(NR)(UR);B-1
	state after	(0, 1, 0, 0, 8, 2, 0, 0)
	fielder	4,8 (2nd baseman,center field on the home team)

Summary 1415 This play, in game 1 of a doubleheader, had 2 errors by 2 different fielders. The final score was 6-0.
In the top of the 8th inning, with a runner on 1st, ATL batter Austin Riley hit a ground ball into a fielder's choice at right center field. An error by the center fielder, then an error by the second baseman, allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 1st.

	game	ATL vs NYN on 20230813
	state before	(2, 1, 1, 1, 5, 3, 4, 1)
53.	error	play,5,1,stewd002,11,BSX,C/E2.3-H;2-3;1-2;B-1
	state after	(2, 1, 1, 1, 5, 3, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1416 *In the bottom of the 5th inning, with bases loaded, NYN batter DJ Stewart had a 1-1 count. Catcher's interference allowed the batter to take 1st. The runner on 3rd scored (1 RBI). The runners on 1st and 2nd base each advanced one base. The bases are still loaded.*

54.	game	PIT vs NYN on 20230814
	state before	(2, 1, 1, 0, 3, 2, 2, 0)
	error	play,3,0,rodre006,12,FBFX,E1/G23.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 3, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1417 *In the top of the 3rd inning, with runners on 1st and 2nd, PIT batter Endy Rodriguez hit a ground ball to shallow 3rd. An error by the pitcher allowed the runner on 2nd to advance to 3rd, the runner on 1st to advance to 2nd, and the batter to reach 1st. The bases are loaded.*

55.	game	PIT vs NYN on 20230815
	state before	(0, 0, 0, 0, 9, 7, 4, 0)
	error	play,9,0,trioj001,22,.CCBBX,E6/G56.B-1
	state after	(0, 1, 0, 0, 9, 7, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1418 *In the top of the 9th inning, with bases empty, PIT batter Jared Triolo hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.*

56.	game	PIT vs NYN on 20230816
	state before	(0, 0, 0, 0, 6, 3, 6, 1)
	error	play,6,1,orter001,01,CX,E5/G6.B-1
	state after	(0, 1, 0, 0, 6, 3, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1419 This game had 2 errors. The final score was 3-8.
In the bottom of the 6th inning, with bases empty, NYN batter Rafael Ortega hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.

	game	PIT vs NYN on 20230816
	state before	(2, 1, 0, 0, 7, 3, 7, 1)
57.	error	play,7,1,narvo001,01,F1X,E3/G3.1-3;B-1
	state after	(2, 1, 0, 1, 7, 3, 7, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1420 *In the bottom of the 7th inning, with a runner on 1st, NYN batter Omar Narvaez hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	ANA vs NYN on 20230825
	state before	(0, 0, 1, 0, 7, 2, 1, 1)
58.	error	play,7,1,arauj001,11,CBX,FC/SH/BG2-.2-3;B-1(E2)
	state after	(0, 1, 0, 1, 7, 2, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1421 *In the bottom of the 7th inning, with a runner on 2nd, NYN batter Jonathan Arauz hit a bunt into a fielder's choice at very shallow infield. The runner on 2nd advanced to 3rd. An error by the catcher allowed the batter to reach 1st.*

	game	ANA vs NYN on 20230826
	state before	(1, 1, 1, 0, 4, 5, 0, 1)
59.	error	play,4,1,voged001,00,C,SB3;SB2.2-H(E3/TH) (NR) ;1-3
	state after	(1, 0, 0, 1, 4, 5, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1422 *In the bottom of the 4th inning, with runners on 1st and 2nd, NYN batter Daniel Vogelbach had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd and the runner on 1st stole 2nd. A throwing error by the first baseman allowed the runner from 2nd (now on 3rd) to score (no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Vogelbach is still at bat, now with a 0-1 count.*

	game	ANA vs NYN on 20230827
	state before	(2, 1, 0, 0, 3, 0, 0, 0)
60.	error	play,3,0,ohtas001,11,C1B1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 3, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1423 *In the top of the 3rd inning, with a runner on 1st, ANA batter Shohei Ohtani has a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Ohtani is still at bat with a 1-1 count.*

	game	TEX vs NYN on 20230830
	state before	(2, 1, 0, 0, 9, 5, 5, 0)
61.	error	play,9,0,lowen001,00,1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 9, 5, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1424 *In the top of the 9th inning, with a runner on 1st, TEX batter Nathaniel Lowe has a 0-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Lowe is still at bat with a 0-0 count.*

21.2.6 September

	game	SEA vs NYN on 20230902
	state before	(2, 0, 0, 1, 6, 6, 3, 0)
62.	error	play,6,0,rodrj007,10,BX,E5/TH/G56S.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 7, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1425 *In the top of the 6th inning, with a runner on 3rd, SEA batter Julio Rodriguez hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	SEA vs NYN on 20230903
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
63.	error	play,1,1,lindf001,02,CFFFFFFX,E4/G4.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1426 *In the bottom of the 1st inning, with bases empty, NYN batter Francisco Lindor hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	ARI vs NYN on 20230911
	state before	(1, 1, 1, 0, 4, 2, 1, 1)
64.	error	play,4,1,maurr002,02,CFX,D9/L89D+.2-H;1-H;B-3(E9/TH)
	state after	(1, 0, 0, 1, 4, 2, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1427 *In the bottom of the 4th inning, with runners on 1st and 2nd, NYN batter Ronny Mauricio hit a hard line drive double to right center field. The runners on 1st and 2nd both scored (2 RBIs). A throwing error by the right fielder allowed the batter to reach 3rd.*

	game	ARI vs NYN on 20230912
	state before	(0, 1, 0, 0, 6, 1, 6, 0)
65.	error	play,6,0,walkc002,10,B1,P01(E1/TH).1-2
	state after	(0, 0, 1, 0, 6, 1, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 1428 *In the top of the 6th inning, with a runner on 1st, ARI batter Christian Walker has a 1-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Walker is still at bat with a 1-0 count.*

	game	ARI vs NYN on 20230913
	state before	(1, 1, 1, 1, 6, 0, 6, 1)
66.	error	play,6,1,alonp001,01,CX,FC1/G1.3-H(E1)(NR)(UR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 6, 0, 7, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1429 *This game had 2 errors. The final score was 1-7. In the bottom of the 6th inning, with bases loaded, NYN batter Pete Alonso hit a ground ball into a fielder's choice at shallow infield. An error by the pitcher allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are still loaded.*

	game	ARI vs NYN on 20230913
	state before	(0, 1, 0, 0, 8, 0, 7, 0)
67.	error	play,8,0,martk001,02,FFX,FC6/G6.1-3(E6);B-1
	state after	(0, 1, 0, 1, 8, 0, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 1430 *In the top of the 8th inning, with a runner on 1st, ARI batter Ketel Marte hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	ARI vs NYN on 20230914
	state before	(1, 1, 0, 0, 7, 0, 7, 1)
68.	error	play,7,1,mcnej002,02,SCX,3/G3.1X2(3E6)
	state after	(2, 0, 1, 0, 7, 0, 7, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1431 *In the bottom of the 7th inning, with a runner on 1st, NYN batter Jeff McNeil hit a ground ball to the first baseman and put out the batter. An error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to safely advance to 2nd.*

	game	CIN vs NYN on 20230915
	state before	(1, 1, 0, 0, 5, 0, 0, 0)
69.	error	play,5,0,badeh001,12,CBCX,FC5/G6D.1-3(E5);B-1
	state after	(1, 1, 0, 1, 5, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1432 *This game had 2 errors. The final score was 5-3. In the top of the 5th inning, with a runner on 1st, CIN batter Harrison Bader hit a ground ball into a fielder's choice at deep short. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	CIN vs NYN on 20230915
	state before	(1, 1, 0, 1, 5, 0, 0, 0)
70.	error	play,5,0,mail1001,02,ML1B,WP.3-H(NR)(UR);1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1433 *In the top of the 5th inning, with runners on 1st and 3rd, CIN batter Luke Maile had a 0-2 count. The next pitch was a wild pitch, which allowed the runner from 3rd to score (unearned, no RBI). A throwing error by the catcher allowed the runner on 1st to advance to 3rd. Maile is still at bat, now with a 1-2 count.*

	game	CIN vs NYN on 20230916
	state before	(2, 1, 0, 1, 2, 0, 1, 0)
71.	error	play,2,0,delae003,02,CF>B,SB2.3-H(E2/TH)(NR)(UR);1-3
	state after	(2, 0, 0, 1, 2, 1, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1434 This game had 2 errors, both with Elly De La Cruz at-bat . The final score was 3-2.

In the top of the 2nd inning, with runners on 1st and 3rd, CIN batter Elly De La Cruz had a 0-2 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (no RBI) and the runner on 1st to advance to 3rd. Berti is still at bat, now with a 1-2 count.

	game	CIN vs NYN on 20230916
	state before	(2, 1, 1, 0, 8, 3, 2, 0)
72.	error	play,8,0,delae003,32,BCBF.B>X,E4/G4.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 8, 3, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1435 *In the top of the 8th inning, with runners on 1st and 2nd, (the same CIN batter as in the last error) Elly De La Cruz hit a ground ball to 2nd base. An error by the second baseman allowed the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for CIN vs NYN on 2023-09-17

position	innings	player	position	innings	player
1	1	Brandon Williamson	1	1	Jose Quintana
1	5	Carson Spiers	1	7	Drew Smith
1	8	Sam Moll	1	8	Jeff Bringham
2	1	Tyler Stephenson	1	9	Brooks Raley
3	1	Christian Encarnacion-Strand	2	1	Francisco Alvarez
4	1	Spencer Steer	3	1	Pete Alonso
5	1	Noelvi Marte	4	1	Ronny Mauricio
6	1	Elly De La Cruz	5	1	Mark Vientos
7	1	Nick Senzel	5	8	Jonathan Arauz
7	7	Will Benson	6	1	Francisco Lindor
8	1	Harrison Bader	7	1	Tim Locastro
8	7	TJ Friedl	7	8	Jeff McNeil
9	1	Hunter Renfroe	8	1	Jeff McNeil
			8	8	Rafael Ortega
			9	1	DJ Stewart

	game	CIN vs NYN on 20230917
	state before	(2, 0, 0, 0, 1, 1, 0, 1)
73.	error	play,1,1,alongp001,02,FSX,E5/G56.B-1
	state after	(2, 1, 0, 0, 1, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1436 This game had 3 errors. The final score was 4-8.

In the bottom of the 1st inning, with bases empty, NYN batter Pete Alonso hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	CIN vs NYN on 20230917
	state before	(2, 1, 0, 0, 5, 2, 3, 1)
74.	error	play,5,1,mcnej002,01,F.X,E3/G34S.1-2;B-1
	state after	(2, 1, 1, 0, 5, 2, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1437 *In the bottom of the 5th inning, with a runner on 1st, NYN batter Jeff McNeil hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CIN vs NYN on 20230917
	state before	(0, 1, 0, 0, 7, 2, 4, 1)
75.	error	play,7,1,vienm001,01,SX,FC1/G1S.1-3(E1/TH);B-1
	state after	(0, 1, 0, 1, 7, 2, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1438 *In the bottom of the 7th inning, with a runner on 1st, NYN batter Mark Vientos hit a ground ball into a fielder's choice at very shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	MIA vs NYN on 20230927
	state before	(1, 0, 0, 1, 7, 1, 6, 0)
76.	error	play,7,0,solej001,00,X,E9/L89/SF.3-H;B-1
	state after	(1, 1, 0, 0, 7, 2, 6, 0)
	fielder	9 (right field on the home team)

Summary 1439 *In the top of the 7th inning, with a runner on 3rd, MIA batter Jorge Soler hit a line drive sacrifice fly to right field. An error by the right fielder allowed the runner on 3rd to score (1 RBI) and the batter to reach 1st.*

	game	MIA vs NYN on 20230927 (1/2)
	state before	(2, 0, 1, 0, 8, 2, 2, 0)
77.	error	play,8,0,hampg001,21,B*BFX,E6/TH/G6.2-3;B-1
	state after	(2, 1, 0, 1, 8, 2, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1440 This 1st game of a doubleheader had 2 errors. The final score was 4-2.

In the top of the 8th inning, with a runner on 2nd, MIA batter Garrett Hampson hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	MIA vs NYN on 20230927 (1/2)
	state before	(0, 1, 1, 1, 9, 2, 2, 0)
78.	error	play,9,0,gurry001,10,.BX,FC5/G5S.3-H(E5) (NR) (UR);2-3;1-2;B-1
	state after	(0, 1, 1, 1, 9, 3, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1441 *In the top of the 9th inning, with bases loaded, MIA batter Yulieski Gurriel hit a ground ball into a fielder's choice at shallow 3rd. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are still loaded.*

	game	MIA vs NYN on 20230928
	state before	(1, 1, 0, 0, 4, 0, 0, 1)
79.	error	play,4,1,vienm001,22,FC.*BBF>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 4, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1442 *In the bottom of the 4th inning, with a runner on 1st, NYN batter Mark Vientos had a 2-2 count. During the next pitch,*

which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Vientos is still at bat, now with a full count.

Visiting/Home team rosters for PHI vs NYN on 2023-09-30 (2/2)

position	innings	player	position	innings	player
1	1	Michael Plassmeyer	1	1	Jose Quintana
1	4	Jeff Hoffman	1	5	Phil Bickford
1	5	Orion Kerker	1	6	Reed Garrett
1	6	Jose Alvarado	1	9	Drew Smith
1	7	Craig Kimbrel	2	1	Francisco Alvarez
1	8	Cristopher Sanchez	3	1	Mark Vientos
2	1	Garrett Stubbs	4	1	Ronny Mauricio
3	1	Weston Wilson	4	9	Luis Guillorme
3	7	Jake Cave	5	1	Brett Baty
4	1	Edmundo Sosa	6	1	Francisco Lindor
4	7	Weston Wilson	6	9	Ronny Mauricio
5	1	Alec Bohm	7	1	Tim Locastro
6	1	Trea Turner	8	1	Rafael Ortega
6	7	Edmundo Sosa	9	1	DJ Stewart
7	1	Cristian Pache			
8	1	Johan Rojas			
9	1	Nick Castellanos			

	game	PHI vs NYN on 20230930 (2/2)
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
80.	error	play,2,1,lindf001,22,BCFFBX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1443 This game 2 of a doubleheader had 3 errors. The final score was 4-11.

In the bottom of the 2nd inning, with bases empty, NYN batter Francisco Lindor hit a ground ball to short. A throwing error by the short-stop allowed the batter to reach 1st base.

	game	PHI vs NYN on 20230930 (2/2)
	state before	(0, 1, 0, 0, 3, 0, 3, 1)
81.	error	play,3,1,maurr002,00,X,D7/L7LD.1-H(E7/TH) (NR)
	state after	(0, 0, 1, 0, 3, 0, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 1444 *In the bottom of the 3rd inning, with a runner on 1st, NYN batter Ronny Mauricio hit a line drive double to deep left field. A throwing error by the left fielder allowed the runner on 1st to score (unearned, no RBI) The batter took 2nd.*

	game	PHI vs NYN on 20230930 (2/2)
	state before	(1, 0, 0, 0, 4, 0, 8, 0)
82.	error	play,4,0,stubg001,12,CFBX,E3/G3.B-1
	state after	(1, 1, 0, 0, 4, 0, 8, 0)
	fielder	3 (1st baseman on the home team)

Summary 1445 *In the top of the 4th inning, with bases empty, PHI batter Garrett Stubbs hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	PHI vs NYN on 20231001
	state before	(0, 1, 0, 0, 4, 1, 1, 1)
83.	error	play,4,1,alonp001,11,BS1,P01(E1/TH) .1-2
	state after	(0, 0, 1, 0, 4, 1, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1446 *In the bottom of the 4th inning, with a runner on 1st, NYN batter Pete Alonso has a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Alonso is still at bat with a 1-1 count.*

Chapter 22

New York Yankees (NYA)

The New York Yankees compete in the American League East Division. The team was founded in 1903 and was renamed the Yankees in 1913. Since that year, they have won 27 World Series titles (an MLB record), the last in 2009.

This chapter is devoted to listing the errors in each regular season game of 2023 played in New York (in the Bronx) against the Yankees. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 NYA team is 0.704 compared with the MLB average¹ of 0.691 that year [BR].

As we will see in the next section, there were 77 errors in the New York Yankees home games, 47 by the Yankees and the rest (30) by visiting teams. The total number of errors by the team (home and away) over the regular season of 2023 was 96, well above the MLB average of 83.9. In 2023, the Yankees error total, home and away, was about one standard deviation larger than the MLB average (which is bad).

¹The standard deviation of these DER values is $\sigma = 0.011 \dots$

22.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

22.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and error-to-run conversion teams.

Runs and errors	totals
Errors by vt:	30
Runs by ht because of errors by vt:	6
Errors by ht:	47
Runs by vt because of errors by ht:	14

This is a fair error-to-run conversion rate: $6/30 = 20\%$ on the part of the Yankees (compared to the better rate $14/47 = 29.7\ldots\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 NYA season (home and away) has about $R/PA = 0.112\ldots$ (11.2%) change of resulting in a run. Therefore, a randomly selected error is almost twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 77, so the expected number of errors (by either team) in a randomly selected NYA 2023 home game is about 0.95.

Errors by position	vt	ht
errors due to pitcher:	5	7
to catcher:	3	6
to 1st baseman:	4	6
to 2nd baseman:	2	8
to 3rd baseman:	9	2
to shortstop:	4	8
to left fielder:	0	1
to center fielder:	2	6
to right fielder:	1	3

22.1.2 Error total by state

This section tabulates the totals by game state.

A table of top frequencies of fielder positions for errors by the Yankees in NYA 2023 home games:

Errors by state	vt	ht	totals
bases empty	15	18	33
(lone) runner on 1st	9	18	27
(lone) runner on 2nd	1	4	5
(lone) runner on 3rd	1	1	2
two runners on base	3	5	8
bases loaded	1	1	2
total	30	47	77

A table of frequencies of most common errors by the Yankees in NYA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	14.9%
(1, 0, 0, 0)	7	14.9%
(2, 0, 0, 0)	4	8.5%
(0, 1, 0, 0)	5	10.6%
(1, 1, 0, 0)	9	19.1%
(2, 1, 0, 0)	4	8.5%
(1, 0, 1, 0)	2	4.3%
(1, 1, 1, 0)	1	2.1%
(1, 1, 1, 1)	1	2.1%

A table of frequencies of most common errors by a visiting team in NYA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	20.0%
(1, 0, 0, 0)	5	16.7%
(2, 0, 0, 0)	4	13.3%
(0, 1, 0, 0)	2	6.7%
(1, 1, 0, 0)	5	16.7%
(2, 1, 0, 0)	2	6.7%
(2, 0, 1, 0)	1	3.3%
(1, 1, 0, 1)	1	3.3%
(2, 1, 0, 1)	1	3.3%
(1, 1, 1, 1)	1	3.3%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Yankees homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `NYA2023.eva`):

state	NYA2023 percentages
(0, 0, 0, 0)	25.6%
(1, 0, 0, 0)	18.5%
(2, 0, 0, 0)	15.3%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.6%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.9%
(1, 0, 1, 0)	2.3%
(2, 0, 1, 0)	3.2%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.6%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.2%
(1, 1, 1, 0)	2.0%
(2, 1, 1, 0)	2.5%

22.1.3 Some interesting errors

Some of the most interesting error plays in the NYA 2023 home games:

- The game SDN vs NYA on 2023-05-28 had 3 errors and a double error play.
- The game SEA vs NYA on 2023-06-22 had 3 errors and a double error play.
- The game MIL vs NYA on 2023-09-09 had 3 errors, including a triple in which the batter scored.

22.1.4 Relative team rankings

We have 81 home games between NYA and some visiting MLB team (vt). A number of teams did not play NYA in the regular season. However, of those that did, NYA *lost* more games than won against²:

CHA (2/3), CHN (2/3), MIL (2/3), WAS (2/3),
TOR (4/6), BAL (4/7), TBA (4/7), BOS (5/6) .

The teams that *split* their series were:

NYN (1/2), HOU (2/4), MIN (2/4) .

The teams that NYA *won* more than lost against

KCA (0/3), OAK (0/3), ANA (1/3), (1, 3, 'ARI'), (1, 3, 'CLE'),
DET (1/3), PHI (1/3), SDN (1/3), SEA (1/3), SFN (1/3), TEX (1/3) .

The teams that did not play in any of the 81 home games in TBA 2023 were:

ATL, CIN, COL, LAN, MIA, PIT, SLN .

22.2 Game-by-game totals

The Official Scorer shall charge an **error** against any fielder whose misplay (fumble, muff or wild throw) prolongs the time at bat of a batter, prolongs the presence on the bases of a runner or permits a runner to advance one or more bases, unless, in the judgment of the Official Scorer, such fielder deliberately permits a foul fly to fall safe with a runner on third base before two are out in order that the runner on third shall not score after the catch.

– **MLB Rule Book** 9.12(a)(1)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for the 2023 New York Yankees is as follows [BA].

²We recorded that vt's W/T record in parentheses (W = numbers of games vt won against NYA, T = Total home games vt played vs the Yankees at New York.

position	player	games played	OBP
1 (C)	Kyle Higashioka	90	.274
3 (1B)	Anthony Rizzo	93	.328
4 (2B)	Gleyber Torres	146	.347
5 (3B)	DJ LeMahieu	69	.327
6 (SS)	Anthony Volpe	158	.283
7 (LF)	Oswaldo Cabrera	52	.275
8 (CF)	Harrison Bader	83	.278
9 (RF)	Aaron Judge	54	.406

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 NYA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

22.2.1 April

	game	SFN vs NYA on 20230401
	state before	(1, 1, 1, 0, 1, 0, 0, 1)
1.	error	play,1,1,stanm004,32,CCBF*BBFX,FC1/G1S.2-H(NR)(UR);1-3(E1/TH);B-1
	state after	(1, 1, 0, 1, 1, 0, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1447 This game had 2 errors. The final score was 7-5.

In the bottom of the 1st inning, with runners on 1st and 2nd, NYA batter Giancarlo Stanton hit a ground ball into a fielder's choice at shallow infield. The runner from 2nd scored (unearned, no RBI). A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	SFN vs NYA on 20230402
	state before	(0, 1, 0, 0, 6, 0, 4, 0)
2.	error	play,6,0,wadel001,12,CSB1,0A.1-2(E3)
	state after	(0, 1, 0, 0, 6, 0, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1448 *In the top of the 6th inning, with a runner on 1st, SFN batter LaMonte Wade had a 1-2 count. The pitcher throw to 1st base in a pickoff attempt. An error by the first baseman allowed the runner on 1st to advance to 2nd. Wade is still at bat with the same count.*

	game	MIN vs NYA on 20230415
	state before	(1, 0, 0, 0, 2, 0, 0, 1)
3.	error	play,2,1, cordf003,12,BFSX,E1/G15.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1449 *In the bottom of the 2nd inning, with bases empty, NYA batter Franchy Cordero hit a ground ball to left infield. An error by the pitcher allowed the batter to reach 1st base.*

	game	ANA vs NYA on 20230418
	state before	(0, 0, 0, 0, 5, 4, 2, 0)
4.	error	play,5,0, ohtas001,12,.SBSX,C/E2.B-1
	state after	(0, 1, 0, 0, 5, 4, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1450 This game had 2 errors. The final score was 5-2. *In the top of the 5th inning, with bases empty, ANA batter Shohei Ohtani had a 1-2 count. Catcher's interference allowed the batter to reach 1st base.*

	game	ANA vs NYA on 20230418
	state before	(1, 1, 0, 0, 5, 4, 2, 0)
5.	error	play,5,0, renda001,10,B>C,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 4, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1451 *In the top of the 5th inning, with a runner on 1st, ANA batter Anthony Rendon had a 1-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rendon is still at bat, with a 1-1 count.*

	game	TOR vs NYA on 20230422
	state before	(1, 0, 0, 0, 2, 0, 0, 1)
6.	error	play,2,1,calhw001,10,BF,FLE2
	state after	(1, 0, 0, 0, 2, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1452 This play had an *infinity play*. (An ∞ play is a foul ball fly error. The game state does not change. The play could theoretically be repeated over and over ad infinitum and the game would never end. The final score was 2-3.

In the bottom of the 2nd inning, with bases empty and a 0-1 count, NYA batter Willie Calhoun hit a fly ball foul behind home plate. An error by the catcher allowed the batter to remain safely at bat. Calhoun now has a 1-1 count.

	game	TOR vs NYA on 20230423
	state before	(1, 0, 0, 0, 6, 0, 0, 0)
7.	error	play,6,0,sprig001,10,BX,E6/G6D.B-1
	state after	(1, 1, 0, 0, 6, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1453 *In the top of the 6th inning, with bases empty, TOR batter George Springer hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

22.2.2 May

	game	CLE vs NYA on 20230501
	state before	(1, 1, 1, 1, 9, 0, 2, 0)
8.	error	play,9,0,naylj001,20,BBX,S9/G34.3-H;2-H;1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 9, 2, 2, 0)
	fielder	9 (right field on the home team)

Summary 1454 *In the top of the 9th inning, with bases loaded, CLE batter Josh Naylor hit a ground bal single to the 1st/2nd base gap. The runners on 2nd and 3rd scored (2 RBI) and the runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	CLE vs NYA on 20230502
	state before	(2, 0, 0, 0, 5, 2, 0, 1)
9.	error	play,5,1,cabro002,31,BFBX,S1/L3+.B-2(E1/TH)
	state after	(2, 0, 1, 0, 5, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1455 *This game had 2 errors. The final score was 2-4. In the bottom of the 5th inning, with bases empty, NYA batter Oswaldo Cabrera hit a hard line drive single to 1st. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	CLE vs NYA on 20230502
	state before	(2, 0, 0, 0, 6, 2, 0, 0)
10.	error	play,6,0,brenw002,11,FBX,S1/G3S-.B-2(E1/TH)
	state after	(2, 0, 1, 0, 6, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1456 *In the top of the 6th inning, with bases empty, CLE batter Will Brennan hit a soft ground ball single to shallow right infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	CLE vs NYA on 20230503
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
11.	error	play,1,0,ramij003,01,CX,E3/G3.1-2;B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1457 *In the top of the 1st inning, with a runner on 1st, CLE batter Jose Ramirez hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	OAK vs NYA on 20230509
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
12.	error	play,1,1,rizza001,00,X,E8/F89D.B-2
	state after	(2, 0, 1, 0, 1, 0, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1458 *This game had 2 errors. The final score was 5-10. In the bottom of the 1st inning, with bases empty, NYA batter Anthony Rizzo hit a fly ball to deep right center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	OAK vs NYA on 20230509
	state before	(1, 1, 0, 1, 3, 1, 0, 1)
13.	error	play,3,1,judga001,21,BBC1X,FC5/G56.3-H;1-2(E5);B-1
	state after	(1, 1, 1, 0, 3, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1459 *In the bottom of the 3rd inning, with runners on 1st and 3rd, NYA batter Aaron Judge hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	TBA vs NYA on 20230511
	state before	(1, 0, 0, 0, 5, 0, 0, 0)
14.	error	play,5,0,lowej002,22,BBSFX,E3/G3.B-1
	state after	(1, 1, 0, 0, 5, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1460 *In the top of the 5th inning, with bases empty, TBA batter Josh Lowe hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	TBA vs NYA on 20230512
	state before	(2, 1, 0, 0, 2, 2, 1, 0)
15.	error	play,2,0,franw002,11,CBX,E6/G4.1-2;B-1
	state after	(2, 1, 1, 0, 2, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1461 *In the top of the 2nd inning, with a runner on 1st, TBA batter Wander Franco hit a ground ball to 2nd base. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TBA vs NYA on 20230514
	state before	(0, 0, 0, 0, 4, 3, 4, 0)
16.	error	play,4,0,lowej002,01,SX,E4/G34.B-1
	state after	(0, 1, 0, 0, 4, 3, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1462 This game had 2 errors. The final score was 8-7. *In the top of the 4th inning, with bases empty, TBA batter Josh Lowe hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	TBA vs NYA on 20230514
	state before	(1, 0, 0, 0, 6, 8, 4, 0)
17.	error	play,6,0,sirij001,31,BBCBX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 6, 8, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1463 *In the top of the 6th inning, with bases empty, TBA batter Jose Siri hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	BAL vs NYA on 20230524
	state before	(1, 1, 0, 1, 7, 6, 5, 0)
18.	error	play,7,0,mounr001,00,.X,9/SF/F9.3-H;1-2(E9/TH)
	state after	(2, 0, 1, 0, 7, 7, 5, 0)
	fielder	9 (right field on the home team)

Summary 1464 This game had 2 errors. The final score was 9-6. *In the top of the 7th inning, with runners on 1st and 3rd, BAL batter Ryan Mountcastle hit a sacrifice fly ball to right field. The runner on 3rd scored. The right fielder putout the batter but a throwing error allowed the runner on 1st to safely reach 2nd.*

	game	BAL vs NYA on 20230524
	state before	(2, 0, 1, 0, 7, 7, 5, 0)
19.	error	play,7,0,santa003,11,CBX,S8/L8.2-H;B-2(E8/TH)
	state after	(2, 0, 1, 0, 7, 8, 5, 0)
	fielder	8 (center field on the home team)

Summary 1465 *In the top of the 7th inning, with a runner on 2nd, BAL batter Anthony Santander hit a line drive single to center field. The runner on 2nd scored (1 RBI). A throwing error by the center fielder allowed the batter to reach 2nd.*

	game	BAL vs NYA on 20230525
	state before	(1, 0, 1, 0, 6, 1, 0, 0)
20.	error	play,6,0,ohear001,22,S1BF>B.X,3E1/G34.2-3;B-1
	state after	(1, 1, 0, 1, 6, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1466 This game had 2 errors. The final score was 3-1.
In the top of the 6th inning, with a runner on 2nd, BAL batter Ryan O'Hearn hit a ground ball to the 1st/2nd base gap. An error by the pitcher (on an assit from the first baseman) allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	BAL vs NYA on 20230525
	state before	(1, 0, 0, 0, 7, 1, 0, 0)
21.	error	play,7,0,fraza001,32,BBCCBX,3E1/G3.B-1
	state after	(1, 1, 0, 0, 7, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1467 *In the top of the 7th inning, with bases empty, BAL batter Adam Frazier hit a ground ball to 1st. An error by the pitcher (on an assit from the first baseman) allowed the batter to reach 1st base.*

	game	SDN vs NYA on 20230527
	state before	(2, 1, 0, 0, 7, 1, 1, 0)
22.	error	play,7,0,grist001,01,.CX,E4/G34.1-2;B-1
	state after	(2, 1, 1, 0, 7, 1, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1468 *In the top of the 7th inning, with a runner on 1st, SDN batter Trent Grisham hit a ground ball to the 1st/2nd base base. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for SDN vs NYA on 2023-05-28

position	innings	player	position	innings	player
1	1	Yu Darvish	1	1	Gerrit Cole
1	3	Drew Carlton	1	7	Jimmy Cordero
1	6	Domingo Tapia	1	8	Ron Marinaccio
1	7	Brent Honeywell	1	9	Clay Holmes
2	1	Austin Nola	2	1	Kyle Higashioka
2	7	Brett Sullivan	3	1	Anthony Rizzo
3	1	Jake Cronenworth	3	7	DJ LeMahieu
4	1	Rougned Odor	4	1	Gleyber Torres
5	1	Ha-Seong Kim	5	1	DJ LeMahieu
6	1	Xander Bogaerts	5	7	Isiah Kiner-Falefa
7	1	Jose Azocar	6	1	Anthony Volpe
8	1	Trent Grisham	7	1	Isiah Kiner-Falefa
9	1	Fernando Tatis Jr.	7	7	Greg Allen
			8	1	Harrison Bader
			9	1	Aaron Judge

	game	SDN vs NYA on 20230528
	state before	(2, 0, 1, 0, 2, 1, 1, 0)
23.	error	play,2,0,azocj001,01,FX,S8/L8.2-H;B-H(E8/TH) (E2/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 2, 3, 1, 0)
	fielder	8, (center field, catcher on the home team)

Summary 1469 This game had 3 errors. The final score was 7-10. *In the top of the 2nd inning, with a runner on 2nd, SDN batter Jose Azocar hit a line single to center field. The runner on 2nd scored (1 RBI). A throwing error by the center fielder **and** a throwing error by the catcher allowed the batter to score as well (unearned, no RBI).*

	game	SDN vs NYA on 20230528
	state before	(2, 1, 0, 0, 8, 6, 9, 1)
24.	error	play,8,1,higak001,12,CBF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 6, 9, 1)
	fielder	2 (catcher on the visiting team)

Summary 1470 *In the bottom of the 8th inning, with a runner on 1st, NYA batter Kyle Higashioka had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Higashioka is still at bat, now with a 2-2 count.*

22.2.3 June

	game	CHA vs NYA on 20230608 (1/2)
	state before	(0, 1, 0, 0, 9, 0, 3, 0)
25.	error	play,9,0,jimee001,01,FX,64(1)/FO/G6.B-2(E4/TH)
	state after	(1, 0, 1, 0, 9, 0, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1471 This was the 1st game of a doubleheader. The final score was 0-3.

In the top of the 9th inning, with a runner on 1st, CHA batter Eloy Jimenez hit a ground ball to short. The second baseman (with an assist from the shortstop) forced out the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.

	game	CHA vs NYA on 20230608 (2/2)
	state before	(2, 0, 0, 0, 4, 4, 2, 0)
26.	error	play,4,0,benia002,00,X,E6/G6.B-1
	state after	(2, 1, 0, 0, 4, 4, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1472 This was the 2nd game of a doubleheader. The final score was 6-5.

In the top of the 4th inning, with bases empty, CHA batter Andrew Benintendi hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	BOS vs NYA on 20230609
	state before	(1, 1, 0, 0, 7, 3, 1, 1)
27.	error	play,7,1,torrg001,22,.CBB1S>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 3, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1473 *In the bottom of the 7th inning, with a runner on 1st, NYA batter Gleyber Torres had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Torres is still at bat, with a full count.*

	game	BOS vs NYA on 20230610
	state before	(2, 1, 0, 0, 3, 0, 0, 0)
28.	error	play,3,0,dever001,01,S>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 3, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1474 This game had 2 errors. The final score was 1-3.

In the top of the 3rd inning, with a runner on 1st, BOS batter Rafael Devers had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Devers is still at bat, now with a 1-1 count.

	game	BOS vs NYA on 20230610
	state before	(2, 0, 1, 0, 7, 1, 2, 1)
29.	error	play,7,1,higak001,00,.X,S4/G6M.2-H;B-2(E4/TH)
	state after	(2, 0, 1, 0, 7, 1, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1475 *In the bottom of the 7th inning, with a runner on 2nd, NYA batter Kyle Higashioka hit a ground ball single to short. The runner on 2nd scored (1 RBI). A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	BOS vs NYA on 20230611
	state before	(0, 0, 0, 0, 5, 1, 2, 1)
30.	error	play,5,1,trevj001,11,BCX,E5/TH/G56S.B-1
	state after	(0, 1, 0, 0, 5, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1476 *This game had 2 errors. The final score was 3-2. In the bottom of the 5th inning, with bases empty, NYA batter Jose Trevino hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	BOS vs NYA on 20230611
	state before	(0, 0, 0, 0, 8, 1, 2, 0)
31.	error	play,8,0,herne001,12,BSC.X,S7/G56.B-2(E4)
	state after	(0, 0, 1, 0, 8, 1, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1477 *In the top of the 8th inning, with bases empty, BOS batter Enrique Hernandez hit a ground ball single to 3rd. An error by the second baseman allowed the batter to reach 2nd.*

	game	SEA vs NYA on 20230620
	state before	(1, 0, 0, 0, 7, 1, 3, 1)
32.	error	play,7,1,trevj001,02,CCX,E3/G34.B-1
	state after	(1, 1, 0, 0, 7, 1, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1478 *In the bottom of the 7th inning, with bases empty, NYA batter Jose Trevino hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	SEA vs NYA on 20230621
	state before	(1, 1, 0, 0, 8, 1, 4, 1)
33.	error	play,8,1,donaj001,02,CSX,54(1)/F0/G5.B-2(E4/TH)
	state after	(2, 0, 1, 0, 8, 1, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1479 *In the bottom of the 8th inning, with a runner on 1st, NYA batter Josh Donaldson hit a ground ball to 3rd. The second baseman (with an assist from the third baseman) putout the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.*

Visiting/Home team rosters for SEA vs NYA on 2023-06-22

position	innings	player	position	innings	player
1	1	Bryan Woo	1	1	Domingo German
1	6	Ty Adcock	1	4	Nick Ramirez
1	8	Chris Flexen	1	7	Albert Abreu
2	1	Cal Raleigh	1	9	Isiah Kiner-Falefa
3	1	Ty France	2	1	Kyle Higashioka
4	1	Kolten Wong	3	1	Anthony Rizzo
5	1	Eugenio Suarez	4	1	DJ LeMahieu
6	1	Jose Caballero	5	1	Josh Donaldson
7	1	Jarred Kelenic	5	8	Isiah Kiner-Falefa
7	7	Dylan Moore	5	9	Oswaldo Cabrera
8	1	Julio Rodriguez	6	1	Anthony Volpe
9	1	Teoscar Hernandez	7	1	Billy McKinney
9	7	Jarred Kelenic	8	1	Harrison Bader
			9	1	Giancarlo Stanton
			9	7	Oswaldo Cabrera
			9	9	Jake Bauers

	game	SEA vs NYA on 20230622
	state before	(1, 1, 0, 0, 3, 6, 0, 0)
34.	error	play,3,0,fordm002,00,X,FC5/G56.1XH(5E1)(E5)(NR)(UR);B-3
	state after	(1, 0, 0, 1, 3, 7, 0, 0)
	fielder	1,5 (pitcher, 3rd baseman on the home team)

Summary 1480 This game had 3 errors, including a double-error play. The final score was 10-2.

*In the top of the 3rd inning, with a runner on 1st, SEA batter Mike Ford hit a ground ball into a fielder's choice at 3rd. An error by the pitcher (on an assist from the third baseman) **and** an error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	SEA vs NYA on 20230622
	state before	(0, 1, 0, 0, 3, 7, 0, 1)
35.	error	play,3,0,kelej001,11,FBX,E3/G3S.1-3;B-1
	state after	(0, 1, 0, 1, 3, 7, 0, 1)
	fielder	3 (1st baseman on the home team)

Summary 1481 *In the bottom of the 3rd inning, with a runner on 1st, NYA batter Jarred Kelenic hit a ground ball to shallow 1st base. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	TEX vs NYA on 20230623
	state before	(2, 1, 0, 0, 4, 0, 1, 0)
36.	error	play,4,0,tavel001,12,CS.*BX,S8/F8S.1-H(E8)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 1, 1, 0)
	fielder	8 (center field on the home team)

Summary 1482 *In the top of the 4th inning, with a runner on 1st, TEX batter Leody Taveras hit a fly ball single to shallow center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

22.2.4 July

	game	BAL vs NYA on 20230703
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
37.	error	play,2,0,ohear001,21,BBFX,S8/L78D+.B-2(E8)
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1483 *In the top of the 2nd inning, with bases empty, BAL batter Ryan O’Hearn hit a hard line drive single to deep left center field. An error by the center fielder allowed the batter to reach 2nd.*

	game	BAL vs NYA on 20230704
	state before	(1, 0, 0, 1, 8, 3, 7, 1)
38.	error	play,8,1,trevj001,12,BCFX,E1/G1S.B-1
	state after	(1, 1, 0, 1, 8, 3, 7, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1484 *This game had 2 errors. The final score was 4-8. In the bottom of the 8th inning, with a runner on 3rd, NYA batter Jose Trevino hit a ground ball to the pitcher. An error by the pitcher allowed the batter to reach 1st base.*

	game	BAL vs NYA on 20230704
	state before	(0, 0, 0, 0, 9, 3, 8, 0)
39.	error	play,9,0,westj002,01,.CX,3E1/G4.B-1
	state after	(0, 1, 0, 0, 9, 3, 8, 0)
	fielder	1 (pitcher on the home team)

Summary 1485 *In the top of the 9th inning, with bases empty, BAL batter Jordan Westburg hit a ground ball to 2nd. An error by the pitcher (on an assist from the first baseman) allowed the batter to reach 1st base.*

	game	BAL vs NYA on 20230705
	state before	(0, 0, 0, 0, 5, 0, 0, 0)
40.	error	play,5,0,fraza001,22,BSS.BX,C/E2.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1486 This game had 2 errors. The final score was 6-3.
In the top of the 5th inning, with bases empty, BAL batter Adam Frazier had a 2-2 count. Catcher's interference allowed the batter to reach 1st base.

	game	BAL vs NYA on 20230705
	state before	(1, 1, 0, 0, 5, 0, 1, 1)
41.	error	play,5,1,volpa001,20,BBX,FC4/G4.1X2(46);B-2(E6/TH)
	state after	(2, 1, 0, 0, 5, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1487 *In the bottom of the 5th inning, with a runner on 1st, NYA batter Anthony Volpe hit a ground ball into a fielder's choice at 2nd. The runner from 1st was putout at 2nd by the shortstop (with an assist from the second baseman). A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	CHN vs NYA on 20230708
	state before	(0, 0, 0, 0, 5, 1, 4, 1)
42.	error	play,5,1,torrg001,21,.FBBX,S5/G56.B-2(E5/TH)
	state after	(0, 0, 1, 0, 5, 1, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1488 *In the bottom of the 5th inning, with bases empty, NYA batter Gleyber Torres hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	CHN vs NYA on 20230709
	state before	(1, 1, 1, 0, 7, 1, 4, 0)
43.	error	play,7,0,morec001,11,FBX,FC4/G4.2-3;1-2(E4);B-1
	state after	(1, 1, 1, 1, 7, 1, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1489 *In the top of the 7th inning, with runners on 1st and 2nd, CHN batter Christopher Morel hit a ground ball into a fielder's choice at 2nd. The runner on 2nd reached 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	KCA vs NYA on 20230721
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
44.	error	play,1,1,rizza001,02,SFFF.1X,FC3/G34.1-2(E3/TH);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1490 *In the bottom of the 1st inning, with a runner on 1st, NYA batter Anthony Rizzo hit a ground ball into a fielder's choice at the 1st/2nd base gap. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	KCA vs NYA on 20230723
	state before	(1, 1, 1, 1, 8, 3, 5, 1)
45.	error	play,8,1,perao002,21,BC*BX,S3/G34S.3-H;2-H(E3/TH)(NR);1-3;B-2
	state after	(1, 0, 1, 1, 8, 3, 7, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1491 *In the bottom of the 8th inning, with bases loaded, NYA batter Oswald Peraza hit a ground ball single to shallow 1st base. The runner on 3rd scored (1 RBI). A throwing error by the first baseman allowed the runner on 2nd to also score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 2nd.*

Visiting/Home team rosters for NYN vs NYA on 20230-7-25

position	innings	player	position	innings	player
1	1	Justin Verlander	1	1	Domingo German
1	7	Brooks Raley	1	7	Nick Ramirez
1	7	Dominic Leone	1	9	Albert Abreu
1	8	David Robertson	1	9	Ron Marinaccio
1	8	Grant Hartwig	2	1	Kyle Higashioka
1	9	Adam Ottavino	2	9	Ben Rortvedt
2	1	Francisco Alvarez	3	1	Anthony Rizzo
2	9	Omar Narvaez	4	1	Gleyber Torres
3	1	Pete Alonso	5	1	DJ LeMahieu
4	1	Jeff McNeil	6	1	Anthony Volpe
4	9	Danny Mendick	7	1	Billy McKinney
5	1	Brett Baty	7	8	Isiah Kiner-Falefa
6	1	Francisco Lindor	8	1	Harrison Bader
7	1	Mark Canha	9	1	Jake Bauers
8	1	Brandon Nimmo			
9	1	DJ Stewart			
9	9	Jeff McNeil			

	game	NYN vs NYA on 20230725
	state before	(0, 0, 0, 0, 5, 4, 0, 1)
46.	error	play,5,1,volpa001,11,BSX,E5/G56.B-1
	state after	(0, 1, 0, 0, 5, 4, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1492 This game had 3 errors. The final score was 9-3.

In the bottom of the 5th inning, with bases empty, NYA batter Anthony Volpe hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	NYN vs NYA on 20230725
	state before	(2, 0, 0, 0, 7, 6, 0, 0)
47.	error	play,7,0,lindf001,00,X,S6/G6.B-2(E6/TH)
	state after	(2, 0, 1, 0, 7, 6, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1493 *In the top of the 7th inning, with bases empty, NYN batter Francisco Lindor hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	NYN vs NYA on 20230725
	state before	(1, 1, 0, 0, 9, 7, 3, 0)
48.	error	play,9,0,lindf001,12,CFFFFFF*BFX,S8/L8D+.1-3(E8)
	state after	(1, 1, 0, 1, 9, 7, 3, 0)
	fielder	8 (center field on the home team)

Summary 1494 *In the top of the 9th inning, with a runner on 1st, NYN batter Francisco Lindor hit a hard line drive single to deep center field. An error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	NYN vs NYA on 20230726
	state before	(0, 0, 0, 0, 4, 1, 2, 1)
49.	error	play,4,1,badeh001,12,CBS.FX,S5/G5S-.B-2(E5/TH)
	state after	(0, 0, 1, 0, 4, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1495 This game had 2 errors. The final score was 1-3. *In the bottom of the 4th inning, with bases empty, NYA batter Harrison Bader hit a soft ground ball single to 3rd base. A throwing error by the third baseman allowed batter to reach 2nd.*

	game	NYN vs NYA on 20230726
	state before	(1, 0, 0, 0, 5, 1, 3, 0)
50.	error	play,5,0,nimmb001,32,BSBF.BFFX,E4/G4MD.B-1
	state after	(1, 1, 0, 0, 5, 1, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1496 *In the top of the 5th inning, with bases empty, NYN batter Brandon Nimmo hit a ground ball to deep 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

22.2.5 August

	game	TBA vs NYA on 20230801
	state before	(2, 1, 0, 1, 9, 5, 1, 1)
51.	error	play,9,1,lemad001,12,CBFFX,S5/G56S.3-H;1-3(E5/TH);B-2
	state after	(2, 0, 1, 1, 9, 5, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1497 *In the bottom of the 9th inning, with runners on 1st and 3rd, NYA batter DJ LeMahieu hit a ground ball single to shallow 3rd base. The runner on 3rd scored. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	HOU vs NYA on 20230805
	state before	(2, 0, 0, 0, 5, 1, 2, 1)
52.	error	play,5,1,torrg001,22,BCSBX,S5/G6S.B-2(E5/TH)
	state after	(2, 0, 1, 0, 5, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1498 *In the bottom of the 5th inning, with bases empty, NYA batter Gleyber Torres hit a ground ball single to shallow short. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	HOU vs NYA on 20230806
	state before	(1, 0, 0, 0, 6, 9, 5, 1)
53.	error	play,6,1,bauej001,31,CBBBX,E6/G6.B-1
	state after	(1, 1, 0, 0, 6, 9, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1499 *In the bottom of the 6th inning, with bases empty, NYA batter Jake Bauers hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs NYA on 20230818
	state before	(1, 0, 0, 0, 2, 4, 0, 0)
54.	error	play,2,0,reyep001,11,CBX,E4/G4.B-1
	state after	(1, 1, 0, 0, 2, 4, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1500 *In the top of the 2nd inning, with bases empty, BOS batter Pablo Reyes hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	BOS vs NYA on 20230819
	state before	(0, 0, 0, 0, 3, 4, 0, 1)
55.	error	play,3,1,alleg002,31,BBBCX,E3/G3.B-1
	state after	(0, 1, 0, 0, 3, 4, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1501 This game had 2 errors. The final score was 8-1. *In the bottom of the 3rd inning, with bases empty, NYA batter Greg Allen hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	BOS vs NYA on 20230819
	state before	(1, 1, 0, 0, 8, 6, 1, 0)
56.	error	play,8,0,duraj001,00,X,C/E2.1-2;B-1
	state after	(1, 1, 1, 0, 8, 6, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1502 *In the top of the 8th inning, with a runner on 1st, BOS batter Jarren Duran had a 0-0 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	BOS vs NYA on 20230820
	state before	(0, 1, 1, 0, 6, 1, 1, 0)
57.	error	play,6,0,yoshm002,11,S*BX,46(1)/FO/G34.2-H(E6/TH)(NR);B-1
	state after	(1, 1, 0, 0, 6, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1503 *In the top of the 6th inning, with runners on 1st and 2nd, BOS batter Masataka Yoshida hit a ground ball to the 1st/2nd base gap. The shortstop (on an assist from the second baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	WAS vs NYA on 20230822
	state before	(1, 0, 0, 0, 4, 1, 1, 0)
58.	error	play,4,0,garrs003,10,BX,C/E2.B-1
	state after	(1, 1, 0, 0, 4, 1, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1504 *In the top of the 4th inning, with bases empty, WAS batter Stone Garrett had a 1-0 count. Catcher's interference allowed the batter to reach 1st base.*

Visiting/Home team rosters for WAS vs NYA on 2023-08-23

position	innings	player	position	innings	player
1	1	MacKenzie Gore	1	1	Luis Severino
1	5	Robert Garcia	1	7	Ian Hamilton
1	7	Jose A. Ferrer	1	9	Wandy Peralta
1	8	Andres Machado	2	1	Kyle Higashioka
2	1	Keibert Ruiz	3	1	DJ LeMahieu
2	8	Riley Adams	3	9	Jake Bauers
3	1	Dominic Smith	4	1	Gleyber Torres
4	1	Jake Alu	5	1	Oswald Peraza
5	1	Ildemaro Vargas	6	1	Anthony Volpe
6	1	CJ Abrams	7	1	Everson Pereira
7	1	Blake Rutherford	8	1	Harrison Bader
8	1	Lane Thomas	9	1	Giancarlo Stanton
8	7	Alex Call	9	8	Oswaldo Cabrera
9	1	Stone Garrett			
9	7	Lane Thomas			

	game	WAS vs NYA on 20230823
	state before	(1, 1, 0, 0, 2, 0, 2, 1)
59.	error	play,2,1,higak001,22,CSBBFFX,E9/F89XD.1-3;B-2
	state after	(1, 0, 1, 1, 2, 0, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 1505 This game had 3 errors. The final score was 1-9.
In the bottom of the 2nd inning, with a runner on 1st, NYA batter Kyle Higashioka hit a fly ball to extra deep right center field. An error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.

	game	WAS vs NYA on 20230823
	state before	(2, 1, 0, 0, 2, 0, 6, 1)
60.	error	play,2,1,stanm004,31,B*BCBX,FC6/G6.1-2(E6);B-1
	state after	(2, 1, 1, 0, 2, 0, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1506 *In the bottom of the 2nd inning, with a runner on 1st, NYA batter Giancarlo Stanton hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	WAS vs NYA on 20230823
	state before	(0, 0, 0, 0, 6, 0, 6, 0)
61.	error	play,6,0,ruthb001,00,X,E3/G9LS.B-1
	state after	(0, 1, 0, 0, 6, 0, 6, 0)
	fielder	3 (1st baseman on the home team)

Summary 1507 *In the top of the 6th inning, with bases empty, WAS batter Blake Rutherford hit a ground ball to short right field down the 1st base line. An error by the first baseman allowed the batter to reach 1st base.*

	game	WAS vs NYA on 20230824
	state before	(2, 0, 1, 1, 3, 0, 1, 0)
62.	error	play,3,0,menej001,11,CB.X,E6/G6.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 3, 1, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1508 *In the top of the 3rd inning, with runners on 2nd and 3rd, WAS batter Joey Meneses hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

22.2.6 September

63.	game	DET vs NYA on 20230907
	state before	(1, 1, 0, 0, 8, 8, 3, 0)
	error	play,8,0,torks001,00,1,P01(E1/TH) .1-2
	state after	(1, 0, 1, 0, 8, 8, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1509 *In the top of the 8th inning, with a runner on 1st, DET batter Spencer Torkelson has a 0-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 2nd. Torkelson is still at bat with the same count.*

64.	game	MIL vs NYA on 20230908
	state before	(1, 1, 0, 0, 2, 0, 0, 0)
	error	play,2,0,tellr001,11,CB1,P01(E1/TH) .1-3
	state after	(1, 0, 0, 1, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1510 *This game had 2 errors. The final score was 8-2. In the top of the 2nd inning, with a runner on 1st, MIL batter Rowdy Tellez has a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 3rd. Tellez is still at bat with the same count.*

65.	game	MIL vs NYA on 20230908
	state before	(1, 0, 1, 0, 7, 3, 2, 0)
	error	play,7,0,santc002,32,FBB1N.BCX,S9/G34.2-H;B-2(E9/TH)
	state after	(1, 0, 1, 0, 7, 4, 2, 0)
	fielder	9 (right field on the home team)

Summary 1511 *In the top of the 7th inning, with a runner on 2nd, MIL batter Carlos Santana hit a ground ball single to the 1st/2nd base gap. The runner on 2nd scored. A throwing error by the right fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for MIL vs NYA on 2023-09-09

position	innings	player	position	innings	player
1	1	Wade Miley	1	1	Michael King
1	4	Elvis Peguero	1	6	Greg Weissert
1	6	Bryse Wilson	1	7	Wandy Peralta
1	7	Joel Payamps	1	8	Jonathan Loaisiga
1	8	Trevor Megill	1	9	Matt Krook
1	9	Hoby Milner	1	9	Ron Marinaccio
2	1	William Contreras	2	1	Austin Wells
3	1	Carlos Santana	3	1	DJ LeMahieu
4	1	Brice Turang	4	1	Gleyber Torres
5	1	Andruw Monasterio	5	1	Oswald Peraza
6	1	Willy Adames	6	1	Anthony Volpe
7	1	Tyrone Taylor	7	1	Everson Pereira
8	1	Sal Frelick	8	1	Jasson Dominguez
8	9	Joey Wiemer	9	1	Giancarlo Stanton
9	1	Mark Canha			
9	9	Sal Frelick			

	game	MIL vs NYA on 20230909
	state before	(2, 0, 0, 0, 3, 0, 0, 1)
66.	error	play,3,1,lemad001,22,BBCS.X,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 3, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1512 This game had 3 errors. The final score was 9-2. *In the bottom of the 3rd inning, with bases empty, NYA batter DJ LeMahieu hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs NYA on 20230909
	state before	(1, 1, 0, 0, 4, 0, 0, 0)
67.	error	play,4,0,adamw002,10,BX,T9/L9D.1-H;B-H(E3/TH) (NR) (UR)
	state after	(1, 0, 0, 0, 4, 2, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1513 *In the top of the 4th inning, with a runner on 1st, MIL batter Willy Adames hit a line drive triple to deep right field. The runner on 1st scored (1 RBI). A throwing error by the first baseman allowed the batter to score (unearned, no RBI).*

	game	MIL vs NYA on 20230909
	state before	(1, 1, 0, 0, 4, 2, 0, 1)
68.	error	play,4,1,stanm004,32,1FBBF*BX,E5/G56.1-2;B-1
	state after	(1, 1, 1, 0, 4, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1514 *In the bottom of the 4th inning, with a runner on 1st, NYA batter Giancarlo Stanton hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TOR vs NYA on 20230919
	state before	(0, 1, 0, 0, 4, 1, 1, 0)
69.	error	play,4,0,guerv002,22,BSSBX,E5/TH/G56.1-3;B-1
	state after	(0, 1, 0, 1, 4, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1515 This game had 2 errors. The final score was 7-1. *In the top of the 4th inning, with a runner on 1st, TOR batter Vladimir Guerrero hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	TOR vs NYA on 20230919
	state before	(2, 0, 0, 1, 9, 6, 1, 0)
70.	error	play,9,0,guerv002,02,SFX,E6/G6+.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 9, 7, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1516 *In the top of the 9th inning, with a runner on 3rd, TOR batter Vladimir Guerrero hit a hard ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	TOR vs NYA on 20230921
	state before	(2, 1, 0, 1, 9, 2, 5, 0)
71.	error	play,9,0,chapm001,01,.CX,E4/TH/G4.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 9, 3, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1517 *In the top of the 9th inning, with runners on 1st and 3rd, TOR batter Matt Chapman hit a ground ball to 2nd base. A throwing error by the second baseman allowed runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

Visiting/Home team rosters for ARI vs NYA on 2023-09-22

position	innings	player	position	innings	player
1	1	Brandon Pfaadt	1	1	Luke Weaver
1	5	Slade Cecconi	1	6	Jhony Brito
1	8	Joe Mantiply	2	1	Austin Wells
2	1	Gabriel Moreno	3	1	Jake Bauers
3	1	Christian Walker	4	1	Gleyber Torres
4	1	Ketel Marte	5	1	Oswald Peraza
5	1	Jace Peterson	6	1	Anthony Volpe
6	1	Geraldo Perdomo	7	1	Oswaldo Cabrera
7	1	Tommy Pham	8	1	Estevan Florial
7	7	Corbin Carroll	9	1	Aaron Judge
8	1	Alek Thomas			
9	1	Corbin Carroll			
9	7	Jake McCarthy			

	game	ARI vs NYA on 20230922
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
72.	error	play,3,1,perao002,00,X,FC1/G13.1-2(E1);B-1
	state after	(0, 1, 1, 0, 3, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1518 This game had 3 errors. The final score was 1-7.

In the bottom of the 3rd inning, with a runner on 1st, NYA batter Oswald Peraza hit a ground ball into a fielder's choice at shallow infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	ARI vs NYA on 20230922
	state before	(2, 0, 0, 0, 5, 0, 3, 0)
73.	error	play,5,0,moreg001,22,FCFFBFFBFX,E6/TH/G6.B-2
	state after	(2, 0, 1, 0, 5, 0, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1519 *In the top of the 5th inning, with bases empty, ARI batter Gabriel Moreno hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 2nd base.*

	game	ARI vs NYA on 20230922
	state before	(1, 0, 0, 0, 5, 0, 6, 1)
74.	error	play,5,1,wella002,02,FS.X,S8/G4M.B-2(E8)
	state after	(1, 0, 1, 0, 5, 0, 6, 1)
	fielder	8 (center field on the visiting team)

Summary 1520 *In the bottom of the 5th inning, with bases empty, NYA batter Austin Wells hit a ground ball single to deep 2nd. An error by the center fielder allowed the batter to reach 2nd.*

	game	ARI vs NYA on 20230924
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
75.	error	play,1,0,carrc005,22,BFSBX,S8/L8+.1-3;B-2(E8)
	state after	(0, 0, 1, 1, 1, 0, 0, 0)
	fielder	8 (center field on the home team)

Summary 1521 This game had 2 errors. The final score was 7-1. *In the top of the 1st inning, with a runner on 1st, ARI batter Corbin Carroll hit a hard line drive single to center field. The runner on 1st reached 3rd. An error by the center fielder allowed the batter to reach 2nd.*

	game	ARI vs NYA on 20230924
	state before	(1, 1, 0, 0, 7, 4, 0, 0)
76.	error	play,7,0,lawlj001,31,..BBB+1C1X,E7/F7.1-2;B-1
	state after	(1, 1, 1, 0, 7, 4, 0, 0)
	fielder	7 (left field on the home team)

Summary 1522 *In the top of the 7th inning, with a runner on 1st, ARI batter Jordan Lawlar hit a fly ball to left field. An error by the left fielder allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	ARI vs NYA on 20230925
	state before	(0, 0, 0, 0, 4, 2, 2, 1)
77.	error	play,4,1,volpa001,00,X,E5/TH/G5S.B-1
	state after	(0, 1, 0, 0, 4, 2, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1523 *In the bottom of the 4th inning, with bases empty, NYA batter Anthony Volpe hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Chapter 23

Oakland Athletics (OAK)

The **Athletics** have moved several times. Founded in 1901, they were originally the Philadelphia Athletics, then they moved to Kansas City (1955-1967). Since 1968, they have been the Oakland Athletics. However, they plan to temporarily move to Sacramento for the 2025–2027 seasons before their permanent move to Las Vegas, NV. The Athletics compete in the American League West Division. Since 1968, they have won 4 World Series titles, the last being in 1989, and 17 ALWD titles, the last being in 2020.

The team is also famously the subject of the movie *Moneyball* (2011). That movie was based on the 2003 Michael Lewis book **Moneyball** which delved into the 2002 Oakland Athletics season (a year they won an ALWD title) and GM Billy Beane's sabermetric approach to managing his team.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Oakland against the Athletics. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The DER of the 2023 OAK team is 0.678 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the OAK is well below the MLB average.

In OAK 2023, there were 95 errors in the 81 home games, 61 by OAK and the rest (34) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 102. Since 83.9 was

the MLB average, the Athletics were well above average for errors (which is bad).

23.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

23.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	34
Runs by ht because of errors by vt:	3
Errors by ht:	61
Runs by vt because of errors by ht:	20

This is a very poor error-to-run conversion rate: $3/34 = 8.8\%$ on the part of the Athletics (compared to $20/61 = 32.7\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 OAK season (home and away) has about $R/PA = 585/5966$ (about 9.8%) change of resulting in a run. (The MLB average for this in 2023 was $R/PA = 22432/184104 = 12.18\%$.) Therefore, a randomly selected error is (for OAK) *less likely* to result in a run than a randomly selected at-bat, at least in this case. (Usually, errors by the visiting team create about *twice as many* runs on average than a random play.)

Note that the total number of errors (by either team) in a home game is 95, so the expected number of errors (by either team) in a randomly selected OAK 2023 home game is about 1.17.

Errors by position	vt	ht
errors due to pitcher:	1	10
to catcher:	8	9
to 1st baseman:	2	10
to 2nd baseman:	1	4
to 3rd baseman:	8	11
to shortstop:	8	6
to left fielder:	1	2
to center fielder:	1	4
to right fielder:	2	4

23.1.2 Error total by state

If there is a runner, or runners, it is a **balk** when

- (10) The pitcher, after coming to a legal pitching position, removes one hand from the ball other than in an actual pitch, or in throwing to a base;
- (11) The pitcher, while touching his plate, accidentally or intentionally has the ball slip or fall out of his hand or glove;
- (12) The pitcher, while giving an intentional base on balls, pitches when the catcher is not in the catcher's box;
- (13) The pitcher delivers the pitch from Set Position without coming to a stop.

– **MLB Rule Book 6.02(a)(10)-(13)**

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	13	21	34
(lone) runner on 1st	12	20	32
(lone) runner on 2nd	2	7	9
(lone) runner on 3rd	0	2	2
two runners on base	5	7	12
bases loaded	2	4	6
total	34	61	95

A table of frequencies of most common errors by the OAK 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	13.1%
(1, 0, 0, 0)	9	14.8%
(2, 0, 0, 0)	4	6.6%
(0, 1, 0, 0)	6	9.8%
(1, 1, 0, 0)	10	16.4%
(2, 1, 0, 0)	4	6.6%
(0, 0, 1, 0)	3	4.9%
(1, 1, 0, 1)	2	3.3%
(1, 1, 1, 1)	3	4.9%

A table of frequencies of most common errors by a visiting team in OAK 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	17.6%
(1, 0, 0, 0)	3	8.8%
(2, 0, 0, 0)	4	11.8%
(0, 1, 0, 0)	5	14.7%
(1, 1, 0, 0)	3	8.8%
(2, 1, 0, 0)	4	11.8%
(0, 1, 1, 0)	1	2.9%
(1, 1, 1, 0)	3	8.8%
(2, 1, 1, 0)	1	2.9%
(1, 1, 1, 1)	2	5.9%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Athletics homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `OAK2023.eva`):

state	OAK2023 percentages
(0, 0, 0, 0)	24.7%
(1, 0, 0, 0)	17.6%
(2, 0, 0, 0)	14.3%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.4%
(2, 1, 0, 0)	6.0%
(0, 0, 1, 0)	2.3%
(1, 0, 1, 0)	3.4%
(2, 0, 1, 0)	3.5%
(0, 0, 0, 1)	0.3%
(1, 0, 0, 1)	1.0%
(2, 0, 0, 1)	1.7%
(0, 1, 1, 0)	1.6%
(1, 1, 1, 0)	2.7%
(2, 1, 1, 0)	2.6%

23.1.3 Some interesting errors

Some of the most interesting error plays in the OAK 2023 home games:

- The game ANA vs OAK on 2023-04-01 had 4 errors.
- The game TEX vs OAK on 2023-05-14 had 4 errors.
- The game BOS vs OAK on 2023-07-19 had 4 errors.

23.1.4 Relative team rankings

We have 81 home games between OAK and some visiting MLB team. In the lists below, we have ordered the visiting teams (vt) with that team's W/T

record in parentheses¹.

A number of teams did not play OAK in the regular season. However, of those that did, OAK *lost* more games than won against:

ARI (2/3), CIN (2/3), CLE (2/3), NYA (2/3), TOR (2/3)
BAL (3/3), CHN (3/3), MIN (3/3), NYN (3/3), PHI (3/3),
SDN (3/3), TEX (5/7), SEA (6/6), HOU (6/7)

and *won* more than lost against

SFN (0/2), ATL (1/3), BOS (1/3), CHA (1/3),
KCA (1/3), ANA (2/6),

The teams that *split* their series were:

DET (2/4), TBA (2/4).

The teams that did not play in any of the 81 home games in OAK 2023 were:

COL, LAN, MIA, MIL, PIT, SLN, WAS.

23.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for OAK 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Shea Langeliers	123	.268
3 (1B)	Ryan Noda	109	.364
4 (2B)	Zack Gelof	69	.337
5 (3B)	Jace Peterson	77	.313
6 (SS)	Nick Allen	104	.263
7 (LF)	Tony Kemp	65	.303
8 (CF)	Esteury Ruiz	110	.309
9 (RF)	Ramón Laureano	56	.280

¹Where W = numbers of games vt won against OAK, T = Total home games vt played against the Athletics at OAK.

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 OAK home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 95 errors in the 81 home games, 61 by OAK and the rest (34) by visiting teams.

	game	ANA vs OAK on 20230330
	state before	(1, 0, 0, 0, 5, 0, 0, 0)
1.	error	play,5,0,urshg001,22,CFFBFBX,S4/G6M.B-2(E4/TH)
	state after	(1, 0, 1, 0, 5, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1524 *In the top of the 5th inning, with bases empty, ANA batter Giovanny Urshela hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd.*

23.2.1 April

Visiting/Home team rosters for ANA vs OAK on 2023-04-01

position	innings	player	position	innings	player
1	1	Patrick Sandoval	1	1	Shintaro Fujinami
1	6	Tucker Davidson	1	3	Adam Oller
2	1	Logan O'Hoppe	1	8	Jeury's Familia
3	1	Jake Lamb	1	9	Sam Moll
4	1	Luis Rengifo	2	1	Shea Langeliers
5	1	Anthony Rendon	3	1	Jesus Aguilar
5	6	Gio Urshela	3	8	Ryan Noda
6	1	Gio Urshela	4	1	Tony Kemp
6	6	David Fletcher	5	1	Aledmys Diaz
7	1	Taylor Ward	6	1	Nick Allen
8	1	Mike Trout	7	1	Seth Brown
8	7	Brett Phillips	8	1	Esteury Ruiz
9	1	Hunter Renfroe	9	1	Ramon Laureano
			9	8	Conner Capel

game	ANA vs OAK on 20230401
state before	(2, 0, 0, 0, 2, 0, 0, 1)
2. error	play,2,1,laurr001,32,CFBBBX,E5/G56+.B-1
state after	(2, 1, 0, 0, 2, 0, 0, 1)
fielder	5 (3rd baseman on the visiting team)

Summary 1525 This game had 4 errors. The final score was 13-1. *In the bottom of the 2nd inning, with bases empty, OAK batter Ramon Laureano hit a hard ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs OAK on 20230401
	state before	(1, 1, 1, 1, 3, 4, 0, 0)
3.	error	play,3,0,lambj001,11,TBX,S8/G4M.3-H;2-H;1-3(E8)
	state after	(1, 1, 0, 1, 3, 6, 0, 0)
	fielder	8 (center field on the home team)

Summary 1526 *In the top of the 3rd inning, with bases loaded, ANA batter Jake Lamb hit a ground ball single to 2nd. The runners on 2nd and 3rd scored (2 RBIs). An error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	ANA vs OAK on 20230401
	state before	(2, 1, 0, 0, 5, 11, 1, 1)
4.	error	play,5,1,ruize001,32,BSBBF>X,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 5, 11, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1527 *In the bottom of the 5th inning, with a runner on 1st, OAK batter Esteury Ruiz had a full count. Catcher's interference allowed the batter to take 1st. The runner already on 1st advanced one base.*

	game	ANA vs OAK on 20230401
	state before	(1, 0, 0, 0, 7, 12, 1, 0)
5.	error	play,7,0,renl001,11,CBX,E3/TH/G34.B-1
	state after	(1, 1, 0, 0, 7, 12, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1528 *In the top of the 7th inning, with bases empty, ANA batter Luis Rengifo hit a ground ball to 1st/2nd. A throwing error by the first baseman allowed the batter to reach 1st base.*

	game	ANA vs OAK on 20230402
	state before	(0, 0, 0, 0, 4, 0, 0, 0)
6.	error	play,4,0,renfh001,01,CX,S2/G23-.B-2(E2/TH)
	state after	(0, 0, 1, 0, 4, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1529 *In the top of the 4th inning, with bases empty, ANA batter Hunter Renfroe hit a soft ground ball single to the shallow right infield. A throwing error by the the catcher allowed the batter to reach 2nd.*

	game	CLE vs OAK on 20230403
	state before	(0, 1, 0, 0, 6, 5, 8, 0)
7.	error	play,6,0,brenw002,11,CBX,FC4/G34.1-2(E4);B-1
	state after	(0, 1, 1, 0, 6, 5, 8, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1530 This game had 2 errors. The final score was 12-11 (in extra innings).

In the top of the 6th inning, with a runner on 1st, CLE batter Will Brennan hit a ground ball into a fielder's choice at 1st/2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	CLE vs OAK on 20230403
	state before	(1, 0, 0, 0, 7, 7, 8, 1)
8.	error	play,7,1,brows003,00,X,E6/TH/G6.B-2
	state after	(1, 0, 1, 0, 7, 7, 8, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1531 *In the bottom of the 7th inning, with bases empty, OAK batter Seth Brown hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 2nd base.*

	game	CLE vs OAK on 20230404
	state before	(0, 0, 0, 0, 9, 3, 2, 0)
9.	error	play,9,0,ariag002,12,..CBFF,FLE3
	state after	(0, 0, 0, 0, 9, 3, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1532 This play is an example of an *infinity play*, one that can be repeated over and over against without changing the game state. *In the top of the 9th inning, with a 1-2 count and bases empty, CLE batter Gabriel Arias hit a fly ball foul of the 1st base line. An error by the first baseman allowed the batter to remain safely at-bat. Arias still has a 1-2 count.*

	game	CLE vs OAK on 20230405
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
10.	error	play,2,0,ariag002,22,BBCCFX,E9/L9S.1-2;B-1
	state after	(0, 1, 1, 0, 2, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 1533 This game had 2 errors. The final score was 6-4 (in extra innings).

In the top of the 2nd inning, with a runner on 1st, CLE batter Gabriel Arias hit a line drive to shallow right field. An error by the right fielder allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CLE vs OAK on 20230405
	state before	(1, 1, 0, 0, 5, 0, 0, 0)
11.	error	play,5,0,rosaa003,10,B>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1534 *In the top of the 5th inning, with a runner on 1st, CLE batter Amed Rosario had a 1-0 count. During the next pitch,*

which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rosario is still at bat, now with a 2-0 count.

	game	CHN vs OAK on 20230418
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
12.	error	play,1,1,diaza003,12,BCFF1X,6/L6.1-3(E6/TH)
	state after	(2, 0, 0, 1, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1535 This game had 2 errors. The final score was 4-0. *In the bottom of the 1st inning, with a runner on 1st, OAK batter Aledmys Diaz hit a line drive to short. The shortstop put out the batter but his throwing error allowed the runner on 1st to advance to 3rd.*

	game	CHN vs OAK on 20230418
	state before	(2, 1, 0, 0, 8, 3, 0, 0)
13.	error	play,8,0,hoern001,32,BCB1FB>X,E9/L9.1-H(NR) (UR) ;B-2
	state after	(2, 0, 1, 0, 8, 4, 0, 0)
	fielder	9 (right field on the home team)

Summary 1536 *In the top of the 8th inning, with a runner on 1st, CHN batter Nico Hoerner hit a line drive to right field. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for CHN vs OAK on 2023-04-19

position	innings	player	position	innings	player
1	1	Justin Steele	1	1	Mason Miller
1	7	Keegan Thompson	1	5	Chad Smith
1	9	Julian Merryweather	1	6	Jeurys Familia
2	1	Tucker Barnhart	1	6	Sam Moll
3	1	Eric Hosmer	1	8	Adam Oller
4	1	Nico Hoerner	1	9	Zach Jackson
5	1	Nick Madrigal	2	1	Shea Langeliers
6	1	Dansby Swanson	3	1	Jesus Aguilar
7	1	Ian Happ	4	1	Jordan Diaz
8	1	Cody Bellinger	4	8	Tony Kemp
9	1	Patrick Wisdom	5	1	Aledmys Diaz
			6	1	Kevin Smith
			7	1	Tyler Wade
			8	1	Esteury Ruiz
			9	1	Conner Capel

	game	CHN vs OAK on 20230419
	state before	(1, 1, 0, 0, 4, 1, 0, 1)
14.	error	play,4,1,langs001,00,X,36(1)/F0/G34.B-2(E6/TH)
	state after	(2, 0, 1, 0, 4, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1537 This game had 3 errors. The final score was 12-2. *In the bottom of the 4th inning, with a runner on 1st, OAK batter Shea Langeliers hit a ground ball to 1st/2nd. At 2nd, the shortstop (with an assist from the first baseman) forced out the runner from 1st. However, a throwing error by the shortstop allowed the batter to reach 2nd.*

	game	CHN vs OAK on 20230419
	state before	(0, 1, 0, 0, 5, 2, 1, 1)
15.	error	play,5,1,diazj008,21,BBCX,S9/L9+.1-H(E9)(NR);B-2
	state after	(0, 0, 1, 0, 5, 2, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 1538 *In the bottom of the 5th inning, with a runner on 1st, OAK batter Jordan Diaz hit a hard line drive single to right field. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	CHN vs OAK on 20230419
	state before	(1, 1, 1, 1, 9, 7, 2, 0)
16.	error	play,9,0,torr1001,12,*SBFFX,D8/L89XD+.3-H;2-H;1-H(E8/TH)(NR)
	state after	(1, 0, 1, 0, 9, 10, 2, 0)
	fielder	8 (center field on the home team)

Summary 1539 *In the top of the 9th inning, with bases loaded, CHN batter Luis Torrens hit a hard line drive double to extra deep right center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the center fielder allowed the runner on 1st to also score (unearned, no RBI). The batter took 2nd.*

Visiting/Home team rosters for CIN vs OAK on 2023-04-28

position	innings	player	position	innings	player
1	1	Luis Cessa	1	1	Drew Rucinski
1	6	Derek Law	1	6	Sam Long
1	7	Buck Farmer	1	8	Shintaro Fujinami
1	8	Lucas Sims	1	9	Sam Moll
1	9	Fernando Cruz	2	1	Shea Langeliers
2	1	Tyler Stephenson	3	1	Ryan Noda
3	1	Spencer Steer	4	1	Jordan Diaz
4	1	Jonathan India	5	1	Jace Peterson
5	1	Nick Senzel	6	1	Kevin Smith
6	1	Jose Barrero	7	1	Tony Kemp
7	1	Stuart Fairchild	8	1	Esteury Ruiz
8	1	TJ Friedl	9	1	Conner Capel
9	1	Jake Fraley			

	game	CIN vs OAK on 20230428
	state before	(1, 0, 0, 0, 3, 3, 1, 0)
17.	error	play,3,0,stees001,21,BBFX,E5/G7LS.B-2
	state after	(1, 0, 1, 0, 3, 3, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1540 This game had 3 errors. In fact, each game in the Reds-Athletics series had at least 2 errors, for a total of 7. The final score of this particular game was 11-7.

In the top of the 3rd inning, with bases empty, CIN batter Spencer Steer hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 2nd base.

	game	CIN vs OAK on 20230428
	state before	(2, 1, 0, 0, 3, 5, 1, 0)
18.	error	play,3,0,senzn001,00,X,E5/TH/G56.1-3;B-2
	state after	(2, 0, 1, 1, 3, 5, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1541 *In the top of the 3rd inning, with a runner on 1st, CIN batter Nick Senzel hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	CIN vs OAK on 20230428
	state before	(1, 0, 0, 0, 7, 8, 3, 1)
19.	error	play,7,1,ruize001,11,BCX,S9/L9.B-2(E9)
	state after	(1, 0, 1, 0, 7, 8, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1542 *In the bottom of the 7th inning, with bases empty, OAK batter Esteury Ruiz hit a line drive single to right field. An error by the right fielder allowed the batter to reach 2nd.*

	game	CIN vs OAK on 20230429
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
20.	error	play,2,1,perec003,21,BBCX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1543 This game had 2 errors. The final score was 3-2. *In the bottom of the 2nd inning, with bases empty, OAK batter Carlos Perez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CIN vs OAK on 20230429
	state before	(1, 1, 0, 0, 3, 1, 2, 0)
21.	error	play,3,0,fairs001,00,1,P01(E1/TH).1-3
	state after	(1, 0, 0, 1, 3, 1, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1544 *In the top of the 3rd inning, with a runner on 1st, CIN batter Stuart Fairchild has a 0-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to safely advance to 3rd. Fairchild is still atbat with a 0-0 count.*

	game	CIN vs OAK on 20230430
	state before	(2, 1, 1, 0, 6, 1, 3, 0)
22.	error	play,6,0,friet001,11,..FBX,S1/G4.2-H(E1/TH) (NR);1-3;B-2
	state after	(2, 0, 1, 1, 6, 2, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1545 This game had 2 errors. The final score was 4-5. *In the top of the 6th inning, with runners on 1st and 2nd, CIN batter TJ Friedl hit a ground ball single to 2nd. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	CIN vs OAK on 20230430
	state before	(0, 1, 0, 0, 9, 4, 4, 1)
23.	error	play,9,1,smstk003,00,X,FC/SH/BG23-.1-2;B-1(E3)
	state after	(0, 1, 1, 0, 9, 4, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1546 *In the bottom of the 9th inning, with a runner on 1st, OAK batter Kevin Smith hit a soft sacrifice bunt to shallow right infield. The runner on 1st advanced to 2nd. An error by the first baseman allowed the batter to reach 1st.*

23.2.2 May

	game	SEA vs OAK on 20230503
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
24.	error	play,3,0,crawj002,22,CBF.BFX,E4/G4.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1547 *In the top of the 3rd inning, with bases empty, SEA batter J.P. Crawford hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	TEX vs OAK on 20230512
	state before	(0, 0, 0, 0, 2, 0, 1, 0)
25.	error	play,2,0,jungj001,00,X,E5/TH/G5.B-3
	state after	(0, 0, 0, 1, 2, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1548 *In the top of the 2nd inning, with bases empty, TEX batter Josh Jung hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 3rd base.*

	game	TEX vs OAK on 20230513
	state before	(0, 1, 0, 0, 4, 3, 0, 1)
26.	error	play,4,1,nodar001,11,F.1B+1>C,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 4, 3, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1549 *In the bottom of the 4th inning, with a runner on 1st, OAK batter Ryan Noda had a 1-1 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at bat, now with a 1-2 count.*

Visiting/Home team rosters for TEX vs OAK on 2023-05-14

position	innings	player	position	innings	player
1	1	Andrew Heaney	1	1	JP Sears
1	7	Jonathan Hernandez	1	6	Garrett Acton
1	8	Yerry Rodriguez	1	8	Austin Pruitt
1	9	Jose Leclerc	1	8	Sam Moll
2	1	Jonah Heim	1	8	Zach Neal
3	1	Nathaniel Lowe	2	1	Shea Langeliers
4	1	Marcus Semien	3	1	Jesus Aguilar
5	1	Josh Jung	4	1	Jordan Diaz
6	1	Ezequiel Duran	5	1	Aledmys Diaz
7	1	Robbie Grossman	6	1	Nick Allen
8	1	Leody Taveras	7	1	Brent Rooker
9	1	Adolis Garcia	8	1	Esteury Ruiz
			9	1	Ramon Laureano

	game	TEX vs OAK on 20230514
	state before	(1, 0, 0, 0, 3, 2, 1, 0)
27.	error	play,3,0,semim001,01,CX,E3/G3.B-1
	state after	(1, 1, 0, 0, 3, 2, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1550 This game had 4 errors. The final score was 11-3.
In the top of the 3rd inning, with bases empty, TEX batter Marcus Semien hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.

	game	TEX vs OAK on 20230514
	state before	(1, 0, 0, 0, 4, 3, 1, 0)
28.	error	play,4,0,durae002,12,SF.BX,E8/F89.B-2
	state after	(1, 0, 1, 0, 4, 3, 1, 0)
	fielder	8 (center field on the home team)

Summary 1551 *In the top of the 4th inning, with bases empty, TEX batter Ezequiel Duran hit a fly ball to right center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	TEX vs OAK on 20230514
	state before	(0, 0, 0, 0, 6, 3, 1, 0)
29.	error	play,6,0,jungj001,12,FFBX,E6/G6D.B-1
	state after	(0, 1, 0, 0, 6, 3, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1552 *In the top of the 6th inning, with bases empty, TEX batter Josh Jung hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	TEX vs OAK on 20230514
	state before	(1, 0, 1, 0, 8, 3, 3, 0)
30.	error	play,8,0,heimj001,21,BFBX,S8/L89D.2-H;B-3(E8)
	state after	(1, 0, 0, 1, 8, 4, 3, 0)
	fielder	8 (center field on the home team)

Summary 1553 *In the top of the 8th inning, with a runner on 2nd, TEX batter Jonah Heim hit a line drive single to deep right center field. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the batter to reach 3rd.*

	game	ARI vs OAK on 20230515
	state before	(2, 0, 0, 0, 3, 4, 0, 1)
31.	error	play,3,1,allen002,22,CSBFBX,E5/TH/G5.B-2
	state after	(2, 0, 1, 0, 3, 4, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1554 *In the bottom of the 3rd inning, with bases empty, OAK batter Nick Allen hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	ARI vs OAK on 20230516
	state before	(2, 1, 1, 0, 7, 8, 4, 1)
32.	error	play,7,1,perec003,21,*BBFX,FC5/G5.2-3(E5);1-2;B-1
	state after	(2, 1, 1, 1, 7, 8, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1555 This game had 2 errors, **both** are from a fielder choices by the same fielder with runners on 1st and 2nd. The final score was 8-9 (in extra innings).

In the bottom of the 7th inning, with runners on 1st and 2nd, OAK batter Carlos Perez hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

	game	ARI vs OAK on 20230516
	state before	(1, 1, 1, 0, 10, 8, 8, 1)
33.	error	play,10,1,allen002,00,X,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 10, 8, 8, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1556 *In the bottom of the 10th inning, with runners on 1st and 2nd, OAK batter Nick Allen hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	HOU vs OAK on 20230526
	state before	(1, 1, 0, 0, 5, 5, 0, 0)
34.	error	play,5,0,tuckk001,01,CX,C/E2.1-2;B-1
	state after	(1, 1, 1, 0, 5, 5, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1557 *In the top of the 5th inning, with a runner on 1st, HOU batter Kyle Tucker had a 0-1 count. Catcher's interference allowed the batter to take 1st. The runner on 1st base advanced one base.*

	game	HOU vs OAK on 20230527
	state before	(2, 0, 1, 0, 1, 2, 0, 0)
35.	error	play,1,0,abrej003,10,BX,E6/TH/G6.2-3;B-1
	state after	(2, 1, 0, 1, 1, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1558 *In the top of the 1st inning, with a runner on 2nd, HOU batter Jose Abreu hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	HOU vs OAK on 20230528
	state before	(2, 1, 0, 0, 2, 1, 1, 1)
36.	error	play,2,1,allen002,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1559 *In the bottom of the 2nd inning, with a runner on 1st, OAK batter Nick Allen had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Allen is still at-bat, now with a 0-1 count.*

	game	ATL vs OAK on 20230529
	state before	(1, 1, 1, 0, 5, 1, 0, 1)
37.	error	play,5,1,ruize001,00,X,S7/G56.2-H;1-3;B-2(E7/TH)
	state after	(1, 0, 1, 1, 5, 1, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 1560 *In the bottom of the 5th inning, with runners on 1st and 2nd, OAK batter Esteury Ruiz hit a ground ball single to 3rd. The runner on 2nd scored (1 RBI) and the runner from 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	ATL vs OAK on 20230530
	state before	(1, 1, 1, 1, 9, 1, 1, 1)
38.	error	play,9,1,bridj001,10,BX,FC5/G56.3-H(E5)(NR)(UR);2-3;1-2;B-1
	state after	(1, 1, 1, 1, 9, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1561 *In the bottom of the 9th inning, with bases loaded, OAK batter Jonah Bride hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	ATL vs OAK on 20230531
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
39.	error	play,1,1,ruize001,02,FF.X,E6/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1562 *In the bottom of the 1st inning, with bases empty, OAK batter Esteury Ruiz hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.*

23.2.3 June

	game	TBA vs OAK on 20230613
	state before	(1, 0, 0, 0, 5, 1, 0, 1)
40.	error	play,5,1,ruize001,22,CBBFX,C/E2.B-1
	state after	(1, 1, 0, 0, 5, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1563 *In the bottom of the 5th inning, with bases empty, OAK batter Esteury Ruiz had a 2-2 count. Catcher's interference allowed the batter to take 1st.*

	game	TBA vs OAK on 20230614
	state before	(1, 1, 0, 0, 5, 1, 3, 0)
41.	error	play,5,0,margm001,10,B>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 5, 1, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1564 This game had 2 errors. The final score was 6-3.
In the top of the 5th inning, with a runner on 1st, TBA batter Manuel Margot had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Margot is still at-bat, with a 2-0 count.

	game	TBA vs OAK on 20230614
	state before	(0, 0, 1, 0, 7, 4, 3, 0)
42.	error	play,7,0,diazy001,20,BB2,P02(E1/TH).2-3
	state after	(0, 0, 0, 1, 7, 4, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1565 *In the top of the 7th inning, with a runner on 2nd, TBA batter Yandy Diaz has a 2-0 count. The pitcher attempted to pickoff the runner on 2nd. A throwing error by the pitcher allowed that runner to reach 3rd. Diaz is still at-bat with a 2-0 count.*

	game	TBA vs OAK on 20230615
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
43.	error	play,2,0,ramih004,00,X,E3/G3.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1566 This game had 2 errors. The final score was 4-3.
In the top of the 2nd inning, with bases empty, TBA batter Harold Ramirez hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.

	game	TBA vs OAK on 20230615
	state before	(1, 1, 1, 0, 5, 3, 0, 1)
44.	error	play,5,1,ruize001,12,CBCX,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 5, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1567 See also the errors in the game against the Diamondbacks on 2023-05-16, especially the Nick Allen at-bat.

In the bottom of the 5th inning, with runners on 1st and 2nd, OAK batter Esteury Ruiz hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	PHI vs OAK on 20230616
	state before	(1, 1, 0, 1, 9, 4, 1, 0)
45.	error	play,9,0,bohma001,12,FFBF1,P03(E1/TH).3-H(NR);1-3
	state after	(1, 0, 0, 1, 9, 5, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1568 *In the top of the 9th inning, with runners on 1st and 3rd, PHI batter Alec Bohm has a 1-2 count. The pitcher attempted to pickoff the runner on 3rd. A throwing error by the pitcher allowed the runner on 3rd to score (unearned, no RBI) and the runner on 1st to advance to 3rd. Bohm is still at-bat with a 1-2 count.*

	game	PHI vs OAK on 20230618
	state before	(2, 0, 0, 0, 1, 1, 0, 0)
46.	error	play,1,0,realj001,31,BBCBX,C/E2.B-1
	state after	(2, 1, 0, 0, 1, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1569 *In the top of the 1st inning, with bases empty, PHI batter J.T. Realmuto had a 3-1 count. Catcher's interference allowed the batter to take 1st.*

Visiting/Home team rosters for NYA vs OAK on 2023-06-28

position	innings	player	position	innings	player
1	1	Domingo German	1	1	JP Sears
2	1	Kyle Higashioka	1	5	Shintaro Fujinami
3	1	Anthony Rizzo	1	7	Austin Pruitt
4	1	DJ LeMahieu	1	7	Yacksel Rios
5	1	Josh Donaldson	1	9	Sam Long
6	1	Anthony Volpe	2	1	Shea Langeliers
7	1	Isiah Kiner-Falefa	3	1	Ryan Noda
8	1	Harrison Bader	4	1	Tony Kemp
9	1	Giancarlo Stanton	5	1	Jonah Bride
9	7	Oswaldo Cabrera	6	1	Aledmys Diaz
			7	1	Brent Rooker
			8	1	Esteury Ruiz
			9	1	Seth Brown

	game	NYA vs OAK on 20230628
	state before	(0, 0, 1, 0, 5, 2, 0, 0)
47.	error	play,5,0,volpa001,00,X,S1/BG34S.2-H(E1/TH) (NR) ;B-2
	state after	(0, 0, 1, 0, 5, 3, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1570 This game had 3 errors. The final score was 11-0.
In the top of the 5th inning, with a runner on 2nd, NYA batter Anthony Volpe hit a bunt single to shallow right infield. A throwing error by the pitcher allowed the runner on 2nd to score and the batter to reach 2nd.

	game	NYA vs OAK on 20230628
	state before	(0, 1, 1, 0, 5, 4, 0, 0)
48.	error	play,5,0,rizza001,12,FBFFF.X,FC3/G3.2-3;1-2(E3) ;B-1
	state after	(0, 1, 1, 1, 5, 4, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1571 *In the top of the 5th inning, with runners on 1st and 2nd, NYA batter Anthony Rizzo hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	NYA vs OAK on 20230628
	state before	(1, 1, 1, 0, 9, 8, 0, 0)
49.	error	play,9,0,badeh001,22,C*BFBX,FC5/G6.2-H(E5/TH)(NR);1-2;B-1
	state after	(1, 1, 1, 0, 9, 9, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1572 *In the top of the 9th inning, with runners on 1st and 2nd, NYA batter Harrison Bader hit a ground ball into a fielder's choice at short. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	NYA vs OAK on 20230629
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
50.	error	play,1,1,nodar001,11,CB+1>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1573 This game had 2 errors. The final score was 10-4. *In the bottom of the 1st inning, with a runner on 1st, OAK batter Ryan Noda had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at-bat, now with a 2-1 count.*

	game	NYA vs OAK on 20230629
	state before	(1, 1, 0, 0, 6, 4, 3, 0)
51.	error	play,6,0,trevj001,01,.1F1,P01(1E3).1-2
	state after	(1, 0, 1, 0, 6, 4, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1574 *In the top of the 6th inning, with a runner on 1st, NYA batter Jose Trevino has a 0-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the first baseman allowed that runner to reach 2nd. Trevino is still at-bat with a 0-1 count.*

23.2.4 July

	game	CHA vs OAK on 20230701
	state before	(2, 1, 0, 0, 8, 6, 6, 1)
52.	error	play,8,1,bledj001,02,FFX,FC4/G4.1-2(E4);B-1
	state after	(2, 1, 1, 0, 8, 6, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1575 This game had 2 errors. The final score was 6-7 (in extra innings).

In the bottom of the 8th inning, with a runner on 1st, OAK batter JJ Bleday hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	CHA vs OAK on 20230701
	state before	(2, 0, 1, 0, 10, 6, 6, 1)
53.	error	play,10,1,bledj001,10,BX,E4/G34.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 10, 6, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1576 *In the bottom of the 10th inning, with a runner on 2nd, OAK batter JJ Bleday hit a ground ball to 1st/2nd. An error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	CHA vs OAK on 20230702
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
54.	error	play,1,0,robel002,01,C>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1577 This game had 2 errors. The final score was 8-7. *In the top of the 1st inning, with a runner on 1st, CHA batter Luis Robert had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Robert is still at-bat, now with a 1-1 count.*

	game	CHA vs OAK on 20230702
	state before	(0, 0, 0, 0, 6, 5, 3, 0)
55.	error	play,6,0,remiz001,11,.BTX,D7/L78XD+.B-3(E7)
	state after	(0, 0, 0, 1, 6, 5, 3, 0)
	fielder	7 (left fielder on the home team)

Summary 1578 *In the top of the 6th inning, with bases empty, CHA batter Zach Remillard hit a line drive double to extra deep left center field. An error by the left fielder allowed the batter to reach 3rd.*

Visiting/Home team rosters for BOS vs OAK on 2023-07-18

position	innings	player	position	innings	player
1	1	Joe Jacques	1	1	Luis Medina
1	2	Chris Murphy	1	6	Sam Long
1	7	Joely Rodriguez	1	7	Shintaro Fujinami
1	8	Richard Bleier	1	8	Sam Moll
2	1	Jorge Alfaro	1	9	Trevor May
3	1	Triston Casas	2	1	Shea Langeliers
4	1	Christian Arroyo	3	1	Ryan Noda
5	1	Justin Turner	4	1	Zack Gelof
6	1	Yu Chang	5	1	Aledmys Diaz
6	8	Enrique Hernandez	5	9	Jace Peterson
7	1	Rob Refsnyder	6	1	Nick Allen
8	1	Jarren Duran	7	1	Tony Kemp
8	8	Adam Duvall	8	1	JJ Bleday
9	1	Alex Verdugo	9	1	Brent Rooker

56.	game	BOS vs OAK on 20230718
	state before	(0, 0, 0, 0, 2, 0, 1, 1)
	error	play,2,1,diaza003,10,BX,S6/G6MS.B-2(E6/TH)
	state after	(0, 0, 1, 0, 2, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1579 This game had 3 errors. The final score was 0-3.

In the bottom of the 2nd inning, with bases empty, OAK batter Aledmys Diaz hit a ground ball single to shallow short. A throwing error by the shortstop allowed the batter to reach 2nd.

	game	BOS vs OAK on 20230718
	state before	(0, 0, 0, 0, 2, 0, 3, 1)
57.	error	play,2,1,langs001,10,BX,E5/G5D.B-1
	state after	(0, 1, 0, 0, 2, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1580 *In the bottom of the 2nd inning, with bases empty, OAK batter Shea Langeliers hit a ground ball to deep 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	BOS vs OAK on 20230718
	state before	(2, 0, 0, 0, 6, 0, 3, 1)
58.	error	play,6,1,langs001,31,BFBBX,E5/TH/G5+.B-1
	state after	(2, 1, 0, 0, 6, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1581 *In the bottom of the 6th inning, with bases empty, OAK batter Shea Langeliers (as batter as in the last error) hit a hard ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs OAK on 2023-07-19

position	innings	player	position	innings	player
1	1	Brayan Bello	1	1	Ken Waldichuk
1	5	Joely Rodriguez	1	5	Lucas Erceg
1	6	Josh Winckowski	1	6	Angel Felipe
1	8	Kenley Jansen	1	8	Sam Long
2	1	Connor Wong	1	9	Trevor May
3	1	Justin Turner	2	1	Shea Langeliers
3	6	Triston Casas	3	1	Tyler Soderstrom
4	1	Enrique Hernandez	3	9	Aledmys Diaz
4	6	Justin Turner	4	1	Jordan Diaz
5	1	Rafael Devers	5	1	Jace Peterson
6	1	Yu Chang	6	1	Nick Allen
6	6	Enrique Hernandez	7	1	Tony Kemp
7	1	Rob Refsnyder	8	1	JJ Bleday
8	1	Adam Duvall	9	1	Cody Thomas
9	1	Alex Verdugo			

	game	BOS vs OAK on 20230719
	state before	(2, 1, 0, 0, 1, 2, 0, 0)
59.	error	play,1,0,verda001,12,CCB1F>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 0, 1, 1, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1582 This game had 4 errors. The final score was 5-6. *In the top of the 1st inning, with a runner on 1st, BOS batter Alex Verdugo had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Verdugo is still at-bat, now with a 2-2 count.*

	game	BOS vs OAK on 20230719
	state before	(0, 1, 0, 0, 5, 2, 6, 0)
60.	error	play,5,0,chany001,22,BBCF1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 5, 2, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 1583 *In the top of the 5th inning, with a runner on 1st, BOS batter Yu Chang has a 2-2 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to reach 3rd. Chang is still at-bat with a 2-2 count.*

	game	BOS vs OAK on 20230719
	state before	(0, 1, 0, 0, 6, 4, 6, 0)
61.	error	play,6,0,verda001,00,1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 6, 4, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 1584 *In the top of the 6th inning, with a runner on 1st, BOS batter Alex Verdugo has a 0-0 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to reach 3rd. Verdugo is still at-bat with a 0-0 count.*

	game	BOS vs OAK on 20230719
	state before	(1, 0, 0, 1, 6, 4, 6, 0)
62.	error	play,6,0,herne001,11,FBX,E5/TH/G56.3-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 6, 5, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1585 *In the top of the 6th inning, with a runner on 3rd, BOS batter Enrique Hernandez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	HOU vs OAK on 20230722
	state before	(1, 0, 0, 0, 3, 0, 0, 0)
63.	error	play,3,0,penaj004,12,FBFX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 3, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1586 This game had 2 errors. The final score was 1-4.
In the top of the 3rd inning, with bases empty, HOU batter Jeremy Pena hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	HOU vs OAK on 20230722
	state before	(0, 1, 0, 0, 4, 0, 0, 0)
64.	error	play,4,0,mccoc001,11,CBX,54(1)/F0/G56.B-2(E4/TH)
	state after	(1, 0, 1, 0, 4, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1587 *In the top of the 4th inning, with a runner on 1st, HOU batter Chas McCormick hit a ground ball to 3rd. The second baseman (with an assist from the third baseman) forced out the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	HOU vs OAK on 20230723
	state before	(0, 1, 1, 0, 3, 0, 1, 1)
65.	error	play,3,1,kempt001,01,LX,FC3/G34.2-3;1-2(E3);B-1
	state after	(0, 1, 1, 1, 3, 0, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1588 This game had 2 errors. The final score was 3-2.
In the bottom of the 3rd inning, with runners on 1st and 2nd, OAK batter Tony Kemp hit a ground ball into a fielder's choice at 1st/2nd. The runner on 2nd advanced to 3rd. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	HOU vs OAK on 20230723
	state before	(0, 1, 0, 0, 6, 1, 1, 1)
66.	error	play,6,1,brows003,12,BCFFF>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 6, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1589 *In the bottom of the 6th inning, with a runner on 1st, OAK batter Seth Brown had a 1-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

23.2.5 August

	game	SFN vs OAK on 20230805
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
67.	error	play,2,0,davij006,12,SCBX,C/E2.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1590 *In the top of the 2nd inning, with bases empty, SFN batter J.D. Davis had a 1-2 count. Catcher's interference allowed the batter to take 1st.*

Visiting/Home team rosters for SFN vs OAK on 2023-08-06

position	innings	player	position	innings	player
1	1	Alex Cobb	1	1	Luis Medina
1	6	Luke Jackson	1	4	Sam Long
1	7	Ryan Walker	1	5	Austin Pruitt
1	7	Scott Alexander	1	6	Kirby Sneed
1	8	Jakob Junis	1	7	Lucas Erceg
2	1	Patrick Bailey	1	8	Angel Felipe
3	1	LaMonte Wade Jr.	1	9	Trevor May
4	1	Thairo Estrada	2	1	Shea Langeliers
5	1	J.D. Davis	3	1	Seth Brown
6	1	Brandon Crawford	4	1	Zack Gelof
7	1	Joc Pederson	5	1	Jordan Diaz
7	6	Austin Slater	6	1	Nick Allen
7	7	Blake Sabol	7	1	JJ Bleday
8	1	Luis Matos	7	8	Tony Kemp
8	7	Austin Slater	8	1	Esteury Ruiz
9	1	Michael Conforto	9	1	Brent Rooker
			9	8	JJ Bleday

	game	SFN vs OAK on 20230806
	state before	(0, 1, 0, 0, 3, 2, 3, 0)
68.	error	play,3,0,davij006,00,X,E5/G6.1-2;B-1
	state after	(0, 1, 1, 0, 3, 2, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1591 This game had 3 errors. The final score was 6-8.
In the top of the 3rd inning, with a runner on 1st, SFN batter J.D. Davis hit a ground ball to short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	SFN vs OAK on 20230806
	state before	(1, 0, 1, 0, 6, 6, 4, 1)
69.	error	play,6,1,rookb001,32,.FC*B*B*B*BX,E6/TH/G6.2-3;B-1
	state after	(1, 1, 0, 1, 6, 6, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1592 *In the bottom of the 6th inning, with a runner on 2nd, OAK batter Brent Rooker hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SFN vs OAK on 20230806
	state before	(2, 0, 0, 0, 8, 6, 8, 1)
70.	error	play,8,1,allen002,00,X,S6/G4M.B-2(E6/TH)
	state after	(2, 0, 1, 0, 8, 6, 8, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1593 *In the bottom of the 8th inning, with bases empty, OAK batter Nick Allen hit a ground ball single to 2nd. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	TEX vs OAK on 20230807
	state before	(2, 0, 1, 1, 7, 2, 3, 0)
71.	error	play,7,0,lowen001,01,CX,3E1/G34.3-H(NR)(UR);2-3;B-1
	state after	(2, 1, 0, 1, 7, 3, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1594 *In the top of the 7th inning, with runners on 2nd and 3rd, TEX batter Nathaniel Lowe hit a ground ball to 1st/2nd. An error by the pitcher (with an assist from the first baseman) allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	BAL vs OAK on 20230818
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
72.	error	play,1,0,santa003,31,*BBC1*BX,S15/G1.1-3(E5/TH)
	state after	(1, 1, 0, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1595 This game had 2 errors. The final score was 9-6.

In the top of the 1st inning, with a runner on 1st, BAL batter Anthony Santander hit a ground ball single to shallow center infield.. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	BAL vs OAK on 20230818
	state before	(0, 0, 1, 0, 4, 5, 2, 0)
73.	error	play,4,0,fraza001,10,BX,E3/G3.2-3;B-1
	state after	(0, 1, 0, 1, 4, 5, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1596 *In the top of the 4th inning, with a runner on 2nd, BAL batter Adam Frazier hit a ground ball to 1st base. An error by the first baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	BAL vs OAK on 20230819
	state before	(1, 1, 0, 1, 10, 4, 2, 0)
74.	error	play,10,0,haysa001,12,BCF1.X,FC6/G6S.3-H;1-3(E6/TH);B-1
	state after	(1, 1, 0, 1, 10, 5, 2, 0)
	fielder	6 (1st baseman on the home team)

Summary 1597 *In the top of the 10th inning, with runners on 1st and 3rd, BAL batter Austin Hays hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	KCA vs OAK on 20230821
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
75.	error	play,1,1,ruize001,10,BX,C/E2.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1598 *In the bottom of the 1st inning, with bases empty, OAK batter Esteury Ruiz had a 1-0 count. Catcher's interference allowed the batter to take 1st.*

	game	KCA vs OAK on 20230822
	state before	(2, 0, 0, 0, 3, 2, 3, 0)
76.	error	play,3,0,peres002,21,BBSX,E6/G6.B-1
	state after	(2, 1, 0, 0, 3, 2, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1599 This game had 2 errors. The final score was 4-5. *In the top of the 3rd inning, with bases empty, KCA batter Salvador Perez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	KCA vs OAK on 20230822
	state before	(1, 0, 0, 0, 4, 2, 5, 0)
77.	error	play,4,0,wated001,12,CBFX,E1/G25-.B-1
	state after	(1, 1, 0, 0, 4, 2, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1600 *In the top of the 4th inning, with bases empty, KCA batter Drew Waters hit a soft ground ball to the shallow left infield. An error by the pitcher allowed the batter to reach 1st base.*

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78.	game	ANA vs OAK on 20230901
	state before	(2, 0, 0, 0, 5, 0, 5, 0)
	error	play,5,0,parik002,22,BST.FBX,E5/G5.B-1
	state after	(2, 1, 0, 0, 5, 0, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1601 *In the top of the 5th inning, with bases empty, ANA batter Kyren Paris hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for ANA vs OAK on 2023-09-03

position	innings	player	position	innings	player
1	1	Tyler Anderson	1	1	Kyle Muller
1	6	Andrew Wantz	1	5	Adrian Martinez
1	7	Jose Marte	1	7	Francisco Perez
1	7	Jose Soriano	1	8	Dany Jimenez
1	8	Kolton Ingram	1	9	Kirby Sneed
2	1	Chad Wallach	1	9	Trevor May
3	1	Brandon Drury	2	1	Shea Langeliers
4	1	Michael Stefanic	3	1	Ryan Noda
5	1	Eduardo Escobar	4	1	Zack Gelof
6	1	Kyren Paris	5	1	Jordan Diaz
7	1	Randal Grichuk	5	7	Aledmys Diaz
8	1	Mickey Moniak	6	1	Nick Allen
9	1	Luis Rengifo	7	1	Aledmys Diaz
			7	7	Seth Brown
			8	1	Esteury Ruiz
			8	8	Lawrence Butler
			9	1	Brent Rooker

	game	ANA vs OAK on 20230903
	state before	(2, 1, 0, 0, 4, 2, 0, 0)
79.	error	play,4,0,wallc001,21,B*BF1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 4, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1602 This game had 3 errors. The final score was 6-10.
In the top of the 4th inning, with a runner on 1st, ANA batter Chad Wallach has a 2-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to advance to 3rd. Wallach is still at-bat with a 2-1 count.

	game	ANA vs OAK on 20230903
	state before	(1, 1, 0, 0, 7, 4, 4, 1)
80.	error	play,7,1,nodar001,10,BX,FC1/G1.1-3(E1/TH);B-1
	state after	(1, 1, 0, 1, 7, 4, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1603 *In the bottom of the 7th inning, with a runner on 1st, OAK batter Ryan Noda hit a ground ball into a fielder's choice at shallow center infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	ANA vs OAK on 20230903
	state before	(1, 1, 1, 1, 7, 4, 4, 1)
81.	error	play,7,1,kempt001,01,.SX,S8/G4M.3-H(UR);2-H;1-H(E8/TH)(NR);B-3
	state after	(1, 0, 0, 1, 7, 4, 7, 1)
	fielder	8 (center field on the visiting team)

Summary 1604 *In the bottom of the 7th inning, with bases loaded, OAK batter Tony Kemp hit a ground ball single to 2nd. The runners on 2nd and 3rd both scored (unearned, 2 RBIs). A throwing error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	TOR vs OAK on 20230905
	state before	(1, 0, 0, 0, 8, 6, 1, 0)
82.	error	play,8,0,kierk001,01,CX,D7/L7LS.B-3(E7/TH)
	state after	(1, 0, 0, 1, 8, 6, 1, 0)
	fielder	7 (left field on the home team)

Summary 1605 *In the top of the 8th inning, with bases empty, TOR batter Kevin Kiermaier hit a line drive double to shallow left field. A throwing error by the left fielder allowed the batter to reach 3rd.*

Visiting/Home team rosters for SDN vs OAK on 2023-09-16

position	innings	player	position	innings	player
1	1	Matt Waldron	1	1	Mason Miller
1	6	Scott Barlow	1	2	Luis Medina
1	7	Robert Suarez	1	5	Easton Lucas
1	9	Josh Hader	1	7	Lucas Erceg
2	1	Brett Sullivan	1	9	Dany Jimenez
3	1	Ji Man Choi	2	1	Carlos Perez
3	7	Garrett Cooper	3	1	Ryan Noda
4	1	Ha-Seong Kim	4	1	Zack Gelof
5	1	Matthew Batten	5	1	Aledmys Diaz
6	1	Xander Bogaerts	6	1	Nick Allen
7	1	Juan Soto	7	1	Tony Kemp
8	1	Trent Grisham	8	1	Lawrence Butler
9	1	Fernando Tatis Jr.	9	1	Seth Brown

	game	SDN vs OAK on 20230916
	state before	(2, 1, 1, 0, 5, 3, 1, 0)
83.	error	play,5,0,sullb001,12,.BFFX,S3/G34.2-H(E3/TH)(NR)(UR);1-H(NR)(UR);B-3
	state after	(2, 0, 0, 1, 5, 5, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1606 This game had 3 errors. The final score was 5-2.

In the top of the 5th inning, with runners on 1st and 2nd, SDN batter Brett Sullivan hit a ground ball single to 1st/2nd. A throwing error by the first baseman allowed the runners on 1st and 2nd to each score (unearned, no RBIs) and the batter to reach 3rd.

84.	game	SDN vs OAK on 20230916
	state before	(2, 1, 0, 0, 7, 5, 2, 1)
	error	play,7,1,nodar001,01,C1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 5, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1607 *In the bottom of the 7th inning, with a runner on 1st, OAK batter Ryan Noda had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at-bat, now with a 1-1 count.*

85.	game	SDN vs OAK on 20230916
	state before	(2, 0, 0, 0, 8, 5, 2, 0)
	error	play,8,0,tatif002,11,BCX,6E3/G6.B-1
	state after	(2, 1, 0, 0, 8, 5, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1608 *In the top of the 8th inning, with bases empty, SDN batter Fernando Tatis hit a ground ball to short. An error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st base.*

86.	game	SEA vs OAK on 20230919
	state before	(1, 1, 0, 0, 7, 2, 0, 0)
	error	play,7,0,moord002,02,.SF1>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1609 *In the top of the 7th inning, with a runner on 1st, SEA batter Dylan Moore had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Moore is still at-bat, now with a 1-2 count.*

	game	SEA vs OAK on 20230920
	state before	(2, 0, 1, 0, 1, 0, 0, 0)
87.	error	play,1,0,hernt002,22,SBS.BFX,E5/G5.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1610 *This game had 2 errors. The final score was 6-3. In the top of the 1st inning, with a runner on 2nd, SEA batter Teoscar Hernandez hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SEA vs OAK on 20230920
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
88.	error	play,2,0,frant002,02,FF.FFX,E5/G5.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1611 *In the top of the 2nd inning, with bases empty, SEA batter Ty France hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for DET vs OAK on 2023-09-21

position	innings	player	position	innings	player
1	1	Tarik Skubal	1	1	Luis Medina
1	8	Brenan Hanifree	1	6	Mason Miller
2	1	Jake Rogers	1	8	Easton Lucas
3	1	Spencer Torkelson	1	9	Trevor May
4	1	Andy Ibanez	2	1	Shea Langeliers
5	1	Matt Vierling	3	1	Jordan Diaz
5	8	Zach McKinstry	3	9	Ryan Noda
6	1	Zach McKinstry	4	1	Zack Gelof
6	8	Zack Short	5	1	Kevin Smith
7	1	Akil Baddoo	6	1	Nick Allen
7	8	Matt Vierling	7	1	Aledmys Diaz
8	1	Parker Meadows	8	1	Esteury Ruiz
9	1	Kerry Carpenter	9	1	Brent Rooker

	game	DET vs OAK on 20230921
	state before	(1, 0, 0, 1, 1, 0, 0, 0)
89.	error	play,1,0,torks001,00,X,FC6/G6.3-H;B-2(E6/TH)
	state after	(1, 0, 1, 0, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1612 This game had 3 errors. This particular play is a little unusual since it is an example of a fielder's choice with the play made to the runner at 1st (usually a FC is indicated when the fielder chooses to put out a runner other than the batter). Each game in this Tigers-Athletics series, for which this is the first, had at least 1 error, for a total of 6. The final score for this game was 7-3.

In the top of the 1st inning, with a runner on 3rd, DET batter Spencer Torkelson hit a ground ball into a fielder's choice at short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 2nd.

	game	DET vs OAK on 20230921
	state before	(1, 1, 0, 0, 5, 3, 0, 0)
90.	error	play,5,0,mckiz001,00,X,D9/L9LD.1-H(E9/TH)(NR);B-3
	state after	(1, 0, 0, 1, 5, 4, 0, 0)
	fielder	9 (right field on the home team)

Summary 1613 *In the top of the 5th inning, with a runner on 1st, DET batter Zach McKinstry hit a line drive double to deep right field. A throwing error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and rgw batter to reach 3rd. to advance to*

	game	DET vs OAK on 20230921
	state before	(0, 1, 1, 1, 8, 6, 0, 0)
91.	error	play,8,0,shorz001,21,.SBBX,FC3/G3S.3-H(E3)(NR)(UR);2-3;1-2;B-1
	state after	(0, 1, 1, 1, 8, 7, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1614 *In the top of the 8th inning, with bases loaded, DET batter Zack Short hit a ground ball into a fielder's choice at 1st. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	DET vs OAK on 20230922
	state before	(0, 0, 0, 0, 6, 2, 2, 1)
92.	error	play,6,1,kempt001,21,SBBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 2, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1615 *In the bottom of the 6th inning, with bases empty, OAK batter Tony Kemp hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

93.	game	DET vs OAK on 20230923
	state before	(2, 0, 1, 0, 3, 0, 0, 0)
	error	play,3,0,torks001,10,BX,E5/G56.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 3, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1616 This game had 2 errors. The final score was 1-4.
In the top of the 3rd inning, with a runner on 2nd, DET batter Spencer Torkelson hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

94.	game	DET vs OAK on 20230923
	state before	(1, 1, 0, 0, 9, 1, 4, 0)
	error	play,9,0,mckiz001,12,SSFBX,E6/TH/G6M.1-2;B-1
	state after	(1, 1, 1, 0, 9, 1, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1617 *In the top of the 9th inning, with a runner on 1st, DET batter Zach McKinstry hit a ground ball to short. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

95.	game	DET vs OAK on 20230924
	state before	(1, 1, 1, 1, 3, 0, 0, 0)
	error	play,3,0,torks001,21,BBFX,S9/F89S.3-H;2-H;1-3(E9/TH);B-2
	state after	(1, 0, 1, 1, 3, 2, 0, 0)
	fielder	9 (right field on the home team)

Summary 1618 *In the top of the 3rd inning, with bases loaded, DET batter Spencer Torkelson hit a fly ball single to shallow right center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

Chapter 24

Philadelphia Phillies (PHI)

The Phillies franchise was founded in Philadelphia over 140 years ago, in 1883. They didn't move or change their name in that time, making them one of the oldest one-name, one-city franchise in professional sports. The Phillies have won two World Series championships, in 1980 and 2008, and eight NL pennants, the latest in 2022.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Philadelphia against the Phillies. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (DER or DefEff) is one measure of the fielding performance of a team (see §3.2 for more details). The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 PHI team is a little better average: 0.695 compared with the MLB average of 0.691 that year [BR] (which is good).

The number of errors (home and away) by the PHI team in 2023 is 91, also above average (the MLB average of 83.9, but 91 is about a half of a standard deviation from the mean (which is bad).

They had a 0.556 winning percentage in the 2023 regular season. That year, there were 88 errors in the 81 home games, 47 by the Phillies and the rest (41) by visiting teams.

24.1 Season totals

Here, `vt` denotes the visiting team and `ht` denotes the home team.

24.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals	RBI	no RBI
Errors by <code>vt</code> :	41	-	-
Runs by <code>ht</code> while an error by <code>vt</code> :	13	3	10
Errors by <code>ht</code> :	47	-	-
Runs by <code>vt</code> while error by <code>ht</code> :	6	2	4

There were about 0.127 opponent runs per error (by the Phillies) and about 0.317 Phillies runs per per error (by the visiting team). In other words, the visiting teams were about 42% less efficient at converting errors to runs than the Phillies. Over the regular season, the chance of a play (with out without and error) creating a run is $R/PA = 796/6207 = 12.8\%$, so the Phillies are over twice as likely to score on an error by the visiting team than in a randomly selected play.

Note that the total number of errors 88 is about the number of games played over the season, so the expected number of errors (by both sides) in a randomly selected home game is about 1.08.

Errors by position	vt	ht	total
errors due to pitcher:	3	8	11
to catcher:	4	5	9
to 1st baseman:	2	7	9
to 2nd baseman:	4	3	7
to 3rd baseman:	12	7	19
to shortstop:	9	10	19
to left fielder:	2	5	7
to center fielder:	4	1	5
to right fielder:	1	1	2
	41	47	88

24.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	29	18	47
(lone) runner on 1st	6	16	22
(lone) runner on 2nd	3	5	8
(lone) runner on 3rd	0	2	2
two runners on base	3	6	9
bases loaded	0	0	0
total	41	47	88

We can use the table above to compare the number of Phillies errors with bases empty verses a (lone) runner on 1st. They are about the same, which is unusual since the bases empty state is typically 2 or 3 times as frequent as the runner on 1st state. On the other hand, the number of visitor errors with bases empty is relatively high when compared to those with a runner on 1st. Why is this?

A table of frequencies of errors by the Phillies in PHI 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	$8/47 = 17.0\%$
(1, 0, 0, 0)	4	8.5%
(2, 0, 0, 0)	6	12.8%
(0, 1, 0, 0)	5	10.6%
(1, 1, 0, 0)	5	10.6%
(2, 1, 0, 0)	6	12.8%
(1, 0, 1, 0)	3	6.4%
(2, 0, 1, 0)	2	4.3%
(0, 1, 1, 0)	2	4.3%
(1, 1, 1, 0)	2	4.3%

A table of frequencies of most common errors by a typical visiting team in PHI 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	15	$4/35 = 36.6\%$
(1, 0, 0, 0)	8	19.5%
(2, 0, 0, 0)	6	14.6%
(0, 1, 0, 0)	0	0%
(1, 1, 0, 0)	3	7.3%
(2, 1, 0, 0)	3	7.3%
(0, 0, 1, 0)	3	7.3%
(1, 1, 1, 0)	1	2.4%
(2, 1, 1, 0)	1	2.4%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Phillies homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `PHI2023.evn`):

state	PHI2023 percentages
(0, 0, 0, 0)	24.2%
(1, 0, 0, 0)	15.8%
(2, 0, 0, 0)	13.3%
(0, 1, 0, 0)	6.2%
(1, 1, 0, 0)	6.9%
(2, 1, 0, 0)	7.6%
(0, 0, 1, 0)	2.4%
(1, 0, 1, 0)	2.6%
(2, 0, 1, 0)	3.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.8%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.5%
(1, 1, 1, 0)	2.3%
(2, 1, 1, 0)	3.1%

24.1.3 Some interesting errors

Several interesting errors occurred in the PHI 2023 homegames.

- For example, in the 2023-05-09 game against Toronto's Blue Jays, in the bottom of the 5th inning, with a runner on 1st, PHI batter Trea Turner had 2 outs and a 0-2 count when he hit a single to 2nd. Incredibly, the runner on 1st scored without help from the error. Of course, the runner will have taken off at batter's contact because of the count on 2 outs. A throwing error by the second baseman allowed Turner to reach 2nd base.
- Another interesting play was in the game against Miami on 2023-09-09. This game had 3 errors, two of which were what I call *infinity* (∞) *plays*. In this simple play, a catchable pop fly is not caught. Retrosheet calls this an **FLE** and the error is described in MLB Rule 9.12(a)(2). What's cool about such errors is that such a play does not change the

game state, so can be, in theory, repeated a infinite number of times without affecting the game.

24.1.4 Relative team rankings

We have 81 home games between the Phillies and some visiting MLB team (vt). A number of teams did not play the Angles in the regular season. However, of those that did, PHI *lost* more games than won against¹:

ARI (2/3), BOS (2/3), MIL (2/3), MIN (2/3), MIA (4/6), ATL (5/6)

and *won* more than lost against

TOR (0/2), DET (0/3), SLN (0/3), ANA (1/3), BAL (1/3), CHN (1/3), CIN
(1/3),
KCA (1/3), LAN (1/3), PIT (1/3), SEA (1/3), SFN (1/3), COL (1/4), SDN
(1/4), NYN (1/7), WAS (3/7) .

The teams that split their series were:

(none)

The teams that did not play in any of the 81 home games in PHI 2023 were:

CHA, CLE, HOU, NYA, OAK, TBA, TEX .

24.2 Game-by-game totals

To save space, we only give the team rosters in special cases. The typical starting lineup for the 2023 Philadelphia Phillies included the following players:

- Kyle Schwarber (OBP = .343) - Left Field
- Trea Turner (OBP = .320) - Shortstop

¹We record the vt W/T record in parentheses (W = numbers of games vt won against PHI at Philadelphia, T = Total number of home games vt played vs PHI played at Philadelphia.

- Bryce Harper (OBP = .401) - First Base/DH
- J.T. Realmuto (OBP = .310) - Catcher
- Nick Castellanos (OBP = .311) - Right Field
- Alec Bohm (OBP = .327) - Third Base
- Bryson Stott (OBP = .329) - Second Base
- Brandon Marsh (OBP = .372) - Center Field

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 Phillies home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

24.2.1 April

	game	CIN vs PHI on 20230408
	state before	(1, 1, 0, 0, 3, 1, 0, 1)
1.	error	play,3,1,pachc001,00,X,FC/SH/BG1S-.1-2;B-1(E1/TH)
	state after	(1, 1, 1, 0, 3, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1619 This game had 2 errors. The Reds lost to the Phillies 2-3.

In the bottom of the 3rd inning, with a runner on 1st, PHI batter Cristian Pache hit a soft bunt into a fielder's choice² at shallow infield. The runner on 1st advanced to 2nd. A throwing error by the pitcher allowed the batter to reach 1st base safely.

²A fielder's choice where the fielder chooses to get the *batter* out is unusual.

	game	CIN vs PHI on 20230408
	state before	(0, 0, 0, 0, 5, 1, 0, 1)
2.	error	play,5,1,sosae001,00,X,E5/G7S.B-1
	state after	(0, 1, 0, 0, 5, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1620 *In the bottom of the 5th inning, with bases empty, PHI batter Edmundo Sosa hit a grounder to short left field. An error by the third baseman allowed reached 1st base.*

	game	MIA vs PHI on 20230411
	state before	(1, 0, 1, 0, 3, 1, 0, 0)
3.	error	play,3,0,coopg002,32,F1>B.B*BCX,E5/G56.B-1
	state after	(1, 1, 1, 0, 3, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1621 *In the top of the 3rd inning, with a runner on 2nd, MIA batter Garrett Cooper hit a ground ball to 3rd. The batter reached 1st base on an error by the third baseman.*

	game	COL vs PHI on 20230421
	state before	(2, 0, 0, 0, 2, 3, 0, 1)
4.	error	play,2,1,bohma001,01,CX,S6/G6M.B-2(E6/TH)
	state after	(2, 0, 1, 0, 2, 3, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1622 This game had 2 errors. The Phillies beat the Rockies 3-4.

In the bottom of the 2nd inning, with bases empty, PHI batter Alec Bohm hit a ground ball single to shallow left field. A throwing error by the shortstop allowed the batter to reach 2nd base.

	game	COL vs PHI on 20230421
	state before	(0, 0, 0, 0, 4, 3, 0, 1)
5.	error	play,4,1,castn001,12,CBFX,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1623 *In the bottom of the 4th inning, with bases empty, PHI batter Nick Castellanos hit a ground ball to 3rd. An error by the third baseman allowed the batter to reach 1st base.*

	game	COL vs PHI on 20230422
	state before	(1, 0, 0, 0, 5, 3, 3, 1)
6.	error	play,5,1,marsb002,22,CBFBX,E4/G4.B-1
	state after	(1, 1, 0, 0, 5, 3, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1624 *In the bottom of the 5th inning, with bases empty, PHI batter Brandon Marsh hit a ground ball to 2nd. An error by the second baseman allowed the batter to reach 1st base.*

	game	COL vs PHI on 20230423
	state before	(2, 1, 0, 0, 7, 3, 4, 0)
7.	error	play,7,0,profj001,00,1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 7, 3, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 1625 This game had 2 errors. The Rockies loost to the Phillies 3-9.

In the top of the 7th inning, with a runner on 1st, COL batter Julio Profar had a 0-0 count. The pitcher attempted a pick off at 1st. Due to a throwing error by the pitcher, the runner advanced from 1st to 3rd. Profar is still at bat with a 0-0 count.

	game	COL vs PHI on 20230423
	state before	(0, 0, 0, 0, 9, 3, 9, 0)
8.	error	play,9,0,casth001,12,..CSBX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 9, 3, 9, 0)
	fielder	6 (shortstop on the home team)

Summary 1626 *In the top of the 9th inning, with bases empty, COL batter Harold Castro hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	SEA vs PHI on 20230425
	state before	(2, 1, 0, 0, 5, 1, 0, 0)
9.	error	play,5,0,crawj002,01,1C>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1627 *In the top of the 5th inning, with a runner on 1st, SEA batter P.J. Crawford had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner to reach 3rd. Crawford is at bat, now with a 1-1 count.*

	game	SEA vs PHI on 20230427
	state before	(0, 0, 0, 0, 6, 0, 1, 1)
10.	error	play,6,1,stotb001,02,CFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1628 *In the bottom of the 6th inning, with bases empty, PHI batter Bryson Stott hit a ground ball but reached 1st base on a throwing error by the shortstop.*

24.2.2 May

	game	BOS vs PHI on 20230505
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
11.	error	play,1,0,tapir001,01,CX,E7/F78D.B-2
	state after	(0, 0, 1, 0, 1, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1629 This game had 2 errors. The Red Sox beat the Phillies 5-3.

In the top of the 1st inning, with bases empty, BOS batter Raimel Tapia hit a fly ball to deep left field. The batter advanced to 2nd base on an error by the left fielder.

	game	BOS vs PHI on 20230505
	state before	(2, 0, 0, 0, 7, 5, 3, 1)
12.	error	play,7,1,turnt001,02,CSX,E5/G56.B-1
	state after	(2, 1, 0, 0, 7, 5, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1630 *In the bottom of the 7th inning, with bases empty, PHI batter Trea Turner hit a ground ball to 3rd. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TOR vs PHI on 20230509
	state before	(2, 1, 0, 0, 5, 1, 2, 1)
13.	error	play,5,1,turnt001,02,F1FX,S8/G4.1-H;B-2(E4/TH)
	state after	(2, 0, 1, 0, 5, 1, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1631 This game had 2 errors. The Blue Jays lost to the Phillies 4-8.

In the bottom of the 5th inning, with a runner on 1st, PHI batter Trea Turner hit a single to 2nd. Incredibly, the runner on 1st scored (1

RBI) without further assist³. A throwing error by the second baseman allowed the batter to advance to 2nd base.

	game	TOR vs PHI on 20230509
	state before	(1, 1, 1, 0, 7, 2, 5, 0)
14.	error	play,7,0,bichb001,12,STBX,3E1/G3.2-H(NR) (UR);1-3;B-2
	state after	(1, 0, 1, 1, 7, 3, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1632 *In the top of the 7th inning, with runners on 1st and 2nd, TOR batter Bo Bichette hit a ground ball to 1st. An error by the pitcher allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd base.*

	game	TOR vs PHI on 20230510
	state before	(1, 1, 1, 0, 10, 1, 1, 1)
15.	error	play,10,1,harpb003,00,X,16(1)/FO/G13.2-H(E6/TH) (NR) (UR);B-1
	state after	(2, 1, 0, 0, 10, 1, 2, 1)
	fielder	6 (shortstop on the home team)

Summary 1633 *In the bottom of the 10th inning with runners on 1st and 2nd, PHI batter Bryce Harper hit a ground ball to the pitcher who assisted the shortstop in forcing out at 2nd the runner from 1st. A throwing error by the shortstop allowed the runner from 2nd to score (unearned, no RBI) and the batter to reach 1st.*

³Confirmed by the [mlb.com](https://www.mlb.com) video recap. Of course, the runner will have taken off at the pitch as there were 2 outs and a 0-2 count on the batter.

Visiting/Home team rosters for game CHN vs PHI on 2023-05-19

position	innings	player	position	innings	player
	1	Marcus Stroman	1	1	Ranger Suarez
1	7	Julian Merryweather	1	3	Jeff Hoffman
1	9	Nick Burdi	1	5	Erich Uelmen
2	1	Yan Gomes	1	6	Andrew Vasquez
3	1	Trey Mancini	1	8	Craig Kimbrel
3	5	Matt Mervis	1	9	Andrew Bellatti
4	1	Nico Hoerner	1	9	Kody Clemens
5	1	Nick Madrigal	2	1	J.T. Realmuto
6	1	Dansby Swanson	3	1	Alec Bohm
7	1	Ian Happ	4	1	Bryson Stott
7	9	Christopher Morel	5	1	Edmundo Sosa
8	1	Christopher Morel	6	1	Trea Turner
8	9	Mike Tauchman	7	1	Kyle Schwarber
9	1	Seiya Suzuki	8	1	Brandon Marsh
			9	1	Nick Castellanos

16.	game	CHN vs PHI on 20230519
	state before	(2, 0, 0, 0, 4, 4, 1, 1)
	error	play,4,1,realj001,21,BBCX,E9/F9S.B-1
	state after	(2, 1, 0, 0, 4, 4, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 1634 This game had 3 errors. The Phillies lost to the Cubs, 10-1.

In the bottom of the 4th inning, with bases empty, PHI batter J.T. Realmuto hit a fly ball to short right field. The batter reached 1st base on an error by the right fielder.

	game	CHN vs PHI on 20230519
	state before	(2, 0, 0, 1, 5, 5, 1, 0)
17.	error	play,5,0,morec001,01,SX,S6/G56.3-H;B-2(E6)
	state after	(2, 0, 1, 0, 5, 6, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1635 *In the top of the 5th inning, with a runner on 3rd, CHN batter Christopher Morel hit a ground ball single to short. The runner on 3rd scored (1 RBI) An error by the shortstop allowed the batter to advance to 2nd base.*

	game	CHN vs PHI on 20230519
	state before	(0, 0, 0, 0, 5, 8, 1, 1)
18.	error	play,5,1,marsb002,21,..CBBX,E3/G3.B-1
	state after	(0, 1, 0, 0, 5, 8, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1636 *In the bottom of the 5th inning, with bases empty, PHI batter Brandon Marsh hit a ground ball to 1st. An error by the first baseman allowed the batter to reach 1st base.*

	game	CHN vs PHI on 20230520
	state before	(1, 0, 0, 0, 1, 0, 4, 1)
19.	error	play,1,1,bohma001,21,CBBX,E5/G56.B-1
	state after	(1, 1, 0, 0, 1, 0, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1637 *In the bottom of the 1st inning with bases empty, PHI batter Alex Bohm hit a ground ball to 3rd. An error by the third baseman allowed the batter to reach 1st base.*

	game	CHN vs PHI on 20230521
	state before	(1, 1, 0, 0, 6, 0, 0, 0)
20.	error	play,6,0,swand001,01,.S1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1638 *In the top of the 6th inning, with a runner on 1st, CHN batter Dansby Swanson had a 0-1 count. During an attempted pick off at 1st, a throwing error by the pitcher allowed the runner on 1st to advance to 2nd. Swanson is still at bat with a 0-1 count.*

	game	ARI vs PHI on 20230522
	state before	(0, 1, 1, 0, 2, 1, 0, 0)
21.	error	play,2,0,moreg001,32,CBBCF*BX,FC4/G4.2-3;1-2(E4);B-1
	state after	(0, 1, 1, 1, 2, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1639 This game had 2 errors. Arizona Diamondbacks beats the Phillies 6-3.

In the top of the 2nd inning, with runners on 1st and 2nd, ARI batter Gabriel Moreno hit a ground ball to a fielder's choice at 2nd base. The runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to advance to 2nd the batter to reach 1st. The bases are now loaded.

	game	ARI vs PHI on 20230522
	state before	(1, 0, 0, 1, 9, 5, 3, 0)
22.	error	play,9,0,martk001,12,SC*BX,E3/G3.3-H(NR)(UR);B-1
	state after	(1, 1, 0, 0, 9, 6, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1640 *In the top of the 9th inning, with a runner on 3rd, ARI batter Ketel Marte hit a ground ball to 1st. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st base safely.*

	game	ARI vs PHI on 20230524
	state before	(2, 1, 1, 0, 6, 5, 0, 1)
23.	error	play,6,1,bohma001,20,.BBX,S8/L8S.2-H;1-3;B-2(E8/TH)
	state after	(2, 0, 1, 1, 6, 5, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 1641 *In the bottom of the 6th inning, with runners on 1st and 2nd, PHI batter Alec Bohm hit a line drive single to shallow center field. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

24.2.3 June

	game	DET vs PHI on 20230605
	state before	(0, 1, 0, 0, 7, 0, 5, 0)
24.	error	play,7,0,baezj001,21,BCBX,FC5/G56.1-2(E5);B-1
	state after	(0, 1, 1, 0, 7, 0, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1642 *In the top of the 7th inning, with a runner on 1st, DET batter Javier Baez hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to made it safely to 1st.*

	game	DET vs PHI on 20230606
	state before	(0, 0, 0, 0, 4, 0, 1, 1)
25.	error	play,4,1,castn001,11,.CBX,E5/G5.B-1
	state after	(0, 1, 0, 0, 4, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1643 *In the bottom of the 4th inning, with bases empty, PHI batter Nick Castellanos hit a ground ball to 3rd. An error by the third baseman allowed the batter to reach 1st.*

	game	DET vs PHI on 20230608
	state before	(1, 0, 0, 0, 8, 1, 1, 1)
26.	error	play,8,1,harrj002,22,BCBFX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 8, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1644 This game had 2 errors. The Phillies beat the Tigers 2-3.

In the bottom of the 8th inning, with bases empty, PHI batter Josh Harrison hit a ground ball to the shortstop, who made a throwing error. As a result, the batter made it safely to 1st.

	game	DET vs PHI on 20230608
	state before	(1, 1, 0, 0, 9, 1, 1, 0)
27.	error	play,9,0,maton001,12,CC.B>F1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 9, 1, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1645 *In the top of the 9th inning, with a runner on 1st, DET batter Nick Maton had a 1-2 count. During an attempted pick off at 1st, a throwing error by the pitcher allowed the runner to advance to 2nd. Maton is still at bat with a 1-2 count.*

	game	ATL vs PHI on 20230622
	state before	(2, 0, 1, 1, 10, 1, 0, 0)
28.	error	play,10,0,rilea001,01,FX,E7/L78.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 10, 3, 0, 0)
	fielder	7 (left field on the home team)

Summary 1646 *In the top of the 10th inning, with runners on 2nd and 3rd, ATL batter Austin Riley hit a single to left center field. An error by the left fielder allowed both runners to score (unearned, no RBIs), and the batter to reach 1st base.*

	game	NYN vs PHI on 20230623
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
29.	error	play,1,1,schw001,32,BCC.BBX,E8/F8S.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1647 *In the bottom of the 1st inning, with bases empty, PHI batter Kyle Schwarber hit a fly ball to shallow center field. An error by the center fielder allowed the batter to advance to 1st.*

	game	NYN vs PHI on 20230625
	state before	(0, 0, 1, 0, 5, 4, 2, 1)
30.	error	play,5,1,castn001,20,B1>B.>B,SB3.2-H(E2/TH) (NR)
	state after	(0, 0, 0, 0, 5, 4, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1648 This game had 2 errors. The Phillies beat the Mets 6-7.

In the bottom of the 5th inning, with a runner on 2nd, PHI batter Nick Castellanos had a 2-0 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed the runner to score (no RBI) Castellanos is at bat, now with a 3-0 count.

	game	NYN vs PHI on 20230625
	state before	(0, 1, 1, 1, 8, 6, 3, 1)
31.	error	play,8,1,boh001,10,.BX,FC5/G56.3-H;2-3;1-2(E5/TH);B-1
	state after	(0, 1, 1, 1, 8, 6, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1649 *In the bottom of the 8th inning, with bases loaded, PHI batter Alec Bohm hit a ground ball to a fielder's choice at 3rd. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to*

3rd. A throwing error by the third baseman allowed the runner on 1st advanced to 2nd and the batter to reach 1st base safely. The bases are still loaded.

	game	WAS vs PHI on 20230630
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
32.	error	play,2,0,ruizk001,11,CBX,E6/TH/G56D.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1650 *In the top of the 2nd inning, with bases empty, WAS batter Keibert Ruiz hit a ground ball to shallow left field. A throwing error by the shortstop allowed the batter to make it safely to 1st.*

24.2.4 July

	game	WAS vs PHI on 20230702
	state before	(1, 0, 0, 0, 3, 0, 3, 0)
33.	error	play,3,0,hilld002,21,BBTX,S8/L78D.B-2(E8)
	state after	(1, 0, 1, 0, 3, 0, 3, 0)
	fielder	8 (center field on the home team)

Summary 1651 *In the top of the 3rd inning, with bases empty, WAS batter Derek Hill hit a line drive single to deep left center field. An error by the center fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for game SDN vs PHI on 2023/07/14

position	innings	player	position	innings	player
	1	Yu Darvish	1	1	Cristopher Sanchez
1	7	Tom Cosgrove	1	6	Jeff Hoffman
1	8	Nick Martinez	1	7	Andrew Vasquez
1	9	Josh Hader	1	7	Dylan Covey
1	9	Luis Garcia	1	9	Andrew Bellatti
2	1	Gary Sanchez	2	1	J.T. Realmuto
3	1	Jake Cronenworth	3	1	Darick Hall
4	1	Ha-Seong Kim	3	8	Drew Ellis
5	1	Manny Machado	4	1	Bryson Stott
6	1	Xander Bogaerts	5	1	Alec Bohm
7	1	Juan Soto	6	1	Trea Turner
8	1	Trent Grisham	7	1	Kyle Schwarber
9	1	Fernando Tatis Jr.	8	1	Brandon Marsh
			9	1	Nick Castellanos

	game	SDN vs PHI on 20230714
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
34.	error	play,1,1,castn001,11,SBX,S4/G4M.B-2(E4/TH)
	state after	(2, 0, 1, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1652 This game had 3 errors. The Phillies lost to the Padres 8-3.

In the bottom of the 1st inning, with bases empty, PHI batter Nick Castellanos hit a ground ball single to shallow center. A throwing error by the second baseman allowed the batter to reach 2nd.

	game	SDN vs PHI on 20230714
	state before	(0, 0, 0, 0, 5, 3, 1, 0)
35.	error	play,5,0,sancg002,12,CBF.X,E3/G3S-.B-1
	state after	(0, 1, 0, 0, 5, 3, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1653 *In the top of the 5th inning, with bases empty, SDN batter Gary Sanchez hit a soft ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	SDN vs PHI on 20230714
	state before	(1, 1, 0, 0, 9, 8, 2, 1)
36.	error	play,9,1,ellid002,22,FSBBX,E7/F78XD+.1-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 9, 8, 3, 1)
	fielder	7 (left field on the visiting team)

Summary 1654 *In the bottom of the 9th inning, with a runner on 1st, PHI batter Drew Ellis hit a fly ball to deep left center field. An error by the left fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	SDN vs PHI on 20230715 (1/2)
	state before	(0, 0, 0, 0, 8, 4, 3, 1)
37.	error	play,8,1,stotb001,12,.CC.BX,E5/G56.B-1
	state after	(0, 1, 0, 0, 8, 4, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1655 This is game 1 of a double header. The Phillies beat the Padres 4-9.

In the bottom of the 8th inning, with bases empty, PHI batter Bryson Stott hit a ground ball to 3rd. A throwing error by the third baseman allowed the batter to reach 1st.

	game	SDN vs PHI on 20230715 (2/2)
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
38.	error	play,1,0,sotoj001,32,FCF1BFB1B>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1656 This game had 2 errors. The Phillies beat the Padres 4-6.

In the top of the 1st inning, with a runner on 1st, SDN batter Juan Soto had a full count. During the next pitch, which was a strike (so the batter struck out) the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner to reach 3rd.

	game	SDN vs PHI on 20230715 (2/2)
	state before	(0, 0, 0, 0, 5, 1, 1, 0)
39.	error	play,5,0,odorr001,22,CBSBX,E5/G6MD.B-2
	state after	(0, 0, 1, 0, 5, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1657 *In the top of the 5th inning, with bases empty, SDN batter Rouned Odor hit a ground ball to short left field. An error by the third baseman allowed the batter to reach 2nd.*

	game	SDN vs PHI on 20230716
	state before	(0, 0, 0, 0, 2, 1, 0, 0)
40.	error	play,2,0,cronj001,22,FS.FBBX,E3/G3.B-1
	state after	(0, 1, 0, 0, 2, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1658 *In the top of the 2nd inning, with bases empty, SDN batter Jake Cronenworth hit a ground ball to 1st. An error by the first baseman allowed the batter to reach 1st.*

Visiting/Home team rosters for game MIL vs PHI on 2023-07-18

position	innings	player	position	innings	player
1	1	Julio Teheran	1	1	Aaron Nola
1	5	Trevor Megill	1	8	Gregory Soto
1	6	Abner Uribe	1	9	Craig Kimbrel
1	8	J.C. Mejia	2	1	J.T. Realmuto
2	1	William Contreras	3	1	Darick Hall
3	1	Owen Miller	3	9	Alec Bohm
4	1	Brice Turang	4	1	Bryson Stott
5	1	Andruw Monasterio	5	1	Alec Bohm
6	1	Willy Adames	5	9	Edmundo Sosa
7	1	Christian Yelich	6	1	Trea Turner
8	1	Joey Wiemer	7	1	Kyle Schwarber
9	1	Raimel Tapia	7	9	Brandon Marsh
			8	1	Brandon Marsh
			8	9	Johan Rojas
			9	1	Nick Castellanos

	game	MIL vs PHI on 20230718
	state before	(2, 0, 1, 0, 5, 1, 3, 0)
41.	error	play,5,0,turab002,00,X,4E3/G4.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 2, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1659 This game had 3 errors. The Phillies beat the Brewers 3-4.

In the top of the 5th inning, with a runner on 2nd, MIL batter Brice Turang hit a ground ball to 2nd. An error by the first baseman (with an assist from the second baseman) allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st base.

	game	MIL vs PHI on 20230718
	state before	(1, 0, 0, 0, 9, 3, 4, 0)
42.	error	play,9,0,winkj002,02,CCX,E4/G34D.B-1
	state after	(1, 1, 0, 0, 9, 3, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1660 *In the top of the 9th inning, with bases empty, MIL batter Jesse Winker hit a ground ball to shallow right field. An error by the second baseman allowed the batter to reach 1st.*

	game	MIL vs PHI on 20230718
	state before	(1, 1, 0, 0, 9, 3, 4, 0)
43.	error	play,9,0,millo002,12,.1FBT1,P01(E3).1-2
	state after	(1, 0, 1, 0, 9, 3, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 1661 This is an example of a *consecutive error*, as the batter just before Owen Miller was Jesse Winker.

In the top of the 9th inning with a runner on 1st, MIL batter Owen Miller had a 1-2 count. During an attempted pick off at 1st, an error by the first baseman allowed the runner to advance to 2nd base. Miller is still at bat with a 1-2 count.

	game	MIL vs PHI on 20230719
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
44.	error	play,2,0,jonej007,32,BBCCBFX,E5/G7S.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1662 This game had 2 errors. The Phillies lost to the Brewers 5-3.

In the top of the 2nd inning, with bases empty, MIL batter Jahmai Jones hit a ground ball to short left field. An error by the third baseman allowed the batter to reach 1st.

	game	MIL vs PHI on 20230719
	state before	(0, 0, 0, 0, 5, 3, 1, 1)
45.	error	play,5,1,bohama001,02,FCX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 5, 3, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1663 *In the bottom of the 5th inning, with bases empty, PHI batter Alec Bohm hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	BAL vs PHI on 20230724
	state before	(0, 0, 0, 0, 4, 1, 0, 0)
46.	error	play,4,0,santa003,32,BFF.BFBFFX,E6/G56.B-1
	state after	(0, 1, 0, 0, 4, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1664 This game had 2 errors. The Phillies lost to the Orioles, 3-2.

In the top of the 4th inning, with bases empty, BAL batter Anthony Santander hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base. error by the shortstop.

	game	BAL vs PHI on 20230724
	state before	(2, 1, 0, 0, 5, 1, 0, 0)
47.	error	play,5,0,matej003,10,BX,FC6/G6M.1-2(E6);B-1
	state after	(2, 1, 1, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1665 *In the top of the 5th inning, with a runner on 1st, BAL batter Jorge Mateo hit a ground ball to fielder's choice in shallow left field. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	BAL vs PHI on 20230726
	state before	(2, 0, 0, 0, 5, 3, 4, 0)
48.	error	play,5,0,haysa001,10,BX,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 5, 3, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1666 *In the top of the 5th inning, with bases empty, BAL batter Austin Hayes reached 1st base on an throwing error by the short-stop.*

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	game	KCA vs PHI on 20230805
	state before	(1, 0, 1, 0, 5, 5, 3, 0)
49.	error	play,5,0,duffm002,00,>B,SB3.2-H(E2/TH) (NR)
	state after	(1, 0, 0, 0, 5, 6, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1667 *In the top of the 5th inning, with a runner on 1st, KCA batter Matt Duffy had a 0-0 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed the runner on 1st to score (no RBI). Duffy is at bat, now with a 1-0 count.*

	game	KCA vs PHI on 20230806
	state before	(2, 0, 0, 0, 8, 4, 8, 0)
50.	error	play,8,0,beatm001,00,X,E3/TH/G3.B-1
	state after	(2, 1, 0, 0, 8, 4, 8, 0)
	fielder	3 (1st baseman on the home team)

Summary 1668 *In the top of the 8th inning, with bases empty, KCA batter Matt Beaty hit a ground ball to 1st. A throwing error by the first baseman allowed the batter to reach 1st.*

Visiting/Home team rosters for game WAS vs PHI on 2023-08-08

position	innings	player	position	innings	player
1	1	Trevor Williams	1	1	Zack Wheeler
1	5	Amos Willingham	1	7	Matt Strahm
1	7	Joe La Sorsa	1	8	Luis F. Ortiz
2	1	Keibert Ruiz	2	1	J.T. Realmuto
3	1	Dominic Smith	3	1	Bryce Harper
4	1	Jake Alu	4	1	Bryson Stott
5	1	Ildemaro Vargas	5	1	Alec Bohm
6	1	CJ Abrams	6	1	Trea Turner
7	1	Blake Rutherford	7	1	Jake Cave
8	1	Alex Call	8	1	Johan Rojas
9	1	Lane Thomas	9	1	Nick Castellanos

51.	game	WAS vs PHI on 20230808
	state before	(1, 1, 0, 0, 2, 1, 0, 0)
	error	play,2,0,vargi001,11,BCX,E1/TH/G3S.1-2;B-1
	state after	(1, 1, 1, 0, 2, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1669 This game, the first of a double header, had 3 errors. The Phillies lost this game to the Nationals 5-4 but won the second one 4-8.

In the top of the 2nd inning, with a runner on 1st, WAS batter Ildemaro Vargas hit a ground ball to short 3rd base, A throwing error by the pitcher allowed the runner on 1st to reach 2nd and the batter to reach 1st base.

	game	WAS vs PHI on 20230808
	state before	(1, 0, 0, 0, 3, 3, 0, 1)
52.	error	play,3,1,rojaj003,32,BLF.BBX,S5/G5.B-2(E5/TH)
	state after	(1, 0, 1, 0, 3, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1670 *In the bottom of the 3rd inning, with bases empty, PHI batter Johan Rojas hit a ground ball single to 3rd base. The batter reached 2nd base on a throwing error by the third baseman.*

	game	WAS vs PHI on 20230808
	state before	(2, 0, 0, 0, 7, 4, 4, 0)
53.	error	play,7,0,garrs003,32,S.BSBBFX,E5/TH/G5.B-1
	state after	(2, 1, 0, 0, 7, 4, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1671 *In the top of the 7th inning, with bases empty, WAS batter Stone Garrett hit a ground ball to 3rd base. The batter reached 1st base on a throwing error by the third baseman.*

Visiting/Home team rosters for game WAS vs PHI on 2023-08-10

position	innings	player	position	innings	player
1	1	Patrick Corbin	1	1	Aaron Nola
1	6	Andres Machado	1	6	Matt Strahm
1	6	Joe La Sorsa	1	7	Seranthony Dominguez
1	8	Jose A. Ferrer	1	8	Yunior Marte
2	1	Keibert Ruiz	1	9	Craig Kimbrel
3	1	Dominic Smith	2	1	J.T. Realmuto
4	1	Jake Alu	3	1	Bryce Harper
5	1	Ildemaro Vargas	3	5	Alec Bohm
6	1	CJ Abrams	4	1	Bryson Stott
7	1	Blake Rutherford	5	1	Alec Bohm
7	6	Stone Garrett	5	5	Rodolfo Castro
8	1	Alex Call	6	1	Trea Turner
9	1	Lane Thomas	7	1	Kyle Schwarber
			8	1	Johan Rojas
			9	1	Weston Wilson
			9	7	Jake Cave

	game	WAS vs PHI on 20230810
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
54.	error	play,2,1,stotb001,22,BCBFX,E6/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1672 This game had 4 errors. The Phillies beat the Nats, 2-6.

In the bottom of the 2nd inning, with bases empty, PHI batter Bryson Stott hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	WAS vs PHI on 20230810
	state before	(2, 1, 0, 1, 2, 0, 0, 1)
55.	error	play,2,1,rojaj003,02,CF.1,P0CS2(1E3)
	state after	(2, 0, 1, 1, 2, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1673 *In the bottom of the 2nd inning, with runners on 1st and 3rd, PHI batter Johan Rojas had a 0-2 count. During a pick off attempt at 1st, the runner on 1st attempted to steal 2nd. An error by the first baseman allowed the runner to successfully reach 2nd (this is not a stolen base, by MLB Rule 10). Rojas is still at bat with a 0-2 count.*

	game	WAS vs PHI on 20230810
	state before	(2, 1, 0, 0, 4, 0, 0, 0)
56.	error	play,4,0,alu-j001,00,X,E9/L89D+.1-3;B-2
	state after	(2, 0, 1, 1, 4, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 1674 *In the top of the 4th inning, with a runner on 1st, WAS batter Jake Alu hit a hard line drive to deep center/right field. The right fielder made a error, allowing the runner on 1st to advance to 3rd and the batter to safely reach 2nd.*

	game	WAS vs PHI on 20230810
	state before	(0, 0, 0, 0, 6, 1, 0, 1)
57.	error	play,6,1,stotb001,12,.BCC.X,3E1/G3.B-1
	state after	(0, 1, 0, 0, 6, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1675 *In the bottom of the 6th inning, with bases empty, PHI batter Bryson Stott hit a grounder to 1st base. An error by the pitcher allowed the batter to reach 1st base.*

58.	game	MIN vs PHI on 20230812
	state before	(2, 1, 0, 0, 9, 8, 1, 1)
	error	play,9,1,castr006,11,*BFX,S6/G6.1-3(E6/TH)
	state after	(2, 1, 0, 1, 9, 8, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1676 *In the bottom of the 9th inning, with a runner on 1st, PHI batter Nick Castellanos hit a ground ball single to short. The runner on 1st advanced to 3rd on a throwing error by the shortstop. The batter is on 1st.*

59.	game	SFN vs PHI on 20230821
	state before	(1, 0, 0, 0, 1, 1, 0, 1)
	error	play,1,1,turnt001,10,BX,S87/L78D.B-2(E8)
	state after	(1, 0, 1, 0, 1, 1, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1677 This game had 2 errors. The Phillies beat the Giants 4-10.

In the bottom of the 1st inning, with bases empty, PHI batter Trea Turner hit a line drive single to deep left center field. The batter advanced to 2nd on an error by the center fielder.

60.	game	SFN vs PHI on 20230821
	state before	(0, 0, 1, 0, 7, 2, 6, 1)
	error	play,7,1,realj001,12,FBFX,S5/G56.2-3;B-2(E5/TH)
	state after	(0, 0, 1, 1, 7, 2, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1678 *In the bottom of the 7th inning, with a runner on 2nd, PHI batter J.T. Realmuto hit a ground ball single to 3rd. The runner on 2nd advanced to 3rd. A throwing error by the third baseman allowed the batter to reach 2nd base.*

Visiting/Home team rosters for game SFN vs PHI on 2023-08-22

position	innings	player	position	innings	player
1	1	Kyle Harrison	1	1	Taijuan Walker
1	4	Ryan Walker	1	6	Seranthony Dominguez
1	6	Luke Jackson	1	7	Jose Alvarado
1	7	Tyler Rogers	1	8	Gregory Soto
1	8	Taylor Rogers	1	9	Craig Kimbrel
1	9	Camilo Doval	2	1	J.T. Realmuto
2	1	Patrick Bailey	3	1	Alec Bohm
3	1	LaMonte Wade Jr.	4	1	Bryson Stott
4	1	Thairo Estrada	5	1	Edmundo Sosa
5	1	J.D. Davis	6	1	Trea Turner
6	1	Casey Schmitt	7	1	Kyle Schwarber
6	6	Johan Camargo	8	1	Johan Rojas
7	1	Joc Pederson	8	8	Brandon Marsh
7	7	Wade Meckler	9	1	Nick Castellanos
8	1	Wade Meckler			
8	7	Austin Slater			
9	1	Michael Conforto			
9	8	Luis Matos			

	game	SFN vs PHI on 20230822
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
61.	error	play,1,0,wadel001,32,BCFBBX,E3/TH/G34.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1679 This game had 3 errors. The Giants lost to the Phillies 3-4.

In the top of the 1st inning, with bases empty, SFN batter LaMonte Wade hit a ground ball to the 1st/2nd base gap. The batter reached 1st

base on a throwing error by the first baseman.

	game	SFN vs PHI on 20230822
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
62.	error	play,1,0,estrt001,10,BX,S7/L7D+.1-3(E7)
	state after	(0, 1, 0, 1, 1, 0, 0, 0)
	fielder	7 (left field on the home team)

Summary 1680 This is an example of a consecutive error, as the batter just before Thairo Estrada was LaMonte Wade.

In the top of the 1st inning, with a runner on 1st, SFN batter Thairo Estrada hit a line drive single to deep left field. A throwing error by the left fielder allowed the runner on 1st to advance to 3rd. The batter is on 1st base.

	game	SFN vs PHI on 20230822
	state before	(1, 0, 0, 0, 6, 3, 2, 1)
63.	error	play,6,1,realj001,10,BX,S5/G5.B-2(E5/TH)
	state after	(1, 0, 1, 0, 6, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1681 *In the bottom of the 6th inning, with bases empty, PHI batter J.T. Realmuto hit a ground ball single to 3rd. A throwing error by the third baseman allowed the batter to advance to 2nd base.*

	game	SLN vs PHI on 20230825
	state before	(0, 0, 0, 0, 5, 2, 4, 1)
64.	error	play,5,1,turnt001,10,BX,E5/G56S.B-1
	state after	(0, 1, 0, 0, 5, 2, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1682 *In the bottom of the 5th inning, with bases empty, PHI batter Trea Turner hit a ground ball to shallow 3rd. The batter advanced to 1st base on an error by the third baseman.*

	game	SLN vs PHI on 20230826
	state before	(0, 1, 0, 0, 8, 1, 12, 0)
65.	error	play,8,0,palar002,12,.CS*BX,FC4/G34.1-2(E4);B-1
	state after	(0, 1, 1, 0, 8, 1, 12, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1683 *In the top of the 8th inning, with a runner on 1st, St. Louis Cardinals batter Richie Palacios hit a ground ball to a fielder's choice in the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to advance to 2nd base and the batter to safely reach 1st.*

	game	SLN vs PHI on 20230827
	state before	(0, 0, 0, 0, 1, 0, 1, 1)
66.	error	play,1,1,turnt001,22,BSTBX,S8/L89D+.B-2(E8)
	state after	(0, 0, 1, 0, 1, 0, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 1684 *In the bottom of the 1st inning, with bases empty, PHI batter Trea Turner hit a single line drive to center field. The batter advanced to 2nd base on an error by the center fielder.*

Visiting/Home team rosters for game ANA vs PHI on 2023-08-29

position	innings	player	position	innings	player
1	1	Tyler Anderson	1	1	Michael Lorenzen
1	6	Jaime Barria	1	7	Jose Alvarado
1	6	Jose Soriano	1	8	Seranthony Dominguez
2	1	Logan O'Hoppe	1	9	Andrew Bellatti
3	1	Nolan Schanuel	2	1	J.T. Realmuto
4	1	Brandon Drury	3	1	Alec Bohm
5	1	Mike Moustakas	4	1	Rodolfo Castro
6	1	Luis Rengifo	4	7	Bryson Stott
6	7	Andrew Velazquez	5	1	Edmundo Sosa
7	1	Randal Grichuk	6	1	Trea Turner
8	1	Mickey Moniak	7	1	Kyle Schwarber
8	7	Luis Rengifo	7	9	Brandon Marsh
9	1	Hunter Renfroe	8	1	Johan Rojas
			9	1	Nick Castellanos
			9	9	Jake Cave

	game	ANA vs PHI on 20230829
	state before	(1, 0, 1, 0, 3, 1, 1, 0)
67.	error	play,3,0,ohtas001,10,BX,S1/G2-.2-H(E1/TH) (NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 3, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1685 This game had 4 errors. The Angels lost to the Phillies 7-12.

In the top of the 3rd inning, with a runner on 2nd, Los Angeles Angels batter Shohei Ohtani hit a soft ground ball single to the shallow infield, A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd base.

	game	ANA vs PHI on 20230829
	state before	(0, 0, 0, 0, 6, 4, 2, 1)
68.	error	play,6,1,castn001,11,CBX,S7/L7D+.B-2(E7)
	state after	(0, 0, 1, 0, 6, 4, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 1686 *In the bottom of the 6th inning, with bases empty, PHI batter Nick Castellanos hit a single line drive to left field. The batter reached 2nd base on an error by the left fielder.*

	game	ANA vs PHI on 20230829
	state before	(2, 1, 0, 0, 9, 5, 12, 0)
69.	error	play,9,0,schan001,12,CF.*BX,S7/L78.1-H(E7/TH)(NR);B-2
	state after	(2, 0, 1, 0, 9, 6, 12, 0)
	fielder	7 (left field on the home team)

Summary 1687 *In the top of the 9th inning, with a runner on 1st, Los Angeles Angels batter Nolan Schanuel hit a line drive single to left field. Due to a throwing error by the left fielder, the runner on 1st scored (no RBI) and the batter advanced to 2nd.*

	game	ANA vs PHI on 20230829
	state before	(2, 0, 1, 0, 9, 7, 12, 0)
70.	error	play,9,0,drurb001,01,SX,E6/TH/G6.2-3;B-1
	state after	(2, 1, 0, 1, 9, 7, 12, 0)
	fielder	6 (shortstop on the home team)

Summary 1688 *In the top of the 9th inning, with a runner on 2nd, ANA batter Brandon Drury hit a ground ball to the shortstop. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

71.	game	ANA vs PHI on 20230830
	state before	(0, 1, 0, 1, 5, 2, 2, 0)
	error	play,5,0,velaa001,12,FL*BX,FC/SH/BG2.3-H;1-2;B-1(E1)
	state after	(0, 1, 1, 0, 5, 3, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1689 *In the top of the 5th inning, with runners on 1st and 3rd, ANA batter Andrew Velazquez bunted to a fielder's choice to shallow infield. The runner on 3rd scored (1 RBI) and the runner on 1st advanced to 2nd. An error by the pitcher allowed the batter to reach 1st base.*

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72.	game	MIA vs PHI on 20230909
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
	error	play,2,0,delab001,22,BCBS.FX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1690 This game had 3 errors. The Phillies beat the Marlins 4-8.

In the top of the 2nd inning, with bases empty, MIA batter Bryan De La Cruz hit a ground ball to the shortstop. The batter reached 1st base on a throwing error by the shortstop.

73.	game	MIA vs PHI on 20230909
	state before	(2, 1, 0, 0, 2, 0, 5, 1)
	error	play,2,1,harpb003,11,1BFF,FLE5
	state after	(2, 1, 0, 0, 2, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1691 It’s an “∞ play.” These rarely occur, and note the state remains unchanged.

In the bottom of the 2nd inning, with a runner on 1st and a 1-1 count, PHI batter Bryce Harper hit a fly ball to foul left field. An error by the third baseman allowed the batter remain safe. Harper is at bat, now with a 1-2 count.

74.	game	MIA vs PHI on 20230909
	state before	(1, 0, 0, 0, 3, 0, 5, 1)
	error	play,3,1,stotb001,01,CF,FLE2
	state after	(1, 0, 0, 0, 3, 0, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1692 Another ∞ play in the *same game*!

In the bottom of the 3rd inning, with bases empty and a 0-1 count, PHI batter Bryson Stott hit a fly ball to infield foul territory. An error by the catcher allowed the batter to remain safe. Stott is at bat, now with a 0-2 count.

Visiting/Home team rosters for game ATL vs PHI on 2023-09-11

position	innings	player	position	innings	player
1	1	Charlie Morton	1	1	Taijuan Walker
1	10	Kirby Yates	1	10	Jose Alvarado
1	7	Dylan Lee	1	6	Andrew Bellatti
1	7	Joe Jimenez	1	8	Dylan Covey
1	8	A.J. Minter	2	1	Garrett Stubbs
1	9	Raisel Iglesias	3	1	Bryce Harper
2	1	Travis d'Arnaud	4	1	Bryson Stott
3	1	Matt Olson	5	1	Alec Bohm
4	1	Ozzie Albies	6	1	Trea Turner
5	1	Austin Riley	7	1	Jake Cave
6	1	Orlando Arcia	7	9	Cristian Pache
7	1	Eddie Rosario	8	1	Brandon Marsh
7	10	Kevin Pillar	8	9	Johan Rojas
8	1	Michael Harris II	9	1	Nick Castellanos
9	1	Ronald Acuna Jr.			

75.	game	ATL vs PHI on 20230911
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
	error	play,1,0,rilea001,21,SBBX,T7/F78D.B-H(E7)(NR)(UR)
	state after	(2, 0, 0, 0, 1, 1, 0, 0)
	fielder	7 (left field on the home team)

Summary 1693 This game, the first of a double header, had 3 errors. The Phillies beat the Braves in the 1st game 5-7 and lost the 2nd game 10-8.

In the top of the 1st inning, with bases empty, ATL batter Austin Riley hit a triple to deep left center field. The batter reached home on an error by the left fielder.

	game	ATL vs PHI on 20230911
	state before	(1, 1, 0, 0, 3, 2, 0, 0)
76.	error	play,3,0,albio001,20,1B1B>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1694 *In the top of the 2nd inning with a runner on 1st, ATL batter Ozzie Albies had a 2-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner to reach 3rd. Albies is still at bat, now with a 3-0 count.*

	game	ATL vs PHI on 20230911
	state before	(2, 0, 0, 0, 9, 8, 6, 1)
77.	error	play,9,1,turnt001,11,CBX,E4/TH/G34.B-1
	state after	(2, 1, 0, 0, 9, 8, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1695 *In the bottom of the 9th inning, with bases empty, PHI batter Trea Turner hit a ground ball to gap between 1st base and 2nd base. The batter reached 1st base on a throwing error by the second baseman.*

	game	ATL vs PHI on 20230911
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
78.	error	play,1,0,harrm004,12,CBFF1,P01(E1/TH).1-3
	state after	(0, 0, 0, 1, 1, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1696 This second game of a double header, also had 3 errors.

In the top of the 1st inning, with a runner on 1st, ATL batter Michael Harris had a 1-2 count. The pitcher attempted a pick off at 1st. A throwing error by the pitcher allowed the runner to advance to 3rd. Harris was still at bat with a 0-0 count.

	game	ATL vs PHI on 20230911
	state before	(2, 0, 0, 0, 6, 4, 7, 0)
79.	error	play,6,0,arcio002,01,CX,E5/TH/G6.B-2
	state after	(2, 0, 1, 0, 6, 4, 7, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1697 *In the top of the 6th inning, with bases empty, ATL batter Orlando Arcia hit a ground ball to 3rd base. The batter reached 2nd base on a throwing error by the third baseman.*

	game	ATL vs PHI on 20230911
	state before	(1, 0, 0, 0, 7, 5, 7, 1)
80.	error	play,7,1,realj001,00,X,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 7, 5, 7, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1698 *In the bottom of the 7th inning, with bases empty, PHI batter J.T. Realmuto reached 1st base on a throwing error by the shortstop.*

	game	ATL vs PHI on 20230913
	state before	(2, 1, 0, 0, 7, 4, 1, 0)
81.	error	play,7,0,albio001,02,SF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 4, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1699 *In the top of the 7th inning, with a runner on 1st, ATL batter Ozzie Albies had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd base. A throwing error by the catcher allowed the runner to reach 3rd. Albies is at bat, now with a 1-2 count.*

	game	NYN vs PHI on 20230921
	state before	(2, 0, 0, 0, 5, 3, 4, 0)
82.	error	play,5,0,alonp001,21,BCBX,E6/TH/G6D.B-1
	state after	(2, 1, 0, 0, 5, 3, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1700 *In the top of the 5th inning, with bases empty, NYN batter Pete Alonso hit a ground ball to shallow left field. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	NYN vs PHI on 20230922
	state before	(1, 1, 0, 0, 7, 2, 3, 1)
83.	error	play,7,1,schw001,11,B1C>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 2, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1701 *In the bottom of the 7th inning, with a runner on 1st, PHI batter Kyle Schwarber had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd base. A throwing error by the catcher allowed the runner to reach 3rd. Schwarber is at bat, now with a 2-1 count.*

	game	NYN vs PHI on 20230923
	state before	(0, 1, 1, 0, 2, 0, 0, 0)
84.	error	play,2,0,batyb001,31,BBCBX,FC6/G78S+.2-H(NR)(UR);1-H(E6)(NR)(UR);B-1
	state after	(0, 1, 0, 0, 2, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1702 This game had 2 errors. The Phillies beat the Mets, 5-7.

In the top of the 2nd inning, with runners on 1st and 2nd, NYN batter Brett Baty hit a hard ground ball to a fielder's choice the left center field. The runner on 2nd scored (unearned, no RBI). An error by the shortstop allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 1st.

	game	NYN vs PHI on 20230923
	state before	(0, 0, 0, 0, 2, 2, 1, 1)
85.	error	play,2,1,realj001,02,CSX,E6/G6.B-1
	state after	(0, 1, 0, 0, 2, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1703 *In the bottom of the 5th inning, with bases empty, PHI batter J.T. Realmuto reached 1st base on a throwing error by the shortstop.*

	game	PIT vs PHI on 20230926
	state before	(0, 0, 0, 0, 7, 1, 2, 1)
86.	error	play,7,1,stotb001,22,..BBCFX,3E1/G3.B-1
	state after	(0, 1, 0, 0, 7, 1, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1704 *In the bottom of the 7th inning, with bases empty, PHI batter Bryson Stott hit a ground ball to 1st base. The batter reached 1st base on an error by the pitcher (assisted by the first baseman).*

	game	PIT vs PHI on 20230928
	state before	(2, 0, 0, 0, 7, 3, 1, 1)
87.	error	play,7,1,pachc001,11,BFX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 7, 3, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1705 This game had 2 errors. The Phillies lost to the Pirates 3-2.

In the top of the 7th inning, with bases empty, PHI batter Cristian Pache reached 1st base on a throwing error by the shortstop.

	game	PIT vs PHI on 20230928
	state before	(0, 0, 1, 0, 8, 3, 1, 1)
88.	error	play,8,1,castr006,22,CSBB>B,SB3.2-H(E2/TH) (NR)
	state after	(0, 0, 0, 0, 8, 3, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1706 *In the bottom of the 5th inning, with a runner on 2nd, PHI batter Rudolfo Castro had a 2-2 count. During the next pitch, which was a ball, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (no RBI). Castro is at bat, now with a full count.*

Chapter 25

Pittsburgh Pirates (PIT)

One of the oldest teams in the nation, the Pittsburgh Pirates compete in the National League Central Division. The club joined the National League in 1887 but competed in the National League East from 1969 through 1993. The Pirates have won five World Series, the last in 1979. They have won no NLCD titles.

On September 21, 1963, the Pirates became the first MLB team to have an African-American manager in Gene Baker.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Pittsburgh against the Pirates. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The DER of the 2023 PIT team is 0.686 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the Pirates is slightly below the MLB average.

There were 87 errors in the 81 PIT 2023 home games, 45 by the Pirates and the rest (42) by visiting teams. The total number of errors by the team (home and away) over the regular season is 91. Note $\mu = 83.9$ was the MLB average, with standard deviation of $\sigma = 13.87$, so the Pirates were slightly above average for errors (which is not good).

25.1 Season totals

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

25.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	42
Runs by ht because of errors by vt:	14
Errors by ht:	45
Runs by vt because of errors by ht:	21

This is a very good error-to-run conversion rate: $14/42 = 33.3\%$ on the part of the Pirates (compared to the excellent $21/45 = 46.6\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 PIT season (home and away) has about $R/PA = 692/6089$ (about 11.3%) change of resulting in a run. Therefore, a randomly selected error is (for PIT) nearly three times as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 87, so the expected number of errors (by either team) in a randomly selected PIT 2023 home game is about 1.07.

Errors by position	vt	ht
errors due to pitcher:	7	6
to catcher:	3	6
to 1st baseman:	6	2
to 2nd baseman:	7	7
to 3rd baseman:	6	5
to shortstop:	8	11
to left fielder:	0	2
to center fielder:	1	2
to right fielder:	2	4

25.1.2 Error total by state

If there is a runner, or runners, it is a **balk** when

- (10) The pitcher, after coming to a legal pitching position, removes one hand from the ball other than in an actual pitch, or in throwing to a base;
- (11) The pitcher, while touching his plate, accidentally or intentionally has the ball slip or fall out of his hand or glove;
- (12) The pitcher, while giving an intentional base on balls, pitches when the catcher is not in the catcher's box;
- (13) The pitcher delivers the pitch from Set Position without coming to a stop.

– **MLB Rule Book** 6.02(a)(10)-(13)

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	13	18	31
(lone) runner on 1st	11	7	18
(lone) runner on 2nd	6	5	11
(lone) runner on 3rd	1	1	2
two runners on base	7	12	18
bases loaded	4	2	6
total	42	45	86

A table of frequencies of the most common errors by PIT in a PIT 2023 home game:

state	frequency	percentage
(0, 0, 0, 0)	6	13.3%
(1, 0, 0, 0)	7	15.6%
(2, 0, 0, 0)	5	11.1%
(0, 1, 0, 0)	2	4.4%
(1, 1, 0, 0)	4	8.9%
(2, 1, 0, 0)	1	2.2%
(2, 0, 1, 0)	3	6.7%
(2, 1, 1, 0)	5	11.1%
(1, 0, 1, 1)	2	4.4%

A table of frequencies of most common errors by a visiting team in PIT 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	9	21.4%
(1, 0, 0, 0)	3	7.1%
(2, 0, 0, 0)	1	2.4%
(0, 1, 0, 0)	3	7.1%
(1, 1, 0, 0)	5	11.9%
(2, 1, 0, 0)	3	7.1%
(1, 1, 1, 0)	2	4.8%
(1, 1, 0, 1)	2	4.8%
(1, 0, 1, 1)	2	4.8%
(2, 1, 1, 1)	3	7.1%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Pirates homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `PIT2023.evn`):

state	PIT2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	16.5%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	5.5%
(1, 1, 0, 0)	6.1%
(2, 1, 0, 0)	6.7%
(0, 0, 1, 0)	1.7%
(1, 0, 1, 0)	2.9%
(2, 0, 1, 0)	3.8%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.9%
(2, 0, 0, 1)	1.3%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.8%
(2, 1, 1, 0)	3.0%

25.1.3 Some interesting errors

Some of the most interesting error plays in the PIT 2023 home games:

- The game ARI vs PIT on 2023-05-21 had 4 errors including a consecutive error.
- The game NYN vs PIT on 2023-06-10 had 3 errors including a consecutive error.
- The game WAS vs PIT on 2023-09-13 had only 2 errors but one was in infinity play.
- The game MIA vs PIT on 2023-09-29 had only 2 errors but each game in this Marlins-Pirates series (which started on September 29th) had at least one error, for a total of 5.

25.1.4 Relative team rankings

We have 81 home games between PIT and some visiting MLB team. A number of teams did not play PIT in the regular season. However, of those that did, PIT *lost* more games than won against¹:

ARI (2/3), CLE (2/3), COL (2/3), HOU (2/3), MIA (2/3), NYA (2/3),
OAK (2/3), TEX (2/3), SFN (3/3), TOR (3/3), CHN (6/7)

and won more than lost against

SDN (0/3), CHA (1/3), LAN (1/3), NYN (1/3),
PHI (1/3), WAS (1/4), SLN (1/6), CIN (2/7),

The teams that split their series were:

DET (1/2), ATL (2/4), MIL (3/6).

The teams that did not play in any of the 81 home games in PIT 2023 were:

ANA, BAL, BOS, KCA, MIN, SEA, TBA.

25.2 Game-by-game totals

The errors are listed chronologically. As a matter of shorthand, we use *short* for the area midway between 2nd and 3rd base, and *1st/2nd* for the area midway between 1st and 2nd base.

To save space, we only give the team rosters in special cases. A typical roster for PIT 2023 is as follows [BA].

¹For the reader's convenience, we record that team's (vt) W/G record in parentheses (W = numbers of games vt Won against PIT, G = total (home) Games against vt played at PIT).

position	player	games played	OBP
1 (C)	Jason Delay	68	.319
3 (1B)	Carlos Santana	85	.321
4 (2B)	Ji Hwan Bae	64	.296
5 (3B)	Ke'Bryan Hayes	122	.309
6 (SS)	Tucupita Marcano	58	.276
7 (LF)	Bryan Reynolds	119	.330
8 (CF)	Jack Suwinski	120	.339
9 (RF)	Henry Davis		.302

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 PIT home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

As we will see below, there were 87 errors in the 81 home games, 45 by PIT and the rest (42) by visiting teams.

25.2.1 April

game	CHA vs PIT on 20230407
state before	(1, 1, 1, 1, 5, 7, 8, 1)
1. error	play,5,1,reynb001,01,.CX,T9/L9L.3-H;2-H;1-H;B-H(E4/TH) (NR) (UR)
state after	(1, 0, 0, 1, 5, 7, 11, 1)
fielder	4 (2nd baseman on the visiting team)

Summary 1707 *In the bottom of the 5th inning, with bases loaded, PIT batter Bryan Reynolds hit a triple to right field. The runners on 1st, 2nd and 3rd all scored (3 RBIs). A throwing error by the second baseman allowed the batter to also score (unearned, no RBI).*

	game	CHA vs PIT on 20230408
	state before	(1, 0, 1, 0, 3, 5, 1, 1)
2.	error	play,3,1,reynb001,21,1*B>B.S2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 3, 5, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1708 *In the bottom of the 3rd inning, with a runner on 2nd, PIT batter Bryan Reynolds has a 2-1 count. The pitcher attempted to pickoff the runner on 2nd. An error by the pitcher allowed that runner to reach 3rd. Reynolds is still at-bat with a 2-1 count.*

	game	HOU vs PIT on 20230410
	state before	(2, 0, 1, 0, 4, 7, 1, 0)
3.	error	play,4,0,julkc001,22,F*BCBX,E6/G6.2-3;B-1
	state after	(2, 1, 0, 1, 4, 7, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1709 *In the top of the 4th inning, with a runner on 2nd, HOU batter Corey Julks hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	HOU vs PIT on 20230411
	state before	(1, 0, 0, 0, 9, 2, 4, 0)
4.	error	play,9,0,dubom001,01,CX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 9, 2, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1710 *In the top of the 9th inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CIN vs PIT on 20230420
	state before	(0, 0, 0, 0, 8, 1, 4, 0)
5.	error	play,8,0,garcj007,22,.CBBCX,S1/G25-.B-2(E1/TH)
	state after	(0, 0, 1, 0, 8, 1, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 1711 *In the top of the 8th inning, with bases empty, CIN batter Jose Barrero hit a soft ground ball single to left shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

Visiting/Home team rosters for CIN vs PIT on 2023-04-21

position	innings	player	position	innings	player
1	1	Graham Ashcraft	1	1	Mitch Keller
1	6	Derek Law	1	7	Robert Stephenson
1	7	Ian Gibaut	1	8	Colin Holderman
1	7	Reiver Sanmartin	1	9	David Bednar
2	1	Tyler Stephenson	2	1	Austin Hedges
3	1	Jason Vosler	3	1	Carlos Santana
4	1	Jonathan India	4	1	Mark Mathias
5	1	Spencer Steer	4	9	Tucupita Marcano
6	1	Jose Barrero	5	1	Ke'Bryan Hayes
7	1	TJ Friedl	6	1	Rodolfo Castro
8	1	Nick Senzel	7	1	Bryan Reynolds
9	1	Wil Myers	8	1	Jack Suwinski
			8	8	Ji Hwan Bae
			9	1	Connor Joe
			9	8	Jack Suwinski

	game	CIN vs PIT on 20230421
	state before	(2, 0, 0, 1, 6, 2, 2, 1)
6.	error	play,6,1,mathm002,10,BX,E6/G6.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 2, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1712 This game had 3 errors. The final score was 2-4.
In the bottom of the 6th inning, with a runner on 3rd, PIT batter Mark Mathias hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.

	game	CIN vs PIT on 20230421
	state before	(1, 0, 0, 0, 7, 2, 3, 0)
7.	error	play,7,0,senzn001,12,SBS.X,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 7, 2, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1713 *In the top of the 7th inning, with bases empty, CIN batter Nick Senzel hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	CIN vs PIT on 20230421
	state before	(0, 0, 0, 0, 8, 2, 4, 0)
8.	error	play,8,0,friet001,10,...BX,S4/G4.B-2(E4/TH)
	state after	(0, 0, 1, 0, 8, 2, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1714 *In the top of the 8th inning, with bases empty, CIN batter TJ Friedl hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	CIN vs PIT on 20230422
	state before	(1, 1, 0, 0, 5, 0, 2, 0)
9.	error	play,5,0,fairs001,22,B1CBFX,S8/L8.1-H(E8)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 5, 1, 2, 0)
	fielder	8 (center field on the home team)

Summary 1715 *In the top of the 5th inning, with a runner on 1st, CIN batter Stuart Fairchild hit a line drive single to center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for LAN vs PIT on 2023-04-25

position	innings	player	position	innings	player
1	1	Noah Syndergaard	1	1	Johan Oviedo
1	5	Victor Gonzalez	1	6	Robert Stephenson
1	6	Evan Phillips	1	7	Jose Hernandez
1	7	Yency Almonte	1	8	Colin Holderman
1	8	Caleb Ferguson	1	9	Duane Underwood Jr.
1	9	Shelby Miller	2	1	Austin Hedges
2	1	Austin Barnes	2	9	Jason Delay
3	1	Freddie Freeman	3	1	Carlos Santana
4	1	Miguel Vargas	4	1	Tucupita Marcano
5	1	Chris Taylor	5	1	Ke'Bryan Hayes
6	1	Mookie Betts	6	1	Rodolfo Castro
7	1	David Peralta	7	1	Jack Suwinski
8	1	James Outman	8	1	Ji Hwan Bae
9	1	Jason Heyward	9	1	Connor Joe
9	7	Trayce Thompson			

	game	LAN vs PIT on 20230425
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
10.	error	play,1,0,heywj001,11,BF>X,C/E2.1-2;B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1716 This game had 3 errors. The final score was 8-7.
In the top of the 1st inning, with a runner on 1st, LAN batter Jason Heyward had a 1-1 count. Catcher's interference allowed the batter to take 1st. The runner on 1st advances one base.

	game	LAN vs PIT on 20230425
	state before	(1, 0, 0, 0, 2, 2, 1, 0)
11.	error	play,2,0,taylc001,22,CSBBFX,E8/F89S.B-2
	state after	(1, 0, 1, 0, 2, 2, 1, 0)
	fielder	8 (center field on the home team)

Summary 1717 *In the top of the 2nd inning, with bases empty, LAN batter Chris Taylor hit a fly ball to shallow right center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	LAN vs PIT on 20230425
	state before	(2, 0, 0, 1, 5, 2, 7, 0)
12.	error	play,5,0,freef001,22,FFBBX,S6/G6.3-H;B-2(E6/TH)
	state after	(2, 0, 1, 0, 5, 3, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 1718 *In the top of the 5th inning, with a runner on 3rd, LAN batter Freddie Freeman hit a ground ball single to short. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 2nd.*

13.	game	LAN vs PIT on 20230427
	state before	(0, 0, 0, 0, 1, 2, 0, 1)
	error	play,1,1,marct001,12,FBSX,S2/BG2-.B-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1719 *In the bottom of the 1st inning, with bases empty, PIT batter Tucupita Marcano hit a soft bunt single to the catcher. A throwing error by the the catcher allowed the batter to reach 3rd.*

25.2.2 May

14.	game	TOR vs PIT on 20230506
	state before	(2, 1, 1, 0, 3, 4, 0, 0)
	error	play,3,0,biggc002,32,CBBC.*B>X,S4/G34-.2-H(E4/TH)(NR);1-3;B-2
	state after	(2, 0, 1, 1, 3, 5, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1720 This game had 2 errors. The final score was 8-2. *In the top of the 3rd inning, with runners on 1st and 2nd, TOR batter Cavan Biggio hit a soft ground ball single to 1st/2nd. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

15.	game	TOR vs PIT on 20230506
	state before	(2, 0, 1, 0, 6, 7, 1, 1)
	error	play,6,1,santc002,00,2,P02(E1/TH).2-3
	state after	(2, 0, 0, 1, 6, 7, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1721 *In the bottom of the 6th inning, with a runner on 2nd, PIT batter Carlos Santana has a 0-0 count. The pitcher attempted to pickoff the runner on 2nd. An error by the pitcher allowed that runner to reach 3rd. Santana is still at-bat with a 0-0 count.*

	game	TOR vs PIT on 20230507
	state before	(1, 1, 0, 0, 8, 5, 0, 0)
16.	error	play,8,0,kierk001,32,SBFBBX,FC6/G6M.1-2(E6);B-1
	state after	(1, 1, 1, 0, 8, 5, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1722 *In the top of the 8th inning, with a runner on 1st, TOR batter Kevin Kiermaier hit a ground ball into a fielder's choice at short. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	COL vs PIT on 20230509
	state before	(0, 1, 1, 0, 4, 1, 1, 0)
17.	error	play,4,0,mcmar001,00,X,FC3/G3.2-H(NR);1X2(3E6);B-1
	state after	(0, 1, 1, 0, 4, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1723 *In the top of the 4th inning, with runners on 1st and 2nd, COL batter Ryan McMahon hit a ground ball into a fielder's choice at 1st. The runner on 2nd scored (unearned, no RBI)but an error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to advance to 2nd.*

	game	COL vs PIT on 20230510
	state before	(2, 0, 1, 0, 4, 1, 3, 0)
18.	error	play,4,0,mcmar001,22,CBFBX,E1/G23-.2-3;B-1
	state after	(2, 1, 0, 1, 4, 1, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1724 *In the top of the 4th inning, with a runner on 2nd, COL batter Ryan McMahon hit a soft ground ball to shallow right in-field. An error by the pitcher allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	ARI vs PIT on 20230519
	state before	(2, 0, 0, 0, 4, 1, 1, 0)
19.	error	play,4,0,smitp002,00,X,E7/L78D+.B-2
	state after	(2, 0, 1, 0, 4, 1, 1, 0)
	fielder	7 (left field on the home team)

Summary 1725 This game had 2 errors. The final score was 3-13.
In the top of the 4th inning, with bases empty, ARI batter Pavin Smith hit a hard line drive to deep left center field. An error by the left fielder allowed the batter to reach 2nd base.

	game	ARI vs PIT on 20230519
	state before	(1, 1, 0, 0, 4, 1, 3, 1)
20.	error	play,4,1,hedga001,01,FX,FC5/G56.1-2(E5);B-1
	state after	(1, 1, 1, 0, 4, 1, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1726 *In the bottom of the 4th inning, with a runner on 1st, PIT batter Austin Hedges hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Merrill Kelly	1	1	Roansy Contreras
1	6	Scott McGough	1	6	Dauri Moreta
1	8	Kevin Ginkel	1	6	Rob Zastryzny
2	1	Jose Herrera	1	7	Robert Stephenson
3	1	Christian Walker	1	8	Yohan Ramirez
4	1	Ketel Marte	2	1	Jason Delay
5	1	Josh Rojas	3	1	Carlos Santana
5	6	Emmanuel Rivera	4	1	Rodolfo Castro
6	1	Geraldo Perdomo	5	1	Ke'Bryan Hayes
7	1	Lourdes Gurriel Jr.	6	1	Tucupita Marcano
8	1	Dominic Fletcher	7	1	Bryan Reynolds
9	1	Corbin Carroll	8	1	Jack Suwinski
			9	1	Connor Joe

	game	ARI vs PIT on 20230521
	state before	(2, 1, 0, 0, 2, 1, 0, 1)
21.	error	play,2,1,castr006,01,C1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 2, 1, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1727 This game had 4 errors, including a consecutive error. The final score was 8-3.

In the bottom of the 2nd inning, with a runner on 1st, PIT batter Rodolfo Castro has a 0-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to reach 3rd. Castro is still at-bat with a 0-1 count.

	game	ARI vs PIT on 20230521
	state before	(1, 0, 0, 0, 6, 3, 3, 0)
22.	error	play,6,0,walkc002,12,SBSX,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 6, 3, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1728 *In the top of the 6th inning, with bases empty, ARI batter Christian Walker hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	ARI vs PIT on 20230521
	state before	(1, 0, 1, 0, 6, 3, 3, 0)
23.	error	play,6,0,gurrl001,11,CBX,S1/G2-.2-H(E1/TH) (NR) (UR)
	state after	(1, 1, 0, 0, 6, 4, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1729 This play is an example of a *consecutive error* since Gurriel bats immediately after Walker (in the previous error) in the Pirates lineup for this game.

In the top of the 6th inning, with a runner on 2nd, ARI batter Lourdes Gurriel hit a soft ground ball single to very shallow infield. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.

	game	ARI vs PIT on 20230521
	state before	(2, 1, 0, 0, 9, 7, 3, 0)
24.	error	play,9,0,walkc002,01,C1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 9, 7, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1730 *In the top of the 9th inning, with a runner on 1st, ARI batter Christian Walker had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Walker is still at bat, now with a 1-1 count.*

	game	TEX vs PIT on 20230523
	state before	(0, 0, 0, 0, 4, 2, 1, 1)
25.	error	play,4,1,santc002,00,X,E3/G3.B-1
	state after	(0, 1, 0, 0, 4, 2, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1731 *In the bottom of the 4th inning, with bases empty, PIT batter Carlos Santana hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for TEX vs PIT on 2023-05-24

position	innings	player	position	innings	player
1	1	Martin Perez	1	1	Johan Oviedo
1	8	Jose Leclerc	1	6	Robert Stephenson
1	8	Will Smith	1	7	Dauri Moreta
2	1	Jonah Heim	1	7	Rob Zastryzny
3	1	Nathaniel Lowe	1	9	Colin Holderman
4	1	Marcus Semien	2	1	Jason Delay
5	1	Josh Jung	3	1	Carlos Santana
6	1	Josh H. Smith	3	9	Connor Joe
7	1	Robbie Grossman	4	1	Rodolfo Castro
7	8	Bubba Thompson	5	1	Ke'Bryan Hayes
8	1	Leody Taveras	6	1	Chris Owings
9	1	Adolis Garcia	7	1	Bryan Reynolds
			8	1	Jack Suwinski
			8	9	Ji Hwan Bae
			9	1	Connor Joe
			9	9	Jack Suwinski

	game	TEX vs PIT on 20230524
	state before	(0, 0, 0, 0, 5, 3, 2, 1)
26.	error	play,5,1,delaj004,20,BBX,S4/G6M+.B-2(E4/TH)
	state after	(0, 0, 1, 0, 5, 3, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1732 This game had 3 errors, but in only 2 plays. The final score was 3-2.

In the bottom of the 5th inning, with bases empty, PIT batter Jason Delay hit a hard ground ball single to short. A throwing error by the second baseman allowed the batter to reach 2nd.

	game	TEX vs PIT on 20230524
	state before	(0, 1, 0, 0, 6, 3, 2, 0)
27.	error	play,6,0,heimj001,11,FBX,E3/G3.1-3(E4);B-1
	state after	(0, 1, 0, 1, 6, 3, 2, 0)
	fielder	3,4 (1st,2nd basemen on the home team)

Summary 1733 *In the top of the 6th inning, with a runner on 1st, TEX batter Jonah Heim hit a ground ball to 1st base. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. An error by the second baseman allowed the runner from 1st (now on 2nd) to reach 3rd.*

25.2.3 June

	game	SLN vs PIT on 20230603
	state before	(0, 0, 0, 0, 6, 1, 1, 1)
28.	error	play,6,1,mathm002,01,CX,E5/G56.B-1
	state after	(0, 1, 0, 0, 6, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1734 This game had 2 errors. The final score was 3-4.
In the bottom of the 6th inning, with bases empty, PIT batter Mark Mathias hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	SLN vs PIT on 20230603
	state before	(2, 0, 1, 0, 7, 1, 4, 1)
29.	error	play,7,1,mccua001,11,CB2,P02(E1/TH).2-3
	state after	(2, 0, 0, 1, 7, 1, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1735 *In the bottom of the 7th inning, with a runner on 2nd, PIT batter Andrew McCutchen has a 1-1 count. The pitcher attempted to pickoff the runner on 2nd. An error by the pitcher allowed that runner to reach 3rd. McCutchen is still at-bat with a 1-1 count.*

	game	OAK vs PIT on 20230605
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
30.	error	play,1,0,ruize001,10,BX,C/E2.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1736 *In the top of the 1st inning, with bases empty, OAK batter Esteury Ruiz had a 1-0 count. Catcher's interference allowed the batter to take 1st.*

	game	OAK vs PIT on 20230606
	state before	(1, 1, 1, 0, 1, 0, 0, 1)
31.	error	play,1,1,suwij001,00,X,FC4/G34.2-H(NR)(UR);1-3(E4);B-1
	state after	(1, 1, 0, 1, 1, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1737 This game had 2 errors. The final score was 11-2. *In the bottom of the 1st inning, with runners on 1st and 2nd, PIT batter Jack Suwinski hit a ground ball into a fielder's choice at 1st/2nd. The runner on 2nd scored (unearned, no RBI). An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	OAK vs PIT on 20230606
	state before	(1, 1, 1, 0, 6, 4, 2, 0)
32.	error	play,6,0,ruize001,00,X,FC3/G3.2-3;1-2(E3/TH);B-1
	state after	(1, 1, 1, 1, 6, 4, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1738 *In the top of the 6th inning, with runners on 1st and 2nd, OAK batter Esteury Ruiz hit a ground ball into a fielder's choice at 1st. The runner on 2nd advanced to 3rd. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for NYN vs PIT on 2023-06-09

position	innings	player	position	innings	player
1	1	Tylor Megill	1	1	Rich Hill
1	4	Zach Muckenhirn	1	8	Rob Zastryzny
1	7	Tommy Hunter	1	9	David Bednar
1	8	Tommy Hunter	2	1	Austin Hedges
2	1	Francisco Alvarez	3	1	Carlos Santana
3	1	Mark Canha	3	9	Connor Joe
3	8	Mark Vientos	4	1	Ji Hwan Bae
4	1	Jeff McNeil	5	1	Ke'Bryan Hayes
4	8	Eduardo Escobar	5	9	Rodolfo Castro
5	1	Eduardo Escobar	6	1	Tucupita Marcano
5	8	Brett Baty	7	1	Bryan Reynolds
6	1	Francisco Lindor	7	8	Josh Palacios
6	8	Luis Guillorme	8	1	Jack Suwinski
7	1	Tommy Pham	9	1	Josh Palacios
7	8	Jeff McNeil	9	8	Canaan Smith-Njigba
8	1	Brandon Nimmo			
8	8	Tommy Pham			
9	1	Starling Marte			
9	8	Mark Canha			

	game	NYN vs PIT on 20230609
	state before	(1, 1, 0, 0, 3, 2, 2, 1)
33.	error	play,3,1,santc002,22,CBBSX,FC6/G6M.1-2(E6);B-1
	state after	(1, 1, 1, 0, 3, 2, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1739 This game had 3 errors. The final score was 7-14.
In the bottom of the 3rd inning, with a runner on 1st, PIT batter Carlos Santana hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter

to reach 1st.

	game	NYN vs PIT on 20230609
	state before	(1, 1, 0, 1, 3, 2, 4, 1)
34.	error	play,3,1,bae-j001,11,SBX,S5/BG25-.3-H;1-H(E5/TH)(NR);B-3
	state after	(1, 0, 0, 1, 3, 2, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1740 *In the bottom of the 3rd inning, with runners on 1st and 3rd, PIT batter Ji Hwan Bae hit a soft bunt single to shallow 3rd base. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	NYN vs PIT on 20230609
	state before	(1, 1, 1, 0, 9, 2, 12, 0)
35.	error	play,9,0,vienm001,12,TS*BX,FC4/G4D.2-H(NR)(UR);1-3(E4);B-1
	state after	(1, 1, 0, 1, 9, 3, 12, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1741 *In the top of the 9th inning, with runners on 1st and 2nd, NYN batter Mark Vientos hit a ground ball into a fielder's choice at deep 2nd. The runner on 2nd scored (unearned, no RBI). An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

Visiting/Home team rosters for NYN vs PIT on 2023-06-10

position	innings	player	position	innings	player
1	1	Kodai Senga	1	1	Johan Oviedo
1	8	David Robertson	1	7	Dauri Moreta
1	9	Adam Ottavino	1	7	Jose Hernandez
2	1	Francisco Alvarez	1	8	Colin Holderman
3	1	Mark Canha	1	9	Angel Perdomo
4	1	Jeff McNeil	2	1	Jason Delay
5	1	Brett Baty	3	1	Carlos Santana
6	1	Luis Guillorme	4	1	Ji Hwan Bae
7	1	Tommy Pham	5	1	Ke'Bryan Hayes
8	1	Brandon Nimmo	6	1	Tucupita Marcano
9	1	Starling Marte	7	1	Bryan Reynolds
			8	1	Jack Suwinski
			9	1	Connor Joe

	game	NYN vs PIT on 20230610
	state before	(1, 1, 0, 0, 4, 1, 0, 1)
36.	error	play,4,1,hayek001,00,X,FC3/G3S.1X3(3E6);B-1
	state after	(2, 1, 0, 0, 4, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1742 This game had 3 errors, including a consecutive error. The final score was 5-1.

In the bottom of the 4th inning, with a runner on 1st, PIT batter Ke'Bryan Hayes hit a ground ball into a fielder's choice at shallow 1st. An error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.

37.	game	NYN vs PIT on 20230610
	state before	(2, 1, 0, 0, 4, 1, 0, 1)
	error	play,4,1,bae-j001,00,X,FC4/G4.3-H(UR);1-3(E4);B-1
	state after	(2, 1, 0, 0, 4, 1, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1743 This play is an example of a *consecutive error* since Bae hits immediately after Hayes in the PIT lineup that game.

In the bottom of the 4th inning, with a runner on 1st, PIT batter Ji Hwan Bae hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI). An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

38.	game	NYN vs PIT on 20230610
	state before	(1, 0, 0, 0, 6, 1, 0, 1)
	error	play,6,1,suwij001,20,BBX,E3/G34.B-1
	state after	(1, 1, 0, 0, 6, 1, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1744 *In the bottom of the 6th inning, with bases empty, PIT batter Jack Suwinski hit a ground ball to 1st/2nd. An error by the first baseman allowed the batter to reach 1st base.*

39.	game	CHN vs PIT on 20230619
	state before	(2, 1, 1, 0, 7, 5, 0, 1)
	error	play,7,1,santc002,02,FFB,0A.2X3(2E6/TH);1-2
	state after	(2, 0, 1, 1, 7, 5, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1745 This game had 2 errors. The final score was 8-0.

In the bottom of the 7th inning, with runners on 1st and 2nd, PIT batter Carlos Santana had a 0-2 count. During the next pitch, which was a ball, the runner on 2nd attempted to steal 3rd. A throwing error

by the shortstop (with an assist from the catcher) allowed the runners on 1st and 2nd to each advance one base, though the runner on 2nd was not credited with a stolen base.. Santana is still at-bat, now with a 1-2 count.

game	CHN vs PIT on 20230619
state before	(1, 1, 1, 0, 8, 5, 0, 0)
40. error	play,8,0,taucm001,32,CCFBBBX,FC4/G34.2-H(NR) (UR);1-3(E4/TH);B-1
state after	(1, 1, 0, 1, 8, 6, 0, 0)
fielder	4 (2nd baseman on the home team)

Summary 1746 *In the top of the 8th inning, with runners on 1st and 2nd, CHN batter Mike Tauchman hit a ground ball into a fielder's choice at 1st/2nd. The runner on 2nd scored (unearned, no RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

game	CHN vs PIT on 20230621
state before	(2, 0, 0, 0, 6, 3, 1, 0)
41. error	play,6,0,morec001,22,SFBBX,E5/TH/G5.B-1
state after	(2, 1, 0, 0, 6, 3, 1, 0)
fielder	5 (3rd baseman on the home team)

Summary 1747 This game had 2 errors. The final score was 8-3.
In the top of the 6th inning, with bases empty, CHN batter Christopher Morel hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

game	CHN vs PIT on 20230621
state before	(2, 1, 1, 1, 7, 6, 1, 1)
42. error	play,7,1,marct001,20,.BBX,E3/TH/G3.3-H(NR) (UR);2-H(NR) (UR);1-3;B-1
state after	(2, 1, 0, 1, 7, 6, 3, 1)
fielder	3 (1st baseman on the visiting team)

Summary 1748 *In the bottom of the 7th inning, with bases loaded, PIT batter Tucupita Marcano hit a ground ball to 1st base. A throwing error by the first baseman allowed the runners on 2nd and 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	SDN vs PIT on 20230627
	state before	(0, 0, 0, 0, 4, 3, 5, 1)
43.	error	play,4,1,hedga001,00,X,C/E2.B-1
	state after	(0, 1, 0, 0, 4, 3, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1749 *In the bottom of the 4th inning, with bases empty, PIT batter Austin Hedges had a 0-0 count. Catcher's interference allowed the batter to take 1st.*

	game	SDN vs PIT on 20230629
	state before	(1, 0, 1, 1, 7, 4, 2, 1)
44.	error	play,7,1,suwij001,12,CCBX,S1/G3S-.3-H;2-H(E1/TH)(NR);B-2
	state after	(1, 0, 1, 0, 7, 4, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1750 *In the bottom of the 7th inning, with runners on 2nd and 3rd, PIT batter Jack Suwinski hit a soft ground ball single to shallow right infield. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 2nd.*

25.2.4 July

Visiting/Home team rosters for MIL vs PIT on 2023-07-02

position	innings	player	position	innings	player
1	1	Colin Rea	1	1	Rich Hill
1	7	Hoby Milner	1	6	Angel Perdomo
1	8	Elvis Peguero	1	7	Carmen Mlodzinski
1	9	Devin Williams	1	8	Ryan Borucki
2	1	William Contreras	1	9	Dauri Moreta
3	1	Owen Miller	2	1	Austin Hedges
4	1	Brice Turang	3	1	Carlos Santana
5	1	Brian Anderson	4	1	Nick Gonzales
6	1	Willy Adames	5	1	Jared Triolo
7	1	Blake Perkins	6	1	Tucupita Marcano
8	1	Joey Wiemer	7	1	Bryan Reynolds
9	1	Raimel Tapia	8	1	Jack Suwinski
			8	8	Josh Palacios
			9	1	Josh Palacios
			9	8	Connor Joe

	game	MIL vs PIT on 20230702
	state before	(2, 0, 0, 0, 5, 3, 2, 0)
45.	error	play,5,0,contw002,12,CC.BFFX,E5/G56.B-1
	state after	(2, 1, 0, 0, 5, 3, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1751 This game had 3 errors. The final score was 6-3.
In the top of the 5th inning, with bases empty, MIL batter William Contreras hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	MIL vs PIT on 20230702
	state before	(0, 0, 0, 0, 5, 3, 2, 1)
46.	error	play,5,1,trioj001,21,BBSX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 5, 3, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1752 *In the bottom of the 5th inning, with bases empty, PIT batter Jared Triolo hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	MIL vs PIT on 20230702
	state before	(2, 1, 0, 1, 8, 3, 2, 0)
47.	error	play,8,0,turab002,02,.CS>X,E1/TH/G25-.3-H(NR)(UR);1-H(NR)(UR);BX2(426)
	state after	(0, 0, 0, 0, 8, 5, 2, 1)
	fielder	1 (pitcher on the home team)

Summary 1753 *In the top of the 8th inning, with 2 outs and runners on 1st and 3rd, MIL batter Brice Turang hit a soft ground ball to shallow left infield. A throwing error by the pitcher allowed the runners on 1st and 3rd to score (unearned, no RBI) and the batter to reach 1st. However, the batter attempted to advance further was tagged out at 2nd by the shortstop (with assists from the catcher and the second baseman). This is the 3rd out and so ends thae half-inning.*

	game	SFN vs PIT on 20230714
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
48.	error	play,2,0,bailp001,00,X,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1754 *This game had 2 errors. The final score was 6-4. In the top of the 2nd inning, with bases empty, SFN batter Patrick Bailey hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	SFN vs PIT on 20230714
	state before	(1, 0, 1, 1, 7, 3, 4, 0)
49.	error	play,7,0,bailp001,21,B.BFX,S9/G34.3-H;2-H(E9) (NR) (UR)
	state after	(1, 1, 0, 0, 7, 5, 4, 0)
	fielder	9 (right field on the home team)

Summary 1755 *In the top of the 7th inning, with runners on 2nd and 3rd, SFN batter Patrick Bailey hit a ground ball single to 1st/2nd. The runner on 3rd scored (1 RBI). An error by the right fielder allowed the runner on 2nd to also score (unearned, no RBI). The batter took 1st.*

	game	SFN vs PIT on 20230715
	state before	(2, 1, 1, 0, 8, 1, 1, 0)
50.	error	play,8,0,confm001,12,SFBX,S9/L9D.2-H;1-H(E2/TH) (NR) (UR);B-2
	state after	(2, 0, 1, 0, 8, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1756 *In the top of the 8th inning, with runners on 1st and 2nd, SFN batter Michael Conforto hit a line drive single to deep right field. The runner on 2nd scored (1 RBI). A throwing error by the catcher allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 2nd.*

	game	SFN vs PIT on 20230716
	state before	(0, 1, 1, 1, 3, 0, 1, 0)
51.	error	play,3,0,davij006,12,BSCX,S9/L89S.3-H;2-H(E9) (NR);1-2
	state after	(0, 1, 1, 0, 3, 2, 1, 0)
	fielder	9 (right field on the home team)

Summary 1757 *In the top of the 3rd inning, with bases loaded, SFN batter J.D. Davis hit a line drive single to shallow right center field. The runner on 3rd scored (1 RBI). An error by the right fielder allowed the runner on 2nd to also score (unearned, no RBI) and the runner on 1st to advance to 2nd. The batter took 1st.*

	game	CLE vs PIT on 20230717
	state before	(0, 0, 0, 0, 6, 7, 0, 1)
52.	error	play,6,1,reynb001,10,..BX,E6/G56.B-1
	state after	(0, 1, 0, 0, 6, 7, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1758 *In the bottom of the 6th inning, with bases empty, PIT batter Bryan Reynolds hit a ground ball to 3rd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs PIT on 20230718
	state before	(2, 0, 1, 0, 1, 4, 0, 0)
53.	error	play,1,0,stram002,02,CCX,S9/L9LS.2-H;B-2(E9)
	state after	(2, 0, 1, 0, 1, 5, 0, 0)
	fielder	9 (right field on the home team)

Summary 1759 *In the top of the 1st inning, with a runner on 2nd, CLE batter Myles Straw hit a line drive single to shallow right field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the batter to reach 2nd.*

	game	PHI vs PIT on 20230728
	state before	(2, 1, 1, 0, 9, 2, 1, 0)
54.	error	play,9,0,turnt001,00,X,E6/G6.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 9, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1760 Each game in this Phillies-Pirates series, which started on July 28th, had at least 1 error, for a total of 5.

In the top of the 9th inning, with runners on 1st and 2nd, PHI batter Trea Turner hit a ground ball to short. An error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

	game	PHI vs PIT on 20230729
	state before	(1, 1, 1, 0, 5, 4, 3, 1)
55.	error	play,5,1,davih001,32,BBCB+2CX,FC3/G4.2-3;1-2(E3);B-1
	state after	(1, 1, 1, 1, 5, 4, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1761 *In the bottom of the 5th inning, with runners on 1st and 2nd, PIT batter Henry Davis hit a ground ball into a fielder's choice at 2nd. The runner on 2nd to advanced to 3rd. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

Visiting/Home team rosters for PHI vs PIT on 2023-07-30

position	innings	player	position	innings	player
1	1	Cristopher Sanchez	1	1	Rich Hill
1	10	Andrew Vasquez	1	10	Angel Perdomo
1	6	Jeff Hoffman	1	6	Dauri Moreta
1	6	Seranthony Dominguez	1	7	Carmen Mlodzinski
1	7	Gregory Soto	1	7	Jose Hernandez
1	8	Yunior Marte	1	9	David Bednar
1	9	Craig Kimbrel	2	1	Austin Hedges
2	1	Garrett Stubbs	2	8	Endy Rodriguez
3	1	Alec Bohm	3	1	Connor Joe
4	1	Josh Harrison	4	1	Nick Gonzales
5	1	Edmundo Sosa	5	1	Jared Triolo
6	1	Trea Turner	6	1	Liover Peguero
7	1	Kyle Schwarber	6	9	Alika Williams
7	9	Brandon Marsh	7	1	Bryan Reynolds
8	1	Johan Rojas	8	1	Joshua Palacios
9	1	Nick Castellanos	9	1	Henry Davis

	game	PHI vs PIT on 20230730
	state before	(1, 0, 0, 0, 7, 4, 2, 1)
56.	error	play,7,1,joe-c001,00,X,E6/P6S.B-1
	state after	(1, 1, 0, 0, 7, 4, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1762 This game had 3 errors. The final score was 4-6 (in extra innings).

In the bottom of the 7th inning, with bases empty, PIT batter Connor Joe hit a pop up to shallow short. An error by the shortstop allowed the batter to reach 1st base.

	game	PHI vs PIT on 20230730
	state before	(1, 1, 0, 0, 7, 4, 2, 1)
57.	error	play,7,1,reynb001,10,BX,S8/L8D.1-H(E8)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 7, 4, 3, 1)
	fielder	8 (center field on the visiting team)

Summary 1763 *In the bottom of the 7th inning, with a runner on 1st, PIT batter Bryan Reynolds hit a line drive single to deep center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	PHI vs PIT on 20230730
	state before	(0, 0, 1, 0, 10, 4, 4, 0)
58.	error	play,10,0,bohma001,02,..FFF.FFFX,S5/G56.2-3(E5/TH);B-2
	state after	(0, 0, 1, 1, 10, 4, 4, 0)
	fielder	5 (center field on the visiting team)

Summary 1764 *In the top of the 10th inning, with a runner on 2nd, PHI batter Alec Bohm hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd.*

25.2.5 August

Visiting/Home team rosters for DET vs PIT on 2023-08-02

position	innings	player	position	innings	player
1	1	Eduardo Rodriguez	1	1	Osvaldo Bido
1	7	Jose Cisnero	1	4	Ryan Borucki
1	8	Jason Foley	1	6	Dauri Moreta
1	9	Alex Lange	1	7	Jose Hernandez
1	9	Tyler Holton	1	8	Yerry De Los Santos
2	1	Eric Haase	2	1	Jason Delay
3	1	Spencer Torkelson	2	8	Endy Rodriguez
4	1	Nick Maton	3	1	Connor Joe
5	1	Zach McKinstry	4	1	Nick Gonzales
5	4	Zack Short	4	9	Liover Peguero
6	1	Javier Baez	5	1	Ke'Bryan Hayes
7	1	Akil Baddoo	6	1	Alika Williams
7	4	Zach McKinstry	7	1	Bryan Reynolds
8	1	Riley Greene	8	1	Jack Suwinski
9	1	Matt Vierling	9	1	Henry Davis

	game	DET vs PIT on 20230802
	state before	(2, 0, 0, 0, 1, 1, 0, 0)
59.	error	play,1,0,carpk001,11,CBX,E4/G9S.B-1
	state after	(2, 1, 0, 0, 1, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1765 This game had 3 errors. The final score was 6-3.
In the top of the 1st inning, with bases empty, DET batter Kerry Carpenter hit a ground ball to shallow right field. An error by the second baseman allowed the batter to reach 1st base.

	game	DET vs PIT on 20230802
	state before	(0, 0, 0, 0, 7, 5, 2, 1)
60.	error	play,7,1,willla005,12,.FS.B*S,K.BX1(2E3)
	state after	(0, 1, 0, 0, 7, 5, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1766 *In the bottom of the 7th inning, with bases empty, PIT batter Alika Williams had a 1-2 count. During the next pitch, whcih was a swinging strike, the catcher failed to catch strike 3. An error by the first baseman (with an assist fom the catcher) allowed the batter to safely reach 1st.*

	game	DET vs PIT on 20230802
	state before	(1, 0, 0, 0, 9, 6, 2, 0)
61.	error	play,9,0,mckiz001,12,BCFFX,3E1/G3.B-1
	state after	(1, 1, 0, 0, 9, 6, 2, 0))
	fielder	1 (pitcher on the home team)

Summary 1767 *In the top of the 9th inning, with runners on 1st and 2nd, DET batter Zach McKinstry hit a ground ball to 1st base. An error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	ATL vs PIT on 20230807
	state before	(1, 1, 1, 0, 4, 2, 6, 0)
62.	error	play,4,0,ozunm001,00,X,S9/L9+.2-H;1-H(E9)(NR)(UR);B-3
	state after	(1, 0, 0, 1, 4, 4, 6, 0)
	fielder	9 (right field on the home team)

Summary 1768 *In the top of the 4th inning, with runners on 1st and 2nd, ATL batter Marcell Ozuna hit a hard line drive single to right field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 3rd.*

63.	game	ATL vs PIT on 20230809
	state before	(2, 1, 0, 0, 8, 6, 5, 1)
	error	play,8,1,trioj001,22,B1SSBX,E5/G5.1-2;B-1
	state after	(2, 1, 1, 0, 8, 6, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1769 *In the bottom of the 8th inning, with a runner on 1st, PIT batter Jared Triolo hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

64.	game	CIN vs PIT on 20230813 (1/2)
	state before	(0, 0, 0, 0, 5, 1, 0, 0)
	error	play,5,0,bensw001,00,X,E6/P78S.B-1
	state after	(0, 1, 0, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1770 This 1st game of a doubleheader had 2 errors. The final score was 2-4.

In the top of the 5th inning, with bases empty, CIN batter Will Benson hit a pop up to shallow left center field. An error by the shortstop allowed the batter to reach 1st base.

65.	game	CIN vs PIT on 20230813 (1/2)
	state before	(1, 1, 0, 0, 6, 2, 1, 0)
	error	play,6,0,encac001,11,FB1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1771 *In the top of the 6th inning, with a runner on 1st, CIN batter Christian Encarnacion-Strand has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd. Encarnacion-Strand is still at-bat with a 1-1 count.*

	game	SLN vs PIT on 20230821
	state before	(1, 0, 1, 0, 1, 0, 0, 0)
66.	error	play,1,0,goldp001,00,X,C/E2.B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1772 This game had 2 errors. The final score was 1-11.
In the top of the 1st inning, with a runner on 2nd, SLN batter Paul Goldschmidt had a 0-0 count. Catcher's interference allowed the batter to take 1st.

	game	SLN vs PIT on 20230821
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
67.	error	play,1,1,joe-c001,11,CBX,E4/TH/G6.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1773 *In the bottom of the 1st inning, with a runner on 2nd, PIT batter Connor Joe hit a ground ball to short. A throwing error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SLN vs PIT on 20230823
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
68.	error	play,1,0,burla001,00,X,S2/BG2-.B-2(E2/TH)
	state after	(1, 0, 1, 0, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1774 *In the top of the 1st inning, with bases empty, SLN batter Alec Burleson hit a soft bunt single to the very shallow infield. A throwing error by the the catcher allowed the batter to reach 2nd.*

Visiting/Home team rosters for CHN vs PIT on 2023-08-24

position	innings	player	position	innings	player
1	1	Justin Steele	1	1	Rob Zastryzny
1	10	Adbert Alzolay	1	10	Thomas Hatch
1	7	Julian Merryweather	1	2	Andre Jackson
1	8	Michael Fulmer	1	8	Colin Holderman
1	9	Daniel Palencia	1	9	David Bednar
2	1	Yan Gomes	2	1	Endy Rodriguez
3	1	Jeimer Candelario	3	1	Alfonso Rivas
4	1	Nico Hoerner	4	1	Vinny Capra
5	1	Nick Madrigal	4	10	Ji Hwan Bae
6	1	Dansby Swanson	5	1	Ke'Bryan Hayes
7	1	Ian Happ	6	1	Alika Williams
8	1	Cody Bellinger	6	10	Liover Peguero
9	1	Mike Tauchman	7	1	Joshua Palacios
			8	1	Ji Hwan Bae
			8	10	Jack Suwinski
			9	1	Connor Joe

	game	CHN vs PIT on 20230824
	state before	(0, 1, 0, 0, 2, 3, 0, 1)
69.	error	play,2,1,caprv001,00,X,FC/SH/BG13.1-2;B-1(E1)
	state after	(0, 1, 1, 0, 2, 3, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1775 This game had 3 errors. This particular play was a fielder's choice with an attempt to force the batter out at 1st. The final score was 5-4 (in extra innings).

In the bottom of the 2nd inning, with a runner on 1st, PIT batter Vinny Capra hit a bunt to right shalow infield. The runner on 1st to advanced to 2nd An error by the pitcher allowed the batter to reach 1st.

	game	CHN vs PIT on 20230824
	state before	(2, 0, 0, 0, 6, 3, 3, 1)
70.	error	play,6,1,rivaa001,11,BCX,D9/L9LD.B-3(E9)
	state after	(2, 0, 0, 1, 6, 3, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 1776 *In the bottom of the 6th inning, with bases empty, PIT batter Alfonso Rivas hit a line drive double to deep right field. A throwing error by the right fielder allowed the batter to reach 1st.*

	game	CHN vs PIT on 20230824
	state before	(1, 0, 1, 1, 10, 3, 3, 0)
71.	error	play,10,0,happi001,12,B.SFX,S9/F9.3-H(UR);2-H(E9)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 10, 5, 3, 0)
	fielder	9 (right field on the visiting team)

Summary 1777 *In the top of the 10th inning, with runners on 2nd and 3rd, CHN batter Ian Happ hit a fly ball single to right field. The runner on 3rd scored (1 RBI). An error by the right fielder allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 2nd.*

25.2.6 September

	game	MIL vs PIT on 20230904
	state before	(0, 0, 0, 0, 3, 2, 0, 1)
72.	error	play,3,1,rivaa001,22,BBCF.X,S5/G56S-.B-2(E4/TH)
	state after	(0, 0, 1, 0, 3, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1778 *In the bottom of the 3rd inning, with bases empty, PIT batter Alfonso Rivas hit a soft ground ball single to shallow 3rd base. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	MIL vs PIT on 20230905
	state before	(1, 0, 1, 1, 5, 5, 0, 0)
73.	error	play,5,0,canhm001,22,BFBCFX,S7/L7S.3-H;2-H(E7)(NR)(UR);BX2(743)
	state after	(2, 0, 0, 0, 5, 7, 0, 0)
	fielder	7 (left field on the home team)

Summary 1779 *In the top of the 5th inning, with runners on 2nd and 3rd, MIL batter Mark Canha hit a line drive single to shallow left field. The runner on 3rd scored (1 RBI). An error by the left fielder allowed the runner on 2nd to also score (unearned, no RBI). However, the batter attempted to advance to 2nd and was tagged out at 2nd by the first baseman (with assists from the second baseman and the left fielder).*

	game	MIL vs PIT on 20230906
	state before	(1, 0, 0, 0, 4, 3, 1, 1)
74.	error	play,4,1,suwij001,20,BBX,E3/G34.B-1
	state after	(1, 1, 0, 0, 4, 3, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1780 *In the bottom of the 4th inning, with bases empty, PIT batter Jack Suwinski hit a ground ball to 1st/2nd. An error by the first baseman allowed the batter to reach 1st base.*

	game	WAS vs PIT on 20230913
	state before	(1, 1, 0, 0, 4, 1, 5, 1)
75.	error	play,4,1,reynb001,10,1B>C,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 4, 1, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1781 *This game had 2 errors. The final score was 6-7. In the bottom of the 4th inning, with a runner on 1st, PIT batter Bryan Reynolds had a 1-0 count. During the next pitch, which was a called*

strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Reynolds is still at bat, now with a 1-1 count.

	game	WAS vs PIT on 20230913
	state before	(0, 0, 0, 0, 9, 5, 7, 0)
76.	error	play,9,0,blant002,00,...F,FLE6
	state after	(0, 0, 0, 0, 9, 5, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 1782 This error is an example of an *infinity play*, since it can be repeated infinitely often without changing the game state. *In the top of the 9th inning, with bases empty and a 0-0 count, WAS batter Travis Blankenhorn hit a foul fly ball. An error by the shortstop allowed the batter to avoid the out. Blankenhorn is still at-bat, now with a 0-1 count.*

Visiting/Home team rosters for WAS vs PIT on 2023-09-14

position	innings	player	position	innings	player
1	1	Josiah Gray	1	1	Mitch Keller
1	7	Jordan Weems	1	9	David Bednar
1	8	Jose A. Ferrer	2	1	Jason Delay
2	1	Keibert Ruiz	3	1	Alfonso Rivas
3	1	Dominic Smith	4	1	Ji Hwan Bae
4	1	Luis Garcia	5	1	Jared Triolo
5	1	Jake Alu	6	1	Liover Peguero
6	1	CJ Abrams	7	1	Joshua Palacios
7	1	Travis Blankenhorn	8	1	Jack Suwinski
8	1	Jacob Young	9	1	Connor Joe
9	1	Lane Thomas			

	game	WAS vs PIT on 20230914
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
77.	error	play,1,0,thoml002,00,X,FC6/G56.1-2(E6);B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1783 This game had 3 errors, but in only 2 plays. The final score was 0-2.

In the top of the 1st inning, with a runner on 1st, WAS batter Lane Thomas hit a ground ball into a fielder's choice at 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	WAS vs PIT on 20230914
	state before	(0, 1, 0, 0, 7, 0, 2, 1)
78.	error	play,7,1,palaj001,01,CX,S9/G34.1-3(E9);B-2(E9/TH)
	state after	(0, 0, 1, 1, 7, 0, 2, 1)
	fielder	9,9 (right field on the visiting team)

Summary 1784 This play had 2 errors by the same fielder.

In the bottom of the 7th inning, with a runner on 1st, PIT batter Josh Palacios hit a ground ball single to 1st/2nd. A throwing error by the right fielder allowed the batter to reach 2nd.

	game	NYA vs PIT on 20230915
	state before	(2, 1, 1, 1, 6, 3, 3, 1)
79.	error	play,6,1,hayek001,32,BBCBF>X,E4/G8S.3-H(NR)(UR);2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 6, 3, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1785 This game had 2 errors. The final score was 7-5.

In the bottom of the 6th inning, with bases loaded, PIT batter Ke'Bryan

Hayes hit a ground ball to shallow center field. An error by the second baseman allowed the runners on 2nd and 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.

80.	game	NYA vs PIT on 20230915
	state before	(1, 1, 1, 1, 9, 4, 5, 0)
	error	play,9,0,volpa001,01,CX,64(1)/FO/G6.3-H;2-H(E4/TH)(NR);B-1
	state after	(2, 1, 0, 0, 9, 6, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1786 *In the top of the 9th inning, with bases loaded, NYA batter Anthony Volpe hit a ground ball to short. The runner at 1st was forced out by the second baseman (with an assist from the shortstop). The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 1st.*

81.	game	NYA vs PIT on 20230916
	state before	(2, 0, 0, 0, 3, 4, 2, 0)
	error	play,3,0,perao002,00,X,E6/TH/G6D.B-1
	state after	(2, 1, 0, 0, 3, 4, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1787 *In the top of the 3rd inning, with bases empty, NYA batter Oswald Peraza hit a ground ball to deep short. A throwing error by the shortstop allowed the batter to reach 1st base.*

82.	game	NYA vs PIT on 20230917
	state before	(2, 0, 1, 0, 6, 1, 2, 1)
	error	play,6,1,andum001,10,BX,E6/G6D.2-3;B-1
	state after	(2, 1, 0, 1, 6, 1, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1788 *In the bottom of the 6th inning, with a runner on 2nd, PIT batter Miguel Andujar hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

Visiting/Home team rosters for MIA vs PIT on 2023-09-29

position	innings	player	position	innings	player
1	1	Edward Cabrera	1	1	Osvaldo Bido
1	4	Steven Okert	1	3	Hunter Stratton
1	6	JT Chargois	1	5	Kyle Nicolas
1	7	Huascar Brazoban	1	7	Ryan Borucki
1	8	David Robertson	1	8	Carmen Mlodzinski
1	9	Tanner Scott	1	8	Colin Selby
2	1	Jacob Stallings	1	9	Cody Bolton
2	8	Nick Fortes	2	1	Endy Rodriguez
3	1	Josh Bell	3	1	Connor Joe
3	8	Yuli Gurriel	4	1	Jared Triolo
4	1	Jon Berti	5	1	Ke'Bryan Hayes
4	8	Xavier Edwards	6	1	Liover Peguero
5	1	Jake Burger	7	1	Bryan Reynolds
6	1	Garrett Hampson	8	1	Jack Suwinski
7	1	Bryan De La Cruz	9	1	Joshua Palacios
7	8	Jon Berti			
8	1	Jazz Chisholm Jr.			
9	1	Jesus Sanchez			
9	8	Bryan De La Cruz			

game	MIA vs PIT on 20230929
state before	(2, 1, 1, 1, 4, 0, 2, 1)
83. error	play,4,1,reynb001,12,. *ST.BFX,E6/TH/G6M.3-H(NR) (UR);2-3;1-2;B-1
state after	(2, 1, 1, 1, 4, 0, 3, 1)
fielder	6 (shortstop on the visiting team)

Summary 1789 This game had 2 errors. Each game in this Marlins-Pirates series had at least one error, for a total of 5. The final score was 4-3.

In the bottom of the 4th inning, with bases loaded, PIT batter Bryan Reynolds hit a ground ball to short. A throwing error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.

	game	MIA vs PIT on 20230929
	state before	(0, 0, 0, 0, 9, 4, 3, 0)
84.	error	play,9,0,hampg001,22,.CBBS.FX,E5/TH/G56S.B-1
	state after	(0, 1, 0, 0, 9, 4, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1790 *In the top of the 9th inning, with bases empty, MIA batter Garrett Hampson hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	MIA vs PIT on 20230930
	state before	(1, 1, 0, 1, 2, 1, 0, 1)
85.	error	play,2,1,gonzn001,11,SBX,FC/SH/BG2.3-H(NR)(UR);1-3;B-2(E1/TH)
	state after	(1, 0, 1, 1, 2, 1, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1791 *In the bottom of the 2nd inning, with runners on 1st and 3rd, PIT batter Nick Gonzales hit a bunt into a fielder's choice² at very shallow infield. The runner on 3rd scored (unearned, no RBI) and the runner on 1st advance to 3rd. A throwing error by the pitcher allowed the batter to reach 2nd.*

²This fielder's choice is unusual in that the play here is to put the batter out at 1st, which is at odds with the usual fielder's choice where the play is to put out a runner *other* than the batter.

	game	MIA vs PIT on 20231001
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
86.	error	play,3,1,delaj004,21,BBC1X,FC5/G56.1-2(E5);B-1
	state after	(0, 1, 1, 0, 3, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1792 This game had 2 errors. The final score was 0-3.
In the bottom of the 3rd inning, with a runner on 1st, PIT batter Jason Delay hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	MIA vs PIT on 20231001
	state before	(1, 0, 0, 0, 4, 0, 0, 0)
87.	error	play,4,0,sancj003,00,X,E4/G34+.B-1
	state after	(1, 1, 0, 0, 4, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1793 *In the top of the 4th inning, with bases empty, MIA batter Jesus Sanchez hit a hard ground ball to 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.*

Chapter 26

San Diego Padres (SDN)

The San Diego Padres were founded as a National League expansion franchise in 1969. They have only played in the NL championships 3 times, in 1984, 1998, and 2022. The Padres have not (yet) won a World Series.

This chapter is devoted to listing the errors in each regular season game of 2023 played in San Diego against the Padres. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 SDN team is .702, which is “high” compared with the MLB average of 0.691 that year [BR], with a standard deviation of $\sigma = 0.011$.

That year, there were 76 errors in the 81 home games, 31 by the Padres and the rest (45) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 73. This is “low” compared to the MLB average of 83.9.

26.1 Season totals

The Official Scorer shall credit the batter with a run batted in for every run that scores by reason of the batter becoming a

runner with the bases full (because of a base on balls, an award of first base for being touched by a pitched ball or for interference or obstruction).

– **MLB Rule Book** 9.04(a)(2)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

26.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	45
Runs by ht because of errors by vt:	16
Errors by ht:	31
Runs by vt because of errors by ht:	6

This is a very good error-to-run conversion rate: $16/45 = 35.5\ldots\%$ on the part of the Padres (compared to $6/31 = 19.3\ldots\%$ on the part of the visiting teams).

Note that the total number of errors is 76, so the expected number of errors (by either team) in a randomly selected SDN2023 home game is about 0.94.

Errors by position	vt	ht
errors due to pitcher:	5	6
to catcher:	8	2
to 1st baseman:	3	4
to 2nd baseman:	6	3
to 3rd baseman:	9	6
to shortstop:	7	4
to left fielder:	3	3
to center fielder:	2	0
to right fielder:	2	3

26.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	17	15	32
(lone) runner on 1st	8	9	17
(lone) runner on 2nd	8	2	10
(lone) runner on 3rd	3	0	3
two runners on base	7	5	12
bases loaded	2	0	2
total	45	31	75

A table of frequencies of most common errors by the Padres in SDN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	6	6/31 = 19%
(1, 0, 0, 0)	4	13%
(2, 0, 0, 0)	5	16%
(0, 1, 0, 0)	2	6.5%
(1, 1, 0, 0)	4	13%
(2, 1, 0, 0)	3	9.7%
(1, 0, 1, 0)	1	3.2%
(1, 1, 1, 0)	2	6.5%
(1, 0, 1, 1)	2	6.5%

A table of frequencies of most common errors by a visiting team in SDN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	$8/45 = 18\%$
(1, 0, 0, 0)	4	8.9%
(2, 0, 0, 0)	5	11%
(0, 1, 0, 0)	2	4.4%
(1, 1, 0, 0)	0	0%
(2, 1, 0, 0)	6	13%
(1, 0, 1, 0)	3	6.7%
(2, 0, 1, 0)	4	8.9%
(2, 0, 0, 1)	3	6.7%
(1, 1, 0, 1)	3	6.7%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Padres homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `SDN2023.evn`):

state	SDN2023 percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	16.1%
(2, 0, 0, 0)	12.9%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.9%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	2.2%
(1, 0, 1, 0)	2.8%
(2, 0, 1, 0)	3.8%
(0, 0, 0, 1)	0.1%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.4%
(0, 1, 1, 0)	1.6%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.2%

26.1.3 Some interesting errors

Some of the most interesting error plays in the SDN 2019 home games:

- The game Colorado Rockies vs Padres on 2023-03-30 had 4 errors, including 2 catcher's interference.
- The game Tampa Bay Rays vs Padres on 2023-06-18 had 3 errors, but 2 were on 1 player in one play.
- Angels vs Padres 2023-07-04/05. Each game had 2 errors, including an ∞ play (that is, an FLE).

- Another Rockies vs Padres game, this one on 2023-09-18, had 3 errors, including a consecutive error and a play where a runner stole home.

26.1.4 Relative team rankings

We have 81 home games between SDN and some visiting MLB team (vt). A number of teams did not play SDN in the regular season. However, of those that did, SDN lost more games than won against¹:

ATL (2/3), BOS (2/3), KCA (2/3), PHI (2/3), PIT (2/3), TBA (2/3), WAS
(2/3),
MIL (3/4), ARI (4/6), LAN (5/7)

and *won* more than lost against

ANA (0/3), TEX (0/3), BAL (1/3), CIN (1/3), CLE (1/3), MIA (1/3),
NYN (1/3), SLN (1/3), SFN (1/6), COL (2/7) .

The teams that *split* their series were:

SEA (1/2), CHN (2/4) .

The teams that did not play in any of the 81 home games in SDN 2023 were:

CHA, DET, HOU, MIN, NYA, OAK, TOR .

26.2 Game-by-game totals

The batter becomes a runner and is entitled to first base without liability to be put out (provided he advances to and touches first base) when the catcher or any fielder interferes with him.

– **MLB Rule Book** 6.01(c)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for SDN 2023 is as follows [BA], [BR].

¹That team's W/T record in parentheses (W = numbers of (home) games vt won against SDN, T = Total home games vt played vs Padres at San Diego.

position	player	games played	OBP
1 (C)	Gary Sánchez	63	.292
3 (1B)	Jake Cronenworth	106	.312
4 (2B)	Ha-Seong Kim	106	.351
5 (3B)	Manny Machado	105	.319
6 (SS)	Xander Bogaerts	146	.350
7 (LF)	Juan Soto	154	.410
8 (CF)	Trent Grisham	153	.315
9 (RF)	Fernando Tatis Jr.	137	.322

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 Padres home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

26.2.1 March

Visiting/Home team rosters for COL vs SDN on 2023-03-30

position	innings	player	position	innings	player
1	1	German Marquez	1	1	Blake Snell
1	7	Dinelson Lamet	1	5	Nabil Crismatt
1	8	Justin Lawrence	1	7	Domingo Tapia
1	9	Jake Bird	1	9	Steven Wilson
2	1	Elias Diaz	2	1	Austin Nola
3	1	C.J. Cron	3	1	Jake Cronenworth
4	1	Ryan McMahon	4	1	Ha-Seong Kim
5	1	Elehuris Montero	5	1	Manny Machado
5	9	Alan Trejo	6	1	Xander Bogaerts
6	1	Ezequiel Tovar	7	1	Juan Soto
7	1	Harold Castro	8	1	Trent Grisham
8	1	Yonathan Daza	9	1	David Dahl
9	1	Kris Bryant			

	game	COL vs SDN on 20230330
	state before	(0, 0, 0, 0, 2, 1, 1, 1)
1.	error	play,2,1,carpm002,10,BX,C/E2.B-1
	state after	(0, 1, 0, 0, 2, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 1794 This game had 4 errors. The final score was 7-2.
In the bottom of the 2nd inning, with bases empty, SDN batter Matt Carpenter had a 1-0 count. Catcher's interference allowed the batter to reach 1st base.

	game	COL vs SDN on 20230330
	state before	(0, 0, 0, 0, 3, 1, 1, 0)
2.	error	play,3,0,bryak001,11,BFX,E4/G4M.B-1
	state after	(0, 1, 0, 0, 3, 1, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1795 *In the top of the 3rd inning, with bases empty, COL batter Kris Bryant hit a ground ball to deep 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	COL vs SDN on 20230330
	state before	(1, 1, 0, 1, 4, 1, 1, 1)
3.	error	play,4,1,carpm002,11,BCX,FC3/G3S.3-H;1-2(E3/TH);B-1
	state after	(1, 1, 1, 0, 4, 1, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1796 *In the bottom of the 4th inning, with runners on 1st and 3rd, SDN batter Matt Carpenter hit a ground ball into a fielder's choice at 1st. The runner from 3rd scored (1 RBI). A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	COL vs SDN on 20230330
	state before	(2, 1, 0, 0, 6, 5, 2, 1)
4.	error	play,6,1,cronj001,01,FX,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 6, 5, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1797 *In the bottom of the 6th inning, with a runner on 1st, SDN batter Jake Cronenworth had a 0-1 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

26.2.2 April

	game	ARI vs SDN on 20230404
	state before	(1, 1, 0, 1, 4, 0, 2, 1)
5.	error	play,4,1,nolaa002,31,M1BB*B3>B,W.3-H(E2/TH)(NR)(UR);1-3
	state after	(1, 1, 0, 1, 4, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1798 This game had 2 errors. The final score was 8-6. *In the bottom of the 4th inning, with runners on 1st and 3rd, SDN batter Austin Nola walks and the runner on 1st advances to 2nd, loading the bases. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI) and the runner now on 2nd reach 3rd.*

	game	ARI vs SDN on 20230404
	state before	(1, 1, 0, 0, 8, 6, 5, 0)
6.	error	play,8,0,rojaj001,10,B1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 8, 6, 5, 0)
	fielder	1 (pitcher on the home team)

Summary 1799 *In the top of the 8th inning, with a runner on 1st, ARI batter Josh Rojas has a 1-0 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd. Rojas is still at bat with a 1-0 count.*

	game	MIL vs SDN on 20230414
	state before	(2, 1, 0, 0, 5, 7, 0, 0)
7.	error	play,5,0,millo002,01,CX,S5/G5S-.1-3(E5/TH)
	state after	(2, 1, 0, 1, 5, 7, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1800 *In the top of the 5th inning, with a runner on 1st, MIL batter Owen Miller hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	MIL vs SDN on 20230415
	state before	(0, 0, 0, 0, 8, 3, 8, 1)
8.	error	play,8,1,azocj001,12,.....BCFFX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 8, 3, 8, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1801 *In the bottom of the 8th inning, with bases empty, SDN batter Jose Azocar hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs SDN on 20230416
	state before	(2, 0, 0, 0, 7, 1, 0, 1)
9.	error	play,7,1,nolaa002,31,BBBCX,E5/TH/G6.B-1
	state after	(2, 1, 0, 0, 7, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1802 *In the bottom of the 7th inning, with bases empty, SDN batter Austin Nola hit a ground ball to short. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	SFN vs SDN on 20230429
	state before	(2, 1, 1, 0, 1, 0, 0, 1)
10.	error	play,1,1,cruzn002,21,C*BBX,S15/L1+.2-H;1-H(E5/TH) (NR) (UR) ;B-3
	state after	(2, 0, 0, 1, 1, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1803 *In the bottom of the 1st inning, with runners on 1st and 2nd, SDN batter Nelson Cruz hit a hard line drive single to shallow infield. The runner on 2nd scored (1 RBI). A throwing error by the third baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	SFN vs SDN on 20230430
	state before	(1, 0, 0, 0, 7, 4, 3, 0)
11.	error	play,7,0,wadel001,21,BBFX,E6/G8S.B-1
	state after	(1, 1, 0, 0, 7, 4, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1804 *In the top of the 7th inning, with bases empty, SFN batter LaMonte Wade hit a ground ball to shallow center field. An error by the shortstop allowed the batter to reach 1st base.*

26.2.3 May

	game	CIN vs SDN on 20230502
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
12.	error	play,1,0,indij001,32,CBFBBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1805 This game had 2 errors. The final score was 2-1. *In the top of the 1st inning, with bases empty, CIN batter Jonathan India hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	CIN vs SDN on 20230502
	state before	(2, 1, 0, 0, 4, 0, 1, 1)
13.	error	play,4,1,nolaa002,00,1,P01(E3).1-2
	state after	(2, 0, 1, 0, 4, 0, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1806 *In the bottom of the 4th inning, with a runner on 1st, SDN batter Austin Nola has a 0-0 count. The pitcher attempted to pick off the runner on 1st. An error by the first baseman allowed that runner to reach 2nd.*

	game	LAN vs SDN on 20230505
	state before	(1, 1, 0, 0, 7, 1, 4, 0)
14.	error	play,7,0,heywj001,11,S*BX,D9/L9L.1-H(E9)(NR)(UR);B-3
	state after	(1, 0, 0, 1, 7, 2, 4, 0)
	fielder	9 (right field on the home team)

Summary 1807 *In the top of the 7th inning, with a runner on 1st, LAN batter Jason Heyward hit a line drive double to right field. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.*

	game	LAN vs SDN on 20230507
	state before	(0, 0, 0, 0, 6, 0, 2, 0)
15.	error	play,6,0,freef001,01,FX,E7/F7LD.B-2
	state after	(0, 0, 1, 0, 6, 0, 2, 0)
	fielder	7 (left field on the home team)

Summary 1808 *In the top of the 6th inning, with bases empty, LAN batter Freddie Freeman hit a fly ball to deep left field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	KCA vs SDN on 20230515
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
16.	error	play,1,1,sotoj001,11,BCX,E1/G1S-.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1809 *In the bottom of the 1st inning, with a runner on 2nd, SDN batter Juan Soto hit a soft ground ball to shallow infield. An error by the pitcher allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	KCA vs SDN on 20230516
	state before	(2, 0, 0, 0, 5, 5, 0, 1)
17.	error	play,5,1,sullb001,22,CBBFX,S9/G4.B-2(E9)
	state after	(2, 0, 1, 0, 5, 5, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 1810 *This game had 2 errors. The final score was 5-4. In the bottom of the 5th inning, with bases empty, SDN batter Brett Sullivan hit a ground ball single to 2nd. An error by the right fielder allowed the batter to reach 2nd.*

	game	KCA vs SDN on 20230516
	state before	(2, 0, 1, 0, 5, 5, 0, 1)
18.	error	play,5,1,tatif002,12,FFBX,E4/TH/G34.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 5, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1811 *In the bottom of the 5th inning, with a runner on 2nd, SDN batter Fernando Tatis hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	BOS vs SDN on 20230519
	state before	(2, 0, 0, 0, 6, 6, 1, 0)
19.	error	play,6,0,duraj001,10,BX,E3/L3.B-1
	state after	(2, 1, 0, 0, 6, 6, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1812 *In the top of the 6th inning, with bases empty, BOS batter Jarren Duran hit a line drive to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	BOS vs SDN on 20230521
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
20.	error	play,1,1,cronj001,21,*BSBX,E1/G1.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1813 This game had 2 errors. The final score was 0-7.
In the bottom of the 1st inning, with a runner on 1st, SDN batter Jake Cronenworth hit a ground ball to shallow infield. An error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	BOS vs SDN on 20230521
	state before	(2, 0, 0, 0, 5, 0, 6, 1)
21.	error	play,5,1,sotoj001,12,CFBX,E4/TH/G34.B-2
	state after	(2, 0, 1, 0, 5, 0, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1814 *In the bottom of the 5th inning, with bases empty, SDN batter Juan Soto hit a ground ball to 1st/2nd base gap base. A throwing error by the second baseman allowed the batter to reach 2nd base.*

26.2.4 June

	game	CHN vs SDN on 20230602
	state before	(0, 0, 0, 0, 8, 2, 1, 1)
22.	error	play,8,1,odorr001,01,TX,S8/L89XD.B-2(E8)
	state after	(0, 0, 1, 0, 8, 2, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 1815 *In the bottom of the 8th inning, with bases empty, SDN batter Rougned Odor hit a line drive single to extra deep right center field. An error by the center fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for CHN vs SDN on 2023-06-03

position	innings	player	position	innings	player
1	1	Drew Smyly	1	1	Yu Darvish
1	6	Michael Fulmer	1	8	Nick Martinez
1	8	Jeremiah Estrada	1	9	Brent Honeywell
2	1	Tucker Barnhart	2	1	Gary Sanchez
3	1	Matt Mervis	3	1	Jake Cronenworth
4	1	Nico Hoerner	3	8	Alfonso Rivas
5	1	Miles Mastrobuoni	4	1	Ha-Seong Kim
6	1	Dansby Swanson	4	8	Jake Cronenworth
7	1	Ian Happ	5	1	Manny Machado
8	1	Mike Tauchman	6	1	Xander Bogaerts
9	1	Seiya Suzuki	6	8	Ha-Seong Kim
			7	1	Juan Soto
			8	1	Trent Grisham
			9	1	Fernando Tatis Jr.

	game	CHN vs SDN on 20230603
	state before	(2, 0, 0, 1, 2, 0, 1, 1)
23.	error	play,2,1,grist001,00,X,D8/L89D.3-H;B-3(E8)
	state after	(2, 0, 0, 1, 2, 0, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 1816 This game (and the next one) had 2 errors. The final score was 0-6.

In the bottom of the 2nd inning, with a runner on 3rd, SDN batter Trent Grisham hit a line drive double to deep right center field. The runner on 3rd scored (1 RBI). An error by the center fielder allowed the batter to reach 3rd.

	game	CHN vs SDN on 20230603
	state before	(2, 0, 0, 0, 3, 0, 1, 0)
24.	error	play,3,0,mastm001,11,BCX,S15/G1.B-2(E5/TH)
	state after	(2, 0, 1, 0, 3, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1817 *In the top of the 3rd inning, with bases empty, CHN batter Miles Mastrobuoni hit a ground ball single to shallow infield. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	CHN vs SDN on 20230604
	state before	(1, 0, 1, 0, 1, 0, 0, 0)
25.	error	play,1,0,happi001,22,BTFBX,S5/G56S-.2-H(E5/TH) (NR) (UR);B-2
	state after	(1, 0, 1, 0, 1, 1, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1818 This game had 2 errors. The final score was 7-1.

In the top of the 1st inning, with a runner on 2nd, CHN batter Ian Happ hit a soft ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.

	game	CHN vs SDN on 20230604
	state before	(0, 0, 0, 0, 3, 7, 0, 1)
26.	error	play,3,1,grist001,00,X,E4/G34D.B-1
	state after	(0, 1, 0, 0, 3, 7, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1819 *In the bottom of the 3rd inning, with bases empty, SDN batter Trent Grisham hit a ground ball to the deep 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	CHN vs SDN on 20230605
	state before	(1, 0, 1, 0, 2, 0, 2, 1)
27.	error	play,2,1,kim-h002,31,BBCCX,E5/G56+.2-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 2, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1820 *In the bottom of the 2nd inning, with a runner on 2nd, SDN batter Ha-Seong Kim hit a hard ground ball to 3rd base. An error by the 3rd baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	SEA vs SDN on 20230607
	state before	(2, 1, 0, 0, 4, 0, 4, 1)
28.	error	play,4,1,sotoj001,02,C1S1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1821 *In the bottom of the 4th inning, with a runner on 1st, SDN batter Juan Soto had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Soto is still at bat, now with a 1-2 count.*

	game	TBA vs SDN on 20230617
	state before	(2, 0, 0, 0, 3, 0, 0, 1)
29.	error	play,3,1,machm001,20,BBX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 3, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1822 *In the bottom of the 3rd inning, with bases empty, SDN batter Manny Machado hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for TBA vs SDN on 2023-06-18

position	innings	player	position	innings	player
1	1	Yonny Chirinos	1	1	Joe Musgrove
1	5	Kevin Kelly	1	7	Steven Wilson
1	7	Robert Stephenson	1	8	Nick Martinez
1	8	Jake Diekman	1	9	Josh Hader
2	1	Christian Bethancourt	2	1	Austin Nola
3	1	Yandy Diaz	2	9	Gary Sanchez
4	1	Taylor Walls	3	1	Jake Cronenworth
4	8	Vidal Brujan	4	1	Rougned Odor
5	1	Isaac Paredes	4	9	Ha-Seong Kim
6	1	Wander Franco	5	1	Manny Machado
7	1	Randy Arozarena	6	1	Xander Bogaerts
8	1	Manuel Margot	7	1	Juan Soto
9	1	Josh Lowe	8	1	Trent Grisham
			9	1	Fernando Tatis Jr.

	game	TBA vs SDN on 20230618
	state before	(1, 1, 0, 1, 3, 1, 0, 1)
30.	error	play,3,1,sotoj001,31,VBBCX,FC2/G2-.3-H(E6) (NR) (UR) ;1-H(E6/TH) (NR) (UR) ;B-3
	state after	(1, 1, 0, 1, 3, 1, 0, 1)
	fielder	6, 6 (shortstop on the visiting team)

Summary 1823 This game had 3 errors, 2 on one fielder. The final score was 4-5.

In the bottom of the 3rd inning, with runners on 1st and 3rd, SDN batter Juan Soto hit a soft ground ball into a fielder's choice at shallow infield. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI). A throwing error by the shortstop allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 3rd.

	game	TBA vs SDN on 20230618
	state before	(0, 0, 0, 0, 4, 1, 1, 0)
31.	error	play,4,0,ralel001,00,X,S1/BG25-.B-2(E1/TH)
	state after	(0, 0, 1, 0, 4, 1, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1824 *In the top of the 4th inning, with bases empty, TBA batter Luke Raley hit a soft bunt single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	WAS vs SDN on 20230625
	state before	(2, 0, 0, 0, 1, 1, 0, 1)
32.	error	play,1,1,sotoj001,10,BF,FLE7
	state after	(2, 0, 0, 0, 1, 1, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 1825 This game had 3 errors. The final score was 8-3.

In the bottom of the 1st inning, with 2 outs and bases empty, SDN batter Juan Soto had a 1-0 count. The next pitch he hit a fly ball foul of the left field line. An error by the left fielder in catching the ball² allowed the batter to remain safe. Soto is still at bat, now with a 1-1 count.

²See MLB Rule 9.04(a)(1).

	game	WAS vs SDN on 20230625
	state before	(1, 0, 0, 0, 7, 3, 1, 0)
33.	error	play,7,0,hilld002,10,BX,E1/TH/BG23.B-1
	state after	(1, 1, 0, 0, 7, 3, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1826 *In the top of the 7th inning, with bases empty, WAS batter Derek Hill hit a bunt to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

	game	WAS vs SDN on 20230625
	state before	(1, 0, 1, 1, 7, 4, 1, 0)
34.	error	play,7,0,garc1006,01,CX,E1/TH/G23-.B-1
	state after	(1, 1, 1, 1, 7, 4, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1827 *In the top of the 7th inning, with runners on 2nd and 3rd, WAS batter Luis Garcia hit a soft ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base. The bases are loaded.*

26.2.5 July

	game	ANA vs SDN on 20230704
	state before	(1, 1, 0, 0, 3, 0, 0, 0)
35.	error	play,3,0,velaa001,12,1CCBF,FLE1
	state after	(1, 1, 0, 0, 3, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1828 This game had 2 errors. The final score was 5-8. *In the top of the 3rd inning, with a runner on 1st, ANA batter Andrew Velazquez has a 1-2 count. The next pitch he hit a pop fly ball foul. An error by the pitcher in catching the ball allowed the batter to remain safe. Velazquez is still at bat, now with a 1-1 count.*

	game	ANA vs SDN on 20230704
	state before	(2, 0, 1, 0, 4, 0, 2, 1)
36.	error	play,4,1,sancg002,11,BCX,E5/TH/G56.2-3;B-1
	state after	(2, 1, 0, 1, 4, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1829 *In the bottom of the 4th inning, with a runner on 2nd, SDN batter Gary Sanchez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	ANA vs SDN on 20230705
	state before	(2, 1, 1, 1, 1, 0, 0, 1)
37.	error	play,1,1,cronj001,21,BCB2,P03(E1/TH) .3-H(NR) (UR)
	state after	(2, 1, 0, 0, 1, 0, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1830 This game had 2 errors. The final score was 3-5. *In the bottom of the 1st inning, with bases loaded, SDN batter Jake Cronenworth has a 2-1 count. The pitcher attempted to pick off the runner on 3rd. A throwing error by the pitcher allowed the runner on 3rd to score (unearned, no RBI).*

	game	ANA vs SDN on 20230705
	state before	(2, 1, 1, 0, 2, 0, 1, 0)
38.	error	play,2,0,rengl001,10,BX,S9/G34.2-H(E9) (NR) (UR);1-2
	state after	(2, 1, 1, 0, 2, 1, 1, 0)
	fielder	9 (right field on the home team)

Summary 1831 *In the top of the 2nd inning, with runners on 1st and 2nd, ANA batter Luis Rengifo hit a ground ball single to the 1st/2nd base gap. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to reach 2nd. The batter took 1st.*

	game	NYN vs SDN on 20230707
	state before	(2, 0, 1, 0, 1, 1, 1, 1)
39.	error	play,1,1,bogax001,32,BBCBSX,E5/G56.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 1, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1832 *In the bottom of the 1st inning, with a runner on 2nd, SDN batter Xander Bogaerts hit a ground ball to 3rd base. An error by the 3rd baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

Visiting/Home team rosters for NYN vs SDN on 2023-07-08

position	innings	player	position	innings	player
1	1	David Peterson	1	1	Blake Snell
1	6	Dominic Leone	1	7	Luis Garcia
1	7	Grant Hartwig	1	8	Nick Martinez
1	8	Trevor Gott	1	9	Josh Hader
2	1	Francisco Alvarez	2	1	Gary Sanchez
3	1	Pete Alonso	3	1	Jake Cronenworth
4	1	Jeff McNeil	4	1	Matthew Batten
5	1	Brett Baty	5	1	Manny Machado
6	1	Francisco Lindor	6	1	Xander Bogaerts
7	1	Tommy Pham	7	1	Juan Soto
8	1	Brandon Nimmo	8	1	Trent Grisham
9	1	Starling Marte	9	1	Fernando Tatis Jr.

	game	NYN vs SDN on 20230708
	state before	(2, 0, 0, 0, 5, 0, 3, 0)
40.	error	play,5,0,batyb001,22,CF.BBFX,C/E2.B-1
	state after	(2, 1, 0, 0, 5, 0, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1833 This game had 3 errors. The final score was 1-3.

In the top of the 5th inning, with bases empty, NYN batter Brett Baty had a 2-2 count. Catchers interference allowed the batter to reach 1st base.

	game	NYN vs SDN on 20230708
	state before	(0, 0, 0, 0, 6, 0, 3, 1)
41.	error	play,6,1,sancg002,11,CBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 6, 0, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1834 *In the bottom of the 6th inning, with bases empty, SDN batter Gary Sanchez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	NYN vs SDN on 20230708
	state before	(1, 0, 0, 0, 8, 1, 3, 1)
42.	error	play,8,1,bogax001,12,BCCX,E6/L6MD.B-1
	state after	(1, 1, 0, 0, 8, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1835 *In the bottom of the 8th inning, with bases empty, SDN batter Xander Bogaerts hit a line drive to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	NYN vs SDN on 20230709
	state before	(0, 0, 0, 0, 6, 0, 5, 1)
43.	error	play,6,1,sancg002,22,.BCSBF,FLE2
	state after	(0, 0, 0, 0, 6, 0, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1836 *In the bottom of the 6th inning, with bases empty, SDN batter Gary Sanchez had a 2-2 count. The next pitch he hit a pop fly ball foul. An error by the catcher in catching the ball allowed the batter to remain safe. Sanchez is still at bat with a 2-2 count.*

	game	PIT vs SDN on 20230726
	state before	(1, 0, 0, 0, 9, 3, 1, 0)
44.	error	play,9,0,davih001,01,SX,E4/P3D.B-2
	state after	(1, 0, 1, 0, 9, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1837 *In the top of the 9th inning, with bases empty, PIT batter Henry Davis hit a pop up to deep 1st base. An error by the second baseman allowed the batter to reach 2nd base.*

	game	TEX vs SDN on 20230730
	state before	(2, 0, 1, 0, 4, 0, 1, 0)
45.	error	play,4,0,tavel001,10,>B.X,S1/G2-.2-H(E1/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 4, 1, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1838 This game had 2 errors. The final score was 3-5.
In the top of the 4th inning, with a runner on 2nd, TEX batter Leody Taveras hit a soft ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.

	game	TEX vs SDN on 20230730
	state before	(1, 0, 1, 0, 8, 3, 4, 1)
46.	error	play,8,1,tatif002,10,BX,S7/L7L+.2-H;B-2(E7)
	state after	(1, 0, 1, 0, 8, 3, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 1839 *In the bottom of the 8th inning, with a runner on 2nd, SDN batter Fernando Tatis hit a hard line drive single to left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the batter to reach 2nd.*

26.2.6 August

Visiting/Home team rosters for LAN vs SDN on 2023-08-05

position	innings	player	position	innings	player
1	1	Michael Grove	1	1	Blake Snell
1	2	Ryan Yarbrough	1	6	Steven Wilson
1	6	Brusdar Graterol	1	7	Nick Martinez
1	8	Alex Vesia	1	9	Ray Kerr
1	8	Caleb Ferguson	2	1	Gary Sanchez
1	8	Yency Almonte	3	1	Jake Cronenworth
2	1	Will Smith	4	1	Ha-Seong Kim
3	1	Freddie Freeman	5	1	Manny Machado
4	1	Amed Rosario	6	1	Xander Bogaerts
4	8	Enrique Hernandez	7	1	Juan Soto
5	1	Chris Taylor	8	1	Trent Grisham
6	1	Miguel Rojas	9	1	Fernando Tatis Jr.
7	1	Enrique Hernandez			
7	8	David Peralta			
8	1	James Outman			
9	1	Mookie Betts			

	game	LAN vs SDN on 20230805
	state before	(1, 0, 0, 0, 5, 2, 1, 0)
47.	error	play,5,0,bettm001,22,CCBBX,D7/L7LS+.B-3(E7)
	state after	(1, 0, 0, 1, 5, 2, 1, 0)
	fielder	7 (left field on the home team)

Summary 1840 This game had 3 errors. The final score was 3-8. *In the top of the 5th inning, with bases empty, LAN batter Mookie Betts hit a hard line drive double to shallow left field. An error by the left fielder allowed the batter to reach 3rd.*

	game	LAN vs SDN on 20230805
	state before	(1, 1, 1, 1, 8, 3, 1, 1)
48.	error	play,8,1,sotoj001,20,. *BBX,S4/G34S-.3-H;2-H(E4/TH) (NR);1-3;B-2
	state after	(1, 0, 1, 1, 8, 3, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1841 *In the bottom of the 8th inning, with bases loaded, SDN batter Juan Soto hit a soft ground ball single to shallow infield. The runner on 3rd scored (1 RBI). An error by the second baseman allowed the runner on 2nd to score, the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	LAN vs SDN on 20230805
	state before	(2, 1, 0, 0, 8, 3, 5, 1)
49.	error	play,8,1,cronj001,12,CF.BF1,POCS2(E1/TH).1-3
	state after	(2, 0, 1, 0, 8, 3, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1842 *In the bottom of the 8th inning, with a runner on 1st, SDN batter Jake Cronenworth has a 1-2 count. The pitcher tried to pick off the runner from 1st as that runner tried to steal 2nd. A throwing error by the pitcher allowed the runner to reach 3rd. Cronenworth is still at bat with a 1-2 count.*

	game	LAN vs SDN on 20230807
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
50.	error	play,1,1,tatif002,12,SSBX,E5/G5.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1843 This game had 2 errors. The final score was 13-7. *In the bottom of the 1st inning, with bases empty, SDN batter Fernando Tatis hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

51.	game	LAN vs SDN on 20230807
	state before	(0, 1, 0, 0, 6, 9, 5, 0)
	error	play,6,0,rojam002,22,1CFBB>FFX,E4/G89D.1-3;B-2
	state after	(0, 0, 1, 1, 6, 9, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1844 *In the top of the 6th inning, with a runner on 1st, LAN batter Miguel Rojas hit a ground ball to deep right center field. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

52.	game	BAL vs SDN on 20230814
	state before	(1, 0, 0, 0, 2, 1, 0, 1)
	error	play,2,1,bogax001,32,CBF.BBFX,S7/G56.B-2(E7)
	state after	(1, 0, 1, 0, 2, 1, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 1845 *In the bottom of the 2nd inning, with bases empty, SDN batter Xander Bogaerts hit a ground ball single to short. An error by the left fielder allowed the batter to reach 2nd.*

53.	game	BAL vs SDN on 20230815
	state before	(0, 0, 0, 0, 7, 1, 10, 1)
	error	play,7,1,bogax001,00,...X,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 7, 1, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1846 *In the bottom of the 7th inning, with bases empty, SDN batter Xander Bogaerts hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

54.	game	BAL vs SDN on 20230816
	state before	(2, 1, 0, 0, 7, 2, 4, 1)
	error	play,7,1,sotoj001,00,.11,SB2.1-3(E1/TH)
	state after	(2, 0, 0, 1, 7, 2, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1847 *In the bottom of the 7th inning, with a runner on 1st, SDN batter Juan Soto had a 0-0 count. During the next pitch, which was a “no pitch,” the runner on 1st stole 2nd. A throwing error by the pitcher allowed that runner to advance to 3rd. Soto is still at bat, with a 0-0 count.*

55.	game	ARI vs SDN on 20230818
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
	error	play,1,1,kim-h002,12,CBC.FX,E6/TH/G56D.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1848 *In the bottom of the 1st inning, with bases empty, SDN batter Ha-Seong Kim hit a ground ball to deep 3rd base. A throwing error by the shortstop allowed the batter to reach 1st base.*

56.	game	ARI vs SDN on 20230819 (2/2)
	state before	(1, 0, 1, 1, 2, 0, 0, 0)
	error	play,2,0,moreg001,22,BFFFBX,63/G6S.3-H;2-H(E3/TH) (NR)
	state after	(2, 0, 1, 1, 2, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1849 This game, the 2nd of a doubleheader, had 2 errors. The final score was 8-1.

In the top of the 2nd inning, with runners on 2nd and 3rd, ARI batter Gabriel Moreno hit a ground ball to shallow short and was put out at 1st (with an assist by the shortstop). The runner on 3rd scored (1 RBI). A throwing error by the first baseman allowed the runner on 2nd to score (unearned, no RBI).

	game	ARI vs SDN on 20230819 (2/2)
	state before	(0, 0, 0, 0, 7, 2, 1, 0)
57.	error	play,7,0,phamt001,22,CBBSX,D7/L7LD.B-3(E7)
	state after	(0, 0, 0, 1, 7, 2, 1, 0)
	fielder	7 (left field on the home team)

Summary 1850 *In the top of the 7th inning, with bases empty, ARI batter Tommy Pham hit a line drive double to deep left field. An error by the left fielder allowed the batter to reach 3rd.*

Visiting/Home team rosters for MIA vs SDN on 2023-08-22

position	innings	player	position	innings	player
1	1	Jesus Luzardo	1	1	Blake Snell
1	7	Andrew Nardi	1	7	Scott Barlow
1	8	Tanner Scott	1	8	Nick Martinez
1	9	David Robertson	1	9	Luis Garcia
2	1	Nick Fortes	2	1	Gary Sanchez
3	1	Yuli Gurriel	3	1	Garrett Cooper
4	1	Luis Arraez	4	1	Jake Cronenworth
5	1	Jake Burger	5	1	Ha-Seong Kim
6	1	Jon Berti	6	1	Xander Bogaerts
6	7	Joey Wendle	7	1	Juan Soto
7	1	Avisail Garcia	8	1	Trent Grisham
7	3	Bryan De La Cruz	8	9	Jose Azocar
8	1	Jazz Chisholm Jr.	9	1	Fernando Tatis Jr.
9	1	Jorge Soler			
9	9	Jesus Sanchez			

	game	MIA vs SDN on 20230822
	state before	(2, 0, 0, 0, 2, 0, 0, 0)
58.	error	play,2,0,gurry001,11,CBX,S9/G3.B-2(E9/TH)
	state after	(2, 0, 1, 0, 2, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 1851 This game had 3 errors. The Miami Marlins shutout the Padres 3-0.

In the top of the 2nd inning, with bases empty, MIA batter Yulieski Gurriel hit a ground ball single to 1st. A throwing error by the right fielder allowed the batter to reach 2nd.

	game	MIA vs SDN on 20230822
	state before	(1, 1, 1, 0, 3, 1, 0, 0)
59.	error	play,3,0,burgj001,12,*SFBFX,S6/G6.2-H(E6/TH)(NR);1-3
	state after	(1, 1, 0, 1, 3, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1852 *In the top of the 3rd inning, with runners on 1st and 2nd, MIA batter Jake Burger hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to reach 3rd.*

	game	MIA vs SDN on 20230822
	state before	(2, 0, 0, 0, 8, 3, 0, 0)
60.	error	play,8,0,delab001,00,X,C/E2.B-1
	state after	(2, 1, 0, 0, 8, 3, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1853 *In the top of the 8th inning, with bases empty, MIA batter Bryan De La Cruz had a 0-0 count. Catcher's interference allowed the batter to reach 1st base.*

	game	MIA vs SDN on 20230823
	state before	(2, 1, 0, 0, 2, 0, 1, 0)
61.	error	play,2,0,sancj003,20,BBX,E3/G34.1-2;B-1
	state after	(2, 1, 1, 0, 2, 0, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1854 *In the top of the 2nd inning, with a runner on 1st, MIA batter Jesus Sanchez hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for SFN vs SDN on 2023-08-31

position	innings	player	position	innings	player
1	1	Jakob Junis	1	1	Pedro Avila
1	5	Sean Manaea	1	4	Matt Waldron
1	8	Ryan Walker	1	9	Ray Kerr
1	9	Scott Alexander	2	1	Gary Sanchez
2	1	Blake Sabol	3	1	Matthew Batten
3	1	LaMonte Wade Jr.	4	1	Ha-Seong Kim
4	1	Thairo Estrada	5	1	Manny Machado
5	1	J.D. Davis	6	1	Xander Bogaerts
6	1	Casey Schmitt	7	1	Juan Soto
7	1	Mitch Haniger	8	1	Trent Grisham
7	9	Wade Meckler	9	1	Fernando Tatis Jr.
8	1	Wade Meckler			
8	9	Austin Slater			
9	1	Mike Yastrzemski			
9	9	Mitch Haniger			

	game	SFN vs SDN on 20230831
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
62.	error	play,1,0,pedej001,12,CBFFFX,S5/G56.1-3(E5/TH);B-2
	state after	(2, 0, 1, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1855 This game had 3 errors. The Giants beat the Padres 7-0.

In the top of the 1st inning, with a runner on 1st, SFN batter Joc Pederson hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	SFN vs SDN on 20230831
	state before	(0, 0, 0, 0, 3, 0, 0, 0)
63.	error	play,3,0,wadel001,00,X,E3/G3.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1856 *In the top of the 3rd inning, with bases empty, SFN batter LaMonte Wade hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	SFN vs SDN on 20230831
	state before	(1, 1, 1, 0, 3, 2, 0, 0)
64.	error	play,3,0,davij006,00,X,E5/TH/G56S-.2-H(NR)(UR);1-3;B-1
	state after	(1, 1, 0, 1, 3, 3, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1857 *In the top of the 3rd inning, with runners on 1st and 2nd, SFN batter J.D. Davis hit a soft ground ball to shallow 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.*

26.2.7 September

	game	SFN vs SDN on 20230901
	state before	(2, 0, 0, 1, 6, 2, 6, 1)
65.	error	play,6,1,sotoj001,22,CB.CBFFX,E4/G34.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 6, 2, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1858 *In the bottom of the 6th inning, with a runner on 3rd, SDN batter Juan Soto hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	SFN vs SDN on 20230902
	state before	(0, 1, 0, 0, 8, 0, 6, 0)
66.	error	play,8,0,florw001,12,CF1FFFBX,FC6/G6.1-2(E6);B-1
	state after	(0, 1, 1, 0, 8, 0, 6, 0)
	fielder	6 (shortstop on the home team)

Summary 1859 *In the top of the 8th inning, with a runner on 1st, SFN batter Wilmer Flores hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	SFN vs SDN on 20230903
	state before	(1, 0, 1, 0, 1, 0, 3, 1)
67.	error	play,1,1,caml002,02,CTX,E3/G3.2-3;B-1
	state after	(1, 1, 0, 1, 1, 0, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1860 This game had 2 errors. The Giants lost to the Padres 0-4.

In the bottom of the 1st inning, with a runner on 2nd, SDN batter Luis

Campusano hit a ground ball to 1st base. An error by the first baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	SFN vs SDN on 20230903
	state before	(2, 1, 0, 0, 5, 0, 4, 1)
68.	error	play,5,1,sancg002,12,C1SB>B,SB2.1XH(2)(E2/TH)
	state after	(0, 0, 0, 0, 6, 0, 4, 0)
	fielder	2 (catcher on the visiting team)

Summary 1861 *In the bottom of the 5th inning, with a runner on 1st, SDN batter Gary Sanchez had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd but when the runner tried to score he was tagged out at home by the catcher. Gary Sanchez is still at bat, now with a 2-2 count.*

	game	PHI vs SDN on 20230904
	state before	(1, 0, 0, 0, 3, 8, 1, 1)
69.	error	play,3,1,kim-h002,32,BFBBCFFFX,E6/TH/G56.B-2
	state after	(1, 0, 1, 0, 3, 8, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1862 *In the bottom of the 3rd inning, with bases empty, SDN batter Ha-Seong Kim hit a ground ball to 3rd base. A throwing error by the shortstop allowed the batter to reach 2nd base.*

	game	PHI vs SDN on 20230905
	state before	(1, 0, 1, 1, 4, 0, 1, 1)
70.	error	play,4,1,caml002,00,X,FC4/G34.3-H;2-H(E4/TH)(NR)(UR);BX2(16)
	state after	(2, 0, 0, 0, 4, 0, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1863 *In the bottom of the 4th inning, with runners on 2nd and 3rd, SDN batter Luis Campusano hit a ground ball into a fielder's choice at the 1st/2nd base gap. The runner on 3rd scored (1 RBI). A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st. However, the batter (now on 1st) tried to advance to 2nd but was tagged out by the shortstop (assisted by the pitcher).*

71.	game	COL vs SDN on 20230918
	state before	(1, 1, 0, 0, 2, 0, 0, 0)
	error	play,2,0,diaze005,22,CBFBX,E6/G6-.1-2;B-1
	state after	(1, 1, 1, 0, 2, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 1864 This game had 3 errors. The Rockies lost to the Padres 9-11.

In the top of the 2nd inning, with a runner on 1st, COL batter Elias Diaz hit a soft ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

72.	game	COL vs SDN on 20230918
	state before	(2, 1, 1, 0, 5, 3, 8, 1)
	error	play,5,1,tatif002,32,2CFFBBB.>X,S5/G56.2-H(E5/TH)(NR);1-3
	state after	(2, 1, 0, 1, 5, 3, 9, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1865 *In the bottom of the 5th inning, with runners on 1st and 2nd, SDN batter Fernando Tatis hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to reach 3rd. The batter took 1st.*

	game	COL vs SDN on 20230918
	state before	(2, 1, 0, 1, 5, 3, 9, 1)
73.	error	play,5,1,sotoj001,22,1C*BBF>B,SB2;SBH.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 3, 9, 1)
	fielder	2 (catcher on the visiting team)

Summary 1866 This is a *consecutive error* in the sense that Soto was next in the batting lineup that game, behind Tatis who was at bat for the previous error.

In the bottom of the 5th inning, with runners on 1st and 3rd, SDN batter Juan Soto had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd and the runner on 3rd stole home. A throwing error by the catcher allowed the runner from 1st (now on 2nd) to advance to 3rd. Juan Soto is still at bat, now with a 3-2 count³.

	game	COL vs SDN on 20230920
	state before	(0, 1, 0, 0, 1, 2, 0, 1)
74.	error	play,1,1,tatif002,11,FB1>S,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 1, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1867 *In the bottom of the 1st inning, with a runner on 1st, SDN batter Fernando Tatis had a 1-1 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Tatis is still at bat, now with a 1-2 count.*

	game	SLN vs SDN on 20230924
	state before	(0, 0, 1, 0, 1, 0, 0, 1)
75.	error	play,1,1,kim-h002,22,MBBCX,E9/F9LD.2-3;B-2
	state after	(0, 0, 1, 1, 1, 0, 0, 1)
	fielder	9 (right field on the visiting team)

³Later in this same at-bat, Soto hit a single that brought the runner, now on 3rd, into score (no RBI).

Summary 1868 *In the bottom of the 1st inning, with a runner on 2nd, SDN batter Ha-Seong Kim hit a fly ball to deep right field. An error by the right fielder allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd base.*

	game	SLN vs SDN on 20230924
	state before	(2, 0, 0, 1, 2, 0, 4, 1)
76.	error	play,2,1,kim-h002,22,BBCFFX,E5/TH/G5.3-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 2, 0, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1869 *In the bottom of the 2nd inning, with a runner on 3rd, SDN batter Ha-Seong Kim hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter reach 2nd.*

Chapter 27

San Francisco Giants (SFN)

Since their San Francisco origin in 1958, the Giants made nine playoff appearances and won three NL pennants. They won the World Series in 2010, 2012, and 2014 – three championships in five years.

This chapter is devoted to listing the errors in each regular season game of 2023 played in San Francisco against the Giants. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 SFN team is over one standard deviation¹ below the mean: 0.679 compared with the MLB average of 0.691 that year [BR].

The total number of errors by the Giants (home and away) over the regular season in 2023 was 117, the highest in the MLB. and well above the MLB average of 83.9. That year, there were 90 errors in the 81 home games at SFN 2023, 55 by the Giants and the rest (35) by visiting teams.

27.1 Season totals

...the Official Scorer shall charge an infielder with an error when a ground ball passes to either side of such infielder if, in

¹The standard deviation of the 30 DefEff values from the 2023 MLB teams is 0.011.

the Official Scorer's judgment, a fielder at that position making ordinary effort would have fielded such ground ball and retired a runner.

– MLB Rule 9.12(a)(1) Comment

Let **vt** denote the visiting team and **ht** the home team.

27.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals	RBI	no RBI
Errors by vt :	35	-	-
Runs by ht because of errors by vt :	9	4	5
Errors by ht :	55	-	-
Runs by vt because of errors by ht :	17	7	10

This is a good error-to-run conversion rate: $9/35 = 25.7\%$ on the part of the Cubs (compared to $17/55 = 30.9\%$ on the part of the visiting teams, which is bad for the Giants!). A typical *play* (not necessarily with an error) in the 2023 SFN season (home and away) has about $R/PA = 674/6095$ (about 11%) change of resulting in a run. Therefore, for the Giants, an error is over twice as likely to score a run than a randomly selected play.

From the table above we can see that the 2023 Giant's error rate in a *home* games was $53/81 = 0.65$. As we remarked above, **teamrankings.com** gave the error rate for all games (home and away) played by the Giants in their regular season was about 72%.

Note that the total number of errors 90 is about the number of games played over the season, so the expected number of errors (by both sides) in a randomly selected home game is about 1.11.

There were about 0.18 opponent runs per error (by the Giants) and 0.1 Giants runs per per error (by the visiting team). In other words, the visiting teams were about 1.8 times better at converting errors to runs than the Giants.

Errors by position	vt	ht
errors due to pitcher:	8	8
to catcher:	5	12
to 1st baseman:	2	7
to 2nd baseman:	5	2
to 3rd baseman:	5	9
to shortstop:	3	7
to left fielder:	0	5
to center fielder:	5	2
to right fielder:	2	1
subtotals:	35	55

27.1.2 Error total by state

Any runner is **out** when he runs more than three feet away from his base path to avoid being tagged unless his action is to avoid interference with a fielder fielding a batted ball. A runner's base path is established when the tag attempt occurs and is a straight line from the runner to the base he is attempting to reach safely.

– **MLB Rule Book** 5.09(b)(1)

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	11	18	29
(lone) runner on 1st	12	19	31
(lone) runner on 2nd	3	3	6
(lone) runner on 3rd	0	0	0
two runners on base	8	13	21
bases loaded	1	2	3
	35	55	90

Of the 90 errors (by both teams), 9 were fielder's choices, 3 were from stolen bases, and 6 were from attempted pick offs (usually to 1st base).

We tabulate the states that occurred more than once in the list of errors given in the next section. This can be compared with the frequency table of game states for all plays (not just those with an error), as discussed in §2.1.4. For the Giants' errors in the home games, the number of those with bases empty is about the same as the number with a lone runner on 1st.

A table of frequencies of errors by the Giants in SFN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	14.5%
(1, 0, 0, 0)	5	9.1%
(2, 0, 0, 0)	5	9.1%
(0, 1, 0, 0)	4	7.3%
(1, 1, 0, 0)	7	12.7%
(2, 1, 0, 0)	8	14.5%
(2, 0, 1, 0)	3	5.4%
(1, 1, 1, 0)	2	3.3%

A table of frequencies of most common errors by a typical visiting team in SFN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	3	$4/35 = 8.6\%$
(1, 0, 0, 0)	5	14.3%
(2, 0, 0, 0)	4	11.4%
(0, 1, 0, 0)	3	8.6%
(1, 1, 0, 0)	6	17.1%
(2, 1, 0, 0)	3	8.6%
(0, 1, 1, 0)	3	8.6%
(1, 1, 1, 0)	2	5.7%
(2, 1, 1, 0)	2	5.7%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Giants homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `SFN2023.evn`):

state	SFN2023 percentages
(0, 0, 0, 0)	25.0%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.6%
(0, 1, 0, 0)	5.8%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	1.8%
(1, 0, 1, 0)	2.6%
(2, 0, 1, 0)	4.0%
(0, 0, 0, 1)	0.3%
(1, 0, 0, 1)	0.6%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.3%
(2, 1, 1, 0)	3.1%

27.1.3 Some interesting errors

There were several interesting errors committed that season. There were two errors in each of the following plays:

- Against the Los Angeles Dodgers on 2023-04-10 (the Giants lost 9-1):
play,3,0,freef001,01,F1X,E1/G4S.1-3(E6/TH);B-1
- Against the St Louis Cardinals on 2023-04-25 (the Giants won 4-5):
play,2,1,sabob001,11,SB>B,SB2.1-H(E2/TH)(E8)(NR)
- Against the Chicago Cubs on 2023-06-11 (the Giants won 3-13):
play,2,0,taucm001,32,CBBBF>X,E6/G6.3-H(NR)(UR);1-3(E6/TH);B-2
That is not a typo: the shortstop was indeed credited with 2 errors on the same play.

These are given in more detail below.

27.1.4 Relative team rankings

We have 81 home games between SFN and some visiting MLB team (vt). A number of teams did not play SFN in the regular season. However, of those that did, SFN *lost* more games than won against²:

ATL (2/3), BAL (2/3), CHN (2/3), KCA (2/3), PIT (2/3), SEA (2/3),
TBA (2/3), TEX (2/3), WAS (2/3), LAN (4/6)

and *won* more than lost against

OAK (0/2), PHI (0/3), BOS (1/3), CIN (1/3), CLE (1/3), MIA (1/3),
MIL (1/3), SLN (1/4), COL (1/6), ARI (2/7), SDN (3/7) .

The teams that *split* their series were:

NYN (2/4) .

The teams that did not play in any of the 81 home games in SFN 2023 were:

ANA, CHA, DET, HOU, MIN, NYA, TOR .

27.2 Game-by-game totals

To save space, we only give the team rosters in special cases.

The typical starting lineup for the 2023 San Francisco Giants included the following players (according to [BA], [BR]):

- Patrick Bailey (Catcher, OBP .285)
- LaMonte Wade Jr. (First Base, OBP .373)
- Thairo Estrada (Second Base, OBP .315)
- J.D. Davis (Third Base, OBP .325)
- Brandon Crawford (Shortstop, OBP .273)
- Michael Conforto (Right Field, OBP .334)
- Mike Yastrzemski (Center Field, OBP .330)
- Mitch Haniger (Left Field, OBP .266).

²The vt's W/T record in parentheses (W = number of (home) games vt won against the Giants, T = Total home games vt vs SFN played at San Fransisco).

27.2.1 April

	game	KCA vs SFN on 20230407
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
1.	error	play,1,0,massm001,12,CBFX,E3/G9L.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1870 *In the top of the 1st inning, with bases empty, KCA batter Michael Massey hit a ground ball to right field and reached 1st base on an error by the first baseman.*

	game	KCA vs SFN on 20230408
	state before	(2, 0, 0, 0, 5, 0, 4, 1)
2.	error	play,5,1,confm001,01,FX,S4/G4.B-2(E4/TH)
	state after	(2, 0, 1, 0, 5, 0, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1871 *In the bottom of the 5th inning, with bases empty, SFN batter Michael Conforto hit a ground ball single to 2nd. A throwing error by the second baseman allowed the batter to advance to 2nd base.*

	game	KCA vs SFN on 20230409
	state before	(0, 0, 0, 0, 9, 1, 3, 0)
3.	error	play,9,0,pasqv001,02,..CSX,C/E2.B-1
	state after	(0, 1, 0, 0, 9, 1, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1872 *In the top of the 9th inning, with bases empty, KCA batter Vinnie Pasquantino reached 1st on catcher's interference.*

	game	LAN vs SFN on 20230410
	state before	(1, 1, 0, 0, 3, 1, 0, 0)
4.	error	play,3,0,freef001,01,F1X,E1/G4S.1-3(E6/TH);B-1
	state after	(1, 1, 0, 1, 3, 1, 0, 0)
	fielder	1, 6 (pitcher/shortstop on the home team)

Summary 1873 This play had 2 errors. The Dodgers beats the Giants 9-1.

In the top of the 3rd inning, with a runner on 1st, LAN batter Freddie Freeman hit a ground ball short of 2nd base. An error by the pitcher allowed the batter to reach 1st. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd. (Video shows that the ball bounced off the glove of the pitcher, and then was fielded by the shortstop who made a throwing error to 1st.)

	game	LAN vs SFN on 20230411
	state before	(0, 1, 1, 0, 6, 0, 2, 0)
5.	error	play,6,0,vargm001,00,X,C/E2.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 6, 0, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1874 *In the top of the 6th inning, with runners on 1st and 2nd, LAN batter Miguel Vargas reached first on catcher's interference. The runners on 1st and 2nd advanced to 2nd and 3rd, respectively.*

	game	LAN vs SFN on 20230412
	state before	(1, 0, 0, 0, 2, 0, 2, 0)
6.	error	play,2,0,vargm001,32,CBBBCX,S1/G25.B-2(E1/TH)
	state after	(1, 0, 1, 0, 2, 0, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1875 *In the top of the 2nd inning, with bases empty, LAN batter Miguel Vargas (same as last play, lucky guy!) hit a ground ball single to the pitcher. A throwing error by the pitcher allowed the batter to advance to 2nd.*

	game	LAN vs SFN on 20230412
	state before	(1, 1, 0, 0, 2, 0, 2, 1)
7.	error	play,2,1,ramoh002,12,CSBX,FC4/G4MD.1-2(E4);B-1
	state after	(1, 1, 1, 0, 2, 0, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1876 *In the bottom of the 2nd inning, with a runner on 1st, SFN batter Heliot Ramos hit a ground ball to a fielder's choice at deep 2nd base. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	NYN vs SFN on 20230421
	state before	(2, 0, 1, 0, 5, 1, 0, 0)
8.	error	play,5,0,mcnej002,01,FX,S2/G2.2-H(E2/TH)(NR)
	state after	(2, 1, 0, 0, 5, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1877 *In the top of the 5th inning, with a runner on 2nd, NYN batter Jeff McNeil hit a ground ball single to the catcher. A throwing error by the catcher allowed the runner on 2nd to score (unearned, no RBI). The batter was safe at 1st.*

	game	NYN vs SFN on 20230422
	state before	(1, 0, 0, 0, 4, 2, 6, 1)
9.	error	play,4,1,estrt001,10,BX,E5/G7S.B-2
	state after	(1, 0, 1, 0, 4, 2, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1878 *In the bottom of the 4th inning, with bases empty, SFN batter Thairo Estrada hit a ground ball to 3rd and reached 2nd base on an error by the third baseman.*

10.	game	NYN vs SFN on 20230422
	state before	(0, 1, 0, 0, 8, 2, 7, 0)
	error	play,8,0,nidot001,11,BFX,S1/BG56S.1-3(E1/TH)
	state after	(0, 1, 0, 1, 8, 2, 7, 0)
	fielder	1 (pitcher on the home team)

Summary 1879 *In the top of the 8th inning, with a runner on 1st, NYN batter Tomas Nido hit a bunt single to the pitcher. An error by the pitcher allowed the batter to reach 1st and the runner on 1st to advance to 3rd.*

Visiting/Home team rosters for NYN vs SFN on 2023-04-23

position	innings	player	position	innings	player
1	1	Tylor Megill	1	1	Ross Stripling
1	5	Jeff Brigham	1	4	Taylor Rogers
1	7	Brooks Raley	1	5	Tyler Rogers
1	8	Drew Smith	1	7	John Brebbia
2	1	Francisco Alvarez	1	8	Scott Alexander
3	1	Pete Alonso	1	9	Camilo Doval
4	1	Luis Guillorme	2	1	Blake Sabol
4	4	Jeff McNeil	2	8	Joey Bart
5	1	Brett Baty	3	1	LaMonte Wade Jr.
6	1	Francisco Lindor	3	8	Wilmer Flores
7	1	Jeff McNeil	4	1	Thairo Estrada
7	4	Mark Canha	5	1	J.D. Davis
8	1	Brandon Nimmo	6	1	Brandon Crawford
9	1	Starling Marte	7	1	Mike Yastrzemski
			7	8	LaMonte Wade Jr.
			8	1	Brett Wisely
			8	8	Mike Yastrzemski
			9	1	Michael Conforto

	game	NYN vs SFN on 20230423
	state before	(1, 1, 1, 0, 2, 0, 1, 0)
11.	error	play,2,0,batyb001,11,CBX,E1/G23-.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 2, 0, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 1880 *This game had 3 errors. The Giants beats the Met 4-5.*

In the top of the 2nd inning, with runners on 1st and 2nd, NYN batter Brett Baty hit a soft infield ground ball towards 1st base. The batter reached 1st on an error by the pitcher, loading the bases.

	game	NYN vs SFN on 20230423
	state before	(1, 1, 1, 1, 4, 1, 2, 0)
12.	error	play,4,0,canhm001,12,..CFBFX,E9/F9L/SF.3-H;2-3;1-2;B-1
	state after	(1, 1, 1, 1, 4, 2, 2, 0)
	fielder	9 (right field on the home team)

Summary 1881 *In the top of the 4th inning, with the bases loaded, NYN batter Mark Canha hit a sacrafic fly ball to right field. An error by the right fielder allowed the runner on 3rd to score, the runners on 1st and 2nd to advance one base each, and the batter to safely reach 1st. The bases are still loaded.*

	game	NYN vs SFN on 20230423
	state before	(2, 1, 0, 0, 4, 3, 4, 1)
13.	error	play,4,1,wadel001,11,FB1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 4, 3, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1882 *In the bottom of the 4th inning, with a runner on 1st, SFN batter LaMonte Wade had a 1-1 count. A throwing error by the pitcher in a pick off attempt allowed the runner on 1st to advance to 2nd. Wade is at bat, still with a 1-1 count.*

	game	SLN vs SFN on 20230424
	state before	(0, 0, 0, 0, 7, 0, 0, 1)
14.	error	play,7,1,yastm001,20,BBX,E4/G34.B-1
	state after	(0, 1, 0, 0, 7, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1883 *In the bottom of the 7th inning, with bases empty, SFN batter Mike Yastrzemski hit a ground ball between 1st and 2nd. An error by the second baseman allowed the batter to reach 1st.*

	game	SLN vs SFN on 20230425
	state before	(1, 1, 0, 0, 2, 0, 0, 1)
15.	error	play,2,1,sabob001,11,SB>B,SB2.1-H(E2/TH) (E8) (NR)
	state after	(1, 0, 0, 0, 2, 0, 1, 1)
	fielder	2, 8 (catcher, center fielder on the visiting team)

Summary 1884 *This game had 3 errors and 2 errors in this one play. In the bottom of the 2nd inning, with a runner on 1st, SFN batter Blake Sabol had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. An error by the center fielder and a throwing error by the catcher allowed the runner on 1st to score (unearned, no RBI). Now the bases are empty. Sabol is still at bat, now with a 2-1 count.*

	game	SLN vs SFN on 20230425
	state before	(0, 0, 0, 0, 4, 1, 1, 0)
16.	error	play,4,0,arenn001,22,CCB.BFF,FLE3
	state after	(0, 0, 0, 0, 4, 1, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1885 *In the top of the 4th inning, with bases empty and a 2-2 count, SLN batter Nolan Arenado hit a foul fly ball that was dropped by the first baseman. Arenado was still at bat with a 2-2 count. (This “∞ play” is relatively rare, as it results in no change at all to the game state.*

	game	SLN vs SFN on 20230426
	state before	(0, 0, 0, 0, 2, 1, 0, 0)
17.	error	play,2,0,burla001,00,X,C/E2.B-1
	state after	(0, 1, 0, 0, 2, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1886 *In the top of the 2nd inning, with bases empty, SLN batter Alec Burleson reached 1st on catcher's interference.*

	game	SLN vs SFN on 20230426
	state before	(2, 1, 0, 0, 5, 2, 2, 1)
18.	error	play,5,1,florw001,21,BBC1X,E8/F89XD.1-3;B-1
	state after	(2, 1, 0, 1, 5, 2, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 1887 *In the bottom of the 5th inning, with a runner on 1st, SFN batter Wilmer Flores hit a fly ball to deep right-center field. An error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

27.2.2 May

Visiting/Home team rosters for MIL vs SFN on 2023-05-05

position	innings	player	position	innings	player
1	1	Corbin Burnes	1	1	Sean Manaea
1	7	Bryse Wilson	1	6	John Brebbia
1	8	Elvis Peguero	1	7	Tyler Rogers
1	8	Hoby Milner	1	9	Camilo Doval
2	1	Victor Caratini	2	1	Joey Bart
3	1	Luke Voit	3	1	LaMonte Wade Jr.
4	1	Owen Miller	4	1	Brett Wisely
5	1	Brian Anderson	5	1	J.D. Davis
6	1	Willy Adames	6	1	Thairo Estrada
7	1	Christian Yelich	7	1	Mitch Haniger
8	1	Joey Wiemer	8	1	Cal Stevenson
9	1	Tyrone Taylor	9	1	Michael Conforto

	game	MIL vs SFN on 20230505
	state before	(1, 1, 0, 0, 2, 1, 2, 0)
19.	error	play,2,0,taylt002,01,CX,FC6/G6S.1-2(E6);B-1
	state after	(1, 1, 1, 0, 2, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1888 *This game had 3 errors. The Brewers lost to the Giants 4-6.*

In the top of the 2nd inning, with a runner on 1st and 2nd, MIL batter Tyrone Taylor hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	MIL vs SFN on 20230505
	state before	(1, 0, 1, 1, 5, 1, 2, 0)
20.	error	play,5,0,millo002,12,BFS+3,0A.3-H(E2/TH)(NR);2-3
	state after	(1, 0, 0, 1, 5, 2, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1889 *In the top of the 5th inning, with runners on 2nd and 3rd, MIL batter Owen Miller had a 1-2 count. A throwing error by the catcher on the throw to third (presumably a pick off attempt) allowed the runner on 3rd to score (no RBI) and the runner on 2nd to advance to 3rd. Miller is still at bat with a 1-2 count.*

	game	MIL vs SFN on 20230505
	state before	(1, 0, 1, 0, 5, 3, 2, 1)
21.	error	play,5,1,stevc001,32,1*BC1FB>B.X,E6/TH/G6.2-3;B-2
	state after	(1, 0, 1, 1, 5, 3, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1890 *In the bottom of the 5th inning, with a runner on 2nd, SFN batter Cal Stevenson hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to safely reach 2nd.*

	game	MIL vs SFN on 20230507
	state before	(0, 0, 0, 0, 9, 5, 2, 0)
22.	error	play,9,0,wiemj001,01,..CX,E6/G56D.B-1
	state after	(0, 1, 0, 0, 9, 5, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1891 *In the top of the 9th inning, with bases empty, MIL batter Joey Wiemer hit a ground ball to deep 3rd base. The batter reached 1st on an error by the shortstop.*

	game	WAS vs SFN on 20230509
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
23.	error	play,1,1,slata001,11,BCX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1892 *In the bottom of the 1st inning, with bases empty, SFN batter Austin Slater hit a ground ball to shortstop. The batter reached 1st on an error by the shortstop.*

	game	WAS vs SFN on 20230509
	state before	(0, 1, 0, 0, 2, 0, 2, 0)
24.	error	play,2,0,calla002,00,X,S5/BG25.1-3(E3);B-2
	state after	(0, 0, 1, 1, 2, 0, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 1893 *In the top of the 2nd inning, with a runner on 1st, WAS batter Alex Call hit a bunt single to 3rd base. An error by the first baseman allowed the batter to reach 2nd and the runner on 1st to advance to 3rd.*

	game	WAS vs SFN on 20230510
	state before	(1, 1, 0, 0, 3, 4, 0, 0)
25.	error	play,3,0,candj002,02,CSX,E5/G6.1-2;B-1
	state after	(1, 1, 1, 0, 3, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1894 *In the top of the 3rd inning, with a runner on 1st, WAS batter Jeimer Candelario hit a ground ball to shortstop. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to safely reach 1st.*

	game	PHI vs SFN on 20230515
	state before	(1, 1, 1, 0, 2, 2, 0, 1)
26.	error	play,2,1,estr001,10,BX,FC6/G6.2-3;1X2(6E4);B-1
	state after	(1, 1, 1, 1, 2, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1895 *In the bottom of the 2nd inning, with runners on 1st and 2nd, SFN batter Thairo Estrada hit a ground ball to a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the second baseman allowed the runner on 1st to safely reach 2nd and the batter to reach 1st.*

	game	PHI vs SFN on 20230516
	state before	(1, 1, 0, 0, 3, 0, 0, 1)
27.	error	play,3,1,wade1001,00,X,S8/L8D+.1-3;B-2(E8/TH)
	state after	(1, 0, 1, 1, 3, 0, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1896 *In the bottom of the 3rd inning, with a runner on 1st, SFN batter LaMonte Wade hit a hard line drive single to center field. The runner on 1st to advance to 3rd. A throwing error by the center fielder allowed the batter to advance to 2nd.*

	game	PHI vs SFN on 20230517
	state before	(1, 0, 0, 0, 3, 0, 4, 1)
28.	error	play,3,1,sabob001,00,X,S1/BG2-.B-2(E1/TH)
	state after	(1, 0, 1, 0, 3, 0, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1897 *In the bottom of the 3rd inning, with bases empty, SFN batter Blake Sabol hit a soft infield ground ball. The batter reached 2nd base on a throwing error by the pitcher.*

29.	game	MIA vs SFN on 20230519
	state before	(0, 1, 0, 0, 6, 2, 1, 1)
	error	play,6,1,estrt001,00,X,S1/G2.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 6, 2, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1898 *In the bottom of the 6th inning, with a runner on 1st, SFN batter Thairo Estrada hit an infield ground ball. The batter reached 2nd base and the runner on 1st advanced to 3rd on a throwing error by the pitcher.*

Visiting/Home team rosters for MIA vs SFN on 2023-05-20

position	innings	player	position	innings	player
1	1	Braxton Garrett	1	1	Logan Webb
1	7	JT Chargois	1	7	Jakob Junis
1	8	Dylan Floro	1	8	Tyler Rogers
1	8	Tanner Scott	1	9	Camilo Doval
2	1	Nick Fortes	2	1	Patrick Bailey
3	1	Garrett Cooper	3	1	LaMonte Wade Jr.
4	1	Luis Arraez	4	1	Thairo Estrada
5	1	Jon Berti	5	1	J.D. Davis
6	1	Joey Wendle	6	1	Casey Schmitt
7	1	Bryan De La Cruz	7	1	Mitch Haniger
8	1	Xavier Edwards	8	1	Bryce Johnson
8	8	Garrett Hampson	9	1	Michael Conforto
9	1	Garrett Hampson			
9	8	Peyton Burdick			

	game	MIA vs SFN on 20230520
	state before	(1, 0, 0, 0, 7, 0, 0, 1)
30.	error	play,7,1,davij006,02,.CCX,C/E2.B-1
	state after	(1, 1, 0, 0, 7, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1899 *This game had 3 errors. The Giants lost to the Marlins 1-0.*

In the bottom of the 7th inning, with bases empty, SFN batter Jody Davis reached 1st on catcher's interference.

	game	MIA vs SFN on 20230520
	state before	(0, 0, 0, 0, 8, 0, 0, 0)
31.	error	play,8,0,hampg001,31,.BBBCX,E5/G56.B-1
	state after	(0, 1, 0, 0, 8, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1900 *In the top of the 8th inning, with bases empty, SFN batter Garrett Hampson hit a ground ball to 3rd. The batter reached 1st on an error by the third baseman.*

	game	MIA vs SFN on 20230520
	state before	(0, 0, 0, 0, 9, 1, 0, 0)
32.	error	play,9,0,delab001,32,.FBBC.FBX,6E3/G6.B-1
	state after	(0, 1, 0, 0, 9, 1, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1901 *In the top of the 9th inning, with bases empty, MIA batter Bryan De La Cruz hit a ground ball to shortstop and reached 1st base on an error by the first baseman.*

	game	MIA vs SFN on 20230521
	state before	(2, 0, 1, 0, 9, 4, 7, 0)
33.	error	play,9,0,bertj001,21,>B.S.BX,S2/G2.2-H(E2/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 9, 5, 7, 0)
	fielder	2 (catcher on the home team)

Summary 1902 *In the top of the 9th inning, with a runner on 2nd, MIA batter Jon Berti hit an infield ground ball single. A throwing error by the catcher allowed the runner on 2nd to score (unearned, no RBI). The batter is safe at 1st.*

	game	PIT vs SFN on 20230529
	state before	(0, 0, 1, 1, 2, 0, 0, 1)
34.	error	play,2,1,bailp001,01,CX,D9/L9L.3-H;2-H;B-3(E9)
	state after	(0, 0, 0, 1, 2, 0, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 1903 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, SFN batter Patrick Bailey hit a line drive double to right field. The runners on 2nd and 3rd scored (2 RBIs). An error by the right fielder allowed the batter to advance to 3rd.*

	game	PIT vs SFN on 20230529
	state before	(0, 1, 0, 0, 8, 3, 14, 1)
35.	error	play,8,1,villd001,00,X,E5/G5.1-2;B-1
	state after	(0, 1, 1, 0, 8, 3, 14, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1904 *In the bottom of the 8th inning, with a runner on 1st, SFN batter David Villar hit a ground ball to 3rd. The runner on 1st advanced to 2nd and the batter reached 1st on an error by the third baseman.*

	game	PIT vs SFN on 20230530
	state before	(1, 1, 0, 0, 5, 1, 1, 0)
36.	error	play,5,0,delaj004,01,TX,S7/L78D.1-3(E7);B-2
	state after	(1, 0, 1, 1, 5, 1, 1, 0)
	fielder	7 (left field on the home team)

Summary 1905 *In the top of the 5th inning, with a runner on 1st, PIT batter Jason Delay hit a line drive single to deep left field. An error by the left fielder allowed the batter to advance to 2nd and the runner on 1st to advance to 3rd.*

	game	PIT vs SFN on 20230531
	state before	(2, 1, 1, 0, 3, 1, 1, 0)
37.	error	play,3,0,castr006,11,S.BX,S7/G56.2-H;1-3(E7/TH);B-2
	state after	(2, 0, 1, 1, 3, 2, 1, 0)
	fielder	7 (left field on the home team)

Summary 1906 *This game had 3 errors. The Pirates beat the Giants 9-4.*

In the top of the 3rd inning, with runners on 1st and 2nd, PIT batter Rodolfo Castro hit a ground ball single to left field. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to advance to 2nd.

	game	PIT vs SFN on 20230531
	state before	(2, 1, 0, 0, 4, 6, 2, 1)
38.	error	play,4,1,wisec001,00,B,WP.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 6, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 1907 *In the bottom of the 4th inning, with a runner on first, SFN batter Brett Wisely has a 0-0 count. The next pitch, which was a ball ruled a wild pitch (which is not counted as an error), allowed the runner on 1st to advance to 3rd due to a throwing error by the catcher. Wisely is still at bat, now with a 1-0 count.*

39.	game	PIT vs SFN on 20230531
	state before	(1, 1, 0, 1, 6, 6, 3, 0)
	error	play,6,0,reynb001,01,SX,54(1)/F0/G56.3-H;B-2(E4/TH)
	state after	(2, 0, 0, 1, 6, 6, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1908 *In the top of the 6th inning, with runners on 1st and 3rd, PIT batter Reynolds hit a ground ball to 3rd base, which forced out the runner from 1st. The runner from 3rd scored (1 RBI). A throwing error by the second baseman on the throw allowed the batter to reach 2nd.*

27.2.3 June

40.	game	BAL vs SFN on 20230602
	state before	(1, 0, 0, 0, 5, 2, 2, 0)
	error	play,5,0,matej003,00,X,C/E2.B-1
	state after	(1, 1, 0, 0, 5, 2, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1909 *In the top of the 5th inning, with bases empty, BAL batter Jorge Mateo reached 1st on catcher's interference.*

41.	game	BAL vs SFN on 20230603
	state before	(1, 1, 0, 0, 5, 0, 3, 1)
	error	play,5,1,sabob001,11,LBX,FC1/G1S.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 5, 0, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1910 *In the bottom of the 5th inning, with a runner on 1st, SFN batter Blake Sabol hit a soft ground ball to a fielder's choice in shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach first.*

	game	BAL vs SFN on 20230604
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
42.	error	play,2,0,hicka001,32,BFCFBBX,S5/G56.B-2(E5/TH)
	state after	(1, 0, 1, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1911 *In the top of the 2nd inning, with bases empty, BAL batter Aaron Hicks hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to advance to 2nd.*

	game	BAL vs SFN on 20230604
	state before	(2, 1, 1, 1, 3, 3, 0, 0)
43.	error	play,3,0,lestj002,32,CBCFFFB*B>X,S8/L89XD+.3-H;2-H;1-H(E8/TH)(NR)(UR)
	state after	(2, 1, 0, 0, 3, 6, 0, 0)
	fielder	8 (center field on the home team)

Summary 1912 *In the top of the 3rd inning, with the bases loaded, BAL batter Josh Lester hit a line drive single to deep center field. The runners on 2nd and 3rd both scored (2 RBIs). A throwing error by the center fielder allowed the runner on 1st to score (unearned, no RBI). The batter safely reached 1st.*

Visiting/Home team rosters for CHN vs SFN on 2023-06-11

position	innings	player	position	innings	player
1	1	Hayden Wesneski	1	1	John Brebbia
1	4	Michael Fulmer	1	2	Tristan Beck
1	5	Brandon Hughes	1	4	Ryan Walker
1	5	Michael Rucker	1	7	Luke Jackson
1	6	Jeremiah Estrada	1	8	Taylor Rogers
1	7	Javier Assad	1	9	Brandon Crawford
2	1	Yan Gomes	2	1	Blake Sabol
2	7	Miguel Amaya	3	1	Wilmer Flores
3	1	Matt Mervis	4	1	Thairo Estrada
4	1	Nico Hoerner	5	1	J.D. Davis
4	7	Christopher Morel	6	1	Casey Schmitt
5	1	Nick Madrigal	7	1	Mitch Haniger
6	1	Dansby Swanson	8	1	Mike Yastrzemski
6	7	Nico Hoerner	9	1	Michael Conforto
7	1	Ian Happ			
8	1	Mike Tauchman			
9	1	Christopher Morel			
9	7	Trey Mancini			

44.	game	CHN vs SFN on 20230611
	state before	(2, 1, 0, 1, 2, 0, 2, 0)
	error	play,2,0,taucm001,32,CBBBF>X,E6/G6.3-H(NR)(UR);1-3(E6/TH);B-2
	state after	(2, 0, 1, 1, 2, 1, 2, 0)
	fielder	6 (shortstop on the home team, twice)

Summary 1913 *This game had 4 errors, all by the Giants, and 2 errors were in this one play by the same fielder³.*

³According to [BR], the SS Casey Schmitt was indeed credited with 2 errors. The

In the top of the 2nd inning, with runners on 1st and 3rd, CHN batter Mike Tauchman hit a ground ball to the shortstop. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI). Furthermore, a throwing error by the shortstop allowed the runner from 1st to advance to 3rd and the batter to advance to 2nd.

	game	CHN vs SFN on 20230611
	state before	(0, 0, 0, 0, 6, 2, 8, 0)
45.	error	play,6,0,mervm001,12,BCS.X,3E1/G34.B-1
	state after	(0, 1, 0, 0, 6, 2, 8, 0)
	fielder	1 (pitcher on the home team)

Summary 1914 *In the top of the 6th inning, with bases empty, CHN batter Matt Mervis hit a ground ball toward 1st and reached 1st base safely on an error by the pitcher.*

	game	CHN vs SFN on 20230611
	state before	(1, 1, 1, 0, 6, 2, 8, 0)
46.	error	play,6,0,taucm001,22,BFCBX,E7/F7L.2-3;1-2;B-1
	state after	(1, 1, 1, 1, 6, 2, 8, 0)
	fielder	7 (left field on the home team)

Summary 1915 *In the top of the 6th inning, with runners on 1st and 2nd, CHN batter Mike Tauchman hit a fly ball to left field. An error by the left fielder allowed the batter to reach 1st and the runners to advance one base each. The bases are now loaded.*

	game	SDN vs SFN on 20230619
	state before	(0, 0, 0, 0, 3, 1, 0, 0)
47.	error	play,3,0,tatif002,00,X,E6/G78S.B-1
	state after	(0, 1, 0, 0, 3, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Giants beats the Cubs 3-13.

Summary 1916 *In the top of the 3rd inning, with bases empty, SDN batter Fernando Tatis hit a ground ball to short left field. The batter reached 1st base due to an error by the shortstop.*

	game	SDN vs SFN on 20230620
	state before	(2, 1, 0, 0, 9, 3, 3, 0)
48.	error	play,9,0,machm001,12,SBSFFFFX,E6/TH/G6.1-2;B-1
	state after	(2, 1, 1, 0, 9, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1917 *In the top of the 9th inning, with a runner on 1st, SDN batter Manny Machado hit a ground ball to shortstop. The runner on 1st reaches 2nd and the batter advanced to 1st base due to a throwing error by the shortstop.*

	game	ARI vs SFN on 20230623
	state before	(2, 0, 1, 0, 4, 1, 3, 0)
49.	error	play,4,0,thoma004,01,CX,E2/TH/G2-.2-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 2, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1918 *In the top of the 4th inning, with a runner on 2nd, ARI batter Alek Thomas hit an infield ground ball. A throwing error by the catcher allowed the runner on second to score (unearned, no RBI) and the batter to reach 2nd.*

	game	ARI vs SFN on 20230624
	state before	(0, 0, 1, 0, 1, 1, 0, 1)
50.	error	play,1,1,estrt001,10,BX,E8/F89.2-3;B-1
	state after	(0, 1, 0, 1, 1, 1, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1919 *In the bottom of the 1st inning, with a runner on 2nd, SFN batter Thairo Estrada hit a fly ball to right center field. The runner on 2nd advanced to 3rd and the batter reached 1st on an error by the center fielder.*

27.2.4 July

51.	game	SEA vs SFN on 20230704
	state before	(0, 0, 1, 1, 2, 1, 0, 0)
	error	play,2,0,murpt002,10,BX,C/E2.B-1
	state after	(0, 1, 1, 1, 2, 1, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1920 *In the top of the 2nd inning, with runners on 2nd and 3rd, SEA batter Tom Murphy reached 1st on catcher's interference.*

52.	game	SEA vs SFN on 20230705
	state before	(1, 1, 1, 0, 5, 0, 1, 1)
	error	play,5,1,wadel001,02,FFX,E8/F8XD.2-H(NR)(UR);1-3;B-1
	state after	(1, 1, 0, 1, 5, 0, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 1921 *In the bottom of the 5th inning, with runners on 1st and 2nd, SFN batter LaMonte Wade hit a fly ball to deep center field. An error by the center fielder allowed the runner on 2nd to score (un-earned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

53.	game	OAK vs SFN on 20230725
	state before	(0, 1, 1, 0, 5, 0, 0, 1)
	error	play,5,1,wiseb001,00,X,FC1/BG1S-.2-3(E1/TH);1-2;B-1
	state after	(0, 1, 1, 1, 5, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1922 *In the bottom of the 5th inning, with runners on 1st and 2nd, SFN batter Brett Wisely hit a bunt ground ball to a fielder's choice at shallow infield. A throwing error by the pitcher allowed the batter to reach 1st and the runners on 1st and 2nd to advance to 2nd and 3rd, respectively.*

	game	OAK vs SFN on 20230726
	state before	(2, 1, 1, 0, 4, 0, 3, 0)
54.	error	play,4,0,petej002,01,.CX,S8/L8S.2-H;1-H(E8/TH)(NR);B-3
	state after	(2, 0, 0, 1, 4, 2, 3, 0)
	fielder	8 (center field on the home team)

Summary 1923 *In the top of the 4th inning, with runners on 1st and 2nd, OAK batter Jace Peterson hit a line drive single to short center field. The runner on 2nd scored (1 RBI). A throwing error by the center fielder allowed the batter to advance to 3rd and the runner on 1st to score (unearned, no RBI).*

	game	OAK vs SFN on 20230726
	state before	(1, 1, 0, 0, 6, 3, 5, 1)
55.	error	play,6,1,bailp001,22,CCFB*B1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 6, 3, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1924 *In the bottom of the 6th inning, with a runner on 1st, SFN batter Patrick Bailey had a 2-2 count. A throwing error by the pitcher in a pick off attempt allowed the runner on on 1st to advance to 2nd. Bailey is at bat, still with a 2-2 count.*

	game	BOS vs SFN on 20230730
	state before	(0, 1, 0, 0, 5, 0, 1, 1)
56.	error	play,5,1,crawb001,21,BBFX,FC5/G56.1-3(E5);B-1
	state after	(0, 1, 0, 1, 5, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1925 *In the bottom of the 5th inning, with a runner on 1st, SFN batter Brandon Crawford hit a ground ball to a fielder's choice at 3rd base. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

27.2.5 August

	game	ARI vs SFN on 20230802
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
57.	error	play,1,0,martk001,32,*BBF1C.F*B>C,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1926 *In the top of the 1st inning, with a runner on 1st, ARI batter Ketel Marte had a full count. During the next pitch, which was a called strike (so Marte struck out), the runner on 1st attempted to steal 2nd. A throwing error by the catcher allowed the runner to advance to 3rd.*

	game	ARI vs SFN on 20230802
	state before	(1, 0, 0, 0, 7, 2, 4, 1)
58.	error	play,7,1,slata001,21,CBBX,E4/TH/G2.B-1
	state after	(1, 1, 0, 0, 7, 2, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1927 *In the bottom of the 7th inning, with bases empty, SFN batter Austin Slater hit a ground ball chopper to the second baseman. The batter reached 1st on a throwing error by the second baseman.*

	game	ARI vs SFN on 20230803
	state before	(2, 1, 0, 0, 6, 0, 1, 0)
59.	error	play,6,0,rivee001,00,.1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 1928 *In the top of the 1st inning, with a runner on 1st, ARI batter Emmanuel Riva had a 0-0 count. During the next pitch, which was a ball, the runner on 1st attempted to steal 2nd. A throwing error by the catcher allowed the runner to advance to 3rd. Rivera is still at bat, now with a 1-0 count.*

	game	TEX vs SFN on 20230811
	state before	(0, 0, 0, 0, 9, 2, 0, 1)
60.	error	play,9,1,ramoh002,22,..CFBBX,D8/L8XD+.B-3(E8/TH)
	state after	(0, 0, 0, 1, 9, 2, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 1929 *In the bottom of the 9th inning, with bases empty, SFN batter Heliot Ramos hit a line drive double to deep center field. A throwing error by the center fielder allowed the batter to advance to 3rd.*

	game	TEX vs SFN on 20230812
	state before	(0, 1, 0, 0, 9, 7, 2, 0)
61.	error	play,9,0,lowen001,00,X,S1/G1S-.1-3(E1/TH);B-2
	state after	(0, 0, 1, 1, 9, 7, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 1930 *In the top of the 9th inning, with a runner on 1st, TEX batter Lowe hit a soft ground ball single to the pitcher. A throwing error by the pitcher allowed the batter to advance to 2nd and the runner on 1st to advance to 3rd.*

	game	TEX vs SFN on 20230813
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
62.	error	play,2,0,tavel001,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1931 *In the top of the 1st inning, with a runner on 1st, ARI batter Leody Tavaras Riva had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st attempted to steal 2nd. A throwing error by the catcher allowed the runner to advance to 3rd. Rivera is still at bat, now with a 0-1 count.*

63.	game	TBA vs SFN on 20230814
	state before	(2, 1, 0, 0, 5, 3, 0, 0)
	error	play,5,0,meadc001,01,CX,S1/G13.1-3(E1/TH)
	state after	(2, 1, 0, 1, 5, 3, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 1932 *In the top of the 5th inning, with a runner on 1st, TBA batter Curtis Mead hit a ground ball single to the pitcher. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd. The batter is on 1st base.*

64.	game	TBA vs SFN on 20230814
	state before	(2, 0, 0, 0, 7, 6, 1, 1)
	error	play,7,1,crawb001,00,X,E3/G9S.B-1
	state after	(2, 1, 0, 0, 7, 6, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1933 *In the bottom of the 7th inning, with bases empty, SFN batter Brandon Crawford hit a ground ball to short right field. An error by the first baseman allowed the batter to reach 1st.*

Visiting/Home team rosters for TBA vs SFN on 2023-08-15

position	innings	player	position	innings	player
1	1	Zack Littell	1	1	Jakob Junis
1	6	Kevin Kelly	1	5	Sean Manaea
1	8	Erasmo Ramirez	1	8	Tyler Rogers
2	1	Rene Pinto	1	9	Luke Jackson
3	1	Yandy Diaz	2	1	Patrick Bailey
4	1	Brandon Lowe	3	1	LaMonte Wade Jr.
5	1	Isaac Paredes	3	8	Wilmer Flores
6	1	Osleivis Basabe	4	1	Thairo Estrada
7	1	Randy Arozarena	5	1	Wilmer Flores
8	1	Jose Siri	5	8	Johan Camargo
9	1	Josh Lowe	6	1	Brandon Crawford
			7	1	Blake Sabol
			7	8	Heliot Ramos
			7	9	Austin Slater
			8	1	Wade Meckler
			9	1	Michael Conforto
			9	9	Heliot Ramos

	game	TBA vs SFN on 20230815
	state before	(2, 1, 0, 0, 3, 0, 0, 0)
65.	error	play,3,0,parei001,22,SFB1BFX,E5/G56.1-2;B-1
	state after	(2, 1, 1, 0, 3, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1934 *This game had 3 errors. The Giants shut out the Tampa Bay Rays 0-7.*

In the top of the 3rd inning, with a runner on 1st, TBA batter Isaac Paredes hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to reach 2nd and the batter advanced to 1st.

	game	TBA vs SFN on 20230815
	state before	(0, 1, 1, 0, 7, 0, 3, 1)
66.	error	play,7,1,sabob001,00,X,E3/G2.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 7, 0, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1935 *In the bottom of the 7th inning, with runners on 1st and 2nd, SFN batter Blake Sabol hit an infield ground ball. An error by the first baseman allowed the runner on 2nd to advance to 3rd, the runner on 1st to advance to 2nd, and the batter to safely reach 1st. The bases are now loaded with no outs.*

	game	TBA vs SFN on 20230815
	state before	(2, 1, 1, 1, 7, 0, 3, 1)
67.	error	play,7,1,wadel001,01,C+3,P03(E2/TH).3-H(NR)(UR);2-H(NR)(UR);1-2
	state after	(2, 0, 1, 0, 7, 0, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 1936 *In the bottom of the 7th inning, with bases loaded, SFN batter LaMonte Wade had a 0-1 count. A throwing error by the catcher in a pick off attempt allowed the runner on 3rd to score (unearned, no RBI), the runner 2nd to score (unearned, no RBI), and the runner on 1st to advance to 2nd. Wade is at bat, still with a 0-1 count.*

	game	TBA vs SFN on 20230816
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
68.	error	play,1,0,arozr001,00,X,FC5/G5.1-2(E5);B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1937 *In the bottom of the 1st inning, with a runner on 1st, TBA batter Randy Arozarena hit a ground ball to a fielder's choice at 3rd. The runner on 1st advanced to 2nd and the batter reached 1st on an error by the third baseman.*

	game	TBA vs SFN on 20230816
	state before	(1, 0, 1, 0, 5, 4, 0, 0)
69.	error	play,5,0,parei001,01,CX,S7/L7.2-H(E7)(NR);B-2
	state after	(1, 0, 1, 0, 5, 5, 0, 0)
	fielder	7 (left field on the home team)

Summary 1938 *In the top of the 5th inning, with a runner on 2nd, TBA batter Paredes hit a line drive single to left field. An error by the left fielder allowed the batter to advance to 2nd and the runner on 2nd to score (no RBI).*

	game	CIN vs SFN on 20230828
	state before	(2, 1, 0, 0, 8, 0, 4, 0)
70.	error	play,8,0,delae003,10,BX,D8/G6+.1-H;B-3(E6/TH)
	state after	(2, 0, 0, 1, 8, 1, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 1939 *In the top of the 8th inning, with a runner on first, CIN batter Elly De La Cruz hit a ground ball double to center field. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to advance to 3rd.*

	game	CIN vs SFN on 20230829
	state before	(2, 0, 0, 0, 3, 0, 2, 0)
71.	error	play,3,0,senzn001,11,SBX,E5/TH/G5.B-1
	state after	(2, 1, 0, 0, 3, 0, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1940 *In the top of the 3rd inning, with bases empty, CIN batter Nick Senzel hit a ground ball to 3rd. The batter reached 1st on a throwing error by the third baseman.*

	game	CIN vs SFN on 20230829
	state before	(2, 0, 0, 0, 3, 0, 3, 1)
72.	error	play,3,1,hanim001,22,CBFB.FX,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 3, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1941 *In the bottom of the 3rd inning, with bases empty, SFN batter Mitch Haniger hit a ground ball to 3rd. The batter reached 1st on an error by the third baseman.*

	game	CIN vs SFN on 20230830
	state before	(2, 1, 0, 0, 2, 0, 0, 0)
73.	error	play,2,0,bensw001,12,CFB1,SB2.1-3(E3/TH)
	state after	(2, 0, 0, 1, 2, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 1942 *This game had 3 errors. The Giants lost to the Reds 4-1.*

In the top of the 2nd inning, with a runner on 1st, CIN batter Will Benson had a count of 1-2. During a pick off attempt by the pitcher, the runner stole 2nd base. A throwing error by the first baseman allowed the runner from 1st to advance to 3rd base.

	game	CIN vs SFN on 20230830
	state before	(1, 1, 0, 1, 6, 2, 0, 1)
74.	error	play,6,1,florw001,00,.1,P01(E1/TH).3-H(NR)(UR);1-2
	state after	(1, 0, 1, 0, 6, 2, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1943 *In the bottom of the 6th inning, with a runner on 1st, SFN batter Wilmer Flores had a 0-0 count. A throwing error by the pitcher in a pick off attempt allowed the runner on 3rd to score (unearned, no RBI) and the runner on 1st to advance to 2nd. Flores is at bat, still with a 0-0 count.*

	game	CIN vs SFN on 20230830
	state before	(2, 0, 0, 0, 7, 2, 1, 0)
75.	error	play,7,0,maill001,00,X,S7/L78.B-2(E7)
	state after	(2, 0, 1, 0, 7, 2, 1, 0)
	fielder	7 (left field on the home team)

Summary 1944 *In the top of the 7th inning, with bases empty, CIN batter Luke Maile hit a line drive single to left field. An error by the left fielder allowed the batter to advance to 2nd.*

	game	CIN vs SFN on 20230830
	state before	(1, 1, 0, 0, 7, 2, 1, 1)
76.	error	play,7,1,meckw001,22,CCBBX,FC1/G1S-.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 7, 2, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1945 *In the bottom of the 7th inning, with a runner on 1st, SFN batter Wade Meckler hit a soft ground ball to a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

27.2.6 September

	game	COL vs SFN on 20230908
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
77.	error	play,2,0,rodgb002,11,SBX,E5/G56.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1946 *In the top of the 2nd inning, with bases empty, COL batter Brendan Rodgers hit a ground ball to 3rd. The batter reached 1st on an error by the third baseman.*

Visiting/Home team rosters for CLE vs SFN on 2023-09-11

position	innings	player	position	innings	player
1	1	Gavin Williams	1	1	Alex Cobb
1	10	Emmanuel Clase	1	10	Luke Jackson
1	7	Reynaldo Lopez	1	6	Taylor Rogers
1	7	Sam Hentges	1	7	John Brebbia
1	8	Enyel De Los Santos	1	7	Tyler Rogers
1	8	Matt Moore	1	8	Camilo Doval
2	1	Bo Naylor	2	1	Joey Bart
3	1	Josh Naylor	2	8	Blake Sabol
4	1	Andres Gimenez	3	1	LaMonte Wade Jr.
5	1	Jose Ramirez	4	1	Thairo Estrada
6	1	Jose Tena	5	1	J.D. Davis
7	1	Steven Kwan	5	8	Wilmer Flores
8	1	Myles Straw	6	1	Brandon Crawford
9	1	Will Brennan	7	1	Mitch Haniger
			8	1	Luis Matos
			8	8	Austin Slater
			9	1	Mike Yastrzemski
			9	8	Luis Matos

	game	CLE vs SFN on 20230911
	state before	(2, 0, 0, 0, 3, 0, 1, 0)
78.	error	play,3,0,ramij003,21,CBBX,E6/G4MD.B-1
	state after	(2, 1, 0, 0, 3, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 1947 This game had 3 errors. The Cleveland Guardians lost to the Giants 4-5.

In the top of the 3rd inning, with bases empty, CLE batter Jose Ramirez

hit a ground ball to the middle of the infield. The batter reached 1st base due to an error by the shortstop.

	game	CLE vs SFN on 20230911
	state before	(2, 1, 0, 0, 4, 2, 3, 0)
79.	error	play,4,0,stram002,00,11,P01(E3).1-2
	state after	(2, 0, 1, 0, 4, 2, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 1948 *In the top of the 4th inning, with a runner on 1st, CLE batter Miles Straw had a 1-1 count. During a pick off attempt, an error by the first baseman allowed the runner on 1st to advance to 2nd. Straw is at bat, still with a 1-1 count.*

	game	CLE vs SFN on 20230911
	state before	(1, 1, 0, 0, 8, 3, 3, 0)
80.	error	play,8,0,gimea001,10,.BX,S1/G15.1-3(E1/TH)
	state after	(1, 1, 0, 1, 8, 3, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 1949 *In the top of the 8th inning, with a runner on 1st, CLE batter Andres Gimenez hit a ground ball single to the pitcher. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd. The batter is safe at 1st.*

	game	CLE vs SFN on 20230912
	state before	(2, 1, 1, 0, 6, 1, 1, 0)
81.	error	play,6,0,calhk001,10,BX,E3/G34.2-3;1-2;B-1
	state after	(2, 1, 1, 1, 6, 1, 1, 0)
	fielder	3 (1st baseman on the home team)

Summary 1950 *In the top of the 6th inning, with runners on 1st and 2nd, CLE batter Kole Calhoun hit a ground ball to 1st. An error by the first baseman allowed the runner on 2nd to reach 3rd, the runner on 1st to advance to 2nd, and the batter to safely reach 1st. The bases are now full.*

82.	game	CLE vs SFN on 20230913
	state before	(2, 0, 1, 1, 1, 2, 0, 0)
	error	play,1,0,ariag002,22,C1B>S.BFX,E5/TH/G5.3-H(NR)(UR);2-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 1, 4, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1951 *In the top of the 1st inning, with runners on 2nd and 3rd, CLE batter Gabriel Arias hit a ground ball to 3rd. A throwing error by the third baseman allowed the runners on 2nd and 3rd to score (no RBIs), and the batter to reach 2nd.*

83.	game	CLE vs SFN on 20230913
	state before	(2, 1, 1, 0, 1, 4, 0, 1)
	error	play,1,1,matol002,10,BX,S9/L9S.2-H;1-3(E9);B-2
	state after	(2, 0, 1, 1, 1, 4, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 1952 *In the bottom of the 1st inning, with runners on 1st and 2nd, SFN batter Luis Matos hit a line drive single to short right field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the batter to advance to 2nd and the runner on 1st to advance to 3rd.*

84.	game	SDN vs SFN on 20230926
	state before	(1, 1, 0, 0, 3, 1, 0, 0)
	error	play,3,0,bogax001,10,BX,S4/G4S.1-3(E4/TH)
	state after	(1, 1, 0, 1, 3, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1953 *In the top of the 3rd inning, with a runner on 1st, SDN batter Xander Bogaerts hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd. The batter is safe at 1st.*

	game	SDN vs SFN on 20230926
	state before	(1, 0, 0, 0, 3, 2, 0, 1)
85.	error	play,3,1,sabob001,11,BFX,E2/TH/G2-.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1954 *In the bottom of the 3rd inning, with bases empty, SFN batter Blake Sabol hit a soft infield ground ball. The batter reached 1st on a throwing error by the catcher.*

	game	SDN vs SFN on 20230927
	state before	(2, 0, 0, 0, 3, 0, 1, 0)
86.	error	play,3,0,bogax001,01,CX,S5/G56.B-2(E5/TH)
	state after	(2, 0, 1, 0, 3, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1955 *This game had 3 errors. The Padres beat the Giants 5-2.*

In the top of the 3rd inning, with bases empty, SDN batter Xander Bogaerts hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to advance to 2nd.

	game	SDN vs SFN on 20230927
	state before	(2, 0, 0, 0, 10, 3, 2, 0)
87.	error	play,10,0,tatif002,12,BSFFFFX,E6/TH/G6.B-2
	state after	(2, 0, 1, 0, 10, 3, 2, 0)
	fielder	6 (3rd baseman on the home team)

Summary 1956 *In the top of the 10th inning, with bases empty, SDN batter Fernando Tatis hit a ground ball to shortstop. The batter reached 2nd base due to an error by the shortstop.*

	game	SDN vs SFN on 20230927
	state before	(2, 1, 1, 0, 10, 3, 2, 0)
88.	error	play,10,0,machm001,22,CFFB*B+2,P02(E2/TH).2-3;1-2
	state after	(2, 0, 1, 1, 10, 3, 2, 0)
	fielder	2 (3rd baseman on the home team)

Summary 1957 *In the top of the 4th inning, with runners on 1st and 2nd, SDN batter Manny Machado had a 1-0 count. During a pick off attempt, a throwing error by the catchern allowed the runner on 1st to advance to 2nd and the runner on 2nd to advance to 3rd. Machado is at bat, still with a 1-0 count.*

	game	LAN vs SFN on 20230930
	state before	(1, 1, 0, 1, 6, 1, 1, 1)
89.	error	play,6,1,florw001,02,SFX,FC5/G56.3-H;1-2(E5);B-1
	state after	(1, 1, 1, 0, 6, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1958 *In the bottom of the 6th inning, with runners on 1st and 3rd, SFN batter Wilmer Flores hit a ground ball to a fielder's choice at 3rd base. The runner on 3rd scored (1 RBI). An error by the third baseman allowed the batter to reach 1st and the runner on 1st to advance to 2nd.*

	game	LAN vs SFN on 20231001
	state before	(2, 0, 0, 0, 6, 5, 1, 1)
90.	error	play,6,1,estrt001,32,CBBBFFX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 6, 5, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1959 *In the bottom of the 6th inning, with bases empty, SFN batter Thairo Estrada hit a ground ball to shortstop. A throwing error by the shortstop allowed the batter to safely reach 1st.*

Chapter 28

Seattle Mariners (SEA)

Founded in 1977, the Seattle Mariners compete in the American League West Division. They have never won a World Series title or an AL Pennant, but they have won 3 division titles. Their last ALWD title was won in 2001.

This chapter is devoted to listing the errors in each regular season game of 2023 played at Seattle against the Mariners. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The DER of the 2023 SEA team is .700 compared with the MLB average of 0.691 that year [BR]. By the way, the standard deviation of these DER values in 2023 is $\sigma = 0.011$, so the defensive efficiency ratio of the SEA is well above the MLB average (which is good).

The total number of errors by the Mariners (home and away) over the regular season of 2023 is 73, while 83.9 was the MLB average ([BR]), well below average for errors (which is good).

As we will see below, there were 90 errors in the 81 SEA 2023 home games, 36 by Seattle and the rest (44) by visiting teams.

28.1 Season totals

Throughout, `vt` denotes the visiting team and `ht` denotes the home team.

28.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

Runs and errors	totals
Errors by vt:	44
Runs by ht because of errors by vt:	8
Errors by ht:	36
Runs by vt because of errors by ht:	6

This is a fair error-to-run conversion rate: $8/44 = 18.2\%$ on the part of the Mariners (compared to the even lower rate $6/36 = 16.7\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 SEA season (home and away) has about $R/PA = 758/6201$ (about 12.2%) change of resulting in a run. Therefore, a randomly selected error is (for SEA) slightly more likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 80, so the expected number of errors (by either team) in a randomly selected SEA 2023 home game is about 1.

Errors by position	vt	ht
errors due to pitcher:	2	4
to catcher:	4	7
to 1st baseman:	7	1
to 2nd baseman:	7	4
to 3rd baseman:	7	4
to shortstop:	6	8
to left fielder:	5	2
to center fielder:	0	0
to right fielder:	3	3

28.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	16	12	28
(lone) runner on 1st	24	13	37
(lone) runner on 2nd	6	4	10
(lone) runner on 3rd	1	1	2
two runners on base	5	5	10
bases loaded	2	1	3
total	54	36	90

A table of frequencies of most common errors by the SEA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	4	11.1%
(1, 0, 0, 0)	4	11.1%
(2, 0, 0, 0)	4	11.1%
(0, 1, 0, 0)	3	8.3%
(1, 1, 0, 0)	4	11.1%
(2, 1, 0, 0)	6	16.7%
(1, 0, 1, 0)	3	8.3%
(2, 1, 1, 0)	2	5.6%
(0, 1, 1, 1)	1	2.8%

A table of frequencies of most common errors by a visiting team in SEA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	15.9%
(1, 0, 0, 0)	5	11.4%
(2, 0, 0, 0)	4	9.1%
(0, 1, 0, 0)	4	9.1%
(1, 1, 0, 0)	7	15.9%
(2, 1, 0, 0)	3	6.8%
(0, 1, 1, 0)	2	4.6%
(2, 1, 1, 0)	1	2.3%
(1, 1, 1, 1)	1	2.3%
(2, 1, 1, 1)	1	2.3%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Mariners homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `SEA2023.evn`):

state	SEA2023 percentages
(0, 0, 0, 0)	24.4%
(1, 0, 0, 0)	16.7%
(2, 0, 0, 0)	12.9%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	6.9%
(2, 1, 0, 0)	6.6%
(0, 0, 1, 0)	2.1%
(1, 0, 1, 0)	3.2%
(2, 0, 1, 0)	3.7%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.8%
(2, 0, 0, 1)	1.5%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	2.5%
(2, 1, 1, 0)	2.8%

28.1.3 Some interesting errors

Some of the most interesting error plays in the SEA 2023 home games:

- The game CLE vs SEA on 2023-03-31 had 3 errors, but in only 2 plays.
- The game MIA vs SEA on 2023-06-12 had 3 errors. This was the start of a three game series Marlins vs Mariners. Each game in this Marlins-Mariners series had at least 2 errors, for a total of 8.
- The game ANA vs SEA on 2023-04-04 had 3 errors, including an ∞ play.

28.1.4 Relative team rankings

We have 81 home games between SEA (the ht) and some visiting MLB team (vt). A number of teams did not play SEA in the regular season. However, of those that did, SEA lost more games than won against¹:

BAL (2/3), DET (2/3), NYA (2/3), WAS (2/3)
LAN (3/3), MIL (3/3)

and won more than lost against

SDN (0/2), COL (0/3), KCA (0/3), BOS (1/3), CHA (1/3), MIA (1/3),
PIT (1/3), SLN (1/3), TBA (1/3), TOR (1/3), OAK (1/7), TEX (3/7) .

The teams that split their series were:

MIN (2/4), ANA (3/6), HOU (3/6).

The teams that did not play in any of the 81 home games in SEA 2023 were:

ARI, ATL, CHN, CIN, NYN, PHI, SFN.

28.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases. A typical roster for SEA 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Cal Raleigh	128	.306
3 (1B)	Ty France	158	.337
4 (2B)	Kolten Wong	65	.241
5 (3B)	Eugenio Suárez	159	.323
6 (SS)	J.P. Crawford	144	.380
7 (LF)	Jarred Kelenic	79	.327
8 (CF)	Julio Rodríguez	152	.333
9 (RF)	Teoscar Hernández	135	.305

¹They are written with that team's W/T record in parentheses (W = numbers of games vt won against SEA, T = Total home games vt vs Mariners played at Seattle).

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 SEA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

28.2.1 March

Visiting/Home team rosters for CLE vs SEA on 2023-03-31

position	innings	player	position	innings	player
1	1	Hunter Gaddis	1	1	Robbie Ray
1	4	Nick Sandlin	1	4	Trevor Gott
1	6	Eli Morgan	1	5	Penn Murfee
1	7	Enyel De Los Santos	1	6	Chris Flexen
1	8	Trevor Stephan	2	1	Cal Raleigh
1	9	Emmanuel Clase	3	1	Ty France
2	1	Mike Zunino	4	1	Kolten Wong
3	1	Josh Bell	5	1	Eugenio Suarez
3	8	Gabriel Arias	6	1	J.P. Crawford
4	1	Andres Gimenez	7	1	Jarred Kelenic
5	1	Jose Ramirez	8	1	Julio Rodriguez
6	1	Amed Rosario	9	1	Teoscar Hernandez
7	1	Steven Kwan			
8	1	Myles Straw			
9	1	Oscar Gonzalez			

	game	CLE vs SEA on 20230331
	state before	(0, 1, 1, 1, 2, 0, 1, 0)
1.	error	play,2,0,stram002,01,FX,9/F9LS.3-H(E9/TH) (NR) ; 2-H(E1/TH) (NR) (UR) ; 1
	state after	(1, 0, 0, 1, 2, 2, 1, 0)
	fielder	9,1 (right field,pitcher on the home team)

Summary 1960 This game had 3 errors, but in only 2 plays. The final score was 7-4.

In the top of the 2nd inning, with bases loaded, CLE batter Myles Straw hit a fly ball to shallow right field. The right fielder put out the batter. A throwing error by the right fielder allowed the runner on 3rd to score (unearned, no RBI). A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI). These errors also allowed the runner on 1st to advance to 3rd.

	game	CLE vs SEA on 20230331
	state before	(1, 1, 0, 0, 5, 5, 4, 0)
2.	error	play,5,0,gimea001,01,CX,FC6/G6.1X2(6E4);B-1
	state after	(1, 1, 1, 0, 5, 5, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 1961 *In the top of the 5th inning, with a runner on 1st, CLE batter Andres Gimenez hit a ground ball into a fielder's choice at short. An error by the second baseman (with an assist from the shortstop) allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

28.2.2 April

	game	CLE vs SEA on 20230401
	state before	(0, 0, 0, 0, 6, 1, 0, 1)
3.	error	play,6,1,crawj002,11,CBX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 6, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1962 *In the bottom of the 6th inning, with bases empty, SEA batter J.P. Crawford hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CLE vs SEA on 20230402
	state before	(1, 0, 0, 1, 5, 4, 4, 1)
4.	error	play,5,1,ralec001,32,BFFB.B.X,E3/G3.3-H(NR) (UR) ;B-1
	state after	(1, 1, 0, 0, 5, 4, 5, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1963 This game had 2 errors. The final score was 6-5 (in extra innings).

In the bottom of the 5th inning, with a runner on 3rd, SEA batter Cal Raleigh hit a ground ball to 1st base. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.

	game	CLE vs SEA on 20230402
	state before	(1, 1, 1, 1, 10, 5, 5, 0)
5.	error	play,10,0,naylj001,01,..FX,12(3)/FO/G1.2-H(E2/TH) (NR) (UR) ;1-3;B-1
	state after	(2, 1, 0, 1, 10, 6, 5, 0)
	fielder	2 (1st baseman on the visiting team)

Summary 1964 *In the top of the 10th inning, with bases loaded, CLE batter Josh Naylor hit a ground ball to shallow center infield. At home, the catcher (with an assist from the pitcher) forced out the runner from 3rd. A throwing error by the catcher allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.*

	game	ANA vs SEA on 20230403
	state before	(0, 0, 0, 0, 3, 1, 1, 1)
6.	error	play,3,1,rodrj007,01,SX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 3, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1965 *In the bottom of the 3rd inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for ANA vs SEA on 2023-04-04

position	innings	player	position	innings	player
1	1	Jose Suarez	1	1	Luis Castillo
1	5	Jaime Barria	1	6	Matt Brash
2	1	Matt Thaiss	1	7	Trevor Gott
3	1	Jake Lamb	1	8	Paul Sewald
4	1	Brandon Drury	1	9	Gabe Speier
5	1	Luis Rengifo	2	1	Cal Raleigh
6	1	David Fletcher	3	1	Ty France
7	1	Taylor Ward	4	1	Sam Haggerty
8	1	Mike Trout	5	1	Eugenio Suarez
9	1	Hunter Renfroe	6	1	J.P. Crawford
			7	1	AJ Pollock
			8	1	Julio Rodriguez
			9	1	Teoscar Hernandez

	game	ANA vs SEA on 20230404
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
7.	error	play,1,1,suare001,02,CSX,E5/TH/G5.2-3;B-1
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1966 This game had 3 errors, including an ∞ play. The final score was 2-11.

In the bottom of the 1st inning, with a runner on 2nd, SEA batter Eugenio Suarez hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

	game	ANA vs SEA on 20230404
	state before	(2, 0, 0, 0, 3, 0, 1, 1)
8.	error	play,3,1,frant002,01,CF,FLE3
	state after	(2, 0, 0, 0, 3, 0, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1967 This play is an example of an *infinity play*, as it can be repeated infinitely often without changing the game state.

In the bottom of the 3rd inning, with a 0-1 count and bases empty, SEA batter Ty France hit a fly ball foul of the 1st base line. An error by the first baseman allowed the batter to avoid the out. France is still at-bat, now with a 0-2 count.

	game	ANA vs SEA on 20230404
	state before	(2, 1, 0, 0, 6, 0, 8, 0)
9.	error	play,6,0,renfh001,02,.CSX,S6/G6.1-3(E6/TH)
	state after	(2, 1, 0, 1, 6, 0, 8, 0)
	fielder	6 (shortstop on the home team)

Summary 1968 *In the top of the 6th inning, with a runner on 1st, ANA batter Hunter Renfroe hit a ground ball single to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	COL vs SEA on 20230415
	state before	(1, 1, 0, 0, 6, 1, 5, 1)
10.	error	play,6,1,wongk001,32,VCFB1*BX,FC3/G34.1X3(3E6);B-1
	state after	(1, 1, 0, 1, 6, 1, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1969 *In the bottom of the 6th inning, with a runner on 1st, SEA batter Kolten Wong hit a ground ball into a fielder's choice at 1st/2nd. An error by the shortstop (with an assist from the first baseman) allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st.*

	game	COL vs SEA on 20230416
	state before	(0, 1, 0, 0, 5, 0, 0, 1)
11.	error	play,5,1,polla001,22,CF>FF1B>FB>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 5, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 1970 *In the bottom of the 5th inning, with a runner on 1st, SEA batter A.J. Pollock had a 2-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Pollock is still at bat, now with a full count.*

	game	MIL vs SEA on 20230417
	state before	(1, 1, 0, 0, 7, 5, 2, 0)
12.	error	play,7,0,adamw002,12,1BTS1.>S,K+SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 5, 2, 0)
	fielder	2 (catcher on the home team)

Summary 1971 This game had 2 errors. The final score was 7-3. *In the top of the 7th inning, with a runner on 1st, MIL batter Willy Adames had a 1-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	MIL vs SEA on 20230417
	state before	(2, 1, 1, 0, 9, 6, 3, 0)
13.	error	play,9,0,voit1001,02,CSX,E6/G7.2-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 9, 7, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1972 *In the top of the 9th inning, with runners on 1st and 2nd, MIL batter Luke Voit hit a ground ball to left field. An error by the shortstop allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	SLN vs SEA on 20230423
	state before	(2, 1, 1, 1, 2, 1, 1, 1)
14.	error	play,2,1,frant002,11,BFX,S7/G56+.3-H;2-H;1-3(E7/TH);B-2
	state after	(2, 0, 1, 1, 2, 1, 3, 1)
	fielder	7 (left field on the visiting team)

Summary 1973 *In the bottom of the 2nd inning, with bases loaded, SEA batter Ty France hit a hard ground ball single to 3rd. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the left fielder allowed the runner on 1st to safely advance to 3rd and the batter to reach 2nd.*

28.2.3 May

	game	HOU vs SEA on 20230506
	state before	(1, 0, 1, 0, 5, 3, 0, 1)
15.	error	play,5,1,rodrj007,10,*BX,6/L6.2-3(E6/TH)
	state after	(2, 0, 0, 1, 5, 3, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1974 This game had 2 errors. The final score was 5-7. *In the bottom of the 5th inning, with a runner on 2nd, SEA batter Julio Rodriguez hit a line drive to short. The batter was put out by the shortstop. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd.*

	game	HOU vs SEA on 20230506
	state before	(2, 0, 0, 0, 6, 3, 0, 0)
16.	error	play,6,0,penaj004,00,X,D7/L7LD.B-3(E7)
	state after	(2, 0, 0, 1, 6, 3, 0, 0)
	fielder	7 (left field on the home team)

Summary 1975 *In the top of the 6th inning, with bases empty, HOU batter Jeremy Pena hit a line drive double to deep left field. An error by the left fielder allowed the batter to reach 3rd.*

	game	HOU vs SEA on 20230507
	state before	(2, 0, 1, 0, 3, 0, 1, 1)
17.	error	play,3,1,ralec001,01,FX,E3/G3.2-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 3, 0, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 1976 *In the bottom of the 3rd inning, with a runner on 2nd, SEA batter Cal Raleigh hit a ground ball to 1st base. An error by the first baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	TEX vs SEA on 20230509
	state before	(0, 0, 0, 0, 7, 0, 2, 1)
18.	error	play,7,1,polla001,10,BX,E5/TH/G5.B-1
	state after	(0, 1, 0, 0, 7, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 1977 *In the bottom of the 7th inning, with bases empty, SEA batter A.J. Pollock hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	TEX vs SEA on 20230510
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
19.	error	play,1,1,frant002,11,BFX,S7/L7LD+.1-H(E7/TH)(NR);BX2(34)
	state after	(1, 0, 0, 0, 1, 0, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 1978 *In the bottom of the 1st inning, with a runner on 1st, SEA batter Ty France hit a hard line drive single to deep left field. A throwing error by the left fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 1st but was tagged out by the second baseman (with an assist from the first baseman) when he attempted to advance to 2nd.*

	game	OAK vs SEA on 20230523
	state before	(2, 1, 0, 0, 8, 2, 3, 1)
20.	error	play,8,1,suare001,20,B1B>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 8, 2, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1979 *In the bottom of the 8th inning, with a runner on 1st, SEA batter Eugenio Suarez had a 2-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Suarez is still at bat, now with a 2-1 count.*

	game	OAK vs SEA on 20230525
	state before	(1, 0, 0, 0, 3, 2, 1, 1)
21.	error	play,3,1,crawj002,32,BBBCC.X,E6/G4.B-1
	state after	(1, 1, 0, 0, 3, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1980 *In the bottom of the 3rd inning, with bases empty, SEA batter J.P. Crawford hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs SEA on 20230526
	state before	(1, 1, 0, 0, 4, 2, 2, 0)
22.	error	play,4,0,marct001,00,X,46(1)/F0/G4.B-2(E6/TH)
	state after	(2, 0, 1, 0, 4, 2, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 1981 *In the top of the 4th inning, with a runner on 1st, PIT batter Tucupita Marcano hit a ground ball to 2nd. At 2nd the shortstop (with an assist from the second baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.*

Visiting/Home team rosters for PIT vs SEA on 2023-05-27

position	innings	player	position	innings	player
1	1	Vince Velasquez	1	1	Luis Castillo
1	3	Jose Hernandez	1	7	Gabe Speier
1	6	Dauri Moreta	1	7	Matt Brash
1	6	Rob Zastryzny	1	8	Trevor Gott
1	8	David Bednar	1	9	Justin Topa
2	1	Jason Delay	2	1	Cal Raleigh
3	1	Connor Joe	3	1	Ty France
4	1	Ji Hwan Bae	4	1	Kolten Wong
4	7	Rodolfo Castro	5	1	Jose Caballero
5	1	Ke'Bryan Hayes	6	1	J.P. Crawford
6	1	Tucupita Marcano	7	1	Jarred Kelenic
7	1	Bryan Reynolds	8	1	Julio Rodriguez
8	1	Jack Suwinski	9	1	Teoscar Hernandez
9	1	Josh Palacios			
9	7	Andrew McCutchen			

	game	PIT vs SEA on 20230527
	state before	(2, 1, 0, 0, 2, 0, 3, 0)
23.	error	play,2,0,palaj001,11,BF>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 3, 0)
	fielder	2 (catcher on the home team)

Summary 1982 This game had 3 errors. The final score was 0-5.

In the top of the 2nd inning, with a runner on 1st, PIT batter Josh Palacios had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Palacios is still at bat, now with a 2-1 count.

	game	PIT vs SEA on 20230527
	state before	(0, 1, 0, 0, 6, 0, 4, 1)
24.	error	play,6,1,wongk001,21,M1BBX,54(1)/F0/G56S.B-2(E4/TH)
	state after	(1, 0, 1, 0, 6, 0, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1983 *In the bottom of the 6th inning, with a runner on 1st, SEA batter Kolten Wong hit a ground ball to shallow 3rd. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	PIT vs SEA on 20230527
	state before	(0, 1, 0, 0, 7, 0, 4, 1)
25.	error	play,7,1,kelej001,12,BFT>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 0, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 1984 *In the bottom of the 7th inning, with a runner on 1st, SEA batter Jarred Kelenic had a 1-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.*

	game	PIT vs SEA on 20230528
	state before	(0, 0, 0, 0, 8, 1, 3, 0)
26.	error	play,8,0,mccua001,22,.SBBCX,S6/G6S.B-2(E6/TH)
	state after	(0, 0, 1, 0, 8, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 1985 *In the top of the 8th inning, with bases empty, PIT batter Andrew McCutchen hit a ground ball single to shallow short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	NYA vs SEA on 20230529
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
27.	error	play,2,0,bauej001,00,>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 2, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 1986 *In the top of the 2nd inning, with a runner on 1st, NYA batter Jake Bauers had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Bauers is still at bat, now with a 1-0 count.*

Visiting/Home team rosters for NYA vs SEA on 2023-05-30

position	innings	player	position	innings	player
1	1	Nestor Cortes	1	1	Logan Gilbert
1	6	Michael King	1	5	Trevor Gott
1	8	Ryan Weber	1	6	Tayler Saucedo
2	1	Jose Trevino	1	7	Darren McCaughan
3	1	Jake Bauers	2	1	Tom Murphy
4	1	Gleyber Torres	2	7	Cal Raleigh
5	1	DJ LeMahieu	3	1	Ty France
6	1	Anthony Volpe	4	1	Jose Caballero
7	1	Isiah Kiner-Falefa	5	1	Eugenio Suarez
8	1	Greg Allen	6	1	J.P. Crawford
9	1	Aaron Judge	7	1	AJ Pollock
9	9	Oswaldo Cabrera	7	8	Taylor Trammell
			8	1	Julio Rodriguez
			9	1	Teoscar Hernandez

	game	NYA vs SEA on 20230530
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
28.	error	play,1,0,lemad001,12,BFFFF2FX,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 1, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1987 This game had 3 errors. The final score was 10-2. *In the top of the 1st inning, with runners on 1st and 2nd, NYA batter DJ LeMahieu hit a ground ball into a fielder's choice at 3rd. The runner on 2nd to advance to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	NYA vs SEA on 20230530
	state before	(2, 0, 0, 0, 1, 3, 0, 1)
29.	error	play,1,1,rodrj007,22,TBBFX,E6/G6.B-1
	state after	(2, 1, 0, 0, 1, 3, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 1988 *In the bottom of the 1st inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	NYA vs SEA on 20230530
	state before	(1, 0, 0, 0, 9, 10, 2, 1)
30.	error	play,9,1,frant002,00,X,S7/L7D+.B-2(E7/TH)
	state after	(1, 0, 1, 0, 9, 10, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 1989 *In the bottom of the 9th inning, with bases empty, SEA batter Ty France hit a hard line drive single to deep left field. A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	NYA vs SEA on 20230531
	state before	(0, 0, 1, 0, 10, 0, 0, 0)
31.	error	play,10,0,lemad001,11,..SBX,E6/G6.2-3;B-1
	state after	(0, 1, 0, 1, 10, 0, 0, 0)
	fielder	6 (left field on the visiting team)

Summary 1990 *In the top of the 10th inning, with a runner on 2nd, NYA batter DJ LeMahieu hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

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Visiting/Home team rosters for MIA vs SEA on 2023-06-12

position	innings	player	position	innings	player
1	1	Jesus Luzardo	1	1	Bryce Miller
1	5	Huascar Brazoban	1	7	Ty Adcock
1	7	Bryan Hoeing	1	9	Gabe Speier
2	1	Nick Fortes	2	1	Tom Murphy
3	1	Garrett Cooper	3	1	Ty France
4	1	Luis Arraez	4	1	Jose Caballero
5	1	Jean Segura	5	1	Eugenio Suarez
6	1	Jon Berti	6	1	J.P. Crawford
7	1	Bryan De La Cruz	7	1	Dylan Moore
8	1	Jonathan Davis	8	1	Julio Rodriguez
9	1	Jesus Sanchez	9	1	Teoscar Hernandez

	game	MIA vs SEA on 20230612
	state before	(2, 0, 1, 0, 1, 0, 0, 1)
32.	error	play,1,1,hernt002,00,2,P02(E1/TH).2-3
	state after	(2, 0, 0, 1, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1991 This game had 3 errors. Each game in this Marlins-Mariners series had at least 2 errors, for a total of 8. The final score for this starting game of the series was 1-8.

In the bottom of the 1st inning, with a runner on 2nd, SEA batter Teoscar Hernandez has a 0-0 count. The pitcher attempted to pickoff the runner on 2nd. An error by the pitcher allowed that runner to reach 3rd. Hernandez is still at-bat with a 0-0 count.

	game	MIA vs SEA on 20230612
	state before	(2, 0, 1, 0, 1, 0, 1, 1)
33.	error	play,1,1,suare001,12,BCFX,E7/L7D+.2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 1, 0, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 1992 *In the bottom of the 1st inning, with a runner on 2nd, SEA batter Eugenio Suarez hit a hard line drive to deep left field. An error by the left fielder allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	MIA vs SEA on 20230612
	state before	(2, 0, 0, 0, 7, 1, 7, 0)
34.	error	play,7,0,seguj002,11,CBX,E5/G5.B-1
	state after	(2, 1, 0, 0, 7, 1, 7, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1993 *In the top of the 7th inning, with bases empty, MIA batter Jean Segura hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	MIA vs SEA on 20230613
	state before	(1, 1, 0, 0, 4, 0, 3, 1)
35.	error	play,4,1,ralec001,11,1BS>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 4, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 1994 This game had 2 errors. The final score was 3-9.

In the bottom of the 4th inning, with a runner on 1st, SEA batter Cal Raleigh had a 1-1 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Raleigh is still at bat, now with a 1-2 count.

36.	game	MIA vs SEA on 20230613
	state before	(2, 1, 0, 0, 6, 1, 5, 0)
	error	play,6,0,delab001,12,FBSX,E5/G6S.1-2;B-1
	state after	(2, 1, 1, 0, 6, 1, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 1995 *In the top of the 6th inning, with a runner on 1st, MIA batter Bryan De La Cruz hit a ground ball to short. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for MIA vs SEA on 2023-06-14

position	innings	player	position	innings	player
1	1	Eury Perez	1	1	Luis Castillo
1	7	Dylan Floro	1	6	Matt Brash
1	8	Tanner Scott	1	7	Andres Munoz
1	9	A.J. Puk	1	8	Justin Topa
2	1	Nick Fortes	1	9	Paul Sewald
3	1	Garrett Cooper	2	1	Cal Raleigh
4	1	Luis Arraez	3	1	Ty France
5	1	Jean Segura	4	1	Jose Caballero
5	8	Jon Berti	5	1	Eugenio Suarez
6	1	Jon Berti	6	1	J.P. Crawford
6	8	Joey Wendle	7	1	Jarred Kelenic
7	1	Bryan De La Cruz	8	1	Julio Rodriguez
8	1	Jonathan Davis	9	1	Teoscar Hernandez
9	1	Jesus Sanchez			

	game	MIA vs SEA on 20230614
	state before	(0, 0, 0, 0, 8, 4, 0, 1)
37.	error	play,8,1,ralec001,22,...CB.BCX,S1/G25.B-2(E1/TH)
	state after	(0, 0, 1, 0, 8, 4, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 1996 This game had 3 errors. The final score was 4-1.
In the bottom of the 8th inning, with bases empty, SEA batter Cal Raleigh hit a ground ball single to shallow left infield. A throwing error by the pitcher allowed the batter to reach 2nd.

	game	MIA vs SEA on 20230614
	state before	(1, 1, 0, 0, 9, 4, 0, 1)
38.	error	play,9,1,hernt002,11,FBX,FC4/G4.1-2(E4);B-1
	state after	(1, 1, 1, 0, 9, 4, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1997 *In the bottom of the 9th inning, with a runner on 1st, SEA batter Teoscar Hernandez hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	MIA vs SEA on 20230614
	state before	(1, 1, 1, 0, 9, 4, 0, 1)
39.	error	play,9,1,kelej001,10,BX,FC4/G4.2-3;1-2(E4/TH);B-1
	state after	(1, 1, 1, 1, 9, 4, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 1998 Remarkably, this is not only a *consecutive error* but also a fielder's choice error by the *same* fielder as in the previous play.
In the bottom of the 9th inning, with runners on 1st and 2nd, SEA batter Jarred Kelenic hit a ground ball into a fielder's choice at 2nd.

The runner on 2nd advanced to 3rd. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	CHA vs SEA on 20230616
	state before	(1, 0, 0, 0, 8, 2, 3, 1)
40.	error	play,8,1,cabaj001,12,CBCX,E5/L56.B-2(E6/TH)
	state after	(1, 0, 1, 0, 8, 2, 3, 1)
	fielder	5,6 (3rd baseman,shortstop on the visiting team)

Summary 1999 This play had 2 errors by different fielders. The final score was 2-3.

In the bottom of the 8th inning, with bases empty, SEA batter Jose Caballero hit a line drive to 3rd base. An error by the third baseman allowed the batter to reach 2nd.

Visiting/Home team rosters for CHA vs SEA on 2023-06-17

position	innings	player	position	innings	player
1	1	Lucas Giolito	1	1	Logan Gilbert
1	11	Jesse Scholtens	1	10	Gabe Speier
1	6	Gregory Santos	1	11	Tayler Saucedo
1	8	Keynan Middleton	1	11	Ty Adcock
1	9	Aaron Bummer	1	6	Matt Brash
2	1	Seby Zavala	1	7	Justin Topa
2	9	Yasmani Grandal	1	8	Andres Munoz
3	1	Andrew Vaughn	1	9	Paul Sewald
4	1	Elvis Andrus	2	1	Tom Murphy
4	4	Zach Remillard	3	1	Ty France
5	1	Jake Burger	4	1	Jose Caballero
6	1	Tim Anderson	5	1	Eugenio Suarez
6	4	Elvis Andrus	6	1	J.P. Crawford
7	1	Andrew Benintendi	7	1	Dylan Moore
8	1	Luis Robert Jr.	7	10	AJ Pollock
9	1	Gavin Sheets	8	1	Julio Rodriguez
9	10	Clint Frazier	9	1	Jarred Kelenic
			9	11	Teoscar Hernandez

	game	CHA vs SEA on 20230617
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
41.	error	play,1,0,benia002,00,X,E4/G34D.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2000 This game had 3 errors. The final score was 4-3 (in extra innings).

In the top of the 1st inning, with bases empty, CHA batter Andrew Benintendi hit a ground ball to deep 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.

	game	CHA vs SEA on 20230617
	state before	(1, 0, 1, 0, 4, 1, 2, 0)
42.	error	play,4,0,vauga001,00,X,E6/TH/G6.B-1
	state after	(1, 1, 1, 0, 4, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2001 *In the top of the 4th inning, with a runner on 2nd, CHA batter Andrew Vaughn hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs SEA on 20230617
	state before	(1, 0, 1, 0, 11, 4, 3, 1)
43.	error	play,11,1,polla001,11,CBX,E6/G6.B-1
	state after	(1, 1, 1, 0, 11, 4, 3, 1)
	fielder	6 (shortstop on the home team)

Summary 2002 *In the bottom of the 11th inning, with a runner on 2nd, SEA batter A.J. Pollock hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	WAS vs SEA on 20230626
	state before	(0, 0, 0, 0, 3, 2, 1, 0)
44.	error	play,3,0,abrac001,02,CFX,S6/G4M.B-2(E6/TH)
	state after	(0, 0, 1, 0, 3, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2003 This game had 2 errors. The final score was 4-8. *In the top of the 3rd inning, with bases empty, WAS batter C. J. Abrams hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	WAS vs SEA on 20230626
	state before	(0, 0, 0, 0, 5, 3, 3, 1)
45.	error	play,5,1,hernt002,01,.CX,S13/G1.B-2(E3/TH)
	state after	(0, 0, 1, 0, 5, 3, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2004 *In the bottom of the 5th inning, with bases empty, SEA batter Teoscar Hernandez hit a ground ball single to right shallow infield. A throwing error by the first baseman allowed the batter to reach 2nd.*

	game	WAS vs SEA on 20230627
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
46.	error	play,1,0,menej001,22,BSBTF1,SB2.1-H(E1/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 1, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2005 *In the top of the 1st inning, with a runner on 1st, WAS batter Joey Meneses had a 2-2 count. During the next pitch, which was a throw to 1st, the runner on 1st stole 2nd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI). Meneses is still at-bat with a 2-2 count.*

	game	TBA vs SEA on 20230630
	state before	(2, 0, 0, 0, 1, 0, 0, 0)
47.	error	play,1,0,ralel001,00,X,E4/TH/G34+.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2006 *In the top of the 1st inning, with bases empty, TBA batter Luke Raley hit a hard ground ball to 1st/2nd. A throwing error by the second baseman allowed the batter to reach 1st base.*

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	game	TBA vs SEA on 20230701
	state before	(1, 1, 0, 0, 8, 2, 6, 0)
48.	error	play,8,0,diazy001,32,SBBBBF>F1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 8, 2, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 2007 *In the top of the 8th inning, with a runner on 1st, TBA batter Yandy Diaz has a full count. The pitcher attempted to pickoff the runner on 1st. An error by the pitcher allowed that runner to reach 2nd. Diaz is still at bat with a full count.*

Visiting/Home team rosters for TBA vs SEA on 2023-07-02

position	innings	player	position	innings	player
1	1	Taj Bradley	1	1	Luis Castillo
1	4	Kevin Kelly	1	7	Andres Munoz
1	6	Colin Poche	1	8	Matt Brash
1	6	Shawn Armstrong	1	9	Paul Sewald
1	7	Jason Adam	2	1	Tom Murphy
1	8	Zack Littell	3	1	Ty France
2	1	Francisco Mejia	4	1	Jose Caballero
3	1	Luke Raley	5	1	Eugenio Suarez
4	1	Vidal Brujan	6	1	J.P. Crawford
5	1	Isaac Paredes	7	1	Jarred Kelenic
5	3	Taylor Walls	8	1	Julio Rodriguez
6	1	Wander Franco	9	1	Teoscar Hernandez
7	1	Randy Arozarena			
8	1	Manuel Margot			
9	1	Josh Lowe			

	game	TBA vs SEA on 20230702
	state before	(0, 0, 0, 0, 3, 2, 1, 0)
49.	error	play,3,0,brujv001,12,FBFX,E4/TH/G4.B-1
	state after	(0, 1, 0, 0, 3, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2008 This game had 3 errors. The final score was 6-7.
In the top of the 3rd inning, with bases empty, TBA batter Vidal Brujan hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.

	game	TBA vs SEA on 20230702
	state before	(1, 0, 1, 0, 3, 5, 1, 0)
50.	error	play,3,0,ramih004,02,SSFX,E6/TH/G6MS.2-3;B-1
	state after	(1, 1, 0, 1, 3, 5, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2009 *In the top of the 3rd inning, with a runner on 2nd, TBA batter Harold Ramirez hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	TBA vs SEA on 20230702
	state before	(1, 0, 0, 0, 7, 6, 6, 0)
51.	error	play,7,0,franw002,32,BBSCBFFX,E5/G6D.B-1
	state after	(1, 1, 0, 0, 7, 6, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2010 *In the top of the 7th inning, with bases empty, TBA batter Wander Franco hit a ground ball to deep short. An error by the third baseman allowed the batter to reach 1st base.*

	game	DET vs SEA on 20230714
	state before	(1, 0, 0, 0, 7, 3, 2, 0)
52.	error	play,7,0,baezj001,22,FCBBX,6E3/G6.B-1
	state after	(1, 1, 0, 0, 7, 3, 2, 0)
	fielder	3 (1st baseman on the home team)

Summary 2011 *In the top of the 7th inning, with bases empty, DET batter Javier Baez hit a ground ball to short. An error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st.*

	game	MIN vs SEA on 20230718
	state before	(2, 1, 0, 1, 3, 4, 3, 0)
53.	error	play,3,0,gallj002,11,F1B>B,SB2.3-H(E2/TH) (NR) (UR)
	state after	(2, 0, 1, 0, 3, 5, 3, 0)
	fielder	2 (catcher on the home team)

Summary 2012 *In the top of the 3rd inning, with runners on 1st and 3rd, MIN batter Joey Gallo had a 1-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). Gallo is still at-bat, now with a 2-1 count.*

	game	TOR vs SEA on 20230722
	state before	(2, 1, 0, 0, 7, 6, 4, 0)
54.	error	play,7,0,kirka001,11,.BFX,S9/G3.1-H(E9) (NR);B-2
	state after	(2, 0, 1, 0, 7, 7, 4, 0)
	fielder	9 (right field on the home team)

Summary 2013 *In the top of the 7th inning, with a runner on 1st, TOR batter Alejandro Kirk hit a ground ball single to 1st. An error by the right fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	TOR vs SEA on 20230723
	state before	(1, 0, 0, 1, 9, 4, 3, 0)
55.	error	play,9,0,varsd001,12,CSBFX,E1/TH/G1-.B-1
	state after	(1, 1, 0, 1, 9, 4, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 2014 *In the top of the 9th inning, with a runner on 3rd, TOR batter Daulton Varsho hit a soft ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

Visiting/Home team rosters for BOS vs SEA on 2023-07-31

position	innings	player	position	innings	player
1	1	Nick Pivetta	1	1	George Kirby
1	8	Joe Jacques	1	6	Gabe Speier
2	1	Connor Wong	1	7	Matt Brash
3	1	Triston Casas	1	8	Andres Munoz
3	8	Yu Chang	1	8	Tayler Saucedo
4	1	Pablo Reyes	1	9	Isaiah Campbell
4	8	Christian Arroyo	2	1	Tom Murphy
5	1	Rafael Devers	3	1	Ty France
6	1	Yu Chang	4	1	Jose Caballero
6	8	Pablo Reyes	5	1	Eugenio Suarez
7	1	Masataka Yoshida	6	1	J.P. Crawford
8	1	Jarren Duran	7	1	Cade Marlowe
9	1	Alex Verdugo	8	1	Julio Rodriguez
9	8	Rob Refsnyder	9	1	Teoscar Hernandez

	game	BOS vs SEA on 20230731
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
56.	error	play,1,0,turnj001,10,B>C,SB2.1-H(E2/TH) (E8/TH) (NR)
	state after	(0, 0, 0, 0, 1, 1, 0, 0)
	fielder	2,8 (catcher,center field on the home team)

Summary 2015 This game had 3 errors, 2 of them in 1 play. In fact, this play has 2 errors by 2 different fielders. The final score was 2-6.

In the top of the 1st inning, with a runner on 1st, BOS batter Justin Turner had a 1-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. A throwing error by the center fielder allowed that runner (now on 3rd) to score (unearned, no RBI). Turner is still at bat, now with a 1-1 count.

	game	BOS vs SEA on 20230731
	state before	(1, 1, 1, 0, 8, 1, 5, 1)
57.	error	play,8,1,ralec001,12,BFCX,FC6/G4M.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 8, 1, 5, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2016 *In the bottom of the 8th inning, with runners on 1st and 2nd, SEA batter Cal Raleigh hit a ground ball into a fielder's choice at 2nd. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

28.2.6 August

	game	BOS vs SEA on 20230801
	state before	(0, 1, 1, 0, 1, 0, 0, 1)
58.	error	play,1,1,suare001,00,X,S7/G56.2-H;1-3;B-2(E7/TH)
	state after	(0, 0, 1, 1, 1, 0, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 2017 This game had 2 errors. The final score was 6-4.

In the bottom of the 1st inning, with runners on 1st and 2nd, SEA batter Eugenio Suarez hit a ground ball single to 3rd. The runner on 2nd scored (1 RBI) and the runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd.

	game	BOS vs SEA on 20230801
	state before	(1, 1, 0, 0, 8, 6, 4, 1)
59.	error	play,8,1,frant002,01,FX,S3/G3.1-3;B-2(E3/TH)
	state after	(1, 0, 1, 1, 8, 6, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2018 *In the bottom of the 8th inning, with a runner on 1st, SEA batter Ty France hit a ground ball single to 1st base. The runner on 1st advanced to 3rd. A throwing error by the first baseman allowed the batter to reach 2nd.*

	game	SDN vs SEA on 20230809
	state before	(0, 0, 0, 0, 3, 1, 0, 1)
60.	error	play,3,1,marlc001,01,CX,S9/L9LS.B-2(E9/TH)
	state after	(0, 0, 1, 0, 3, 1, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 2019 This game had 2 errors. The final score was 1-6. *In the bottom of the 3rd inning, with bases empty, SEA batter Cade Marlowe hit a line drive single to shallow right field. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	SDN vs SEA on 20230809
	state before	(2, 1, 0, 0, 8, 1, 5, 1)
61.	error	play,8,1,rojaj001,12,FTBX,E3/G34.1-2;B-1
	state after	(2, 1, 1, 0, 8, 1, 5, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2020 *In the bottom of the 8th inning, with a runner on 1st, SEA batter Josh Rojas hit a ground ball to 1st/2nd. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	BAL vs SEA on 20230812
	state before	(1, 1, 0, 0, 10, 1, 0, 0)
62.	error	play,10,0,ohear001,20,BBX,FC6/G4M.1-3(E6);B-1
	state after	(1, 1, 0, 1, 10, 1, 0, 0)
	fielder	6 (1st baseman on the visiting team)

Summary 2021 *In the top of the 10th inning, with a runner on 1st, BAL batter Ryan O'Hearn hit a ground ball into a fielder's choice at 2nd. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	BAL vs SEA on 20230813
	state before	(2, 0, 0, 0, 2, 0, 0, 1)
63.	error	play,2,1,canzd001,22,CBCFBFX,E4/G4.B-1
	state after	(2, 1, 0, 0, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2022 This game had 2 errors. The final score was 5-3 (in extra innings).

In the bottom of the 2nd inning, with bases empty, SEA batter Dominic Canzone hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.

	game	BAL vs SEA on 20230813
	state before	(1, 0, 0, 0, 3, 0, 0, 0)
64.	error	play,3,0,mccaj001,01,CX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 3, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2023 *In the top of the 3rd inning, with bases empty, BAL batter James McCann hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	KCA vs SEA on 20230825
	state before	(1, 1, 0, 1, 2, 1, 2, 0)
65.	error	play,2,0,wated001,01,F1,P03(E1/TH).3-H(NR)
	state after	(1, 1, 0, 0, 2, 2, 2, 0)
	fielder	1 (pitcher on the home team)

Summary 2024 *In the top of the 2nd inning, with runners on 1st and 3rd, KCA batter Drew Waters has a 0-1 count. The pitcher attempted to pickoff the runner on 3rd. An error by the pitcher allowed that runner to score (unearned, no RBI). Waters is still at-bat with a 0-1 count.*

	game	KCA vs SEA on 20230826
	state before	(2, 1, 0, 0, 1, 0, 0, 1)
66.	error	play,1,1,ralec001,21,C*BBX,S9/G34.1-3(E9)
	state after	(2, 1, 0, 1, 1, 0, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 2025 *In the bottom of the 1st inning, with a runner on 1st, SEA batter Cal Raleigh hit a ground ball single to 1st/2nd. An error by the right fielder allowed the runner on 1st to advance to 3. The batter took 1st.*

	game	KCA vs SEA on 20230827
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
67.	error	play,1,1,suare001,11,FBX,64(1)/F0/G6.B-2(E4/TH)
	state after	(2, 0, 1, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2026 *In the bottom of the 1st inning, with a runner on 1st, SEA batter Eugenio Suarez hit a ground ball to short. The second baseman (with an assist from the shortstop) focred out the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	OAK vs SEA on 20230829
	state before	(2, 0, 0, 0, 2, 3, 0, 1)
68.	error	play,2,1,canzd001,02,CCX,E4/G34.B-1
	state after	(2, 1, 0, 0, 2, 3, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2027 *In the bottom of the 2nd inning, with bases empty, SEA batter Dominic Canzone hit a ground ball to 1st base. An error by the second baseman allowed the batter to reach 1st base.*

	game	OAK vs SEA on 20230830
	state before	(2, 1, 0, 0, 2, 2, 0, 0)
69.	error	play,2,0,nodar001,00,1>P,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 2, 0, 0)
	fielder	2 (catcher on the home team)

Summary 2028 *In the top of the 2nd inning, with a runner on 1st, OAK batter Ryan Noda had a 0-0 count. During the next pitch, which was a pitchout (ball), the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Noda is still at bat, now with a 1-0 count.*

28.2.7 September

	game	ANA vs SEA on 20230911
	state before	(2, 1, 1, 0, 11, 6, 5, 0)
70.	error	play,11,0,adamj003,00,X,E5/G7S.2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 11, 7, 5, 0)
	fielder	5 (catcher on the home team)

Summary 2029 Each game in this Angels-Pirates series had at least one error, for a total of 4.

In the top of the 11th inning, with runners on 1st and 2nd, ANA batter

Jordyn Adams hit a ground ball to shallow left field. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 1st.

	game	ANA vs SEA on 20230912
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
71.	error	play,1,1,rodrj007,11,BFX,E5/G56.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2030 This game had 2 errors. The final score was 0-8.
In the bottom of the 1st inning, with bases empty, SEA batter Julio Rodriguez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.

	game	ANA vs SEA on 20230912
	state before	(1, 0, 0, 0, 4, 0, 3, 1)
72.	error	play,4,1,frant002,00,X,E5/G7S.B-1
	state after	(1, 1, 0, 0, 4, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2031 *In the bottom of the 4th inning, with bases empty, SEA batter Ty France hit a ground ball to shallow left field. An error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs SEA on 20230913
	state before	(1, 1, 0, 0, 3, 1, 0, 1)
73.	error	play,3,1,crawj002,12,BF>FX,FC3/G3.1-2(E3/TH);B-1
	state after	(1, 1, 1, 0, 3, 1, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2032 *In the bottom of the 3rd inning, with a runner on 1st, SEA batter J.P. Crawford hit a ground ball into a fielder's choice at 1st. A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

74.	game	LAN vs SEA on 20230915
	state before	(0, 1, 0, 0, 8, 4, 3, 0)
	error	play,8,0,smitw003,02,1CF>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 8, 4, 3, 0)
	fielder	2 (catcher on the home team)

Summary 2033 *In the top of the 8th inning, with a runner on 1st, LAN batter Will Smith had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 3rd. A throwing error by the catcher allowed that runner to advance to 3rd. Smith is still at bat, now with a 1-2 count.*

75.	game	LAN vs SEA on 20230916
	state before	(0, 0, 1, 0, 11, 6, 1, 1)
	error	play,11,1,ralec001,02,..CFX,E1/G23-.2-3;B-1
	state after	(0, 1, 0, 1, 11, 6, 1, 1)
	fielder	1 (catcher on the home team)

Summary 2034 *In the bottom of the 11th inning, with a runner on 2nd, SEA batter Cal Raleigh hit a soft ground ball to shallow right infield. An error by the pitcher allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

76.	game	LAN vs SEA on 20230917
	state before	(2, 0, 0, 0, 1, 1, 0, 0)
	error	play,1,0,perad001,11,FBX,E7/F7LD.B-2
	state after	(2, 0, 1, 0, 1, 1, 0, 0)
	fielder	7 (left field on the home team)

Summary 2035 *In the top of the 1st inning, with bases empty, LAN batter David Peralta hit a fly ball to deep left field. An error by the left fielder allowed the batter to reach 2nd base.*

Visiting/Home team rosters for HOU vs SEA on 2023-09-26

position	innings	player	position	innings	player
1	1	Cristian Javier	1	1	George Kirby
1	5	Bennett Sousa	1	7	Justin Topa
1	5	Rafael Montero	1	7	Matt Brash
1	7	Phil Maton	1	9	Andres Munoz
1	8	Ryne Stanek	2	1	Cal Raleigh
2	1	Martin Maldonado	3	1	Ty France
2	7	Yainer Diaz	4	1	Josh Rojas
3	1	Jose Abreu	5	1	Eugenio Suarez
4	1	Jose Altuve	6	1	J.P. Crawford
5	1	Alex Bregman	7	1	Jarred Kelenic
6	1	Jeremy Pena	8	1	Julio Rodriguez
7	1	Yordan Alvarez	9	1	Teoscar Hernandez
8	1	Chas McCormick			
9	1	Kyle Tucker			

	game	HOU vs SEA on 20230926
	state before	(0, 1, 1, 0, 3, 0, 0, 1)
77.	error	play,3,1,roja,j001,32,BBCCFFX,9/L9D.2-H(E4/TH)(NR);1-3
	state after	(1, 0, 0, 1, 3, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2036 This game had 3 errors. The final score was 2-6.

In the bottom of the 3rd inning, with runners on 1st and 2nd, SEA batter Josh Rojas hit a line drive to deep right field. The right fielder put out the batter. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 3rd.

	game	HOU vs SEA on 20230926
	state before	(0, 0, 0, 0, 4, 0, 1, 1)
78.	error	play,4,1,kelej001,00,X,E5/G56.B-1
	state after	(0, 1, 0, 0, 4, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2037 *In the bottom of the 4th inning, with bases empty, SEA batter Jarred Kelenic hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	HOU vs SEA on 20230926
	state before	(2, 1, 1, 0, 5, 0, 1, 1)
79.	error	play,5,1,ralec001,30,.B*BBX,S9/L9LS.2-H;1-H(E9/TH) (NR);B-3
	state after	(2, 0, 0, 1, 5, 0, 3, 1)
	fielder	9 (right field on the visiting team)

Summary 2038 *In the bottom of the 5th inning, with runners on 1st and 2nd, SEA batter Cal Raleigh hit a line drive single to shallow right field. The runner on 2nd scored (1 RBI. A throwing error by the right fielder allowed the runner on 1st to also score (unearned, no RBI) and the batter to reach 3rd.*

	game	HOU vs SEA on 20230927
	state before	(1, 0, 0, 0, 7, 4, 3, 0)
80.	error	play,7,0,brega001,21,BCBX,S9/L89.B-3(E9)
	state after	(1, 0, 0, 1, 7, 4, 3, 0)
	fielder	9 (right field on the home team)

Summary 2039 *In the top of the 7th inning, with bases empty, HOU batter Alex Bregman hit a line drive single to center right field. An error by the right fielder allowed the batter to reach 3rd.*

Chapter 29

St Louis Cardinals (SLN)

The St Louis Cardinals have won 11 World Series championships, second in MLB only to the New York Yankees. They have been in St Louis Missouri since 1881 and have been known as the St Louis Cardinals since 1900. However, in 2023 the Cardinals finished last in the National League Central, marking the first time they ended a season in last place since 1990. Their last World Series title was won in 2011 and their last NLCD title was in 2022.

This chapter is devoted to listing the errors in each regular season game of 2023 played in St Louis against the Cardinals. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 SLN team is the lowest in the MLB¹, 0.67, almost 2 standard deviations below the MLB average of 0.691 that year [BR] (which is very bad).

In 2023, there were 68 errors (by both sides) in the 81 SLN 2023 home games, 28 by the Cardinals and the rest (40) by visiting teams. The total number of errors by SLN (home and away) over the regular season in 2023 was 67. Since 83.9 was the MLB average, the Cardinals' error total was about 1 standard deviation below average (which is very good).

¹Tied with the Colorado Rockies.

The 2023 Cardinals are a very interesting team in that they have one of the lowest error rates in the MLB and an good average OBP for their starting roster (see §29.2 below), yet their runs-per-game is lower than average and their winning percentage that year was also low (0.438 based on 71 wins).

29.1 Season totals

The Official Scorer shall credit a batter with a base hit when a fair ball that has not been touched by a fielder touches a runner or an umpire, unless a runner is called out for having been touched by an Infield Fly.

– **MLB Rule Book** 9.05(a)(5)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

29.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	40
Runs by ht because of errors by vt:	8
Errors by ht:	28
Runs by vt because of errors by ht:	8

This is a good (not great) error-to-run conversion rate: $8/38 = 20.5\%$ on the part of the Cardinals (compared to the better $8/28 = 28.5\%$ on the part of the visiting teams). A randomly selected *play* (not necessarily with an error) in the 2023 SLN season (home and away) has about 0.1158 (11.5%) likelihood of resulting in a run². Therefore, a randomly selected error is almost twice as likely to result in a run than a randomly selected at-bat, at least in this case.

²Take Runs over Plate Appearances, R/PA, from [BR]. For example for the Cardinals, R/PA = 719/6204 (about 11%).

Note that the total number of errors (by either team) in a home game is 67, so the expected number of errors (by either team) in a randomly selected SLN 2023 home game is about 0.827.

Errors by position	vt	ht
errors due to pitcher:	6	1
to catcher:	4	4
to 1st baseman:	3	1
to 2nd baseman:	7	6
to 3rd baseman:	7	4
to shortstop:	9	8
to left fielder:	1	2
to center fielder:	0	0
to right fielder:	2	2

29.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	20	11	31
(lone) runner on 1st	11	8	19
(lone) runner on 2nd	5	1	6
(lone) runner on 3rd	0	3	3
two runners on base	3	2	5
bases loaded	1	3	4
total	40	28	68

A table of frequencies of most common errors by the Cardinals in SLN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	25%
(1, 0, 0, 0)	4	14.3%
(2, 0, 0, 0)	0	0%
(0, 1, 0, 0)	3	10.7%
(1, 1, 0, 0)	1	3.57%
(2, 1, 0, 0)	4	14.3%
(2, 0, 0, 1)	3	10.7%
(2, 1, 0, 1)	2	7.14%
(2, 1, 1, 1)	2	7.14%

A table of frequencies of most common errors by a visiting team in SLN 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	12	30%
(1, 0, 0, 0)	3	7.5%
(2, 0, 0, 0)	5	12.5%
(0, 1, 0, 0)	2	5%
(1, 1, 0, 0)	8	20%
(2, 1, 0, 0)	1	2.5%
(0, 0, 1, 0)	1	2.5%
(1, 0, 1, 0)	3	7.5%
(2, 0, 1, 0)	1	2.5%
(2, 1, 1, 1)	1	2.5%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Cardinals homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `SLN2023.evn`):

state	SLN2023 percentages
(0, 0, 0, 0)	23.7%
(1, 0, 0, 0)	16.3%
(2, 0, 0, 0)	13.1%
(0, 1, 0, 0)	5.9%
(1, 1, 0, 0)	7.5%
(2, 1, 0, 0)	7.5%
(0, 0, 1, 0)	2.2%
(1, 0, 1, 0)	2.3%
(2, 0, 1, 0)	3.2%
(0, 0, 0, 1)	0.1%
(1, 0, 0, 1)	0.8%
(2, 0, 0, 1)	0.9%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	3.0%
(2, 1, 1, 0)	3.7%

29.1.3 Some interesting errors

Some of the most interesting error plays in the SLN 2023 home games:

- The game DET vs SLN on 2023-05-07 with 5 errors.
- The game CHN vs SLN on 2023-06-25 with 4 errors.
- The game WAS vs SLN on 2023-07-14 had 3 errors and one play with two stolen bases, including stealing home plate.
- The game MIL vs SLN on 2023-09-21 had a play with 2 errors.
- The game CIN vs SLN on 2023-09-29 had 4 errors and a play with 2 errors.

29.1.4 Relative team rankings

We have 81 home games between SLN and some visiting MLB team (vt). In the lists below, we have ordered the visiting teams with that team's W/T

record in parentheses³.

A number of teams did not play SLN in the regular season. However, of those that did, SLN *lost* more games than won against:

ARI (2/3), COL (2/3), DET (2/3), HOU (2/3), MIN (2/3), PHI (2/3), ANA
(3/3),
ATL (3/3), SFN (3/3), NYN (3/4), CHN (4/6), MIL (4/7), PIT (4/7)

and *won* more than lost against

MIA (0/3), NYA (1/3), OAK (1/3), SDN (1/3),
TOR (1/3), WAS (1/3), LAN (1/4) .

The teams that *split* their series were:

KCA (1/2), CIN (3/6) .

The teams that did not play in any of the 81 home games in SLN 2023 were:

BAL, BOS, CHA, CLE, SEA, TBA, TEX .

29.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for SLN 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Willson Contreras	97	.358
3 (1B)	Paul Goldschmidt	133	.363
4 (2B)	Nolan Gorman	76	.328
5 (3B)	Nolan Arenado	128	.315
6 (SS)	Paul DeJong	81	.297
7 (LF)	Tyler O'Neill	58	.312
8 (CF)	Lars Nootbaar	73	.367
9 (RF)	Jordan Walker	93	.342

³Where W = numbers of (home) games vt won against SLN, T = Total home games vt played against the Cardinals at St. Louis.

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 SLN home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

	game	TOR vs SLN on 20230330
	state before	(0, 0, 0, 0, 1, 3, 0, 1)
1.	error	play,1,1,donob001,32,CBBBF.X,S6/G6.B-2(E6/TH)
	state after	(0, 0, 1, 0, 1, 3, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2040 *In the bottom of the 1st inning, with bases empty, SLN batter Brendan Donovan hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

29.2.1 April

	game	TOR vs SLN on 20230401
	state before	(2, 1, 0, 1, 3, 0, 0, 1)
2.	error	play,3,1,arenn001,00,X,E5/TH/G56.3-H(NR)(UR);1-3;B-2
	state after	(2, 0, 1, 1, 3, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2041 *In the bottom of the 3rd inning, with runners on 1st and 3rd, SLN batter Nolan Arenado hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	PIT vs SLN on 20230413
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
3.	error	play,2,1,contw001,32,BFBBCX,E4/G4MD.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2042 *In the bottom of the 2nd inning, with bases empty, SLN batter Willson Contreras hit a ground ball to deep 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	PIT vs SLN on 20230416
	state before	(2, 1, 0, 1, 6, 2, 1, 0)
4.	error	play,6,0,marct001,01,.CX,E4/G34.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 6, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2043 *In the top of the 6th inning, with runners on 1st and 3rd, PIT batter Tucupita Marcano hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	ARI vs SLN on 20230419
	state before	(1, 0, 0, 0, 1, 0, 0, 0)
5.	error	play,1,0,marct001,31,BCBBX,S6/G6MS.B-2(E6/TH)
	state after	(1, 0, 1, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2044 This game had 2 errors and a play with two stolen bases. The final score was 5-14.

In the top of the 1st inning, with bases empty, ARI batter Ketel Marte hit a ground ball single to shallow short. A throwing error by the shortstop allowed the batter to reach 1st.

	game	ARI vs SLN on 20230419
	state before	(1, 1, 1, 0, 1, 0, 1, 1)
6.	error	play,1,1,contw001,00,>C,SB3;SB2.2-H(E2/TH)(NR)
	state after	(1, 1, 0, 0, 1, 0, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 2045 *In the bottom of the 1st inning, with runners on 1st and 2nd, SLN batter Willson Contreras had a 0-0 count. During the next pitch, which was a called strike, the runner on 2nd stole 3rd and the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI). Contreras is still at bat, now with a 0-1 count.*

29.2.2 May

	game	ANA vs SLN on 20230502
	state before	(1, 1, 0, 0, 5, 4, 0, 1)
7.	error	play,5,1,dejop001,12,CTBFX,S9/G34.1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 5, 4, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 2046 *In the bottom of the 5th inning, with a runner on 1st, SLN batter Paul DeJong hit a ground ball single to the 1st/2nd base gap. The runner from 1st reached 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	ANA vs SLN on 20230503
	state before	(0, 0, 0, 0, 2, 1, 1, 1)
8.	error	play,2,1,burla001,22,BCCBX,E4/G34.B-1
	state after	(0, 1, 0, 0, 2, 1, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2047 *In the bottom of the 2nd inning, with bases empty, SLN batter Alec Burleson hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	ANA vs SLN on 20230504
	state before	(1, 0, 0, 0, 4, 10, 2, 0)
9.	error	play,4,0,lambj001,11,BCX,E6/L6M.B-1
	state after	(1, 1, 0, 0, 4, 10, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2048 *In the top of the 4th inning, with bases empty, ANA batter Jake Lamb hit a line drive to short. An error by the shortstop allowed the batter to reach 1st base.*

10.	game	DET vs SLN on 20230505
	state before	(0, 0, 1, 0, 4, 2, 1, 1)
	error	play,4,1,contw001,32,BCBSFBFX,E9/F9.2-3;B-2
	state after	(0, 0, 1, 1, 4, 2, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 2049 *In the bottom of the 4th inning, with a runner on 2nd, SLN batter Willson Contreras hit a fly ball to right field. An error by the right fielder allowed the runner on 2nd to advance to 3rd and the batter to reach 2nd base.*

11.	game	DET vs SLN on 20230506
	state before	(1, 0, 1, 0, 10, 6, 5, 0)
	error	play,10,0,vierm001,00,X,E5/G6.B-1
	state after	(1, 1, 1, 0, 10, 6, 5, 0)
	fielder	5 (right field on the visiting team)

Summary 2050 *In the top of the 10th inning, with a runner on 2nd, DET batter Matt Vierling hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for DET vs SLN on 2023-05-07

position	innings	player	position	innings	player
1	1	Alex Faedo	1	1	Steven Matz
1	5	Tyler Alexander	1	6	James Naile
1	6	Mason Englert	1	6	JoJo Romero
1	6	Tyler Holton	1	7	Drew VerHagen
2	1	Jake Rogers	1	9	Ryan Helsley
3	1	Spencer Torkelson	2	1	Andrew Knizner
4	1	Jonathan Schoop	3	1	Paul Goldschmidt
4	6	Zach McKinsty	4	1	Nolan Gorman
5	1	Andy Ibanez	5	1	Nolan Arenado
5	6	Nick Maton	6	1	Tommy Edman
6	1	Javier Baez	7	1	Brendan Donovan
7	1	Eric Haase	8	1	Dylan Carlson
8	1	Riley Greene	9	1	Lars Nootbaar
9	1	Matt Vierling			

12.	game	DET vs SLN on 20230507
	state before	(0, 0, 0, 0, 2, 0, 1, 1)
	error	play,2,1,contw001,22,CBFBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2051 This game had 5 errors. The final score was 6-12.
In the bottom of the 2nd inning, with bases empty, SLN batter Willson Contreras hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

13.	game	DET vs SLN on 20230507
	state before	(2, 1, 0, 0, 6, 1, 3, 0)
	error	play,6,0,haase001,02,CSX,E6/G6.1-2;B-1
	state after	(2, 1, 1, 0, 6, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 2052 *In the top of the 6th inning, with a runner on 1st, DET batter Eric Haase hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	DET vs SLN on 20230507
	state before	(0, 0, 1, 1, 6, 6, 3, 1)
14.	error	play,6,1,arenn001,02,.>C.FFX,E6/G6.3-H;2-3;B-1
	state after	(0, 1, 0, 1, 6, 6, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2053 *In the bottom of the 6th inning, with runners on 2nd and 3rd, SLN batter Nolan Arenado hit a ground ball to short. The runner on 3rd scored (1 RBI). An error by the shortstop allowed the runner on 2nd advanced to 3rd and the batter to reach 1st.*

	game	DET vs SLN on 20230507
	state before	(1, 1, 0, 0, 6, 6, 7, 1)
15.	error	play,6,1,edmat001,02,CSX,FC5/G56.1-2(E5);B-1
	state after	(1, 1, 1, 0, 6, 6, 7, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2054 *In the bottom of the 6th inning, with a runner on 1st, SLN batter Tommy Edman hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner from 1st to advance to 2nd and the batter to reach 1st.*

	game	DET vs SLN on 20230507
	state before	(1, 1, 0, 0, 6, 6, 10, 1)
16.	error	play,6,1,goldp001,01,FX,FC5/G56.1-2(E5);B-1
	state after	(1, 1, 1, 0, 6, 6, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2055 *In the bottom of the 6th inning, with a runner on 1st, SLN batter Paul Goldschmidt hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner from 1st to advance to 2nd and the batter to reach 1st.*

	game	LAN vs SLN on 20230518
	state before	(1, 0, 0, 0, 3, 2, 0, 0)
17.	error	play,3,0,smitw003,11,BCX,E6/G6.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2056 *This game had 3 errors. The final score was 8-16. In the top of the 3rd inning, with bases empty, LAN batter Will Smith hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	LAN vs SLN on 20230518
	state before	(2, 1, 0, 0, 6, 2, 9, 0)
18.	error	play,6,0,perad001,01,TX,E3/G9LS.1-3;B-1
	state after	(2, 1, 0, 1, 6, 2, 9, 0)
	fielder	3 (1st baseman on the home team)

Summary 2057 *In the top of the 6th inning, with a runner on 1st, LAN batter David Peralta hit a ground ball to shallow right field down the 1st base line. An error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	LAN vs SLN on 20230518
	state before	(2, 0, 0, 0, 6, 7, 9, 1)
19.	error	play,6,1,goldp001,32,BBCCFBFFX,E4/G4.B-1
	state after	(2, 1, 0, 0, 6, 7, 9, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2058 *In the bottom of the 6th inning, with bases empty, SLN batter Paul Goldschmidt hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	LAN vs SLN on 20230519
	state before	(0, 0, 0, 0, 5, 0, 0, 0)
20.	error	play,5,0,bettm001,11,BCX,E5/TH/G56S.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2059 This game had 3 errors. The final score was 5-0.
In the top of the 5th inning, with bases empty, LAN batter Mookie Betts hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	LAN vs SLN on 20230519
	state before	(1, 0, 0, 0, 8, 5, 0, 1)
21.	error	play,8,1,merco003,00,X,E6/G6.B-1
	state after	(1, 1, 0, 0, 8, 5, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2060 *In the bottom of the 8th inning, with bases empty, SLN batter Oscar Mercado hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

29.2.3 June

	game	CIN vs SLN on 20230610
	state before	(2, 0, 1, 0, 4, 5, 0, 0)
22.	error	play,4,0,mclam001,11,BCX,E4/TH/G6M.2-3;B-1
	state after	(2, 1, 0, 1, 4, 5, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2061 *In the top of the 4th inning, with a runner on 2nd, CIN batter Matt McLain hit a ground ball to deep short. A throwing error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	SFN vs SLN on 20230612
	state before	(2, 0, 0, 0, 3, 2, 0, 1)
23.	error	play,3,1,edmat001,31,1>B.FBBX,E3/G34D+.B-1
	state after	(2, 1, 0, 0, 3, 2, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2062 *In the bottom of the 3rd inning, with bases empty, SLN batter Tommy Edman hit a hard ground ball to the deep 1st/2nd base gap. An error by the first baseman allowed the batter to reach 1st base.*

	game	SFN vs SLN on 20230613
	state before	(0, 1, 0, 0, 5, 3, 2, 0)
24.	error	play,5,0,confm001,21,B*BFX,S7/F7L.1-3(E7);B-2
	state after	(0, 0, 1, 1, 5, 3, 2, 0)
	fielder	7 (left field on the home team)

Summary 2063 *In the top of the 5th inning, with a runner on 1st, SFN batter Michael Conforto hit a fly ball single to left field. An error by the left fielder allowed the the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	CHN vs SLN on 20230624
	state before	(0, 0, 0, 0, 9, 7, 1, 0)
25.	error	play,9,0,hoern001,01,.CX,E6/TH/G6.B-2
	state after	(0, 0, 1, 0, 9, 7, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2064 *In the top of the 9th inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 2nd base.*

Visiting/Home team rosters for CHN vs SLN on 2023-06-25

position	innings	player	position	innings	player
1	1	Marcus Stroman	1	1	Matthew Liberatore
1	4	Michael Fulmer	1	3	Jake Woodford
1	6	Mark Leiter Jr.	1	5	Genesis Cabrera
1	7	Hayden Wesneski	1	7	Giovanny Gallegos
1	8	Adbert Alzolay	1	8	Andre Pallante
2	1	Tucker Barnhart	1	9	Jordan Hicks
2	7	Yan Gomes	2	1	Willson Contreras
3	1	Trey Mancini	3	1	Paul Goldschmidt
3	3	Cody Bellinger	4	1	Nolan Gorman
4	1	Nico Hoerner	5	1	Nolan Arenado
5	1	Nick Madrigal	6	1	Paul DeJong
6	1	Dansby Swanson	7	1	Brendan Donovan
7	1	Ian Happ	8	1	Tommy Edman
8	1	Christopher Morel	9	1	Lars Nootbaar
9	1	Seiya Suzuki			

	game	CHN vs SLN on 20230625
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
26.	error	play,1,0,hoern001,00,X,E6/G6.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2065 This game had 4 errors. The final score was 5-7.
In the top of the 1st inning, with bases empty, CHN batter Nico Hoerner hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	CHN vs SLN on 20230625
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
27.	error	play,1,0,morec001,00,1>C,SB2.1-3(E4)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2066 *In the top of the 1st inning, with a runner on 1st, CHN batter Christopher Morel had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. An error by the second baseman allowed that runner to advance to 3rd. Morel is still at bat, now with a 0-1 count.*

	game	CHN vs SLN on 20230625
	state before	(0, 0, 0, 0, 1, 4, 0, 1)
28.	error	play,1,1,donob001,12,CBFX,E6/TH/G6M.B-1
	state after	(0, 1, 0, 0, 1, 4, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2067 *In the bottom of the 1st inning, with bases empty, SLN batter Brendan Donovan hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	CHN vs SLN on 20230625
	state before	(2, 1, 1, 1, 2, 4, 0, 1)
29.	error	play,2,1,edmat001,21,CBBX,4E3/G34.3-H(NR)(UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 2, 4, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2068 *In the bottom of the 2nd inning, with bases loaded, SLN batter Tommy Edman hit a ground ball to the 1st/2nd base gap. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI), the runners on 2nd and 3rd to each advance one base and the batter to reach 1st. The bases are still loaded.*

	game	HOU vs SLN on 20230627
	state before	(0, 0, 0, 0, 1, 0, 0, 0)
30.	error	play,1,0,dubom001,00,X,E4/G4.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2069 This game had 2 errors. The final score was 2-4.
In the top of the 1st inning, with bases empty, HOU batter Mauricio Dubon hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.

	game	HOU vs SLN on 20230627
	state before	(1, 1, 0, 0, 5, 2, 2, 1)
31.	error	play,5,1,arenn001,10,BX,D9/F89XD+.1-H;B-3(E4/TH)
	state after	(1, 0, 0, 1, 5, 2, 3, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2070 *In the bottom of the 5th inning, with a runner on 1st, SLN batter Nolan Arenado hit a hard fly ball double to deep right center field. The runner on 1st scored (1 RBI). A throwing error by the second baseman allowed the batter to reach 3rd.*

29.2.4 July

	game	NYA vs SLN on 20230701 (1/2)
	state before	(1, 1, 0, 0, 3, 0, 3, 1)
32.	error	play,3,1,arenn001,00,X,FC5/G6.1-2(E5/TH);B-1
	state after	(1, 1, 1, 0, 3, 0, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2071 This game had 2 errors. The final score was 4-11.
 (By the way, the final score for game 2 was 6-2.)
In the bottom of the 3rd inning, with a runner on 1st, SLN batter Nolan

Arenado hit a ground ball into a fielder's choice at short. A throwing error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

33.	game	NYA vs SLN on 20230701 (1/2)
	state before	(1, 1, 0, 0, 6, 0, 11, 1)
	error	play,6,1,edmat001,22,*BSSFBX,FC1/G1.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 6, 0, 11, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2072 *In the bottom of the 6th inning, with a runner on 1st, SLN batter Tommy Edman hit a ground ball into a fielder's choice at shallow infield. A throwing error by the first baseman allowed the batter to reach 1st.*

34.	game	NYA vs SLN on 20230702
	state before	(0, 0, 0, 0, 6, 0, 2, 1)
	error	play,6,1,burla001,10,BX,C/E2.B-1
	state after	(0, 1, 0, 0, 6, 0, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 2073 This game had 2 errors. The final score was 1-5.
In the bottom of the 6th inning, with bases empty, SLN batter Alec Burleson had a 1-0 count. Catcher's interference allowed the batter to reach 1st base.

35.	game	NYA vs SLN on 20230702
	state before	(1, 0, 1, 0, 8, 1, 4, 1)
	error	play,8,1,kniza001,32,*BBBC.S>*S,K+SB3.2-H(E2/TH) (NR)
	state after	(2, 0, 0, 0, 8, 1, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 2074 *In the bottom of the 8th inning, with a runner on 2nd, SLN batter Andrew Knizner had a full count. During the next pitch, which was a swinging strike, the runner on 2nd stole 3rd. A throwing error by the catcher allowed that runner to score (unearned, no RBI). The batter struck out.*

	game	WAS vs SLN on 20230714
	state before	(0, 1, 0, 0, 5, 1, 1, 0)
36.	error	play,5,0,dickc002,02,F*S+1,P01(E2/TH).1-2
	state after	(0, 0, 1, 0, 5, 1, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2075 This game had 3 errors and one play with a double steal. (In fact, one of the steals was stealing home.) Two errors involved the same batter. The final score was 7-5.

In the top of the 5th inning, with a runner on 1st, WAS batter Corey Dickerson had a 0-2 count. The catcher tried to pick off the runner at 1st. A throwing error by the catcher allowed that runner to safely reach 2nd. Dickerson is still at bat with a 0-2 count.

	game	WAS vs SLN on 20230714
	state before	(2, 1, 0, 1, 5, 2, 1, 0)
37.	error	play,5,0,thoml002,11,C1*B>B,SBH;SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 5, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2076 *In the top of the 5th inning, with runners on 1st and 3rd, WAS batter Lane Thomas had a 1-1 count. During the next pitch, which was a ball, the runner on 3rd stole home (no RBI) and the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Thomas is still at bat, now with a 2-1 count.*

38.	game	WAS vs SLN on 20230714
	state before	(2, 0, 0, 1, 5, 3, 1, 0)
	error	play,5,0,thom1002,21,C1*B>B.X,E5/TH/G56.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2077 *The play above is continued. In the top of the 5th inning, with a runner on 3rd, WAS batter Lane Thomas hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

39.	game	MIA vs SLN on 20230717
	state before	(1, 1, 0, 0, 5, 2, 2, 1)
	error	play,5,1,contw001,32,BBCC>F1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 5, 2, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2078 *In the bottom of the 5th inning, with a runner on 1st, SLN batter Willson Contreras has a 3-2 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed the runner to safely reach 2nd.*

40.	game	MIA vs SLN on 20230718
	state before	(0, 0, 0, 0, 5, 1, 0, 1)
	error	play,5,1,burla001,22,BBCFFFX,S5/G56S.B-2(E5/TH)
	state after	(0, 0, 1, 0, 5, 1, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2079 *In the bottom of the 5th inning, with bases empty, SLN batter Alec Burleson hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	MIA vs SLN on 20230719
	state before	(1, 0, 1, 0, 7, 2, 4, 1)
41.	error	play,7,1,noot1001,20,BBX,S5/G56S.2-H(E5/TH)(NR);B-3
	state after	(1, 0, 0, 1, 7, 2, 5, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2080 *In the bottom of the 7th inning, with a runner on 2nd, SLN batter Lars Nootbaar hit a ground ball single to shallow 3rd. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 3rd.*

	game	CHN vs SLN on 20230727
	state before	(2, 0, 0, 0, 9, 10, 3, 1)
42.	error	play,9,1,donob001,01,CX,E5/G6S.B-1
	state after	(2, 1, 0, 0, 9, 10, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2081 *In the bottom of the 9th inning, with bases empty, SLN batter Brendan Donovan hit a ground ball to short. An error by the third baseman allowed the batter to reach 1st base.*

	game	CHN vs SLN on 20230730
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
43.	error	play,1,1,noot1001,11,BCX,S7/L78.B-2(E7)
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 2082 *In the bottom of the 1st inning, with bases empty, SLN batter Lars Nootbaar hit a line drive single to left center field. An error by the left fielder allowed the batter to reach 2nd.*

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	game	MIN vs SLN on 20230801
	state before	(2, 0, 0, 1, 5, 0, 0, 0)
44.	error	play,5,0,taylm002,00,X,E6/TH/G6.3-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2083 *In the top of the 5th inning, with a runner on 3rd, MIN batter Michael Taylor hit a ground ball to short. A throwing error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.*

	game	MIN vs SLN on 20230802
	state before	(2, 1, 0, 0, 7, 3, 7, 1)
45.	error	play,7,1,oneit001,02,*SC>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 3, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 2084 *In the bottom of the 7th inning, with a runner on 1st, SLN batter Tyler O'Neill hit a single had a 0-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. O'Neill is still at bat, now with a 1-2 count.*

	game	OAK vs SLN on 20230814
	state before	(2, 1, 0, 0, 3, 0, 1, 0)
46.	error	play,3,0,ruize001,00,X,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 3, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2085 This game had 2 errors. The final score was 5-7. *In the top of the 3rd inning, with a runner on 1st, OAK batter Esteury Ruiz had a 0-0 count. Catcher's interference allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	OAK vs SLN on 20230814
	state before	(0, 0, 0, 0, 9, 5, 7, 0)
47.	error	play,9,0,allen002,10,BX,E1/TH/G15.B-2
	state after	(0, 0, 1, 0, 9, 5, 7, 0)
	fielder	1 (pitcher on the home team)

Summary 2086 *In the top of the 9th inning, with bases empty, OAK batter Nick Allen hit a ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd base.*

	game	OAK vs SLN on 20230816
	state before	(0, 0, 0, 0, 5, 5, 0, 0)
48.	error	play,5,0,rookb001,32,BFSBBFX,E9/F89XD.B-3
	state after	(0, 0, 0, 1, 5, 5, 0, 0)
	fielder	9 (right field on the home team)

Summary 2087 *In the top of the 5th inning, with bases empty, OAK batter Brent Rooker hit a fly ball to extra deep right center field. An error by the right fielder allowed the batter to reach 3rd base.*

	game	NYN vs SLN on 20230817
	state before	(2, 0, 0, 0, 8, 3, 2, 1)
49.	error	play,8,1,oneit001,11,BCX,5E3/G56S.B-1
	state after	(2, 1, 0, 0, 8, 3, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2088 *In the bottom of the 8th inning, with bases empty, SLN batter Tyler O'Neill hit a ground ball to shallow 3rd. An error by the first baseman (on an assist from the shortstop) allowed the batter to reach 1st base.*

Visiting/Home team rosters for NYN vs SLN on 2023-08-19

position	innings	player	position	innings	player
1	1	Kodai Senga	1	1	Miles Mikolas
1	8	Vinny Nittoli	1	5	Andrew Suarez
2	1	Francisco Alvarez	1	9	Guillermo Zuniga
3	1	Pete Alonso	2	1	Andrew Knizner
4	1	Jeff McNeil	3	1	Paul Goldschmidt
5	1	Danny Mendick	3	8	Luken Baker
6	1	Francisco Lindor	4	1	Tommy Edman
6	8	Jonathan Arauz	5	1	Nolan Arenado
7	1	Rafael Ortega	5	8	Taylor Motter
7	8	Tim Locastro	6	1	Masyn Winn
8	1	Brandon Nimmo	7	1	Tyler O'Neill
8	8	Rafael Ortega	8	1	Richie Palacios
9	1	DJ Stewart	9	1	Alec Burleson

50.	game	NYN vs SLN on 20230819
	state before	(1, 0, 0, 0, 1, 2, 0, 1)
	error	play,1,1,goldp001,10,BX,4E1/G4.B-1
	state after	(1, 1, 0, 0, 1, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2089 This game had 3 errors. The final score was 13-2. *In the bottom of the 1st inning, with bases empty, SLN batter Paul Goldschmidt hit a ground ball to 2nd. An error by the pitcher (on an assist from the second baseman) allowed the batter to reach 1st base.*

51.	game	NYN vs SLN on 20230819
	state before	(1, 0, 0, 0, 3, 2, 0, 0)
	error	play,3,0,nimmb001,22,CBFBX,E4/TH/G34.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2090 *In the top of the 3rd inning, with bases empty, NYN batter Brandon Nimmo hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	NYN vs SLN on 20230819
	state before	(2, 1, 1, 1, 8, 10, 1, 0)
52.	error	play,8,0,lindf001,12,F.BFX,S5/G56S.3-H;2-H(E5/TH)(NR)(UR);1-H(NR)(UR);B-2
	state after	(2, 0, 1, 0, 8, 13, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2091 *In the top of the 8th inning, with bases loaded, NYN batter Francisco Lindor hit a ground ball single to shallow 3rd base. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the runners on 1st and 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

	game	NYN vs SLN on 20230820
	state before	(2, 1, 0, 0, 8, 2, 7, 0)
53.	error	play,8,0,alvaf001,22,SCB*BX,S7/L7D+.2-H;1XH(742)(E7);B-2
	state after	(0, 0, 0, 0, 8, 3, 7, 1)
	fielder	7 (left field on the home team)

Summary 2092 *In the top of the 8th inning, with 2 outs and a runner on 1st, NYN batter Francisco Alvarez hit a hard line drive single to deep left field. The runner on 2nd scored (1 RBI). A error by the left fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd. However, the runner on 1st was tagged out at home by the catcher (assists by left fielder and second baseman). This out ended the half-inning.*

	game	SDN vs SLN on 20230828
	state before	(0, 0, 0, 0, 5, 0, 0, 1)
54.	error	play,5,1,winnm001,00,X,E4/G4.B-1
	state after	(0, 1, 0, 0, 5, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2093 *In the bottom of the 5th inning, with bases empty, SLN batter Masyn Winn hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	SDN vs SLN on 20230829
	state before	(0, 1, 0, 0, 1, 0, 0, 0)
55.	error	play,1,0,sotoj001,00,X,S2/BG2-.1-3(E2/TH);B-2
	state after	(0, 0, 1, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 2094 This game had 2 errors. The final score was 5-6.
In the top of the 1st inning, with a runner on 1st, SDN batter Juan Soto hit a soft bunt single to the shallow infield. A throwing error by the catcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	SDN vs SLN on 20230829
	state before	(0, 0, 0, 0, 4, 3, 1, 1)
56.	error	play,4,1,contw001,22,BCTBX,E6/G6D.B-1
	state after	(0, 1, 0, 0, 4, 3, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2095 *In the bottom of the 4th inning, with bases empty, SLN batter Willson Contreras hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

29.2.6 September

	game	PIT vs SLN on 20230901
	state before	(1, 0, 0, 0, 9, 1, 1, 1)
57.	error	play,9,1,edmat001,21,BSBX,S1/G1.B-2(E1/TH)
	state after	(1, 0, 1, 0, 9, 1, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2096 *In the bottom of the 9th inning, with bases empty, SLN batter Tommy Edman hit a ground ball single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	PIT vs SLN on 20230903
	state before	(0, 1, 0, 0, 4, 3, 3, 1)
58.	error	play,4,1,nootl001,11,B1CX,FC6/G6.1-2(E6);B-1
	state after	(0, 1, 1, 0, 4, 3, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2097 This game had 2 errors. The final score was 4-6.
In the bottom of the 4th inning, with a runner on 1st, SLN batter Lars Nootbaar hit a ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to reach 2nd and the batter to reach 1st.

	game	PIT vs SLN on 20230903
	state before	(2, 0, 1, 0, 7, 3, 5, 1)
59.	error	play,7,1,palar002,32,CBFFFFBFFFBX,S4/G4.2-H(E4/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 7, 3, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2098 *In the bottom of the 7th inning, with a runner on 2nd, SLN batter Richie Palacios (after an 11 pitch at-bat!) hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

Visiting/Home team rosters for PHI vs SLN 2023-09-15

position	innings	player	position	innings	player
1	1	Aaron Nola	1	1	Zack Thompson
1	5	Matt Strahm	1	6	Casey Lawrence
1	6	Jeff Hoffman	1	6	Matthew Liberatore
1	7	Seranthony Dominguez	1	8	Drew VerHagen
1	8	Craig Kimbrel	1	9	Andre Pallante
1	9	Jose Alvarado	2	1	Willson Contreras
2	1	J.T. Realmuto	3	1	Paul Goldschmidt
3	1	Bryce Harper	4	1	Tommy Edman
3	3	Alec Bohm	5	1	Nolan Arenado
4	1	Bryson Stott	6	1	Masyn Winn
5	1	Alec Bohm	7	1	Richie Palacios
5	3	Edmundo Sosa	7	6	Tyler O'Neill
6	1	Trea Turner	8	1	Lars Nootbaar
7	1	Cristian Pache	9	1	Jordan Walker
7	9	Brandon Marsh			
8	1	Johan Rojas			
9	1	Nick Castellanos			

	game	PHI vs SLN on 20230915
	state before	(0, 0, 0, 0, 1, 4, 0, 1)
60.	error	play,1,1,noot1001,10,BX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 1, 4, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2099 This game had 3 errors. The final score was 5-4.
In the bottom of the 1st inning, with bases empty, SLN batter Lars Nootbaar hit a ground ball to shortstop. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	PHI vs SLN on 20230915
	state before	(1, 1, 0, 0, 4, 4, 2, 1)
61.	error	play,4,1,edmat001,10,*BX,E1/TH/G23-.1-3;B-1
	state after	(1, 1, 0, 1, 4, 4, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2100 *In the bottom of the 4th inning, with a runner on 1st, SLN batter Tommy Edman hit a soft ground ball to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	PHI vs SLN on 20230915
	state before	(2, 0, 0, 0, 6, 5, 3, 1)
62.	error	play,6,1,winnm001,31,BBBCX,E6/L6MD.B-1
	state after	(2, 1, 0, 0, 6, 5, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2101 *In the bottom of the 6th inning, with bases empty, SLN batter Masyn Winn hit a line drive to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	PHI vs SLN on 20230917
	state before	(1, 1, 1, 1, 4, 0, 3, 0)
63.	error	play,4,0,castn001,00,X,64(1)/FO/G6.3-H;2-H(E4/TH)(NR);B-2
	state after	(2, 0, 1, 1, 4, 0, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2102 *In the top of the 4th inning, with bases loaded, PHI batter Nick Castellanos hit a ground ball to short. The runner from 1st was forced out at 2nd by the second baseman with an assist by the shortstop. The runner on 3rd scored. A throwing error by the second baseman allowed the runner on 2nd to also score (unearned, no RBI) and the batter to reach 2nd.*

	game	MIL vs SLN on 20230921
	state before	(0, 0, 0, 0, 5, 1, 0, 0)
64.	error	play,5,0,turab002,12,CBCX,E9/L89XD.B-3(E9/TH)
	state after	(0, 0, 0, 1, 5, 1, 0, 0)
	fielder	9, 9 (right field on the home team)

Summary 2103 This play had 2 errors, both by the right fielder.
In the top of the 5th inning, with bases empty, MIL batter Brice Turang hit a line drive to extra deep right center field. An error by the right fielder allowed the batter to reach 3rd.

Visiting/Home team rosters for CIN vs SLN on 2023-09-29

position	innings	player	position	innings	player
1	1	Brandon Williamson	1	1	Jake Woodford
1	7	Carson Spiers	1	2	Casey Lawrence
1	8	Carson Spiers	1	4	Andrew Suarez
2	1	Tyler Stephenson	1	8	Jacob Barnes
3	1	Christian Encarnacion-Strand	1	9	Matthew Liberatore
4	1	Jonathan India	2	1	Andrew Knizner
5	1	Noelvi Marte	3	1	Paul Goldschmidt
6	1	Elly De La Cruz	3	6	Juniel Querecuto
7	1	Spencer Steer	4	1	Tommy Edman
7	8	Nick Martini	4	9	Irving Lopez
8	1	TJ Friedl	5	1	Jose Fermin
8	7	Stuart Fairchild	6	1	Masyn Winn
9	1	Will Benson	7	1	Richie Palacios
9	8	Nick Senzel	8	1	Lars Nootbaar
			8	6	Michael Siani
			9	1	Jordan Walker

	game	CIN vs SLN on 20230929
	state before	(0, 0, 0, 0, 2, 7, 0, 1)
65.	error	play,2,1,bakel001,11,CBX,E4/TH/G4S-.B-1
	state after	(0, 1, 0, 0, 2, 7, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2104 This game had 4 errors, including one play with a double error. The final score was 19-2. Note: On 6/24/2023-06-24, the (visiting) Angels beat the Rockies 25-1, so this is not a record run differential.

In the bottom of the 2nd inning, with bases empty, SLN batter Luken Baker hit a soft ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.

	game	CIN vs SLN on 20230929
	state before	(2, 1, 1, 1, 8, 14, 2, 0)
66.	error	play,8,0,stept001,10,BX,E6/G6.3-H(NR)(UR);2-H(E6/TH)(NR)(UR);1-3;B-2
	state after	(2, 0, 1, 1, 8, 16, 2, 0)
	fielder	6, 6 (shortstop on the home team)

Summary 2105 *In the top of the 8th inning, with bases loaded, CIN batter Tyler Stephenson hit a ground ball to short. The **first** error by the shortstop allowed the runner on 3rd to score (unearned, no RBI). A throwing error by the shortstop (this was the **second** error) allowed the runner on 2nd to also score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	CIN vs SLN on 20230929
	state before	(2, 0, 0, 1, 8, 18, 2, 0)
67.	error	play,8,0,martn004,12,CFBX,S5/G6S.3-H(UR);B-2(E5/TH)
	state after	(2, 0, 1, 0, 8, 19, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2106 *In the top of the 8th inning, with a runner on 3rd, CIN batter Noelvi Marte hit a ground ball single to shallow short. The runner on 3rd scored (1 RBI). A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	CIN vs SLN on 20230930
	state before	(0, 1, 0, 0, 7, 6, 11, 1)
68.	error	play,7,1,winnm001,11,BC111,P01(E1/TH).1-2
	state after	(0, 0, 1, 0, 7, 6, 11, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2107 *In the bottom of the 7th inning, with a runner on 1st, SLN batter Masyn Winn had a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 2nd. Winn is still at bat with a 1-1 count.*

Chapter 30

Tampa Bay Rays (TBA)

The Tampa Bay Rays, founded in 1998 as the Tampa Bay Devil Rays, made their breakthrough in 2008, when they changed their name to simply the Tampa Bay Rays and they won their first-ever American League pennant and advanced to the World Series. They lost to the Phillies. The Rays are one of 5 MLB teams that have never won a World Series title¹. In 2023, the Rays won 99 games, with a .611 winning percentage, but did not get past the first round of the post-season playoffs².

This chapter is devoted to listing the errors in each regular season game of 2023 played in Tampa Bay against the Rays. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 TBA team is 0.703 compared with the MLB average of 0.691 that year [BR]. The standard deviation of these 2023 DER values is $\sigma = 0.011$, so the Rays in 2023 were about one standard deviation above the MLB average. This is a “high” DER (which is good).

In TBA 2023, there were 73 errors in the 81 home games, 35 by the Rays and the rest (38) by visiting teams. The total number of errors by the team

¹The Rays other AL pennant win was in 2020, but they lost the World Series to the Dodgers.

²Unfortunately, they lost all their games in their Wild Card series against the Rangers.

(home and away) over the regular season in 2023 was 75. Since 83.9 was the MLB average, the Rays had a error rate in the lower 30% of all MLB teams (which is also good).

30.1 Season totals

The Official Scorer's judgment must determine whether a **run batted in** shall be credited for a run that scores when a fielder holds the ball or throws to a wrong base. Ordinarily, if the runner keeps going, the Official Scorer should credit a run batted in; if the runner stops and takes off again when the runner notices the misplay, the Official Scorer should credit the run as scored on a fielder's choice.

– **MLB Rule Book** 9.04(c)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

30.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	38
Runs by ht because of errors by vt:	17
Errors by ht:	35
Runs by vt because of errors by ht:	8

This is an excellent error-to-run conversion rate: $17/38 = 44.7\%$ on the part of the Rays (compared to the merely good rate $8/35 = 22.8\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 TBA season (home and away) has about $R/PA = 860/6164$ (about 13.9%) chance of resulting in a run. Therefore, a randomly selected visitor's error is over three times as likely to result in a (TBA) run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 73, so the expected number of errors (by either team) in a randomly selected TBA 2023 home game is about 0.9.

Errors by position	vt	ht
errors due to pitcher:	4	3
to catcher:	6	6
to 1st baseman:	4	2
to 2nd baseman:	8	7
to 3rd baseman:	4	7
to shortstop:	6	7
to left fielder:	3	0
to center fielder:	1	2
to right fielder:	2	1

30.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	13	13	26
(lone) runner on 1st	14	13	27
(lone) runner on 2nd	0	2	2
(lone) runner on 3rd	2	0	2
two runners on base	7	7	14
bases loaded	2	0	2
total	38	35	73

A table of frequencies of most common errors by the Rays in TBA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	4	11.4%
(1, 0, 0, 0)	5	14.3%
(2, 0, 0, 0)	4	11.4%
(0, 1, 0, 0)	3	8.6%
(1, 1, 0, 0)	7	20%
(2, 1, 0, 0)	3	8.6%
(1, 0, 1, 0)	1	2.9%
(0, 1, 1, 0)	3	8.6%
(0, 0, 1, 1)	1	2.9%

A table of frequencies of most common errors by a visiting team in TBA 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	21.1%
(1, 0, 0, 0)	2	5.3%
(2, 0, 0, 0)	3	7.9%
(0, 1, 0, 0)	3	7.9%
(1, 1, 0, 0)	9	23.7%
(2, 1, 0, 0)	2	5.3%
(1, 1, 1, 0)	3	7.9%
(2, 1, 0, 1)	1	2.6%
(1, 0, 1, 1)	2	5.3%
(1, 1, 1, 1)	2	5.3%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Rays homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file TBA2023.eva):

state	TBA2023 percentages
(0, 0, 0, 0)	24.4%
(1, 0, 0, 0)	16.2%
(2, 0, 0, 0)	12.0%
(0, 1, 0, 0)	5.8%
(1, 1, 0, 0)	7.7%
(2, 1, 0, 0)	7.0%
(0, 0, 1, 0)	2.1%
(1, 0, 1, 0)	3.4%
(2, 0, 1, 0)	4.2%
(0, 0, 0, 1)	0.3%
(1, 0, 0, 1)	1.1%
(2, 0, 0, 1)	1.4%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	2.9%

30.1.3 Some interesting errors

Some of the most interesting error plays in the TBA 2023 home games:

- The game CLE vs TBA on 2023-08-11 has 4 errors.
- The game BOS vs TBA on 2023-09-05 had 4 errors.

30.1.4 Relative team rankings

We have 81 home games between TBA and some visiting MLB team (vt). A number of teams did not play TBA in the regular season. However, of those that did, TBA *lost* more games than won against³:

ATL (2/3), HOU (2/3), SLN (2/3), BAL (4/6) .

³We give the vt's W/T record in parentheses (W = numbers of games vt won against TBA, T = Total home games vt played vs the Rays at Tampa Bay.

The teams that *split* their series were:

MIA (1/2), KCA (2/4) .

The teams that TBA *won* more than lost against

CHA (0/3), COL (0/3), DET (0/3), MIN (0/3), OAK (0/3), PIT (0/3), MIA
(1/2),
ANA (1/3), CLE (1/3), LAN (1/3), MIL (1/3), TEX (1/3), SEA (1/4), BOS
(1/7) .

The teams that did not play in any of the 81 home games in TBA 2023 were:

ARI, CHN, CIN, NYN, SDN, SFN, WAS .

30.2 Game-by-game totals

Any runner is out when he runs more than three feet away from his base path to avoid being tagged unless his action is to avoid interference with a fielder fielding a batted ball. A runner's base path is established when the tag attempt occurs and is a straight line from the runner to the base he is attempting to reach safely.

– **MLB Rule Book** 5.09(b)(1)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for TBA 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Christian Bethancourt	103	.254
3 (1B)	Yandy Díaz	118	.410
4 (2B)	Brandon Lowe	106	.328
5 (3B)	Isaac Paredes	117	.352
6 (SS)	Wander Franco	111	.344
7 (LF)	Randy Arozarena	139	.364
8 (CF)	Jose Siri	101	.267
9 (RF)	Josh Lowe	101	.335

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 TBA home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

30.2.1 April

Visiting/Home team rosters for DET vs TBA on 2023-04-01

position	innings	player	position	innings	player
1	1	Spencer Turnbull	1	1	Zach Eflin
1	3	Trey Wingenter	1	6	Kevin Kelly
1	3	Tyler Alexander	1	8	Garrett Cleavinger
1	7	Garrett Hill	1	9	Calvin Faucher
1	8	Jose Cisnero	2	1	Francisco Mejia
2	1	Eric Haase	3	1	Yandy Diaz
2	8	Jake Rogers	4	1	Brandon Lowe
3	1	Spencer Torkelson	4	5	Taylor Walls
4	1	Zach McKinstry	5	1	Taylor Walls
5	1	Nick Maton	5	5	Isaac Paredes
6	1	Javier Baez	6	1	Wander Franco
6	7	Ryan Kreidler	7	1	Randy Arozarena
7	1	Austin Meadows	8	1	Jose Siri
7	8	Eric Haase	9	1	Josh Lowe
8	1	Riley Greene			
9	1	Kerry Carpenter			

	game	DET vs TBA on 20230401
	state before	(2, 1, 0, 0, 2, 0, 1, 0)
1.	error	play,2,0,cabrm001,00,X,E5/G56.1-2;B-1
	state after	(2, 1, 1, 0, 2, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2108 This game had 3 errors. The final score was 2-12.
In the top of the 2nd inning, with a runner on 1st, DET batter Miguel Cabrera hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	DET vs TBA on 20230401
	state before	(1, 1, 1, 0, 3, 0, 3, 1)
2.	error	play,3,1,lowej002,01,C+2,P02(E2/TH).2-3;1-2
	state after	(1, 0, 1, 1, 3, 0, 3, 1)
	fielder	2 (catcher on the visiting team)

Summary 2109 *In the bottom of the 3rd inning, with runners on 1st and 2nd, TBA batter Josh Lowe has a 0-1 count. The catcher attempted to pickoff the runner on 2nd. A throwing error by the catcher allowed the runners on 1st and 2nd to each advance one base. Lowe is still at bat with a 0-1 count.*

	game	DET vs TBA on 20230401
	state before	(1, 1, 0, 0, 6, 2, 10, 0)
3.	error	play,6,0,torks001,00,X,FC5/G56.1-2(E5);B-1
	state after	(1, 1, 1, 0, 6, 2, 10, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2110 *In the top of the 6th inning, with a runner on 1st, DET batter Spencer Torkelson hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	OAK vs TBA on 20230408
	state before	(1, 0, 0, 0, 4, 0, 0, 0)
4.	error	play,4,0,aguij001,01,CX,C/E2.B-1
	state after	(1, 1, 0, 0, 4, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 2111 *In the top of the 4th inning, with bases empty, OAK batter Jesus Aguilar had a 0-1 count. Catcher's interference allowed the batter to take 1st.*

	game	OAK vs TBA on 20230409
	state before	(0, 1, 0, 0, 6, 0, 7, 1)
5.	error	play,6,1,franw002,11,SBX,4/P6MD.1-3(E4/TH)
	state after	(1, 0, 0, 1, 6, 0, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2112 *In the bottom of the 6th inning, with a runner on 1st, TBA batter Wander Franco hit a pop up to deep short. The second baseman put out the batter. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd.*

	game	BOS vs TBA on 20230411
	state before	(1, 0, 1, 1, 7, 1, 6, 1)
6.	error	play,7,1,arozr001,31,BB*BSX,FC3/G34.3-H(E3)(NR)(UR);2-3;B-1
	state after	(1, 1, 0, 1, 7, 1, 7, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2113 *In the bottom of the 7th inning, with runners on 2nd and 3rd, TBA batter Randy Arozarena hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the first baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 2nd to advance to 3rd, and the batter to reach 1st.*

	game	BOS vs TBA on 20230412
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
7.	error	play,1,1,franw002,00,X,E6/G6.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2114 *In the bottom of the 1st inning, with bases empty, TBA batter Wander Franco hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs TBA on 20230422
	state before	(1, 1, 0, 0, 5, 2, 3, 1)
8.	error	play,5,1,loweb001,00,1,P01(E1/TH) .1-3
	state after	(1, 0, 0, 1, 5, 2, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2115 *In the bottom of the 5th inning, with a runner on 1st, TBA batter Brandon Lowe has a 0-0 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely advance to 3rd. Lowe is still at bat with a 0-0 count.*

	game	CHA vs TBA on 20230423
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
9.	error	play,1,0,vauga001,00,>B,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 1, 0, 0, 0)
	fielder	2 (catcher on the home team)

Summary 2116 *In the top of the 1st inning, with a runner on 1st, CHA batter Andrew Vaughn had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Vaughn is still at bat, with a 0-0 count.*

	game	HOU vs TBA on 20230426
	state before	(1, 1, 0, 1, 1, 0, 0, 0)
10.	error	play,1,0,brega001,21,BBSX,E6/G6.3-H;1-2;B-1
	state after	(1, 1, 1, 0, 1, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2117 *In the top of the 1st inning, with runners on 1st and 3rd, HOU batter Alex Bregman hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (1 RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

30.2.2 May

Visiting/Home team rosters for PIT vs TBA on 2023-05-03

position	innings	player	position	innings	player
1	1	Mitch Keller	1	1	Shane McClanahan
1	6	Jose Hernandez	1	7	Chase Anderson
1	7	Cody Bolton	2	1	Francisco Mejia
2	1	Austin Hedges	3	1	Yandy Diaz
3	1	Carlos Santana	4	1	Brandon Lowe
4	1	Ji Hwan Bae	5	1	Taylor Walls
5	1	Ke'Bryan Hayes	6	1	Wander Franco
6	1	Rodolfo Castro	7	1	Luke Raley
7	1	Connor Joe	8	1	Jose Siri
8	1	Bryan Reynolds	9	1	Josh Lowe
9	1	Miguel Andujar			

	game	PIT vs TBA on 20230503
	state before	(2, 1, 0, 1, 3, 1, 0, 1)
11.	error	play,3,1,ramih004,01,CX,E6/G6.3-H(NR)(UR);1-2;B-1
	state after	(2, 1, 1, 0, 3, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2118 This game had 3 errors, including a play with a double error. The final score was 1-8.

In the bottom of the 3rd inning, with runners on 1st and 3rd, TBA batter Harold Ramirez hit a ground ball to short. An error by the shortstop

allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.

12.	game	PIT vs TBA on 20230503
	state before	(1, 0, 0, 1, 5, 1, 3, 1)
	error	play,5,1,franw002,12,BFCX,E4/G34.3-H;B-3(E4/TH)
	state after	(1, 0, 0, 1, 5, 1, 4, 1)
	fielder	4,4 (2nd baseman on the visiting team)

Summary 2119 This play had 2 errors, but the same fielder. The final score was 1-8.

In the bottom of the 5th inning, with a runner on 3rd, TBA batter Wander Franco hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed runner on 3rd scored (1 RBI). Another (throwing) error by the second baseman also allowed the batter to reach 3rd.

13.	game	PIT vs TBA on 20230504
	state before	(1, 1, 0, 0, 9, 0, 3, 0)
	error	play,9,0,reybn001,00,X,FC4/G34.1-2(E4);B-1
	state after	(1, 1, 1, 0, 9, 0, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2120 *In the top of the 9th inning, with a runner on 1st, PIT batter Bryan Reynolds hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

14.	game	NYA vs TBA on 20230505
	state before	(2, 1, 0, 0, 7, 4, 4, 0)
	error	play,7,0,rizza001,01,C1>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 4, 4, 0)
	fielder	2 (catcher on the home team)

Summary 2121 This game had 2 errors. The final score was 4-5.

In the top of the 7th inning, with a runner on 1st, NYA batter Anthony Rizzo had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Rizzo is still at bat, with a 1-1 count.

	game	NYA vs TBA on 20230505
	state before	(1, 1, 0, 0, 7, 4, 4, 1)
15.	error	play,7,1,franw002,12,BFFX,D7/L7D.1-H(E7)(NR)
	state after	(1, 0, 1, 0, 7, 4, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 2122 *In the bottom of the 7th inning, with a runner on 1st, TBA batter Wander Franco hit a line drive double to deep left field. An error by the left fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 2nd.*

	game	NYA vs TBA on 20230506
	state before	(1, 1, 0, 0, 2, 0, 2, 0)
16.	error	play,2,0,bauej001,10,BX,E4/L89.1-2;B-1
	state after	(1, 1, 1, 0, 2, 0, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2123 This game had 2 errors. The final score was 3-2.

In the top of the 2nd inning, with a runner on 1st, NYA batter Jake Bauers hit a line drive to right center field. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	NYA vs TBA on 20230506
	state before	(1, 1, 0, 0, 6, 0, 2, 0)
17.	error	play,6,0,rizza001,10,BX,46(1)/F0/G34.B-2(E6/TH)
	state after	(2, 0, 1, 0, 6, 0, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2124 *In the top of the 6th inning, with a runner on 1st, NYA batter Anthony Rizzo hit a ground ball to the 1st/2nd base gap. At 2nd, the shortstop (with an assist from the second baseman) forced out the runner from 1st. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	NYA vs TBA on 20230507
	state before	(1, 1, 0, 0, 5, 6, 1, 1)
18.	error	play,5,1,franw002,21,CBBX,S4/G3.1-H(E4/TH) (NR) (UR) ;B-2
	state after	(1, 0, 1, 0, 5, 6, 2, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2125 *In the bottom of the 5th inning, with a runner on 1st, TBA batter Wander Franco hit a ground ball single to 1st base. A throwing error by the second baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	MIL vs TBA on 20230519
	state before	(1, 1, 0, 0, 8, 0, 0, 1)
19.	error	play,8,1,mejif001,00,>S,SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 8, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2126 *In the bottom of the 8th inning, with a runner on 1st, TBA batter Francisco Mejia had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Mejia is still at bat, now with a 0-1 count.*

	game	TOR vs TBA on 20230522
	state before	(2, 0, 0, 1, 2, 0, 0, 1)
20.	error	play,2,1,bethc001,00,X,4E3/G4M.3-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 2, 0, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2127 This game had 2 errors. The final score was 4-6.

In the bottom of the 2nd inning, with a runner on 3rd, TBA batter Christian Bethancourt hit a ground ball to 2nd. An error by the first baseman (with an assist from the second baseman) allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.

	game	TOR vs TBA on 20230522
	state before	(0, 1, 0, 0, 3, 0, 3, 1)
21.	error	play,3,1,franw002,11,CBX,31E3/G3+.1-2;B-1
	state after	(0, 1, 1, 0, 3, 0, 3, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2128 *In the bottom of the 3rd inning, with a runner on 1st, TBA batter Wander Franco hit a hard ground ball to 1st base. An error by the first baseman (with an assist from the pitcher) allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	TOR vs TBA on 20230523
	state before	(2, 0, 0, 0, 5, 9, 0, 0)
22.	error	play,5,0,bichb001,32,BSFFFB.BX,E5/TH/G5.B-2
	state after	(2, 0, 1, 0, 5, 9, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2129 *In the top of the 5th inning, with bases empty, TOR batter Bo Bichette hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	TOR vs TBA on 20230525
	state before	(1, 1, 1, 0, 3, 1, 3, 1)
23.	error	play,3,1,margm001,11,BF>B,SB3;SB2.2-H(E2/TH)(NR);1-3
	state after	(1, 0, 0, 1, 3, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2130 This game had 2 errors. The final score was 3-6.

In the bottom of the 3rd inning, with runners on 1st and 2nd, TBA batter Manuel Margot had a 1-1 count. During the next pitch, which was a ball, the runners on 1st and 2nd each stole the next base. A throwing error by the catcher allowed the runner from 2nd (now on 3rd) to score (unearned, no RBI) and the runner from 1st (now on 2nd) to advance to 3rd. Margot is at bat, now with a 2-1 count.

	game	TOR vs TBA on 20230525
	state before	(0, 0, 0, 0, 4, 1, 5, 0)
24.	error	play,4,0,belbt001,12,BCFX,E3/G3.B-1
	state after	(0, 1, 0, 0, 4, 1, 5, 0)
	fielder	3 (1st baseman on the home team)

Summary 2131 *In the top of the 4th inning, with bases empty, TOR batter Brandon Belt hit a ground ball to 1st base. An error by the first baseman allowed the batter to reach 1st base.*

	game	LAN vs TBA on 20230528
	state before	(0, 1, 1, 0, 3, 4, 7, 0)
25.	error	play,3,0,muncm001,11,BCX,FC4/G34.2-H(NR);1-3(E4/TH);B-2
	state after	(0, 0, 1, 1, 3, 5, 7, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2132 This game had 2 errors. The final score was 10-11.

In the top of the 3rd inning, with runners on 1st and 2nd, LAN batter Max Muncy hit a ground ball into a fielder's choice at the 1st/2nd base gap. The runner on 2nd scored (unearned, no RBI). A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	LAN vs TBA on 20230528
	state before	(1, 0, 0, 0, 7, 9, 10, 0)
26.	error	play,7,0,rojam002,20,BBX,E8/L8D.B-2
	state after	(1, 0, 1, 0, 7, 9, 10, 0)
	fielder	8 (center field on the home team)

Summary 2133 *In the top of the 7th inning, with bases empty, LAN batter Miguel Rojas hit a line drive to deep center field. An error by the center fielder allowed the batter to reach 2nd base.*

30.2.3 June

	game	MIN vs TBA on 20230608
	state before	(1, 1, 1, 0, 6, 1, 3, 1)
27.	error	play,6,1,ralel001,11,.BTN,SB3.2-H(E1/TH) (NR) (UR);1-2
	state after	(1, 1, 1, 0, 6, 1, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2134 *In the bottom of the 6th inning, with runners on 1st and 2nd, TBA batter Luke Raley had a 1-1 count. During the next pitch, which was a “no pitch”, the runner on 2nd stole 3rd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI) and the runner on 1st to advance to 2nd. Raley is still at bat with a 1-1 count.*

	game	TEX vs TBA on 20230609
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
28.	error	play,1,1,ramih004,12,CSBFX,1E4/G34.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2135 *In the bottom of the 1st inning, with bases empty, TBA batter Harold Ramirez hit a ground ball to the 1st/2nd base gap. An error by the second baseman (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	TEX vs TBA on 20230610
	state before	(0, 1, 0, 0, 2, 0, 0, 0)
29.	error	play,2,0,jankt001,00,X,FC4/G34.1-H(E4/TH) (NR) (UR);B-2
	state after	(0, 0, 1, 0, 2, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2136 *In the top of the 2nd inning, with a runner on 1st, TEX batter Travis Jankowski hit a ground ball into a fielder's choice at 1st. A throwing error by the second baseman allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	TEX vs TBA on 20230611
	state before	(0, 1, 0, 0, 3, 1, 4, 0)
30.	error	play,3,0,semim001,21,S*BBX,D8/F78XD+.1-H;B-3(E8)
	state after	(0, 0, 0, 0, 1, 3, 2, 4, 0)
	fielder	8 (center field on the home team)

Summary 2137 *In the top of the 3rd inning, with a runner on 1st, TEX batter Marcus Semien hit a hard fly ball double to extra deep left center field. The runner on 1st score (1 RBI). An error by the center fielder allowed the batter to reach 3rd.*

	game	BAL vs TBA on 20230621
	state before	(0, 0, 0, 0, 2, 0, 2, 1)
31.	error	play,2,1,margm001,21,BBFX,E1/G1-.B-1
	state after	(0, 1, 0, 0, 2, 0, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2138 This game had 2 errors. The final score was 2-7. *In the bottom of the 2nd inning, with bases empty, TBA batter Manuel Margot hit a soft ground ball to shallow infield. An error by the pitcher allowed the batter to reach 1st base.*

	game	BAL vs TBA on 20230621
	state before	(1, 0, 1, 1, 2, 0, 2, 1)
32.	error	play,2,1,sirij001,22,SS.BFFBX,S1/G2-.3-H(UR);2-H(E1/TH)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 2, 0, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2139 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, TBA batter Jose Siri hit a soft ground ball single to very shallow infield. The runner on 3rd scored (1 RBI). A throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the batter reach 2nd.*

Visiting/Home team rosters for KCA vs TBA on 2023-06-22

position	innings	player	position	innings	player
1	1	Jose Cuas	1	1	Shane McClanahan
1	2	Austin Cox	1	4	Kevin Kelly
1	6	Carlos Hernandez	1	6	Shawn Armstrong
1	7	Nick Wittgren	1	7	Robert Stephenson
1	7	Taylor Clarke	1	8	Jason Adam
1	8	Aroldis Chapman	1	9	Pete Fairbanks
1	9	Scott Barlow	2	1	Francisco Mejia
2	1	Salvador Perez	3	1	Yandy Diaz
3	1	Matt Duffy	4	1	Vidal Brujan
3	8	Nick Pratto	5	1	Isaac Paredes
4	1	Samad Taylor	6	1	Taylor Walls
4	8	Nicky Lopez	7	1	Randy Arozarena
5	1	Maikel Garcia	8	1	Jose Siri
6	1	Bobby Witt Jr.	9	1	Luke Raley
7	1	Dairon Blanco			
8	1	Drew Waters			
9	1	MJ Melendez			

	game	KCA vs TBA on 20230622
	state before	(0, 1, 1, 0, 1, 0, 0, 1)
33.	error	play,1,1,arozr001,11,CBX,S9/L9.2-H;1-3;B-2(E9/TH)
	state after	(0, 0, 1, 1, 1, 0, 1, 1)
	fielder	9 (right field on the visiting team)

Summary 2140 This game had 3 errors. The final score was 6-5.

In the bottom of the 1st inning, with runners on 1st and 2nd, TBA batter Randy Arozarena hit a line drive single to right field. The runner on 2nd scored (1 RBI) and the runner from 1st advanced to 3rd. A throwing error by the right fielder allowed the batter reached 2nd.

	game	KCA vs TBA on 20230622
	state before	(1, 1, 0, 0, 2, 0, 2, 1)
34.	error	play,2,1,mejif001,22,.C1SB*BX,E9/L9.1-2;B-1
	state after	(1, 1, 1, 0, 2, 0, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 2141 *In the bottom of the 2nd inning, with a runner on 1st, TBA batter Francisco Mejia hit a line drive to right field. An error by the right fielder allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	KCA vs TBA on 20230622
	state before	(0, 0, 0, 0, 4, 2, 2, 0)
35.	error	play,4,0,tayls003,11,FBX,E5/G5.B-1
	state after	(0, 1, 0, 0, 4, 2, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2142 *In the top of the 4th inning, with bases empty, KCA batter Samad Taylor hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	KCA vs TBA on 20230623
	state before	(2, 0, 0, 0, 3, 3, 3, 0)
36.	error	play,3,0,melem001,12,BCSX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 3, 3, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 2143 *In the top of the 3rd inning, with bases empty, KCA batter MJ Melendez hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	KCA vs TBA on 20230624
	state before	(2, 1, 0, 0, 4, 7, 4, 1)
37.	error	play,4,1,franw002,00,1>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 4, 7, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2144 *In the bottom of the 4th inning, with a runner on 1st, TBA batter Wander Franco had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Franco is still at bat, with a 0-1 count.*

	game	KCA vs TBA on 20230625
	state before	(1, 1, 0, 0, 7, 1, 1, 1)
38.	error	play,7,1,diazy001,01,1CX,S16/G4MS.1-3(E6/TH);B-2
	state after	(1, 0, 1, 1, 7, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2145 *In the bottom of the 7th inning, with a runner on 1st, TBA batter Yandy Diaz hit a ground ball single to shallow 2nd. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

30.2.4 July

	game	PHI vs TBA on 20230704
	state before	(1, 1, 0, 0, 3, 1, 0, 1)
39.	error	play,3,1,diazy001,22,CBBCX,FC6/G6+.1-3(E6);B-1
	state after	(1, 1, 0, 1, 3, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2146 This game had 2 errors. The final score was 3-1.

In the bottom of the 3rd inning, with a runner on 1st, TBA batter Yandy Diaz hit a soft ground ball into a fielder's choice at short. An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	PHI vs TBA on 20230704
	state before	(1, 0, 1, 0, 7, 1, 0, 0)
40.	error	play,7,0,realj001,21,SBBX,E6/G6.2-3;B-1
	state after	(1, 1, 0, 1, 7, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2147 *In the top of the 7th inning, with a runner on 2nd, PHI batter J.T. Realmuto hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	PHI vs TBA on 20230706
	state before	(1, 0, 0, 0, 6, 1, 1, 0)
41.	error	play,6,0,turnt001,00,X,3E1/G34S-.B-1
	state after	(1, 1, 0, 0, 6, 1, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 2148 *In the top of the 6th inning, with bases empty, PHI batter Trea Turner hit a soft ground ball to shallow right infield. A throwing error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.*

	game	ATL vs TBA on 20230707
	state before	(1, 0, 0, 0, 4, 0, 1, 0)
42.	error	play,4,0,olsom001,32,CFFBBBX,E4/G34.B-1
	state after	(1, 1, 0, 0, 4, 0, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2149 *In the top of the 4th inning, with bases empty, ATL batter Matt Olson hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.*

	game	ATL vs TBA on 20230708
	state before	(1, 1, 0, 0, 8, 4, 0, 1)
43.	error	play,8,1,rale1001,00,X,53/G6.1-H(E3/TH)(NR)(UR)
	state after	(2, 0, 0, 0, 8, 4, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2150 This game had 2 errors. The final score was 6-1.
In the bottom of the 8th inning, with a runner on 1st, TBA batter Luke Raley hit a ground ball to short. The first baseman (with an assist from the third baseman) put out the batter. A throwing error by the first baseman allowed the runner on 1st to score (unearned, no RBI).

	game	ATL vs TBA on 20230708
	state before	(1, 1, 0, 0, 9, 4, 0, 0)
44.	error	play,9,0,harrm004,10,BX,FC6/G6.1XH(6E4)(NR)(UR);B-2
	state after	(2, 1, 0, 0, 9, 4, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2151 *In the top of the 9th inning, with a runner on 1st, ATL batter Michael Harris hit a ground ball into a fielder's choice at short. An error by the second baseman (with an assist from the short-stop) allowed the runner on 1st to safely score (unearned, no RBI) and the batter to reach 2nd.*

	game	ATL vs TBA on 20230709
	state before	(0, 1, 0, 0, 5, 2, 7, 1)
45.	error	play,5,1,aranj001,01,C>S,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 5, 2, 7, 1)
	fielder	2 (catcher on the visiting team)

Summary 2152 *In the bottom of the 5th inning, with a runner on 1st, TBA batter Jonathan Aranda had a 0-1 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Aranda is still at bat, now with a 0-2 count.*

	game	BAL vs TBA on 20230720
	state before	(2, 0, 1, 0, 4, 2, 1, 0)
46.	error	play,4,0,hicka001,00,X,S7/G6.2XH(7E2)(NR)(UR);B-2
	state after	(2, 0, 1, 0, 4, 3, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2153 *In the top of the 4th inning, with a runner on 2nd, BAL batter Aaron Hicks hit a ground ball single to short. An error by the catcher (with an assist from the left fielder) allowed the runner on 2nd to safely score (unearned, no RBI) and the batter to reach 2nd.*

	game	BAL vs TBA on 20230723
	state before	(0, 0, 0, 0, 7, 4, 3, 0)
47.	error	play,7,0,cowsc001,22,.BCBCX,E5/G56.B-1
	state after	(0, 1, 0, 0, 7, 4, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2154 *In the top of the 7th inning, with bases empty, BAL batter Colton Cowser hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	MIA vs TBA on 20230725
	state before	(1, 0, 0, 0, 3, 1, 3, 1)
48.	error	play,3,1,arozr001,12,CSBX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 3, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2155 *In the bottom of the 3rd inning, with bases empty, TBA batter Randy Arozarena hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	MIA vs TBA on 20230726
	state before	(0, 0, 0, 0, 4, 5, 1, 1)
49.	error	play,4,1,franw002,01,FX,E5/TH/G56S.B-1
	state after	(0, 1, 0, 0, 4, 5, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2156 *In the bottom of the 4th inning, with bases empty, TBA batter Wander Franco hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

30.2.5 August

	game	SLN vs TBA on 20230810
	state before	(0, 1, 1, 0, 7, 3, 0, 0)
50.	error	play,7,0,nootl001,11,CBX,36(1)/FO/G3.2-H(E6/TH)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 7, 4, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2157 *In the top of the 7th inning, with runners on 1st and 2nd, SLN batter Lars Nootbaar hit a ground ball to 1st. At 2nd, the shortstop forced out the runner from 1st (with an assist from the first baseman). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.*

Visiting/Home team rosters for CLE vs TBA on 2023-08-11

position	innings	player	position	innings	player
1	1	Xzavion Curry	1	1	Aaron Civale
1	6	Daniel Norris	1	6	Jake Diekman
1	7	Michael Kelly	1	7	Colin Poche
1	9	Nick Sandlin	1	8	Jason Adam
2	1	Bo Naylor	1	9	Pete Fairbanks
3	1	Kole Calhoun	1	9	Robert Stephenson
4	1	Andres Gimenez	2	1	Christian Bethancourt
5	1	Jose Ramirez	3	1	Yandy Diaz
6	1	Brayan Rocchio	4	1	Brandon Lowe
7	1	Steven Kwan	5	1	Isaac Paredes
8	1	Myles Straw	6	1	Wander Franco
9	1	Ramon Laureano	7	1	Randy Arozarena
			8	1	Josh Lowe
			8	7	Manuel Margot
			9	1	Luke Raley
			9	7	Josh Lowe

	game	CLE vs TBA on 20230811
	state before	(1, 0, 0, 0, 2, 0, 0, 0)
51.	error	play,2,0,roccb001,01,FX,3E1/G34.B-1
	state after	(1, 1, 0, 0, 2, 0, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2158 This game had 4 errors. The final score was 8-9.
In the top of the 2nd inning, with bases empty, CLE batter Brayan Rocchio hit a ground ball to the 1st/2nd base gap. A throwing error by the pitcher (with an assist from the first baseman) allowed the batter to reach 1st base.

	game	CLE vs TBA on 20230811
	state before	(0, 1, 1, 0, 6, 2, 3, 0)
52.	error	play,6,0,strom002,10,B+2,0A.2-3(E2/TH)
	state after	(0, 1, 0, 1, 6, 2, 3, 0)
	fielder	2 (catcher on the home team)

Summary 2159 *In the top of the 6th inning, with runners on 1st and 2nd, CLE batter Myles Straw has a 1-0 count. The catcher attempted a pickoff at 2nd. A throwing error by the catcher allowed the runner on 2nd to advance to 3rd. Straw is still at bat with a 1-0 count.*

	game	CLE vs TBA on 20230811
	state before	(1, 1, 1, 0, 6, 4, 3, 0)
53.	error	play,6,0,ramij003,01,FX,64(1)/F0/G6.2-3;BX1(4E3)
	state after	(2, 1, 0, 1, 6, 4, 3, 0)
	fielder	3 (1st baseman on the home team)

Summary 2160 *In the top of the 6th inning, with runners on 1st and 2nd, CLE batter Jose Ramirez hit a ground ball to short. The second baseman (with an assist from the shortstop) focred out the runner from 1st. The runner from 2nd advanced to 3rd. An error by the first baseman (with an assist from the second baseman) allowed the batter to safely reach 1st.*

	game	CLE vs TBA on 20230811
	state before	(1, 1, 1, 1, 7, 4, 5, 1)
54.	error	play,7,1,arozr001,00,X,FC4/G4.3-H;2-H(E4) (NR) (UR);1-2;B-1
	state after	(1, 1, 1, 0, 7, 4, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2161 *In the bottom of the 7th inning, with bases loaded, TBA batter Randy Arozarena hit a ground ball into a fielder's choice at 2nd. The runner on 3rd scored (1 RBI). An error by the second baseman allowed the runner on 2nd to safely score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	CLE vs TBA on 20230813
	state before	(0, 1, 0, 0, 4, 6, 1, 0)
55.	error	play,4,0,gimea001,01,FX,S1/G15-.1-H(NR);B-3(E1/TH)
	state after	(0, 0, 0, 1, 4, 7, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 2162 *In the top of the 4th inning, with a runner on 1st, CLE batter Andres Gimenez hit a soft ground ball single to shallow left infield. The runner on 1st scored (no RBI) while a throwing error by the pitcher allowed the batter to reach 3rd.*

	game	COL vs TBA on 20230822
	state before	(1, 1, 1, 1, 8, 4, 5, 1)
56.	error	play,8,1,arozr001,32,.F*BF*BFBFFX,E7/L7/SF.3-H;2-H(NR)(UR);1-2;B-1
	state after	(1, 1, 1, 0, 8, 4, 7, 1)
	fielder	7 (left field on the visiting team)

Summary 2163 *In the bottom of the 8th inning, with bases loaded, TBA batter Randy Arozarena hit a line drive to left field. An error by the left fielder allowed the runner on 3rd to score (1 RBI), the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

	game	NYA vs TBA on 20230825
	state before	(0, 0, 0, 0, 8, 6, 1, 1)
57.	error	play,8,1,basao001,10,BX,E5/TH/G2.B-1
	state after	(0, 1, 0, 0, 8, 6, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2164 *In the bottom of the 8th inning, with bases empty, TBA batter Osleivis Basabe hit a ground ball to very shallow infield. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	NYA vs TBA on 20230826
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
58.	error	play,2,1,parei001,22,CBC.BX,E5/G56.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2165 *In the bottom of the 2nd inning, with bases empty, TBA batter Isaac Paredes hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	NYA vs TBA on 20230827
	state before	(1, 1, 0, 0, 1, 0, 0, 1)
59.	error	play,1,1,loweb001,01,C1>S,SB2.1-H(E2/TH)(E8/TH)(NR)
	state after	(1, 0, 0, 0, 1, 0, 1, 1)
	fielder	2,8 (catcher, center fielder on the visiting team)

Summary 2166 This play has 2 errors by 2 different fielders. The final score was 4-7.

In the bottom of the 1st inning, with a runner on 1st, TBA batter Brandon Lowe had a 0-1 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. A throwing error by the center fielder allowed that runner to score (unearned, no RBI). Lowe is still at bat, with a 0-2 count.

30.2.6 September

Visiting/Home team rosters for BOS vs TBA on 2023-09-05

position	innings	player	position	innings	player
1	1	Kutter Crawford	1	1	Zach Eflin
1	11	Kenley Jansen	1	10	Chris Devenski
1	4	Joe Jacques	1	11	Erasmo Ramirez
1	6	Josh Winckowski	1	6	Colin Poche
1	7	Brennan Bernardino	1	6	Kevin Kelly
1	8	Chris Martin	1	7	Shawn Armstrong
1	9	John Schreiber	1	8	Jake Diekman
2	1	Connor Wong	1	9	Pete Fairbanks
2	11	Reese McGuire	2	1	Rene Pinto
3	1	Triston Casas	2	9	Christian Bethancourt
4	1	Enmanuel Valdez	3	1	Yandy Diaz
4	11	Trevor Story	4	1	Brandon Lowe
4	8	Luis Urias	5	1	Isaac Paredes
5	1	Rafael Devers	6	1	Vidal Brujan
6	1	Ceddanne Rafaela	7	1	Randy Arozarena
7	1	Masataka Yoshida	8	1	Jose Siri
8	1	Wilyer Abreu	9	1	Josh Lowe
8	6	Rob Refsnyder			
9	1	Adam Duvall			

	game	BOS vs TBA on 20230905
	state before	(2, 0, 0, 0, 1, 0, 0, 1)
60.	error	play,1,1,parei001,32,BFBS.BX,E6/G6.B-1
	state after	(2, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2167 This game had 4 errors. The final score was 6-8.
In the bottom of the 1st inning, with bases empty, TBA batter Isaac

Paredes hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

61.	game	BOS vs TBA on 20230905
	state before	(0, 0, 1, 1, 6, 2, 5, 0)
	error	play,6,0,turnj001,32,SBBFBX,S6/G56.3-H;2-3;B-2(E6/TH)
	state after	(0, 0, 1, 1, 6, 3, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2168 *In the top of the 6th inning, with runners on 2nd and 3rd, BOS batter Justin Turner hit a ground ball single to 3rd. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to 3rd. A throwing error by the shortstop allowed the batter to reach 2nd.*

62.	game	BOS vs TBA on 20230905
	state before	(0, 0, 0, 0, 6, 3, 5, 1)
	error	play,6,1,pintr003,12,..CSFBX,E4/G4D.B-1
	state after	(0, 1, 0, 0, 6, 3, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2169 *In the bottom of the 6th inning, with bases empty, TBA batter Rene Pinto hit a ground ball to deep 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

63.	game	BOS vs TBA on 20230905
	state before	(1, 1, 0, 0, 7, 3, 5, 0)
	error	play,7,0,rafac001,00,X,FC4/G34.1-2(E4);B-1
	state after	(1, 1, 1, 0, 7, 3, 5, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2170 *In the top of the 7th inning, with a runner on 1st, BOS batter Ceddanne Rafaela hit a ground ball into a fielder's choice at the 1st/2nd base gap. An error by the second baseman allowed the runner on 1st to safely advance to 2nd and the batter to reach 1st.*

	game	SEA vs TBA on 20230907
	state before	(2, 0, 0, 0, 7, 1, 0, 1)
64.	error	play,7,1,diazy001,00,X,E4/G8.B-1
	state after	(2, 1, 0, 0, 7, 1, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2171 *In the bottom of the 7th inning, with bases empty, TBA batter Yandy Diaz hit a ground ball to center field. An error by the second baseman allowed the batter to reach 1st base.*

	game	SEA vs TBA on 20230909
	state before	(2, 1, 0, 0, 1, 3, 0, 1)
65.	error	play,1,1,parei001,02,CCX,D7/L7LD.1-H(E7) (NR) (UR)
	state after	(2, 0, 1, 0, 1, 3, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 2172 This game had 2 errors. The final score was 5-7.
In the bottom of the 1st inning, with a runner on 1st, TBA batter Isaac Paredes hit a line drive double to deep left field. A throwing error by the left fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 2nd.

	game	SEA vs TBA on 20230909
	state before	(2, 0, 0, 0, 8, 4, 5, 0)
66.	error	play,8,0,suare001,11,BCX,E5/G56.B-1
	state after	(2, 1, 0, 0, 8, 4, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2173 *In the top of the 8th inning, with bases empty, SEA batter Eugenio Suarez hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	ANA vs TBA on 20230919
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
67.	error	play,1,1,diazy001,10,BX,D8/L78XD.B-3(E8)
	state after	(0, 0, 0, 1, 1, 0, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 2174 *In the bottom of the 1st inning, with bases empty, TBA batter Yandy Diaz hit a line drive double to extra deep left center field. An error by the ?? allowed the batter to reach 3rd.*

	game	ANA vs TBA on 20230920
	state before	(0, 0, 0, 0, 3, 6, 0, 1)
68.	error	play,3,1,bethc001,11,BFX,E5/G7L.B-2
	state after	(0, 0, 1, 0, 3, 6, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2175 This game had 2 errors. The final score was 8-3. *In the bottom of the 3rd inning, with bases empty, TBA batter Christian Bethancourt hit a ground ball to left field. An error by the third baseman allowed the batter to reach 2nd base.*

	game	ANA vs TBA on 20230920
	state before	(0, 0, 0, 0, 4, 6, 1, 1)
69.	error	play,4,1,loweb001,22,CFBBFX,E4/G4D.B-1
	state after	(0, 1, 0, 0, 4, 6, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2176 *In the bottom of the 4th inning, with bases empty, TBA batter Brandon Lowe hit a ground ball to deep 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

70.	game	ANA vs TBA on 20230921
	state before	(2, 1, 1, 0, 4, 0, 1, 0)
	error	play,4,0,fletd002,11,CBX,S9/L9S.2-H;1-H(E9) (NR)
	state after	(2, 1, 0, 0, 4, 2, 1, 0)
	fielder	9 (right field on the home team)

Summary 2177 *In the top of the 4th inning, with runners on 1st and 2nd, ANA batter David Fletcher hit a line drive single to shallow right field. The runner on 2nd scored (1 RBI). An error by the right fielder allowed the runner on 1st to score (unearned, no RBI). The batter took 1st.*

Visiting/Home team rosters for TOR vs TBA on 2023-09-23

position	innings	player	position	innings	player
1	1	Hyun Jin Ryu	1	1	Zack Littell
1	5	Trevor Richards	1	6	Shawn Armstrong
1	6	Chad Green	1	7	Colin Poche
1	7	Genesis Cabrera	1	8	Andrew Kittredge
1	8	Yimi Garcia	1	8	Chris Devenski
1	9	Jordan Romano	1	8	Pete Fairbanks
2	1	Tyler Heineman	2	1	Christian Bethancourt
2	8	Alejandro Kirk	2	9	Rene Pinto
3	1	Vladimir Guerrero Jr.	3	1	Yandy Diaz
4	1	Davis Schneider	4	1	Isaac Paredes
5	1	Matt Chapman	5	1	Curtis Mead
6	1	Bo Bichette	6	1	Taylor Walls
7	1	Daulton Varsho	7	1	Harold Ramirez
7	7	Whit Merrifield	8	1	Manuel Margot
8	1	Kevin Kiermaier	9	1	Josh Lowe
9	1	Cavan Biggio			

	game	TOR vs TBA on 20230923
	state before	(2, 0, 0, 0, 6, 0, 5, 0)
71.	error	play,6,0,kierk001,02,CFX,E5/TH/G6S.B-1
	state after	(2, 1, 0, 0, 6, 0, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2178 This game had 3 errors. The final score was 6-7.
In the top of the 6th inning, with bases empty, TOR batter Kevin Kiermaier hit a ground ball to shallow short. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	TOR vs TBA on 20230923
	state before	(2, 1, 0, 0, 6, 0, 5, 0)
72.	error	play,6,0,heint001,12,FFBS,K+WP.1-3(E2/TH);B-1
	state after	(2, 1, 0, 1, 6, 0, 5, 0)
	fielder	2 (catcher on the home team)

Summary 2179 *In the top of the 6th inning, with a runner on 1st, TOR batter Tyler Heineman had a 1-2 count. The next pitch was a wild pitch and a swinging strike. A throwing error by the catcher allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	TOR vs TBA on 20230923
	state before	(0, 0, 0, 0, 7, 4, 5, 0)
73.	error	play,7,0,merrw001,12,..CFBX,1E6/TH/G6S.B-2
	state after	(0, 0, 1, 0, 7, 4, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2180 *In the top of the 7th inning, with bases empty, TOR batter Whit Merrifield hit a ground ball to shallow short. A throwing error by the shortstop (with an assist from the pitcher) allowed the batter to reach 2nd base.*

Chapter 31

Texas Rangers (TEX)

The Texas Rangers were founded as a National League expansion franchise based in Arlington¹ in 1972. The Texas Rangers won their first-ever World Series title in franchise history in 2023, defeating the Arizona Diamondbacks. This victory marked a historic moment for the team, which had never won a championship in their 62-year history (including time as the Washington Senators). In a remarkable feat, the Rangers won every road game they played during the 2023 postseason, including the World Series.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Arlington against the Rangers. The errors shall be organized first by home team (in alphabetial order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 TEX team is .703 compared with the MLB average of 0.691 that year [BR]. The standard deviation of these DER values is $\sigma = 0.011$, so the Rangers are about one standard deviation above average (which is good).

That year, there were 69 errors in the 81 home games, 26 by the Rangers²

¹The Washington Senators moved from Washington DC to Arlington TX in 1971 and renamed themselves in honor of a historic law enforcement agency operating in Texas.

²For comparison, the SF Giants had 55 errors at home that year, over double the number the Rangers had.

and the rest (43) by visiting teams. The total number of errors by the Rangers (home and away) over the regular season in 2023 was 57, well below the average of 83.9 (which is good)³.

31.1 Season totals

The Official Scorer shall credit the batter with a run batted in for every run that scores by reason of the batter becoming a runner with the bases full (because of a base on balls, an award of first base for being touched by a pitched ball or for interference or obstruction).

– **MLB Rule Book** 9.04(a)(2)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

31.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	43
Runs by ht because of errors by vt:	16
Errors by ht:	26
Runs by vt because of errors by ht:	8

This is a very good error-to-run conversion rate: $16/43 = 37.3\%$ on the part of the Rangers (compared to $8/26 = 30.7\ldots\%$ on the part of the visiting teams, which is also pretty good). A typical *play* (not necessarily with an error) in the 2023 TEX season (home and away) has about a 14% of resulting in a run⁴. Therefore, a randomly selected error is over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

³It is the second lowest in the MLB and about half the home and away error total for the Giants.

⁴Take Runs over Plate Appearances, R/PA, from [BR]. For example for the Rangers, $R/PA = 881/6306 = 0.139\ldots$, which is about 14%.

Note that the total number of errors (by either team) in a home game is 67, so the expected number of errors (by either team) in a randomly selected TEX2023 home game is about 0.83.

Errors by position	vt	ht
errors due to pitcher:	4	2
to catcher:	2	1
to 1st baseman:	1	1
to 2nd baseman:	6	1
to 3rd baseman:	7	8
to shortstop:	14	6
to left fielder:	4	1
to center fielder:	4	0
to right fielder:	1	4

31.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	21	10	31
(lone) runner on 1st	6	6	12
(lone) runner on 2nd	5	1	6
(lone) runner on 3rd	2	0	2
two runners on base	6	8	14
bases loaded	3	1	4
total	43	26	69

A table of frequencies of most common errors by the Rangers in TEX 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	7	26.9%
(1, 0, 0, 0)	1	3.8%
(2, 0, 0, 0)	2	7.7%
(0, 1, 0, 0)	0	0%
(1, 1, 0, 0)	4	15.4%
(2, 1, 0, 0)	2	7.7%
(1, 0, 1, 0)	1	3.8%
(1, 1, 1, 0)	6	23.1%
(0, 1, 1, 1)	1	3.8%

A table of frequencies of most common errors by a visiting team in TEX 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	9	20.9%
(1, 0, 0, 0)	4	9.3%
(2, 0, 0, 0)	8	18.6%
(0, 1, 0, 0)	3	6.9%
(1, 1, 0, 0)	1	2.3%
(2, 1, 0, 0)	2	4.6%
(1, 0, 1, 0)	4	9.3%
(2, 0, 0, 1)	2	4.6%
(1, 0, 1, 1)	2	4.6%
(1, 1, 1, 1)	3	6.9%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department:
the 2023 Rangers homegames, has a similar frequency table (compiled with
the aid of some Python programming using the Retrosheet data file `TEX2023.evn`):

state	TEX2023 percentages
(0, 0, 0, 0)	24.4%
(1, 0, 0, 0)	16.0%
(2, 0, 0, 0)	12.3%
(0, 1, 0, 0)	5.8%
(1, 1, 0, 0)	7.5%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	2.1%
(1, 0, 1, 0)	3.1%
(2, 0, 1, 0)	3.3%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.4%
(0, 1, 1, 0)	1.3%
(1, 1, 1, 0)	3.1%
(2, 1, 1, 0)	3.4%

31.1.3 Some interesting errors

Some of the most interesting error plays in the TEX 2023 home games:

- The game ANA vs TEX on 2023-06-12 had 3 errors but 2 of them were in the 12th inning when the Angels went ahead (for the first time in the game) and won.
- The game MIL vs TEX on 2023-08-20 had 5 errors. Very unusual for the Rangers.

31.1.4 Relative team rankings

We have 81 home games between TEX and some visiting MLB team (vt). In the lists below, we have ordered the visiting teams with that team's W/T record in parentheses⁵.

⁵Where W = numbers of games vt won against TEX, T = Total home games vt played against the Rangers at Arlington.

A number of teams did not play TEX in the regular season. However, of those that did, TEX *lost* more games than won against:

ATL (2/3), BAL (2/3), LAN (2/3), MIN (2/3),
MIL (3/3), ANA (4/7), HOU (6/7),

and *won* more than lost against

CHA (0/3), CLE (0/3), COL (0/3), MIA (0/3), PHI (0/3), TBA (0/3), SEA
(0/6),
BOS (1/3), KCA (1/3), SLN (1/3), TOR (1/3), NYA (1/4) .

The teams that *split* their series were:

ARI (1/2), DET (2/4) .

The teams that did not play in any of the 81 home games in TEX 2023 were:

CHN, CIN, NYN, PIT, SDN, SFN, TEX, WAS .

31.2 Game-by-game totals

The batter becomes a runner and is entitled to first base without liability to be put out (provided he advances to and touches first base) when the catcher or any fielder interferes with him.

– **MLB Rule Book** 6.01(c)

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for TEX 2023 is as follows [BA].

position	player	games played	OBP
1 (C)	Jonah Heim	124	.317
3 (1B)	Nathaniel Lowe	161	.360
4 (2B)	Marcus Semien	162	.348
5 (3B)	Josh Jung	121	.315
6 (SS)	Corey Seager	112	.390
7 (LF)	Travis Jankowski	78	.357
8 (CF)	Leody Taveras	140	.312
9 (RF)	Adolis García	135	.328

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 Rangers home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

31.2.1 April

Visiting/Home team rosters for

position	innings	player	position	innings	player
1	1	Zack Wheeler	1	1	Nathan Eovaldi
1	5	Yunior Marte	1	6	Dane Dunning
1	6	Jose Alvarado	1	9	Taylor Hearn
1	7	Andrew Bellatti	2	1	Mitch Garver
1	7	Seranthony Dominguez	3	1	Nathaniel Lowe
1	8	Craig Kimbrel	4	1	Marcus Semien
1	8	Josh Harrison	5	1	Josh Jung
2	1	J.T. Realmuto	6	1	Corey Seager
2	8	Garrett Stubbs	6	9	Josh H. Smith
3	1	Darick Hall	7	1	Josh H. Smith
3	8	Alec Bohm	7	9	Ezequiel Duran
4	1	Bryson Stott	8	1	Adolis Garcia
5	1	Alec Bohm	8	8	Travis Jankowski
5	8	Edmundo Sosa	9	1	Robbie Grossman
5	8	Josh Harrison	9	8	Bubba Thompson
6	1	Trea Turner			
7	1	Kyle Schwarber			
8	1	Brandon Marsh			
9	1	Jake Cave			

	game	PHI vs TEX on 20230401
	state before	(2, 0, 0, 1, 3, 0, 3, 1)
1.	error	play,3,1,garvm001,21,BFBX,E6/G6.3-H(NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 3, 0, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2181 There were 2 errors in this game. The final score was 3-16.

In the bottom of the 3rd inning, with a runner on 3rd, TEX batter Mitch Garver hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 1st.

	game	PHI vs TEX on 20230401
	state before	(0, 1, 1, 0, 7, 3, 10, 1)
2.	error	play,7,1,semim001,22,*BBSCX,FC5/G56.2-3;1-2(E5) ;B-1
	state after	(0, 1, 1, 1, 7, 3, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2182 *In the bottom of the 7th inning, with runners on 1st and 2nd, TEX batter Marcus Semien hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	BAL vs TEX on 20230403
	state before	(1, 0, 0, 0, 5, 2, 0, 1)
3.	error	play,5,1,millb002,32,BCSBBX,E6/G4M.B-1
	state after	(1, 1, 0, 0, 5, 2, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2183 *In the bottom of the 5th inning, with bases empty, TEX batter Brad Miller hit a ground ball to deep 2nd base. An error by the shortstop allowed the batter to reach 1st base.*

	game	BAL vs TEX on 20230404
	state before	(1, 1, 1, 0, 2, 0, 0, 0)
4.	error	play,2,0,mcker001,12,CBFX,S7/L7L.2-H;1-3(E7/TH)
	state after	(1, 1, 0, 1, 2, 1, 0, 0)
	fielder	7 (left field on the home team)

Summary 2184 *In the top of the 2nd inning, with runners on 1st and 2nd, BAL batter Ryan McKenna hit a line drive single to left field. The runner on 2nd scored (1 RBI). A throwing error by the left fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.*

	game	BAL vs TEX on 20230405
	state before	(0, 0, 0, 0, 3, 0, 2, 1)
5.	error	play,3,1,semim001,00,X,S5/G5S-.B-2(E5/TH)
	state after	(0, 0, 1, 0, 3, 0, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2185 There were 2 errors in this game. The final score was 2-5.

In the bottom of the 3rd inning, with bases empty, TEX batter Marcus Semien hit a soft ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st.

	game	BAL vs TEX on 20230405
	state before	(1, 1, 0, 1, 5, 0, 2, 0)
6.	error	play,5,0,haysa001,11,FBX,FC4/G34.3-H;1-2(E4);B-1
	state after	(1, 1, 1, 0, 5, 1, 2, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2186 *In the top of the 5th inning, with runners on 1st and 3rd, BAL batter Austin Hays hit a ground ball into a fielder's choice at the 1st/2nd base gap. The runner on 3rd score (1 RBI). An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	KCA vs TEX on 20230410
	state before	(1, 1, 1, 0, 1, 0, 0, 0)
7.	error	play,1,0,peres002,31,B*BBFX,E9/F9LD.2-H(NR)(UR);1-3;B-2
	state after	(1, 0, 1, 1, 1, 1, 0, 0)
	fielder	9 (right field on the home team)

Summary 2187 There were 2 errors in this game. The final score was 2-5.

In the top of the 1st inning, with runners on 1st and 2nd, KCA batter Salvador Perez hit a fly ball to deep right field. An error by the right fielder allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.

	game	KCA vs TEX on 20230410
	state before	(0, 0, 1, 0, 3, 1, 1, 1)
8.	error	play,3,1,semim001,22,F*BC*BX,E6/G6S.B-1
	state after	(0, 1, 1, 0, 3, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2188 *In the bottom of the 3rd inning, with a runner on 2nd, TEX batter Marcus Semien hit a ground ball to shallow short. An error by the shortstop allowed the batter to reach 1st base.*

	game	OAK vs TEX on 20230421
	state before	(0, 0, 0, 0, 5, 2, 4, 0)
9.	error	play,5,0,smstk003,10,BX,S5/G5.B-2(E5/TH)
	state after	(0, 0, 1, 0, 5, 2, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2189 *In the top of the 5th inning, with bases empty, OAK batter Kevin Smith hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	OAK vs TEX on 20230423
	state before	(0, 0, 0, 0, 5, 0, 4, 0)
10.	error	play,5,0,diazj008,11,BSX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 5, 0, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 2190 *In the top of the 5th inning, with bases empty, OAK batter Jordan Diaz hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	NYA vs TEX on 20230430
	state before	(0, 0, 0, 0, 7, 1, 14, 1)
11.	error	play,7,1,tavel001,10,BX,E1/G15.B-1
	state after	(0, 1, 0, 0, 7, 1, 14, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2191 *In the bottom of the 7th inning, with bases empty, TEX batter Leody Taveras hit a ground ball to shallow infield. An error by the pitcher allowed the batter to reach 1st base.*

31.2.2 May

	game	ARI vs TEX on 20230503
	state before	(2, 0, 0, 0, 9, 12, 7, 1)
12.	error	play,9,1,garca005,10,BX,E4/G4S.B-1
	state after	(2, 1, 0, 0, 9, 12, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2192 *In the bottom of the 9th inning, with bases empty, TEX batter Adolis Garcia hit a ground ball to shallow 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	ATL vs TEX on 20230515
	state before	(2, 0, 0, 0, 9, 12, 0, 1)
13.	error	play,9,1,semim001,31,BBBCX,S8/L78D.B-2(E8)
	state after	(2, 0, 1, 0, 9, 12, 0, 1)
	fielder	8 (center field on the visiting team)

Summary 2193 *In the bottom of the 9th inning, with bases empty, TEX batter Marcus Semien hit a line drive single to deep left center field. An error by the center fielder allowed the batter to reach 2nd base.*

	game	ATL vs TEX on 20230516
	state before	(1, 0, 1, 0, 7, 1, 4, 1)
14.	error	play,7,1,semim001,12,1C>C.BFFX,S7/G6.2-H;B-2(E2)
	state after	(1, 0, 1, 0, 7, 1, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 2194 *In the bottom of the 7th inning, with a runner on 2nd, TEX batter Marcus Semien hit a ground ball single to short. The runner on 2nd scored (1 RBI). An error by the catcher (on a relay from the left fielder) allowed the batter to reach 2nd base.*

	game	COL vs TEX on 20230519
	state before	(0, 0, 0, 0, 5, 0, 2, 1)
15.	error	play,5,1,tavel001,10,BX,E6/TH/G6MD.B-1
	state after	(0, 1, 0, 0, 5, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2195 *In the bottom of the 5th inning, with bases empty, TEX batter Leody Taveras hit a ground ball to deep short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	COL vs TEX on 20230520
	state before	(1, 1, 0, 1, 2, 0, 5, 1)
16.	error	play,2,1,garca005,31,BBBCX,FC5/G56.3-H;1-2(E5);B-1
	state after	(1, 1, 1, 0, 2, 0, 6, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2196 *In the bottom of the 2nd inning, with runners on 1st and 3rd, TEX batter Adolis Garcia hit a ground ball into a fielder's choice at 3rd. The runner on 3rd scored (1 RBI). An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	COL vs TEX on 20230521
	state before	(1, 1, 1, 0, 6, 0, 11, 0)
17.	error	play,6,0,mousm001,00,X,FC3/G3.2-H(NR)(UR);1-2(E3/TH);B-1
	state after	(1, 1, 1, 0, 6, 1, 11, 0)
	fielder	3 (1st baseman on the home team)

Summary 2197 *In the top of the 6th inning, with runners on 1st and 2nd, COL batter Mike Moustakas hit a ground ball into a fielder's choice at 1st. The runner on 2nd scored (unearned, no RBI). A throwing error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

31.2.3 June

	game	SLN vs TEX on 20230605
	state before	(0, 0, 0, 0, 2, 0, 0, 0)
18.	error	play,2,0,contw001,02,CFF.FX,E9/F89S.B-2
	state after	(0, 0, 1, 0, 2, 0, 0, 0)
	fielder	9 (right field on the home team)

Summary 2198 *In the top of the 2nd inning, with bases empty, SLN batter Willson Contreras hit a fly ball to shallow right right field. An error by the right fielder allowed the batter to reach 2nd base.*

19.	game	SLN vs TEX on 20230606
	state before	(0, 0, 0, 0, 2, 2, 0, 0)
	error	play,2,0,walkj003,32,CBBC.BFX,E6/TH/G6M.B-1
	state after	(0, 1, 0, 0, 2, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2199 There were 2 errors in this game. The final score was 4-6.

In the top of the 2nd inning, with bases empty, SLN batter Jordan Walker hit a ground ball to deep short. A throwing error by the shortstop allowed the batter to reach 1st base.

20.	game	SLN vs TEX on 20230606
	state before	(1, 1, 1, 1, 4, 2, 2, 1)
	error	play,4,1,semim001,12,CBFX,D9/F89XD+.3-H;2-H;1-H(E4/TH) (NR) (UR);B-3
	state after	(1, 0, 0, 1, 4, 2, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2200 *In the bottom of the 4th inning, with bases loaded, TEX batter Marcus Semien hit a hard fly ball double to extra deep right center field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the second baseman allowed the runner on 1st to also score (unearned, no RBI). and the batter to reach 3rd base.*

Visiting/Home team rosters for ANA vs TEX on 2023-06-12

position	innings	player	position	innings	player
1	1	Tyler Anderson	1	1	Dane Dunning
1	10	Sam Bachman	1	11	Cole Ragans
1	12	Aaron Loup	1	12	Yerry Rodriguez
1	7	Chris Devenski	1	6	Grant Anderson
1	8	Jacob Webb	1	7	Josh Sborz
1	9	Carlos Estevez	1	9	Will Smith
2	1	Matt Thaiss	2	1	Jonah Heim
2	10	Chad Wallach	2	11	Sandy Leon
3	1	Jared Walsh	3	1	Nathaniel Lowe
3	10	Taylor Ward	4	1	Marcus Semien
3	8	Brandon Drury	5	1	Josh Jung
4	1	Brandon Drury	6	1	Corey Seager
4	8	Luis Rengifo	7	1	Ezequiel Duran
5	1	Anthony Rendon	7	7	Travis Jankowski
5	10	Gio Urshela	8	1	Leody Taveras
6	1	Zach Neto	9	1	Adolis Garcia
7	1	Taylor Ward			
7	10	Mickey Moniak			
8	1	Mike Trout			
9	1	Hunter Renfroe			

	game	ANA vs TEX on 20230612
	state before	(1, 1, 1, 0, 5, 2, 5, 0)
21.	error	play,5,0,renda001,00,X,9/F9D.2-H(E9/TH)(NR);1-2
	state after	(2, 1, 1, 0, 5, 2, 5, 0)
	fielder	9 (right field on the home team)

Summary 2201 There were 3 errors in this game. The final score was 9-6.

In the top of the 5th inning, with runners on 1st and 2nd, ANA batter

Anthony Rendon hit a fly ball to deep right field. The right fielder put out the batter. A throwing error by the right fielder allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to reach 2nd.

	game	ANA vs TEX on 20230612
	state before	(2, 0, 0, 0, 12, 8, 5, 0)
22.	error	play,12,0,renfh001,00,.X,E6/G6.B-1
	state after	(2, 1, 0, 0, 12, 8, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2202 *In the top of the 12th inning, with bases empty, ANA batter Hunter Renfroe hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	ANA vs TEX on 20230612
	state before	(1, 0, 1, 0, 12, 8, 5, 1)
23.	error	play,12,1,garvm001,02,CCX,E6/TH/G6.2-H(NR)(UR);B-1
	state after	(1, 1, 0, 0, 12, 8, 6, 1)
	fielder	6 (shortstop on the home team)

Summary 2203 *In the bottom of the 12th inning, with a runner on 2nd, TEX batter Mitch Garver hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	ANA vs TEX on 20230613
	state before	(1, 1, 1, 0, 6, 1, 3, 0)
24.	error	play,6,0,renda001,02,CFX,46(1)/FO/G4.2-H(E6/TH)(NR);B-1
	state after	(2, 1, 0, 0, 6, 2, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 2204 There were 2 errors in this game. The final score was 7-3.

In the top of the 6th inning, with runners on 1st and 2nd, ANA batter Anthony Rendon hit a ground ball to 2nd. The runner from 1st was forced out by the shortstop (assisted by the second baseman). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	ANA vs TEX on 20230613
	state before	(1, 1, 0, 0, 9, 7, 3, 1)
25.	error	play,9,1,lowen001,01,FX,FC1/G13.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 9, 7, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2205 *In the bottom of the 9th inning, with a runner on 1st, TEX batter Nathaniel Lowe hit a ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	ANA vs TEX on 20230614
	state before	(1, 0, 1, 0, 8, 1, 4, 1)
26.	error	play,8,1,heimj001,02,CCFX,E6/TH/G6M.2-3;B-1
	state after	(1, 1, 0, 1, 8, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2206 *In the bottom of the 8th inning, with a runner on 2nd, TEX batter Jonah Heim hit a ground ball to short. A throwing error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	TOR vs TEX on 20230616
	state before	(1, 0, 0, 0, 4, 0, 1, 0)
27.	error	play,4,0,guerv002,00,X,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 4, 0, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2207 *In the top of the 4th inning, with bases empty, TOR batter Vladimir Guerrero hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	TOR vs TEX on 20230617
	state before	(2, 0, 0, 0, 8, 2, 4, 1)
28.	error	play,8,1,durae002,20,BBX,E6/G6.B-1
	state after	(2, 1, 0, 0, 8, 2, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2208 *In the bottom of the 8th inning, with bases empty, TEX batter Ezequiel Duran hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for TOR vs TEX on 2023-06-18

position	innings	player	position	innings	player
1	1	Chris Bassitt	1	1	Jon Gray
1	4	Nate Pearson	1	3	John King
1	5	Tim Mayza	1	6	Grant Anderson
1	6	Yimi Garcia	1	8	Jose Leclerc
1	7	Adam Cimber	1	9	Joe Barlow
1	7	Mitch White	2	1	Mitch Garver
2	1	Alejandro Kirk	2	7	Sam Huff
2	2	Danny Jansen	3	1	Nathaniel Lowe
3	1	Vladimir Guerrero Jr.	4	1	Marcus Semien
4	1	Whit Merrifield	5	1	Josh Jung
5	1	Santiago Espinal	6	1	Corey Seager
6	1	Bo Bichette	7	1	Ezequiel Duran
7	1	Daulton Varsho	7	9	Travis Jankowski
8	1	Kevin Kiermaier	8	1	Leody Taveras
9	1	George Springer	9	1	Adolis Garcia

	game	TOR vs TEX on 20230618
	state before	(2, 0, 0, 0, 3, 6, 0, 1)
29.	error	play,3,1,lowen001,00,X,E3/TH/G3.B-1
	state after	(2, 1, 0, 0, 3, 6, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2209 There were 3 errors in this game. The final score was 7-11.

In the bottom of the 3rd inning, with bases empty, TEX batter Nathaniel Lowe hit a ground ball to 1st base. A throwing error by the first baseman allowed the batter to reach 1st base.

	game	TOR vs TEX on 20230618
	state before	(2, 1, 1, 0, 3, 6, 0, 1)
30.	error	play,3,1,jungj001,12,BCC.X,S8/G4M.2-H(UR);1-H(E8/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 3, 6, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 2210 *In the bottom of the 3rd inning, with runners on 1st and 2nd, TEX batter Josh Jung hit a ground ball single to deep 2nd base. The runner from 2nd scored (1 RBI). A throwing error by the center fielder allowed the runner from 1st to score (unearned, no RBI). The batter took 1st.*

	game	TOR vs TEX on 20230618
	state before	(2, 1, 0, 0, 6, 7, 8, 0)
31.	error	play,6,0,merrw001,10,1B>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 6, 7, 8, 0)
	fielder	2 (catcher on the home team)

Summary 2211 *In the top of the 6th inning, with a runner on 1st, TOR batter Whit Merrifield had a 1-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Merrifield is at bat, now with a 2-0 count.*

	game	DET vs TEX on 20230627
	state before	(2, 1, 0, 0, 4, 2, 0, 1)
32.	error	play,4,1,durae002,11,CB1,P01(E1/TH) .1-2
	state after	(2, 0, 1, 0, 4, 2, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2212 *In the bottom of the 4th inning, with a runner on 1st, TEX batter Ezequiel Duran has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed the runner on 1st to safely reach 2nd. Duran is still at bat with a 1-1 count.*

	game	DET vs TEX on 20230628
	state before	(1, 0, 1, 0, 1, 0, 1, 1)
33.	error	play,1,1,garca005,10,BX,E5/TH/G56.B-1
	state after	(1, 1, 1, 0, 1, 0, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2213 *In the bottom of the 1st inning, with a runner on 2nd, TEX batter Adolis Garcia hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	DET vs TEX on 20230628
	state before	(2, 0, 0, 0, 3, 0, 6, 1)
34.	error	play,3,1,tavel001,00,X,E6/G6M.B-1
	state after	(2, 1, 0, 0, 3, 0, 6, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2214 *In the bottom of the 3rd inning, with bases empty, TEX batter Leody Taveras hit a ground ball to deep short. An error by the shortstop allowed the batter to reach 1st base.*

	game	HOU vs TEX on 20230630
	state before	(0, 1, 0, 0, 2, 1, 1, 1)
35.	error	play,2,1,heimj001,22,BBFF.X,4/L34+.1-2(E4/TH)
	state after	(1, 0, 1, 0, 2, 1, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2215 *In the bottom of the 2nd inning, with a runner on 1st, TEX batter Jonah Heim hit a hard line drive to 2nd . The second baseman put the batter out. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd.*

31.2.4 July

	game	HOU vs TEX on 20230701
	state before	(1, 1, 1, 0, 8, 0, 4, 1)
36.	error	play,8,1,lowen001,32,BBSFBX,8/F8D.2-H(E8/TH) (NR) (UR) ;1-3
	state after	(2, 1, 1, 0, 8, 0, 4, 1)
	fielder	8 (center field on the visiting team)

Summary 2216 *In the bottom of the 8th inning, with runners on 1st and 2nd, TEX batter Nathaniel Lowe hit a fly ball to deep center field. The center fielder put out the batter. A throwing error by the center fielder allowed the runner from 2nd to score (unearned, no RBI) and the runner from 1st to reach 3rd.*

Visiting/Home team rosters for HOU vs TEX on 2023-07-03

position	innings	player	position	innings	player
1	1	Cristian Javier	1	1	Martin Perez
1	5	Phil Maton	1	2	Glenn Otto
1	6	Ryne Stanek	1	5	Cody Bradford
1	7	Hector Neris	1	8	Aroldis Chapman
1	8	Bryan Abreu	1	9	Will Smith
1	9	Ryan Pressly	2	1	Jonah Heim
2	1	Martin Maldonado	3	1	Nathaniel Lowe
3	1	Jose Abreu	4	1	Marcus Semien
4	1	Jose Altuve	5	1	Josh Jung
5	1	Alex Bregman	6	1	Corey Seager
6	1	Mauricio Dubon	7	1	Travis Jankowski
7	1	Corey Julks	8	1	Leody Taveras
8	1	Chas McCormick	9	1	Adolis Garcia
9	1	Kyle Tucker			

	game	HOU vs TEX on 20230703
	state before	(1, 0, 0, 0, 3, 6, 2, 1)
37.	error	play,3,1,semim001,12,BCFX,E6/TH/G6S.B-1
	state after	(1, 1, 0, 0, 3, 6, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2217 There were 3 errors in this game. The final score was 12-11.

In the bottom of the 3rd inning, with bases empty, TEX batter Marcus Semien hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	HOU vs TEX on 20230703
	state before	(1, 1, 1, 1, 5, 10, 6, 1)
38.	error	play,5,1,jankt001,12,BFFX,S76/P5D.3-H;2-H;1-3(E7)
	state after	(1, 1, 0, 1, 5, 10, 8, 1)
	fielder	7 (left field on the visiting team)

Summary 2218 *In the bottom of the 5th inning, with bases loaded, TEX batter Travis Jankowski hit a pop up single to deep 3rd base. The runners from 2nd and 3rd scored (2 RBIs). An error by the left fielder allowed the runner from 1st to reach 3rd. The batter took 1st.*

	game	HOU vs TEX on 20230703
	state before	(1, 0, 1, 0, 8, 10, 10, 1)
39.	error	play,8,1,semim001,21,B*BCX,FC5/G6S.2-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 8, 10, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2219 *In the bottom of the 8th inning, with a runner on 2nd, TEX batter Marcus Semien hit a ground ball into a fielder's choice at short. A throwing error by the third baseman allowed the runner on 2nd to advance to 3rd base and the batter to reach 1st.*

	game	CLE vs TEX on 20230715
	state before	(1, 1, 1, 1, 2, 0, 1, 1)
40.	error	play,2,1,semim001,12,BFSX,9/SF/F9LSF/FL.3-H;2-3;1-2(E9/TH)
	state after	(2, 0, 1, 1, 2, 0, 2, 1)
	fielder	9 (right field on the visiting team)

Summary 2220 *In the bottom of the 2nd inning, with bases loaded, TEX batter Marcus Semien hit a sacrifice fly to right field. The runner on 3rd scored (1 RBI) and the runner on 2nd advanced to 3rd. A throwing error by the right fielder allowed the runner on 1st to reach 2nd.*

	game	LAN vs TEX on 20230722
	state before	(0, 0, 0, 0, 2, 0, 1, 0)
41.	error	play,2,0,muncm001,12,BCCX,S9/L89.B-2(E9)
	state after	(0, 0, 1, 0, 2, 0, 1, 0)
	fielder	9 (right field on the home team)

Summary 2221 There were 2 errors in this game. The final score was 16-3.

In the top of the 2nd inning, with bases empty, LAN batter Max Muncy hit a line drive single to right center field. The error by the right fielder allowed the batter to reach 2nd.

	game	LAN vs TEX on 20230722
	state before	(0, 1, 0, 0, 2, 3, 1, 1)
42.	error	play,2,1,heimj001,21,BFBX,36(1)/F0/G34.BX1(6E1)
	state after	(1, 1, 0, 0, 2, 3, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2222 *In the bottom of the 2nd inning, with a runner on 1st, TEX batter Jonah Heim hit a ground ball to the 1st/2nd base gap. The runner on 1st was forced out at 2nd by the shortstop. An error by the pitcher (with assist from the shortstop) allowed the batter to safely reach 1st.*

	game	LAN vs TEX on 20230723
	state before	(2, 1, 0, 0, 4, 4, 7, 1)
43.	error	play,4,1,durae002,02,F1FX,D7/L7D.1-H;B-3(E6/TH)
	state after	(2, 0, 0, 1, 4, 4, 8, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2223 *In the bottom of the 4th inning, with a runner on 1st, TEX batter Ezequiel Duran hit a line drive double to deep left field. The runner from 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 3rd.*

31.2.5 August

	game	MIA vs TEX on 20230804
	state before	(2, 0, 0, 0, 7, 2, 6, 1)
44.	error	play,7,1,garca005,21,BCBX,D7/L78XD.B-3(E7)
	state after	(2, 0, 0, 1, 7, 2, 6, 1)
	fielder	7 (left field on the visiting team)

Summary 2224 There were 2 errors in this game. The final score was 2-6.

In the bottom of the 7th inning, with bases empty, TEX batter Adolis Garcia hit a extra deep line drive double to left center field. An error by the left fielder allowed the batter to reach 3rd.

	game	MIA vs TEX on 20230804
	state before	(1, 1, 0, 0, 9, 2, 6, 0)
45.	error	play,9,0,burgj001,12,BSF1,P01(E1/TH).1-2
	state after	(1, 0, 1, 0, 9, 2, 6, 0)
	fielder	1 (pitcher on the home team)

Summary 2225 *In the top of the 9th inning, with a runner on 1st, MIA batter Jake Burger has a 1-2 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to reach 2nd. Burger is still at bat with a 1-2 count.*

	game	MIA vs TEX on 20230806
	state before	(0, 1, 0, 0, 2, 0, 0, 1)
46.	error	play,2,1,jungj001,22,CBBC.FX,FC6/G6.1-2(E6/TH);B-1
	state after	(0, 1, 1, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2226 There were 2 errors in this game. The Rangers shut out the Marlins 0-6.

In the bottom of the 2nd inning, with a runner on 1st, TEX batter Josh Jung hit a ground ball into a fielder's choice at short. A throwing error by the shortstop allowed the runner on 1st advance to 2nd and the batter to reach 1st.

	game	MIA vs TEX on 20230806
	state before	(1, 0, 0, 0, 3, 0, 3, 1)
47.	error	play,3,1,garca005,00,X,E6/G6.B-1
	state after	(1, 1, 0, 0, 3, 0, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2227 *In the bottom of the 3rd inning, with bases empty, TEX batter Adolis Garcia hit a ground ball to shortstop. An error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for ANA vs TEX on 2023-08-14

position	innings	player	position	innings	player
1	1	Patrick Sandoval	1	1	Max Scherzer
1	3	Griffin Canning	1	8	Martin Perez
1	7	Aaron Loup	2	1	Mitch Garver
1	8	Eduardo Escobar	2	8	Austin Hedges
2	1	Matt Thaiss	3	1	Nathaniel Lowe
3	1	Mike Moustakas	4	1	Marcus Semien
3	8	Chad Wallach	5	1	Ezequiel Duran
4	1	Brandon Drury	6	1	Corey Seager
5	1	Luis Rengifo	6	8	Josh H. Smith
6	1	Andrew Velazquez	7	1	J.P. Martinez
7	1	Randal Grichuk	8	1	Leody Taveras
8	1	Mickey Moniak	9	1	Adolis Garcia
9	1	Hunter Renfroe	9	8	Travis Jankowski
9	8	Jordyn Adams			

	game	ANA vs TEX on 20230814
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
48.	error	play,1,1,semim001,20,BBX,E6/TH/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2228 There were 3 errors in this game. The Rangers shut out the Angels 0-12.

In the bottom of the 1st inning, with bases empty, TEX batter Marcus Semien hit a ground ball to 3rd base. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	ANA vs TEX on 20230814
	state before	(1, 0, 1, 1, 2, 0, 0, 1)
49.	error	play,2,1,semim001,11,CBX,S8/L8.3-H;2-H(E8) (NR) (UR)
	state after	(1, 1, 0, 0, 2, 0, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 2229 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, TEX batter Marcus Semien hit a line drive single to center field. The runner from 3rd scored (1 RBI). An error by the center fielder allowed the runner on 2nd to also score (unearned, no RBI). The batter took 1st.*

	game	ANA vs TEX on 20230814
	state before	(0, 0, 0, 0, 8, 0, 11, 1)
50.	error	play,8,1,grosr001,12,...BCFX,E6/TH/G6MD.B-1
	state after	(0, 1, 0, 0, 8, 0, 11, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2230 *In the bottom of the 8th inning, with bases empty, TEX batter Robbie Grossman hit a ground ball to deep short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	ANA vs TEX on 20230815
	state before	(1, 1, 0, 0, 2, 0, 0, 0)
51.	error	play,2,0,renfh001,12,FCBX,E5/G5D.1-2;B-1
	state after	(1, 1, 1, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2231 There were 2 errors in this game. The Rangers beat the Angels 3-7.

In the top of the 2nd inning, with a runner on 1st, ANA batter Hunter Renfroe hit a ground ball to 3rd base. An error by the 3rd baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	ANA vs TEX on 20230815
	state before	(2, 0, 0, 1, 5, 1, 4, 1)
52.	error	play,5,1,garvm001,32,CB*BFBX,C/E2.B-1
	state after	(2, 1, 0, 1, 5, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2232 *In the bottom of the 5th inning, with a runner on 3rd, TEX batter Mitch Garver had a full count. Catcher's interference allowed the batter to reach 1st base.*

	game	ANA vs TEX on 20230816
	state before	(1, 0, 0, 0, 2, 1, 0, 1)
53.	error	play,2,1,durae002,01,CX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 2, 1, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2233 *In the bottom of the 2nd inning, with bases empty, TEX batter Ezequiel Duran hit a ground ball to shortstop. A throwing error by the shortstop allowed the batter to reach 1st base.*

Visiting/Home team rosters for MIL vs TEX on 2023-08-20

position	innings	player	position	innings	player
1	1	Adrian Houser	1	1	Max Scherzer
1	6	Trevor Megill	1	4	Grant Anderson
1	7	Elvis Peguero	1	6	Brock Burke
1	9	Bryse Wilson	1	7	Jose Leclerc
1	9	Devin Williams	1	8	Chris Stratton
2	1	Victor Caratini	1	8	Will Smith
3	1	Carlos Santana	2	1	Mitch Garver
4	1	Brice Turang	3	1	Nathaniel Lowe
5	1	Andruw Monasterio	4	1	Marcus Semien
6	1	Willy Adames	5	1	Ezequiel Duran
7	1	Christian Yelich	6	1	Corey Seager
8	1	Tyrone Taylor	7	1	Travis Jankowski
8	7	Joey Wiemer	8	1	Leody Taveras
9	1	Mark Canha	9	1	J.P. Martinez
9	7	Tyrone Taylor			

54.	game	MIL vs TEX on 20230820
	state before	(0, 0, 0, 0, 6, 3, 1, 0)
	error	play,6,0,carav001,00,.X,E5/G5.B-1
	state after	(0, 1, 0, 0, 6, 3, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2234 There were 5 errors in this game. The Rangers lost to the Brewers 6-2.
In the top of the 6th inning, with bases empty, MIL batter Victor Caratini hit a ground ball to 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.

	game	MIL vs TEX on 20230820
	state before	(1, 1, 0, 0, 7, 3, 1, 0)
55.	error	play,7,0,adamw002,22,B1FFBX,FC5/G56S.1-2(E5);B-1
	state after	(1, 1, 1, 0, 7, 3, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2235 *In the top of the 7th inning, with a runner on 1st, MIL batter Willy Adames hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st advance to 2nd and the batter to reach 1st.*

	game	MIL vs TEX on 20230820
	state before	(0, 0, 0, 0, 8, 3, 1, 0)
56.	error	play,8,0,carav001,21,.CBBX,S5/G56S.B-2(E5/TH)
	state after	(0, 0, 1, 0, 8, 3, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2236 *In the top of the 8th inning, with bases empty, MIL batter Victor Caratini hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	MIL vs TEX on 20230820
	state before	(0, 0, 0, 0, 9, 6, 1, 1)
57.	error	play,9,1,durae002,02,.FFF.X,S4/G34.B-2(E4/TH)
	state after	(0, 0, 1, 0, 9, 6, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2237 *In the bottom of the 9th inning, with bases empty, TEX batter Ezequiel Duran hit a ground ball single to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	MIL vs TEX on 20230820
	state before	(2, 0, 0, 0, 9, 6, 2, 1)
58.	error	play,9,1,semim001,02,CFX,E5/TH/G56S.B-1
	state after	(2, 1, 0, 0, 9, 6, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2238 *In the bottom of the 9th inning, with bases empty, TEX batter Marcus Semien hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

31.2.6 September

	game	HOU vs TEX on 20230904
	state before	(2, 1, 0, 0, 5, 0, 3, 0)
59.	error	play,5,0,altuj001,11,BCX,S1/G1S-.1-3(E1/TH);B-2
	state after	(2, 0, 1, 1, 5, 0, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 2239 There were 2 errors in this game. The final score was 13-6.

In the top of the 5th inning, with a runner on 1st, HOU batter Jose Altuve hit a soft ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.

	game	HOU vs TEX on 20230904
	state before	(0, 1, 1, 1, 7, 6, 5, 0)
60.	error	play,7,0,mccoc001,01,FX,FC6/G6.3-H;2-H(NR);1-3(E6);B-1
	state after	(0, 1, 0, 1, 7, 8, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2240 *In the top of the 7th inning, with bases loaded, HOU batter Chas McCormick hit a ground ball into a fielder's choice at short.*

The runners on 2nd and 3rd scored (no RBI and 1 RBI, resp.). An error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st.

	game	HOU vs TEX on 20230906
	state before	(0, 0, 0, 0, 5, 8, 1, 1)
61.	error	play,5,1,jankt001,21,FBBX,E4/G34.B-1
	state after	(0, 1, 0, 0, 5, 8, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2241 There were 2 errors in this game. The final score was 12-3.

In the bottom of the 5th inning, with bases empty, TEX batter Travis Jankowski hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the batter to reach 1st base.

	game	HOU vs TEX on 20230906
	state before	(1, 1, 0, 0, 7, 8, 2, 0)
62.	error	play,7,0,dubom001,10,BX,FC5/G56.1-3(E5/TH);B-1
	state after	(1, 1, 0, 1, 7, 8, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2242 *In the top of the 7th inning, with a runner on 1st, HOU batter Mauricio Dubon hit a ground ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	OAK vs TEX on 20230908
	state before	(0, 0, 0, 0, 4, 2, 2, 1)
63.	error	play,4,1,semim001,11,.BCX,D7/L7LD.B-3(E7)
	state after	(0, 0, 0, 1, 4, 2, 2, 1)
	fielder	7 (left field on the visiting team)

Summary 2243 *In the bottom of the 4th inning, with bases empty, TEX batter Marcus Semien hit a line drive double to deep left field. An error by the left fielder allowed the batter to reach 2nd.*

	game	OAK vs TEX on 20230910
	state before	(2, 1, 1, 0, 3, 3, 3, 0)
64.	error	play,3,0,butl1001,01,.SX,E5/G56.2-H(NR)(UR);1-3;B-2
	state after	(2, 0, 1, 1, 3, 4, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2244 There were 2 errors in this game. The final score was 4-9.

In the top of the 3rd inning, with runners on 1st and 2nd, OAK batter Lawrence Butler hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 2nd.

	game	OAK vs TEX on 20230910
	state before	(1, 0, 1, 1, 5, 4, 3, 1)
65.	error	play,5,1,lowen001,01,FX,S7/L7.3-H;2-H;B-2(E7/TH)
	state after	(1, 0, 1, 0, 5, 4, 5, 1)
	fielder	7 (left field on the visiting team)

Summary 2245 *In the bottom of the 5th inning, with runners on 2nd and 3rd, TEX batter Nathaniel Lowe hit a line drive single to left field. The runners on 2nd and 3rd scored (2 RBIs). A throwing error by the left fielder allowed the batter to reach 2nd.*

	game	BOS vs TEX on 20230919
	state before	(1, 1, 1, 0, 7, 3, 4, 0)
66.	error	play,7,0,verda001,12,.BCCX,FC6/G6.2-H(NR)(UR);1-3(E6/TH);B-2
	state after	(1, 1, 1, 0, 7, 3, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 2246 *In the top of the 7th inning, with runners on 1st and 2nd, BOS batter Alex Verdugo hit a ground ball into a fielder's choice at short. The runner on 2nd scored (unearned, no RBI). A throwing error by the shortstop allowed the runner on 1st to reach 3rd and the batter to reach 2nd.*

	game	BOS vs TEX on 20230919
	state before	(2, 0, 0, 0, 8, 3, 6, 1)
67.	error	play,8,1,seagc001,00,X,E4/TH/G34.B-2
	state after	(2, 0, 1, 0, 8, 3, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2247 *In the bottom of the 8th inning, with bases empty, TEX batter Corey Seager hit a ground ball to the 1st/2nd base gap. A throwing error by the second baseman allowed the batter to reach 2nd base.*

	game	BOS vs TEX on 20230920
	state before	(0, 0, 0, 0, 5, 4, 10, 1)
68.	error	play,5,1,semim001,11,BCX,E5/G6S.B-1
	state after	(0, 1, 0, 0, 5, 4, 10, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2248 *In the bottom of the 5th inning, with bases empty, TEX batter Marcus Semien hit a ground ball to shallow short. An error by the 3rd baseman allowed the batter to reach 1st base.*

	game	SEA vs TEX on 20230922
	state before	(2, 0, 0, 0, 7, 4, 8, 0)
69.	error	play,7,0,hernt002,10,BX,E5/G56S.B-1
	state after	(2, 1, 0, 0, 7, 4, 8, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2249 *In the top of the 7th inning, with bases empty, SEA batter Teoscar Hernandez hit a ground ball to shallow 3rd base. An error by the 3rd baseman allowed the batter to reach 1st base.*

Chapter 32

Toronto Blue Jays (TOR)

Founded in 1977, the Toronto Blue Jays compete in the American League East Division. They have won 2 World Series titles, the last in 1993, and 6 ALED titles, the last in 2015.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Toronto against the Blue Jays. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The defensive efficiency ratio (DER or DefEff) is one measure of the fielding performance of a team (see §3.2 for more details). The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 TOR team is 0.693, which is very close to the MLB average of 0.691 that year [BR]. The standard deviation of these 2023 DER values is $\sigma = 0.011$, so the Blue Jays in 2023 were less than half a standard deviation from the MLB average. This is neither a “high” or a “low” DER.

In TOR 2023, there were 73 errors in the 81 home games, 27 by the Blue Jays and the rest (46) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 71, much lower than 83.9, the MLB average.

32.1 Season totals

Throughout, `vt` denotes the visiting team and `ht` denotes the home team.

32.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	48
Runs by ht because of errors by vt:	8
Errors by ht:	27
Runs by vt because of errors by ht:	5

This is a poor error-to-run conversion rate: $8/48 = 16.7\%$ on the part of the Jays (compared to $5/27 = 18.5\ldots\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 TOR season (home and away) has $R/PA = 746/6220$ (about 12%) chance of resulting in a run. For *most* MLB teams, a randomly selected error is over twice as likely to result in a run than a randomly selected at-bat, but not in this case.

Errors by position	vt	ht
errors due to pitcher:	4	3
to catcher:	5	3
to 1st baseman:	3	3
to 2nd baseman:	8	5
to 3rd baseman:	9	9
to shortstop:	8	3
to left fielder:	3	0
to center fielder:	4	1
to right fielder:	2	0

32.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	15	11	26
(lone) runner on 1st	16	11	27
(lone) runner on 2nd	5	1	6
(lone) runner on 3rd	0	0	0
two runners on base	11	4	15
bases loaded	2	0	2
total	49	27	76

A table of frequencies of most common errors by the Blue Jays in TOR 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	4	14.8%
(1, 0, 0, 0)	4	14.8%
(2, 0, 0, 0)	3	11.1%
(0, 1, 0, 0)	3	11.1%
(1, 1, 0, 0)	2	7.4%
(2, 1, 0, 0)	6	22.2%
(2, 0, 1, 0)	1	3.7%
(0, 1, 1, 0)	2	7.4%
(2, 0, 1, 1)	1	3.7%

A table of frequencies of most common errors by a visiting team in TOR 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	11	22.9%
(1, 0, 0, 0)	3	6.2%
(2, 0, 0, 0)	1	2.1%
(0, 1, 0, 0)	5	10.4%
(1, 1, 0, 0)	7	14.6%
(2, 1, 0, 0)	4	8.3%
(0, 0, 1, 0)	2	4.2%
(1, 0, 1, 0)	2	4.2%
(2, 1, 1, 0)	3	6.2%
(2, 1, 1, 1)	1	2.1%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Jays homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file TOR2023.eva):

state	TOR2023 percentages
(0, 0, 0, 0)	24.1%
(1, 0, 0, 0)	16.3%
(2, 0, 0, 0)	12.9%
(0, 1, 0, 0)	5.7%
(1, 1, 0, 0)	7.0%
(2, 1, 0, 0)	7.1%
(0, 0, 1, 0)	2.3%
(1, 0, 1, 0)	2.8%
(2, 0, 1, 0)	3.2%
(0, 0, 0, 1)	0.1%
(1, 0, 0, 1)	0.6%
(2, 0, 0, 1)	1.4%
(0, 1, 1, 0)	1.6%
(1, 1, 1, 0)	2.9%
(2, 1, 1, 0)	3.4%

32.1.3 Some interesting errors

Some of the most interesting error plays in the TOR 2023 home games:

- The game ATL vs TOR on 2023-05-14 had 5 errors.
- The game NYA vs TOR on 2023-05-17 had 3 errors but each game in the Yankees-Blue Jays series that started on 2023-05-15 had at least 1 error, for a total of 6.

32.1.4 Relative team rankings

We have 81 home games between TOR and some visiting MLB team (vt). A number of teams did not play TOR in the regular season. However, of those that did, TOR *lost* more games than won against¹:

CHN (2/3), CLE (2/3), MIN (2/3), SDN (2/3),
TEX (4/4), NYA (5/7), BAL (6/7) .

The teams that *split* their series were:

BOS (3/6), PHI (1/2), TBA (3/6) .

The teams that TOR *won* more than lost against

ARI (0/3), ATL (0/3), CHA (0/3), KCA (0/3), ANA (1/3), DET (1/3),
MIL (1/3), OAK (1/3), SEA (1/3), SFN (1/3), WAS (1/3), HOU (1/4) .

The teams that did not play in any of the 81 home games in TOR 2023 were:

CIN, COL, LAN, MIA, NYN, PIT, SLN.

32.2 Game-by-game totals

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for TOR 2023 is as follows [BA], [BR].

position	player	games played	OBP
1 (C)	Alejandro Kirk	99	.334
3 (1B)	Vladimir Guerrero Jr.	121	.345
4 (2B)	Whit Merrifield	84	.318
5 (3B)	Matt Chapman	137	.330
6 (SS)	Bo Bichette	130	.339
7 (LF)	Daulton Varsho	117	.285
8 (CF)	Kevin Kiermaier	127	.322
9 (RF)	George Springer	131	.327

¹The vt's W/T is given in parentheses (W = numbers of (home) games vt won against TOR, T = Total home games vt played vs the Jays at Toronto.

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 TOR home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database [Re1].

32.2.1 April

	game	DET vs TOR on 20230413
	state before	(2, 0, 1, 0, 4, 1, 1, 1)
1.	error	play,4,1,espis001,02,CCX,E4/G4.2-3;B-1
	state after	(2, 1, 0, 1, 4, 1, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2250 *In the bottom of the 4th inning, with a runner on 2nd, TOR batter Santiago Espinal hit a ground ball to 2nd base. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	TBA vs TOR on 20230414
	state before	(2, 1, 0, 0, 2, 0, 1, 1)
2.	error	play,2,1,sprig001,11,CB+1,P01(E3).1-2
	state after	(2, 0, 1, 0, 2, 0, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2251 This game had 2 errors. The final score was 3-6. *In the bottom of the 2nd inning, with a runner on 1st, TOR batter George Springer has a 1-1 count. The pitcher attempted to pickoff the runner on 1st. An error by the first baseman allowed that runner to safely advance to 2nd. Springer is still at-bat with a 1-1 count.*

	game	TBA vs TOR on 20230414
	state before	(1, 1, 1, 1, 5, 1, 4, 1)
3.	error	play,5,1,jansd001,12,CBCX,FC6/G6S.3-H;2-H(NR)(UR);1X3(6E4);B-1
	state after	(1, 1, 0, 1, 5, 1, 6, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2252 *In the bottom of the 5th inning, with bases loaded, TOR batter Danny Jansen hit a ground ball into a fielder's choice at shallow short. The runner on 2nd score (1 RBI). The runner on 2nd also score (unearned, no RBI). An error by the second baseman (with an assist from the shortstop) allowed the runner on 1st to safely advance to 3rd and the batter to reach 1st. The bases are loaded.*

	game	CHA vs TOR on 20230424
	state before	(1, 0, 0, 0, 7, 2, 4, 0)
4.	error	play,7,0,sosal001,21,BFBX,E5/TH/G56.B-2
	state after	(1, 0, 1, 0, 7, 2, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2253 *In the top of the 7th inning, with bases empty, CHA batter Lenyn Sosa hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	SEA vs TOR on 20230430
	state before	(0, 0, 0, 0, 2, 4, 2, 1)
5.	error	play,2,1,varsd001,01,LX,E3/TH/G3.B-1
	state after	(0, 1, 0, 0, 2, 4, 2, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2254 This game had 2 errors. The final score was 10-8 (in extra innings).

In the bottom of the 2nd inning, with bases empty, TOR batter Daulton Varsho hit a ground ball to 1st base. A throwing error by the first baseman allowed the batter to reach 1st base.

	game	SEA vs TOR on 20230430
	state before	(1, 0, 0, 0, 8, 5, 8, 0)
6.	error	play,8,0,kelej001,01,S.X,E4/G34D.B-1
	state after	(1, 1, 0, 0, 8, 5, 8, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2255 *In the top of the 8th inning, with bases empty, SEA batter Jarred Kelenic hit a ground ball to deep 1st/2nd. An error by the second baseman allowed the batter to reach 1st base.*

32.2.2 May

Visiting/Home team rosters for ATL vs TOR on 2023-05-14

position	innings	player	position	innings	player
1	1	"Collin McHugh"	1	1	"Yusei Kikuchi"
1	2	"Michael Tonkin"	1	5	"Anthony Bass"
1	6	"Jesse Chavez"	1	6	"Jay Jackson"
1	7	"Kirby Yates"	1	7	"Trevor Richards"
1	8	"Nick Anderson"	1	9	"Nate Pearson"
1	9	"Raisel Iglesias"	2	1	"Danny Jansen"
2	1	"Travis d'Arnaud"	3	1	"Vladimir Guerrero Jr."
3	1	"Matt Olson"	4	1	"Santiago Espinal"
4	1	"Ozzie Albies"	5	1	"Matt Chapman"
5	1	"Austin Riley"	6	1	"Bo Bichette"
6	1	"Orlando Arcia"	7	1	"Whit Merrifield"
7	1	"Kevin Pillar"	8	1	"Daulton Varsho"
8	1	"Michael Harris II"	9	1	"George Springer"
9	1	"Ronald Acuna Jr."			

	game	ATL vs TOR on 20230514
	state before	(0, 0, 0, 0, 2, 2, 0, 1)
7.	error	play,2,1,merrw001,22,CFBBX,E4/P34D.B-2
	state after	(0, 0, 1, 0, 2, 2, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2256 This game had 5 errors. The final score was 5-6.
In the bottom of the 2nd inning, with bases empty, TOR batter Whit

Merrifield hit a pop up to deep 1st/2nd. An error by the second baseman allowed the batter to reach 2nd base.

	game	ATL vs TOR on 20230514
	state before	(0, 0, 0, 0, 3, 2, 3, 0)
8.	error	play,3,0,acunr001,22,BCCFBX,E5/G56.B-1
	state after	(0, 1, 0, 0, 3, 2, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2257 *In the top of the 3rd inning, with bases empty, ATL batter Ronald Acuna hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	ATL vs TOR on 20230514
	state before	(0, 0, 0, 0, 4, 5, 4, 1)
9.	error	play,4,1,bichb001,01,CX,E3/P3D.BX2(96)
	state after	(1, 0, 0, 0, 4, 5, 4, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2258 *In the bottom of the 4th inning, with bases empty, TOR batter Bo Bichette hit a pop up to deep 1st. An error by the first baseman allowed the batter to reach 1st. However, when the batter/runner attempted to advance, the shortstop (with an assist from the right fielder) tagged him out at 2nd.*

	game	ATL vs TOR on 20230514
	state before	(2, 1, 0, 0, 7, 5, 4, 0)
10.	error	play,7,0,pillk001,11,BCX,E5/TH/G56.1-3;B-1
	state after	(2, 1, 0, 1, 7, 5, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2259 *In the top of the 7th inning, with a runner on 1st, ATL batter Kevin Pillar hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	ATL vs TOR on 20230514
	state before	(0, 0, 0, 0, 7, 5, 4, 1)
11.	error	play,7,1,chapm001,21,.CBBX,E7/F8.B-2
	state after	(0, 0, 1, 0, 7, 5, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 2260 *In the bottom of the 7th inning, with bases empty, TOR batter Matt Chapman hit a fly ball to center field. An error by the left fielder allowed the batter to reach 2nd base.*

	game	NYA vs TOR on 20230515
	state before	(0, 1, 0, 0, 8, 7, 1, 1)
12.	error	play,8,1,beltb001,10,BX,FC4/G4.1-2(E4/TH);B-1
	state after	(0, 1, 1, 0, 8, 7, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2261 This game had 2 errors. Each game in the Yankees-Blue Jays series that started with this game had at least 1 error, for a total of 6. The final score was 7-4.

In the bottom of the 8th inning, with a runner on 1st, TOR batter Brandon Belt hit a ground ball into a fielder's choice at 2nd. A throwing error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.

	game	NYA vs TOR on 20230515
	state before	(2, 0, 0, 0, 9, 7, 4, 0)
13.	error	play,9,0,volpa001,12,CSBX,E4/TH/G4M.B-1
	state after	(2, 1, 0, 0, 9, 7, 4, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2262 *In the top of the 9th inning, with bases empty, NYA batter Anthony Volpe hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	NYA vs TOR on 20230516
	state before	(1, 1, 0, 0, 9, 6, 3, 1)
14.	error	play,9,1,bichb001,00,X,E6/G6.1-2;B-1
	state after	(1, 1, 1, 0, 9, 6, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2263 *In the bottom of the 9th inning, with a runner on 1st, TOR batter Bo Bichette hit a ground ball to short. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

Visiting/Home team rosters for NYA vs TOR on 2023-05-17

position	innings	player	position	innings	player
1	1	"Gerrit Cole"	1	1	"Chris Bassitt"
1	10	"Wandy Peralta"	1	8	"Tim Mayza"
1	7	"Clay Holmes"	1	8	"Yimi Garcia"
1	8	"Jimmy Cordero"	1	9	"Jordan Romano"
1	9	"Michael King"	2	1	"Alejandro Kirk"
2	1	"Jose Trevino"	3	1	"Brandon Belt"
2	10	"Kyle Higashioka"	3	9	"Cavan Biggio"
3	1	"Anthony Rizzo"	4	1	"Santiago Espinal"
4	1	"Gleyber Torres"	5	1	"Matt Chapman"
5	1	"DJ LeMahieu"	6	1	"Bo Bichette"
6	1	"Oswaldo Cabrera"	7	1	"Whit Merrifield"
6	8	"Anthony Volpe"	8	1	"Daulton Varsho"
7	1	"Jake Bauers"	9	1	"George Springer"
7	8	"Oswaldo Cabrera"			
8	1	"Harrison Bader"			
9	1	"Aaron Judge"			

	game	NYA vs TOR on 20230517
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
15.	error	play,1,1,sprig001,00,X,E6/TH/G56D.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2264 This game had 3 errors. The final score was 0-3 (in extra innings).

In the bottom of the 1st inning, with bases empty, TOR batter George Springer hit a ground ball to deep 3rd. A throwing error by the shortstop allowed the batter to reach 1st base.

	game	NYA vs TOR on 20230517
	state before	(0, 1, 0, 0, 1, 0, 0, 1)
16.	error	play,1,1,bichb001,22,*BB>FCF1X,13E1/G34.1-2;B-1
	state after	(0, 1, 1, 0, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2265 *In the bottom of the 1st inning, with a runner on 1st, TOR batter Bo Bichette hit a ground ball. An error by the pitcher (with an assist from the first baseman) allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	NYA vs TOR on 20230517
	state before	(0, 0, 1, 0, 10, 0, 0, 1)
17.	error	play,10,1,merrw001,10,...BX,E6/G6M.2-3;B-1
	state after	(0, 1, 0, 1, 10, 0, 0, 1)
	fielder	6 (pitcher on the visiting team)

Summary 2266 *In the bottom of the 10th inning, with a runner on 2nd, TOR batter Whit Merrifield hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	BAL vs TOR on 20230519
	state before	(2, 1, 0, 0, 8, 4, 1, 0)
18.	error	play,8,0,haysa001,02,FC1,P01(E1/TH).1-2
	state after	(2, 0, 1, 0, 8, 4, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 2267 *In the top of the 8th inning, with a runner on 1st, BAL batter Austin Hays has a 0-2 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 2nd. Hays is still at-bat with a 0-2 count.*

	game	MIL vs TOR on 20230530
	state before	(0, 1, 1, 0, 2, 2, 4, 1)
19.	error	play,2,1,sprig001,32,TCBBBX,E5/G56S.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 2, 2, 4, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2268 *In the bottom of the 2nd inning, with runners on 1st and 2nd, TOR batter George Springer hit a ground ball to shallow 3rd base. An error by the third baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	MIL vs TOR on 20230531
	state before	(0, 0, 0, 0, 5, 2, 0, 1)
20.	error	play,5,1,merrw001,10,BX,E5/TH/G56S.B-1
	state after	(0, 1, 0, 0, 5, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2269 *In the bottom of the 5th inning, with bases empty, TOR batter Whit Merrifield hit a ground ball to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

32.2.3 June

	game	HOU vs TOR on 20230605
	state before	(2, 1, 0, 0, 3, 6, 1, 0)
21.	error	play,3,0,meyej002,12,C1*BFX,S8/G4M.1-3(E8)
	state after	(2, 1, 0, 1, 3, 6, 1, 0)
	fielder	8 (center field on the home team)

Summary 2270 This game had 2 errors. The final score was 11-4.
In the top of the 3rd inning, with a runner on 1st, HOU batter Jake Meyers hit a ground ball single to center field. An error by the center fielder allowed the runner on 1st to advance to 3rd. The batter took 1st.

	game	HOU vs TOR on 20230605
	state before	(2, 1, 0, 0, 8, 10, 3, 0)
22.	error	play,8,0,brega001,00,X,S7/G5.1-H(E4/TH) (NR)
	state after	(2, 1, 0, 0, 8, 11, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2271 *In the top of the 8th inning, with a runner on 1st, HOU batter Alex Bregman hit a ground ball single to 3rd. A throwing error by the second baseman allowed the runner on 1st to score (unearned, no RBI). The batter took 1st.*

	game	HOU vs TOR on 20230606
	state before	(0, 1, 0, 0, 8, 1, 4, 1)
23.	error	play,8,1,beltb001,21,CBB+1,P01(E2/TH).1-2
	state after	(0, 0, 1, 0, 8, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2272 This game had 2 errors. The final score was 1-5.
In the bottom of the 8th inning, with a runner on 1st, TOR batter

Brandon Belt has a 2-1 count. The catcher attempted to pickoff the runner on 1st but a throwing error allowed that runner to safely reach 2nd. Belt is still at-bat with a 2-1 count.

	game	HOU vs TOR on 20230606
	state before	(1, 0, 1, 0, 8, 1, 4, 1)
24.	error	play,8,1,merrw001,01,SF,FLE9
	state after	(1, 0, 1, 0, 8, 1, 4, 1)
	fielder	9 (right field on the visiting team)

Summary 2273 This is an example of an *infinity* play.

In the bottom of the 8th inning, with a runner on 2nd and a 0-1 count, TOR batter Whit Merrifield hit a fly ball into foul territory. An error by the rightfielder allowed the batter to avoid the out. Merrifield is still at-bat, now with a 1-1 count.

	game	HOU vs TOR on 20230608
	state before	(1, 0, 0, 0, 3, 2, 0, 0)
25.	error	play,3,0,altuj001,22,BBCCFX,E6/G6.B-1
	state after	(1, 1, 0, 0, 3, 2, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2274 This game had 2 errors. The final score was 2-3.

In the top of the 3rd inning, with bases empty, HOU batter Jose Altuve hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.

	game	HOU vs TOR on 20230608
	state before	(0, 0, 0, 0, 4, 2, 0, 1)
26.	error	play,4,1,kirka001,01,CX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 4, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2275 *In the bottom of the 4th inning, with bases empty, TOR batter Alejandro Kirk hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	MIN vs TOR on 20230609
	state before	(1, 0, 0, 0, 7, 2, 2, 1)
27.	error	play,7,1,guerv002,11,BFX,E5/TH/G6S.B-1
	state after	(1, 1, 0, 0, 7, 2, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2276 This game had 2 errors. The final score was 3-2 (in extra innings).

In the bottom of the 7th inning, with bases empty, TOR batter Vladimir Guerrero hit a ground ball to shallow short. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	MIN vs TOR on 20230609
	state before	(1, 0, 1, 1, 10, 2, 2, 0)
28.	error	play,10,0,taylm002,11,FBX,7/SF/L78.3-H(UR);2-3(E7/TH)
	state after	(2, 0, 0, 1, 10, 3, 2, 0)
	fielder	7 (3rd baseman on the visiting team)

Summary 2277 *In the top of the 10th inning, with runners on 2nd and 3rd, MIN batter Michael Taylor hit a line drive to left center. The left fielder put out the batter. The runner on 1rd scored (1 RBI). A throwing error by the left fielder allowed the runner on 2nd to advance to 3rd.*

	game	MIN vs TOR on 20230610
	state before	(2, 0, 1, 0, 9, 9, 3, 0)
29.	error	play,9,0,larnt001,11,BCX,E4/G34.2-3;B-1
	state after	(2, 1, 0, 1, 9, 9, 3, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2278 *In the top of the 9th inning, with a runner on 2nd, MIN batter Trevor Larnach hit a ground ball to 1st/2nd. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	MIN vs TOR on 20230611
	state before	(1, 1, 0, 1, 5, 6, 1, 1)
30.	error	play,5,1,guerv002,01,FX,46(1)/F0/G4.3-H;B-2(E6/TH)
	state after	(2, 0, 1, 0, 5, 6, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2279 This game had 2 errors. The final score was 6-7.
In the bottom of the 5th inning, with runners on 1st and 3rd, TOR batter Vladimir Guerrero hit a ground ball to 2nd. At 2nd, the shortstop (with an assist from the second baseman) forced out the runner from 1st. The runner on 3rd scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 2nd.

	game	MIN vs TOR on 20230611
	state before	(1, 0, 0, 0, 7, 6, 3, 1)
31.	error	play,7,1,guerv002,10,BX,E5/TH/G56.B-1
	state after	(1, 1, 0, 0, 7, 6, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2280 *In the bottom of the 7th inning, with bases empty, TOR batter Vladimir Guerrero (same batter as the previous error in this game!) hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	OAK vs TOR on 20230624
	state before	(2, 1, 0, 0, 6, 2, 5, 0)
32.	error	play,6,0,langs001,12,FF.*B>X,E6/TH/G56S.1-2;B-1
	state after	(2, 1, 1, 0, 6, 2, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2281 *In the top of the 6th inning, with a runner on 1st, OAK batter Shea Langeliers hit a ground ball to shallow 3rd. A throwing error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	OAK vs TOR on 20230625
	state before	(1, 1, 0, 0, 1, 0, 1, 1)
33.	error	play,1,1,guerv002,10,*BX,S8/L8D+.1-H(E8) (NR) (UR);B-2
	state after	(1, 0, 1, 0, 1, 0, 2, 1)
	fielder	8 (center field on the visiting team)

Summary 2282 *In the bottom of the 1st inning, with a runner on 1st, TOR batter Vladimir Guerrero hit a hard line drive single to deep center field. An error by the center fielder allowed the runner on 1st to score (unearned, no RBI) and the batter to reach 2nd.*

	game	SFN vs TOR on 20230629
	state before	(1, 1, 0, 0, 2, 0, 0, 1)
34.	error	play,2,1,varsd001,10,BX,7/F5D.1-2(E7/TH)
	state after	(2, 0, 1, 0, 2, 0, 0, 1)
	fielder	7 (left field on the visiting team)

Summary 2283 This game had 2 errors. The final score was 1-2.
In the bottom of the 2nd inning, with a runner on 1st, TOR batter Daulton Varsho hit a fly ball to deep 3rd. The left fielder put out the batter. A throwing error by the left fielder allowed the runner on 1st to advance to 2nd.

	game	SFN vs TOR on 20230629
	state before	(2, 0, 0, 0, 9, 1, 2, 0)
35.	error	play,9,0,matol002,00,X,E5/TH/G5.B-1
	state after	(2, 1, 0, 0, 9, 1, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2284 *In the top of the 9th inning, with bases empty, SFN batter Luis Matos hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	BOS vs TOR on 20230630
	state before	(0, 0, 0, 0, 9, 5, 0, 1)
36.	error	play,9,1,sprig001,00,.X,E5/P4S.B-1
	state after	(0, 1, 0, 0, 9, 5, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2285 *In the bottom of the 9th inning, with bases empty, TOR batter George Springer hit a pop up to shallow 2nd. An error by the third baseman allowed the batter to reach 1st base.*

32.2.4 July

	game	BOS vs TOR on 20230701
	state before	(0, 0, 0, 0, 7, 6, 2, 0)
37.	error	play,7,0,dever001,22,FSFBBF,FLE2
	state after	(0, 0, 0, 0, 7, 6, 2, 0)
	fielder	2 (catcher on the home team)

Summary 2286 This game had 2 errors and an *infinity* play. The final score was 7-6.

In the top of the 7th inning, with bases empty and a 2-2 count, BOS batter Rafael Devers hit a fly ball into foul territory. An error by the catcher allowed the batter to avoid the out. Devers is still at-bat with a 2-2 count.

	game	BOS vs TOR on 20230701
	state before	(2, 0, 1, 1, 7, 6, 2, 1)
38.	error	play,7,1,kierk001,32,BFBSBF2,P03(E1/TH).3-H(NR)(UR)
	state after	(2, 1, 0, 0, 7, 6, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2287 *In the bottom of the 7th inning, with runners on 2nd and 3rd, TOR batter Kevin Kiermaier has a full count. The pitcher attempted to pickoff the runner on 3rd. A throwing error by the pitcher allowed that runner to score (unearned, no RBI). Kiermaier is still at-bat with a full count.*

	game	BOS vs TOR on 20230702
	state before	(0, 1, 0, 0, 7, 3, 4, 0)
39.	error	play,7,0,verda001,32,BSBBC>X,53/G56S.1-H(E3/TH) (NR)
	state after	(1, 1, 0, 0, 7, 3, 4, 0)
	fielder	3 (1st baseman on the home team)

Summary 2288 *In the top of the 7th inning, with a runner on 1st, BOS batter Alex Verdugo hit a ground ball to shallow 3rd. The first baseman (with an assist from the third baseman) put out the batter. A throwing error by the first baseman allowed the runner on 1st to score (unearned, no RBI).*

	game	ARI vs TOR on 20230714
	state before	(0, 1, 1, 0, 2, 0, 1, 1)
40.	error	play,2,1,varsd001,11,*BFX,E4/G34D.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 2, 0, 1, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2289 This game had 2 errors. The final score was 2-7. *In the bottom of the 2nd inning, with runners on 1st and 2nd, TOR batter Daulton Varsho hit a ground ball to deep 1st/2nd. An error by the second baseman allowed the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	ARI vs TOR on 20230714
	state before	(2, 1, 1, 0, 7, 2, 6, 1)
41.	error	play,7,1,jansd001,10,.BX,E4/TH/G34.2-H(NR) (UR);1-3;B-2
	state after	(2, 0, 1, 1, 7, 2, 7, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2290 *In the bottom of the 7th inning, with runners on 1st and 2nd, TOR batter Danny Jansen hit a ground ball to 1st/2nd. A throwing error by the second baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	ARI vs TOR on 20230715
	state before	(1, 1, 0, 0, 2, 0, 0, 1)
42.	error	play,2,1,kirka001,22,1C>FBBFX,S9/G34.1-3;B-2(E9/TH)
	state after	(1, 0, 1, 1, 2, 0, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 2291 *In the bottom of the 2nd inning, with a runner on 1st, TOR batter Alejandro Kirk hit a ground ball single to 1st/2nd. The runner on 1st advanced to 3rd. A throwing error by the right fielder allowed the batter to reach 2nd.*

	game	ARI vs TOR on 20230716
	state before	(2, 1, 1, 0, 1, 0, 0, 0)
43.	error	play,1,0,longe001,32,BCBS.FB2,P02(E1/TH).2-H(E8)(NR)(UR);1-2
	state after	(2, 0, 1, 0, 1, 1, 0, 0)
	fielder	1,8 (pitcher,center field on the home team)

Summary 2292 This game had 2 errors, but in only 1 play and by 2 different fielders. The final score was 5-7.

In the top of the 1st inning, with runners on 1st and 2nd, ARI batter Evan Longoria has a 3-2 count. The pitcher attempted to pickoff the runner on 2nd. An error by the center fielder and a throwing error by the pitcher allowed the runner on 2nd to score (unearned, no RBI) and the runner on 1st to advance to 2nd.

	game	SDN vs TOR on 20230720
	state before	(1, 1, 0, 0, 2, 0, 0, 0)
44.	error	play,2,0,sancg002,11,BCX,FC4/G4M.1-3(E4);B-1
	state after	(1, 1, 0, 1, 2, 0, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2293 *In the top of the 2nd inning, with a runner on 1st, SDN batter Gary Sanchez hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st.*

	game	ANA vs TOR on 20230728
	state before	(0, 1, 0, 0, 6, 1, 2, 1)
45.	error	play,6,1,bichb001,00,>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 6, 1, 2, 1)
	fielder	2 (catcher on the visiting team)

Summary 2294 This game had 2 errors. The final score was 1-4. *In the bottom of the 6th inning, with a runner on 1st, TOR batter Bo Bichette had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Bichette is still at bat, now with a 1-0 count.*

	game	ANA vs TOR on 20230728
	state before	(1, 1, 1, 0, 6, 1, 3, 1)
46.	error	play,6,1,sprig001,00,X,FC6/G6.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 6, 1, 3, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2295 *In the bottom of the 6th inning, with runners on 1st and 2nd, TOR batter George Springer hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.*

	game	ANA vs TOR on 20230730
	state before	(1, 0, 1, 0, 5, 1, 0, 1)
47.	error	play,5,1,merrw001,02,FFX,S7/L7L.2-H;B-2(E7)
	state after	(1, 0, 1, 0, 5, 1, 1, 1)
	fielder	7 (left field on the visiting team)

Summary 2296 This game had 2 errors. The final score was 3-2 (in extra innings).

In the bottom of the 5th inning, with a runner on 2nd, TOR batter Whit Merrifield hit a line drive single to left field. The runner on 2nd scored (1 RBI). An error by the left fielder allowed the batter to reach 2nd.

	game	ANA vs TOR on 20230730
	state before	(2, 1, 1, 0, 8, 1, 1, 1)
48.	error	play,8,1,jansd001,11,. *BCX,FC5/G5.2-3(E5);1-2;B-1
	state after	(2, 1, 1, 1, 8, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2297 *In the bottom of the 8th inning, with runners on 1st and 2nd, TOR batter Danny Jansen hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

32.2.5 August

	game	BAL vs TOR on 20230801
	state before	(0, 0, 0, 0, 7, 4, 3, 0)
49.	error	play,7,0,matej003,11,.BFX,E5/G5.B-1
	state after	(0, 1, 0, 0, 7, 4, 3, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2298 *In the top of the 7th inning, with bases empty, BAL batter Jorge Mateo hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	BAL vs TOR on 20230802
	state before	(0, 0, 0, 0, 3, 0, 0, 1)
50.	error	play,3,1,varsd001,10,BX,C/E2.B-1
	state after	(0, 1, 0, 0, 3, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2299 *This game had 2 errors. The final score was 1-4. In the bottom of the 3rd inning, with bases empty, TOR batter Daulton Varsho had a 1-0 count. Catcher's interference allowed the batter to take 1st.*

	game	BAL vs TOR on 20230802
	state before	(2, 1, 1, 1, 6, 1, 3, 1)
51.	error	play,6,1,varsd001,01,CX,E6/G6.3-H(NR) (UR);2-3;1-2;B-1
	state after	(2, 1, 1, 1, 6, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2300 *In the bottom of the 6th inning, with bases loaded, TOR batter Daulton Varsho hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI), the runners on 1st and 2nd to each advance one base, and the batter to reach 1st. The bases are loaded.*

	game	BAL vs TOR on 20230803
	state before	(0, 1, 1, 0, 8, 3, 1, 0)
52.	error	play,8,0,mcker001,00,X,FC/SH/BG25-.2-H(E5/TH) (NR) (UR);1-3;B-1
	state after	(0, 1, 0, 1, 8, 4, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2301 *In the top of the 8th inning, with runners on 1st and 2nd, BAL batter Ryan McKenna hit a soft sacrifice bunt to shallow 3rd. A throwing error by the third baseman allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 3rd and the batter to reach 1st.*

53.	game	CHN vs TOR on 20230811
	state before	(0, 0, 0, 0, 4, 3, 0, 0)
	error	play,4,0,swand001,00,X,6E3/G6MS.B-1
	state after	(0, 1, 0, 0, 4, 3, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 2302 *In the top of the 4th inning, with bases empty, CHN batter Dansby Swanson hit a ground ball to shallow short. A throwing error by the first baseman (with an assist from the shortstop) allowed the batter to reach 1st base.*

Visiting/Home team rosters for CHN vs TOR on 2023-08-13

position	innings	player	position	innings	player
1	1	"Jameson Taillon"	1	1	"Hyun Jin Ryu"
1	4	"Hayden Wesneski"	1	6	"Genesis Cabrera"
1	6	"Drew Smyly"	1	7	"Jay Jackson"
1	7	"Daniel Palencia"	1	8	"Erik Swanson"
1	8	"Caleb Kilian"	1	9	"Nate Pearson"
2	1	"Miguel Amaya"	2	1	"Danny Jansen"
3	1	"Patrick Wisdom"	3	1	"Brandon Belt"
4	1	"Nico Hoerner"	4	1	"Cavan Biggio"
5	1	"Nick Madrigal"	5	1	"Santiago Espinal"
6	1	"Dansby Swanson"	6	1	"Paul DeJong"
7	1	"Ian Happ"	7	1	"Whit Merrifield"
8	1	"Cody Bellinger"	8	1	"Daulton Varsho"
9	1	"Seiya Suzuki"	9	1	"George Springer"

	game	CHN vs TOR on 20230813
	state before	(1, 1, 0, 0, 1, 0, 0, 0)
54.	error	play,1,0,happi001,12,BCFFX,E3/G34.1-2;B-1
	state after	(1, 1, 1, 0, 1, 0, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 2303 This game had 3 errors. The final score was 4-11.
In the top of the 1st inning, with a runner on 1st, CHN batter Ian Happ hit a ground ball to 1st/2nd. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.

	game	CHN vs TOR on 20230813
	state before	(2, 0, 0, 0, 3, 2, 5, 1)
55.	error	play,3,1,espis001,02,CCFX,E4/TH/G4M.B-1
	state after	(2, 1, 0, 0, 3, 2, 5, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2304 *In the bottom of the 3rd inning, with bases empty, TOR batter Santiago Espinal hit a ground ball to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

	game	CHN vs TOR on 20230813
	state before	(2, 1, 0, 0, 5, 2, 8, 1)
56.	error	play,5,1,guerv002,11,CBX,E5/G56+.1-2;B-1
	state after	(2, 1, 1, 0, 5, 2, 8, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2305 *In the bottom of the 5th inning, with a runner on 1st, TOR batter Vladimir Guerrero hit a hard ground ball to 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

57.	game	PHI vs TOR on 20230816
	state before	(2, 0, 1, 1, 5, 3, 4, 0)
	error	play,5,0,stotb001,12,BC2CX,E5/TH/G5.3-H(NR)(UR);2-H(NR)(UR);B-1
	state after	(2, 1, 0, 0, 5, 5, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2306 *In the top of the 5th inning, with runners on 2nd and 3rd, PHI batter Bryson Stott hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runners on 2nd and 3rd to each score (unearned, no RBI) and the batter to reach 1st.*

58.	game	CLE vs TOR on 20230825
	state before	(1, 0, 0, 0, 9, 5, 2, 1)
	error	play,9,1,varsd001,22,BBFSFX,34E1/G3.B-1
	state after	(1, 1, 0, 0, 9, 5, 2, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2307 *In the bottom of the 9th inning, with bases empty, TOR batter Daulton Varsho hit a ground ball to 1st. An error by the pitcher (with an assist from the 2nd baseman) allowed the batter to reach 1st base.*

59.	game	CLE vs TOR on 20230826
	state before	(0, 1, 0, 0, 6, 2, 5, 0)
	error	play,6,0,ramij003,11,*BCX,E5/G56S.1-2;B-1
	state after	(0, 1, 1, 0, 6, 2, 5, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2308 This game had 2 errors. The final score was 3-8. *In the top of the 6th inning, with a runner on 1st, CLE batter Jose Ramirez hit a ground ball to shallow 3rd base. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st base.*

	game	CLE vs TOR on 20230826
	state before	(0, 1, 1, 0, 6, 2, 5, 0)
60.	error	play,6,0,gonzo001,00,X,E6/G6.2-3;1-2;B-1
	state after	(0, 1, 1, 1, 6, 2, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2309 *In the top of the 6th inning, with runners on 1st and 2nd, CLE batter Oscar Gonzalez hit a ground ball to short. An error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	WAS vs TOR on 20230828
	state before	(0, 0, 0, 0, 3, 1, 4, 1)
61.	error	play,3,1,varsd001,10,.BF,FLE6
	state after	(0, 0, 0, 0, 3, 1, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2310 This game had 2 errors and this particular play is an example of an *infinity play*. The final score was 3-6.
In the bottom of the 3rd inning, with bases empty and a 1-0, TOR batter Daulton Varsho hit a fly ball into foul territory. An error by the shortstop allowed the batter to avoid the out. Varsho is still at-bat, now with a 1-1 count.

	game	WAS vs TOR on 20230828
	state before	(1, 1, 0, 0, 3, 1, 5, 1)
62.	error	play,3,1,kierk001,22,CBFBX,FC1/G1S.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 3, 1, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2311 *In the bottom of the 3rd inning, with a runner on 1st, TOR batter Kevin Kiermaier hit a ground ball into a fielder's choice at shallow center infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

32.2.6 September

	game	KCA vs TOR on 20230908
	state before	(2, 1, 0, 0, 8, 3, 5, 1)
63.	error	play,8,1,sprig001,00,X,C/E2.1-2;B-1
	state after	(2, 1, 1, 0, 8, 3, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 2312 *In the bottom of the 8th inning, with a runner on 1st, TOR batter George Springer had a 0-0 count. Catcher's interference allowed the batter to take 1st.*

	game	KCA vs TOR on 20230909
	state before	(2, 0, 0, 0, 6, 1, 3, 0)
64.	error	play,6,0,peres002,11,CBX,C/E2.B-1
	state after	(2, 1, 0, 0, 6, 1, 3, 0)
	fielder	2 (catcher on the home team)

Summary 2313 *In the top of the 6th inning, with bases empty, KCA batter Salvador Perez had a 1-1 count. Catcher's interference allowed the batter to take 1st.*

	game	BOS vs TOR on 20230915
	state before	(0, 0, 0, 0, 2, 0, 0, 1)
65.	error	play,2,1,schnd001,01,TX,E6/TH/G6.B-1
	state after	(0, 1, 0, 0, 2, 0, 0, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2314 *In the bottom of the 2nd inning, with bases empty, TOR batter Davis Schneider hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	BOS vs TOR on 20230916
	state before	(2, 1, 0, 0, 7, 2, 0, 0)
66.	error	play,7,0,wongc001,01,S1,P01(E1/TH).1-3
	state after	(2, 0, 0, 1, 7, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2315 *In the top of the 7th inning, with a runner on 1st, BOS batter Connor Wong has a 0-1 count. The pitcher attempted to pickoff the runner on 1st. A throwing error by the pitcher allowed that runner to reach 3rd. Wong is still at-bat with a 0-1 count.*

	game	BOS vs TOR on 20230917
	state before	(1, 0, 0, 0, 4, 0, 1, 0)
67.	error	play,4,0,stort001,22,CF.BFBX,E5/G5D.B-1
	state after	(1, 1, 0, 0, 4, 0, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2316 *In the top of the 4th inning, with bases empty, BOS batter Trevor Story hit a ground ball to deep 3rd. An error by the third baseman allowed the batter to reach 1st base.*

	game	NYA vs TOR on 20230926
	state before	(2, 1, 0, 0, 7, 0, 0, 1)
68.	error	play,7,1,kierk001,01,F11>B,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 7, 0, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2317 *In the bottom of the 7th inning, with a runner on 1st, TOR batter Kevin Kiermaier had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Kiermaier is still at bat, now with a 1-1 count.*

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Visiting/Home team rosters for TBA vs TOR on 2023-09-29

position	innings	player	position	innings	player
1	1	"Aaron Civale"	1	1	"Yusei Kikuchi"
1	2	"Zack Littell"	1	6	"Chad Green"
1	4	"Erasmus Ramirez"	1	7	"Yimi Garcia"
1	7	"Taj Bradley"	1	8	"Jay Jackson"
1	8	"Colin Poche"	2	1	"Alejandro Kirk"
2	1	"Rene Pinto"	3	1	"Vladimir Guerrero Jr."
3	1	"Isaac Paredes"	4	1	"Cavan Biggio"
3	8	"Harold Ramirez"	5	1	"Matt Chapman"
4	1	"Curtis Mead"	6	1	"Bo Bichette"
5	1	"Osleivis Basabe"	6	8	"Santiago Espinal"
5	8	"Isaac Paredes"	7	1	"Daulton Varsho"
6	1	"Junior Caminero"	7	9	"Cam Eden"
7	1	"Randy Arozarena"	8	1	"Kevin Kiermaier"
8	1	"Manuel Margot"	8	9	"Daulton Varsho"
9	1	"Harold Ramirez"	9	1	"George Springer"
9	8	"Raimel Tapia"			

	game	TBA vs TOR on 20230929
	state before	(1, 1, 0, 0, 3, 0, 4, 1)
69.	error	play,3,1,chapm001,00,X,E8/L89XD.1-3;B-2
	state after	(1, 0, 1, 1, 3, 0, 4, 1)
	fielder	8 (center field on the visiting team)

Summary 2318 This game had 3 errors. The final score was 4-11. *In the bottom of the 3rd inning, with a runner on 1st, TOR batter Matt Chapman hit a line drive to extra deep right center field. An error by the center fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	TBA vs TOR on 20230929
	state before	(1, 1, 0, 0, 4, 0, 5, 1)
70.	error	play,4,1,biggc002,12,CBCX,S8/L8D.1-H(E8)(NR)(UR);B-H(NR)(UR)
	state after	(1, 0, 0, 0, 4, 0, 7, 1)
	fielder	8 (center field on the visiting team)

Summary 2319 *In the bottom of the 4th inning, with a runner on 1st, TOR batter Cavan Biggio hit a line drive single to deep center field. An error by the center fielder allowed the runner on 1st and the batter to both score (unearned, no RBI).*

	game	TBA vs TOR on 20230929
	state before	(0, 0, 1, 0, 6, 4, 8, 1)
71.	error	play,6,1,bichb001,01,FX,S5/G56S.2-3;B-2(E5/TH)
	state after	(0, 0, 1, 1, 6, 4, 8, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2320 *In the bottom of the 6th inning, with a runner on 2nd, TOR batter Bo Bichette hit a ground ball single to shallow 3rd base. The runner on 2nd advanced to 3rd. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	TBA vs TOR on 20230930
	state before	(1, 1, 1, 0, 4, 2, 1, 1)
72.	error	play,4,1,merrw001,02,CSX,FC6/G56.2-3(E6);1-2;B-1
	state after	(1, 1, 1, 1, 4, 2, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2321 *In the bottom of the 4th inning, with runners on 1st and 2nd, TOR batter Whit Merrifield hit a ground ball into a fielder's choice at 3rd. An error by the shortstop allowed the runners on 1st and 2nd to each advance one base and the batter to reach 1st. The bases are loaded.*

	game	TBA vs TOR on 20230930
	state before	(0, 1, 0, 0, 6, 4, 4, 0)
73.	error	play,6,0,wallt003,01,LB,PB.1-3(E2/TH)
	state after	(0, 0, 0, 1, 6, 4, 4, 0)
	fielder	2 (catcher on the home team)

Summary 2322 *In the top of the 6th inning, with a runner on 1st, TBA batter Davis Taylor Walls had a 0-1 count. The next pitch was a passed ball. A throwing error by the catcher allowed the runner on 1st to advance to 3rd. Walls is still at-bat, now with a 1-1 count.*

	game	TBA vs TOR on 20231001
	state before	(0, 1, 0, 0, 6, 11, 4, 1)
74.	error	play,6,1,espis001,00,X,FC4/G4.1-2(E4);B-1
	state after	(0, 1, 1, 0, 6, 11, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2323 *In the bottom of the 6th inning, with a runner on 1st, TOR batter Santiago Espinal hit a ground ball into a fielder's choice at 2nd. An error by the second baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	TBA vs TOR on 20231001
	state before	(2, 1, 1, 0, 7, 12, 6, 1)
75.	error	play,7,1,varsd001,00,X,E8/F78XD.2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 7, 12, 7, 1)
	fielder	8 (center field on the visiting team)

Summary 2324 *In the bottom of the 7th inning, with runners on 1st and 2nd, TOR batter Daulton Varsho hit a fly ball to extra deep left center field. An error by the center fielder allowed the runner on 2nd to score (unearned, no RBI), the runner on 1st to advance to 2nd, and the batter to reach 1st.*

Chapter 33

Washington Nationals (WAS)

In 2005, the MLB established the Washington Nationals (formerly the Montreal Expos) as a franchise of the NL located in Washington D.C. . In 2019, the team won the World Series as a Wild Card team defeating the Houston Astros.

This chapter is devoted to listing the errors in each regular season game of 2023 played in Washington DC against the Nationals. The errors shall be organized first by home team (in alphabetical order), then chronologically.

The index at the end of the report tabulates a few interesting stats, such as the errors with a fielder's choice or an attempted pick off.

The **defensive efficiency ratio** (usually denoted DER or DefEff) is one measure of the fielding performance of a team. The total number of errors E is another metric. High DER and low E is ideal.

The DER of the 2023 WAS team is 0.682 compared with the MLB average¹ of 0.691 that year [BR]. Therefore, the Nationals were below average, in terms of their defensive efficiency (which is bad).

In 2023, there were 96 errors in the 81 home games, 51 by the Nationals and the rest (45) by visiting teams. The total number of errors by the team (home and away) over the regular season in 2023 was 90. Since 83.9 was the MLB average, the Nationals have a slightly higher than average error total (which is also bad).

¹The standard deviation of these DER values is $\sigma = 0.011$.

33.1 Season totals

A box score shall *balance* (or *is proven*) when the total of the team's times at bat, bases on balls received, hit batters, sacrifice bunts, sacrifice flies and batters awarded first base because of interference or obstruction equals the total of that team's runs, players left on base and the opposing team's putouts.

– **MLB Rule Book** 9.03(c)

Throughout, **vt** denotes the visiting team and **ht** denotes the home team.

33.1.1 Run conversion and error total by position

This section tabulates the totals by fielder position and by error-to-run rate.

Runs and errors	totals
Errors by vt:	45
Runs by ht because of errors by vt:	12
Errors by ht:	51
Runs by vt because of errors by ht:	14

This is a good error-to-run conversion rate: $12/45 = 26.6\ldots\%$ on the part of the Nationals (compared to $14/51 = 27.4\ldots\%$ on the part of the visiting teams). A typical *play* (not necessarily with an error) in the 2023 WAS season (home and away) has about $R/PA = 700/6085$ (11.5%) change of resulting in a run. Therefore, a randomly selected WAS 2023 error is over twice as likely to result in a run than a randomly selected at-bat, at least in this case.

Note that the total number of errors (by either team) in a home game is 96, so the expected number of errors (by either team) in a randomly selected WAS 2023 home game is about 1.18.

As we will see below, there were 14 games with at least 3 errors, for a total of 45 errors. That's 46.8% of the errors in 17.3% of the games.

Errors by position	vt	ht
errors due to pitcher:	5	8
to catcher:	7	5
to 1st baseman:	3	3
to 2nd baseman:	4	5
to 3rd baseman:	14	7
to shortstop:	7	15
to left fielder:	1	3
to center fielder:	3	3
to right fielder:	1	2

33.1.2 Error total by state

This section tabulates the totals by game state.

A table of frequencies of most common errors categorized by runner configuration.

Errors by state	vt	ht	totals
bases empty	23	15	38
(lone) runner on 1st	10	18	28
(lone) runner on 2nd	4	6	10
(lone) runner on 3rd	2	0	2
two runners on base	3	11	14
bases loaded	3	1	4
total	45	51	96

A table of frequencies of most common errors by the Nationals in WAS 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	8	15.7%
(1, 0, 0, 0)	1	1.9%
(2, 0, 0, 0)	6	11.8%
(0, 1, 0, 0)	4	7.8%
(1, 1, 0, 0)	8	15.7%
(2, 1, 0, 0)	6	11.8%
(1, 0, 1, 0)	2	3.9%
(1, 1, 1, 0)	6	11.8%
(1, 1, 1, 1)	1	1.9%

A table of frequencies of most common errors by a visiting team in WAS 2023 home games:

state	frequency	percentage
(0, 0, 0, 0)	12	26.7%
(1, 0, 0, 0)	9	20%
(2, 0, 0, 0)	2	4.4%
(0, 1, 0, 0)	6	13.3%
(1, 1, 0, 0)	3	6.6%
(2, 1, 0, 0)	1	2.2%
(1, 0, 1, 0)	2	4.4%
(1, 0, 0, 1)	2	4.4%
(0, 1, 1, 1)	2	4.4%
(2, 1, 1, 1)	1	2.2%

For comparison, here is a part of Lindsey's table for frequency of game states in all plays (with and without an error) in a large number of baseball games. See §2.1.4 for more details.

state	Lindsey's percentages
(0, 0, 0, 0)	24.3%
(1, 0, 0, 0)	17.3%
(2, 0, 0, 0)	13.7%
(0, 1, 0, 0)	6.4%
(1, 1, 0, 0)	7.6%
(2, 1, 0, 0)	7.8%
(0, 0, 1, 0)	1.1%
(1, 0, 1, 0)	2.4%
(2, 0, 1, 0)	2.9%
(0, 0, 0, 1)	0.2%
(1, 0, 0, 1)	0.7%
(2, 0, 0, 1)	1.2%
(0, 1, 1, 0)	1.4%
(1, 1, 1, 0)	2.6%
(2, 1, 1, 0)	3.3%

From the More-Things-Change-The-More-They-Stay-The-Same Department: the 2023 Nationals homegames, has a similar frequency table (compiled with the aid of some Python programming using the Retrosheet data file `WAS2023.evn`):

state	WAs2023 percentages
(0, 0, 0, 0)	24.2%
(1, 0, 0, 0)	16.6%
(2, 0, 0, 0)	13.0%
(0, 1, 0, 0)	5.8%
(1, 1, 0, 0)	6.7%
(2, 1, 0, 0)	6.5%
(0, 0, 1, 0)	2.1%
(1, 0, 1, 0)	2.7%
(2, 0, 1, 0)	3.5%
(0, 0, 0, 1)	0.5%
(1, 0, 0, 1)	1.2%
(2, 0, 0, 1)	1.7%
(0, 1, 1, 0)	1.7%
(1, 1, 1, 0)	2.4%
(2, 1, 1, 0)	2.6%

33.1.3 Some interesting errors

Some of the most interesting error plays in the WAS 2023 home games:

- The game ATL vs WAS on 2023-03-30 which had 5 errors.
- The game MIA vs WAS on 2023-09-03 which had 4 errors.
- The game MIL vs WAS on 2023-07-71 had 3 errors and an ∞ play.
- The game MIA vs WAS on 2023-09-03 had 4 errors.
- The game LAN vs WAS on 2023-09-09 had 3 errors, including 2 involving a stolen base.

33.1.4 Relative team rankings

We have 81 home games between WAS and some visiting MLB team (vt). In the lists below, the visiting teams with that team's W/T record in paren-

theses².

A number of teams did not play WAS in the regular season. However, of those that did, WAS *lost* more games than won against:

BAL (2/2), CLE (2/3), LAN (2/3), PIT (2/3), SDN (2/3), SLN (2/3),
ARI (3/3), TBA (3/3), CIN (4/4), ATL (5/7), MIA (7/7)

and *won* more than lost against

OAK (0/3), SFN (0/3), BOS (1/3), CHA (1/3), COL (1/3), DET (1/3),
MIL (1/3), TEX (1/3), CHN (1/4) .

The teams that *split* their series were:

NYN (3/6), PHI (3/6) .

The teams that did not play in any of the 81 home games in WAS 2023 were:

ANA, HOU, KCA, MIN, NYA, SEA, TOR .

33.2 Game-by-game totals

Where's my English teacher when you need her:

“The Official Scorer shall not credit a base hit when a (sic) is called out for interference with a fielder attempting to field a batted ball.”

– **MLB Rule Book 9.05(b)(5)**

The errors are listed chronologically. To save space, we only give the team rosters in special cases.

A typical roster for WAS 2023 is as follows [BA].

²Where W = numbers of home games vt won against WAS, T = Total home games vt vs Nationals at WAS.

position	player	games played
1 (C)	Keibert Ruiz	119
3 (1B)	Dominic Smith	153
4 (2B)	Luis García Jr.	123
5 (3B)	Jeimer Candelario	98
6 (SS)	CJ Abrams	153
7 (LF)	Stone Garrett	78
8 (CF)	Alex Call	82
9 (RF)	Lane Thomas	153

In general, the rosters were found using my Python function `team_roster`. Moreover, the tabulation of all plays with an error in 2023 WAS home games is obtained from my Python program `errors_in_homegames`. Each program pulls data from a locally installed Retrosheet database.

The first home game of the Washington Nationals 2023 season had 5 errors. Braves' ace Max Fried started the game, but left after 3.1 innings due to a hamstring injury. He allowed one earned run and struck out two batters. Also, it marked the return of Ronald Acuña Jr. in full health for the start of the season, as well as the debut of young talent, with CJ Abrams.

33.2.1 March

Visiting/Home team rosters for ATL vs WAS on 2023-03-30

position	innings	player	position	innings	player
1	1	Max Fried	1	1	Patrick Corbin
1	4	Lucas Luetge	1	4	Erasmo Ramirez
1	5	Jesse Chavez	1	6	Mason Thompson
1	6	Dylan Lee	1	8	Hunter Harvey
1	7	Collin McHugh	1	9	Kyle Finnegan
1	9	A.J. Minter	2	1	Keibert Ruiz
2	1	Sean Murphy	3	1	Dominic Smith
3	1	Matt Olson	4	1	Luis Garcia
4	1	Ozzie Albies	5	1	Jeimer Candelario
5	1	Austin Riley	6	1	CJ Abrams
6	1	Orlando Arcia	7	1	Alex Call
7	1	Marcell Ozuna	7	9	Corey Dickerson
7	7	Sam Hilliard	8	1	Victor Robles
8	1	Michael Harris II	9	1	Lane Thomas
9	1	Ronald Acuna Jr.			

	game	ATL vs WAS on 20230330
	state before	(1, 1, 1, 0, 2, 1, 0, 0)
1.	error	play,2,0,acunr001,12,BCFX,FC6/G6+.2-3;1-2(E6);B-1
	state after	(1, 1, 1, 1, 2, 1, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2325 This game had 5 errors. The final score was 7-2.

In the top of the 2nd inning, with runners on 1st and 2nd, ATL batter Ronald Acuna hit a hard ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. An error by the shortstop allowed the runner on 1st to safely reach 2nd and the batter to reach 1st. The bases are loaded.

	game	ATL vs WAS on 20230330
	state before	(0, 0, 1, 1, 2, 3, 0, 1)
2.	error	play,2,1,calla002,11,CBX,8/SF/F89.3-H;2-3(E8/TH)
	state after	(1, 0, 0, 1, 2, 3, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 2326 *In the bottom of the 2nd inning, with runners on 2nd and 3rd, WAS batter Alex Call hit a sacrifice fly ball to right center field. The batter is put out by the center fielder. The runner on 3rd scored (1 RBI). A throwing error by the center fielder allowed the runner on 2nd to safely reach 3rd.*

	game	ATL vs WAS on 20230330
	state before	(2, 0, 0, 0, 5, 4, 1, 0)
3.	error	play,5,0,arcio002,22,SSBBX,E6/TH/G6+.B-1
	state after	(2, 1, 0, 0, 5, 4, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2327 *In the top of the 5th inning, with bases empty, ATL batter Orlando Arcia hit a hard ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	ATL vs WAS on 20230330
	state before	(0, 0, 0, 0, 6, 4, 2, 1)
4.	error	play,6,1,smitd008,22,CBBSX,S5/G6.B-2(E5/TH)
	state after	(0, 0, 1, 0, 6, 4, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2328 *In the bottom of the 6th inning, with bases empty, WAS batter Dominic Smith hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd base.*

	game	ATL vs WAS on 20230330
	state before	(1, 0, 1, 0, 9, 6, 2, 0)
5.	error	play,9,0,arcio002,10,BX,9/L9LD+.2-H(E6/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 9, 7, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2329 *In the top of the 9th inning, with a runner on 2nd, ATL batter Orlando Arcia hit a hard line drive to deep right field. The batter was put out by the right fielder. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI).*

33.2.2 April

	game	TBA vs WAS on 20230403
	state before	(1, 1, 0, 0, 2, 2, 0, 0)
6.	error	play,2,0,margm001,22,L1LF*BBFFX,S1/G23-.1-3(E1/TH);B-2
	state after	(1, 0, 1, 1, 2, 2, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2330 This game had 3 errors. The final score was 6-2. *In the top of the 2nd inning, with a runner on 1st, TBA batter Manuel Margot hit a soft ground ball single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	TBA vs WAS on 20230403
	state before	(0, 0, 0, 0, 7, 4, 0, 1)
7.	error	play,7,1,candj002,11,.BCX,E4/G4.B-1
	state after	(0, 1, 0, 0, 7, 4, 0, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2331 *In the bottom of the 7th inning, with bases empty, WAS batter Jeimer Candelario hit a ground ball to 2nd base. An error by the second baseman allowed the batter to reach 1st base.*

	game	TBA vs WAS on 20230403
	state before	(1, 0, 1, 0, 9, 6, 1, 0)
8.	error	play,9,0,mejif001,11,FB2,P02(E1/TH).2-3
	state after	(1, 0, 0, 1, 9, 6, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 2332 *In the top of the 9th inning, with a runner on 2nd, TBA batter Francisco Mejia has a 1-1 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to safely reach 3rd. Mejia is still at bat with a 1-1 count.*

	game	TBA vs WAS on 20230405
	state before	(2, 0, 1, 0, 6, 5, 2, 0)
9.	error	play,6,0,sirij001,22,B*BSFFX,S9/L89D+.2-H;BX3(56)(E9/TH)
	state after	(0, 0, 0, 0, 6, 6, 2, 1)
	fielder	9 (right field on the home team)

Summary 2333 *In the top of the 6th inning, with 2 outs and a runner on 2nd, TBA batter Jose Siri hit a hard line drive single to deep right center field. The runner from 2nd scored (1 RBI). A throwing error by the right fielder allowed the batter to advance to 2nd but was tagged out at 3rd by the shortstop when he tried to reach 3rd. This marked the 3rd out so the half-inning is over.*

	game	CLE vs WAS on 20230414
	state before	(1, 0, 0, 0, 6, 1, 3, 1)
10.	error	play,6,1,abrac001,12,FFBX,E5/P1.B-1
	state after	(1, 1, 0, 0, 6, 1, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2334 *This game had 2 errors. The final score was 4-3. In the bottom of the 6th inning, with bases empty, WAS batter C.J. Abrams hit a pop up to shallow infield. An error by the third baseman allowed the batter to reach 1st base.*

	game	CLE vs WAS on 20230414
	state before	(0, 0, 0, 0, 8, 3, 3, 0)
11.	error	play,8,0,gimea001,12,LFFBX,E1/G4S-.B-1
	state after	(0, 1, 0, 0, 8, 3, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 2335 *In the top of the 8th inning, with bases empty, CLE batter Andres Gimenez hit a soft ground ball to 2nd base. An error by the pitcher allowed the batter to reach 1st base.*

	game	CLE vs WAS on 20230415
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
12.	error	play,1,1,calla002,21,BCBX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2336 *In the bottom of the 1st inning, with bases empty, WAS batter Alex Call hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

Visiting/Home team rosters for CLE vs WAS on 2023-04-16

position	innings	player	position	innings	player
1	1	Shane Bieber	1	1	Patrick Corbin
1	7	Tim Herrin	1	7	Hobie Harris
1	8	Nick Sandlin	1	8	Mason Thompson
1	8	Trevor Stephan	1	9	Kyle Finnegan
2	1	Mike Zunino	2	1	Keibert Ruiz
3	1	Josh Bell	3	1	Dominic Smith
4	1	Andres Gimenez	4	1	Luis Garcia
5	1	Jose Ramirez	5	1	Jeimer Candelario
6	1	Gabriel Arias	6	1	CJ Abrams
7	1	Will Brennan	7	1	Stone Garrett
8	1	Myles Straw	7	9	Alex Call
9	1	Oscar Gonzalez	8	1	Victor Robles
			9	1	Lane Thomas

13.	game	CLE vs WAS on 20230416
	state before	(0, 0, 1, 0, 3, 1, 2, 0)
	error	play,3,0,strom002,12,*BFMX,E6/G6.2-3;B-1
	state after	(0, 1, 0, 1, 3, 1, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2337 This game had 3 errors. The final score was 6-7.
In the top of the 3rd inning, with a runner on 2nd, CLE batter Myles Straw hit a ground ball to short. An error by the shortstop allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.

14.	game	CLE vs WAS on 20230416
	state before	(0, 1, 0, 1, 3, 2, 2, 0)
	error	play,3,0,ariag002,12,AS*B1>B,SB2.3-H(E5)(NR)(UR)
	state after	(0, 0, 1, 0, 3, 3, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2338 *In the top of the 3rd inning, with runners on 1st and 3rd, CLE batter Gabriel Arias had a 1-2 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI). Arias is still at bat, now with a 2-2 count.*

	game	CLE vs WAS on 20230416
	state before	(1, 1, 0, 0, 4, 4, 3, 0)
15.	error	play,4,0,brenw002,00,X,FC1/G1S-.1-2(E1/TH);B-1
	state after	(1, 1, 1, 0, 4, 4, 3, 0)
	fielder	1 (pitcher on the home team)

Summary 2339 *In the top of the 4th inning, with a runner on 1st, CLE batter Will Brennan hit a soft ground ball into a fielder's choice at shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	BAL vs WAS on 20230418
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
16.	error	play,3,1,roblv001,32,>L1B+11BLB>FX,E5/TH/G56S.1-3;B-2
	state after	(0, 0, 1, 1, 3, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2340 *In the bottom of the 3rd inning, with a runner on 1st, WAS batter Victor Robles hit a ground ball to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

	game	BAL vs WAS on 20230419
	state before	(1, 0, 0, 0, 5, 3, 0, 1)
17.	error	play,5,1,thoml002,31,BCBBX,6E3/G6.B-2
	state after	(1, 0, 1, 0, 5, 3, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2341 This game had 2 errors. The final score was 4-0.
In the bottom of the 5th inning, with bases empty, WAS batter Lane Thomas hit a ground ball the batter to reach 2nd base.

	game	BAL vs WAS on 20230419
	state before	(2, 0, 0, 0, 9, 3, 0, 0)
18.	error	play,9,0,uriar001,10,BX,E6/TH/G6.B-1
	state after	(2, 1, 0, 0, 9, 3, 0, 0)
	fielder	6 (shortstop on the home team)

Summary 2342 *In the top of the 9th inning, with bases empty, BAL batter Ramon Urias hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	PIT vs WAS on 20230429 (1/2)
	state before	(0, 0, 0, 0, 4, 2, 0, 1)
19.	error	play,4,1,garrs003,11,BFX,C/E2.B-1
	state after	(0, 1, 0, 0, 4, 2, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2343 This game (game 1 of a doubleheader) had 2 errors. The final score was 16-1.

In the bottom of the 4th inning, with bases empty, WAS batter Stone Garrett had a 1-1 count. Catcher's interference allowed the batter to reach 1st base.

	game	PIT vs WAS on 20230429 (1/2)
	state before	(1, 1, 1, 1, 4, 4, 0, 0)
20.	error	play,4,0,reynb001,10,*BX,S9/G34.3-H;2-H(E3)(NR);1-3;B-2
	state after	(1, 0, 1, 1, 4, 6, 0, 0)
	fielder	3 (1st baseman on the home team)

Summary 2344 *In the top of the 4th inning, with bases loaded, PIT batter Bryan Reynolds hit a ground ball single to the 1st/2nd base gap. The runner on 3rd scored (1 RBI). An error by the first baseman allowed the runner on 2nd to also score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd. right field.*

33.2.3 May

21.	game	CHN vs WAS on 20230501
	state before	(2, 1, 0, 0, 1, 1, 0, 0)
	error	play,1,0,manct001,11,BC1,P01(E1/TH) .1-3
	state after	(2, 0, 0, 1, 1, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2345 *In the top of the 1st inning, with a runner on 1st, CHN batter Trey Mancini had a 1-1 count. The pitcher attempted to pick off the runner on 1st. A throwing error by the pitcher allowed that runner to safely reach 3rd.*

22.	game	CHN vs WAS on 20230502
	state before	(0, 1, 0, 0, 3, 0, 0, 1)
	error	play,3,1,roblv001,10,BX,FC/SH/BG23-.1-2;B-1(E3)
	state after	(0, 1, 1, 0, 3, 0, 0, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2346 *In the bottom of the 3rd inning, with a runner on 1st, WAS batter Victor Robles hit a soft bunt into a fielder's choice at right shallow infield. The runner on 1st advanced to 2nd. An error by the first baseman allowed the batter to reach 1st.*

23.	game	NYN vs WAS on 20230512
	state before	(1, 0, 0, 1, 4, 0, 1, 1)
	error	play,4,1,alu-j001,00,X,E6/G6.3-H(NR)(UR);B-2
	state after	(1, 0, 1, 0, 4, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2347 *In the bottom of the 4th inning, with a runner on 3rd, WAS batter Jake Alu hit a ground ball to short. An error by the shortstop allowed the runner on 3rd to score (unearned, no RBI) and the batter to reach 2nd.*

24.	game	NYN vs WAS on 20230514
	state before	(2, 1, 0, 1, 5, 6, 1, 0)
	error	play,5,0,canhm001,01,S>B,SB2.3-H(E2/TH) (NR) ; 1-3
	state after	(2, 0, 0, 1, 5, 7, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2348 *In the top of the 5th inning, with runners on 1st and 3rd, NYN batter Mark Canha had a 0-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI) and the runner now on 2nd to safely reach 3rd. Canha is still at bat, now with a 1-1 count.*

Visiting/Home team rosters for NYN vs WAS on 2023-05-15

position	innings	player	position	innings	player
1	1	David Peterson	1	1	Patrick Corbin
1	6	Tommy Hunter	1	7	Andres Machado
2	1	Francisco Alvarez	1	8	Carl Edwards Jr.
3	1	Mark Canha	1	9	Hobie Harris
4	1	Jeff McNeil	2	1	Keibert Ruiz
5	1	Eduardo Escobar	3	1	Dominic Smith
6	1	Francisco Lindor	4	1	Luis Garcia
6	8	Luis Guillorme	5	1	Jeimer Candelario
7	1	Tommy Pham	6	1	CJ Abrams
8	1	Brandon Nimmo	7	1	Stone Garrett
9	1	Starling Marte	8	1	Alex Call
			9	1	Lane Thomas

	game	NYN vs WAS on 20230515
	state before	(2, 1, 0, 0, 2, 0, 1, 0)
25.	error	play,2,0,mcnej002,00,>C,SB2.1-3(E2/TH)
	state after	(2, 0, 0, 1, 2, 0, 1, 0)
	fielder	2 (catcher on the home team)

Summary 2349 This game had 3 errors. The final score was 3-10.
In the top of the 2nd inning, with a runner on 1st, NYN batter Jeff McNeil had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. McNeil is still at bat, now with a 0-1 count.

	game	NYN vs WAS on 20230515
	state before	(2, 1, 0, 1, 4, 1, 3, 1)
26.	error	play,4,1,thoml002,22,BCBC>B,SB2.3-H(E2/TH) (NR)
	state after	(2, 0, 1, 0, 4, 1, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2350 *In the bottom of the 4th inning, with runners on 1st and 3rd, WAS batter Lane Thomas had a 2-2 count. During the next pitch, which was a bal, the runner on 1st stole 2nd. A throwing error by the catcher allowed the runner on 3rd to score (unearned, no RBI). Thomas is still at bat, now with a full count.*

	game	NYN vs WAS on 20230515
	state before	(2, 1, 0, 0, 6, 2, 6, 0)
27.	error	play,6,0,marts002,22,1C*BS*BX,S5/G56S.1-3(E5/TH)
	state after	(2, 1, 0, 1, 6, 2, 6, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2351 *In the top of the 6th inning, with a runner on 1st, NYN batter Starling Marte hit a ground ball single to shallow 3rd base. A throwing error by the third baseman allowed the runner on 1st to reach 3rd. The batter took 1st.*

	game	DET vs WAS on 20230519
	state before	(0, 0, 0, 0, 1, 1, 0, 0)
28.	error	play,1,0,baezj001,22,FBBC.X,E1/G13.B-1
	state after	(0, 1, 0, 0, 1, 1, 0, 0)
	fielder	1 (pitcher on the home team)

Summary 2352 This game had 2 errors. The final score was 8-6.
In the top of the 1st inning, with bases empty, DET batter Javier Baez hit a ground ball to right shallow infield. An error by the pitcher allowed the batter to reach 1st base.

	game	DET vs WAS on 20230519
	state before	(1, 1, 0, 0, 3, 3, 0, 0)
29.	error	play,3,0,ibana001,01,1FX,64(1)/F0/G6S.B-2(E4/TH)
	state after	(2, 0, 1, 0, 3, 3, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2353 *In the top of the 3rd inning, with a runner on 1st, DET batter Andy Ibanez hit a ground ball to shallow short. At 2nd, the second baseman (with an assist from the shortstop) forced out the runner from 1st. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	DET vs WAS on 20230520
	state before	(2, 0, 0, 0, 3, 2, 0, 0)
30.	error	play,3,0,greer003,32,CBFBBX,E7/F7S.B-1
	state after	(2, 1, 0, 0, 3, 2, 0, 0)
	fielder	7 (left field on the home team)

Summary 2354 *In the top of the 3rd inning, with bases empty, DET batter Riley Greene hit a fly ball to shallow left field. An error by the left fielder allowed the batter to reach 1st base.*

Visiting/Home team rosters for DET vs WAS on 2023-05-21

position	innings	player	position	innings	player
1	1	Joey Wentz	1	1	Josiah Gray
1	3	Jose Cisnero	1	6	Andres Machado
1	4	Mason Englert	1	6	Kyle Finnegan
1	6	Tyler Alexander	1	8	Carl Edwards Jr.
1	8	Jason Foley	1	9	Hunter Harvey
2	1	Jake Rogers	2	1	Riley Adams
3	1	Spencer Torkelson	3	1	Dominic Smith
4	1	Nick Maton	4	1	Michael Chavis
5	1	Zach McKinstry	5	1	Jeimer Candelario
6	1	Javier Baez	6	1	Ildemaro Vargas
7	1	Akil Baddoo	7	1	Stone Garrett
8	1	Riley Greene	8	1	Alex Call
9	1	Matt Vierling	9	1	Lane Thomas

	game	DET vs WAS on 20230521
	state before	(1, 1, 0, 0, 2, 0, 4, 1)
31.	error	play,2,1,menej001,12,SBFX,S7/L78.1-3;B-2(E7/TH)
	state after	(1, 0, 1, 1, 2, 0, 4, 1)
	fielder	7 (left field on the visiting team)

Summary 2355 This game had 3 errors. The final score was 4-6.

In the bottom of the 2nd inning, with a runner on 1st, WAS batter Joey Meneses hit a line drive single to left center field. The runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd.

	game	DET vs WAS on 20230521
	state before	(1, 1, 0, 0, 6, 3, 6, 1)
32.	error	play,6,1,garrs003,11,BSX,S48/G6M.1-3;B-2(E8)
	state after	(1, 0, 1, 1, 6, 3, 6, 1)
	fielder	8 (center field on the visiting team)

Summary 2356 *In the bottom of the 6th inning, with a runner on 1st, WAS batter Stone Garrett hit a ground ball single to deep 2nd base. The runner on 1st advanced to 3rd. A throwing error by the center fielder allowed the batter to reach 2nd.*

	game	DET vs WAS on 20230521
	state before	(0, 1, 0, 0, 7, 3, 6, 0)
33.	error	play,7,0,maton001,00,X,FC3/G3.1-3(E3/TH);B-2
	state after	(0, 0, 1, 1, 7, 3, 6, 0)
	fielder	3 (1st baseman on the home team)

Summary 2357 *In the top of the 7th inning, with a runner on 1st, DET batter Nick Maton hit a ground ball into a fielder's choice at 1st. A throwing error by the first baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SDN vs WAS on 20230524
	state before	(0, 1, 0, 0, 2, 0, 0, 1)
34.	error	play,2,1,smitd008,00,X,S9/G34.1-3(E9);B-2
	state after	(0, 0, 1, 1, 2, 0, 0, 1)
	fielder	9 (right field on the visiting team)

Summary 2358 This game had 3 errors. The final score was 3-5. *In the bottom of the 2nd inning, with a runner on 1st, WAS batter Dominic Smith hit a ground ball single to the 1st/2nd base gap. An error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

	game	SDN vs WAS on 20230524
	state before	(1, 0, 0, 0, 4, 2, 4, 1)
35.	error	play,4,1,calla002,00,X,S9/L89D.B-3(E4/TH)
	state after	(1, 0, 0, 1, 4, 2, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2359 *In the bottom of the 4th inning, with bases empty, WAS batter Alex Call hit a line drive single to deep right center field. A throwing error by the second baseman allowed the batter to reach 3rd.*

	game	SDN vs WAS on 20230524
	state before	(0, 1, 0, 0, 7, 3, 5, 1)
36.	error	play,7,1,thoml002,21,BBC>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 7, 3, 5, 1)
	fielder	2 (catcher on the visiting team)

Summary 2360 *In the bottom of the 7th inning, with a runner on 1st, WAS batter Lane Thomas had a 2-1 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Thomas is still at bat, now with a 3-1 count.*

Visiting/Home team rosters for SDN vs WAS on 2023-05-25

position	innings	player	position	innings	player
1	1	Blake Snell	1	1	Jake Irvin
1	6	Brent Honeywell	1	5	Andres Machado
1	7	Nick Martinez	1	6	Mason Thompson
1	7	Tim Hill	1	7	Thaddeus Ward
1	8	Drew Carlton	1	8	Kyle Finnegan
1	9	Josh Hader	1	9	Hunter Harvey
2	1	Brett Sullivan	2	1	Riley Adams
2	8	Austin Nola	2	8	Keibert Ruiz
3	1	Jake Cronenworth	3	1	Dominic Smith
3	2	Brandon Dixon	4	1	Luis Garcia
4	1	Rougned Odor	5	1	Jeimer Candelario
4	2	Jake Cronenworth	6	1	CJ Abrams
5	1	Ha-Seong Kim	7	1	Stone Garrett
5	2	Rougned Odor	7	8	Corey Dickerson
6	1	Xander Bogaerts	8	1	Alex Call
7	1	Juan Soto	9	1	Lane Thomas
8	1	Trent Grisham			
9	1	Fernando Tatis Jr.			

37.	game	SDN vs WAS on 20230525
	state before	(0, 0, 0, 0, 1, 0, 0, 1)
	error	play,1,1,thom1002,31,BBFBX,S5/L56S+.B-2(E5/TH)
	state after	(0, 0, 1, 0, 1, 0, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2361 This game had 3 errors. The final score was 8-6. *In the bottom of the 1st inning, with bases empty, WAS batter Lane Thomas hit a hard line drive single to shallow 3rd base. A throwing error by the third baseman allowed the batter to reach 2nd.*

	game	SDN vs WAS on 20230525
	state before	(0, 1, 1, 0, 3, 2, 1, 0)
38.	error	play,3,0,sotoj001,20,BB2,P02(E1/TH) .2-3;1-2
	state after	(0, 0, 1, 1, 3, 2, 1, 0)
	fielder	1 (pitcher on the home team)

Summary 2362 *In the top of the 3rd inning, with runners on 1st and 2nd, SDN batter Juan Soto has a 2-0 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed the runners on 1st and 2nd to each advance one base. Soto is still at bat with a 2-0 count.*

	game	SDN vs WAS on 20230525
	state before	(0, 1, 1, 0, 7, 5, 4, 1)
39.	error	play,7,1,calla002,00,X,S1/BG2.2-H(E1/TH) (NR) ;1-3;B-2
	state after	(0, 0, 1, 1, 7, 5, 5, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2363 *In the bottom of the 7th inning, with runners on 1st and 2nd, WAS batter Alex Call hit a bunt single to shallow infield. A throwing error by the pitcher allowed the runner on 1st to advance to 3rd and the batter to reach 2nd.*

33.2.4 June

	game	PHI vs WAS on 20230602
	state before	(1, 1, 1, 0, 8, 6, 7, 0)
40.	error	play,8,0,schwk001,22,..BFCFBX,6(1)/FO/G6M.2-H(E6/TH) (NR) (UR) ;B-1
	state after	(2, 1, 0, 0, 8, 7, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 2364 *In the top of the 8th inning, with runners on 1st and 2nd, PHI batter Kyle Schwarber hit a ground ball to deep short. The*

runner from 1st was forced out by the shortstop. A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.

	game	PHI vs WAS on 20230604
	state before	(2, 1, 0, 0, 5, 2, 1, 0)
41.	error	play,5,0,turnt001,22,CC*BBX,FC6/G56D.1-2(E6);B-1
	state after	(2, 1, 1, 0, 5, 2, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2365 *In the top of the 5th inning, with a runner on 1st, PHI batter Trea Turner hit a ground ball into a fielder's choice at deep 3rd. An error by the shortstop allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	ARI vs WAS on 20230606
	state before	(0, 0, 0, 0, 4, 1, 4, 0)
42.	error	play,4,0,rojaj001,01,CX,S1/BG23.B-2(E1/TH)
	state after	(0, 0, 1, 0, 4, 1, 4, 0)
	fielder	1 (pitcher on the home team)

Summary 2366 *This game had 2 errors. The final score was 10-5. In the top of the 4th inning, with bases empty, ARI batter Josh Rojas hit a bunt single to shallow infield. A throwing error by the pitcher allowed the batter to reach 2nd.*

	game	ARI vs WAS on 20230606
	state before	(0, 0, 0, 0, 6, 6, 5, 0)
43.	error	play,6,0,martk001,00,.X,E6/L6.B-1
	state after	(0, 1, 0, 0, 6, 6, 5, 0)
	fielder	6 (shortstop on the home team)

Summary 2367 *In the top of the 6th inning, with bases empty, ARI batter Ketel Marte hit a line drive to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	ARI vs WAS on 20230607
	state before	(2, 0, 0, 0, 1, 3, 0, 1)
44.	error	play,1,1,candj002,11,BCX,E5/TH/G56.B-1
	state after	(2, 1, 0, 0, 1, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2368 *In the bottom of the 1st inning, with bases empty, WAS batter Jeimer Candelario hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	MIA vs WAS on 20230617
	state before	(1, 1, 0, 0, 4, 1, 0, 0)
45.	error	play,4,0,davij007,01,CX,FC4/G4.1-3(E4/TH);B-2
	state after	(1, 0, 1, 1, 4, 1, 0, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2369 *In the top of the 4th inning, with a runner on 1st, MIA batter Jonathan Davis hit a ground ball into a fielder's choice at 2nd. A throwing error by the second baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	MIA vs WAS on 20230618
	state before	(0, 1, 0, 0, 4, 2, 2, 0)
46.	error	play,4,0,amayj001,10,B1X,S7/G56.1-3;B-2(E7/TH)
	state after	(0, 0, 1, 1, 4, 2, 2, 0)
	fielder	7 (left field on the home team)

Summary 2370 *In the top of the 4th inning, with a runner on 1st, MIA batter Jacob Amaya hit a ground ball single to left field. The runner on 1st advanced to 3rd. A throwing error by the left fielder allowed the batter to reach 2nd base.*

	game	SLN vs WAS on 20230620
	state before	(2, 1, 0, 0, 9, 9, 2, 0)
47.	error	play,9,0,donob001,12,C*BSX,E6/TH/G6.1-3;B-1
	state after	(2, 1, 0, 1, 9, 9, 2, 0)
	fielder	6 (shortstop on the home team)

Summary 2371 *In the top of the 9th inning, with a runner on 1st, SLN batter Brendan Donovan hit a ground ball to short. A throwing error by the shortstop allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

	game	ARI vs WAS on 20230622
	state before	(2, 1, 0, 0, 1, 0, 0, 0)
48.	error	play,1,0,walkc002,00,X,S8/G6M.1-H(E8)(NR)(UR)
	state after	(2, 1, 0, 0, 1, 1, 0, 0)
	fielder	8 (center field on the home team)

Summary 2372 *This game had 2 errors. The final score was 5-3. In the top of the 1st inning, with a runner on 1st, ARI batter Christian Walker hit a ground ball single to deep short. An error by the center fielder allowed the runner on 1st to core (unearned, no RBI). The batter took 1st.*

	game	ARI vs WAS on 20230622
	state before	(0, 0, 0, 0, 7, 2, 1, 0)
49.	error	play,7,0,thoma004,01,.FX,E4/TH/L4+.B-1
	state after	(0, 1, 0, 0, 7, 2, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2373 *In the top of the 7th inning, with bases empty, ARI batter Alek Thomas hit a hard line drive to 2nd base. A throwing error by the second baseman allowed the batter to reach 1st base.*

33.2.5 July

	game	CIN vs WAS on 20230705
	state before	(2, 0, 0, 0, 2, 1, 1, 0)
50.	error	play,2,0,stees001,02,CCX,E5/TH/G5.B-1
	state after	(2, 1, 0, 0, 2, 1, 1, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2374 This game had 2 errors. The final score was 9-2.
In the top of the 2nd inning, with bases empty, CIN batter Spencer Steer hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.

	game	CIN vs WAS on 20230705
	state before	(2, 0, 1, 0, 2, 3, 1, 0)
51.	error	play,2,0,friet001,00,X,E4/G34.2-3;B-1
	state after	(2, 1, 0, 1, 2, 3, 1, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2375 *In the top of the 2nd inning, with a runner on 2nd, CIN batter TJ Friedl hit a ground ball to the 1st/2nd base gap. An error by the second baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	TEX vs WAS on 20230708
	state before	(1, 1, 0, 0, 3, 0, 7, 0)
52.	error	play,3,0,seagc001,00,X,46(1)/F0/G3.B-2(E6/TH)
	state after	(2, 0, 1, 0, 3, 0, 7, 0)
	fielder	6 (shortstop on the home team)

Summary 2376 This game had 2 errors. The final score was 3-8.
In the top of the 3rd inning, with a runner on 1st, TEX batter Corey Seager hit a ground ball to 1st. The runner from 1st was forced out by the shortstop (with an assist from the second baseman). A throwing error by the shortstop allowed the batter to reach 2nd base.

	game	TEX vs WAS on 20230708
	state before	(2, 0, 0, 0, 3, 0, 7, 1)
53.	error	play,3,1,garc1006,11,BFX,T9/L9LD+.B-H(E4/TH) (NR) (UR)
	state after	(2, 0, 0, 0, 3, 0, 8, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2377 *In the bottom of the 3rd inning, with bases empty, WAS batter Luis Garcia hit a hard line drive triple to deep right field. A throwing error by the second baseman allowed the batter to score (unearned, no RBI). The bases are empty.*

	game	TEX vs WAS on 20230709
	state before	(1, 0, 0, 0, 1, 0, 0, 1)
54.	error	play,1,1,thom1002,12,CFBX,E1/TH/G1S.B-1
	state after	(1, 1, 0, 0, 1, 0, 0, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2378 *In the bottom of the 1st inning, with bases empty, WAS batter Lane Thomas hit a ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.*

Visiting/Home team rosters for COL vs WAS on 2023-07-24

position	innings	player	position	innings	player
1	1	Jake Bird	1	1	Patrick Corbin
1	3	Karl Kauffmann	1	7	Rico Garcia
1	7	Brad Hand	1	8	Cory Abbott
1	8	Gavin Hollowell	2	1	Keibert Ruiz
1	9	Matt Koch	3	1	Dominic Smith
2	1	Elias Diaz	4	1	Luis Garcia
3	1	Michael Toglia	5	1	Jeimer Candelario
4	1	Harold Castro	6	1	CJ Abrams
5	1	Alan Trejo	7	1	Corey Dickerson
6	1	Ezequiel Tovar	7	8	Stone Garrett
7	1	Jurickson Profar	7	9	Ildemaro Vargas
8	1	Brenton Doyle	8	1	Alex Call
9	1	Nolan Jones	9	1	Lane Thomas
			9	9	Stone Garrett

	game	COL vs WAS on 20230724
	state before	(1, 1, 1, 0, 4, 0, 0, 0)
55.	error	play,4,0,gricr001,00,X,D8/L89D.2-H;1-H(E8) (NR)
	state after	(1, 0, 1, 0, 4, 2, 0, 0)
	fielder	8 (center field on the home team)

Summary 2379 This game had 3 errors. The final score was 10-6. *In the top of the 4th inning, with runners on 1st and 2nd, COL batter Randal Grichuk hit a line drive double to deep right center field. The runner on 2nd scored (1 RBI). An error by the center fielder allowed the runner on 1st to also score (unearned, no RBI). The batter took 2nd.*

	game	COL vs WAS on 20230724
	state before	(2, 0, 1, 0, 7, 5, 0, 0)
56.	error	play,7,0,tovae001,32,FB.BFBFFFX,S5/G5.2-H(E5/TH) (NR) (UR)
	state after	(2, 1, 0, 0, 7, 6, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2380 *In the top of the 7th inning, with a runner on 2nd, COL batter Ezequiel Tovar hit a ground ball single to 3rd base. An error by the third baseman allowed the runner on 2nd to score (unearned, no RBI). The batter took 1st.*

	game	COL vs WAS on 20230724
	state before	(0, 0, 0, 0, 9, 10, 4, 1)
57.	error	play,9,1,ruizk001,12,.CBCFX,S4/G4.B-2(E4/TH)
	state after	(0, 0, 1, 0, 9, 10, 4, 1)
	fielder	4 (2nd baseman on the visiting team)

Summary 2381 *In the bottom of the 9th inning, with bases empty, WAS batter Keibert Ruiz hit a ground ball single to 2nd base. A throwing error by the second baseman allowed the batter to reach 2nd.*

	game	COL vs WAS on 20230725
	state before	(0, 0, 0, 0, 2, 2, 0, 1)
58.	error	play,2,1,menej001,02,CC.FX,E5/TH/G56.B-1
	state after	(0, 1, 0, 0, 2, 2, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2382 This game had 2 errors. The final score was 5-6. *In the bottom of the 2nd inning, with bases empty, WAS batter Joey Meneses hit a ground ball to 3rd base. A throwing error by the third baseman allowed the batter to reach 1st base.*

	game	COL vs WAS on 20230725
	state before	(0, 0, 1, 0, 8, 5, 2, 1)
59.	error	play,8,1,thoml002,22,BC*B>C.X,E5/G56.2-3;B-1
	state after	(0, 1, 0, 1, 8, 5, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2383 *In the bottom of the 8th inning, with a runner on 2nd, WAS batter Lane Thomas hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 2nd to advance to 3rd and the batter to reach 1st base.*

	game	COL vs WAS on 20230726
	state before	(1, 1, 0, 0, 3, 3, 0, 1)
60.	error	play,3,1,thoml002,32,B1FB+11SB>F>F>X,S5/G56.1-3(E5/TH);B-2
	state after	(1, 0, 1, 1, 3, 3, 0, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2384 *In the bottom of the 3rd inning, with a runner on 1st, WAS batter Lane Thomas hit a ground ball single to 3rd base. A throwing error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

Visiting/Home team rosters for MIL vs WAS on 2023-07-31

position	innings	player	position	innings	player
1	1	Corbin Burnes	1	1	Jake Irvin
1	7	Elvis Peguero	1	6	Jose A. Ferrer
1	7	Hoby Milner	1	7	Jordan Weems
1	8	Bryse Wilson	1	8	Mason Thompson
2	1	Victor Caratini	1	9	Kyle Finnegan
3	1	Carlos Santana	2	1	Keibert Ruiz
4	1	Brice Turang	3	1	Dominic Smith
4	7	Owen Miller	4	1	Luis Garcia
5	1	Andruw Monasterio	5	1	Ildemaro Vargas
6	1	Willy Adames	6	1	CJ Abrams
7	1	Christian Yelich	7	1	Corey Dickerson
8	1	Joey Wiemer	7	8	Stone Garrett
9	1	Sal Frelick	8	1	Alex Call
			9	1	Lane Thomas

61.	game	MIL vs WAS on 20230731
	state before	(0, 0, 0, 0, 3, 0, 1, 0)
	error	play,3,0,turab002,02,CFF,FLE7
	state after	(0, 0, 0, 0, 3, 0, 1, 0)
	fielder	7 (left field on the home team)

Summary 2385 This game had 3 errors and an ∞ play. The final score was 3-5.

In the top of the 3rd inning, with bases empty, MIL batter Brice Turang had a 0-2 count. The batter hit a foul fly ball to left field. An error by the left fielder allowed the batter to remain safe at bat. Turand still has a 0-2 count. There is no change in the game state.

	game	MIL vs WAS on 20230731
	state before	(0, 0, 0, 0, 5, 1, 1, 1)
62.	error	play,5,1,calla002,22,CF.BBX,E6/G6.B-1
	state after	(0, 1, 0, 0, 5, 1, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2386 *In the bottom of the 5th inning, with bases empty, WAS batter Alex Call hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	MIL vs WAS on 20230731
	state before	(1, 0, 1, 0, 5, 1, 1, 1)
63.	error	play,5,1,thoml002,31,*BF>B.BX,S7/L78.2-H;B-3(E5/TH)
	state after	(1, 0, 0, 1, 5, 1, 2, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2387 *In the bottom of the 5th inning, with a runner on 2nd, WAS batter Lane Thomas hit a line drive single to left center field. The runner on 2nd base scored. An error by the third baseman allowed the batter to reach 3rd.*

33.2.6 August

	game	MIL vs WAS on 20230802
	state before	(0, 0, 0, 0, 2, 1, 0, 1)
64.	error	play,2,1,garrs003,00,X,C/E2.B-1
	state after	(0, 1, 0, 0, 2, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2388 This game had 3 errors. The final score was 2-3. *In the bottom of the 2nd inning, with bases empty, WAS batter Stone Garrett had a 0-0 count. Catcher's interference allowed the batter to reach 1st base.*

	game	MIL vs WAS on 20230802
	state before	(0, 0, 0, 0, 9, 2, 1, 1)
65.	error	play,9,1,dickc002,32,...SS.BFBBX,E3/G9S.B-1
	state after	(0, 1, 0, 0, 9, 2, 1, 1)
	fielder	3 (1st baseman on the visiting team)

Summary 2389 *In the bottom of the 9th inning, with bases empty, WAS batter Corey Dickerson hit a ground ball to shallow right field. An error by the first baseman allowed the batter to reach 1st base.*

	game	MIL vs WAS on 20230802
	state before	(0, 1, 1, 1, 9, 2, 1, 1)
66.	error	play,9,1,calla002,12,.SFBX,FC5/G56.3-H(E5/TH)(NR)(UR);2-H(NR)(UR);1-2;B-1
	state after	(0, 1, 1, 0, 9, 2, 3, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2390 *In the bottom of the 9th inning, with bases loaded, WAS batter Alex Call hit a ground ball into a fielder's choice at 3rd. A throwing error by the third baseman allowed the runners on 2nd and 3rd both scored (no RBIs), The runner on 1st to advance to 3rd, and the batter to reach 1st.*

	game	OAK vs WAS on 20230811
	state before	(1, 1, 0, 0, 9, 2, 8, 0)
67.	error	play,9,0,butl1001,02,SFFX,FC3/G3.1-2(E3);B-1
	state after	(1, 1, 1, 0, 9, 2, 8, 0)
	fielder	3 (1st baseman on the home team)

Summary 2391 *In the top of the 9th inning, with a runner on 1st, OAK batter Lawrence Butler hit a ground ball into a fielder's choice at 1st. An error by the first baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st.*

	game	OAK vs WAS on 20230813
	state before	(1, 1, 1, 0, 9, 7, 2, 0)
68.	error	play,9,0,rookb001,22,CB*S*BX,FC5/G56.2-3;1-2(E5);B-1
	state after	(1, 1, 1, 1, 9, 7, 2, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2392 This game had 2 errors. The final score was 7-8.

In the top of the 9th inning, with runners on 1st and 2nd, OAK batter Brent Rooker hit a ground ball into a fielder's choice at 3rd. The runner on 2nd advanced to 3rd. An error by the third baseman allowed the runner on 1st to advance to 2nd and the batter to reach 1st. The bases are loaded.

	game	OAK vs WAS on 20230813
	state before	(2, 1, 1, 1, 9, 7, 5, 1)
69.	error	play,9,1,calla002,11,BCX,E6/G6.3-H(NR)(UR);2-H(NR)(UR);1-3;B-1
	state after	(2, 1, 0, 1, 9, 7, 7, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2393 *In the bottom of the 9th inning, with bases loaded, WAS batter Alex Call hit a ground ball to short. An error by the shortstop allowed the runners on 2nd and 3rd both scored (no RBIs), the runner on 1st to advance to 3rd, and the batter to reach 1st.*

	game	BOS vs WAS on 20230815
	state before	(0, 0, 1, 1, 4, 3, 4, 0)
70.	error	play,4,0,mcgur002,00,.X,C/E2.B-1
	state after	(0, 1, 1, 1, 4, 3, 4, 0)
	fielder	2 (catcher on the home team)

Summary 2394 *In the top of the 4th inning, with runners on 2nd and 3rd, BOS batter Reese McGuire had a 0-0 count. Catcher's interference allowed the batter to reach 1st base. The bases are loaded.*

71.	game	BOS vs WAS on 20230817
	state before	(0, 1, 0, 0, 3, 1, 0, 1)
	error	play,3,1,thom1002,22,FBCB+1>S,K+SB2.1-3(E2/TH)
	state after	(1, 0, 0, 1, 3, 1, 0, 1)
	fielder	2 (catcher on the visiting team)

Summary 2395 This game had 2 errors. The final score was 7-10.
In the bottom of the 3rd inning, with a runner on 1st, WAS batter Lane Thomas had a 2-2 count. During the next pitch, which was a swinging strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. The batter struck out.

72.	game	BOS vs WAS on 20230817
	state before	(1, 0, 0, 1, 3, 1, 0, 1)
	error	play,3,1,menej001,10,BX,E5/G5S.3-H(UR);B-1
	state after	(1, 1, 0, 0, 3, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2396 *In the bottom of the 3rd inning, with a runner on 3rd, WAS batter Joey Meneses hit a ground ball to shallow 3rd base. An error by the third baseman allowed runner on 3rd to score (1 RBI) and the batter to reach 1st base.*

73.	game	PHI vs WAS on 20230818
	state before	(1, 0, 0, 0, 4, 6, 1, 1)
	error	play,4,1,smitd008,31,CBBBX,E8/F78XD+.B-2
	state after	(1, 0, 1, 0, 4, 6, 1, 1)
	fielder	8 (center field on the visiting team)

Summary 2397 *In the bottom of the 4th inning, with bases empty, WAS batter Dominic Smith hit a fly ball to extra deep left center field. An error by the center fielder allowed the batter to reach 2nd base.*

Visiting/Home team rosters for PHI vs WAS on 2023-08-19

position	innings	player	position	innings	player
1	1	Cristopher Sanchez	1	1	Jake Irvin
1	7	Seranthony Dominguez	1	7	Andres Machado
1	8	Andrew Bellatti	1	7	Jose A. Ferrer
1	9	Luis F. Ortiz	1	8	Cory Abbott
2	1	J.T. Realmuto	1	9	Ildemaro Vargas
3	1	Jake Cave	2	1	Riley Adams
3	9	Alec Bohm	3	1	Dominic Smith
4	1	Bryson Stott	4	1	Michael Chavis
5	1	Alec Bohm	4	8	Jake Alu
5	9	Rodolfo Castro	4	9	Jeter Downs
6	1	Trea Turner	5	1	Ildemaro Vargas
6	9	Edmundo Sosa	5	9	Jake Alu
7	1	Kyle Schwarber	6	1	CJ Abrams
7	9	Weston Wilson	7	1	Stone Garrett
8	1	Johan Rojas	8	1	Alex Call
9	1	Nick Castellanos	9	1	Lane Thomas
9	9	Jake Cave			

74.	game	PHI vs WAS on 20230819
	state before	(2, 0, 0, 0, 2, 0, 0, 0)
	error	play,2,0,realj001,02,SCX,E5/G5.B-1
	state after	(2, 1, 0, 0, 2, 0, 0, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2398 This game had 3 errors. The final score was 12-3. *In the top of the 2nd inning, with bases empty, PHI batter J.T. Realmuto hit a ground ball to 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

	game	PHI vs WAS on 20230819
	state before	(1, 0, 0, 0, 6, 0, 2, 1)
75.	error	play,6,1,menej001,32,TF.BBFBFX,E6/TH/G6.B-1
	state after	(1, 1, 0, 0, 6, 0, 2, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2399 *In the bottom of the 6th inning, with bases empty, WAS batter Joey Meneses hit a ground ball to short. A throwing error by the shortstop allowed the batter to reach 1st base.*

	game	PHI vs WAS on 20230819
	state before	(1, 0, 0, 0, 7, 3, 3, 1)
76.	error	play,7,1,calla002,00,X,S1/BG25-.B-3(E1/TH)
	state after	(1, 0, 0, 1, 7, 3, 3, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2400 *In the bottom of the 7th inning, with bases empty, WAS batter Alex Call hit a bunt single to shallow infield. A throwing error by the pitcher allowed the batter to reach 3rd base.*

	game	MIA vs WAS on 20230831
	state before	(2, 0, 0, 0, 9, 5, 1, 0)
77.	error	play,9,0,chisj001,02,SCX,E6/G6.B-1
	state after	(2, 1, 0, 0, 9, 5, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2401 *In the top of the 9th inning, with bases empty, MIA batter Jazz Chisholm hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

33.2.7 September

	game	MIA vs WAS on 20230902
	state before	(1, 0, 0, 0, 3, 3, 1, 0)
78.	error	play,3,0,chisj001,10,BX,E6/P5D.B-1
	state after	(1, 1, 0, 0, 3, 3, 1, 0)
	fielder	6 (shortstop on the home team)

Summary 2402 This game had 2 errors. The final score was 11-5.
In the top of the 3rd inning, with bases empty, MIA batter Jazz Chisholm hit a pop up to deep 3rd. An error by the shortstop allowed the batter to reach 1st base.

	game	MIA vs WAS on 20230902
	state before	(0, 1, 0, 0, 7, 9, 3, 0)
79.	error	play,7,0,sancj003,02,SSX,S9/L9S.1-3(E9);B-2
	state after	(0, 0, 1, 1, 7, 9, 3, 0)
	fielder	9 (right field on the home team)

Summary 2403 *In the top of the 7th inning, with a runner on 1st, MIA batter Jesus Sanchez hit a line drive single to shallow right field. An error by the right fielder allowed the runner on 1st to advance to 3rd and the batter to reach 2nd base.*

Visiting/Home team rosters for MIA vs WAS on 2023-09-03

position	innings	player	position	innings	player
1	1	Sandy Alcantara	1	1	Josiah Gray
1	9	Tanner Scott	1	5	Robert Garcia
2	1	Jacob Stallings	1	6	Jordan Weems
3	1	Josh Bell	1	7	Jose A. Ferrer
3	9	Yuli Gurriel	1	8	Hunter Harvey
4	1	Luis Arraez	1	9	Kyle Finnegan
5	1	Jon Berti	2	1	Drew Millas
6	1	Joey Wendle	3	1	Dominic Smith
7	1	Bryan De La Cruz	4	1	Jake Alu
8	1	Jazz Chisholm Jr.	5	1	Ildemaro Vargas
9	1	Jesus Sanchez	6	1	CJ Abrams
			7	1	Travis Blankenhorn
			8	1	Jacob Young
			9	1	Lane Thomas

	game	MIA vs WAS on 20230903
	state before	(0, 0, 0, 0, 5, 3, 1, 1)
80.	error	play,5,1,vargi001,21,CBBX,E1/TH/G13.B-1
	state after	(0, 1, 0, 0, 5, 3, 1, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2404 This game had 4 errors. The final score was 6-4.
In the bottom of the 5th inning, with bases empty, WAS batter Ildemaro Vargas hit a ground ball to shallow infield. A throwing error by the pitcher allowed the batter to reach 1st base.

	game	MIA vs WAS on 20230903
	state before	(0, 1, 0, 1, 6, 3, 4, 0)
81.	error	play,6,0,wendj002,21,C1BBX,E5/G56.3-H(NR)(UR);1-3;B-2
	state after	(0, 0, 1, 1, 6, 4, 4, 0)
	fielder	5 (3rd baseman on the home team)

Summary 2405 *In the top of the 6th inning, with runners on 1st and 3rd, MIA batter Joey Wendle hit a ground ball to 3rd base. An error by the third baseman allowed the runner on 3rd to score (unearned, no RBI), the runner on 1st to advance to 3rd, and the batter to reach 2nd.*

	game	MIA vs WAS on 20230903
	state before	(2, 0, 1, 0, 6, 4, 4, 1)
82.	error	play,6,1,smitd008,01,C2,P02(E1/TH).2-3
	state after	(2, 0, 0, 1, 6, 4, 4, 1)
	fielder	1 (pitcher on the visiting team)

Summary 2406 *In the bottom of the 6th inning, with a runner on 2nd, WAS batter Dominic Smith has a 0-1 count. The pitcher attempted to pick off the runner on 2nd. A throwing error by the pitcher allowed that runner to safely reach 3rd. Smith is still at bat with a 0-1 count.*

	game	MIA vs WAS on 20230903
	state before	(1, 1, 1, 0, 9, 4, 4, 0)
83.	error	play,9,0,chisj001,22,CBTDX,36(1)/FO/G3.2-H(E6/TH)(NR);B-1
	state after	(2, 1, 0, 0, 9, 5, 4, 0)
	fielder	6 (shortstop on the home team)

Summary 2407 *In the top of the 9th inning, with runners on 1st and 2nd, MIA batter Jazz Chisholm hit a ground ball to 1st. The runner from 1st was forced out by the shortstop (assist from the first baseman). A throwing error by the shortstop allowed the runner on 2nd to score (unearned, no RBI) and the batter to reach 1st.*

	game	NYN vs WAS on 20230905
	state before	(0, 1, 1, 1, 8, 10, 3, 1)
84.	error	play,8,1,adamr004,00,.X,9/SF/F9S.3-H;2-3;1-2(E2)
	state after	(1, 0, 1, 1, 8, 10, 4, 1)
	fielder	2 (catcher on the visiting team)

Summary 2408 *In the bottom of the 8th inning, with bases loaded, WAS batter Riley Adams hit a sacrifice fly ball to shallow right field. The right fielder put out the batter. The runner from 3rd scored (1 RBI) and the runner from 2nd advanced to 3rd. An error by the catcher allowed the runner on 1st to reach 2nd.*

	game	NYN vs WAS on 20230906
	state before	(0, 0, 0, 0, 6, 2, 0, 0)
85.	error	play,6,0,maurr002,01,.CX,S8/L78.B-2(E8)
	state after	(0, 0, 1, 0, 6, 2, 0, 0)
	fielder	8 (center field on the home team)

Summary 2409 *In the top of the 6th inning, with bases empty, NYN batter Ronny Mauricio hit a line drive single to left center field. An error by the center fielder allowed the batter to reach 2nd.*

Visiting/Home team rosters for LAN vs WAS on 2023-09-09

position	innings	player	position	innings	player
1	1	Bobby Miller	1	1	Jake Irvin
1	10	Ryan Brasier	1	10	Robert Garcia
1	11	Gus Varland	1	11	Andres Machado
1	8	Shelby Miller	1	7	Hunter Harvey
1	9	Brusdar Graterol	1	7	Jordan Weems
2	1	Will Smith	1	9	Kyle Finnegan
3	1	Freddie Freeman	2	1	Keibert Ruiz
4	1	Kolten Wong	3	1	Dominic Smith
4	11	Enrique Hernandez	3	11	Michael Chavis
5	1	Max Muncy	4	1	Luis Garcia
6	1	Enrique Hernandez	4	10	Michael Chavis
6	11	Miguel Rojas	4	11	Ildemaro Vargas
7	1	David Peralta	5	1	Ildemaro Vargas
7	10	Chris Taylor	5	11	Carter Kieboom
8	1	James Outman	6	1	CJ Abrams
9	1	Jason Heyward	7	1	Jake Alu
9	10	David Peralta	8	1	Jacob Young
			9	1	Travis Blankenhorn
			9	8	Alex Call

	game	LAN vs WAS on 20230909
86.	state before	(0, 1, 0, 0, 5, 1, 1, 1)
	error	play,5,1,garc1006,00,>B,SB2.1-3(E2/TH)
	state after	(0, 0, 0, 1, 5, 1, 1, 1)
	fielder	2 (catcher on the visiting team)

Summary 2410 This game had 3 errors. The final score was 6-7.

In the bottom of the 5th inning, with a runner on 1st, WAS batter Luis Garcia had a 0-0 count. During the next pitch, which was a ball, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Garcia is still at bat, now with a 0-1 count.

game	LAN vs WAS on 20230909
state before	(1, 1, 0, 0, 9, 4, 5, 0)
error	play,9,0,wongk001,00,>C,SB2.1-3(E2/TH)
state after	(1, 0, 0, 1, 9, 4, 5, 0)
fielder	2 (catcher on the home team)

Summary 2411 *In the top of the 9th inning, with a runner on 1st, LAN batter Kolten Wong had a 0-0 count. During the next pitch, which was a called strike, the runner on 1st stole 2nd. A throwing error by the catcher allowed that runner to advance to 3rd. Wong is still at bat, now with a 0-1 count.*

game	LAN vs WAS on 20230909
state before	(1, 0, 0, 0, 9, 5, 5, 1)
error	play,9,1,garc1006,00,X,E5/G56S.B-1
state after	(1, 1, 0, 0, 9, 5, 5, 1)
fielder	5 (3rd baseman on the visiting team)

Summary 2412 *In the bottom of the 9th inning, with bases empty, WAS batter Luis Garcia hit a ground ball to shallow 3rd base. An error by the third baseman allowed the batter to reach 1st base.*

game	CHA vs WAS on 20230918
state before	(0, 0, 0, 0, 6, 6, 0, 1)
87. error	play,6,1,abrac001,21,CBBX,E6/G6.B-1
state after	(0, 1, 0, 0, 6, 6, 0, 1)
fielder	6 (shortstop on the visiting team)

Summary 2413 This game had 2 errors. The final score was 6-1. *In the bottom of the 6th inning, with bases empty, WAS batter C. J. Abrams hit a ground ball to short. An error by the shortstop allowed the batter to reach 1st base.*

	game	CHA vs WAS on 20230918
	state before	(1, 1, 0, 0, 7, 6, 0, 0)
88.	error	play,7,0,vauga001,01,CS,WP.1-3(E2/TH)
	state after	(1, 0, 0, 1, 7, 6, 0, 0)
	fielder	2 (catcher on the home team)

Summary 2414 *In the top of the 7th inning, with a runner on 1st, CHA batter Andrew Vaughn had a 0-1 count. During the next pitch, which was a wild pitch, the runner on 1st advanced to 2nd. A throwing error by the catcher allowed that runner to reach 3rd. Vaughn is still at bat, now with a 1-1 count.*

Visiting/Home team rosters for CHA vs WAS on 2023-09-20

position	innings	player	position	innings	player
1	1	Michael Kopech	1	1	Josiah Gray
1	2	Jesse Scholtens	1	6	Jose A. Ferrer
1	2	Tanner Banks	1	7	Andres Machado
1	4	Deivi Garcia	1	8	Thaddeus Ward
1	6	Luis Patino	2	1	Keibert Ruiz
1	8	Lane Ramsey	3	1	Dominic Smith
2	1	Carlos Perez	4	1	Luis Garcia
3	1	Andrew Vaughn	5	1	Ildemaro Vargas
4	1	Lenyn Sosa	5	7	Carter Kieboom
5	1	Yoan Moncada	6	1	CJ Abrams
6	1	Tim Anderson	6	7	Ildemaro Vargas
6	6	Zach Remillard	7	1	Jake Alu
7	1	Andrew Benintendi	8	1	Jacob Young
8	1	Luis Robert Jr.	9	1	Lane Thomas
8	4	Trayce Thompson	9	7	Alex Call
9	1	Gavin Sheets			

89.	game	CHA vs WAS on 20230920
	state before	(1, 0, 1, 0, 1, 0, 0, 1)
	error	play,1,1,menej001,11,>B.SX,FC5/G56.3XH(52E6)(NR)(UR);B-2
	state after	(1, 0, 1, 0, 1, 0, 1, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2415 This game had 3 errors. The final score was 3-13.
In the bottom of the 1st inning, with a runner on 2nd, WAS batter Joey Meneses hit a ground ball into a fielder's choice at 3rd. An error by the shortstop allowed the runner on 3rd to safely score (unearned, no RBI) and the batter to reach 2nd.

90.	game	CHA vs WAS on 20230920
	state before	(2, 1, 0, 0, 2, 1, 1, 1)
	error	play,2,1,younj004,21,BBFX,FC7/G56.1-3(E5);B-1
	state after	(2, 1, 0, 1, 2, 1, 1, 1)
	fielder	5 (3rd baseman on the visiting team)

Summary 2416 *In the bottom of the 2nd inning, with a runner on 1st, WAS batter Jacob Young hit a ground ball into a fielder's choice at 3rd. An error by the third baseman allowed the runner on 1st to advance to 3rd and the batter to reach 1st base.*

91.	game	CHA vs WAS on 20230920
	state before	(1, 1, 1, 0, 8, 4, 11, 0)
	error	play,8,0,moncy001,21,BFBX,FC6/G6.2-3;1X2(6E4);B-1
	state after	(1, 1, 1, 1, 8, 4, 11, 0)
	fielder	4 (2nd baseman on the home team)

Summary 2417 *In the top of the 8th inning, with runners on 1st and 2nd, CHA batter Yoan Moncada hit a ground ball into a fielder's choice at short. The runner on 2nd advanced to 3rd. And error by the sennd baseman allowed the runner from 1st to safely reach 2nd and the batter to reach 1st. The bases are loaded.*

	game	ATL vs WAS on 20230922
	state before	(0, 0, 0, 0, 6, 6, 3, 0)
92.	error	play,6,0,arcio002,12,.CFBFX,S6/G56.B-2(E6/TH)
	state after	(0, 0, 1, 0, 6, 6, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 2418 *In the top of the 6th inning, with bases empty, ATL batter Orlando Arcia hit a ground ball single to short. A throwing error by the shortstop allowed the batter to reach 2nd.*

	game	ATL vs WAS on 20230924 (1/2)
	state before	(0, 1, 0, 0, 4, 0, 3, 0)
93.	error	play,4,0,olsom001,12,CBFX,D98/L89XD+.1-H;B-3(E6/TH)
	state after	(0, 0, 0, 1, 4, 1, 3, 0)
	fielder	6 (shortstop on the home team)

Summary 2419 This game (game 1 of a doubleheader) had 2 errors. The final score was 8-5. (For the sake of completeness, the final score of game 2 was 2-3.)

In the top of the 4th inning, with a runner on 1st, ATL batter Matt Olson hit a hard line drive double to extra deep right center field. The runner on 1st scored (1 RBI). A throwing error by the shortstop allowed the batter to reach 3rd.

	game	ATL vs WAS on 20230924 (1/2)
	state before	(1, 0, 0, 0, 8, 8, 4, 1)
94.	error	play,8,1,smitd008,22,CSBBX,E6/G4.B-1
	state after	(1, 1, 0, 0, 8, 8, 4, 1)
	fielder	6 (shortstop on the visiting team)

Summary 2420 *In the bottom of the 8th inning, with bases empty, WAS batter Dominic Smith hit a ground ball to 2nd base. An error by the shortstop allowed the batter to reach 1st base.*

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